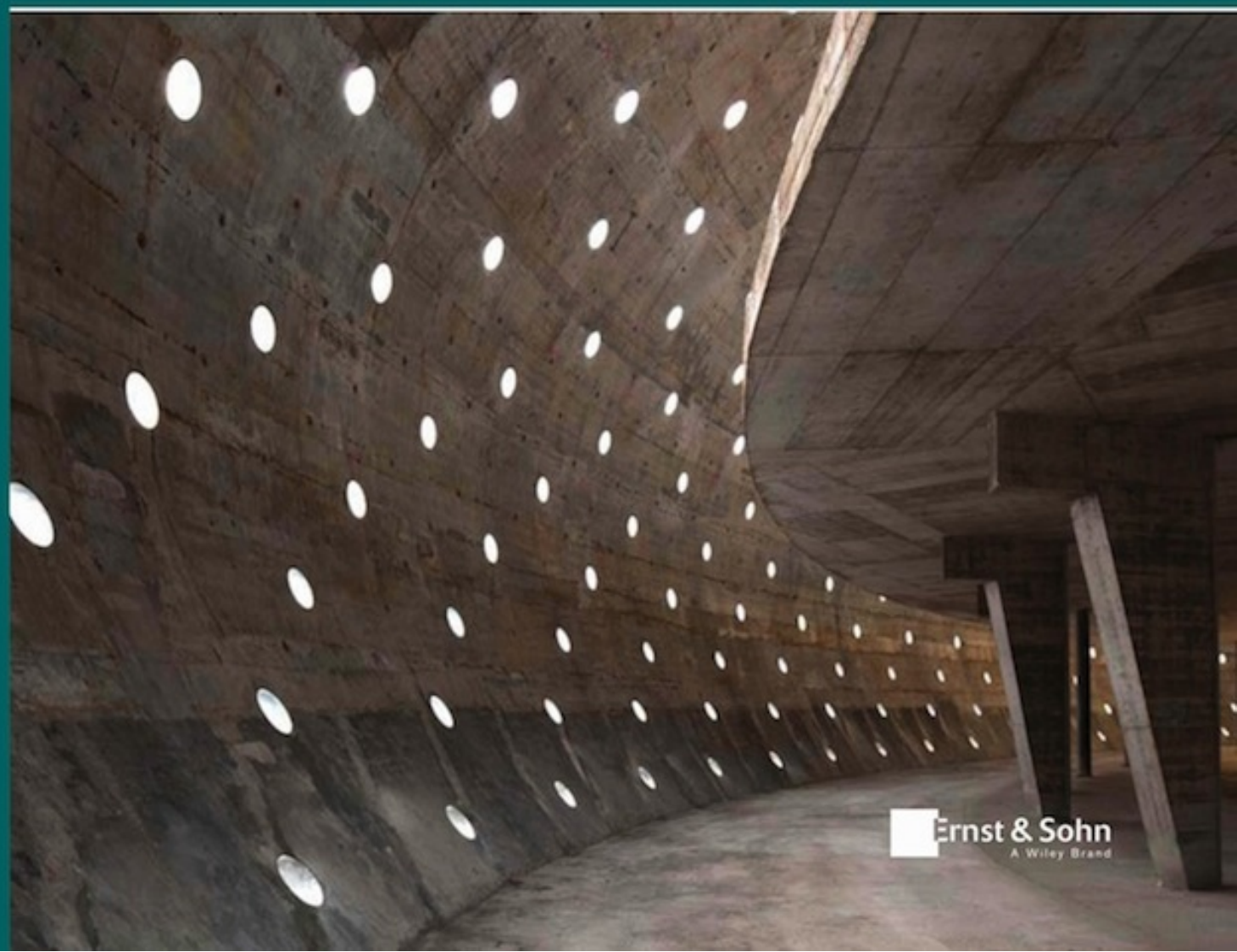


fib Model Code for Concrete Structures 2010



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Notations

Meaning of roman capital letters

A	area
C	torsional moment of inertia; serviceability constraints
D	fatigue damage factor; diffusion coefficient
E	modulus of elasticity; earthquake action; load (action) effect
F	action in general; local loading
G	permanent action; shear modulus
H	horizontal component of a force
I	second moment of a plane area
J	creep function
K	(permeability) coefficient
M	bending moment; coefficient of water absorption; safety margin
N	axial force
P	force
Q	variable action
R	resistance; strength (resisting load effect); reaction at a support; resultant
S	static moment of a plane area
T	torsional moment; temperature
V	shear force, volume
W	modulus of inertia
X	material or soil properties in general; reaction or force in general, parallel to x axis
Y	reaction or force in general, parallel to y axis
Z	reaction or force in general, parallel to z axis
NOTE: Roman capital letters can be used to denote types of material, e.g. C for concrete, LC for lightweight concrete, S for steel, Z for cement.	

Meaning of roman lower case letters

a	deflection; distance; acceleration
b	width
c	concrete cover
d	effective height; diameter (see also h)
e	eccentricity; sets of loads (actions)
f	strength
g	distributed permanent load; acceleration due to gravity; limit state function
h	total height or diameter of a section; thickness
i	radius of gyration
j	number of days
k	all coefficients with dimension
l	span; length of an element
m	bending moment per unit length or width; mass; average value of a sample
n	normal (longitudinal, axial) force per unit length or width
p	prestressing
q	distributed variable load
r	radius; resistance variables; resistance function
s	spacing; standard deviation of a sample
t	time; torsional moment per unit length or width; thickness of thin elements
u	perimeter

v	velocity; shear force per unit length or width
w	width of a crack
x	coordinate; height of compression zone
y	coordinate; height of rectangular diagram coordinate; lever arm

Use of Greek lower case letters

alpha	α	angle; ratio; coefficient
beta	β	angle; ratio; coefficient
gamma	γ	safety factor; density; shear strain (angular strain)
delta	δ	coefficient
epsilon	ϵ	strain
zeta	ζ	coefficient
eta	η	coefficient
theta	θ	rotation
lambda	λ	slenderness ratio; coefficient
mu	μ	relative bending moment; coefficient of friction; mean value of a whole population
nu	ν	relative axial force; Poisson's ratio
xi	ξ	coefficient; ratio
pi	π	mathematical use only
rho	ρ	geometrical ratio of reinforcement; bulk density
sigma	σ	axial stress; standard deviation of a whole population
tau	τ	shear stress
phi	φ	coefficient
chi	χ	coefficient
psi	ψ	coefficient; ratio
omega	ω	mechanical ratio of reinforcement

Mathematical symbols and special symbols

S	sum
Δ	difference; increment (enlargement)
$\emptyset$	nominal diameter of a reinforcing bar or of a cable
'	(single prime) compression (only in a geometrical or locational sense)
e	base of Napierian logarithms
exp	power of the number e
π	ratio of the circumference of a circle to its diameter
n	number of ...
w/c	water/cement ratio
<	smaller than
>	greater than

General subscripts

a	support settlement; additional; accidental load
b	bond; bar; beam
c	concrete; compression; column
d	design value
e	elastic limit of a material
f	forces and other actions; beam flange; bending; friction
g	permanent load
h	horizontal; hook
i	initial
j	number of days

k	characteristic value
l	longitudinal
m	mean value; material; bending moment
n	axial force
o	zero
p	prestressing steel
q	variable load
r	cracking
s	ordinary steel; snow; slab
t	tension;* torsion;* transverse
u	ultimate (limit state)
v	shear; vertical
w	wind; web; wire; wall
x	linear coordinate
y	linear coordinate; yield
z	linear coordinate
1, 2, 3	particular values of quantities
cc	conventional asymptotic value
*	When confusion is possible between tension and torsion, the subscripts tn (tension) and tr (torsion) should be used.

Subscripts for actions and action effects

$a(A)$	support settlement; accidental action
cc	creep of concrete
cs	shrinkage of concrete
ep	earth pressure
ex	explosion; blast
$g(G)$	permanent load
im	impact
lp	liquid pressure
$m(M)$	bending moment
$n(N)$	axial force
$p(P)$	prestress
$q(Q)$	variable load
$s(S)$	snow load
$t(T)$	torsion; temperature
$v(V)$	shear
$w(W)$	wind load

Subscripts obtained by abbreviation

abs	absolute
act	acting
adm	admissible, permissible
cal	calculated, design
$crit$ (or cr)	critical
ef	effective
el (or e)	elastic
est	estimated
exc	exceptional
ext	external
fat	fatigue
inf	inferior
int	internal
lat	lateral
lim	limit
max	maximum
min	minimum
nec	necessary
net	net

nom	nominal
obs	observed
pl	plastic
$prov$ (or pr)	provisional (stage of construction); provided
red	reduced
rel	relative; relaxation
rep	representative
req	required
res	resisting, resistant
ser	serviceability
tot	total
var	variable

Notation list

Roman lower case letters

$1/r$	curvature of a section of an element
$1/r_{(g)}$	curvature due to g
$1/r_{(g+q)}$	curvature due to g and q
$1/r_{0(g+q)}$	instantaneous (elastic) curvature due to g and q
$1/r_1$	curvature of an uncracked concrete section (state I)
$1/r_{1r}$	curvature in state I under cracking moment
$1/r_2$	curvature of a cracked concrete section (state II)
$1/r_{2r}$	curvature in state II under cracking moment
$1/r_{ts}$	tension stiffening correction for curvature
a	geometrical quantity in general; deformation; deflection
a_d	design values of geometrical quantity
a_0	elastic deflection (calculated with rigidity $E_c I_e$)
b	breadth of compression zone or flange, width of concrete section
b_f	width of FRP section; width of flange
b_{red}	reduced width of web
b_x	smaller side dimension of a rectangular section
b_y	greater side dimension of a rectangular section
b_w	width of web
c	concentration of a substance in a volume element; concrete cover; coefficient for shear resistance due to adhesive bond
c_r	coefficient for shear resistance due to aggregate interlock
c_1	column dimension parallel to the eccentricity of the load
c_2	column dimension perpendicular to the eccentricity of the load
c_{min}	minimum concrete cover
c_{nom}	nominal value of concrete cover ($= c_{min} + \text{tolerance}$)
d	effective depth to main tension reinforcement
d'	effective depth to compression reinforcement
d_{max}	maximum aggregate size
e	load eccentricity
e_0	first order eccentricity ($= M_{Ed}/N_{Ed}$)
e_{01}	smaller value of the first order eccentricity at one end of the considered element
e_{02}	greater value of the first order eccentricity at one end of the considered element
e_{tot}	total eccentricity
f	strength
f_{bd}	design bond strength
$f_{bd,0}$	basic design bond strength
f_{bpd}	design bond strength for prestressing tendon

f_c	cylinder compressive strength of concrete	f_{Ftu}	ultimate residual strength (post-cracking strength for ultimate crack opening) for fibre-reinforced concrete
f_{lc}	cylinder compressive strength of lightweight aggregate concrete	f_{Ftud}	design value of post-cracking strength for ultimate crack opening for fibre-reinforced concrete
f_c^*	cylinder compressive strength of concrete under triaxial loading (confined strength), reduced concrete strength due to transverse tension	f_k	characteristic value of material or product property; characteristic value of strength
f_{cc}	cylinder compressive strength of concrete under uniaxial stress	f_L	Limit of Proportionality
f_{cd}^*	design compressive strength of concrete under triaxial loading (confined strength), reduced design concrete strength due to transverse tension	f_{Lk}	characteristic value of Limit of Proportionality
f_{cd}	design value of f_c	f_{lck}	characteristic value of compressive strength of lightweight aggregate concrete
$f_{cd,fat}$	design fatigue reference strength of concrete under compression	f_{lcm}	mean value of compressive strength of lightweight aggregate concrete
$f_{c, imp, k}$	characteristic compressive strength under high rates of loading	$f_{lctk, max}$	upper bound value of the characteristic tensile strength of lightweight aggregate concrete
f_{ck}	characteristic value of compressive strength of concrete	$f_{lctk, min}$	lower bound value of the characteristic tensile strength of lightweight aggregate concrete
$f_{ck,c}$	value of f_{ck} of confined concrete	f_{lctm}	mean value of axial tensile strength of lightweight aggregate concrete
$f_{ck,cube}$	characteristic value of cube compressive strength of concrete	$f_{p0.1}$	0.1% proof strength of prestressing steel
$f_{ck,fat}$	characteristic value of fatigue reference compressive strength	$f_{p0.2}$	0.2% proof strength of prestressing steel
$f_{ck,ft}$	characteristic value of concrete compressive strength after freeze-thaw attack	$f_{p0.1k}$	characteristic 0.1% proof strength of prestressing steel
f_{cm}	mean value of compressive strength of concrete	$f_{p0.2k}$	characteristic 0.2% proof strength of prestressing steel
$f_{cm,sus}(t,t_0)$	mean value of compressive strength of concrete at time t when subjected to a high sustained compressive stress at an age at loading t_0	f_{pt}	tensile strength of prestressing steel; UTS (Ultimate Tensile Strength) of prestressing steel
f_{ct}	axial tensile strength of concrete	f_{ptd}	design tensile strength of prestressing steel
f_{ctd}	design value of f_{ct}	f_{ptk}	characteristic value of tensile strength of prestressing steel; characteristic value of UTS (Ultimate Tensile Strength) of prestressing steel
$f_{ct, imp, k}$	characteristic tensile strength under impact loading	f_{py}	tensile yield stress of prestressing steel
f_{ctk}	characteristic value of f_{ct}	f_{pyd}	design value of tensile yield stress of prestressing steel
$f_{ctk, is}$	characteristic measured in-situ tensile strength	f_{pyk}	characteristic value of tensile yield stress of prestressing steel
$f_{ctk, max}$	upper bound value of the characteristic tensile strength of concrete	f_r	relative (or projected) rib area
$f_{ctk, min}$	lower bound value of the characteristic tensile strength of concrete	$f_{R,j}$	residual flexural tensile strength of fibre reinforced concrete corresponding to Crack Mouth Opening Displacement ($CMOD$) = $CMOD_j$
$f_{ctk, sus}$	characteristic tensile strength of concrete under sustained loading	f_{R1k}	characteristic residual strength of fibre reinforced concrete significant for serviceability conditions
f_{ctm}	mean value of axial tensile strength of concrete	f_{R3k}	characteristic residual strength of fibre reinforced concrete significant for ultimate conditions
$f_{ct,fl}$	flexural tensile strength (at $T = 20^\circ\text{C}$)	$f_{sp,\theta}$	proportional limit of reinforcing steel at temperature θ
$f_{ctm,fl}$	mean flexural tensile strength (at $T = 20^\circ\text{C}$)	$f_{sy,\theta}$	maximum stress of reinforcing steel at temperature θ
$f_{ct,sp}$	splitting tensile strength of concrete	$f_{0.2}$	0.2% proof strength of reinforcing steel
$f_{ctm,sp}$	mean splitting tensile strength of concrete	$f_{0.2k}$	characteristic value of 0.2% proof strength of reinforcing steel
f_d	design value of material or product property; design value of strength	f_t	tensile strength of reinforcing steel
f_f	tensile strength of non-metallic reinforcement	f_{tk}	characteristic value of tensile strength of reinforcing steel
f_{fad}	design anchorage bond strength for non-metallic reinforcement	f_{tm}	mean value of tensile strength of reinforcing steel
f_{fbd}	design value of tensile stress in non-metallic reinforcement limited by bond to concrete	f_y	yield strength of reinforcing steel in tension
f_{fbm}	mean value of tensile stress in the non-metallic reinforcement limited by bond to concrete	$f_{y,act}$	actual yield strength of reinforcing steel in compression
f_{fd}	design tensile strength of non-metallic reinforcement	f_{yc}	yield strength of reinforcing steel in compression
f_{fk}	characteristic value of tensile strength of non-metallic reinforcement	f_{ycd}	design yield strength of reinforcing steel in compression
f_{Fts}	serviceability residual strength (post-cracking strength for serviceability crack opening) for fibre-reinforced concrete	f_{yd}	design yield strength of reinforcing steel in tension
f_{Ftsd}	design value of post-cracking strength for serviceability crack opening for fibre-reinforced concrete	f_{yk}	characteristic value of yield strength of reinforcing steel in tension
		f_{ym}	mean value of yield strength of reinforcing steel in tension
		g_d	design value of distributed permanent load
		h	overall depth of member, total height; notional size of a member ($2 A_c/u$; u : perimeter in contact with the atmosphere)
		h_b	depth of beam

h_f	depth of flange	r	radius
h_{key}	height of shear key in joint interface	s	slip (relative displacement between steel and concrete cross-sections), shear slip (at interfaces); spacing of bars; coefficient which depends on the strength class of cement
h_{sp}	distance between the notch tip and the top of the specimen	s_m	slip at maximum bond stress
Δh_w	height of water column	$s_{n,t}$	slip due to permanent or repeated loading
i	radius of gyration	s_{max}	maximum bar spacing
k	plasticity number; unintentional angular displacement	s_r	distance between cracks; radial spacing of layers of shear reinforcement
k_a	effectiveness coefficient of anchorage system	$s_{r,m}$	mean spacing between cracks
k_b	shape factor	s_t	longitudinal spacing of confining reinforcement
k_{bl}	bond length calibration factor	s_u	ultimate slip
k_c	coefficient	t	time, age, duration; thickness of thin elements
k_d	effectiveness factor dependent on the reinforcement detail	t_0	age at first loading
k_l	stress–strength ratio	t_1	age of the concrete when its temperature returns to ambient temperature
k_m	coefficient of confinement from transverse reinforcement	t_f	thickness of non-metallic reinforcement
k_n	displacement factor for repeated constant amplitude loading	t_{eq}	equivalent time interval for calculation of relaxation losses
k_t	displacement factor for permanent load	t_{p1}	mean duration of a heating cycle
l	design span, effective span, length of an element, thickness of a penetrated section	t_R	reference period
Δl	change in distance between two measuring points	t_s	concrete age at the beginning of shrinkage or swelling
l_0	design lap length, effective length (of columns); distance between measuring points	t_T	temperature adjusted concrete age
l_b	design anchorage length; design lap length	u	length of a perimeter; component of displacement of a point
l_{bp}	basic anchorage length of bonded pretensioned reinforcement	u_0	length of the periphery of the column or distribution area of load
l_{bpd}	design anchorage length of bonded pretensioned reinforcement	u_1	length of the control perimeter for punching
l_{bpt}	transmission length of bonded pretensioned reinforcement	u_{ef}	length of the perimeter of A_{ef}
$l_{b,min}$	minimum anchorage length; minimum lap length	u_n	length of the control perimeter for punching outside a slab zone with shear reinforcement
l_{cs}	characteristic length (fracture parameter)	v	shear force per unit width (out-of-plane loading), component of displacement of a point
l_p	development length for bonded prestressing reinforcement	w	crack width; component of displacement of a point
Δl_{pl}	residual elongation after unloading	w_c	crack width for $\sigma_{ct} = 0$
$l_{p,max}$	length over which the slip between prestressing steel and concrete occurs	w_k	calculated characteristic crack width
$l_{s,max}$	length over which the slip between steel and concrete occurs	w_{lim}	nominal limit value of crack width
l_t	transmission length	w_u	maximum crack opening accepted in structural design: its value depends on the ductility required
m	moment per unit width (out-of-plane loading); mass of substance flowing; degree of hydration; moisture content	x	depth of compression zone; distance; parameter
n	number of bars, number of load cycles; force per unit width (in-plane loading)	$x_c(t)$	carbonation depth at the time t
n_b	number of anchored bars or pairs of lapped bars in the potential splitting surface; number of bars in the bundle	x_d	design value of parameter x
n_{Ri}	number of cycles leading to failure at stress levels $S_{i,min}$ and $S_{i,max}$, respectively	z	internal lever arm
n_{Si}	number of cycles applied at constant minimum and maximum stress levels $S_{i,min}$ and $S_{i,max}$, respectively		
n_t	number of legs of confining reinforcement crossing a potential splitting failure surface at a section		
p	local gas pressure; overall steel ductility parameter		
p_m	mean pressure		
p_{tr}	transverse pressure perpendicular to the bar axis; mean compressive stress perpendicular to the potential splitting failure surface at the ultimate limit state		
q_d	design value of distributed variable load		

Greek lower case letters

α	coefficient; reduction factor; inclination of reinforcement crossing an interface; sum of the angular displacements
α_e	modular ratio ($= E_s/E_c$)
$\alpha_{e,p}$	modular ratio ($= E_p/E_c$)
$\alpha_{e,sec}$	secant modular ratio ($= E_{s,sec}/E_{c,sec}$)
α_{fl}	conversion factor ($= f_{ctm}/f_{ctm,fl}$)
α_i	unintended inclination of compressive members
α_{im}	unintended inclination of group of vertical prestressing members
α_p	coefficient of thermal expansion of prestressing reinforcement
α_{spl}	conversion factor ($= f_{ctm}/f_{ctm,spl}$)
α_{sT}	coefficient of thermal expansion for steel
α_T	coefficient of thermal expansion in general

α_1	coefficient representing the influence of reinforcement provided	$\varepsilon_{cs}(t)$	shrinkage or swelling strain at concrete age t
α_2	coefficient representing the influence of passive confinement from cover	$\varepsilon_{c\zeta}(t)$	stress dependent strain at concrete age t
α_3	coefficient representing the influence of passive confinement from transverse reinforcement	ε_{ct}	concrete tensile strain
β	coefficient characterizing the bond quality of reinforcing bars, coefficient for the compressive strength of a strut across an interface	$\varepsilon_{cT}(t)$	thermal strain at a concrete age t
β_c	coefficient for the compressive strength of a strut across an interface	ε_{clim}	ultimate strain of concrete in compression
$\beta_{bc}(t, t_0)$	coefficient to describe the development of basic creep with time after loading	ε_{pd0}	strain of prestressed reinforcement corresponding to P_{d0}
$\beta_{dc}(t, t_0)$	coefficient to describe the development of drying creep with time after loading	ε_f	strain of non-metallic reinforcement
$\beta_{cc}(t)$	coefficient to describe the development of strength of concrete with time	ε_{fu}	strain of non-metallic reinforcement at maximum force in tension
$\beta_{c,sus}(t, t_0)$	coefficient to describe the decrease of strength with time under sustained load	ε_{fuk}	characteristic value of strain of non-metallic reinforcement at maximum force in tension
$\beta_E(t)$	coefficient to describe the development of modulus of elasticity of concrete with time	ε_{lc1}	lightweight aggregate concrete strain at maximum compressive stress
$\beta_{lcc}(t)$	coefficient to describe the development of strength of lightweight aggregate concrete with time	ε_{lclim}	ultimate strain of lightweight aggregate concrete in compression
γ	safety factor	ε_{pu}	strain of prestressing steel at maximum force
γ_c	partial safety factor for concrete material properties	ε_{puk}	characteristic value of strain of prestressing steel at maximum force
γ_{cb}	partial safety factor for bond	ε_r	strain at the onset of cracking
$\gamma_{c,fat}$	partial safety factor for concrete material properties under fatigue loading	ε_s	steel strain
γ_d	partial safety factor for partial factors for model uncertainties	ε_{s1}	steel strain in uncracked concrete
γ_f	partial safety factor for the tensile strength of non-metallic reinforcement	ε_{s2}	steel strain in crack
γ_F	partial safety factor for actions; partial safety factor for fibre reinforced concrete	ε_{sm}	mean steel strain
γ_G	partial safety factor for permanent actions	$\Delta\varepsilon_{sr}$	increase of steel strain due to crack formation in the section
γ_m	partial safety factor for material properties	ε_{sr1}	steel strain at the point of zero slip under cracking forces
γ_M	partial safety factor for material properties, model uncertainties and geometrical uncertainties	ε_{sr2}	steel strain in the crack under cracking forces (σ_{ct} reaching f_{ctm})
γ_Q	partial safety factor for variable actions	ε_{sT}	thermal strain of steel
γ_{Rd}	partial safety factor associated with the uncertainty of the model and geometrical uncertainties	ε_{su}	strain of reinforcing steel at maximum load
γ_s	partial safety factor for the material properties of reinforcing and prestressing steel	$\Delta\varepsilon_{ts}$	increase of strain by the effect of tension stiffening
$\gamma_{s,fat}$	partial safety factor for the material properties of reinforcing and prestressing steel under fatigue loading	ε_u	limit strain value; strain of reinforcing steel at maximum force
γ_{sd}	partial safety factor accounting for model uncertainty	ε_{uk}	characteristic value of reinforcing steel strain at maximum force
δ	shear displacement	ε_{yd}	design yield strain of reinforcing steel ($= f_{yd}/E_s$)
δ_{jj}	node displacement	ε_v	transverse contraction
ε	strain	ζ	ratio of bond strength of prestressing steel and high-bond reinforcing steel
ε_c	concrete compressive strain	η	viscosity of gas
ε_c^*	concrete compressive strain under triaxial stress	η_1	coefficient representing the type of reinforcing bar being anchored or lapped
ε_{cbs}	concrete basic shrinkage strain	η_2	coefficient representing the position of the bar during casting
ε_{cds}	concrete drying shrinkage strain	η_3	coefficient representing the bar diameter
ε_{cm}	average concrete strain within $l_{s,max}$	η_4	coefficient representing the characteristic strength of steel reinforcement being anchored or lapped
ε_{cl}	concrete strain at maximum compressive stress	η_{p1}	coefficient representing the type of prestressing tendon
$\varepsilon_{cl,imp}$	impact concrete strain at maximum load	η_{p2}	coefficient representing the casting position of the tendon
$\varepsilon_{cc}(t)$	concrete creep strain at concrete age $t > t_0$	θ	angle between web compression and the axis of a member; rotation
$\varepsilon_{ci}(t_0)$	stress dependent initial strain of concrete at the time of first loading	θ_f	angle between inclined compression in a flange and the axis of the member
ε_{cf}	strain at maximum stress due to repeated loads	κ	coefficient
$\varepsilon_{cn}(t)$	stress independent strain at a concrete age t	κ_1	coefficient for axial force in interface connectors
		κ_2	coefficient for dowel action resistance of interface connectors
		λ	slenderness ratio ($= l_0/i$)

μ	coefficient of friction; relative bending moment	χ	aging coefficient in the evaluation of creep structural effects
ν_c	Poisson's ratio of concrete	$\psi(t, t_0)$	relaxation coefficient
ν_s	Poisson's ratio of steel	ω_c	mechanical reinforcement ratio
ξ	creep induced stress redistribution after modification of restraint conditions		
ρ	ratio of (longitudinal) tensile reinforcement (= $A_s/(bd)$); density		
$\rho_{s,ef}$	effective reinforcement ratio (= $A_s/A_{c,ef}$)		
ρ_t	relaxation after t hours	A	total area of a section or part of a section (enclosed within the outer circumference)
$\rho_t(T)$	relaxation after t hours at temperature T	A_I	section area in state I (taking into account the reinforcement)
ρ_{100}	relaxation after 100 hours	A_b	area of single bar
ρ_{1000}	relaxation after 1000 hours	A_c	area of concrete cross section or concrete compression chord
ρ_w	ratio of web reinforcement (= $A_{sw}/(b_w s_w \sin \alpha)$)	$A_{c,ef}$	effective area of concrete in tension
σ	stress	A_{core}	effectively confined area of cross-section in compression
$\sigma_1, \sigma_2, \sigma_3$	principal stresses	A_d	design value of accidental action
σ_c	concrete compressive stress	A_{Ed}	design value of seismic action
σ_{cd}	design concrete compressive stress	A_{Ek}	representative value of seismic action
σ_{ct}	concrete tensile stress	A_k	area enclosed by the centrelines of a shell resisting torsion
$\sigma_{c,c}$	compressive stress of confined concrete	A_p	area of prestressing steel
$\sigma_{c,max}$	maximum compressive stress	A_s	area of reinforcement
$\sigma_{c,min}$	minimum compressive stress	$A_{s'}$	area of compression reinforcement
$\sigma_{ct,max}$	maximum tensile stress	A_{sh}	area of hoop reinforcement for torsion
σ_f	stress in non-metallic reinforcement	A_{sl}	area of longitudinal reinforcement
σ_n	(lowest) compressive stress resulting from normal force acting at the interface	A_{sp}	cross sectional area of prestressing steel
$\sigma_{p0}(x)$	initial stress in prestressing steel at a distance x from anchorage device	A_{st}	area of transverse reinforcement; cross sectional area of one leg of a confining bar
$\sigma_{p0,max}$	maximum tensile stress in prestressing steel at tensioning	A_{sw}	area of shear reinforcement
σ_{pcs}	stress in prestressing steel after all losses (including creep and shrinkage)	$A_{s,cal}$	calculated area of reinforcement required by design
σ_{pd}	tendon stress under design load	$A_{s,min}$	minimum reinforcement area
$\Delta\sigma$	stress range relevant to fatigue of reinforcement	AF	amplification factor
$\Delta\sigma_{Rsk}(n)$	stress range relevant to n cycles obtained from a characteristic fatigue strength function	C	serviceability constraints
σ_s	steel stress	C_0	initial chloride content of concrete
σ_{sd}	steel stress to be anchored by bond over the distance l_b	C_f	aggregate effectivity factor
σ_{s2}	steel stress in the crack	$C_{S,\Delta x}$	chloride content at a depth of Δx
σ_{se}	steel stress at the point of zero slip	D	fatigue damage; diffusion coefficient; deformation
σ_{sr2}	steel stress in the crack under cracking load (σ_{ct} reaching f_{ctm})	D_{app}	apparent diffusion coefficient of a substance in concrete
$\Delta\sigma_{Es}$	steel stress range under the acting loads	D_{eff}	effective diffusion coefficient of a substance in concrete
τ_0	bond stress according to the bond stress-slip curve	D_{lim}	limiting fatigue damage
τ_a	ultimate shear capacity due to adhesion or interlocking	D_{RCM}	rapid chloride migration coefficient
τ_b	local bond stress	E	modulus of elasticity; load (action) effect; cumulative leaching
$\tau_{b,m}$	bond stress modified in case of bar yielding, transverse pressure and cracking parallel to the bar axis and cyclic loading	E_c	modulus of elasticity of concrete
τ_{bmax}	maximum value of bond stress	E_{ci}	tangent modulus of elasticity of concrete at an age of 28 days
τ_{Ed}	design interface shear stress	$E_{ci}(t_0)$	tangent modulus of elasticity of concrete at the time of loading t_0
τ_{Rdi}	design value of interface shear strength	$E_{ci}(t)$	tangent modulus of elasticity of concrete at an age $t \neq 28$ days
τ_u	ultimate shear friction capacity	$E_{c,1}$	secant modulus from the origin to the peak compressive stress
$\tau_{u,split}$	peak value of bond strength in a splitting failure	$E_{c,imp}$	modulus of elasticity of concrete for impact loading
$\varphi(t, t_0)$	creep coefficient	E_d	design action-effect
φ_0	basic creep coefficient	E_f	modulus of elasticity for non-metallic reinforcement
φ_0, dc	drying creep coefficient	E_{lc}	modulus of elasticity for lightweight aggregate concrete
φ_0, k	nonlinear notional creep coefficient	E_{lci}	tangent modulus of elasticity of lightweight aggregate concrete at concrete age of 28 days
φ_l	basic creep coefficient for lightweight aggregate concrete	E_p	modulus of elasticity of prestressing steel
$\Delta\varphi_{T,trans}$	transient thermal creep coefficient which occurs at the time of the temperature increase	E_s	modulus of elasticity of reinforcing steel
		$E_{s,\theta}$	modulus of elasticity of reinforcing steel at temperature θ

$E_{s,sec}$	secant modulus of elasticity of steel	R_a	average roughness
F	action in general; applied load or load effect	R_{AAC}	inverse effective carbonation resistance of dry concrete determined using the accelerated carbonation test ACC
F_b	bond force transmitted along the transmission length	R_d	design value of resistance
F_c	strut force (compression force)	R_k	characteristic value of resistance
F_d	design value of action	R_m	mean value of resistance
$F_{Ed,ef}$	effective concentric load (punching load enhanced to allow for the effects of moments)	R_{NAC}	inverse effective carbonation resistance of dry concrete determined using the normal carbonation test NAC
F_j	load corresponding to crack mouth opening displacement ($CMOD$) = $CMOD_j$	R_t	peak-to-mean line height (derived from sand patch method)
F_{pt}	tensile force in prestressing steel	R_z	mean peak-to-valley height
$F_{p,0,max}$	maximum tensile force in the prestressing steel reinforcement at tensioning	$R(t,t_0)$	relaxation function, representing the stress response to a unit imposed strain
$F_{p0.1}$	characteristic 0.1% proof load	RH	ambient relative humidity
F_{pkT}	characteristic long-term tensile strength of the tendon for declared design life	RH_0	100% relative humidity
F_{rep}	representative value of the actions	S	absorption coefficient
F_t	tie force (tensile force)	ΔS_c	stress range under fatigue loading
F_{ud}	ultimate dowel force	$S_{cd,max}$	design value of maximum compressive stress level (fatigue loading)
G	permanent action	$S_{cd,min}$	design value of minimum compressive stress level (fatigue loading)
G_F	fracture energy of concrete	$S_{c,max}$	maximum compressive stress level (fatigue loading)
G_{inf}	favourable part of permanent action	$S_{c,min}$	minimum compressive stress level (fatigue loading)
G_{sup}	unfavourable part of permanent action	$S_{ct,max}$	maximum tensile stress level (fatigue loading)
H	humidity; horizontal force, horizontal component of a force	S_d	design load effect (M, N, V, T)
I	second moment of area	S_s	slope of the unloading branch of the bond–slip relationship
I_1	second moment of area in state I (including the reinforcement)	T	temperature; torsional moment
I_2	second moment of area in state II (including the reinforcement)	$T(t)$	temperature at time t
I_c	second moment of area of the uncracked concrete cross-section (excluding reinforcement)	ΔT	temperature change
I_e	second moment of area for short-term loading	T_{Ed}	design value of applied torsional moment
$J(t,t_0)$	creep or compliance function representing the total stress dependent strain per unit stress	$T_{Ed,eff}$	effective design value of applied torsional moment
K	orientation factor	T_g	glass transition temperature
K_g	coefficient of gas permeability	T_{max}	maximum temperature of the concrete during heat treatment
K_{tr}	density of transverse reinforcement	T_{Rd}	design value of resistance to torsional moment
K_s	dissociation constant	V	shear force; volume of gas or liquid
K_w	coefficient of water permeability	V_{Ed}	design value of applied shear force
L	span, length of an element	V_{Rd}	design value of resistance to shear force
L_{pl}	plastic length (region in which tensile strain is larger than yield strain)	V_u	ultimate shear force
M	bending moment; maturity of concrete	W_1	section modulus in state I (including the reinforcement)
M_{Ed}	design value of applied moment	W_2	section modulus in state II (including the reinforcement)
M_r	cracking moment	W_c	section modulus of the uncracked concrete cross-section (excluding reinforcement)
M_{Rd}	design value of resistant moment	W_e	external work
M_u	ultimate moment	W_i	internal work
M_w	coefficient of water absorption	X	value of material and soil properties in general
M_y	yielding moment	X_d	design value of material and soil properties in general
N	axial force, number of cycles to failure (fatigue loading)	Others	
N_{Ed}	design value of applied axial force	l_b	length of bonded area
N_r	axial cracking force	$l_{b,max}$	value of l_b that, if exceeded, would not lead to an increase in the force transferred between concrete and non-metallic reinforcement
N_{Rd}	design value of resistance to axial force	$\emptyset$	nominal diameter of bar
P_{d0}	design value of prestressing force (initial force)	$\emptyset_n$	equivalent diameter of bundles containing n bars
$P_{k,inf}$	lower characteristic value of prestressing force	$\emptyset_p$	diameter of prestressing steel (for bundles equivalent diameter)
$P_{k,sup}$	upper characteristic value of prestressing force	$\phi(t,t_0)$	creep coefficient
P_m	mean value of prestressing force	ϕ_0	notional creep coefficient
Q	variable single action; volume of a transported substance (gas or liquid)	Θ_{pl}	plastic rotation capacity
Q_k	characteristic value of variable action	ΣU	total perimeter of reinforcing bars
R	resistance (strength); bending radius; universal gas constant		

Ψ_0	coefficient for the combination value of a variable action
Ψ_1	coefficient for the frequent value of a variable action
Ψ_2	coefficient for the quasi-permanent value of a variable action
Ω_{cr}	factor for modified bond in case of cracking parallel to the bar axis
Ω_{cyc}	factor for modified bond in case of cyclic loading
$\Omega_{p,tr}$	factor for modified bond in case of transverse pressure
Ω_y	factor for modified bond in case of bar yielding
Λ_{cyc}	dissipated energy during cyclic loading
Λ_0	dissipated energy during monotonic loading

Statistical symbols

Roman lower case letters

$f_E(s)$	probability density function of action
$f_x(x)$	probability density function (of normal distribution)
$f_r(r)$	probability density function (of log–normal distribution)
$f_R(r)$	probability density function of resistance
k	normalised variable or fractile factor
m_x	mean (same meaning as $\bar{x}$)
m_R	mean of resistance
m_E	mean of action
$\tilde{x}$	median
$\hat{x}$	modal value
$\bar{x}$	mean (same meaning as m_x)
x_d	design value
x_k	characteristic value
x_p	p% fractile

Greek lower case letters:

α	sensitivity factor
β	reliability index
γ	(partial) safety factor
μ	mean value
σ_E	standard deviation of action
σ_x^2	scattering or variance
σ_x	standard deviation
σ_R	standard deviation of resistance
δ_R	coefficient of variation of the parameter under consideration

Roman capital letters:

$F_r(r)$	probability distribution function (of log–normal distribution)
$F_x(x)$	probability distribution function (of normal distribution)
P_f	failure probability
R	resistance
E	action (load) effect
M	safety margin
V	coefficient of variation

Others

$\Phi(k)$	normalized function
θ	variables which account for the model uncertainties
θ_d	design values of the variables which account for model uncertainties

Acronyms

AAEM	age adjusted effective modulus (for creep calculations)	ISO	International Organization for Standardization
AAR	alkali aggregate reaction	JCSS	Joint Commission on Structural Safety
ACI	American Concrete Institute	JSCE	Japanese Society of Civil Engineers
AFRP	aramide fibre reinforced plastic	JSSC	Japanese Society of Steel Construction
ASR	alkali silica reaction	LC	indication for lightweight concrete strength class
ASTM	American Society for Testing and Materials	LCC	life cycle cost
BCD	birth certificate document	LCF	life cycle file
CCL	condition control level	LCM	life cycle management
CCP	condition control plan	LoA	level of approximation
CEB	Comité Euro-Internationale du Béton / Euro-International Committee for Concrete	LWAC	light weight aggregate concrete
CEN	European Commission for Normalization	MC	Model Code
CEM	indication for cement type	MPa	megapascal
CFRP	carbon fibre reinforced plastic	PC	prestressed concrete
CMOD	crack mode opening displacement	PL	protection level
CTE	coefficient of thermal expansion	PQP	project quality plan
DIN	German Institution for Normalization	QM	quality management
ECE	electrochemical chloride extraction	RC	reinforced concrete
EDC	equivalent durability concept	SIA	social impact assessment, or Swiss Union of Engineers and Architects
EE	embodied energy	SFRC	steel fibre reinforced concrete
EIC	environmental impact calculation	SLD	service life design
EN	European Norm	RH	relative humidity
ETA	European Technical Approval	SETRA	French Road and Motorway Technical Studies Department
ETAG	European Technology Assessment Group	SCA	service criteria agreement
<i>fib</i>	fédération internationale du béton / International Federation for Structural Concrete (created from the merger of CEB and FIP)	SCC	self compacting (consolidating) concrete
FIP	Fédération Internationale de la Précontrainte / International Federation for Prestressing	SLS	serviceability limit state
FRC	fibre reinforced concrete	RILEM	International Union of Laboratories and Experts in Construction Materials, Systems and Structures
FRP	fibre reinforced plastic	UFC	Unified Facilities Criteria (code for military structures)
GFRP	glass fibre reinforced plastic	UHPFRC	ultra high performance fibre reinforced concrete
GHG	green house gas	ULS	ultimate limit state
GWP	global warming potential	UTS	ultimate tensile strength
Hz	hertz		
IABSE	International Association for Bridges and Shell Structures		

Preface

The International Federation for Structural Concrete (*fib*) is a pre-normative organization. “Pre-normative” implies pioneering work in codification. This work has now been realized with the *fib* Model Code for Concrete Structures 2010. Earlier Model Codes from the *fib*’s parent organizations were published as CEB-FIP Model Codes 1978 and 1990. The objectives of the *fib* Model Code for Concrete Structures 2010 are to (a) serve as a basis for future codes for concrete structures, and (b) present new developments with regard to concrete structures, related structural materials and new ideas in order to achieve optimum behaviour.

Structural concrete is more than a continuously developing material. It also represents a remarkable development in design concepts and strategies. Requirements for concrete structures have often been formulated as follows: concrete structures should be resistant, serviceable, durable, economic and aesthetic. Today, several further requirements or expectations regarding concrete structures have to be met; for example, they should be robust enough to avoid progressive collapse, should need only minimal maintenance, should be able to embed waste materials, should provide protection against accidents, should provide barriers against or following hazards, should be reusable or at least recyclable, should support sustainability in all possible ways and, in addition, provide adequate fire and earthquake resistance and be environmentally compatible.

The *fib* Model Code for Concrete Structures 2010 includes the whole life cycle of a concrete structure, from design and construction to conservation (assessment, maintenance, strengthening) and dismantlement, in one code for buildings, bridges and other civil engineering structures. Design is largely based on performance requirements. The chapter on materials is particularly extended with new types of concrete and reinforcement (such as fibres and non-metallic reinforcements).

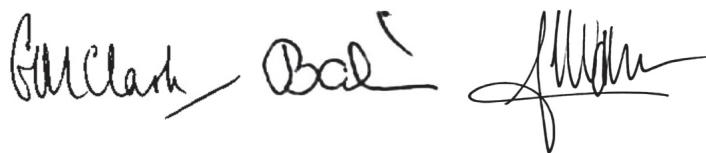
The *fib* Model Code for Concrete Structures 2010 – like the previous Model Codes – not only specifies requirements but also gives the corresponding explanations in a separate column

of the document. Additionally, the *fib* Model Code for Concrete Structures 2010 is supported by background documents that have already been (or will soon be) published in *fib* Bulletins and articles in the *fib* journal Structural Concrete.

The *fib* Model Code for Concrete Structures 2010 was produced during the past ten years through an exceptional effort by 44 countries from five continents: *Argentina, Australia, Austria, Belgium, Belarus, Brazil, Canada, China, Croatia, Cyprus, the Czech Republic, Denmark, Egypt, Estonia, Finland, France, Germany, Greece, Hungary, India, Iran, Israel, Italy, Japan, Luxembourg, the Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Russia, Serbia, Slovakia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Turkey, Ukraine, the UK, the USA*. The General Assembly of *fib* accepted the Model Code 2010 on 29 October 2011 in Lausanne, Switzerland.

On behalf of *fib*, we would like to acknowledge the efforts of all those who contributed to the drafting, correcting or editing of the MC2010 text, including the members of the *fib* Special Activity Group 5, *New Model Code*, and also the *fib* Commissions and Task Groups (see the names listed on the following pages). Special thanks are owed to Agnieszka Bigaj-van Vliet for her work as technical secretary and to Laura Thommen-Vidale for her editorial help.

We believe that the *fib* Model Code for Concrete Structures 2010 provides an extraordinary contribution to the advancement of knowledge and technical developments in the field of design and assessment of concrete structures.



Gordon Clark
President of *fib*

György L. Balázs
Immediate
Past President

Joost Walraven
Convener of SAG5

1 Scope

The *fib* Model Code for Concrete Structures 2010 was an initiative taken by *fib*'s predecessors CEB (Comité Euro-International du Béton) and FIP (Fédération Internationale de la Précontrainte) at a time when there were hardly any international codes. Since, in those days, CEB and FIP were both organisations aiming to synthesize international research and experience, it was regarded as an important step forward to convert this knowledge and experience into practical documents for design, so that national code commissions could take advantage of it. The first code-like recommendations in 1964 and 1970 were used in this way. The Model Code 1978 also contributed to international harmonization. The Model Code 1990 provided confirmation of that intention, by serving as an important basis for the most recent version of Eurocode 2.

The main intention of the *fib* Model Code 2010 is to contribute to the development of improved design methods and the application of improved structural materials. Therefore adequate attention is given to innovative materials such as high-strength concrete, steel fibre concrete and non-metallic reinforcement. Constitutive relations are given for concrete up to strength classes of C120 for normal density concrete and LC80 for lightweight concrete. Moreover design rules are given for fibre reinforced concrete, which apply as well to higher strength classes. An important new aspect is the life cycle concept, which serves as a basis for a holistic design approach. Structures have to be designed for structural safety and serviceability *for a specified period*. This includes design for durability and sustainability. In order to design a structure with a low need for substantial maintenance during its service life, measures have to be taken in the design stage to ensure this and to carry out control when the structure is in service.

Explanations are given on the left-hand side. In this respect, reference is often made to the sources that were used to derive the design recommendations. These sources can be *fib* Bulletins, CEB-FIP Bulletins, and other codes (ISO) or papers in scientific journals.

Level I is reserved for structures where high accuracy is not required. It can also be used for pre-design of structures in a more general sense. Higher level methods can be used in cases where higher accuracy is required. An example of this is the assessment of an existing structure for its bearing capacity, supporting the decision of whether repair is necessary or not.

1.1 Aim of the *fib* Model Code 2010

The *fib* Model Code for Concrete Structures 2010 is intended to serve as the basis for future codes for concrete structures. Whereas existing operational codes are legal documents, based on mature knowledge, the *fib* Model Code also takes into account new developments with respect to concrete structures, the structural material concrete, and new ideas with respect to requirements to be formulated, so that structures achieve optimum behaviour according to new insights and ideas. In this Model Code, those new ideas refer not only to traditional demands with regard to safety and serviceability, but also take into account the increasing significance of design criteria for durability and sustainability.

For those who will be involved in updating existing codes or developing new codes for concrete structures, the *fib* Model Code should be a source of information. Whereas a normal operational code predominantly gives sets of application rules that should be transparent enough to be applied by professional designers while also accurate enough to be economical, the *fib* Model Code also aims to give sufficient background information.

Nevertheless the *fib* Model Code is meant also to be an operational document for everyday design situations and structures.

1.2 Format

The format of this *fib* Model Code follows the earlier CEB-FIP tradition:

- the main provisions are presented on the right-hand side in a logical sequence of topics. Structural requirements are stated, followed by the relevant design criteria, appropriate engineering models and/or design rules: their application is intended to satisfy the relevant structural requirements;
- explanations are given on the left-hand side, with specific diagrams, alternative simplified rules, short justifications of the options found on the right-hand side and references to other sources.

1.3 Levels of approximation

Various levels of approximation are possible for the design and assessment of concrete structures. Therefore in a number of chapters methods are offered with different levels of accuracy. Level I methods generally represent the most simple and straightforward approach, valid for standard cases. Higher levels are presented, which generally require more effort but may lead to more economic solutions.

Part I, Principles: in chapters 2–4 subjects such as terminology, performance requirements and basis of life cycle management are addressed. Design strategies and design methods are subsequently presented.

Part II, Design input data: in chapters 5–6 the properties of the structural materials concrete, reinforcing and prestressing steel are given. Moreover, characteristics are given for interfaces between steel and concrete, and between concrete of different ages.

Part III, Design: in chapter 7 various design methods are addressed in 13 subchapters. A wide range of loads and environmental conditions are considered.

Part IV, Construction: in chapter 8 execution rules are given for concrete, steel and formwork.

Part V, Conservation and dismantlement: chapter 9 deals with conservation strategies, condition survey and assessment, interventions and recording. Finally, chapter 10 completes the life cycle discussion with information about dismantlement.

1.4 Structure of the *fib* Model Code 2010

The *fib* Model Code 2010 is subdivided into five parts. The sequence of the parts reflects the basis of life cycle thinking:

- Part I: Principles
- Part II: Design input data
- Part III: Design
- Part IV: Construction
- Part V: Conservation and dismantlement

2 Terminology

2.1 Definitions

This section defines the various technical terms that appear in the *fib* Model Code 2010. Definitions are based on the sources listed in section 2.2.

Acceptance: Agreement of the stakeholders (i.e. owners, users, contractors, society)¹ to take over the structure or a part of it as its own property.

Accidental action: Design situation involving exceptional conditions of the structure or its exposure, including fire, explosion, impact or local failure.²

Accidental design situation: Design situation taking into account accidental conditions for the structure or its components under consideration.¹

Accompanying action: Action accompanying the leading action considered.¹

Action effect: Effect of action(s) on structural members (e.g. internal force, moment, stress, strain) or on the whole structure (e.g. deflection, rotation).

Actions: a) set of forces (loads) applied to the structure (direct action);
b) set of imposed deformations or accelerations caused, for example, by temperature changes, moisture variation, uneven settlement or earthquake (indirect action).²

Adverse state: State in which the performance criterion is not met.

Aesthetics of structures: Aspects of the appearance of a structure perceived in terms of visual aesthetic considerations.

Examples of the *action effects* are stresses, stress resultants, reactions, deformations, displacements, as well as other effects, depending on the type of structure.¹

Aesthetics of structures is usually associated with the visual sense and, to some extent, the senses of sound and texture, as well as with the perception of the recognized associations and the context.

Although any person's response to the aesthetics of a structure will be unique to that individual, many aesthetic principles can be identified and used by the creator of the structure to achieve specific aesthetic effects. Effects relevant for structures include for instance repetition, symmetry/asymmetry, rhythm, perspective, proportion, harmony, contrast, pattern, ornamentation, texture, colour, granularity and interaction of sunlight and shadows.

In order to derive an *analytical model*, use is made of basic relationships such as equilibrium conditions, constitutive relationships and kinematic conditions.

Availability refers to the probability that a structure is actually available for use during the period of time when it is supposed to be available.³

Analytical model: Mathematical relationship between the forces and imposed deformations exerted on the structure or a structural element and its response to those forces (e.g. deformations, displacements or internal forces).

Assessment: see *Condition assessment*.

Availability: The ability of a structure to operate satisfactorily at any point in time, excluding times when the structure is under repair.³

Basic variable: Part of a specified set of variables representing physical quantities, which characterize actions and environmental influences, geometrical quantities and material properties.⁴

Basis of design: Technical description of the implementation of the service criteria agreement.¹

Bearing: Device to transfer a mainly compressive force for supporting an element.

Biological actions: The aggression of biological organisms (bacteria, insects, fungi, algae) affecting and influencing the structure or its components.

Birth certificate: A document, report or technical file (depending on the size and complexity of the structure concerned) containing engineering information formally defining the form and the condition of the structure after construction.⁵

The *birth certificate* should provide specific details on parameters that are important to the durability and service life of the structure concerned (e.g. cover to reinforcement, concrete permeability, environmental conditions, quality of workmanship achieved) and the basis on which future knowledge of through-life performance should be recorded.⁵

The framework laid down in the *birth certificate* should provide a means of comparing actual behaviour/performance with that anticipated at the time of design of the structure.⁵

The *birth certificate* should offer reference to facilitate ongoing (through-life) evaluation of the service life which is likely to be achieved by the structure.⁵

Collapse may be a sudden occurrence, giving limited warning of the impending calamity.⁵

Composite elements can consist of basically different materials but also of variants of similar materials, such as concretes cast at different times.

Condition evaluation would generally consider whether any subsequent intervention is required to meet the specified performance requirements (original or revised), or the implementation of structure management measures to allow the structure to remain in service, such as a reduction of the permitted imposed loading.

The term *condition assessment* may be used more commonly in connection with damaged or deteriorated structures.⁵

A wide range of parameters might be included in *condition survey*, with data being obtained by activities such as visual inspection and various ways of testing. *Condition survey* would

Capacity design: Method of seismic design with appropriately defined areas of plastic deformations exhibiting adequate ductility, together with other areas of the structure that are provided with increased yielding resistance to ensure elastic behaviour.¹

Characteristic value of a material property: The value of a material property (e.g. structural material or soil) having an a priori specified probability of not being attained in the supply produced within the scope of the relevant material standard.⁶ The characteristic value generally corresponds to a specified fractile of the assumed statistical distribution of the particular property of the material or product. A nominal value is used as the characteristic value in some circumstances.²

Characteristic value of a geometrical property: Value usually corresponding to the dimensions specified in the design.⁶ Where relevant, characteristic values of geometrical quantities may correspond to some prescribed fractiles of the statistical distribution.²

Characteristic value of an action: Principal representative value of an action.⁶

Chemical actions: The reactive transport of chemicals (e.g. salts, acids, alkaline substances and organic compounds) affecting and influencing the structure or its components.

Collapse: Catastrophic physical disruption, giving-way or breakdown of elements or components of a structure, to such an extent that the structure is unable to perform its intended loadbearing function.⁵

Commissioning: Start of planned use.¹

Composite element: Element consisting of at least two different structural materials which cooperate in satisfying the requirements for ULS and/or SLS.

Conception: Identifying, developing and assessing different design alternatives.

Conceptual design: All activities and developments leading from the design criteria to a suitable structural solution.

Condition assessment: A process of reviewing information gathered about the current condition of a structure or its components, its service environment and general circumstances, allowing a prognosis to be made of current and future performance, taking account of active deterioration mechanisms and, if appropriate, predictions of potential future damage.

Condition control: The overall through-life process for conserving the condition of a structure, involving condition survey, condition assessment, condition evaluation, decision-making and the execution of any necessary interventions, performed as a part of the conservation process.

Condition evaluation: Similar to *condition assessment*, but is concerned with establishing the adequacy of the structure for future service, judged by its ability to comply with specified performance requirements comprising a defined set of loadings and environmental circumstances.

Condition survey: The process of acquiring information relating to the current condition of the structure with regard to its appearance, functionality and/or ability to meet specified performance

also seek to gain an understanding of the (previous) circumstances which have led to the development of that state, together with the associated mechanisms causing damage or deterioration.

Conservation activities may involve restoring the current condition of a structure to a satisfactory state, or include preventive measures which aim to ensure that the future condition of a structure remains within satisfactory bounds, or improvements to meet revised performance requirements. For this, the effects of potential future deterioration should be considered.

For comparison, see definition of *Structural materials*.

The *construction* is deemed to include any necessary preparatory works (e. g. excavation, landfill) and finishing works required to be carried out at a particular site or location to facilitate the creation of the desired entity (e. g. bridge).⁵

Construction products are either construction materials or various components, elements and assemblies made of construction materials, which are used during construction.

Cumulative knowledge of through-life performance concerns the evolution of certain properties or parameters relevant to safety, serviceability and/or durability of the structure, the type of loading (especially if fatigue effects are of potential concern), data on the characteristics of the environment(s) affecting the structure, and so on.⁵

Defects may be in-built or may be the result of deterioration or damage.⁷

requirements with the aim of recognizing important limitations, defects and deterioration.

Configuration: Creation of an aesthetic expression by means of spatial arrangement, shaping and choice of structural materials.⁵

Connection: Transition between structural elements able to transmit forces and/or moments.

Conservation: Activities and measures taken which seek to ensure that the condition of a structure remains within satisfactory bounds to meet the performance requirements for a defined period of time, with respect to structural safety, serviceability and sustainability requirements, which may include considerations such as aesthetics.

Conservation plan: The overall plan for controlling and conserving the condition of a structure; that is, condition survey, condition assessment, condition evaluation, decision-making and the execution of any necessary intervention.

Construction: see *Construction process*

Construction documents: Contract documents, construction programmes, minutes of meetings and records of construction inspections, together with the daily record of work carried out.¹

Construction inspection plan: Specifying the type, extent, execution and timing of construction inspections, including information on quality requirements and admissible deviations as well as resolving questions of responsibilities and information flow.¹

Construction inspections: Checking whether the design specifications are implemented correctly during execution.¹

Construction materials: Structural and non-structural materials used in a construction process.

Construction process: The overall process of assembling construction elements or products to create a structure.

Construction product: Any product that is manufactured for erecting a building or infrastructural facility.

Construction work: Carrying out the construction according to contract.¹

Construction works documents: Documents specific to construction works.¹

Control measurement: Measurement to monitor selected physical quantities (e. g. geometrical characteristics or structural deformations).¹

Cumulative knowledge of through-life performance: Information on the performance of a structure, based on systematic gathering and evaluation of data during the service life.⁵

Damage: Physical disruption or change in the condition of a structure or its components, caused by external actions, such that some aspect of either the current or future performance of the structure or its components will be impaired.⁵

Decommissioning: Discontinuation or interruption of use.¹

Degradation: Worsening of condition with time; see also *Deterioration*.

Defect: A specific *deficiency* or inadequacy in the structure or its components which affects their ability to perform according to their intended function at the required level, either now or at some future time.⁵

Deficiency: Imperfection, possibly arising as a result of an error in design or construction, which affects the ability of the

Design of structures (process) may be subdivided into *conceptual design*, *structural analysis* and *dimensioning*.

In the context of performance-based design, sets of *performance requirements* are used as input for the *design of structures*. Therefore performance-based *design of structures* will be preceded by the *conceptual design* including a *requirements development phase* (which may be preceded by a *feasibility study* of the project).

The *design situations* considered will include all foreseeable conditions that can occur during construction and use. The design will demonstrate that the relevant limit states are not exceeded for the identified design situation.

The *design value of a geometrical property* is generally a *nominal value*.

Where relevant, the *design value of a geometrical property* may be equal to the characteristic value, and correspond to some prescribed fractile of the statistical distribution. However, it may be treated differently in cases where the limit state under consideration is very sensitive to the value of the geometrical property.⁸

Alternatively, the *design value of a geometrical property* can be established on a statistical basis, with a value corresponding to a more appropriate fractile (e.g. rarer value) than applies to the characteristic value.⁴

Typically, *deterioration* of a structure or its components will be driven by chemical, mechanical or physical processes or agents, or combinations of those actions.

structure to perform according to its intended function, either now or in the future.⁵

Deformation capacity: (Elastic and/or plastic) deformation of a structure or a structural component reached at failure or at any other defined state of loading.

Demolition: The process of dismantling and removal of existing structures.⁵

Design: Developing a suitable solution, taking due account of functional, environmental and economical requirements.

Design alternatives: Feasible alternatives to solve the design assignment.

Design boundary conditions: Space, time, legal, financial, structural, material-related, execution-related and service-related conditions for design.¹

Design criteria: see *Performance criteria*.

Design of structures: Process of developing a suitable solution, taking due account of safety, functionality and sustainability of a structure during its intended service life.

Design service life: see *Specified (design) service life*

Design situations: Sets of defined actions and physical conditions representing the real situation expected during a specified time interval, for which the design is performed.

Design value of a geometrical property: Specified minimum or maximum value of geometrical dimension, which should not be exceeded.

Design value of an action: Value obtained by multiplying the representative value by the partial safety factor, corresponding to the design situation considered.

Design value of material or product property: Value obtained by dividing the characteristic value of the material or product property considered by a partial safety factor or, in particular circumstances, by direct determination.²

Desired state: State in which the performance criteria should be met.

Destruction: Loss of reliability, serviceability or durability due to damage to a structure that is of such severity that repair is not a practical or viable option.

Detailing: Determining the dimensions of structural components and reinforcement layout and geometry in local areas of the structure and specifying the structural details.

Deterioration: Worsening of condition with time, or a progressive reduction in the ability of a structure or its components to perform according to their intended functional specifications.⁵

Deterioration mechanism: (Scientifically describable) process of the cause and development of deterioration.¹

The term *diagnosis* is typically applied to forms of deterioration and degradation or other mechanisms causing an alteration in the expected or desired behaviour of the structure or its components.⁵

Dimensioning is usually performed in combination with numerical verifications by design equations.¹

In the context of performance-based design of structures, *durability* refers to the fulfilment of the performance requirements within the framework of the planned use and the foreseeable actions, without unforeseen expenditure on maintenance and repair.¹

Environmental influences need to be taken into account during planning of service life, design and construction of a particular structure or asset.⁵

Environmental influences may need to be considered at different scales ranging from macro level (affecting the overall structure), meso level (affecting an individual element or component) down to micro level (localized influences).⁵

In the context of limit state design, *failure* is reached when the criteria of the limit state under consideration are not met.

In the context of performance-based design, a *feasibility study* may be carried out before starting the *requirements development phase* and the design of structure.

Ground can be built on (e.g. foundations to structures), built in (e.g. tunnels, culverts, basements), built with (e.g. roads, runways, embankments, dams) or supported (e.g. retaining walls, quays).

Diagnosis: Identification of the cause or explanation of the mechanism(s) by which a phenomenon affects the behaviour or the condition of a structure or its components, based on an investigation of signs and indications exhibited.⁵

Dimensioning: Determining the dimensions, the structural materials (including their properties) and the detailing of a structure on the basis of structural and execution-related considerations.¹

Dimensioning criteria: see *Design criteria*

Dimensioning situations: see *Design situations*

Dimensioning value: see *Design value*

Disintegration: Severe physical damage and disruption of a structure or its components which results in its (localized) break-up into fragments, with the possibility of gross impairment of their functional capability.⁵

Dismantlement: Demolition of a structure with separation of the structural members and structural materials, fulfilling disposal requirements.¹

Ductility: Plastic deformation capacity characterized by irreversible deformations and energy dissipation, usually referred to quantitatively as the ratio between plastic deformation and the limit of the elastic behaviour.

Durability: The capability of structures, products or materials to fulfil the requirements defined, determined after a specified period of time and usage.³

Economy: Moderate use of financial means and natural resources in relation to the whole period of design, execution and service.¹

Environmental influences: Physical, chemical and biological actions resulting from the atmospheric conditions or characteristics of the surroundings to the structure. (Loads associated with wind or wave effects are classified as mechanical loads.)

Estimate: Estimated mean value of a quantity.¹

Examination: Condition survey and evaluation, including recommendation of remedial measures occasioned by special circumstances.¹

Execution: All the activities and measures involved in the physical creation of a structure, including preparation for construction.¹

Failure: The state where the performance level of a structure or a structural element is inadequate.

Fatigue resistance: Ultimate resistance under frequently repeated actions.¹

Feasibility study: Preliminary analysis of all possible solutions to a problem and a recommendation on the best solution. A *feasibility study* is undertaken to ascertain the likelihood of the project's success.

Fixed action: Action with fixed distribution over the structure or structural member; everywhere the magnitude and the direction follow clearly from the information at a point.¹

Free action: An action whose distribution over the structure is not fixed.¹

Geometrical properties: Planned dimensions and unwanted imperfections of a structure.¹

Ground: Subsurface material (e.g. sand, silt, clay, gravel, boulders or rock) in the area under or adjacent to a structure.

Hazard: An occurrence which has the potential to cause deterioration, damage, harm or loss.⁵

Often the term *ingress* is associated with the entry of substances which cause deterioration (e.g. chlorides into reinforced or prestressed concrete, sulphates and carbon dioxide (CO₂) into concretes).⁵

Interventions may be preventative (applying some form of treatment or taking action to ensure that the condition of a structure remains within satisfactory bounds or that an unsatisfactory performance condition is not reached) or reactive (taking action after damage has become visible, e.g. cracking or spalling of concrete).⁵

Interventions may be planned or unplanned. Planned interventions tend to be classified as maintenance. Unplanned interventions tend to be classified as repairs.

Interventions might be instigated for the purposes of, for example, repair, rehabilitation or remediation of the structure concerned.⁵

Inventory may be established to assist in the management of the structures.⁵

The process of inquiry might employ sampling, testing and various other means of gathering information about the structure, as well as theoretical studies to evaluate the importance of the findings in terms of the performance of the structure.⁵

Limit state represents the transition between the *desired state* and the *adverse state (failure)*.

Maintenance activities involve recurrent or continuous measures which enable the structure to fulfil the requirements for reliability.⁷

The term *maintenance* is commonly applied in the context of building fabric components with a limited life, components associated with water management and rainwater run-off, items where regular intervention is required to maintain their effective operation and so on. The term *maintenance* is commonly applied to ancillary items such as gutters, drains, sealants, movement joints and bearings.

Hazard scenario: Critical situation characterized by a leading hazard and defined circumstances.

Ingress: The entry of substances into structural and/or non-structural components of a structure.⁵

Inspection: A primarily visual examination, often at close range, of a structure or its components with the objective of gathering information about their form, current condition, service environment and general circumstances.⁵

Integration: Adaptation of a structure to the natural and manmade environment.¹

Intervention: A general term relating to an action or series of activities taken to modify or preserve the future performance of a structure or its components.

Inventory: Detailed list or register of items or elements, possibly classified by type, function or some other principal attributes.⁵

Investigation: The process of inquiry into the cause or mechanism associated with some form of deterioration or degradation of the structure and the evaluation of its significance in terms of its current and future performance. The term may also be employed during the assessment of defects and deficiencies.⁵

Irreversible serviceability limit states: Serviceability limit states where some consequences of actions exceeding the specified service requirements will remain when the actions are removed.²

Leading action: Main action in a load case.¹

Leading hazard: Main hazard in a hazard scenario.¹

Limit state: State beyond which the structure no longer satisfies the relevant performance criteria.²

Load: see *Mechanical loading*

Load case: Compatible load arrangements, sets of deformations and imperfections considered simultaneously with fixed variable actions for a particular verification.²

Maintenance: A set of planned (usually periodic) activities performed during the service life of the structure, intended to either prevent or correct the effects of minor deterioration, degradation or mechanical wear of the structure or its components in order to keep their future serviceability at the level anticipated by the designer.⁵

Maintenance plan: Instructions for maintenance specific to the structure considered.¹

Maintainability refers to the probability that an item will be restored to specified conditions within a given period of time when maintenance action is performed in accordance with prescribed procedures and resources.³

Management of structures often involves conflicting requirements and objectives, which invariably requires compromise and judgement about the action to be taken or not taken, due to limitations in the available resources.⁵

Structural *monitoring* typically involves gathering information by a range of possible techniques and procedures to aid the management of an individual structure or class of structures. It often involves the automatic recording of performance data for the structure and possibly some degree of associated data processing.⁵

Monitoring involves similar activities to surveying, but with measurements being undertaken on an ongoing and possibly quasi-continuous basis. Monitoring could involve installed instrumentation. If so, this will introduce ways of measurement and data gathering different from those used for a survey. Under some circumstances, these activities might possibly include various forms of local/global response measurement or testing.

The *nominal value* of a material or a product property is normally used as a characteristic value and established from an appropriate document such as a standard.⁴

The uncertainties in material properties are dealt with by the *partial safety factor* for a material property. The uncertainties of the (resistance) models (including geometric deviations associated with them, if these are not modelled explicitly) are dealt with by the *partial safety factor* for the (resistance) model. The uncertainties in the actions are dealt with by the *partial safety factors* for loads and environmental actions.

In many instances the term *penetration* is used interchangeably with the term *ingress*, but it may also be used in the context of

Maintainability: The ability of a structure to meet service objectives with a minimum expenditure of maintenance effort under service conditions in which maintenance and repair are performed.³

Management (of structures): Processes and procedures adopted for the operation, maintenance, inspection, testing, assessment and repair or other remedial action of structures in order to provide effective control against (predetermined) criteria to ensure the continued safe service of individual structures or wider groupings of structures and related assets.⁵

Material: Metal, non-metallic inorganic or organic material with useful technical properties.¹

Mechanical loading: (External) pressure, force or imposed displacement to which the structure or its components are subjected.

Method of construction: Manner in which the construction is carried out.¹

Modification: Making changes to a structure for the purpose of adapting it to new requirements.¹

Monitoring: To keep watch over, recording progress and changes in materials and/or structural properties with time; possibly also controlling the functioning or working of an associated entity or process (e.g. by using warning alarms based upon parameters such as applied load, element deflection or some other aspect of structural response).⁵

Monitoring plan: Instructions for monitoring specific to the structure.¹

Nominal value: Value fixed on a non-statistical basis, for instance on acquired experience or on physical conditions, or a planned prescribed value.²

Objective of protection: Qualitative and quantitative specification of the requirements of a structure for the case of accidental occurrences and conditions.¹

Observation: Examining the serviceability by simple and regular checks.¹

Observational method: Possible procedure in the case of insufficiently reliable basic information for the design, execution and use of a structure, involving certain acceptable risks, a prediction of behaviour and the specification of associated limit values, together with corresponding monitoring and safety measures.¹

Operational instruction: Instructions for the owners and users on the handling and operation of the technical equipment.¹

Overall stability: State of stable equilibrium for the whole structure as a rigid body.¹

Partial safety factor: A factor employed to deal with the uncertainties in the model variable.

Passive state/passivity: The state in which, by virtue of a protective oxide film, steel does not spontaneously corrode.⁷

Penetration: The entry of substances into structural and/or non-structural components of the fabric of a building or structure.⁵

evaluating the depth to which a deleterious agent has penetrated the component concerned (e.g. chlorides have penetrated to the depth of the reinforcing steel).⁵

The term *penetration* may also be associated with the introduction of agents which will help to extend the service life of the structure (e.g. the introduction of resins or corrosion inhibitors into concrete).⁵

In the context of limit state design, *performance criteria* are the limit values that describe for each limit state the conditions to be fulfilled.⁴

A *performance indicator* is associated with and gives meaning to the *performance criteria* used to define the *performance requirements* for a design, an actual, a potential or an intended intervention option.⁵

Performance requirements are established by means of *performance criteria* and associated *performance indicators* and constraints related to *service life* and *reliability*.

Performance requirements refer to the fulfilment of the essential demands of the stakeholders (i.e. owners, users, contractors, society) during the intended lifetime of structures or structural elements.⁵

Sets of *performance requirements* are used as input into the performance-based design of structures.

Performance requirements are established by means of *performance criteria* and associated *performance indicators* and constraints related to *service life* and *reliability*.

Performance requirements refer to the fulfilment of the essential demands of the stakeholders (i.e. owners, users, contractors, society) during the intended life time of structures or structural elements.⁵

Sets of *performance requirements* are used as input into the performance-based design of structures.

Physical actions are usually caused by change of humidity or temperature (e.g. shrinkage, creep, fire exposure, heating and cooling, freeze–thaw, salt weathering) or movement of agents of wind, water, solid, ice (e.g. water erosion, wind erosion).

Products that are commonly fabricated by precasting, include beams and joists, slab units, wall panels, columns and utility items such as pipes and ducts.³

Performance: The behaviour of a structure or a structural element as a consequence of actions to which it is subjected or which it generates.

Performance aspect: Aspect of the behaviour of a structure or a structural element for a specific action to which it is subjected or which it generates.

Performance criteria: Quantitative limits, associated to a *performance indicator*, defining the border between desired and adverse behaviour.

Performance indicator: A measurable/testable parameter (i.e. characteristic of materials and structures) that quantitatively describes a performance aspect.

Performance level: Qualification of a structure or a structural element, which is established by verifying its behaviour against the performance requirements. A satisfactory performance level is reached when a structure or a structural element has demonstrated a sufficient behaviour to meet the performance requirements. In the opposite case, the performance level of a structure or a structural element is considered to be unsatisfactory.

Performance requirement: A condition for design, or an actual, potential or intended option for intervention, aiming at meeting a specified performance criterion during the service life with appropriate reliability and in a sustainable way.

Performance requirement: A condition for a design, an actual, potential or intended intervention option that the performance criterion must be met during the service life with appropriate reliability and in a sustainable way.

Permanent action: Action almost constant or monotonically approaching a limiting value during a reference period.¹

Persistent design situation: Design situation which is relevant during a period of the same order of magnitude as the design service life.²

Physical actions: Physical phenomena other than mechanical loads (e.g. hydro-thermal processes, weathering, erosion processes) affecting and influencing the behaviour of the structure or its components.

Precast concrete: Concrete that is produced by wet-casting or extruding and cured at a location other than its final position in a structure.³

Precast element: element manufactured in compliance with a specific product standard in a factory, or in a location other than its final position in the structure.

Precast structure: a structure made of precast elements.

Preparation for construction: Invitation to tender, tendering, evaluation of tenders, conclusion of contract for materials and work, as well as preparation of construction work.¹

The situation may include circumstances where the performance requirements have changed over time or where the planned service life has been extended. The treatment or action is taken before deterioration and/or damage become apparent/visible on the structure, for example due to cracking or spalling of concrete.

In the context of the Model Code, the *(owner's) professional team* means those engaged or commissioned by the *stakeholders* to advise and assist through the appropriate provision of technical and related services. Some, possibly all, of the individuals may reside within the entity or organization owning the facility concerned.⁵

Protection involves an action or series of actions undertaken to seek to defend a structure from the effects of further or future deterioration by providing a physical or chemical barrier to aggressive species (e.g. chloride ions) or other deleterious environmental agents and loadings upon the in-service performance and durability of a structure. Typically this will often be provided by surface coatings, impregnation treatments, overlays, membranes, electrochemical treatments, enclosure or surface wrappings applied to the concrete structure, elements or parts thereof.⁵

Typically, the *prudent estimate* is concerned with soil properties.

Typically, *recalculation* is concerned with in-service performance assessment and structural load capacity in particular. The process may utilize similar steps and procedures to design but fundamentally differs from this by seeking to take into account the actual form and condition of the structure as found, including deterioration. This will often include a more realistic consideration of the actual loading regimes, rather than the idealized values used in design. The recalculation process may be used to predict future structural performance, taking into account the influence of ongoing deterioration processes and any remediation actions.⁵

Generally, *reconstruction* is concerned with meeting specific objectives such as strength or future durability requirements.⁵

The aim of *rehabilitation* is in principle similar to the aim of reconstruction, but possibly with greater emphasis upon the serviceability requirements associated with the existing or proposed revised usage of the structure.⁵

Preventive intervention: A proactive conservation activity concerned with applying some form of treatment or taking action that anticipates a change in a material property (e.g. carbonation or chloride ingress causing deterioration) adversely affecting the ability of a structure, or parts of it, to meet the required performance levels.

(Owner's) professional team: A group of persons, generally from one or more organizations, who together are skilled in the various technical aspects and processes required for the design, construction and maintenance of buildings, works and other facilities of public or commercial utility.⁵

Protection: A measure which prevents or reduces the development of defects.⁷

Prudent estimate: A value which, compared to the estimate, is provided with an adequate margin to meet the required reliability.¹

Reactive intervention: A reactive conservation activity, undertaken after deterioration and/or damage has become apparent/visible (e.g. cracking or spalling of concrete) such that, because of the deterioration, it has adversely affected the ability of the structure, or parts thereof, to meet the required performance levels (which may include consideration of issues such as aesthetics).

Re-birth certificate: A document, report or technical file similar to the birth certificate for a structure, but related to the information and circumstances associated with a project for the repair/remediation/ refurbishment of the structure, or a part thereof, to extend its anticipated service life.⁵

Rebuild: To create a new structure or structural component to replace the original damaged, defective or deteriorated entity after its destruction or demolition, without restriction upon the materials or methods employed.⁵

Recalculation: A process of analytical examination using mathematical models or simplified representations of an existing structure or structural elements in order to make an estimate of the performance, taking into account the actual form and condition of the structure as found, including deterioration.

Reconstruction: Restore or reinstating all or part of a structure or component that is in a changed, defective or deteriorated state compared to its original or higher level of performance, without restriction upon the methods or materials employed.⁵

Record of construction: Collection of construction works documents updated according to the state of the execution.¹

Reference period: Chosen period of time used as a basis for assessing statistically variable actions, and possibly for accidental actions.²

Rehabilitation: Intervention to restore the performance of a structure or its components that are in a changed, defective, degraded or deteriorated state to the original level of performance, generally without restriction upon the materials or methods employed.⁵

In some instances, the *rehabilitation* may not be intended to bring the structure or its components back to the original level of serviceability or durability. The work may sometimes be intended simply to reduce the rate of deterioration or degradation, without significantly enhancing the current level of serviceability.⁵

In the context of performance-based design of structures, *reliability* refers to the ability of a structure or a structural member to fulfil the performance requirements during the service life for which it has been designed⁴ at a required failure probability level corresponding to a specified reference period.

Possible *remedial interventions* are wide ranging and may involve structural strengthening through to preventative measures, such as applying surface coatings to provide a barrier to the ingress of deleterious environmental agents (e.g. chloride ions). The situation may include circumstances where the performance requirements have changed over time or where the planned service life has been extended.

The term *remodelling* is often employed where changes principally involve appearance, rather than alteration of the structural components.⁵

In some instances, the *repair* may not be intended to bring the structure or its components back to its original level of serviceability or durability. The work may sometimes be intended simply to reduce the rate of deterioration or degradation, without significantly enhancing the current level of performance.⁷

Replacement may include improvements and strengthening, but does not usually involve a change in function.⁵

The *required service life* is the basis for determining the *specified (design) service life* (for new structures) and the *specified (design) residual service life* (for existing structures).

The *requirements development phase* may be subdivided into gathering the requirements from *stakeholders*, checking for consistency and completeness, definition (writing down descriptive requirements) and specification (creating an initial bridge between requirements and design). The *requirements development phase* may have been preceded by a *feasibility study* of the project.

While *stakeholders* usually believe that they know which performance of a structure they are requesting, it may require skill and experience in structural engineering to recognize incomplete, ambiguous or contradictory requirements.

The *residual service life* is related to the *required service life*, as given by the stakeholders (i.e. owners, users, contractors, society) of the structure and to the other implications of service criteria agreement, for example with regard to structural analysis, maintenance and quality management.

Reliability: Ability of a structure or a structural member to perform its intended function satisfactorily (from the viewpoint of the stakeholder) for its intended life under specified environmental and operating conditions.³ Reliability is usually expressed in probabilistic terms.⁴

Reliability differentiation: Measures intended for socio-economic optimization of the resources to be used to build structures, taking into account all expected consequences of failures and the cost of the structures.²

Remediation: see *Remedial intervention*

Remedial intervention: A conservation activity undertaken after a change in a material property (e.g. that caused by the influence of carbonation or chlorides) has adversely affected the ability of the structure, or parts thereof, to meet the required performance levels because of deterioration.

Remodelling: Changes or alterations to a structure to meet revised functions, performance requirements, usage or occupancy.⁵

Removal: Removing parts from a structure.⁵

Renewal: To reinstate the performance of a damaged or deteriorated component or structure using original methods and materials.⁵

Repair: Intervention taken to reinstate to an acceptable level the current and future performance of a structure or its components which are either defective, deteriorated, degraded or damaged in some way so that their performance level is below that anticipated by the designer; generally without restriction upon the materials or methods employed.

Representative value of an action: The value of an action used for the verification of a limit state. A representative value may be the characteristic value, the combination value, the frequent value and the quasi-permanent value, but it may also be another value of an action.^{2,6}

Replacement: Action to provide substitute new components for ones which have experienced deterioration, damage, degradation or mechanical wear.⁵

Required service life: The demand stated by the stakeholders (i.e. owners, users, contractors, society) for the period in which the required performance has to be achieved.

Requirements development phase: Phase of extracting and describing performance requirements for a structure.

Resistance: Capacity of a member or component, or a cross-section of a member or component of a structure, to withstand actions.⁴

Residual service life: The demand for the remaining period in which the required performance has to be achieved, used in the re-design of existing structures.

Robustness indicates the ability of a structural system to mobilize alternative load paths around an area of local damage. It is related to the strength and form of the structural system, particularly the degree of redundancy (number of potential alternative load paths) within the structural system.⁵

In the context of performance-based design of structures, *safety* is one of the basic performance requirements. For comparison, see the definition of *structural safety*.

For comparison see the definition of *required service life, specified (design) service life, residual service life*.

CEN documents use the term *working life* where this Model Code uses the term *service life*.

Serviceability may be evaluated under various headings, and consideration would normally be given to a number of issues affecting either the whole structure or parts thereof. The issues would typically include various limit state cases (e.g. deflection, vibration, thermal movements, appearance).⁵ In the context of performance-based design of structures, *serviceability* is one of the basic performance requirements.

The *specified (design) service life* is the service life, as required by the stakeholders (i.e. owners, users, contractors, society) and to the other implications of service criteria agreement, such as with regard to structural analysis, maintenance and quality management.

As a rule, the key *stakeholders* would be the founders, the owners, the residents, the users, the neighbours (if construction interferes with them), the contractor, the design and constructing team, the tenancy management team and the maintenance team. Other *stakeholders* may be the government and society.

Restoration: Intervention to bring the structure or its components back to their original condition, not only with regard to function and performance level anticipated by the designer, but also with regard to aesthetic appearance and possibly other (historical) considerations.⁵

Risk: The combination of the likelihood of occurrence of a particular hazard and its consequences.⁵

Robustness: The ability of a structure, subject to accidental or exceptional loading, to sustain local damage to some structural components without experiencing a disproportionate degree of overall distress or collapse.⁵

Safety: Ability of a structure or structural element to ensure that no harm would come to the users and the people in the vicinity of the structure under any (combination of) expected actions.¹⁰

Safety criterion: Performance criterion for the ultimate limit state (ULS).

Service life: The period for which the required performance of a structure or structural element is achieved, when it is used for its intended purpose and under the expected conditions of use.^{4,5}

Serviceability: Ability of a structure or structural element to perform adequately for normal use under all (combinations of) actions expected during the service life.⁶

Serviceability limit: Specified limit of serviceability.¹

Serviceability limit state (SLS): State that corresponds to conditions beyond which specified service requirements for a structure or structural member are no longer met.²

Serviceability criterion: Performance criterion for a serviceability limit state (SLS).²

Service criteria: Requirements for the behaviour of a structure resulting from the planned use.¹

Service criteria agreement: Description of the utilization and protection aims of the stakeholders (i.e. owners, users, contractors, society) as well as the basic conditions and regulations for the design, execution and use of the structure.¹

Service instructions: Instructions for the owner and the operator on the use of the construction works.¹

Service situations: Physical circumstances and conditions during the design service life.¹

Specified (design) service life: The period during which the required performance must be achieved, used in the design of new structures.

Stakeholder: Person or organization that has a legitimate participation in a project.

Strengthening: An intervention made to increase the strength (load resistance/load capacity) and/or possibly the stiffness of a structure or its components, and/or to improve overall structural stability and/or the overall robustness of the structure to a performance level above that adopted by the designer.

Structural integrity: The ability of structural components to act together as a competent single entity.⁵

Construction materials that are used primarily for decoration, insulation or other than structural purposes are not included in group of *structural materials*.³

Structural safety is usually expressed by the ratio (safety factor) between the actions that would cause collapse or other similar forms of structural failure and the actions that are imposed upon it in service.³

Survey is taken to mean the range of activities used to evaluate conformity with the design data for actions and/or material and/or product properties used in the service life design (SLD) on a periodic basis during the service life of the structure. Survey activities would be expected to include a visual inspection undertaken in conjunction with various forms of localized condition testing and measurement (e.g. measurement of depth of cover to reinforcement).

The term *survey* may be applied to the inspection of a number of similar structures/components to obtain an overview. The term *survey* is also used to describe the formal record of inspections, measurements and other related information which describes the form and current condition of a structure and its components.⁵

Various types of *testing* are recognized, their classification being primarily based on the amount of damage or interference caused to the structure. The main divisions are:

- non-destructive testing, which does not cause damage to the structure by the test procedure (e.g. testing with cover meter, radar, acoustic emission, load testing in the elastic range);
- destructive testing, which may cause damage to the structure or marking of the surface finishes (e.g. pull-out tests, material sampling, load testing beyond the elastic range).⁵

Generally, the *ultimate limit state* corresponds to the maximum load-carrying resistance of a structure or structural member.²

Upgrading (retrofitting) relates particularly to the strengthening of structures as a means of minimizing damage during specified loading events.

Structural analysis: Determination of action effects by means of a structural model, if necessary in steps, using different analytical models for the structures as a whole, individual members and local effects.¹

Structural design concept: The basic idea underlying the structural design.¹

Structural materials: Construction materials which, because of their ability to withstand actions, are considered in the design of a structure.³

Structural member: Physically distinguishable part of a structure, such as a column, a beam, a slab or a foundation pile.¹

Structural model: Result of delimiting and idealizing the structural system.¹

Structural safety: Ability of a structure and its members to guarantee the overall stability as well as an adequate ultimate bearing resistance, corresponding to the assumed actions and the required reliability for the specified reference period.¹

Structural system: Arrangement of interacting structural members offering a potential solution to provide bearing resistance to a specified combination of actions.

Structure: Product of human design, intended to fulfil societal functions with adequate reliability with regard to safety, serviceability and sustainability, for a defined period of time.

Substrate: The surface layer in which a protection or repair material has been applied or is to be applied.⁵

Survey: The process, often involving visual examination or utilizing various forms of sampling and testing, aimed at collecting information about the shape and current condition of a structure or its components.⁵

Sustainability: Ability of a structure or structural element to contribute positively to the fulfilment of the present needs of humankind with respect to nature, society, economy and well-being, without compromising the ability of future generations to meet their needs in a similar manner.

Technical report: Explanatory report on design work.¹

Tender documents: Text of the planned contract for materials and work, special conditions, bill of quantities or work description, plans and general conditions.¹

Testing: Procedure aimed at obtaining information about the current condition or performance of a structure or its components.⁵

Tie: Tensile continuous element acting across the structure, horizontally and/or vertically.

Transient design situation: Design situation that is relevant during a period much shorter than the design working life of the structure and which has a high probability of occurrence.²

Ultimate limit state (ULS): State associated with collapse or with other similar forms of structural failure.²

Ultimate resistance: Limit of resistance.¹

Upgrading (retrofitting): Intervention to enhance the functionality or form of a structure or its components so as to improve some aspect of future performance above that defined/achieved during design and construction; typically undertaken to achieve an improved (higher) load resistance against specified loads/actions.

Use: Utilization of a structure as described in the service criteria agreement and in the basis of design.¹

Variable action: Action which is not permanently acting, not constant or not monotonically changing during a reference period.¹

Verification: Confirmation of the fulfilment of a performance requirement.

2.2 References

The definitions given in section 2.1 are based on the following sources:

1. SN 505 260 (SIA 260:2003), Basis of Structural Design, 2003
2. CEN, EN 1990:2002, Eurocode – Basis of Structural Design, 2002
3. McGraw-Hill Encyclopedia of Science and Technology Online, in <http://www.accessscience.com/>, last modified Sept. 2003
4. *fib* Bulletin 34, Model Code for Service Life Design. Fédération Internationale du Béton, 2006
5. *fib* Bulletin 17, Management, maintenance and strengthening of concrete structures. Fédération Internationale du Béton, 2002
6. ISO 2394:1998, General principles on reliability for structures, 1998
7. CEN, ENV 1504:1997: Part 9, Products and systems for the protection and repair of concrete structures – Definitions, requirements, quality control and evaluation of conformity – Part 9: General principles for the use of products and systems, 1997
8. “Probabilistic Model Code”, Joint Committee on Structural Safety (JCSS PMC), 2000
9. SN 505 262 (SIA 262:2003), Concrete Structures, 2003
10. Asian Concrete Model Code, APMC 2006

3 Basic principles

3.1 General

3.1.1 Levels of performance

The performance of a structure or a structural component refers to its behaviour as a consequence of actions to which it is subjected or which it generates.

Structures and structural members must be designed, constructed and maintained in such a way that they perform adequately and in an economically reasonable way during construction, service life and dismantlement.

In general:

- structures and structural members must remain fit for the use for which they have been designed;
- structures and structural members must withstand extreme and/or frequently repeated actions and environmental influences liable to occur during their construction and anticipated use, and must not be damaged by accidental and/or exceptional events to an extent that is disproportional to the triggering event;
- structures and structural members must be able to contribute positively to the needs of humankind with regard to nature, society, economy and well-being.

Durability is an inherent aspect of serviceability and structural safety, and the performance verification must be conducted with proper consideration of the change of performance over time. Accordingly, durability criteria are implicitly involved in the requirement that structures are designed for structural safety and serviceability for a predefined service life, see subsection 3.3.2.

Robustness is a specific aspect of structural safety that refers to the ability of a system subject to accidental or exceptional loadings (such as fire, explosions, impact or the consequences of human error) to sustain local damage to some structural components without experiencing a disproportionate degree of overall distress or collapse.

Accordingly, three categories of performance have to be addressed:

- serviceability, that is the ability of a structure or structural members to perform, with appropriate levels of reliability, adequately for normal use under all (combinations of) actions expected during service life;
- structural safety, that is the ability of a structure and its structural members to guarantee the overall stability, adequate deformability and ultimate bearing resistance, corresponding to the assumed actions (both extreme and/or frequently repeated actions and accidental and/or exceptional events) with appropriate levels of reliability for the specified reference periods. The structural safety must be analysed for all possible damage states and exposure events relevant to the design situation under consideration;
- sustainability, that is the ability of a material, structure or structural members to contribute positively to the fulfilment of the present needs of humankind with respect to nature and human society, without compromising the ability of future generations to meet their needs in a similar manner.

In ISO 15392 (Sustainability in Building Construction – General Principles), sustainability is defined as the state in which components of the ecosystem and their functions are maintained for present and future generations.

3.1.2 Levels-of-approximation approach

All analyses performed for the design of structural members are approximations of reality. These approximations have different levels of accuracy.

A levels-of-approximation (LoA) approach is a design strategy where the accuracy of the estimate of a structural member's response (behaviour or strength) can be, if necessary, progressively refined through a better estimate of the physical parameters involved in the design equations.

The LoA approach is based on the use of rational theories that are based on physical models. The behaviour and strength of structural members are characterized through a series of parameters and a set of design equations. The parameters may involve physical variables (such as crack widths), mechanical properties (such as concrete compressive strength) or geometrical parameters (such as the width of a member).

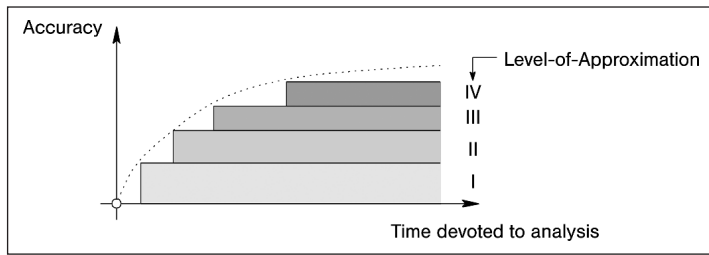


Figure 3.1-1: Accuracy on the estimate of the actual behaviour as a function of time devoted to the analysis for various levels-of-approximation

In the LoA approach, the accuracy in the estimate of the various physical parameters is refined in each new LoA by devoting more time to the analyses, so that the accuracy in the behaviour and strength provided by the design equations is also improved; see Figure 3.1-1.

Building projects typically involve a number of design phases, such as preliminary design, tender design and executive design. The required accuracy of the estimate of the structural behaviour and strength (and the available time to do it) increases as a project evolves. A suitable design strategy consists of using low-order LoAs for the first design phases and higher LoAs for the last design phases. This strategy also applies to assessment of existing structures.

The choice of a suitable LoA depends on the type of analysis performed, on the context of the analysis (preliminary or detailed calculations) and on the potential savings that can be provided if a higher-order LoA is performed.

The first LoA has to provide simple and safe hypotheses for evaluating the physical parameters of design equations. It leads to safe (yet realistic) values of the behaviour and strength of the structural member. This LoA is simple and low time consuming and usually sufficient for preliminary design purposes. Also, the first LoA can be used to check whether a given failure mode cannot be governing (in case a structure shows sufficient strength under the safe assumptions of the first LoA). In such a case, performing further analyses by using higher-order LoAs is not necessary.

The estimate of the first LoA can be refined progressively in successive LoAs by devoting more time to the estimate of the physical parameters involved. This can be done by using analytical or numerical procedures.

For higher LoA (second or third levels), the physical parameters of the design equations are typically evaluated through simplified analytical formulas accounting for the internal forces and other geometrical and mechanical parameters. These LoAs are still low time consuming and are usually sufficient to cover most design cases. Their use is advised for the tender and final design of new structures as well as for the assessment of existing structures.

Numerical procedures typically allow the best estimates of the physical parameters of design equations to be obtained. They are normally used on the highest-order LoAs. The use of such LoAs can however be very time consuming and is only advised for the final design of very complex structures or for the assessment of critical existing structures. This is justified when a more accurate estimate of the physical parameters can lead to significant savings by avoiding or limiting strengthening of the structures.

Further background information on the role of reliability in the performance-based approach, as treated in this section, is given by Bigaj, A., Vrouwenvelder, T. (2013), Reliability in the performance-based concept of the *fib* Model Code 2010, 14. doi: 10.1002/suco.201300053.

Performance requirements must be satisfied in a well-balanced manner throughout the life cycle of the structure.

In the context of limit state design, the term *failure* means failing to fulfil the criteria of the limit state under consideration.

It should be noted that the requirements for existing structures may be different from those for new structures.

As a rule, the key stakeholders would be the founders, the owners, the residents, the users, the neighbours (if construction interferes with them), the contractor, the design and construction team, the tenancy management and maintenance team. Other stakeholders may be the government and the society.

While stakeholders usually believe that they know which performance criteria they should define for a structure, it may require skill and experience in structural engineering to recognize incomplete, ambiguous or contradictory demands.

Specifying performance requirements and associated constraints of service life and reliability relates the needs of the stakeholders to the design or the assessment. Sets of specified performance requirements are used as input for the performance-based design or assessment of structures.

The degree of refinement of the specification of performance requirements depends on the complexity of the project under consideration.

The service life for new structures and the residual service life for existing structures should be defined taking due notice of the implications of the service criteria agreement, for example with regard to maintenance and quality management (QM).

3.2 Performance-based design and assessment

3.2.1 General approach

Using a performance-based approach, a structure or a structural component is designed to perform in a required manner during its entire life cycle. In the case of existing structures, by using a performance-based approach we can assess whether the actual performance of an existing structure or structural members and their performance during the residual life satisfies the demands of the stakeholders.

Performance is evaluated by verifying the behaviour of a structure or a structural component against the specified performance requirements.

An adequate performance is reached when a structure or a structural component has demonstrated satisfactory behaviour to meet the performance requirements. In the opposite case, the performance of a structure or a structural component is considered to be inadequate.

In this Model Code, the state where the performance of a structure or a structural component is inadequate is referred to as failure.

The performance-based design of a new structure or a structural component is completed when it has been shown that the performance requirements are satisfied for all relevant aspects of performance related to serviceability, structural safety and sustainability.

The performance-based assessment of an existing structure or a structural component is completed when it has been identified whether all relevant performance requirements are satisfied or not. In the latter case the performance of a structure or a structural component is qualified as inadequate (failure).

3.2.2 Basis for verification

The stakeholders have to give demands for performance of a structure or a structural component and its required service life.

Those demands reflect the role(s) that a structure or a structural element should play under the intended conditions of construction, service and dismantlement.

For each aspect of performance that is relevant for a structure or structural component under consideration, the performance requirements must be specified. The demands of the stakeholders are the basis for specifying the performance requirements. Accordingly, the performance requirements refer to the fulfilment of the essential demands of the stakeholders.

Performance requirements are established by means of the performance criteria and the associated constraints related to service life and reliability. The performance requirements are satisfied if all relevant performance criteria are met during the service life at the required reliability level.

Performance criteria are quantitative limits defining the border between the desired and the adverse behaviour, relevant for the specific aspect of performance.

Constraints related to service life are given by means of a specified (design) service life (relevant for the design of new structures) or a residual service life (relevant for the re-design of existing structures). The specified (design) service life and the

The target reliability level must be adopted to suit the use of the structure, depending on the type of structure or a structural component and the situation considered in design.

An example of a set of performance requirements, specified on the basis of performance criteria and associated constraints for different performance categories, is given in Table 3.2-1. For further information, see for example EN 1990, Annexes B and C.

Table 3.2-1: Example of performance requirements for the design of a new structure

Performance category	Performance criteria	Constraints
Serviceability	Deformation limit Crack width limit Vibration limit etc.	Specified (design) service life: 50 year Target reliability level: $\beta = 1.5$
Structural safety	Stress limit Capacity limit Progressive collapse limit etc.	Specified (design) service life: 50 year Target reliability level: $\beta = 3.8$
Sustainability	Emission limits Impact on society Aesthetics etc.	

residual service life refer to the period in which the required performance has to be achieved for structures to be designed and for existing structures, respectively.

Constraints related to reliability are specified by means of a target reliability level. A target reliability level refers to an acceptable failure probability corresponding to a specified reference period, which is required to assure the performance of a structure or structural component for which it has been designed. The target reliability level for structures to be designed and for existing structures may adequately be expressed in terms of the target reliability index β or target probability of failure P_f .

The particular choice of performance requirements used in the design depends on the situation that is being modelled.

Considerations regarding the choice of the performance criteria and the associated constraints are found in subsection 3.3.1 (performance requirements for serviceability and structural safety), in subsection 3.3.2 (service life), in subsection 3.3.3 (reliability) and in section 3.4 (performance requirements for sustainability).

The limit states refer to the entire structure, to structural members or to local regions of the members.

In practical design, most of the limit states refer to simplified models for describing the exposure and the structural response. However, limit states may also be introduced which are not directly related to any losses/damages but which are introduced, for example, in order to account for several actual limit states simultaneously.

The durability criteria are implicitly involved in the requirement that structures are designed for structural safety and serviceability for a predefined service life (subsection 3.3.2).

In very particular cases a limit between the serviceability limit states and the ultimate limit states may be defined, a so-called “partial damage limit state” – for example, in the case of earthquake damage of plant structures a “partial damage limit state” is associated with the safe shutdown of the plant. For more details, see section 3.11 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988), and to the JCSS Probabilistic Model Code (JCSS, 2001) [<http://www.jcss.ethz.ch>].

In the cases of irreversible local damage or irreversible unacceptable deformations, the exceedance of a serviceability limit state causes inadequate serviceability of the structure, that is, failure. Some repair may be necessary for the structure to be fit-for-use.

In other cases (such as temporary local damage by, for instance, wide cracks, temporary large deformations or vibrations) the exceedance of a serviceability limit state may be reversible. In those cases failure occurs:

- the first time that the serviceability limit state is exceeded, if exceedance is considered unacceptable;
- if exceedance is acceptable but the time during which the structure is in the undesired state is longer than specified;
- if exceedance is acceptable but the number of times that the serviceability limit state is exceeded is larger than specified.

Frequently exceeding the serviceability limit states may affect the efficient use of a structure, its components (tanks, pipes, canals) or

3.3 Performance requirements for serviceability, structural safety, service life and reliability

In this Model Code, the concept of limit state design is applied to carrying out performance-based design (or re-design) for serviceability and structural safety.

In the context of the performance-based limit state design for structural safety and serviceability, the structural performance of a whole structure or part of it has to be described with reference to a specified set of limit states, which separate desired states of the structure from adverse states.

Limit states are states beyond which the performance requirements are no longer satisfied.

Conceptually, limit states correspond to a discrete representation of the structural response under specified exposure to which specific losses and/or damages can be associated.

Limit states must be related to design situations. They may relate to persistent situations during the service life of the works, transient situations during the execution of the construction works (stage of construction and/or assembling or repair), extreme actions and environmental influences, unintended use or accidents.

Design principles with respect to the performance-based limit state design for structural safety and serviceability are given in chapter 7.

3.3.1 Performance criteria for serviceability and structural safety

In the context of performance-based limit state design, performance criteria for serviceability and structural safety are specified by :

- serviceability limit states criteria (subsection 3.3.1.1);
- ultimate limit states criteria (subsection 3.3.1.2);
- robustness criteria (subsection 3.3.1.3).

3.3.1.1 Serviceability limit states

Serviceability limit states correspond to the states beyond which specified demands for a structure or a structural component related to its normal use or function are no longer met.

its appearance. In many cases, the risk of damage is indirectly excluded by ultimate limit state verifications or by detailing.

Generally, a structure satisfies the operational limit state criteria if all the following conditions are met:

- the facility has suffered practically no damage and can continue serving its original intention with little disruption of use for repairs, supported either by undamaged lifelines or by back-up systems, and any repair that is necessary can be deferred to some future time without disruption of normal use.

Generally, a structure satisfies the immediate use limit state criteria if all of the following conditions apply:

- the structure itself is very lightly damaged (i.e. localized yielding of reinforcement, cracking or local spalling of concrete, without residual drifts or other permanent structural deformations);
- the normal use of the facility is temporarily but safely interrupted (in the case of an industrial plant, after a safe shutdown) and can be restored as soon as utility systems are back in operation;
- risk to life is negligible;
- the structure fully retains its earlier strength and stiffness and its ability to withstand loading;
- the (minor) damage of non-structural components and systems can be easily and economically repaired at a later stage.

The serviceability limit state criteria may refer to, for example:

- unacceptable deformations or deflections which impair the functionality of the structures or their contents, cause damage to non-structural components, cause discomfort to people, affect the appearance of structural or non-structural components or the functioning of equipment (The conditions to be fulfilled with regard to limiting the deformation are associated with the type of building or the civil engineering structure, and are often, for the sake of simplification, substituted by rough approximations);
- excessive vibrations which limit the functional effectiveness of the structures, affect non-structural components, impair the user's comfort or the functioning of equipment (Although such limit states may be characterized by the magnitude of the vibrations, they are commonly indirectly covered by limiting the fundamental period of vibrations of the structure or some of its structural components, in comparison to the expected period of the excitation vibrations);
- local damage (e.g. cracking, slip in connections) which does not affect structural safety but may affect the efficiency or appearance of structural or non-structural components;
- local or global degradation due to environmental actions (e.g. depassivation of reinforcement, weathering) which may affect the efficiency or appearance of structural or non-structural components;
- lack of tightness, or defective sealing, that restrict the functionality or impair the user's comfort.

The serviceability limit states address fitness-for-use of a structure.

Accordingly, the serviceability limit states that should be considered can be described as:

- operational limit states;
- immediate use limit states.

The corresponding serviceability limit state criteria are related to:

- functionality of the structure related to its normal use;
- comfort of using the structure.

The limit values that define the serviceability limit state criteria differ, depending on whether it concerns an operational limit state or an immediate use limit state.

Design principles regarding the formulation of performance criteria for the analysis of the serviceability limit states are given in chapter 4.

The procedures for the verification of the serviceability limit states are given in section 7.6 (for RC and PC structures) and in section 7.7 (for FRC structures).

3.3.1.2 Ultimate limit states

The effect of exceeding an ultimate limit state is almost always irreversible and causes failure the first time it occurs.

Ultimate limit states are limit states associated with the various modes of structural collapse or stages close to structural collapse which, for practical purposes, are also considered as ultimate limit states.

The ultimate limit states address:

- life safety;
- protection of the structure and environment;
- protection of operations.

Accordingly, the ultimate limit states that should be considered can be described as:

- life-safety limit states;

Generally, a life-safety limit state is reached if any of the following conditions are met (but not exceeded):

- the structure is significantly damaged, but does not collapse, not even partly, retaining its integrity;
- the structure does not provide sufficient safety for normal use, although it is safe enough for temporary use;
- secondary or non-structural components are seriously damaged, but do not obstruct emergency use or cause life-threatening injuries by falling down;
- the structure is on the verge of losing capacity, although it retains sufficient loadbearing capacity and sufficient residual strength and stiffness to protect life for the period until repair is completed;
- repair is economically questionable and demolition may be preferable.

Generally, a structure has reached the near-collapse limit state if any of the following conditions are met:

- the structure is heavily damaged and is at the verge of collapse;
- most non-structural components (e.g. partition walls in buildings) have collapsed;
- although life safety is mostly ensured during the loading event, it is not fully guaranteed because there may be life-threatening injury situations due to falling debris;
- the structure is unsafe even for emergency and would probably not survive additional loading;
- the structure presents low residual strength and stiffness but is still able to support the quasi-permanent loads.

- near-collapse limit states.

The ultimate limit states which may require consideration include:

- attainment of the maximum resistance of structures, structural members and sections (regions), for example by:
 - attainment of the maximum resistance by material failure, excessive deformations or settlement;
 - attainment of the maximum resistance resulting from loss of capacity caused by fire;
 - attainment of the maximum resistance resulting from the loss of capacity caused by degradation of structural components due to environmental actions (e.g. corrosion of reinforcement, corrosion induced cracking and spalling, alkali silica reaction);
- attainment of the maximum resistance caused by impact or explosion;
- reduction of residual resistance below a certain limit due to an earthquake;

The corresponding ultimate limit states criteria are related to:

- resistance of critical regions;
- fatigue;
- stability.

- permanent deformations exceeding a certain limit after an earthquake;
- rupture of structural members caused by fatigue under essentially repetitive loading or other time-dependent effects;
- loss of stability of the structure or any part of it, including supports and foundations, for example:
 - sudden change of the assumed structural system to a new system (e.g. transformation into a kinematic mechanism or snap through);
 - buckling of slender structures or structural members, in which second order effects play a role;
 - loss of equilibrium of the structure or of a part of the structure, considered as a rigid body (e.g. overturning);
 - loss of equilibrium caused by impact or explosion;
 - sliding beyond a certain limit or overturning due to an earthquake.

The limit values that define the ultimate limit state criteria vary, depending on whether a life-safety limit or a near-collapse limit applies.

Design principles regarding the formulation of performance criteria for ultimate limit state analysis are given in chapter 4.

The procedures for verification of the ultimate limit states are given in section 7.3 (for predominantly static loading of RC and PC structures), section 7.4 (for non-static loading of RC and PC structures,) and in section 7.7 (for FRC structures).

3.3.1.3 Robustness

By virtue of its robustness, the structural system should be able to continue to fulfil the function for which it was created, modified or preserved, without being damaged to an extent disproportional to the cause of the damage.

Robustness is important for maintaining the ability of the structural system to fulfil its function during events such as accidental loading or due to consequences of human error.

Robustness of the structural system addresses:

- life safety;
- property and environment protection;
- protection of operations.

The limit states which may require consideration are related to:

- disproportional failure of a large part of the structure or the whole structure caused by an accidental load or failure of a structural component (e.g. due to explosion, loads by extremely high water table, flooding, loads due to extreme circumstances such as fire, impact, explosion or earthquake), resulting in:
 - system collapse;
 - life-threatening component collapse.

Accordingly, the robustness criteria are related to:

- resistance of the structural system;
- special functions (e.g. shelter from climatic phenomena, containment of substances, providing fortification, security, shade etc.).

Some specific aspects of verification of robustness in the case of extreme loading are addressed in section 7.4.

The general principles and the procedures for the verification of robustness are given in section 7.9.

3.3.2 Service life

3.3.2.1 Specified service life and residual service life

For the main dimensioning and for reliability verifications, the service life is for practical purposes expressed in terms of a reference period t_R .

The residual service life of an existing structure may be shorter than the specified service life intended for a structure in the original structural design. In such a case it may be necessary to upgrade the structure.

Some examples of the specified (design) service life for different types of structures are given in Table 3.3-1:

For new structures, the specified service life defines the period during which the structure has to satisfy the performance criteria agreed.

For existing structures the specified residual service life defines the period during which the structures have to meet the performance criteria agreed.

The specified (design) service life and the residual service life follow from the required service life as given by the stakeholders and from other implications of the service criteria agreement, for example with regard to structural analysis, maintenance and quality

Table 3.3-1: Example of specified (design) service life for the design of a new structure, according to ISO 2394

Type of structure	Specified (design) service life
Temporary structure	1 to 5 years
Replaceable components of structures, for example gantry girders, bearings	25 years
Buildings and other common structures of average importance	50 years
Structures of greater importance, for example monumental buildings, large bridges, other special or important structures	100 years or more

Table 3.3-1 should be used with care. Some buildings, such as factories, will often have an economical service life corresponding to the installed machinery. On the other hand, structural parts of residential buildings will, as expected by society at large, normally have a service life much longer than 50 years as indicated in the table.

A differentiation between replaceable and non-replaceable components of the structure may be considered when choosing the specified (design) service life for the structure and its components.

Limit states associated to the time-dependent material degradation are, for example, initiation of reinforcement corrosion, cover concrete cracking and spalling due to corrosion.

Due consideration is needed to decide whether limit states related to a change of performance due to material degradation should be regarded as serviceability limit states (which may be a failure to achieve some performance, such as good appearance of the structure) or as ultimate limit states (which may be a failure such as falling of spalled concrete that may diminish the resistance or be harmful to people around the structure).

management. The required service life should be given by the owner in consideration of the interests of other stakeholders (i.e. users, contractors, society).

If the performance requirements are satisfied during the specified (design) service life (in case of structures to be designed) or during the residual service life (in case of existing structures), a structure is considered to be sufficiently durable.

The nominal/formal end of the service life is reached when the performance criteria are no longer met at the required reliability level.

3.3.2.2 Verification of service life

The performance verification must be conducted with proper consideration of the change of performance over time, for instance due to degradation or time-dependent effects. Effects of creep and shrinkage of concrete on the structural performance over time must be evaluated according to the guidelines of subsection 7.2.4. Currently, this proper consideration of the chronological change of performance is not fully possible, at least for the effects of material degradation.

Therefore, a staggered approach is taken with regard to the verification of performance requirements for safety and serviceability. Verification of limit states associated with safety and serviceability is performed without considering a change of performance over time due to degradation. In parallel, verification of limit states associated with time-dependent material degradation is performed by means of service life verification.

Accordingly, the service life verification is performed as a justification of the assumption of time-independence of the structural performance, which is made when verifying safety and serviceability according to the procedures described in sections 7.3 (verification of structural safety for predominantly static loading of RC and PC structures), 7.4 (verification of structural safety for non-static loading), 7.6 (verification of the serviceability for RC and PC structures) and 7.7 (verification of safety and serviceability for FRC structures).

Service life verification demonstrates that during the specified (design) service life (new structures) or the residual service life

Further considerations for the choice of the level of reliability are found in the JCSS Probabilistic Model Code (JCSS, 2001) [<http://www.jcss.ethz.ch>].

The costs involved when upgrading the performance of existing structures (e. g. increasing their safety) are usually high compared to the costs of improving the same performance by a structural design in the case of a new structure. Upgrading existing structures may entail relocation of occupants and disruption of activities or influencing heritage values, which does not play a role in case of the design of new structures. Finally, sustainability requirements (e. g. recycling and reuse, reduction of waste) can usually be better satisfied in the design of new structures.

The relationship between P_f and β -values is given in Table 3.3-2.

Table 3.3-2: β -values related to the failure probability P_f , according to EN 1990:2002

P_f	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-6}
β	1.28	2.32	3.09	3.72	4.75

Reliability management has to be supported by suitable databases of different types of structures and their performance over time, taking into account various degradation processes. Therefore, data have to be collected in order to quantify risk, and hence decide on the target reliability values.

The principles of probabilistic structural limit state design with a possibility for differentiating the reliability level are described in the JCSS Probabilistic Model Code (JCSS, 2001) [<http://www.jcss.ethz.ch>].

It is noted that (design) service life and target β value are two independent requirements on structural performance. For example, the same β value may be required for structures with different (design) service lives and vice versa (ISO 2394).

However, the target reliability is sometimes presented not for the (design) service life but as an equivalent value for different (e. g. 1 year) reference period t_R . In Table 3.3-3 the EN 1990 values are given for a 50-year reference period, which is supposed to be the standard (design) service life. These target β -values are equivalent to the values in Table 3.3-4, which are given for a reference period t_R of 1 year. Note that in both Tables 3.3-3 and 3.3-4 the (design) service life is equal to 50 years. Similar arguments hold for Tables 3.3-5 and 3.3-6.

(existing structures) degradation does not result in violation of the performance criteria.

Design principles and the procedures for service life design are given in chapter 4 and section 7.8 respectively.

3.3.3 Reliability

3.3.3.1 Target reliability level

The choice of the target level of reliability should take into account the possible consequences of failure in terms of risk to life or injury, potential economic losses and the degree of societal inconvenience. The choice of the target level of reliability also takes into account the amount of expense and effort required to reduce the risk of failure.

Because of large differences in the outcome of such considerations, due attention should be given to differentiating the reliability level of structures to be designed and that of existing structures.

Reliability requirements for structures to be designed and for existing structures may adequately be expressed in terms of the reliability index β :

$$\beta = -\Phi^{-1}(P_f) \quad (3.3-1)$$

where

$\Phi(\cdot)$ is the standard normal probability distribution function;

P_f is the failure probability corresponding to a specified reference period.

In order to make the right choice for the target β values, the reference period, the consequences of failure and the cost of safety measures have to be analysed for the specific case considered.

The maximum acceptable failure probability depends on the type of the limit state and considered consequences of failure for the relevant construction work.

A differentiation of the reliability level for different consequences of failure and the cost of safety measures may be done on the basis of well-founded analysis. If such analysis is omitted, this Model Code recommends applying target reliability indices for structures to be designed, as given in Table 3.3-5.

Normally, the specified (design) service life is considered as the reference period for a structure to be designed for serviceability and fatigue, while the residual service life determined at the assessment is often considered as the reference period for an existing structure.

Table 3.3-3: Target β -values related to a reference period of 50 years (examples), according to EN 1990

Relative costs of safety measures	Consequences of failure			
	small	some	moderate	great
High	0	1.5	2.3	3.1
Moderate	1.3	2.3	3.1	3.8
Low	2.3	3.1	3.8	4.3

Table 3.3-4: Target β -values related to a reference period of 1 year (examples), according to EN 1990

Relative costs of safety measures	Consequences of failure			
	small	some	moderate	great
High	2.3	3.0	3.5	4.1
Moderate	2.9	3.5	4.1	4.7
Low	3.5	4.1	4.7	5.1

The target reliability indices given in Table 3.3-5 for serviceability limit states verification correspond approximately to the values recommended in ISO 2394 for *small consequences of failure* and *moderate relative costs of safety measures*. The target reliability indices given in Table 3.3-5 for ultimate limit states verification correspond to those recommended in ISO 2394 for, respectively: *some, moderate and great consequences of failure* and *low relative costs of safety measures*.

The target reliability level for the existing structures may be chosen lower than for new structures, because for existing structures the costs of achieving a higher reliability level are usually high compared to structures under design.

For more details, see ISO 13822 “Bases for design of structures – Assessment of existing structures” and ISO 2394 “General principles on reliability for structures”.

Table 3.3-5: Recommended target reliability indices β for structures to be designed, related to the specified reference periods

Limit states	Target reliability index β	Reference period
Serviceability		
reversible	0.0	Service life
irreversible	1.5	50 years
irreversible	3.0	1 year
Ultimate		
low consequence of failure	3.1	50 years
	4.1	1 year
medium consequence of failure	3.8	50 years
	4.7	1 year
high consequence of failure	4.3	50 years
	5.1	1 year

The β values given in Table 3.3-5 may also be used for the assessment of existing structures, but differentiation of the target reliability level for the new structures and for the existing structures may need to be considered.

A decision to choose a different target reliability level for existing structures may be taken only on the basis of well-founded analysis of consequences of failure and the cost of safety measures for any specific case. Some suggestions for the reliability index for existing structures are given in Table 3.3-6 for the specified reference periods.

Table 3.3-6: Suggested range of target reliability indices β for existing structures, related to the specified reference periods.

Limit states	Target reliability index β	Reference period
Serviceability	1.5	Residual service life
Ultimate	in the range of 3.1–3.8* in the range of 3.4–4.1* in the range of 4.1–4.7*	50 years 15 years 1 year

* depending on costs of safety measures for upgrading the existing structure

For more details, see the JCSS Probabilistic Model Code (JCSS, 2001) [<http://www.jcss.ethz.ch>].

Experience shows that actual reliabilities are often higher than the target values as a result of residual strength effects, not considered in current design models. Such hidden residual capacities can be

The requirements for the reliability of the components of the system will depend on the system characteristics. The target reliability indices given in Tables 3.3-5 and 3.3-6 relate to the structural system or in approximation to the dominant failure mode or structural component dominating system failure. Therefore, structures with multiple, equally important failure modes should be designed for a higher level of reliability per component than recommended in this Model Code.

The target reliability indices given in Tables 3.3-5 and 3.3-6 are valid for ductile structural components or redundant systems for which a collapse is preceded by some kind of warning, which allows

taken into account for the assessment of existing structures on the basis of a careful analysis.

The target reliability index $\beta = 1.5$ corresponds to the value given in Table 3.3-5 for the serviceability limit state verification in the case of irreversible failure and reference period 50 years.

The target reliability index $\beta = 3.1$ corresponds to the value given in Table 3.3-5 for ultimate limit state verification in the case of low consequence of failure and reference period 50 years. Depending on particular consequences of fatigue failure and the possibility of inspection and repair in the case considered, higher or lower values for β for fatigue verification may be appropriate.

The target reliability index $\beta = 3.8$ corresponds to the value given in Table 3.3-5 for ultimate limit state verification in the case of medium consequence of failure and reference period 50 years.

It is noted that Eurocode EN 1990, Annex B also gives partial factors to loads corresponding to β values for other consequences classes.

The fully probabilistic design method as described in section 4.4 may be used for any β value.

Component reliability is the reliability of one single structural component which has one dominating failure mode.

System reliability is the reliability of a structural system composed of a number of components or the reliability of a single component which has several failure modes of nearly equal importance.

A probabilistic approach provides a better platform from which system behaviour can be explored and utilized. For more details see the JCSS Probabilistic Model Code (JCSS, 2001) [<http://www.jcss.ethz.ch>].

measures to be taken to avoid severe consequences. Therefore by explicit requirements or by appropriate detailing it should be assured that brittle failure does not occur. A structural component or structural system that would be likely to collapse suddenly without warning should be designed for a higher level of reliability than is recommended in this Model Code for ductile structural components.

To satisfy performance requirements at the target reliability levels as recommended in Tables 3.3-5 and 3.3-6, one normally proceeds from the safety concepts, as explained in chapter 4.

In this Model Code the partial factor method is calibrated in such a way that when applying the values of partial factors given in section 4.5, the following reliability requirements are satisfied for a defined period of 50 years:

$\beta = 1.5$ for serviceability limit states verification;

$\beta = 3.1$ for fatigue verification

$\beta = 3.8$ for ultimate limit states verification.

For other β values (e.g. applied in the assessment of existing structures), the partial factor format, explained in section 4.5 can also be applied. However, reconsideration of the partial factors and characteristic values of the fundamental basic variables as given in subsections 4.5.1 and 4.5.2 may be required, following from the consideration of actual uncertainties regarding actions, resistances, geometry, structural modelling and the determination of action effects. This is further discussed in subsections 4.5.1.4 and 4.5.2.2.3.

3.3.3.2 Component reliability and system reliability

Structural analysis methods, as described in this Model Code, are primarily concerned with component behaviour with respect to one dominant failure mode. Each limit state equation is, in most cases, related to a single mode of failure of a single component.

However, individual components may also be susceptible to a number of possible failure modes. Therefore, in design, the susceptibility of the individual components to a number of possible failure modes must be checked where relevant, by checking a number of limit state equations.

Furthermore, most structures are an assembly of structural components. System failure is usually the most serious consequence of component failure. Therefore, the likelihood of system failure following an initial component failure should be assessed in relation to robustness with respect to accidental events, redundancy (alternative load paths) and complexity of the structure (multiple failure modes). Accordingly, system analysis must be carried out as a part of the design.

In particular, it is necessary to determine the system characteristics in relation to robustness with respect to accidental and/or exceptional events (section 7.9).

The system analysis requires considerable inventiveness and initiative from the engineer. In general, the system behaviour of structures can be quantified in terms of limit state design by deterministic approach (e.g. progressive collapse analysis) or by a probabilistic approach.

3.4 Performance requirements for sustainability

3.4.1 General

The true nature of global environmental problems is a result of socio-economic systems that came about following the explosion of industrialization during the Industrial Revolution, in which mass production, mass consumption and mass disposal have flourished. Such systems have caused the destruction of ecological systems due to the use of land and natural resources, and energy depletion, as well as water pollution, the emission and diffusion of hazardous substances and greenhouse gases, waste excretions etc. Humankind has realized that these impacts exceed allowable limits.

As a fundamental scheme in socioeconomic activities, therefore, a paradigm shift to sustainable development has become significant. The concept of sustainable development was proposed in the Brundtland Report in 1987 “World Commission on Environment and Development: Our Common Future” (Oxford University Press, 1987). Sustainable development was defined as “development which meets the needs of the present without compromising the ability of future generations to meet their own needs.” The report described three fundamental aspects: environmental protection, economic growth and social equality. After the publication of this report, the term “sustainable development” became firmly established as the final target worldwide.

In general, a concrete structure must be designed so that it can satisfy performance requirements regarding serviceability, safety and sustainability in a well-balanced manner throughout its design service life.

Economic aspects should be satisfied during the first stage, as the most fundamental requirement, or it may change depending on the other factors.

Aesthetics is one of the important aspects to be considered when a structure is constructed. It is considered as a factor of social impact.

The purpose of design for sustainability is to reduce impacts on the environment, society and economy by evaluating and verifying the performance of concrete, concrete components and structures.

The fulfilment of sustainability requirements for a structure presumes that all aspects of design, construction, use, conservation, demolition and recycling and disposal that are relevant for the environment and society are taken into account.

The economic aspects of sustainability are not dealt with as a performance requirement in this Model Code.

Accordingly, the performance requirements for sustainability are related to:

- impact on the environment, which is defined as the influence on the environment of the activities, from the design to disposal;
- impact on society, which is defined as the influence on society of the activities from the design to disposal.

Performance requirements, which are necessary for the verification of sustainability, are determined by a decision-maker on the basis of legislative regulations, particular intents of stakeholders (e.g. specifiers or owners) or international agreements etc.

Performance requirements related to sustainability are formulated in subsection 3.4.2 (impact on environment) and subsection 3.4.3 (impact on society).

The recommended verification methods are given in section 7.10.

Rational evaluation of the sustainability of a structure can be realized by means of life cycle assessment, including cost and risk and other reasonable methods. In general, such assessment of a structure must consider:

- environmental and social aspects of design, construction, use, recycling and disposal, and the costs and so on, arising from them;
- risks and consequences of failure of the structure during its service life and costs of insurance covering these risks;
- costs of inspections, maintenance, planned partial renewal and repair;
- costs of operation and administration.

However, in this Model Code cost and risk are not considered to be part of the performance requirements of a structure.

The relevant impact categories include:

- urban air pollution;
- hazardous substances;
- destruction of the ozone layer;
- global warming;
- eco-toxicity;
- acidification;
- eutrophication;
- photochemical oxidants;
- land use;
- waste material;
- resource consumption.

However, it is generally difficult to set up an appropriate indicator by an end-point approach such as performance requirements. Therefore, inventory items, such as CO₂, NO_x, SO_x, wastes etc., will be used as performance indicators.

More detailed information on the environmental aspects of concrete and concrete structures is available from *fib* Bulletin 18: “Recycling of offshore concrete structures” (*fib*, 2002), *fib* Bulletin 21: “Environmental issues in prefabrication” (*fib*, 2003), *fib* Bulletin 23: “Environmental effects of concrete” (*fib*, 2003), *fib* Bulletin 28: “Environmental design” (*fib*, 2004) and *fib* Bulletin 47: “Environmental design of concrete structures: general principles” (*fib*, 2008).

For sustainable development on Earth, we have to prevent global warming, which is thought to be caused by greenhouse gases such as CO₂. The Kyoto Protocol to the United Nations Framework Convention on Climate Change (UN, 1998) [<http://unfccc.int>] specifies targets for the limitation of emissions of greenhouse gases. In particular, the aggregate anthropogenic carbon dioxide equivalent emissions of the greenhouse gases must not exceed the assigned emission limitation and reduction commitments, which are intended to reduce the overall emissions of such gases by at least 5% below the 1990 levels in the commitment period 2008 to 2012. However, it is becoming important to reduce CO₂ even more drastically, such as 50–80%.

Regarding performance requirements for aesthetics, a structure should be designed in such a way that it has a pleasing aesthetic appearance, with appropriate integration into its surroundings.

When a structure is designed, there are several aspects to be considered. One of the most important aspects in design is safety. The aesthetics are also considered to be part of the structure’s value. On the other hand, it has also been pointed out that the aesthetics of a structure include an element of subjective judgement. In civil engineering structures a structure with a logical and simple flow of forces may be considered beautiful. In case of buildings, the intention of a designer may be emphasized in an extreme shape.

3.4.2 Performance requirements for environmental impact

A structure must be designed in such a way that the impact on the environment is appropriately taken into consideration in the life cycle.

Performance requirements for environmental impact must address, depending on the objects of protection, the following issues:

- impact on human health;
- impact on social property;
- impact on biodiversity;
- impact on primary productivity.

Accordingly, performance requirements for environmental impact can refer to:

- selection of materials;
- structural design;
- execution methods;
- use;
- maintenance procedures;
- demolition and waste disposal;
- recycling procedures;
- energy and resource consumption,
- required limits with regard to CO₂ emissions, water pollution, soil contamination, dust, noise, vibration, chemical substances.

The procedures for verification of environmental impact are given in subsection 7.10.1.

3.4.3 Performance requirements for impact on society

A structure must be designed in such a way that the impact on society is appropriately taken into consideration in the life cycle.

The assessment of impacts on society, addresses the intended and unintended social effects, both positive and negative, of the project and any social change processes caused by the project.

A beautiful structure can only be achieved if, in addition to efficient functioning, the aesthetics are developed from the beginning as an essential part of the global structural concept. Owners and engineers have a responsibility and duty to contribute to the aesthetic aspect of a structure, at a reasonable cost.

Performance requirements for aesthetics address:

- visual appearance of the structure;
- harmony of a structure and its environment.

Performance requirements for aesthetics can refer to:

- choice of shape and composition;
- selection of colours, textures and materials;
- integration into the surroundings.

For a more detailed discussion, see *fib* Bulletin 9: “Guidance for good bridge design” (*fib*, 2000).

The through-life management of a structure, as part of the service life design and conservation processes, is discussed in chapter 9.

The word “economic” may need to be interpreted in the widest socioeconomic sense. This may include not only the direct building costs, but also costs of exploitation, maintenance and repair. Costs of decommissioning, user costs and environmental impact should be taken into account as appropriate.

Optimization involves making trade-offs between competing objectives. Interactions and interdependencies between factors such as cost, profits, risk and quality need to be considered. Accordingly the process of making LCM evaluations should be approached with caution. As an optimization problem, the goal of LCM has been to minimize the expected costs on a net present value basis, but increasingly the expectation is that this should be done in conjunction with minimizing adverse environmental and social impacts.

In contemporary engineering practice a practical approach is to minimize the costs associated with achieving the required performance (i.e. to meet relevant performance criteria during the service life at the required reliability level) while achieving an appropriate (minimum) quality requirement.

Quality management is a comprehensive approach to help all parties involved in design, construction, use and dismantlement/demolition of the structure to ensure that appropriately high standards of quality and service are achieved while systematically seeking to reduce costs and impacts associated with through-life care and conservation of the structure.

Performance requirements for impact on society must be set by using appropriate indicators.

The procedures for verification of social impact are given in 7.10.2.

3.5 Life cycle management

3.5.1 General

Life cycle management (LCM) is the overall strategy to be used in managing a structure through its development and service life, with the aim of improving its efficiency from a business/engineering point of view, ensuring that it meets the associated performance requirements defined at the time of design or as may be subsequently modified during the service life of the structure.

LCM is a way of facilitating choices between various design, construction and conservation options on the basis of economics, sustainability and/or other criteria.

In general, LCM seeks to optimize the balance between factors such as cost, profits, risk and quality, durability, sustainability and so on. The LCM process seeks to consider these items in a coherent and integrated way in the process of design, construction, use and conservation of a structure.

A fully integrated approach to LCM is complex and requires realistic life cycle cost (LCC) calculations, assuming appropriate service lives for the various elements and components making up the structure.

In this Model Code, quality measures and quality requirements are given in subsection 3.5.2 on Quality Management. Specific methods of achieving required performance of structures at different phases of the life cycle are given in chapter 7 for design, in chapter 8 for construction, in chapter 9 for conservation and in chapter 10 for dismantlement, recycle and reuse.

3.5.2 Quality management

3.5.2.1 General

Quality management (QM) is a life cycle process for ensuring that concrete structures achieve the required quality and performance.

The main principle of QM is to address quality issues at their root cause. In order to establish adequate quality in the finished

Communication between parties involved in the development of the project is vital. It is important that the client remains engaged in the process even after his basic needs have been defined. It is essential to monitor progress and to communicate with the client throughout the whole project development.

Communication needs to take place throughout the whole process from project inception to its life-end. The iterative nature of the design process needs to be recognized. For most of the individual phases of the project communication procedures are generally formalized. But at interfaces, communication should get special attention. This is especially the case at the start of the design phase where realistic, feasible and clear requirements and criteria need to be agreed between the client and the designer. This is often an iterative process where the designer should support the client by providing feedback on how various starting points may affect economic and technical feasibility of the scheme and its sustainability and to advise upon alternatives.

Appropriate planning allows the parties involved to ensure alignment between project and quality goals. For proper quality planning, it is necessary to determine quality goals and quality metrics, and to use an agreed set of criteria and a standard methodology for defining the desired levels of quality.

ISO 10005:2005 “Quality management – Guidelines for quality plans” gives further advice on the development, acceptance, application and revision of quality plans.

Requirements for quality assurance and quality control may be defined in terms of parameters such as design supervision levels, execution classes and condition control levels. A systematic approach using these concepts is given in *fib* Bulletin 34: “Model Code for Service Life Design” (*fib*, 2006). Minimum levels for the quality assurance and quality control may be defined in national legislation of some countries.

Reviews are an important aspect of quality assurance and quality control, and therefore of the general management of the overall design and construction process. Reviews should be planned in advance and their timing should be linked to decisive milestones within the overall schedule of activity. It is desirable that the first review is undertaken shortly after completion of the basis of design phase or at the start of the design, in order to have the basis of the design reviewed and, as such, confirmed.

A typical contents list of a PQP is as follows:

- general: description of the project, description of the assignment, quality objectives in general, distribution and revisions of the PQP, abbreviations;
- financial: contract data, change procedure, cost control, invoicing, project evaluation;
- risk management: risk inventory, risk mitigation and management, safety and health plan;
- organizational: project organization, sub-consultants/contractors, interface management, communication procedures (reporting, meetings);

structure, quality issues need to be addressed at an early stage in the overall design and construction process.

QM enables quality improvement through quality planning, that comprises quality assurance and quality control issues, at all stages of the project: design (subsection 3.5.3), construction (subsection 3.5.4), conservation (subsection 3.5.5) and dismantlement (subsection 3.5.6).

To make QM effective, there must be a clear and unambiguous understanding between the owner and the designer about the performance requirements and criteria, along with the strategies to be applied in the design, construction, conservation and dismantlement/demolition phases (including the maintenance strategy of the project).

3.5.2.2 Project quality plan

Quality planning is required to give structure to the measures, to assure coherence between the various disciplines and stages of development and to allow quantitative management of quality. For quality planning, a project quality plan (PQP) is widely used and often required.

The PQP should define the tasks and responsibilities of all parties involved and provide adequate control and checking procedures and the organization and filing of adequate documentation of the building process and its results.

The PQP should cover quality assurance and quality control issues.

The PQP should address or refer to:

- objectives and criteria applicable to the project;
- organizational structure;
- technical and organizational working methods and procedures;
- lines of communication;
- tasks and responsibilities;
- QM measures applicable to the outsourcing/subcontracting of activities;
- key personnel involved;
- handling of non-conformities.

- time schedule: planning schedule, milestones, document planning, review and audit planning schedule;
- information management: document control, acceptance procedures, change management, filing, as-built documentation, confidentiality agreements;
- process quality: overview of applicable procedures, progress reporting, non-conformities, audits, customer satisfaction, project evaluation;
- product quality: functional requirements, boundary conditions, basic data and criteria, codes and practices, verification plan, design validation plan, design and drafting tools.

Checklists may be useful for the implementation of a project quality plan. Examples are given in CEB Bulletin 194: “Modelling of Structural Reinforced and Prestressed Concrete in Computer” (CEB, 1990).

For standard schemes handled by a single source company with a certified company quality plan, a simple reference can be made to such a plan for most of the items to be addressed in the PQP. For more complicated schemes and/or schemes handled by a combination of partners, the PQP will generally be project-specific. In such cases the ISO 9000 series of codes may be a useful support.

There is a crucial interaction with the skills of the individuals involved. Although subjective, requirements for skills and qualifications need to be assessed. Where these are deficient, training and education measures should be instigated or more appropriate staff assigned to the project, or a combination of these measures implemented.

While the ISO 9000 series of standards is accepted worldwide as the model approach for QM, with the focus in contemporary standards upon the concept of the “continuous improvement” of an organization’s management system in order to improve overall performance and customer satisfaction, sole reliance on this concept can present various difficulties in respect of the construction of concrete structures on site. In this context there is a need to prevent the occurrence of nonconformities in the “one-off” circumstances associated with the site placement of concrete in a particular structure or component, especially where these may impact upon the structural capacity, performance or durability of the finished entity. Thus there needs to be a focus – within the practices and procedures for assuring quality – upon preventive measures that minimize the risk of nonconformities occurring. This is compatible with a risk-based approach and related methodology to QM.

For more information upon pre-construction planning, the role of the project specification and of QM during execution of concrete structures, see Annexes F and G of *fib* Bulletin 44: “Concrete structure management – Guide to ownership and good practice” (*fib*, 2008).

The life cycle file should be initiated during the design phase and populated with the first set of the relevant information/documents. Later phases further complete the life cycle file. The life cycle file also serves as an interface document managing the collection and transfer of information from one phase of the project into the next.

The extent of a PQP may differ: depending on the nature and size of the project, type of contract and parties involved, each development phase should have a plan, or the plan may cover a number of phases. Coherence and transfer of information and/or instructions between phases is critical. For non-standard and/or complicated projects, a project specific risk analysis should be conducted to define the issues to be addressed specifically in the PQP.

Quality cannot be assured by procedures and an organizational structure only, so the methods of improving quality practices need to be introduced into the process for potential benefits to be realized.

3.5.2.3 Life cycle file

To allow effective and efficient QM, the project quality status/progress should be documented. Therefore, development of the life cycle file should be integrated with QM activities.

The life cycle file is a living document, which continues to be developed throughout the entire life cycle of the project. Thus data on the quality metrics for the life cycle file are collected throughout the life cycle, through comprehensive verification and validation processes, including process audits, peer reviews, analysis and testing, as appropriate.

During the design stage, the life cycle file will be populated with functional requirements, basic data and boundary conditions, selected engineering approach and applied models, engineering results and applicable criteria, specific instructions for construction (specifications for workmanship and materials, assumed or mandatory construction sequence), risk file, results of tests, certificates etc.

At the beginning of construction, the life cycle file will be populated with requirements for execution of the works and the condition control during the service life of the structure.

At the end of construction, the life cycle file will be populated with as-built information from construction and associated tests. As such, the document will allow owners to develop an optimized maintenance strategy and will provide the factual information needed to develop future modifications of this.

The birth certificate document (BCD) is a component of the overall life cycle file documentation. It contains details about the as-built condition of the structure (subsection 3.5.4.2). The BCD should correspond to the information included in the Design File.

During the service life, actual maintenance and findings must also be included in the life cycle file.

After dismantlement of the structure, essential information from the Dismantlement Document must be included in the life cycle file.

The design process provides a way whereby the initial desire of an owner to get a specific performance realized is interpreted and then developed into the detailed information required by the contractor to actually build the project. An iterative process is employed to take the initial starting points/outline of the owner requirements through to detailed specifications and drawings. Through a series of cycles the plan takes shape, its contents become defined and then refined. The cycles form different stages which create specific outputs that support the owner's decision-making process. Without an iterative design process that engages effectively with the owner's decision-making process, there may be a risk that substantial re-working of the design may be required at a later stage.

Although there are various ways in which progress through the design stages can be organized, clients decision models are generally based on go/no-go milestones, with a requirement for an associated increase in the accuracy of the prediction of the project budget required. Generally, the engineering input is gradually similarly increased through the various stages of design development. The greater the confidence that the project is correctly formulated and is likely to proceed, the greater is the justification for more detailed design effort. It gives an effective model of how to phase the design process. Desired accuracy levels (plus and minus) will typically be about 30% in the scouting phase, about 20% at the basis of design stage, about 10% at project specification stage, about 5% at the final design/detailed design

In the life cycle file, information to manage the project throughout the service life should be available. Therefore, the life cycle file should contain all relevant data, such as relevant engineering documents, engineering instructions, specifications, test results and certificates, such as built documentation, maintenance strategy, factual maintenance data and the decommissioning strategy of the scheme.

The life cycle file should be populated with information extracted from the following documents:

- design file, see subsection 3.5.3.2;
- “as-built documentation”: birth certificate document, see subsection 3.5.4.2;
- service life file, see subsection 3.5.5.2;
- dismantlement document, see subsection 3.5.6.2.

3.5.3 Quality management in design

3.5.3.1 Objectives

To enhance the effectiveness and efficiency of the design process this is generally split into a number of phases. These must be formulated in a way that is compatible with the decision process employed by the owner.

Generally, the following design stages can be distinguished:

- briefing phase, see subsection 3.5.3.3;
- scouting phase, see subsection 3.5.3.4;
- basis of design phase, see subsection 3.5.3.5;
- project specification phase, see subsection 3.5.3.6;
- final design phase, see subsection 3.5.3.7;
- detailed design phase, see subsection 3.5.3.8.

stage. While these values have typically related to project costing, they could be equally applicable to factors such as environmental impact and the evaluation of sustainability parameters.

More specific performance goals are more closely and better defined, which increases their effectiveness. Goals that are not clear and measurable may be open to different interpretations, which is likely to limit their effectiveness.

Setting realistic performance goals involves engineering and economic analysis to determine what is possible and how much it will cost.

It is desirable for the performance requirements of the structure to be established by the owner in consultation with the stakeholders and in conjunction with the project team (owner's) professional team.

The project team (owner's) professional team is a group of persons who are skilled in the various technical aspects and processes required for the design, construction and maintenance of structures. This group will include the designer, who is more generally referred to elsewhere in this Model Code.

The stakeholders must not withdraw from the interaction/communication process once their basic needs have been established. It is important to monitor progress and communicate with the owner during all stages of the project. Communication needs to take place throughout the whole project process, from project inception to its life-end.

In many instances the brief is an evolving document. In the briefing phase the brief does not provide all the answers, but it should pose questions and challenges for the designers. The discussion and clarification of the final client's/owner's requirements comes during the scouting phase (subclause 3.5.3.4).

Key issues to consider when developing an initial brief include:

- type of structure and its location (decided after examination of other means of achieving the general objectives – a process which is undertaken before deciding to build);
- planned function(s) of the structure and its components;
- requirements for appearance/aesthetics (initially and throughout the life of the structure);
- requirements for usable space, dimensions, services and fittings;
- the period of service, what constitutes the end of service life and the requirements for the structure at the end of this period;

3.5.3.2 Design file

The design file of the project must be initiated at the briefing phase. Upon completion of the detailed design phase, all relevant documents from the design must be included in the design file. The design file must contain the following documents:

- client or owner's brief, see subsection 3.5.3.3;
- scouting report, see subsection 3.5.3.4;
- service criteria agreement, see subsection 3.5.3.5;
- project specification document, see subsection 3.5.3.6;
- final design report, see subsection 3.5.3.7;
- calculations report, technical report and design drawings, see subsection 3.5.3.8.

Upon completion of the design, the design files must be included in the life cycle file and handed over to the owner for further development in the next stage of the project.

3.5.3.3 Briefing phase

Objectives

When applying a performance-based approach, general performance goals must be developed during the initial stage of design or assessment. General objective statements must be used to define the global performance requirements for all performance categories.

The stakeholders must define the desired performance of the structure. Minimum performance requirements, such as those specified in applicable national standards, should not be violated.

Client or owner's brief

The client's or owner's requirements must be written down in a formal document called the (initial) client/owner's brief.

The client/owner's brief addresses the relevant needs and aims of the project, resources to be provided by the client/owner, the details of the project and any appropriate design requirements. It sets a framework within which all subsequent briefing (when needed) and design can take place.

- need of future changes of use (e. g. to increase flexibility and minimize the risk of obsolescence);
- time, budget and/or quality limitations.

It is common practice to limit the design effort expenses because the feasibility of the project will usually be uncertain at this stage. The objective of making an initial estimate of the overall project cost with limited staff input (and hence incurred cost) will normally require suitably experienced personnel to develop an outline project concept and to make judgements about potential cost, sustainability impacts and so on. At this stage the target accuracy for the estimate of overall project cost might typically be $\pm 30\%$. However, this requirement could also be applied to other factors such as environmental impact and the evaluation of sustainability parameters.

One approach which is commonly adopted is to review relevant former schemes, adapting them to the specific circumstances and requirements of the new project. To do so effectively, with limited staff effort, the designer needs to be well-experienced and to understand the general cost drivers associated with the new and previous project concepts. The goal is to identify project specific, decisive points of attention and cost drivers that need to be considered in detail during the next phase of the development of the design.

At this stage the target accuracy for the estimate of overall project cost might typically be $\pm 20\%$. However, this requirement could also be applied to other factors such as environmental impact and the evaluation of sustainability parameters.

Examples of relevant basic data include:

- geotechnical data;
- metocean data;
- topographical and bathymetrical data;
- climatological data;
- environmental data (earthquake, hurricanes, the aggressiveness of the service environment);
- material properties.

Goals in the initial brief need to be prioritized into “must haves” and “desirables” in order to guide the project team and help them make compromises when the need arises (e. g. prioritizing of time, cost and quality).

3.5.3.4 Scouting phase

Objectives

The scouting phase is an initial (basic/simplified) feasibility evaluation of the project/scheme. Generally it will be based on an outline project concept established from the global performance requirements defined in the briefing phase.

To support the owner’s decision-making process, it will usually be necessary to prepare an indicative budget.

Document

The output will be in the form of a scouting phase evaluation report upon the feasibility of the project and the project scheme, with global functional requirements, outline concept and budget estimate.

3.5.3.5 Basis of design phase

Objectives

During this phase the functional requirements, basic data and design criteria will be developed and the service criteria will be agreed. A conceptual design (see also section 7.1) will also be developed to support a more accurate budget estimate. Quite some effort is required at this stage as the basis of design should be agreed, fixed and frozen upon completion of this stage. An essential part of this phase is the service criteria agreement.

Service criteria agreement

The service criteria must be clearly specified in the service criteria agreement, which must comprise:

- general aims for the use of the construction works;
- basic relevant data, including third-party interactions;

Operational and maintenance requirements may comprise:

- the use of de-icing salts;
- replacement strategy of components subjected to wear;
- flexibility in terms of space requirements, future extension or loadbearing capacity.

The objectives and the degree of protection should be determined on the basis of a risk evaluation.

Fixing the performance criteria for serviceability and structural safety must follow consideration of:

- the hazards, together with means by which the hazards might be avoided, reduced, mitigated, controlled, managed or resisted;
- the type and consequences of deterioration and failure;
- the resistance and mitigation mechanisms.

Fixing the specified (design) service life for which the structures are to be designed and the residual service life for existing structures should follow consideration of factors such as:

- the required service life of a structure, as given by the owner and/or stakeholders;
- what constitutes the end of service life in individual parts of the structure;
- a need for differentiation of service life for individual parts of the structure (e.g. depending on factors such as their replaceability);
- the implications of other service criteria, for example with regard to structural analysis, maintenance and QM.

Fixing the target reliability level must follow consideration of factors such as:

- type and consequences of failure;
- amount of acceptable damage;
- importance of the structure in dealing with a catastrophe following an accidental event;
- expenditure to reduce the risk;
- possibilities of monitoring, maintenance and repair as well as the corresponding expenditure;
- need for differentiation of target reliability level depending on the limit state and reference period, either for the whole structure or its structural components;
- possible hazard scenarios, which should be considered and evaluated, and suitable measures specified in order to keep the hazards under control or to limit them to an acceptable extent.

The following principles may be applied to mitigate the hazards:

- elimination, prevention or hazard reduction;
- controls or alarm systems;
- choice of structural systems that are less susceptible to the hazards under consideration;
- choice of structural systems that can tolerate local damage as well as the loss of a structural member or a whole part of the structure without failing totally;
- choice of structural systems that do not fail without prior warning;
- limiting the spread of fire by the provision of fire compartments;
- choice of suitable structural materials that, if well maintained, will not substantially degenerate during the required service life;

– operational and maintenance requirements;

- special requirements of the stakeholders;
- objectives for consideration of, protection against and treatment of special risks;
- loadings and loading combinations;
- codes and regulatory requirements.

In particular, the specification in the service criteria must address:

- performance criteria for serviceability and structural safety, see subsection 3.3.1;

– service life constraints, see subsection 3.3.2;

– reliability constraints, see subsection 3.3.3;

- accepting a shorter service life for structural components, which may be replaced one or more times during the specified service life;
- appropriate structural analysis and dimensioning;
- careful detailing;
- dimensioning the structure in a manner that allows for or compensates for deterioration during the specified service life;
- choice of an appropriate execution method;
- execution carried out as planned and with the necessary care;
- planning and applying suitable protective and mitigating systems;
- appropriate monitoring and conservation, including inspections at fixed or condition-dependent intervals and necessary preventive intervention or remedial activities.

Fixing the performance criteria for sustainability should follow consideration of factors such as:

- the importance of the structure to the global, regional and local environments;
- the required achievements with respect to sustainability, as given by the owner and/or stakeholders;
- the type and consequences of not meeting the required achievements with respect to sustainability;
- the flexibility to allow future extensions and/or modifications of the functional requirements;
- the expenditure to reduce the risk of not meeting the required achievements with respect to sustainability;
- the need to differentiate the required achievements with respect to sustainability for individual parts of the structure (e.g. depending on factors such as their replaceability).

Well-defined performance requirements allow evaluation of the achievement of performance goals throughout the design, execution, operation and dismantlement/demolition of the structure.

At this stage the target accuracy for the estimate of overall project cost might typically be $\pm 10\%$. However, this requirement could also be applied to other factors such as environmental impact and the evaluation of sustainability parameters.

The output of this stage can serve as the technical part of the invitation to bid for a design/construct contract.

Factors influencing the constructability/economic feasibility of the project may include:

- accessibility of the site;

- performance requirements for sustainability, see section 3.4.

Progress toward the performance requirements should be traceable.

3.5.3.6 Project specification phase

Objectives

With the basis of design as the starting point, the design will be developed first into a preliminary design. Specifications for workmanship, materials and detailed design will then be developed. Significant effort is generally required at this stage.

At this stage, alternative structural concepts will generally be developed and evaluated against each other (see also section 7.1). Numerous aspects should be included in this judgement, potentially including the following:

- robustness of the concept;
- constructability of the concept;
- planning schedule for the concept;
- economy of the project/overall life cycle cost and its achievements with respect to sustainability parameters;
- feasibility of future extensions;
- reliability of the concept as a whole and critical components especially;
- maintenance and repair considerations;
- dismantling of the structure/demolition aspects.

In order to develop the structural concept issues such as the following need to be taken into account:

- the service criteria agreement;
- constructability/economic feasibility of the scheme;

- bearing capacity of the subsoil at the site with respect to anticipated construction equipment loads;
- lifting capacity at the site;
- minimum/maximum size of structural components;
- clearance between energy units necessary for construction;
- quality, availability and reusability of construction materials;
- restrictions regarding the design and construction times, and the budget limitations;
- legal aspects (laws, ordinances, directives);
- construction methods, transport and assembly techniques;
- equipment and facilities for monitoring and maintenance measures;
- maintaining the use of traffic structures and lifelines (pipelines etc.);
- demolition approach at the end of useful life;
- life cycle cost considerations.

The following deviations should be considered:

- deviation from the assumed values of the actions;
- deviations from the planned values of the ultimate resistances of the structure or the soil;
- eccentricities due to construction tolerances, imperfections in the dimensions of structural members.

A structure can be designed for flexibility, anticipating on possible future changes of its function.

A clear statement must be given, indicating which data are fixed and frozen, which data needs further development, which data have been assumed and what assumptions have been made.

In the context of partial safety factor verification, ensuring the required reliability level requires adequate consideration of the uncertainties regarding actions, structural modelling and the determination of action effects. The differentiation of the partial safety factors depending on the uncertainties in actions, material properties and applied models is addressed in chapter 4.

At this stage, the target accuracy for the estimate of overall project cost might typically be $\pm 5\%$. However, this requirement could also

- the critical actions and action effects, as well as the sensitivity of the concept to deviations from the anticipated values;

- the foreseeable service situations, which should be considered and evaluated and appropriate measures taken to ensure serviceability;
- aspects of sustainability in agreement with the requirements of the owner, stakeholders or governing authorities.

Project specification document

The project specification document needs to include information such as the following:

- the chosen structural system;
- the specified (design) service life;
- the service conditions considered;
- the hazard scenarios considered;
- the requirements for structural safety, serviceability, robustness and sustainability, together with the measures needed to achieve them, including attribution of responsibilities, processes, controls and corrective mechanisms;
- a reliability qualification statement for the data used for design;
- the most important dimensions, construction material properties and construction details;
- the assumed soil conditions;
- the important assumptions in the structural and analytical models;
- the accepted risks;
- advised/required additional investigations;
- other conditions relevant to the design;
- comments on the envisaged methods of construction;
- specifications for detailed design, materials and workmanship.

The extent and content of the project specification document should be adapted to the importance of the structure and the associated hazards and environmental risks.

3.5.3.7 Final design phase

Objectives

At this stage, all primary structural members will be specified and typical details will be designed.

be applied to other factors such as environmental impact and the evaluation of sustainability parameters.

Structural analysis and calculations report

The structural analysis should consider the behaviour of the structure in relation to the envisaged dimensioning situations, taking into account the relevant factors that significantly influence the potential performance of the structure or the structural components concerned.

The methods of structural analysis must be based on established theories – experimentally confirmed if necessary – and engineering practice.

The results of the structural analysis must be checked for credibility; for example, they should be subject to a review utilizing general engineering judgement.

Final design report

The final design report must contain all data used for design, all phases considered, including construction phases, applied static schemes, structural analysis, applied criteria and material properties, construction method considered and a traceable demonstration of compliance with the project specification.

The report must also contain a risk file. The risk file must present the identified risks, how they have been managed and, if any, instructions for the next phases of design and construction.

Drawings must present the overall layout of the project, as well as the geometry, shape and dimensions of primary structural members and typical details.

3.5.3.8 Detailed design phase

Objectives

The output of this stage should allow construction of the project. All calculations that are needed to demonstrate compliance with codes and requirements/ specifications of the project will be prepared during this stage. The level of detail of drawings and specifications/ site instructions must allow unambiguous understanding by the contractor of what is required and how the scheme has to be constructed, as well as how compliance with the documents should be demonstrated. A risk file must be prepared to inform the contractor of the risks involved, how these risks have been handled in the previous stages of design and how the remaining risks are to be handled. Issues which require special attention in this respect must be clearly noted on the construction drawings.

Dimensioning

Dimensioning concerns the determination of the dimensions, the structural materials and the detailing of a structure on the basis of structural and execution-related considerations or numerical verifications.

Detailing, limit measures and special provisions supplement the use of models for various purposes, such as:

- to avoid superfluous calculations;
- to satisfy the minimum performance requirement or comply with deemed-to-satisfy provisions with regard to unidentified or poorly quantified hazards. These measures include provision of a minimum resistance to lateral forces, multiple load paths and ties between structural components (see subsections 2.1 and 3.2.3 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988).
- to ensure the validity of calculation models, for example by minimum ratios of reinforcement;
- to ensure a good standard of execution and/or durability, for example by rules for bar spacing and concrete cover depth.

The dimensioning may be assisted by testing, for example, if:

- actions, structural materials or soil properties are not adequately known;
- no appropriate analytical models are available;
- the structure contains components for which there is limited experience and which have a critical influence on the reliability of the structure.

Calculations report

The basis and the results of the detailed design phase must be documented.

Technical report and design drawings

The dimensions, the structural materials and the detailing of a structure as determined during dimensioning must be documented in the technical report and design drawings.

3.5.4 Quality management in construction

3.5.4.1 Objectives

EN 13670:2009, “Execution of concrete structures” defines a set of minimum requirements for the execution.

The main objective is to meet the minimum requirements for QM in construction, as specified in the execution standard and as assumed in the design.

3.5.4.2 As-built documentation (birth certificate document)

The as-built documentation refers to new structures. For existing structures, which have been repaired or strengthened, as-rebuilt documentation is foreseen. This will be dealt with in subsection 3.5.5.2.

The as-built documentation must be a reliable representation of the project as actually constructed. It must include the results of the initial inspection of the completed work/project. The extent of the inspection of the completed work and the content of the as-built documentation will depend on the nature and size of the project, on the design assumptions and on the verification methods, as well as on the QM and the control measures for the project.

The expected outcome would be that either (a) the conformity evaluation confirmed that the design assumptions had been met or (b) the basis for corrective measures would be given.

The BCD would provide a record of at least the following:

- verification of the as-built condition of the structure and a record of the standard of execution/variability achieved during construction;
- a known benchmark for reference on service life design matters;
- initial data as required for the verification of the limit states (in particular limit states associated with durability).

Information included in as-built documentation must allow a conformity evaluation to be performed upon the completed work/elements of the project.

An extract of the as-built documentation or birth certificate document (BCD), will include the results of an initial inspection of a new structure. The content of the BCD is usually limited to the documentation of the direct input parameters for the future condition control of the structure, such as cover thickness to the reinforcement, diffusion coefficient for the concrete cover and so on.

The data gathered in BCD would also allow:

- a first review of service life predictions based upon the initial measured data;
- assessment of compliance/non-compliance with the design requirements and support for decision-making regarding any interventions or remedial activities required.

The BCD might serve as the basis for monitoring the condition of the structure and for planning conservation activities during its service life. Recommendations on conservation procedures, which depend on the specifics of the project, are given in chapter 9.

3.5.5 Quality management in conservation

3.5.5.1 Objectives

A proper inspection regime during the service life of a structure and documentation of the inspection results will give the owner the possibility to perform condition control during the service life and to apply protective measures when the expectations for the service life design are not met.

The objective of QM in conservation is to control and manage the activities and measures taken, which seek to ensure that the condition of a structure remains within satisfactory limits in order to meet the performance requirements for a defined period of time; this applies to structural safety and functional performance requirements, which may include considerations about aspects such as aesthetics. This is achieved through activities that may involve condition survey, monitoring the performance of the structure

For new structures, recording during conservation would be expected to draw upon information obtained for and detailed in the BCD. For existing structures, there is the expectation that recording during conservation would draw upon/contribute to the preparation of a re-birth certificate document (RCD), depending on whether a previous version had been prepared and was to be updated.

The RCD would provide a record of at least the following:

- verification of the condition of the structure after an intervention (preventative or remedial) has been made and a record of the standard of execution/variability achieved in that process and previously;
- updated (in-service) benchmark for reference on service life design matters;
- updated data as required for revision of verification of the limit states, and in particular, limit states associated to durability.

The data gathered would also allow:

- a review of service life predictions based on updated (in-service) measured data and a revised prognosis on future performance;
- assessment of compliance/non-compliance to design requirements and planning for any future preventative/remedial activities required.

There may be a range of additional activities associated with the dismantlement/demolition works, such as those involved in the cleaning up and/or treatment of the site in order to decontaminate it and/or make it suitable for future use or redevelopment.

For dismantlement, a plan should be made covering at least the following:

- provision of adequate structural and personnel safety in all stages of dismantlement;
- minimization of societal hindrance by dust, dirt and noise;
- minimization of contamination of soil, respecting at least the local regulations;

through-life, condition assessment, condition evaluation, decision-making and the execution of any necessary intervention; the corresponding conservation activities and measures undertaken must be recorded.

3.5.5.2 Service life file

The service life file must document the conservation activities carried out during the life of the structure. It must also include results of inspection of the structure or its components carried out during the service life of the structure. Such a record must include:

- classification of the structure and conservation strategy;
- reference to relevant agencies, drawings, details of the immediate and surrounding environment;
- details concerning inspection and evaluation procedures, including results of inspection and monitoring carried out, results of deterioration, rate estimation and evaluation of the structure;
- details of the plan and actual execution of the preventive or remedial interventions carried out.

An extract of the service life file, called the re-birth certificate document (RCD), includes results of in-service inspection of an existing structure after preventative or remedial action has been undertaken. The content of the RCD usually corresponds to the information included in the birth certificate document.

The service life file must be preserved while a structure remains in service. It may also be desirable to keep such records for an indefinite period for reference purposes for the design, construction and conservation of other similar structures.

The records must be kept in a format which can easily be understood.

3.5.6 Quality management in dismantlement

3.5.6.1 Objectives

The objective of QM in dismantlement is to control and manage the activities and measures taken to allow the safe removal of an existing structure and the clearance of the site as appropriate by means of:

- dismantling the structure into its components;
- demolishing the structure by physically breaking it up;
- or a combination of such measures, facilitating the reuse and/or recycling of the original components parts and materials for new use in a manner that minimizes the associated environmental and social impacts.

- conditioning and removal of operating wastes in such a way that the principles of sustainability, as formulated in section 3.4, are satisfied;
- recycling the appropriate parts of the dismantled material;
- cleaning the site and reintegration in the environment after dismantlement.

3.5.6.2 Dismantlement document

The dismantlement document sets down the activities, measures and procedures which will allow the safe removal of an existing structure and the clearance of the site in a manner that minimizes the associated environmental and social impact.

4 Principles of structural design

For complex structures and structures with a sequential change of the structural system during construction, or in use, which are sensitive to time dependent behaviour, the consideration of load- or deformation history may be necessary. In such a case it may be required to carry out both an initial and a long term reliability assessment.

The various types of design situations are defined in section 3.2.2 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988).

Accidental action is defined as action of usually short duration, that is unlikely to occur with a significant magnitude on a given structure during the design service life, but its consequences might be catastrophic, for example fire, explosions or impact from vehicles. The insensitivity requirement is defined in section 2.1 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988).

Unlike accidental actions, which cannot be associated with a statistical probability of being exceeded, seismic actions can be classified in terms of probability of occurrence and severity.

Construction states can be considered as persistent or transient design situations. Accidental design situations involve either the accidental situation itself or they refer to the situation immediately after the accidental event.

Examples of appropriate length of design service life for new structures are given in subsection 3.3.2 (see also EN 1990, chapter 2).

Reference is made as well to EN 1991-1-6 where for specified nominal durations shorter return periods are considered. For middle size buildings often a reference period shorter than 1 year is taken.

In accidental design the failure probability depends on the occurrence of the particular event considered.

In seismic design the failure probability is found by convoluting the probabilities of occurrence of seismic actions greater or smaller than the design one during the design service life for new structures or the residual one for existing structures.

Failure of the structural components and failure of the system must be analysed for all possible damage states and exposure events relevant for the design situation under consideration.

4.1 Design situations

Actions, environmental influences and structural properties may vary with time. Such variations, which occur throughout the lifetime of the structure, should be considered by selecting design situations, each one representing a certain time interval with associated load cases and other hazards, conditions and relevant structural limit states. The design situations considered must include all foreseeable conditions that can occur during execution and use.

In the design procedures, various design situations should be identified as relevant, by distinguishing:

- persistent situations, which refer to conditions of normal use of the structure and are generally related to the structure’s design service life;
- transient situations, which refer to temporary conditions of the structure, in terms of its use or its exposure;
- accidental situations, which refer to exceptional conditions of the structure or its exposure;
- seismic situations, which refer to conditions of the structure under an earthquake event.

In many cases judgement is necessary to supplement codified provisions, in order to identify those design situations that are to be taken into account for a particular structure.

For persistent situations a reference period t_R is commonly considered equal to the design service life for new structures or the residual service life for existing ones. Usually, for persistent situations in case of new structures a reference period t_R of 50 years is adopted for buildings and 100 years for bridges and tunnels.

For transient situations a reference period t_R of 1 year is normally taken.

Accidental situations are considered to be instantaneous and the corresponding reference period t_R is defined as the duration of the design event.

In the context of seismic situations a reference period t_R is normally taken equal to the design service life for new structures or the residual service life for existing structures.

4.2 Design strategies

Structures must be designed for all relevant design situations (i.e. persistent, transient, accidental and seismic, if relevant).

Depending on the type of action or damage state, the following strategies must be applied in design for different categories of the design situations:

- strategies applied in persistent and transient design situations for limiting the consequences of identified permanent and variable actions, which are:
 - design the structure to sustain the action;
 - design the structure to avoid the action;
 - design the structure for damage limitation;

Section 3.2.3 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988) gives similar guidance on the choice of a design procedure appropriate to limit damage due to identified or unidentified hazards.

The general principles and the procedures for the verification of robustness are given in section 7.9.

The limit states either refer to the entire structure, to structural elements or to local regions of elements.

The assessment of $e(\mathbf{e})$ may be referred to as overall analysis, while the assessment of $r(\mathbf{r})$ may be referred to as local analysis.

The probabilistic safety format (sometimes referred to as fully probabilistic design method) allows us to explicitly include the reliability requirements in terms of the reliability index β and the reference period. This may be used for structures to be designed and for existing structures in cases where such an increased effort is economically justified. However, it will seldom be used for the design of new structures due to lack of statistical data. The probabilistic format is more suited for the assessment of existing structures, in particular for the calculation of residual service life.

The partial safety factor format is the usual way of verifying structural design. It is a simplified verification concept, which is based on past experience and calibrated in such a way that the general reliability requirements are satisfied with a sufficient margin during a defined period of time. In the future, this safety format might also be applicable for the verification of service life, provided that sufficiently long term experience is gained or a sufficient amount of data becomes available for a calibration by the probabilistic method.

- strategies applied in accidental or seismic design situations for limiting the consequences of identified accidental or seismic actions are:
 - design the structure to sustain the action;
 - or
 - design the structure to avoid the action;
 - and
 - design the structure for sufficient robustness.

4.3 Design methods

4.3.1 Limit state design principles

The structural performance of a whole structure or part of it should be described with reference to a specified set of limit states which distinguish desired states of the structure from adverse states.

In general terms, attainment of a limit state can be expressed as:

$$g(\mathbf{e}, \mathbf{r}) = 0 \quad (4.3-1)$$

where:

$g(\mathbf{e}, \mathbf{r})$ is the limit state function, $\mathbf{e}$ represents sets of loads (actions) and $\mathbf{r}$ represents resistance variables.

Conventionally, failure (i.e. an adverse state) is represented as:

$$g(\mathbf{e}, \mathbf{r}) \leq 0 \quad (4.3-2)$$

Although limit state equations representing different limit state conditions are various, the limit state function $g(\mathbf{e}, \mathbf{r})$ can often be subdivided into a resistance function $r(\mathbf{r})$ and a loading (or action effect) function $e(\mathbf{e})$. In such a case, equation (4.3-1) can be expressed as:

$$r(\mathbf{r}) - e(\mathbf{e}) = 0 \quad (4.3-3)$$

Consequently, Eq. (4.3-3) lends itself to the following representation of failure:

$$r(\mathbf{r}) \leq e(\mathbf{e}) \quad (4.3-4)$$

4.3.2 Safety formats

Verification of a structure with respect to a particular limit state is carried out via a model describing the limit state in terms of a function (called the limit state function) whose value depends on all relevant design parameters.

Verification of the limit states must be realised by a probability-based method. This Model Code recommends for verification of the limit states to use one of the following safety formats:

- probabilistic safety format – see section 4.4;

- partial safety factor format – see section 4.5;

In the global resistance format the resistance is considered on a global structural level, as compared to local verification of sections with partial safety factors. It is especially suitable for design based on non-linear analysis, where verification of limit states is performed by numerical simulations.

The deemed-to-satisfy approach includes a set of appropriate values from a set of predetermined alternatives given in a standard. This method is the normal way of verifying service life design of new structures.

Design by avoidance is applicable both for the verification of traditional structural design and design for service life.

The variables pertaining to the various limit states may be time-dependent.

In a component analysis with one dominating failure mode, the limit state condition can normally be described by a single limit state equation. In a system analysis, where more than one failure mode may be governing, several equations may apply.

A probabilistic safety format must be applied in accordance with the principles and recommendations laid down in the JCSS Probabilistic Model Code (JCSS, 2001) [<http://www.jcss.ethz.ch>] and in the RILEM publication “Probabilistic Assessment of Existing Structures – JCSS Report” (RILEM, 2001).

Examples of cases characteristic of existing structures, where reliability of existing structures may need to be assessed, are the following:

- doubts about the performance of the structure;
- the expiration of (design or residual) service life (e. g. granted on the basis of design or an earlier assessment of the structure);
- detection of design or construction errors;
- occurrence of unusual incidents during use, which could have damaged the structure;
- a planned change of the use of the structure.

Examples of design situations that are out of the range of application of this Model Code and therefore must be analysed according to a probabilistic safety format are the following:

- actions and hazards outside the range covered by this Model Code;
- use of structural materials and combinations of structural materials outside the usual range of experience;
- new structural materials with properties outside the range covered by this Model Code;
- service life requirements outside the range covered by this Model Code;
- reliability level not covered by this Model Code;

– global resistance format – see section 4.6;

– deemed-to-satisfy approach – see section 4.7;

– design by avoidance – see section 4.8.

For each specific limit state the relevant basic variables should be identified, that is the variables which characterize actions and environmental influences, properties of materials and soils, geometrical parameters and so on.

The variability of basic variables must be analysed based on the available information. In the case of the probabilistic format the basic variables are treated as random variables, or random fields. In the case of the partial factor format, the basic variables are treated as deterministic quantities. In the case of the global safety format, the global resistance is treated as a random variable.

For each limit state, models should be established that describe the behaviour of a structure. These models include mechanical models, which describe the structural behaviour, as well as other physical or chemical models, which describe the effects of environmental influences on the material properties. The parameters of such models should in principle be treated in the same way as the basic variables and model uncertainties must be regarded.

Models for the verification of the limit states can be either analytical (sections 7.3–7.8) or numerical (section 7.11), possibly supported by testing (section 7.12).

4.4 Probabilistic safety format

4.4.1 General

The main objective of a reliability analysis by the probabilistic approach is a probabilistic assessment of the safety of the structure by estimating the failure probability (or the reliability index β).

The probabilistic safety format is a suitable approach for the assessment of the performance of existing structures.

The probabilistic approach may support the design according to the partial factor format or deemed-to-satisfy approach, for example to ensure an appropriate robustness of structures or to account for specific requirements out of the range of application of this Model Code.

- extraordinary structural systems or extraordinary geometry of a structure;
- cases where failure would lead to serious consequences.

4.4.2 Basic rules for probabilistic approach

The verification of a structure with respect to a particular limit state is carried out via estimation of the probability of occurrence of failure in a specified reference period and its verification against reliability requirements – see subsection 3.3.3.1.

With the failure criteria formulated according to Eq. (4.3-2), the probability of occurrence of failure can be generally expressed as:

$$P_f = \text{Prob}\{g(\mathbf{e}, \mathbf{r}) \leq 0\} = \text{Prob}\{M \leq 0\} \quad (4.4-1)$$

where:

$M = g(\mathbf{e}, \mathbf{r})$ represents the safety margin

If the limit state function is expressed in the form of Eq. (4.3-4) and parameters characterising actions, environmental influences, material and geometry are represented by the random variables $\mathbf{E}$ and $\mathbf{R}$, the probability of occurrence of failure can be expressed as:

$$P_f = \text{Prob}\{r(\mathbf{R}) \leq e(\mathbf{E})\} = \text{Prob}\{R \leq E\} \quad (4.4-2)$$

where

$E = e(\mathbf{E})$ and $R = r(\mathbf{R})$ are the basic random variables associated with loading and resistance, respectively.

A proper choice of the distribution of the basic random variables is of importance, since the results of the reliability analysis can be very sensitive to the type of distribution adopted.

4.5 Partial factor format

4.5.1 General

This separation is theoretically not correct, and in practice not complete, because the various factors are not mutually independent. Hence, constant values given in partial factors should be considered as approximations having limited fields of validity. This approximation of using constant values for partial factors may not apply in the following cases:

- non-linear limit state equations;
- mutually correlated variables;
- design by testing.

For the application of partial factors to non-linear analysis see 7.11.3.

The general method of deriving the updated design values to be used in the partial factor method in the case of existing structures is given in ISO 2394 “General principles on reliability for structures” and ISO 13822 “Basis for design of structures – Assessment of existing structures”.

The partial factor format separates the treatment of uncertainties and variabilities originating from various causes by means of design values assigned to variables. In this Model Code the representative values of the variables and the partial safety factors are chosen in such a way that the reliability requirements for the design of new structures, which are expressed in 3.3.3.1 in terms of β related to the reference period, are met.

In the case of existing structures, the same principles of the partial factor format can be applied as for new structures. However, the design values of the variables (i.e. the characteristic values and the partial factors) for existing structures need to be updated in order to guarantee that the reliability requirements for the assessment of existing structures are satisfied at the level discussed in subsection 3.3.3.1.

4.5.1.1 Basic variables

These reliability margins seem to cover the whole set of uncertainties, but a part of the model uncertainties is commonly directly covered by the codified models themselves.

For basic variables, design values include reliability margins. For other variables, whose dispersion may be neglected or is covered by a set of partial factors, they are normally taken as equal to their most likely values.

In this Model Code the following variables are considered as basic:

- actions (F), unless specified otherwise in particular sections;

This does not exclude the fact that some actions (e.g. shrinkage) can be negligible in particular cases. What is to be considered as one individual action is defined in the corresponding standard and explained in section 4.2.1 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988). For prestress, see subsection 4.5.1.4.2 of this Model Code.

For these basic geometrical quantities, tolerances should be carefully fixed (see subsection 4.5.1.4.4) and controlled. For the other geometrical quantities, tolerances generally reflect usual practice. For all geometrical quantities it would not be realistic to specify tolerances less than twice the mean deviation expected or minimum attainable. As a consequence, tolerances may, according to the case considered, be either the basis for the design or necessary complements to the design.

More information is found in sections 4.1 and 6.1 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988). Identifying and selecting the other relevant basic variables is one of the major responsibilities of a designer who faces a problem involving some unusual aspects.

According to the limit state under consideration, the design conditions may have to be formulated:

- either in the space of internal and external moments and forces and directly presented as in Eq. (4.5-2); or
- in the space of forces, as

$$F_E \leq F_R \quad (4.5-3)$$

(F_R being for example a bearing resistance); or

- in the space of stresses as

$$\sigma \leq \alpha f \quad (4.5-4)$$

where f is the material strength and α is a reduction factor depending on the case considered, with $0 \leq \alpha \leq 1$; or

- in the space of geometrical quantities, as

$$a \leq D \quad (4.5-5)$$

where:

D is, for example, a deflection, a crack width or a plastic rotation.

Typically, the design value x_{di} of any particular variable x_{ki} is given by:

$$x_{di} = \gamma_i x_{ki} \text{ in case of loading variables} \quad (4.5-6a)$$

or

$$x_{di} = x_{ki}/\gamma_i \text{ in case of resistance variables} \quad (4.5-6b)$$

where:

x_{ki} is a characteristic value strictly defined as the value of a random variable which has a prescribed probability of not being exceeded (or of being attained); in time-varying

- material or product properties (X), unless specified otherwise in particular sections (e.g. strengths (f), creep (ϕ) and friction coefficients (μ));
- some geometrical quantities (a);

- variables which account for the model uncertainties (θ).

Occasionally other variables should be considered as basic variables. This may be the case for the numbers of repetitions of loads in fatigue verifications.

4.5.1.2 Design condition

With reference to the representation of failure given in Eq. (4.3-2), the design condition can be expressed in terms of design values of basic variables as:

$$g(F_d, X_d, a_d, \theta_d, C) \geq 0 \quad (4.5-1)$$

where:

F_d are design values of actions;

X_d are design values of material and soil properties;

a_d are design values of geometrical quantities;

θ_d are design values of the variables which account for model uncertainties;

C are serviceability constraints.

The relationship given in Eq. (4.3-4) lends itself to the following representation of the partial factor checking format:

$$e(F_d, \dots) \leq r(X_d, \dots) \quad (4.5-2)$$

4.5.1.3 Design values of basic variables

In this Model Code the design values of the basic variables are expressed as follows:

- loads, a value other than the characteristic value may be introduced; for material properties, a specified or nominal value is often used as a specified characteristic value;
- γ_i is a partial safety factor with a value commonly greater than unity.

Some actions (e.g. non-closely bounded hydraulic actions) should be expressed in another way, as mentioned in section 4.1 of Bulletin 191. Furthermore, for verifications relating to fatigue and vibrations, the format is generally different (see subsection 4.5.2.3 for verifications relating to fatigue and subsection 7.6.6 regarding limitation of vibrations).

For material properties other than strengths (e.g. modulus of elasticity, creep, friction coefficients) see the relevant parts of chapters 5 and 6.

Numerical values of γ_M may be different in various parts of the limit state equation given by Eq. (4.3-4), especially for the calculations of $e(e)$ and $r(r)$; for example (see provisions regarding γ_M factors in subsection 4.5.2.2(b)) γ_M may be reduced for the assessment of $e(e)$ by a non-linear analysis.

For concrete and steel, γ_M usually covers the deviations of structural dimensions not considered as basic variables and includes a conversion factor η converting the strength obtained from test specimens to the strength in the actual structure. For practical applications, see the provisions regarding γ_M in subsection 4.5.2.2.4(b).

Other factors – applied to f_d or implicitly included in design formulas – take into account the variations of strength due to non-standardized loading conditions.

As explained in sections 6.3 and 6.6 of CEB Bulletin 191, γ_M may in some cases be substituted by one or two partial factors γ_{Rd} , applicable to the resistance, and a partial factor γ_m applicable to f_k .

It should be noted that, as an alternative to the use of a partial safety factor γ_{Rd} at the resistance side, it is possible to use a partial safety factor γ_{Ed} at the loading side. Such an approach will, for example, be used in subsection 4.5.2.2, Eq. (4.5-13).

Liquid levels representing hydraulic actions should in some cases be expressed as $a_k + \Delta a$, where a_k is a characteristic level and Δa an additive or reducing reliability margin.

A part of the model uncertainties is commonly directly covered by the codified model itself. Partial factors for model uncertainties γ_d take account of uncertainties of models as far as can be found from measurements or comparative calculations.

For existing structures model uncertainties must be considered in the same way as in the design of new structures, unless previous structural behaviour (especially damage) indicates otherwise. In some cases model factors, coefficients and other design assumptions may be established from measurements on the existing structures (e.g. wind pressure coefficient or effective width values). For more information, see ISO 2394 “General principles on reliability for structures” and ISO 13822 “Basis for design of structures – Assessment of existing structures”.

For a resistance parameter X , the updated design value x_d can be obtained from the following procedure according to ISO 13822:

$$x_d = \mu(1 - \alpha \beta V) \text{ for a normal random variable} \quad (4.5-9a)$$

(a) *Design values of actions:*

$$F_d = \gamma_F F_{rep} \quad (4.5-7)$$

where:

- F_{rep} is the representative value of the action, defined in 4.5.1.4.1;
 γ_F is a partial safety factor.

(b) *Design values of material or product property:*

$$f_d = f_k / \gamma_m \quad (4.5-8a)$$

or in case uncertainty in the design model is taken into account by:

$$f_d = f_k / \gamma_M = f_k / (\gamma_m \cdot \gamma_{Rd}) \quad (4.5-8b)$$

where:

- f_k is the characteristic value of the resistance;
 γ_m is a partial safety factor for a material property;
 γ_{Rd} is a partial safety factor associated with the uncertainty of the (resistance) model plus geometric deviations, if these are not modelled explicitly;
 $\gamma_M = \gamma_m \cdot \gamma_{Rd}$ is a partial safety factor for a material property also accounting for the model uncertainties and dimensional variations.

(c) *Design values of geometrical quantities* to be considered as basic variables are generally directly expressed by their design values a_d .

(d) *Design values of the variables which account for the model uncertainties* are expressed as γ_d or $1/\gamma_d$, where γ_d are partial factors for model uncertainties (e.g. γ_{Rd} associated with the uncertainty of the resistance model).

In the design of new structures the design values of the basic variables should be determined using representative values of the basic variables and partial safety factors given in subsection 4.5.1.4.1 (representation of actions), 4.5.1.4.2 (representation of prestress), 4.5.1.4.3 (representation of material properties), 4.5.1.4.4 (representation of geometrical quantities).

When assessing existing structures, reconsideration of the design values of the basic variables may be required. Guidance is given in subsections 4.5.1.4.1 to 4.5.1.4.4, where relevant.

or

$x_d = \mu \exp(-\alpha \beta \sigma - 0.5\sigma^2)$ for a lognormal random variable (4.5-9b)

where:

x_d is the updated design value of X;

μ is the mean value of the resistance parameter X;

α is a sensitivity factor;

β is the target reliability index for an existing structure;

V is the updated coefficient of variation;

$\sigma^2 = \ln(1 + V^2)$.

The value of β for existing structures is discussed in subsection 3.3.3.1. The values of α can be taken equal to those commonly used for new structures (−0.7 for the dominating parameter at the action side, 0.8 for the dominating parameter at the resistance side and $0.3 \cdot (-0.7)$ for non-dominating parameters at the action side and $0.3 \cdot 0.8$ for non-dominating parameters at the resistance side, according to ISO 2394).

As an alternative procedure, one might also determine first a characteristic value x_k and calculate the design value by applying the appropriate partial factor γ_m .

Here:

$$x_d = x_k / \gamma_m \quad (4.5-10)$$

and

$$x_k = \mu(1 - kV) \text{ for a normal random variable} \quad (4.5-11a)$$

or

$$x_k = \mu \exp(-k \sigma - 0.5\sigma^2) \text{ for a lognormal random variable} \quad (4.5-11b)$$

where:

$k = 1.64$ is generally used

For loads and geomechanical properties, a similar procedure may be applied, but usually other distribution types will be more appropriate. For more information, see ISO 2394 “General principles on reliability for structures” and ISO 13822 “Basis for design of structures – Assessment of existing structures”.

For practical classifications of the most common actions, see the relevant Appendices to ISO 2394 and CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988).

Seismic action is considered to be an accidental action or as a variable one, depending on the geographic location of the structure (see for instance EN 1998-1:2004). In general, seismic action is considered as a variable action wherever the available information is sufficient to quantify the representative values of the seismic action with a prescribed probability of not being exceeded during a reference period t_R . If there is not sufficient information to this end (for instance in regions of very low seismicity), the seismic action is considered as accidental.

Permanent actions, self-weight included, although usually classified as fixed, may have to be considered as partially free where the effects are very sensitive to their variation in space, for example for static equilibrium and analogous verifications.

Soil reactions, for example soil pressure underneath foundation slabs or footings, are strongly influenced by soil–structure interaction. They should be determined by analysis, but the result should commonly be considered widely uncertain, especially the distribution in space.

4.5.1.4 Representative values of basic variables

4.5.1.4.1 Representation of actions

Actions should be classified as:

- direct or indirect;
- permanent, variable or accidental;

- static, quasi-static or dynamic;
- closely bounded or not-closely bounded;
- fixed or free.

Reactions, mainly on supports, should also be distinguished from directly imposed actions. Although they are taken into account like actions for some verifications, they are in reality effects of actions and may need specific reliability measures in design.

For each free action, different load arrangements should be defined.

Load arrangements are sometimes defined in the load standards. If several actions are free, the load cases (fixing the arrangements of all actions by taking into account their compatibility) are sometimes defined in the same documents. More information on load arrangements is given in section 4.2.3 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988). See also EN 1991-2 for the load arrangements due to traffic actions.

When overloading has been observed in the past, it may be appropriate to increase representative values. When some loads have been reduced or removed completely, the representative values of the load magnitudes can be appropriately reduced and/or the partial factors can be adjusted. Guidelines are given in ISO 2394 “General principles on reliability for structures” and ISO 13822 “Basis for design of structures – Assessment of existing structures”.

In the first two cases, G is considered as a mean value and should be calculated from nominal dimensions.

In the third case, it is defined as G_{sup} or G_{inf} .

The difference between G_{sup} or G_{inf} and G_m should not exceed 0.1 G_m . For some types of prestressed structures this maximum acceptable difference may have to be reduced to 0.05 G_m .

This case is mainly applicable to finishes and equipment. G_{sup} and G_{inf} may normally be defined as corresponding to 0.95 and 0.05 fractiles plus (or minus) the expected variation in time of G_m .

For structures to be designed for the most common variable actions these values are given in standards or codes associated with the same γ_F values as in this Model Code.

Ψ values depend on the model of the action – see ISO 2394.

An example of the choice of the coefficients ψ_i according to EN 1990 (Eurocode 0), “Basis of structural design”, is given in Table 4.5-1.

Table 4.5-1: The coefficients ψ_i according to EN 1990

Action	ψ_0	ψ_1	ψ_2
Variable loads in buildings:			
Category A: domestic, residential	0.7	0.5	0.3
Category B: office areas	0.7	0.5	0.3
Category C: congregation areas	0.7	0.7	0.6
Category D: shopping areas	0.7	0.7	0.6
Category E: storage areas	1.0	0.9	0.8
Category F: traffic area, ≤ 30 kN	0.7	0.7	0.6
Category G: traffic area, 30–160 kN	0.7	0.5	0.3
Category H: roofs	0	0	0
Snow load: $H \leq 1000$ m a.s.l.	0.5	0.2	0
Wind loads on buildings	0.6	0.2	0

The representative values of actions to be applied in design of new structures are given below.

When assessing existing structures, the load characteristics should be introduced with values corresponding to the actual situation.

Representative values of permanent actions

Each permanent action is represented by a single representative value G if at least one of the following conditions is satisfied:

- the variability of the action in time and with regard to the design is small;
- the influence of the action on the total effect of the actions is small;
- it is evident that one of the two representative values (the upper or the lower) governs for all parts of the structure.

In the other cases, two representative values (upper and lower, G_{sup} and G_{inf}) should be defined, taking into account variations which can be foreseen.

Nominal numerical values of densities are given in subsection 5.1.3 for plain, reinforced and prestressed concrete, and in ISO 9194 for other materials. For future possible permanent equipment an upper value should be specified.

The representative values of the prestress are defined in subsection 4.5.1.4.2.

Representative values of variable actions

Each variable action may be represented by

- characteristic value Q_k ;
- combination value $\Psi_0 Q_k$;
- frequent value $\Psi_1 Q_k$;
- quasi-permanent value $\Psi_2 Q_k$;

where:

Ψ_0 is the coefficient for the combination value of a variable action, taking into account the reduced probability of simultaneous occurrence of the most unfavourable values of several independent actions;

Ψ_1 is the coefficient for the frequent value of a variable action, generally representing the value that is exceeded 5% of the reference period;

Ψ_2 is the coefficient for the quasi-permanent value of a variable action, generally representing the value that is exceeded 50% of the reference period.

These values are associated with the methods of verification defined in subsection 4.5.2.3.

For structures to be designed, these values are normally defined by the competent public authority or by the client and correspond to the values beyond which a high probability of integrity of the structure can no longer be assured.

For ordinary facilities appropriate multiple representative seismic actions are the following:

- for the serviceability limit states as defined in subsection 3.3.1.1:
 - for the operational limit state: a “frequent” seismic action, expected to be exceeded at least once during the design service life of the structure (i.e. having a mean return period much shorter than the design service life);
 - for the immediate use limit state: an “occasional” earthquake, not expected to be exceeded during the design service life of the structure (e.g. with a mean return period of about twice the design service life);
- for the two ultimate limit states defined in subsection 3.3.1.2:
 - for the life safety limit state: a “rare” seismic action, with a low probability of being exceeded (10%) during the design service life of the structure;
 - for the near collapse limit state: a “very rare” seismic action, with very low probability of being exceeded (2–5%) in the design service life of the structure.

For facilities whose consequences of failure are very high, the “very rare” seismic action may be appropriate for the life safety limit state. For those which are essential for the immediate post-earthquake period a “rare” seismic action may be appropriate for the immediate use limit state or even the operational limit state.

It is not sufficient to define a representative seismic action by scaling standard spectral shapes to a single ground motion parameter, notably the effective or the peak ground acceleration. Instead, the seismic action should be defined in terms of its full spectrum, throughout the full range of structural periods of relevance.

Normally it is sufficient to consider only the two horizontal translational components of the ground motion.

For buildings or similar structures, in general the vertical component may be neglected, with the possible exception (depending on seismicity) of:

- horizontal members with significant concentrated masses along the span;
- long horizontal spans (e.g. over 20 m) or cantilevers (e.g. over 5 m);
- prestressed horizontal members.

In bridges, the vertical component should always be taken into account for the design of prestressed decks or bearings.

Simulated records are produced from mathematical models of the seismic source which dominates the seismic hazard, including the

Besides, for some variable actions, specific representative values are defined for fatigue verifications.

Representative values of accidental actions

Each accidental action can be given by a single representative value, which is usually the design value A_d .

Representative values of seismic actions

A representative seismic action, with a prescribed probability of not being exceeded during a reference period t_R , is defined for each limit state considered.

Depending on the use and importance of the facility, competent authorities will choose how many and which limit states should be verified as a minimum and to which representative seismic action they will be paired off.

The basic definition of each representative seismic action is through its elastic response spectrum for a single-degree-of-freedom oscillator, as a function of viscous damping (the default value being 5% of critical damping). The spectrum applies to the top of the ground under free-field conditions and should be specified taking into account the site's subsoil conditions and the local topography and geology, if relevant.

The elastic response spectrum is the same for the horizontal components of the ground motion, but should be specified separately for the vertical.

The components of the seismic action should be taken to act simultaneously.

Time-histories of the relevant components of the ground motion are needed for response-history analyses of the structure.

Preference should be given to historic or simulated records over artificial ones.

rupture event, the wave propagation through the bedrock to the site and through the subsoil to the ground surface.

Historic records should come from seismic events, with magnitude, fault distance and mechanism of rupture at the source that are consistent with those dominating the seismic hazard for the representative seismic action in question. Their travel path and the subsoil conditions of the recording station should preferably resemble those applying at the site.

Artificial (or “synthetic”) records, mathematically derived from the target elastic response spectrum, are not realistic if they are rich in all frequencies in the same way as the target spectrum. Therefore, perfect matching of the elastic response spectrum should be avoided.

The period range of interest may be taken to extend from twice to 20% of the fundamental period of the structure in the direction of the seismic action component in question.

For the estimation of peak response quantities, a minimum of seven such events is needed if the corresponding results of the analyses are averaged, or a minimum of three, if the most adverse peak response from the analyses is used. Many more seismic events than these minimum numbers are necessary for the estimation of residual deformations through non-linear response-history analyses.

Generally, during prestressing, the external forces are imposed and the associated elongations of the tendons are controlled. The prestressing load is determined at the time of its application

Even where prestress has to be considered as an action, a prestrain $\varepsilon_p(x, t)$ has commonly also to be considered in some parts of the calculations, especially in verifications with regard to the ULS. Where only immediate losses are considered $\varepsilon_p(x, t)$ is deduced from $P(x, t)$ by dividing it by the product $E_p A_p$. Where also long term losses are considered, this simple division may have to be supplemented by a correction transforming the relaxation of the tendon into a variation of strain.

Length and angular deviation may be considered small if the ratio $\Delta P_m(x, t)/P(0, 0)$ is not, at any time t , higher than 0.30.

The significance of these values is shown in section 6.3 of CEB Bulletin 191 “General Principles on Reliability for Structures”. In exceptional cases, where an increase of the strength results in a decrease in reliability, upper characteristic values and specific γ_m values (smaller than 1) should be used.

When the original design documents are available and no serious deterioration, design errors or construction errors are observed or

To conform to the basic definition of the representative seismic action, each individual component time-history should be scaled so that the values of its elastic response spectrum for the default damping are at least 90% of the specified spectrum throughout the period range of interest.

A sufficient number of independent seismic events (in terms of component time-histories) should be used for the derivation of meaningful and robust statistics of the action effects.

4.5.1.4.2 Representation of prestress

Prestressing forces are regarded as actions on the structure.

Representative values

Losses are numerically defined as mean values $\Delta P_m(x, t)$ in the subsections 5.4.5 and 5.4.6, assuming that the structure is submitted to the quasi-permanent combination of actions defined in subsection 4.5.2.

For a given set of tendons, considered in the same calculation of losses, the mean value of the prestressing force is defined as:

$$P_m(x, t) = P(0, 0) - \Delta P_m(x, t) \quad (\Delta P \text{ in absolute value})$$

Two characteristic values of the prestressing force are also defined.

In the cases where the length and angular deviation of the tendons are not exceptionally large, the following formulas, although conservative if the angular deviation is small, may be used as acceptable approximations.

(a) Bonded tendons

$$P_{k \text{ sup}}(x, t) = 1.1 P_m(x, t)$$

$$P_{k \text{ inf}}(x, t) = 0.9 P_m(x, t)$$

(b) Unbonded tendons

$$P_{k \text{ sup}}(x, t) = 1.05 P_m(x, t)$$

$$P_{k \text{ inf}}(x, t) = 0.95 P_m(x, t)$$

The design values of forces in prestressing tendons are discussed in subsection 5.4.7

4.5.1.4.3 Representation of material properties

Representative values

Strengths and other material properties to be considered as basic variables are represented by their characteristic values f_k (strength) or X_k (general properties) or by their mean values.

When assessing existing structures, the material properties must be considered according to the actual state of the structure.

suspected, the characteristic value in accordance with the original design should be used. If appropriate, destructive or non-destructive inspections should be performed and evaluated using statistical methods. For more information, see ISO 2394 “General principles on reliability for structures” and ISO 13822 “Basis for design of structures – Assessment of existing structures”.

Mean and characteristic values for strength properties of concrete and steel are given in subsection 7.2.3.

Where strengths and other material properties are not considered as basic variables in limit state equations, they may be represented by mean values f_m (or X_m) which usually are the most likely values of f , and not by other fractiles taken out of the same statistical populations as f_k values. However, these may generally be substituted by characteristic values f_k , as an acceptable approximation for such verifications.

When the original design documents are available and no change in dimensions has occurred or other evidence of deviations is known, the nominal dimensions in accordance with the original design documents should be used in the analysis. These dimensions must be verified by inspection to an adequate extent. For more information, see ISO 2394 “General principles on reliability for structures” and ISO 13822 “Basis for design of structures – Assessment of existing structures”.

In this section, only geometrical quantities representing the structure are considered. For most of the quantities, their deviations within the specified tolerances should be considered as statistically covered by γ_{Ed} and γ_{Rd} , that is by γ_F and γ_M factors. Only those quantities, which might in some verification be one of the main variables, should, in those verifications only, be taken as basic.

The depths of reinforcement in thin members are taken into account by modifying their nominal values by additive reliability margins.

Dimensions in slabs larger than intended may significantly increase the self-weight, whereas smaller dimensions and/or lever arms of steel bars may significantly reduce the resistance. Similarly, a concrete cover smaller than the nominal value may endanger the durability or the anchorage resistance of steel bars. An unintended inclination of columns may disproportionately increase their action effects.

Because of the complicated nature of the related phenomena, no explicit figure of general validity can be given on the amount of such performance reduction, but it is considerably less than 4%.

In the absence of a more justified set of tolerances, the following limitations may apply:

(a) **Table 4.5-2:** Tolerances for concrete sectional dimensions, according to ISO 22966 (for Tolerance Class 1 and 2)

Elements and dimension (mm)	Permitted deviation Δ (mm)	
	Class 1	Class 2
<i>Beams slabs and columns</i>		
$a < 150$ mm	± 10 mm	± 5 mm
$a = 400$ mm	± 15 mm	± 10 mm
$a \geq 2500$ mm	± 30 mm	± 30 mm

with linear interpolation for intermediate values

4.5.1.4.4 Representation of geometrical quantities

Representative values

The representative values of geometrical quantities to be applied in design of new structures are defined below.

When assessing existing structures the dimensions of the structural elements must be taken according to the actual state of the structure.

Unintentional eccentricities, inclinations and parameters defining curvatures affecting columns and walls and the depth of reinforcement in members thinner than 100 mm, are unique geometrical quantities defined in this Model Code to be taken into account as basic variables, if not specified otherwise. The other geometrical quantities are as specified in the drawings of the design.

The basic geometrical variables are directly fixed as design values in the chapters where the relevant limit states are treated.

Tolerances

The possible deviations in the geometry of the concrete elements, of the cover or of the position of steel, must not alter significantly either the SLS or the ULS performance of the relevant elements.

As a general rule for these geometrical basic variables, the corresponding specified tolerances may be taken equal to their design values of the deviations divided by 1.2 and should be controlled accordingly.

For the other geometrical variables, the values of the material partial safety factors included in this Model Code are meant to cover small reductions of performance (resistances, mainly) which may result from their deviations.

Depending on the quality assurance scheme applicable, relevant tolerance values should be respected for each category of possible deviations under well-specified conditions of measurements and evaluations. Possible foreseen higher deviations should lead to additional design steps, taking into account all the consequences of deviations that exceed the specified tolerances.

(b) **Table 4.5-3:** Tolerances for the location of ordinary and prestressing reinforcement, according to ISO 22966 (for Tolerance Class 1 and 2)

Height of cross-section <i>h</i> (mm)	Permitted deviation Δ (mm)	
	Class 1	Class 2
<i>Ordinary reinforcement</i>		
<i>h</i> ≤ 150 mm	+ 10 mm	+ 5 mm
<i>h</i> = 400 mm	+ 15 mm	+ 10 mm
<i>h</i> ≥ 2500 mm	+ 20 mm	+ 20 mm

with linear interpolation for intermediate values

<i>Prestressing reinforcement</i>	
<i>h</i> ≤ 200 mm	± 0.03 <i>h</i>
<i>h</i> > 200 mm	the smaller of ± 0.03 <i>h</i> or ± 30 mm

(c) *Tolerance of cover:* $c_{nom} - c_{act} < 10\text{ mm}$.

(d) **Table 4.5-4:** Tolerances of unintentional deviations of columns, walls, beams and slabs according to ISO 22966 (for Tolerance Class 1)

Elements and type of deviation	Permitted deviation Δ (mm)
<i>Columns and walls</i>	
– inclination of a column or wall at any level in a single- or multi-storey building	the larger of <i>h</i> /300 or 15 mm where <i>h</i> is free height
– deviation between centre	the larger of <i>t</i> /30 or 15 mm but not more than 30 mm where $t = (t_1 + t_2)/2$
– lateral deviation of a column between adjacent storey levels	the larger of <i>h</i> /300 or 15 mm but not more than 30 mm where <i>h</i> is free height
– location of a column or a wall at any storey level, from a vertical line through its intended centre at base level in a multistorey structure	the smaller of 50 mm or $\Sigma h/(200\ n^{1/2})$, where <i>h</i> is free height and <i>n</i> is the number of storeys and <i>n</i> > 1
<i>Beams and slabs</i>	
– location of a beam-to-column connection measured relative to the column	the larger of ± <i>b</i> /30 or ± 20 mm, where <i>b</i> is dimension of column in the same direction as Δ
– position of bearing axis of support when structural bearings are used	the larger of ± <i>l</i> /20 or ± 15 mm, where <i>l</i> is intended distance from edge

The tolerance values apply to the structure, after compaction and hardening of the concrete.

The basic design rules differ according to the limit state under consideration.

In some cases, defined in other chapters, some limit state calculations may be substituted by detailing rules or special provisions.

4.5.2 Basic rules for partial factor approach
4.5.2.1 General

In design by the partial factor method it should be proven that the structure, given the design values for the basic variables, does not reach the relevant limit states for loads below the design load. The basic design rules given in this section are applicable to the limit states as defined in chapter 3.

In principle, all relevant limit states should be considered, as well as all relevant design situations, load arrangements and load cases and combinations of actions.

Reduced values of γ may be appropriate for the assessment of existing structures, derived from reduced values of β (see subsections 3.3.3.1 and 4.5.1.3). This may be the case if large scale repair would be the consequence of using the γ values for new structures, leading to significant consequences for economy, public safety and environmental impact during repair.

These numerical values are considered to be appropriate in the design of new structures for the socioeconomic conditions in most European countries. In some countries where different conditions prevail (and possibly depending on the type of building or civil engineering works), γ factors for design may be reduced.

The γ_{Gsup} and γ_Q values given in subsection 4.5.2.2 may be reduced in the following cases:

- design of one-storey buildings (ground floor plus roof) with spans not exceeding 9 m, that are only occasionally occupied (storage buildings, sheds, greenhouses, small silos and buildings for agricultural purposes);
- floors resting directly on the ground;
- light partition walls;
- lintels;
- sheeting;
- ordinary lighting masts.

Some γ_M factors may, however, have to be increased in cases where quality measures, considered normal in the actual case, would not be expected, but this is intended to maintain the reliability degree, not to modify it.

The numerical values of γ factors given in subsection 4.5.2.2 are applicable to the design of new structures. For existing structures reduced values may be considered.

In subsection 4.5.1.3 explanations are given with regard to updating the design values of the variables. After the evaluation of the updated design values, one may check the structural reliability of existing structures using the standard procedures for new structures.

The numerical values of γ factors given in subsections 4.5.2.2 are applicable to the design of buildings and civil engineering works not subject to variable actions having an exceptional variability.

In the design of new structures the γ_{Gsup} and γ_Q values given in subsection 4.5.2.2 may be reduced respectively to 1.2 and 1.35 for reliability differentiation, provided that these reductions are not associated with a reduced quality assurance level.

If the basic set of γ factors given in this section is adopted, any increase of the reliability degree is normally limited to the consideration of supplementary hazards or higher values of accidental actions, and more refined analyses.

4.5.2.2 Ultimate limit states

4.5.2.2.1 Design principle

It should be verified that the following condition is satisfied :

$$\varepsilon < \varepsilon_u$$

where:

ε is the generic strain in the structure;

ε_u is its limit value.

For the sake of operational simplicity, the condition becomes:

$E_d < R_d$ if a single-component action-effect is to be considered;

$E_d < R_d^*$ if a multi-component action-effect is to be considered;

where:

E_d denotes a design action-effect;

R_d denotes a design resistance (and R_d^* a design resistance domain).

4.5.2.2.2 Application of partial safety factors

At the action side, at least the following variables should be differentiated:

- self-weight of the structure;
- other permanent loads;
- variable actions;
- prestressing;
- other actions (earthquake, fire, accidental actions etc.).

In operational codes, by justifying the values of the underlying assumptions, a selection of partial safety factors different from those commonly used can be obtained.

Indicative values are $\gamma_{Rd1} = 1.05$ for concrete strength and $\gamma_{Rd1} = 1.025$ for steel strength. In some cases – such as punching in the ULS, where concrete crushing is governing the behaviour – models may be affected by larger uncertainty, which can be accounted for by adding a specific factor in the verification formulas).

For taking into account geometrical uncertainties an indicative value is $\gamma_{Rd2} = 1.05$ (regarding the variability of the size of the concrete section or the position of the reinforcing steel).

For concrete strength this leads to $\gamma_{Rd,c} = \gamma_{Rd1,c} \cdot \gamma_{Rd2,c} = 1.05 \cdot 1.05 = 1.10$ and for steel strength $\gamma_{Rd,s} = \gamma_{Rd1,s} \cdot \gamma_{Rd2,s} = 1.025 \cdot 1.05 = 1.08$.

Moreover:

$$\gamma_m = \frac{R_k}{R_d} = \frac{\mu_R(1 - k \cdot \delta_R)}{\mu_R(1 - \alpha_R \cdot \beta \cdot \delta_R)} = \frac{1 - k \cdot \delta_R}{1 - \alpha_R \cdot \beta \cdot \delta_R}$$

considering a normal distribution, or

$$\gamma_m = \frac{R_k}{R_d} = \frac{\exp(\mu_{\ln R} - k \cdot \delta_{\ln R})}{\exp(\mu_{\ln R} - \alpha_R \cdot \beta \cdot \delta_{\ln R})} = \exp(-k \cdot \delta_{\ln R} + \alpha_R \cdot \beta \cdot \delta_{\ln R})$$

considering a lognormal distribution.

Commonly the 5% fractile is used for the characteristic value, yielding $k = 1.64$. Moreover, most commonly the following values are used:

α_R = 0.8 being the sensitivity factor of the parameter under consideration, based on the simplified level II method as suggested by König and Hosser in CEB Bulletin 147: “Conceptual Preparation of Future Codes – Progress Report” (CEB, 1982).

β = 3.8 for structures of consequence class 2 according to EN 1990.

δ_R = coefficient of variation of the parameter under consideration: for example $\delta_c = 0.15$ is commonly used for normal quality concrete and $\delta_s = 0.05$ for reinforcing steel.

Based on these commonly used values and considering a normal distribution $\gamma_c = 1.39$ and $\gamma_s = 1.08$.

This finally results in:

$$\gamma_c = \gamma_{Rd,c} \cdot \gamma_c = 1.10 \cdot 1.39 = 1.52 \cong 1.50 \text{ and}$$

$$\gamma_s = \gamma_{Rd,s} \cdot \gamma_s = 1.08 \cdot 1.08 = 1.17 \cong 1.15.$$

The commonly used partial safety factors mentioned before can be modified in operational codes, by justifying the values of the underlying assumptions.

An indicative value is $\gamma_{Sd} = 1.05$, in case of a permanent load.

For unfavourable permanent actions, the partial factor γ_g can be derived as:

$$\gamma_{g,\text{sup}} = \frac{G_d}{G_k} = \frac{\mu_G(1 - \alpha_E \cdot \beta \cdot \delta_G)}{\mu_G} = 1 - \alpha_E \cdot \beta \cdot \delta_G$$

At the resistance side, at least the following parameters should be differentiated:

- concrete strength;
- steel strength;
- model uncertainty.

4.5.2.2.3 Determination of partial safety factors

For the sake of simplification, uncertainties related to some variable can be incorporated into the partial factors of another variable (e. g. some geometric uncertainties are incorporated in γ_m).

Materials

For materials the following relations apply:

$$\gamma_{Rd} = \gamma_{Rd1} \cdot \gamma_{Rd2}$$

$$\gamma_M = \gamma_m \cdot \gamma_{Rd}$$

where:

γ_m = partial safety factor for material properties;

γ_{Rd1} = partial safety factor accounting for model uncertainty;

γ_{Rd2} = partial safety factor accounting for geometrical uncertainties.

Permanent loads

For permanent loads, the following relation applies:

$$\gamma_G = \gamma_{Sd} \cdot \gamma_g$$

where γ_{Sd} is partial safety value accounting for model uncertainty.

where most commonly the following values are used:

- α_E = -0.7 being the sensitivity factor of the parameter under consideration, based on the simplified level II method as suggested by König and Hosser in CEB Bulletin 147: “Conceptional Preparation of Future Codes – Progress Report” (CEB, 1982);
- β = 3.8 for structures of consequence class 2 according to EN 1990;
- δ_G is coefficient of variation for permanent loads, for example $\delta_G = 0.05$ or $\delta_G = 0.10$ if no distinction is made between self-weight and other permanent actions.

Based on these commonly used values and considering a normal distribution, the following values are found:

$$\begin{aligned}\gamma_G &= \gamma_{g,\text{sup}} = 1.13 \text{ when } \delta_G = 0.05 \\ \gamma_G &= \gamma_{g,\text{sup}} = 1.27 \text{ when } \delta_G = 0.10 \\ \gamma_G &= \gamma_{Sd,g} \cdot \gamma_{g,\text{sup}} = 1.05 \cdot 1.13 = 1.19 \cong 1.20 \text{ when } \delta_G = 0.05 \\ \gamma_G &= \gamma_{Sd,g} \cdot \gamma_{g,\text{sup}} = 1.05 \cdot 1.27 = 1.33 \cong 1.35 \text{ when } \delta_G = 0.10\end{aligned}$$

Preferably, there should be a distinction between partial safety factors related to *self-weight* (well defined and constant intensity) and *other permanent loads*. Furthermore, it should be noted that some “permanent actions” may vary considerably; then they should be considered as variable actions (e.g. earth coverings or doubling the weight of floor finishing on a slab). Based on the previous formulas, the partial safety factors for self-weight and other permanent actions can be derived as follows.

Considering a coefficient of variation of $\delta_{G,sw} = 0.05$ for self-weight and $\delta_{G,sw} = 0.10$ in the case of other permanent actions, the suggested partial factors in the case of unfavourable permanent actions become:

$$\begin{aligned}\gamma_{G,sw} &= \gamma_{Sd,g} \cdot \gamma_{g,\text{sup}} = 1.05 \cdot 1.13 \cong 1.20 \text{ when } \delta_G = 0.05 \\ \gamma_{G,pa} &= \gamma_{Sd,g} \cdot \gamma_{g,\text{sup}} = 1.05 \cdot 1.29 \cong 1.35 \text{ when } \delta_G = 0.10\end{aligned}$$

However, as noted before, the latter figure might require much higher values for “permanent” actions that can undergo modifications.

The general context of γ -factors for loads is defined in section 6.2.2 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988).

An example of particular actions is that of some hydraulic actions – see CEB Bulletin 201: “Recommendations for Mechanical Splices of Reinforcing Bars – Recommendations for Spacers, Chairs and Tying of Steel Reinforcement – Reliability Considerations for Hydraulic Variables” (CEB, 1991).

Prestressing is in most situations intended to be favourable so that a general value of $\gamma_p = 1.0$ is appropriate. This also applies to tendons in cross-sections which might be considered to act “unfavourably” as a single element but favourably if regarded in combination with other tendons. Therefore in general cases $\gamma_{p,\text{fav}} = \gamma_{p,\text{unfav}} = 1.0$.

In particular cases such as the verification of the ultimate limit state for stability with external prestress, where an increase of the prestressing force can be unfavourable, a value $\gamma_{p,\text{unfav}} > 1.0$ should be used. For global effects $\gamma_{p,\text{unfav}} = 1.3$ is appropriate, whereas for local effects $\gamma_{p,\text{unfav}} = 1.2$ may be considered to be sufficient.

4.5.2.2.4 Common values for partial safety factors

(a) γ_F factors

(a1) *Persistent and transient situations.* The numerical values applicable to non-particular actions for the limit state of static equilibrium are given in the following tables and sections.

Table 4.5-5: Partial safety factors for loads in the limit state of static equilibrium

Actions	Unfavourable effect (γ_{sup})	Favourable effect (γ_{inf})
Permanent (G), γ_G	1.05–1.1	0.9–0.95
Prestress (P), γ_P	1.0	1.0
Leading variable action ($Q_{k,1}$), γ_Q	1.5	usually neglected
Accompanying variable action ($Q_{k,i}$), γ_Q	1.5 $\Psi_{0,i}$	usually neglected

The basic numerical values applicable to the ultimate limit state in case of non-particular actions not involving geotechnical actions are given in the following table and sections.

Tables 4.5-5–7 are basically valid for buildings.

In Tables 4.5-5–8 the design value of the prestress (P) may be based on the mean value of the prestressing force.

The basic values given in Table 4.5-6 are in some cases conservative for the design of new structures. See subsection 3.3.3.1 and subsection 4.5.2 for reliability differentiation.

Table 4.5-6: Partial safety factors for loads in the design of structural members not involving geotechnical actions: basic values

Actions	Unfavourable effect (γ_{sup})	Favourable effect (γ_{inf})
Permanent (G), γ_G	1.35	1.0
Prestress (P), γ_P	1.0	1.0
Leading variable action ($Q_{k,1}$), γ_Q	1.5	usually neglected
Accompanying variable action ($Q_{k,i}$), γ_Q	1.5 $\Psi_{0,i}$	usually neglected

In the most common cases one of γ_G ($\gamma_{G,\text{sup}}$ or $\gamma_{G,\text{inf}}$) may be applied globally to all permanent actions (unfavourable or not), except prestress. The other cases should be identified by judgement.

Alternatively, a more refined approach can be taken in the design of structural members not involving geotechnical actions: the less favourable of the combination of the partial γ_F factors given in the following table (SET1 or SET2) can be used.

Table 4.5-7: Partial safety factors γ_F for loads in the design of structural members not involving geotechnical actions: alternative combination of values

Actions, γ_F	Unfavourable effect (γ_{sup})	Favourable effect (γ_{inf})
SET1		
Permanent (G), γ_G	1.35	1.0
Prestress (P), γ_P	1.0	1.0
Leading variable action ($Q_{k,1}$), γ_Q	1.5 $\Psi_{0,1}$	usually neglected
Accompanying variable action ($Q_{k,i}$), γ_Q	1.5 $\Psi_{0,i}$	usually neglected
SET2		
Permanent (G), γ_G	0.85 · 1.35	1.0
Prestress (P), γ_P	1.0	1.0
Leading variable action ($Q_{k,1}$), γ_Q	1.5	usually neglected
Accompanying variable action ($Q_{k,i}$), γ_Q	1.5 $\Psi_{0,i}$	usually neglected

Safety is normally ensured by the design values of the action or of the other parameters describing the accidental or seismic situation.

The general content of γ_M factors is defined in subsection 6.3.2 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988).

As a simplification a conversion factor η is included in γ_c .

The values of γ_c and γ_s , given in Table 4.5-8 should be increased if the geometrical tolerances given in subsection 4.5.1.4.4 are not fulfilled. Conversely they might be reduced by 0.1 and 0.05 respectively, at the maximum, if these tolerances are reduced by 50% and are strictly controlled (e.g. precast concrete components and structures).

A variation of γ_c or γ_s , according to the degree of control of f_{ck} (without making the control tests more severe), does not seem to be justified, because the variation of the control can more rationally be taken into account by the compliance criteria included in the control itself. In any case, it cannot be numerically fixed independently of the control criteria. In some cases (for instance as a result of very good quality management, (e.g. for precast concrete) the coefficient of variation δ_c considered for the derivation of partial safety factors may be reduced, according to the method described in the subsection 4.5.2.2.3.

(a2). γ_F factors for accidental or seismic situations

The values of γ_F applicable to all actions are equal to 1.

(b) γ_M factors

The numerical values of γ_M to be used for calculating R_d are given in Table 4.5-8.

Table 4.5-8: Partial factors γ_M for structural materials

Basic variable	Design situation	
	Persistent/transient	Accidental
Concrete		
Compressive strength (f_{ck}), γ_c	1.5	1.2
Tensile strength (f_{ctk}), γ_{ct}	*	*
Reinforcing and prestressing steel		
Tensile strength (f_{stk}), γ_{st}	1.15	1.0
Compressive strength (f_{sck}), γ_{sc}	1.15	1.0

* See relevant sections

The γ_M factors applicable to other basic variables are given in the relevant sections.

Strengths may intervene in E_d via stiffness and the spatial distribution throughout the structure. They may generally be favourable as well as unfavourable and are not to be considered as basic variables.

These rules must be amended for accidental situations (see the section regarding general rules for combinations of actions in the sequel) and if possible simplifications or refinements regarding combinations of actions are applied – see Eq. (4.5-17).

Eq. (4.5-13) is the more general. Particular cases are mainly those where:

- E_d is an under-proportional function of the actions (or the principal of them); in these cases Eq. (4.5-12) may be unsafe, or:
- the effects of some actions have a sense opposite to the effects of the other actions and are of the same order of magnitude; in these cases Eq. (4.5-12) may be too conservative (this may be the case for the isostatic effects of prestress).

This rule (not splitting γ_M into γ_m and γ_{Rd}) is not applicable in design by testing.

For the definition of individual actions, see subsections 1.2.1 and 6.2.1 of CEB Bulletin 191: “General Principles on Reliability for Structures – A commentary on ISO 2394 approved by the Plenum of the JCSS” (CEB, 1988).

For the Ψ factors, see the information regarding representative values of variable actions in subsection 4.5.1.4.1.

Ψ factors take account of the reduced probability of simultaneous occurrence of actions. The choice between $\Psi_{1,1}Q_{k,1}$ or $\Psi_{2,1}Q_{k,1}$ depends on the type of accidental design situation, such as impact, fire or survival after an accidental event or situation.

In seismic situations masses are consistent with the gravity loads corresponding to the combination $G_{\text{sup}} + G_{\text{inf}} + \sum_{i \geq 1} \Psi_{2,i}Q_{k,i}$.

Prestressing P should be added, if relevant.

In most cases some variable actions, which obviously are not the leading ones for a given verification, need not be considered as $Q_{k,1}$.

For fire situations, apart from the temperature effect on the material properties, A_d should represent the design value of the indirect thermal action due to fire.

In general, there will be two different levels of A_{Ed} , one for each ultimate limit state introduced in subsection 3.3.1.2.

Whenever strengths intervene in the value of the action-effect S_d , the associated γ_M values should be taken equal to 1. This rule is not applicable to buckling verifications, in which strengths are important favourable basic variables.

(c) Introduction of the partial coefficients into the calculations

In most cases, γ_F factors should be applied globally as follows:

$$E_d = E \left\{ \gamma_G G + \gamma_P P + \gamma_Q \left(Q_{1k} + \sum_{i>1} \Psi_{oi} Q_{ik} \right) \right\} \quad (4.5-12)$$

In particular cases, defined in the relevant sections of other chapters or to be identified by judgement, for persistent or transient situations, this formula may be substituted by:

$$E_d = \gamma_{Sd} E \left\{ \gamma_g G + \gamma_P P + \gamma_q \left(Q_{1k} + \sum_{i>1} \Psi_{oi} Q_{ik} \right) \right\} \quad (4.5-13)$$

where the partial factors should be taken by referring to the preceding section (a1).

These two formulas are partially symbolic and should be applied by following in detail the combination rules given in the sequel.

The use of a sum of permanent actions $\gamma_{G,i} G_{k,i}$ instead of a single permanent load G is allowed.

γ_M factors should generally be applied globally.

Combinations of actions

(a) General rules

The combinations of design values to be taken into account for applying Eqs. (4.5-12) and (4.5-13) are as follows, in symbolic presentation:

- fundamental combinations applicable for persistent and transient situations

$$E_d = \gamma_{G \text{ sup}} G_{\text{sup}} + \gamma_{G \text{ inf}} G_{\text{inf}} + \gamma_P P + \gamma_{Q,1} Q_{k,1} + \sum_{i>1} \gamma_{Q,i} \Psi_{0,i} Q_{k,i} \quad (4.5-14)$$

- accidental combinations, applicable for accidental situations

$$E_d = G_{\text{sup}} + G_{\text{inf}} + P + (A_d \text{ or } 0) + (\Psi_{1,1} \text{ or } \Psi_{2,1}) Q_{k,1} + \sum_{i>1} \Psi_{2,i} Q_{k,i} \quad (4.5-15)$$

- seismic combinations, applicable for seismic situations

$$E_d = G_{\text{sup}} + G_{\text{inf}} + P + A_{Ed} + \sum_{i \geq 1} \Psi_{2,i} Q_{k,i} \quad (4.5-16)$$

In these combinations:

- G_{sup} and G_{inf} refer to the unfavourable and favourable parts of the permanent actions, respectively;
- P refers to prestressing;
- $Q_{k,i}$ refers to any variable action, in succession;

- A_d denotes the unique accidental action associated with the accidental situation, if this situation is due to this action. If it is due to another event or to a past action, A_d is substituted by 0.

- A_{Ed} denotes the design seismic action having a prescribed probability of not being exceeded during the reference period t_d and associated with the ultimate limit state of interest in this specific seismic situation.

The cases of incompatibility or negligible compatibility are very numerous. They are given in the codes or standards on actions or identified by judgement (e.g. snow and maximum climatic temperature).

Other simplifications may be envisaged and discussed, for example by directly giving design combinations for a given set of common variable actions, such as some imposed loads, wind, snow and temperature.

Judgement is necessary because the concept of one action is very blurred. For example, the actions of wind, snow, water and imposed loads should be considered as different actions, but the imposed loads on different floors should be considered as one action.

This simplification is mainly intended for common buildings. The influence of this simplification on the resulting reliability should be carefully analysed.

Attention is drawn to the risk that an accident results in consequences on variable actions; for example many persons may gather in some places in order to escape during or immediately after an accident.

This may be the case, for example, if a failure should be limited to a small part of the structure.

This introduces one more combination. Attention is drawn to the necessity, in this case, to verify more completely and carefully than usual the serviceability limit states, which may be less covered than usually by ultimate limit state verifications.

In many cases, this does not result in important changes of design.

Static actions not repeated more than 10^4 times or for which $\psi_1 = 0$ are considered unable to produce fatigue failure. Examples of actions able to cause fatigue are loads due to vehicles, moving machinery, wind (gusts, turbulence, vortices etc.) and wave action.

This is an indirect verification that the loss of strength will not be significant.

In assessing the stress range, stress variations in opposite senses (due for example to successive arrangements of a moveable load) must be, if relevant, taken into account.

Other design properties associated with tensile stress of concrete (e.g. a formal shear stress) may also have to be considered.

The actions to be included in any combination are only those that are mutually compatible or are considered as such, as an acceptable approximation. Non-simultaneous actions should be considered in the same combination if their effects are simultaneous.

(b) Possible simplifications

As an approximation to be recognized by judgement, it is frequently sufficient to limit the total number of variable actions to a maximum of three in any fundamental combination and to two in any accidental combination.

Fundamental combinations that are obviously identified as non-critical may be omitted in the calculations.

In many cases Ψ_{oi} factors may be merged with γ_Q and E_d may then be calculated, for persistent and transient situations, by

$$E_d = E \left(\gamma_G G + \gamma_Q \sum_{i=1}^n Q_{ik} \right) \quad (4.5-17)$$

where:

$\gamma_G = 1$ or 1.35 (take the more unfavourable);

$\gamma_Q = 1.5$ for $n = 1$, or 1.35 for $n \geq 2$ (take the more unfavourable).

In accidental combinations $\Psi_{1,1}$ may often be substituted by the lower value $\Psi_{2,1}$, for most, or all, variable actions, as a judged approximation or because the occurrence of a greater value during the accidental situation is judged to be very unlikely.

(c) Possible refinements

In cases where the most likely consequences of a failure do not seem to be exceptionally severe, the following reductions of γ_F factors in fundamental combinations are possible.

– reduce $\gamma_{G,\text{sup}}$ to 1.2 or, alternatively, $Q_{k,1}$ to $\Psi_{01}Q_{k,1}$, or

– reduce the γ_Q value applicable to $\Psi_{oi}Q_{k,i}$ ($i > 1$) to 1.2.

4.5.2.3 Fatigue verification

Design principles

Fatigue design must ensure that in any fatigue-endangered cross-section, the expected damage D will not exceed a limiting damage D_{lim} . The verifications of this requirement can be performed according to four methods of increasing refinement.

Level I approximation

This is a qualitative verification that no variable action is able to produce fatigue. If the conclusion of this verification is not positive, a verification according to one of the higher levels must be made.

Level II approximation:

This is a verification by a simplified procedure. It is verified whether the following stresses or stress ranges:

- the maximum design stress range in the steel $\gamma_{Ed} \Delta \sigma_s(G, P, \Psi_1 Q_k)$;
- the maximum design concrete compressive stress $\gamma_{Ed} \sigma_{c,\text{max}}(G, P, \Psi_1 Q_k)$;
- the maximum design tensile stress in plain concrete $\gamma_{Ed} \sigma_{ct,\text{max}}(G, P, \Psi_1 Q_k)$;

do not exceed the limit values given in subsection 7.4.1.

If the stress analysis is sufficiently accurate or conservative, and this fact is verified by in-situ observations, it may be possible to take $\gamma_{Ed} = 1.0$.

In Eq. (4.5-18) the term between the brackets is the static part and the term

Q_{fat} is the dynamic part.

For Q_{fat} , in many cases the frequent value $\Psi_{1,1}Q_k$ may be used as an equivalent or conservative approach.

The load factor γ_{Ed} is assumed to be 1.1.

Level III approximation:

This verification refers to a representation of the variable load dominant for fatigue by a single load level Q associated with a number of repetitions n during the required lifetime.

The stresses in the structural materials, or the stress range, are calculated under the following combination of actions:

$$(G_{sup} + G_{inf} + P + \Psi_{1,1} Q_{k,1} + \sum_{i>1} \Psi_{2,i} Q_{k,i}) + Q_{fat} \quad (4.5-18)$$

where:

Q_{fat} is the relevant fatigue load (e. g. traffic load or other cyclic load).

The stresses found under the load according to Eq. (4.5-18) are multiplied by a factor $\gamma_{Ed} = 1.1$, or 1.0 if accurate stress analysis is possible. At the resistance side the strength of the materials is divided by $\gamma_{s,fat} = 1.15$ for the steel and $\gamma_{c,fat} = 1.5$ for the concrete.

Level IV approximation:

This is a verification based on an assessment of the fatigue damage resulting from various magnitudes of loads. According to this method, the load history during the required life is represented by a spectrum in a discretized form. The accumulation of fatigue damage is calculated on the basis of the Palmgren–Miner summation.

4.5.2.4 Verification of structures subjected to impact and explosion

Impact and explosions are regarded as accidental loads, so Eq. (4.5-15) applies.

4.5.2.5 Serviceability limit states

Design principle

(a) Limit state of cracking and excessive compression

It should be verified that in any cross-section:

$\sigma(F_d) < \alpha f_d$ for crack formation and excessive creep effects;

$w_d(F_d, f_d) < w_{lim}$ for design crack width;

$\sigma(F_d) \leq 0$ for crack re-opening;

where:

σ is a defined stress;

F_d is the design value of action;

f_d is a tensile, shear or compressive design strength;

w_d is a defined crack width;

α is a reduction factor for the case considered, with $0 \leq \alpha \leq 1$.

(b) Limit state of deformations

It should be verified that:

$$a(F_d, f_d) \leq C_d \quad (4.5-19)$$

where:

a is a defined deformation (generally a deflection or a rotation at a member end);

F_d and f_d are values as defined under (a);

C_d is the limit value for the deformation considered.

(c) Limitation of vibrations

In the most common cases, the limitation is ensured by indirect measures, such as limiting the deformations or the periods of vibration of the structure in order to avoid the risk of resonance. In the other cases, a dynamic analysis is necessary.

As mentioned in subsections 7.6.4.6. and 7.6.5.2.4. some of these rules may be substituted by stress limitations, detailing rules or other indirect verifications.

The α -factor (e. g. 0.6 for excessive compression) describes the limit state and is not a reliability factor.

In such equations f_d generally is not to be considered as a basic variable.

This rule may in some cases be substituted by a maximum slenderness ratio.

If not fixed by the Code, C_d should be fixed by the contract or chosen by the designer, possibly depending on non-structural parts.

See subsection 7.6.6.

Pragmatic values smaller than 1 may be envisaged for indirect actions.

Values of partial factors

- (a) γ_F factors are taken equal to 1;
- (b) γ_M factors are taken equal to 1.

Combinations of actions

(a) General rules

The combinations which should be considered depend on the particular limit state under consideration and are identified in the corresponding chapters.

They are defined as follows, in a symbolic presentation:

$$\text{characteristic:} \quad G + P + Q_{k,1} + \sum_{i>1} (\Psi Q_{k,i}) \quad (4.5-20)$$

$$\text{frequent:} \quad G + P + \Psi_{1,1} Q_{k,1} + \sum_{i>1} (\Psi_{2,i} Q_{k,i}) \quad (4.5-21)$$

$$\text{quasi-permanent:} \quad G + P + \sum_{i \geq 1} (\Psi_{2,i} Q_{k,i}) \quad (4.5-22)$$

$$\text{seismic:} \quad G + P + A_{Ek} + \sum_{i \geq 1} (\Psi_{2,i} Q_{k,i}) \quad (4.5-23)$$

In general, there will be two different levels of A_{Ek} – one for each serviceability limit state introduced in subsection 3.3.1.1.

In the seismic situations masses are consistent with the gravity loads corresponding to the combination $G_{\text{sup}} + G_{\text{inf}} + \sum_{i \geq 1} \Psi_{2,i} Q_{k,i}$.

where:

- G is taken according to subsection 4.5.1.4.1;
- P is the mean value of the prestressing load, as defined in subsection 4.5.1.4.2, where the most unfavourable value (with or without losses) should be applied;
- $Q_{k,i}$ refers to any variable action, successively;
- A_{Ek} is the representative seismic action prescribed for the serviceability limit state of interest.

(b) Possible simplification

The first two paragraphs of subsection 4.5.2.2 regarding possible simplifications for combination of actions may be applied to combinations for serviceability limit states.

In common cases for reinforced concrete structures, the characteristic combinations may be simplified by avoiding reference to various Ψ_{oi} factors. They are substituted, in a symbolic presentation, by

$$G + Q_{k,1} \quad (4.5-24)$$

or

$$G + 0.9 \sum_{i=1}^n Q_{k,i} \quad (\text{take the more unfavourable}) \quad (4.5-25)$$

in which $Q_{k,1}$ is the most unfavourable variable action.

4.6 Global resistance format

4.6.1 General

The global resistance approach was initiated by the introduction of non-linear analysis, which is based on a global structural model and offers tools for the safety assessment. It is a general approach, which follows the probabilistic safety concept more rationally than the partial factor method. It is applicable to the safety check on structural level. However, it can be applied also to members or sections as well.

The global safety factor reflects the variability of the structural response due to random properties of basic variables. The effect of random variation of basic variables, such as strength f on resistance R is dependent on the type of limit state function $r(f, \dots)$. The limit state function is represented by non-linear numerical analysis. Thus, for dominating concrete failure the resistance variability is much higher than for steel failure. This also means that the variability of resistance is, in general, not constant for a given set of material parameters and their random variations and depends on the structural model considered.

The global resistance has a general meaning and usually describes the response of an element or a structure to given load actions. The resistance can be described by a scalar, vector or a function, depending on design and limit state formulation. A significant feature of the structural resistance is the integration of various random effects of material properties, dimensions and so on, and their interactions. Unlike in the partial factor design method, the uncertainties are evaluated on a global structural level and not in local material points.

The meaning of global resistance can be illustrated by an example of a simple beam under the action of a force. The global resistance is expressed by the ultimate force, which can be resisted by the beam. This resistance covers all material properties, geometry, reinforcement, boundary conditions and modes of failure. Typically, the beam can fail in bending or in shear and both of these failure modes are described by the same variable – maximum force resisted by the beam. The same calculation model, for example a finite element analysis, is used and the failure mode is detected automatically in the analysis.

The uncertainty of resistance R is described by its random distribution function with its parameters: function type, mean, standard deviation (and possibly others). The parameters of scatter for a given random distribution of resistance can be used to derive the mean, characteristic and design values of resistance R_m , R_k , R_d . The global safety can be expressed either by a global safety factor or by a reliability index.

In contrast, if the same beam is verified by the partial safety factor method, a specific section is considered and local checks are made for specific actions in a cross-section. Two separate verifications are performed in the section, one for the bending failure and another one for shear failure. The global safety is not evaluated, but it is guaranteed by the formulation of partial safety factors.

In many cases it is possible to estimate the mean and the characteristic values of resistance by the values of resistance derived from mean and characteristic values of the basic variables, respectively. When the mean value obtained in this way differs from the mean value obtained by other means (e. g. experiments) special care is advised.

The global resistance format treats the uncertainties of the structural behaviour as described by the limit state condition according to Eq. (4.3-4) on the level of structural resistance. The effects of various uncertainties (of material properties, geometrical dimensions etc.) are integrated in a global design resistance and can also be expressed by a global safety factor. The representative values of the global resistance variables and the global safety factors should be chosen in such a way that the reliability requirements for the design of new structures, which are expressed in subsection 3.3.3.1 in terms of reliability index β related to the reference period, are met.

4.6.2 Basic rules for global resistance approach

4.6.2.1 Representative variables

The representative variable for the global resistance is the structural resistance R .

The uncertainty of resistance is expressed by the following values of resistance:

R_m mean value of resistance;

R_k characteristic value of resistance (corresponding to a 5% fractile);

R_d design value of resistance.

The basic variables, defined for the partial factors in subsection 4.5.2.1, are used for calculating the resistance values. The values of these variables (f , a , ...) should be chosen in accordance with the safety formats described further in this chapter. The value of action F is considered in the same way as in the partial factor method.

It is important to recognize, that in the present formulation the global safety factor γ_R^* is related to the mean variable. To distinguish this from the partial safety factors, which are referring to characteristic values a notation with asterisk superscript is used.

Furthermore, it is useful to introduce a scaling factor for a loading pattern. In general, action F_d and resistance R_d , which appear in design Eq. (4.6-2), may include many components (e. g. vertical and horizontal forces, body forces and temperature) and can be described by a point in a multidimensional space. The resistance scaling factor k_R describes the relation between resistance and action and has the same meaning as a safety factor. In a symbolic form, considering a pair of corresponding components it can be defined as:

$$k_R = \frac{R_m}{F_d} \quad (4.6-4)$$

Then, the design condition formulated in Eq. (4.6-2) can be rewritten as:

$$k_R \geq \gamma_R^* \quad (4.6-5)$$

where γ_R^* is a required global safety factor for resistance. If relevant, the global safety factor can include the model uncertainty. The factor k_R can be used to calculate the relative safety margin m_R for resistance:

$$m_R = k_R - \gamma_R^* \quad (4.6-7)$$

The model uncertainty factor γ_{Rd} should be chosen based on the knowledge of the design conditions of the structure during its service life. The value $\gamma_{Rd} = 1.0$ should be used only in exceptional cases, when an evidence of the model validation in the design conditions is available. An example of such a condition is the case of assessment of an existing structure.

The value $\gamma_{Rd} = 1.06$ should be used for models based on a refined numerical analysis, such as non-linear finite element analysis. The model should be objective (low mesh sensitivity) and validated. The factor 1.06 does not cover the errors due to approximations in the numerical model. It covers the other effects not included in the numerical model, such as time effects and environmental effects. An example of such a case is the usual design according to the partial safety factor method.

The value $\gamma_{Rd} = 1.1$ should be used for models sufficiently validated as in the case above, but with a higher uncertainty of structural conditions due to an unknown design situation. An example of such a case is a design under uncertain load history due to actions imposed by environmental effects.

4.6.2.2 Design condition

The design condition derived from Eq. (4.3-4) for the global format takes the following form:

$$e(F_{d,...}) \leq r(R_{d,...}) \quad (4.6-1)$$

In a simplified force representation, it can take the form:

$$F_d \leq R_d \quad (4.6-2)$$

The design and mean values of resistance are related as:

$$R_d = R_m / \gamma_R^* \quad (4.6-3)$$

where γ_R^* is the global safety factor for mean resistance.

The global safety factor γ_R^* accounts for random uncertainties of model parameters, namely of material properties. An uncertainty due to model formulation, must be treated by a separate safety factor for model uncertainty γ_{Rd} . This can be applied either to the action or to the resistance. In the latter case, the design resistance takes the form:

$$R_d = \frac{R_m}{\gamma_R^* \gamma_{Rd}} \quad (4.6-6)$$

The value of the model uncertainty factor depends on the quality of formulation of the resistance model. The recommended values are:

$\gamma_{Rd} = 1.0$	for no uncertainties;
$\gamma_{Rd} = 1.06$	for models with low uncertainties;
$\gamma_{Rd} = 1.1$	for models with high uncertainties.

The deemed-to-satisfy approach is applicable both for the traditional structural design and for the design associated to durability. The method may comprise sets of predetermined alternatives given in a standard. In most operational standards the design associated with durability is based on the deemed-to-satisfy approach.

Traditionally, durability related deemed-to-satisfy provisions include requirements to the workmanship, concrete composition, possible air entrainment, cover thickness to the reinforcement, crack width limitations and curing of the concrete. However, other provisions may also be relevant.

Examples of the calibration of deemed-to-satisfy criteria based on a probabilistic safety format and data derived from 10–15 years old structures are given in *fib* Bulletin 34: “Model code for Service Life Design” (*fib*, 2006).

Durability related exposure categories in the design situations may be classified in exposure classes. For more information on classification of environmental actions as exposure classes, see ISO 22965-1, “Concrete – Part 1: Methods of specifying and guidance for the specifier”.

In Table 4.7-2 a classification of exposure classes according to ISO 22965-1 is given. The same classification is adopted by the European CEN standards on the design of concrete structures.

Table 4.7-2: Exposure classes related to environmental conditions for concrete with reinforcement or embedded metal according to ISO 22965-1

Class designation	Environmental conditions and examples
<i>No risk of corrosion or attack</i>	
X0	Exposure to very dry environment, for example: components inside buildings with very low air humidity and no risk of corrosion or attack
<i>Corrosion induced by carbonation</i>	
XC1	Exposure to dry or permanently wet environment, for example: interior of buildings with low air humidity, components permanently submerged in water, for example: surfaces exposed to airborne chlorides
XC2	Exposure to wet or rarely dry environment, for example: surfaces subject to long term water contact, like foundations, swimming pools and components exposed to industrial waters containing chlorides
XC3	Exposure to moderate humid or cyclic wet and dry environment, for example: components inside buildings with moderate or high air humidity, exterior of buildings sheltered from rain
XC4	Exposure to cyclic wetting and drying, for example concrete surfaces subjected to water contact, not within exposure class XC2

4.7 Deemed-to-satisfy approach

4.7.1 General

The deemed-to-satisfy approach is a set of rules for

- dimensioning;
- material and product selection; and
- execution procedures

that ensures that the target reliability for not violating the relevant limit state during the design service life is not exceeded when the concrete structure or component is exposed to the design situations.

The specific requirements for design, materials selection and execution for the deemed-to-satisfy approach must be determined in either of two ways:

- on the basis of statistical evaluation of experimental data and field observations according to requirements of section 4.4 regarding the probabilistic safety format;
- on the basis of calibration to a long term experience of building tradition.

The limitations to the validity of the provisions – such as the range of cement types covered by the calibration – must be clearly stated.

4.7.2 Durability related exposure categories

In the absence of a more specific study, the durability related exposure categories related to environmental conditions may be classified for concrete with reinforcement or embedded metal as given in Table 4.7-1.

Table 4.7-1: Durability related exposure categories related to environmental conditions for concrete with reinforcement or embedded metal

Exposure categories	Environmental conditions
<i>No risk of corrosion or attack</i>	Exposure to very dry environment
<i>Corrosion induced by carbonation</i>	Exposure to air and moisture
<i>Corrosion induced by chlorides other than from seawater</i>	Exposure to de-icing agents or airborne chlorides
<i>Corrosion induced by chlorides from seawater</i>	Exposure to seawater
<i>Freezing and thawing attack</i>	Exposure to moisture and freeze-thaw cycles
<i>Chemical attack</i>	Exposure to aggressive chemical environment, for example components exposed to aggressive chemical environment (gas, liquid or solid) or aggressive industrial atmosphere

Class designation	Environmental conditions and examples
<i>Corrosion induced by chlorides other than from seawater</i>	
XD1	Exposure to moderate humid environment and chlorides from sources other than from seawater (e.g. chlorides from de-icing agents), for example: surfaces exposed to airborne chlorides
XD2	Exposure to wet or rarely dry environment and chlorides from sources other than from seawater (e.g. chlorides from de-icing agents)
XD3	Exposure to cyclic wet and dry environment and chlorides from sources other than from seawater (e.g. chlorides from de-icing agents), for example: pavements, car park slabs, components exposed to spray containing chlorides.
<i>Corrosion induced by chlorides from seawater</i>	
XS1	Exposure to airborne salt but not in direct contact with seawater, for example: surfaces near to or on the coast
XS2	Exposure to permanent saturation in seawater, for example: components of marine structures permanently submerged in seawater.
XS3	Exposure to seawater in tidal, splash and spray zones, for example: components of marine structures
<i>Freezing and thawing attack</i>	
XF1	Exposure to freeze-thaw cycles and moderate water saturation without de-icing agent, for example: vertical surfaces exposed to rain and freezing
XF2	Exposure to freeze-thaw cycles moderate water saturation in combination with de-icing agent, for example: vertical surfaces of road structures exposed to freezing and airborne de-icing agents
XF3	Exposure to freeze-thaw cycles and high water saturation without de-icing agent, for example: horizontal surfaces exposed to rain and freezing
XF4	Exposure to freeze-thaw cycles and high water saturation in combination with de-icing agent, for example: road and bridge decks exposed to de-icing agents; surfaces exposed to direct spray containing de-icing agents and freezing; splash zone of marine structures exposed to freezing
<i>Chemical attack</i>	
XA1	Exposure to slightly aggressive chemical environment
XA2	Exposure to moderately aggressive chemical environment
XA3	Exposure to highly aggressive chemical environment

4.8 Design by avoidance

Traditional structural design involving the avoidance method includes a concept based on avoiding or reducing the detrimental effect, for example sheltering the structure from certain loads such as environmental loads, wind, wave loads impact by vehicles or missiles.

In design for durability the avoidance-of-deterioration method implies that the deterioration process will not occur, due to for instance:

- separation of the environmental action from the structure or component, for example by cladding or membranes;
- using non-reactive materials, for example certain stainless steels or alkali-non-reactive aggregates;
- separation of reactants, for example keeping the structure or component below a critical degree of moisture;
- suppressing the harmful reaction, for example by electrochemical methods.

In seismic design, seismic isolation may be introduced at certain horizontal levels:

- between the superstructure of buildings or similar structures and the foundation or the ground;
- between a bridge deck and the tops of the piers and abutments;
- between sensitive equipment, containers of hazardous materials, important artefacts and such like, and the supporting structure or foundation.

The assumed effectiveness of the actual concept must be documented, for instance for products by complying with relevant minimum requirements in product standards.

The specific requirements for design, materials selection and execution for the avoidance-of-deterioration method can in principle be determined in the same way as for the deemed-to-satisfy approach.

The limitations to the validity of the provisions must be clearly stated.

5 Materials

The choice of methods is up to the responsible designer, based on considerations such as time, cost and need for precise estimates. All models and relations given in section 5.1 are physically sound and are based on the evaluation of experimental data as well as available field data.

In the *fib* Bulletin 70 “Code-type models for structural behaviour of concrete – Background of the constitutive relations and material models in *fib* MC 2010”, the background of the models and relations subsequently presented will be given, together with fundamental data as well as relevant references.

The constitutive relations given in these sections are applicable for the entire range of concrete grades dealt with in this Model Code.

Throughout section 5.1 the following sign conventions are maintained which may differ from those used in other parts of this Model Code:

- material properties are positive or to be used in absolute terms, such as compressive strength, $f_{cm} = |f_{cm}|$;
- tensile stresses and tensile strains (elongations) are positive;
- compressive stresses and compressive strains (contractions) are negative;
- where multiaxial stress states are considered, $\sigma_1 > \sigma_2 > \sigma_3$ is valid for the principal stresses.

It is assumed that the concrete complies with ISO 22965-1 “Concrete – Part 1: Methods of specifying and guidance to the specifier” and ISO 22965-2 “Concrete – Part 2: Specification of constituent materials, production of concrete and conformity of concrete”, with the amendments and alterations given in this Model Code.

Green concrete (also known as sustainable or ecological concrete) may be characterized by having a significantly improved sustainability compared to ordinary structural concrete. This holds particularly true, if the CO₂ emission associated with a concrete is significantly reduced and/or the energy necessary to produce the concrete and its constituent materials is considerably lower than for ordinary concrete. So far, no generally accepted limiting values and benchmarks exist.

Green concrete may be produced, for example, by the replacement of cement by chemically reactive or inert fine materials, by a significant reduction of the total binder content and also by the replacement of the aggregates, for example with recycled concrete. Further, environmentally harmful substances possibly contained in concrete making materials – for example, also in additions and admixtures – have to be excluded.

There is no detailed information available on the constitutive and durability behaviour of green concrete. Hence, an expert has to evaluate the structural behaviour in view of the composition of green concrete.

Production control and attestation of conformity of concrete must be in accordance with ISO 22965-2.

5.1 Concrete

Section 5.1 provides the designer with the best possible code-type characterization of the material properties of concrete to be used in their specific design models. Naturally this is best obtained from full-scale testing of in-field exposed structures. As this normally cannot be realized, the alternative is direct testing, while the last option should be to derive material properties from other material characteristics (e. g. tensile strength based on compressive strength or permeability based on strength or water/cement ratio).

5.1.1 General and range of applicability

The subsequent sections apply to structural concrete with normal and lightweight aggregates, composed and compacted so as to retain no appreciable amount of entrapped air other than intentionally entrained air.

Though the relations in principle also apply for heavyweight concrete, special consideration may be necessary for such concretes.

Concerning compressive strength, *fib* Model Code for Concrete Structures 2010 covers concretes up to a characteristic strength of 120 MPa, that is normal strength concrete (NSC, $f_{ck} \leq 50$ MPa) and high strength concrete (HSC, $f_{ck} > 50$ MPa) are dealt with; see subsection 5.1.4.

As a first approximation, the subsequent relations also apply for self-compacting concrete, unless additional information is provided.

The relations given also roughly apply for green concrete, as far as the composition of such concrete deviates from the composition of ordinary structural concrete only by the replacement of a certain amount of cement by fly ash, silica fume, blast furnace slag and natural pozzolans – that is, chemically reactive substitutes.

The information given in subsections 5.1.4, 5.1.5, 5.1.7 and 5.1.11.2 is valid for monotonically increasing compressive stresses or strains at a constant range of approximately $0.2 \text{ MPa/s} < |\dot{\sigma}_c| < 10^7 \text{ MPa/s}$ and $10 \cdot 10^{-6} \text{ s}^{-1} < |\dot{\epsilon}_c| < 3 \cdot 10^2 \text{ s}^{-1}$, respectively.

For tensile stresses or strains, this information is valid approximately for $0.02 \text{ MPa/s} < \dot{\sigma}_{ct} < 10^7 \text{ MPa/s}$ and $1 \cdot 10^{-6} \text{ s}^{-1} < \dot{\epsilon}_{ct} < 3 \cdot 10^2 \text{ s}^{-1}$, respectively.

5.1.2 Classification by strength

In this Model Code concrete is classified on the basis of its compressive strength. Design is based on a grade of concrete which

The specification of concrete given to the concrete producer must include all assumptions made during the design as well as those properties needed to ensure that the needs during transportation and execution on the site are considered.

The dual designation of concrete grades (e. g. C30/37) has been abandoned, as this is a pure European approach whereas both ISO 22965-2 and the former CEB-FIP MC 1990 specify only the cylindrical concrete strength.

However, this Model Code uses the designations Cxx and LCxx, while ISO 22965 uses Bxx and LBxx, respectively.

There are attempts to classify the characteristic values of compressive and tensile strengths according to the strength obtained at a concrete age of 56 days for concretes made of CEM III, CEM IV and CEM V cements. Nevertheless, we should keep in mind that some specifications – for example the requirements defined for the different exposure classes – are based on the 28-day compressive strength.

This classification corresponds to ISO 22965.

Lightweight aggregate concrete with a density $<800 \text{ kg/m}^3$ can usually not be used for structural applications.

With increasing compressive strength, concrete generally contains more cement and less water, resulting in a higher density of HSC compared to NSC. Also, HSC members may contain more reinforcement than NSC members. Nevertheless, the relevant density values may vary within relatively wide limits, depending on mix composition and density of aggregate materials (both may vary between countries), reinforcement ratio and air content.

The values given in Table 5.1-1 assume an air content of 2 %. A change of air content by 1 % results in a density change of 1 %. The values may be used for design purposes in calculating self-weight or imposed permanent loading. Where a higher accuracy is required than is provided by Table 5.1-1, the concrete density may be determined experimentally, for example according to ISO 1920-5.

The values given in Table 5.1-2 are valid for plain and reinforced lightweight aggregate concrete with usual percentages of reinforcement. The values for in-situ density may be used for design purposes in calculating self-weight or imposed permanent loading.

Where a higher accuracy is required than is provided by Table 5.1-2 the concrete density may be determined experimentally, for example according to ISO 1920-5.

In addition to the density class specifications, a further option is the definition of the so-called “target-density” – see for example ISO 22965-1.

For special requirements or in national codes, test specimens other than cylinders 150/300 mm and stored in other environments than those specified in ISO 1920-3 may be used to specify the concrete compressive strength. In such cases conversion factors should either be determined experimentally or, when given in national codes, used accordingly for a given category of testing equipment.

corresponds to a specific value of its characteristic compressive strength f_{ck} as defined in subsection 5.1.4.

Concrete grades for normal weight concrete (C) can be selected from the following series:

C12, C16, C20, C25, C30, C35, C40, C45, C50, C60, C70, C80, C90, C100, C110, C120.

Concrete grades for lightweight aggregate concrete (LC) can be selected from the following series:

LC8, LC12, LC16, LC20, LC25, LC30, LC35, LC40, LC45, LC50, LC55, LC60, LC70, LC80.

The numbers following the symbols C and LC denote the specified characteristic strength f_{ck} in MPa.

Unless specified otherwise, the compressive strength of concrete and the tensile strength of concrete is understood as the strength value obtained at a concrete age of 28 days.

Characteristic compressive strength values for normal weight and lightweight concrete are given in subsection 5.1.4 (Tables 5.1-3 and 5.1-4)

5.1.3 Classification by density

Concrete is classified in three categories of oven-dry density:

- lightweight aggregate concrete ($800\text{--}2000 \text{ kg/m}^3$);
- normal weight concrete ($>2000\text{--}2600 \text{ kg/m}^3$);
- heavyweight concrete ($>2600 \text{ kg/m}^3$).

For ordinary normal weight concrete, both, normal strength (NSC) and high strength concrete (HSC), the in-situ density may be estimated from Table 5.1-1.

Table 5.1-1: In-situ density [kg/m^3] of NSC and HSC, plain and with different steel reinforcement ratios

Reinforcement ratio	C30 (w/c $\approx$ 0.65)	C80 (w/c $\approx$ 0.35)	C120 (w/c $\approx$ 0.25)
0.0%	2350	2450	2500
1.0%	2400	2500	2550
2.0%	2450	2550	2600

The classification of lightweight aggregate concrete according to its oven-dry density is given in Table 5.1-2.

Table 5.1-2: Density classes and corresponding design densities of lightweight aggregate concrete

Density classes		D1.0	D1.2	D1.4	D1.6	D1.8	D2.0
Oven-dry density ρ [kg/m^3]		801–1000	1001–1200	1201–1400	1401–1600	1601–1800	1801–2000
In-situ density [kg/m^3]	Plain concrete	1050	1250	1450	1650	1850	2050
	Reinforced concrete	1150	1350	1550	1750	1950	2150

5.1.4 Compressive strength

The reference compressive strength of the concrete according to this Model Code is measured on cylinders 150/300 mm in accordance with ISO 1920-3; for classification, see subsection 5.1.2.

In the case where concrete cubes of 150 mm size are used, the characteristic strength values given in Table 5.1-3 must be obtained for the various concrete grades of normal weight concrete, whereas Table 5.1-4 gives the corresponding characteristic strength values for lightweight aggregate concrete.

For some verifications in design, or for an estimate of other concrete properties, it is necessary to refer to a mean value of compressive strength f_{cm} (or f_{lcm} for lightweight aggregate concrete) associated with a specific characteristic compressive strength f_{ck} (or f_{lck} for lightweight aggregate concrete). In this case f_{cm} and f_{lcm} may be estimated from Eqs. (5.1-1) and (5.1-2), respectively:

$$f_{cm} = f_{ck} + \Delta f \quad (5.1-1)$$

$$f_{lcm} = f_{lck} + \Delta f \quad (5.1-2)$$

where:

$$\Delta f = 8 \text{ MPa.}$$

Background information on the strength independence of Δf may be found in Müller, H. S., Anders, I., Breiner, R. and Vogel, M., “Concrete: treatment of types and properties in *fib* Model Code 2010” (Structural Concrete, Vol. 14, No. 4, December 2013).

In analysis and design of concrete structures, the characteristic compressive strength f_{ck} [MPa] is applied. This value may be derived from strength tests by the criterion that 5 % of all possible strength measurements for the specified concrete are expected to fall below the value f_{ck} ; see also subsection 5.1.2 and Tables 5.1-3 and 5.1-4.

Table 5.1-3: Characteristic strength values of normal weight concrete [MPa]

Concrete grade	C12	C16	C20	C25	C30	C35	C40	C45	C50
f_{ck}	12	16	20	25	30	35	40	45	50
$f_{ck,cube}$	15	20	25	30	37	45	50	55	60

Concrete grade	C55	C60	C70	C80	C90	C100	C110	C120
f_{ck}	55	60	70	80	90	100	110	120
$f_{ck,cube}$	67	75	85	95	105	115	130	140

Table 5.1-4: Characteristic strength values of lightweight aggregate concrete [MPa]

Concrete grade	LC8	LC12	LC16	LC20	LC25	LC30	LC35
f_{lck}	8	12	16	20	25	30	35
$f_{lck,cube}$	9	13	18	22	28	33	38

Concrete grade	LC40	LC45	LC50	LC55	LC60	LC70	LC80
f_{lck}	40	45	50	55	60	70	80
$f_{lck,cube}$	44	50	55	60	66	77	88

5.1.5 Tensile strength and fracture properties

5.1.5.1 Tensile strength

Although the uniaxial tensile testing is the most appropriate method for determining the tensile strength of concrete, it is rarely used anywhere other than in research because of the experimental difficulties in performing such experiments. Therefore, in many instances the splitting tensile strength or the flexural tensile strength are determined.

When testing tensile strength, special attention should be paid to possible effects of moisture gradients.

Table 5.1-5 gives tensile strength values for normal weight concrete, estimated from the characteristic compressive strength f_{ck} according to Eqs. (5.1-3) to (5.1-5).

Table 5.1-5: Tensile strength in MPa for different concrete grades

Concrete grade	C12	C16	C20	C25	C30	C35	C40	C45	C50
f_{ctm}	1.6	1.9	2.2	2.6	2.9	3.2	3.5	3.8	4.1
$f_{ctk,min}$	1.1	1.3	1.5	1.8	2.0	2.2	2.5	2.7	2.9
$f_{ctk,max}$	2.0	2.5	2.9	3.3	3.8	4.2	4.6	4.8	5.3

Concrete grade	C55	C60	C70	C80	C90	C100	C110	C120
f_{ctm}	4.2	4.4	4.6	4.8	5.0	5.2	5.4	5.6
$f_{ctk,min}$	3.0	3.1	3.2	3.4	3.5	3.7	3.8	3.9
$f_{ctk,max}$	5.5	5.7	6.0	6.3	6.6	6.8	7.0	7.2

The tensile strength of the concrete and the term “tensile strength”, unless stated otherwise in this Model Code, refer to the uniaxial tensile strength f_{ct} determined in related experiments.

In the absence of experimental data, the mean value of tensile strength f_{ctm} in MPa may be estimated for normal weight concrete from the characteristic compressive strength f_{ck} :

$$f_{ctm} = 0.3 \cdot (f_{ck})^{2/3} \quad \text{concrete grades} \leq \text{C50} \quad (5.1-3a)$$

$$f_{ctm} = 2.12 \cdot \ln(1 + 0.1 \cdot (f_{ck} + \Delta f)) \quad \text{concrete grades} > \text{C50} \quad (5.1-3b)$$

where:

f_{ck} is the characteristic compressive strength in MPa according to Table 5.1-3;

$$\Delta f = 8 \text{ MPa.}$$

The lower and upper bound values of the characteristic tensile strength $f_{ctk,min}$ and $f_{ctk,max}$ may be estimated using, respectively:

$$f_{ctk,min} = 0.7 \cdot f_{ctm} \quad (5.1-4)$$

$$f_{ctk,max} = 1.3 \cdot f_{ctm} \quad (5.1-5)$$

If there is no test procedure agreed or given in national guidelines, tests may be performed according to RILEM CPC 7, 1975.

Eq. (5.1-3) was derived by evaluating available data from axial tension and compression tests. The data from splitting and flexural tests were not considered, in order to avoid evident uncertainties resulting from indirect testing – see *fib* Bulletin 42 “Constitutive modelling for high strength/high performance concrete” (*fib*, 2008).

In existing national and international codes and standards values of the conversion factor α_{sp} may be found, which vary from 0.67 to 0.95. However, comprehensive new research results show that this factor is beyond 1.0; see Malárics, V. and Müller, H. S., “Evaluation of the splitting tension test for concrete from a fracture mechanical point of view” (Proceedings of the 7th International Conference on Fracture Mechanics of Concrete and Concrete Structures, Oh, B. H. et al. (eds.), Hanrimwon Co. Ltd., Seoul, Korea, CD: 05-06, pp. 709–716, 2010). The data indicates that for cast specimens, $\alpha_{sp} = 2.08 \cdot (f_{cm})^{-0.16}$ with f_{cm} in MPa. For crushed aggregates the value for α_{sp} may be increased up to 20%.

As a compromise, $\alpha_{sp} = 1.0$ has been chosen.

Eqs. (5.1-8a) and (5.1-8b) were deduced from fracture mechanics considerations. In CEB-FIP MC 1990 the relation given by Eq. (5.1-8b) was given for normal strength concrete. Since the ratio of flexural strength to uniaxial tensile strength of concrete $f_{ctm,fl}/f_{ctm}$ should decrease for a given beam depth if the concrete becomes more brittle, α_{fl} should depend on the brittleness of the concrete and decrease as brittleness increases. This means that in Eq. (5.1-8b) the number 0.06 for high strength concrete and for lightweight aggregate concrete should be replaced by a value lower than 0.06, to be determined by experiments.

The fracture mode of concrete subjected to tension allows the application of fracture mechanics concepts, that is, energy considerations. In those concepts, the fracture energy of concrete G_F is often used as a material characteristic to describe the resistance of concrete subjected to tensile stresses.

G_F should best be determined from uniaxial tension tests. Most frequently, however, indirect tests, first of all three-point bend tests on notched beams are used, which are easier to perform.

For normal weight concrete the fracture energy depends primarily on the water/cement ratio, the maximum aggregate size and the age of concrete. Curing conditions also have a significant effect on experimentally determined G_F values. Further, G_F is affected by the size of a structural member and in particular by the depth of the ligament above a crack or notch. The fracture energy of high strength normal weight concrete is also influenced by the above-mentioned parameters, but not to the same extent as in the case of normal strength concrete. The aggregate type and content seem to affect the

To estimate a mean value of the tensile strength f_{lctm} for lightweight aggregate concrete, f_{ctm} according to Eq. (5.1-3) must be multiplied by a reduction factor η_l :

$$f_{lctm} = \eta_l \cdot f_{ctm} \quad (5.1-6a)$$

$$\eta_l = (0.4 + 0.6 \cdot \rho/2200) \quad (5.1-6b)$$

where:

ρ is the oven-dry density of the lightweight aggregate concrete in kg/m³.

The lower and upper bound values of the characteristic tensile strength $f_{lctk,min}$ and $f_{lctk,max}$ may be estimated for lightweight aggregate concrete using Eqs. (5.1-4) and (5.1-5), respectively, replacing f_{ctm} by f_{lctm} .

If the tensile strength is measured as splitting tensile strength $f_{ct,sp}$ or as flexural tensile strength $f_{ct,fl}$ a conversion factor α should be determined by means of uniaxial tension tests.

If such conversion factors are not available, the mean uniaxial tensile strength f_{ctm} may be estimated from the mean splitting tensile strength $f_{ctm,sp}$ as:

$$f_{ctm} = \alpha_{sp} \cdot f_{ctm,sp} \quad (5.1-7)$$

where:

$f_{ctm,sp}$ is the mean value of splitting tensile strength determined according to ISO 1920-4;

$\alpha_{sp} = 1.0$.

The same conversion factor $\alpha_{sp} = 1.0$ may be used for lightweight aggregate concrete.

In order to estimate the mean uniaxial tensile strength f_{ctm} from the mean flexural tensile strength $f_{ctm,fl}$ we can use:

$$f_{ctm} = \alpha_{fl} \cdot f_{ctm,fl} \quad (5.1-8a)$$

where:

$f_{ctm,fl}$ is the mean flexural tensile strength;

$$\alpha_{fl} = \frac{0.06 \cdot h_b^{0.7}}{1 + 0.06 \cdot h_b^{0.7}} \quad (5.1-8b)$$

h_b is the beam depth [mm].

5.1.5.2 Fracture energy

The fracture energy of concrete G_F [N/m], defined as the energy required to propagate a tensile crack of unit area, should be determined by related tests.

In the absence of experimental data G_F in N/m for ordinary normal weight concrete may be estimated as:

$$G_F = 73 \cdot f_{cm}^{0.18} \quad (5.1-9)$$

where:

f_{cm} is the mean compressive strength in MPa.

fracture energy of concrete much more than the size of aggregate. This phenomenon is caused by the transition from interfacial fracture to trans-aggregate fracture. For high strength concrete, the effect of curing conditions on G_F is somewhat less pronounced than for normal strength concrete, but it is still significant. For further information, see *fib* Bulletin 42 “Constitutive modelling for high strength/high performance concrete” (*fib*, 2008).

The knowledge of fracture mechanisms of lightweight aggregate concrete (LWAC) is still insufficient, and the dependence of fracture energy of LWAC on different parameters (density, types of aggregates, strength etc.) must be addressed by future research. LWAC is notch sensitive (most important for this sensitivity are eigenstresses, because of moisture gradients). The maximum crack opening depends on the kind of matrix and the kind of aggregates, respectively. Thus, tests to determine fracture energy and softening behaviour are mandatory if relevant information on LWAC needs to be used for analysis and design.

This failure criterion is one among several acceptable formulations. It has been chosen since it is not too difficult to use and agrees well with test data. For further details and the range of applicability of Eq. (5.1-11), refer to the CEB Bulletin 156 “Concrete under multiaxial states of stress – constitutive equations for practical design” (CEB, 1983) and to Ottosen, N., “A Failure Criterion for Concrete” (Journal Engineering Mechanics Division, ASCE, Vol. 103, EM4, August 1977).

The criterion applies for monotonic stress increase until failure. An external compressive stress can destroy the structure, whereas for some stress ratios it can be supporting for a destroyed structure. The load capacity remains as long as the stress exists. At unloading or modification of the stress ratio there remains only a low load capacity. This effect occurs especially with lightweight concrete, but also with normal concrete at stress ratios with high hydrostatic stresses. In this case, introducing a cap function may be useful, which closes the open top failure curve. For normal concrete, concerning the intersection point of the cap function with the hydrostatic axis, different information ranging from 1.6 to 2.3 times the uniaxial strength can be found.

The invariants of the stress tensor (I_1) and the stress deviators (J_2 and J_3) used in Eqs. (5.1-11) to (5.1-13) may be calculated as follows:

$$I_1 = \sigma_1 + \sigma_2 + \sigma_3$$

$$J_2 = \frac{1}{6} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]$$

$$J_3 = (\sigma_1 - \sigma_m) \cdot (\sigma_2 - \sigma_m) \cdot (\sigma_3 - \sigma_m)$$

$$\sigma_m = (\sigma_1 + \sigma_2 + \sigma_3) / 3$$

The stress coordinates σ_o and τ_o (octahedron stresses) may be calculated as:

$$\sigma_o = I_1 / 3 \quad \text{and} \quad \tau_o = \sqrt{\frac{2}{3} \cdot J_2}$$

Note that f_c and f_{c2c} are defined as positive values; all other compressive stresses and strengths are negative values.

No standardized test method is available for determining the multiaxial strength. During the test the load has to be applied by special test devices, which follow the deformation of the specimen and prevent parts of the load being introduced through friction into the lateral load application system. Detailed information is

As an approximation for estimating the fracture energy of lightweight aggregate concrete, we can use:

$$G_{F,l} = G_{FoA} + 16 \cdot f_{lctm} \quad (5.1-10)$$

where:

$G_{F,l}$ is obtained in N/m;

$G_{FoA} = 24$ N/m for lightweight aggregate concrete with normal weight sand;

$= 0$ for lightweight aggregate concrete with lightweight sand;

f_{lctm} is the mean value of tensile strength in MPa.

5.1.6 Strength under multiaxial states of stress

The mean value of strength under multiaxial states of stress may be estimated from the failure criterion given by Eq. (5.1-11).

For normal weight and self-compacting concrete we can use:

$$\alpha \frac{J_2}{f_{cm}^2} + \lambda \frac{\sqrt{J_2}}{f_{cm}} + \beta \frac{I_1}{f_{cm}} - 1 = 0 \quad (5.1-11)$$

where:

$$\lambda = c_1 \cdot \cos \left[\frac{1}{3} \cdot \arccos (c_2 \cdot \cos 3\theta) \right] \quad (5.1-12)$$

$$\cos 3\theta = \frac{3\sqrt{3}}{2} \cdot \frac{J_3}{J_2^{3/2}} \quad (5.1-13)$$

The parameters J_2 , J_3 and I_1 in Eqs. (5.1-11) to (5.1-13) represent the invariants of the two stress deviators and the stress tensor, respectively, characterizing the state of stress considered.

For lightweight aggregate concrete f_{cm} in Eq. (5.1-11) has to be replaced by f_{lcm} .

The coefficients α , β , c_1 and c_2 are material parameters which depend on the uniaxial compressive strength f_{cm} (or f_{lcm} for lightweight aggregate concrete), the uniaxial tensile strength f_{ctm} (or f_{lctm}), the biaxial compressive strength f_{c2cm} (or f_{lc2cm}) and the triaxial compressive strength at one point on the compressive meridian ($\sigma_1 = \sigma_2 > \sigma_3$) described by σ_{com} and τ_{com} (or σ_{lcom} and τ_{lcom}). To determine these coefficients, the parameters have to be calculated as:

$$k = \frac{f_{lctm}}{f_{cm}}; f_{2c} = \frac{f_{c2cm}}{f_{cm}}; x = \frac{\sigma_{com}}{f_{cm}}; y = \frac{\tau_{com}}{f_{cm}}; h = -\frac{\sqrt{2} \cdot x + y}{\frac{y}{\sqrt{2}} - \frac{1}{3}} \quad (5.1-14)$$

available in: Gerstle et al., “Behavior of concrete under multiaxial stress” (Journal of the Engineering Mechanics Division, Proceedings of the ASCE, Vol. 106, No. EM6, Dec. 1980, pp. 1383–1403).

In the absence of experimental data the biaxial compressive strength and the triaxial compressive strength at one point on the compressive meridian may be estimated from the uniaxial compressive strength. Note that no consolidated experience exists for a stress level above $\sigma_{com} = -240$ MPa for normal weight concrete and $\sigma_{lcom} = -60$ MPa for lightweight aggregate concrete. No sufficient experimental data exist for self-compacting concrete. Approximately, the relations of normal concrete can be applied.

$$f_{c2c} = \left(1.2 - \frac{f_c}{1000}\right) \cdot f_c$$

where $f_c = f_{cm}$ for $f_{c2c} = f_{c2cm}$;
 $f_c = f_{ck}$ for $f_{c2c} = f_{c2ck}$;
 $f_c = f_{lcm}$ for $f_{c2c} = f_{lcm}$;
 $f_c = f_{lck}$ for $f_{c2c} = f_{lck}$.

$$\tau_{com} = \left[185 - 180 \cdot \frac{f_{cm}}{100} + 260 \cdot \left(\frac{f_{cm}}{100}\right)^2 - 84 \cdot \left(\frac{f_{cm}}{100}\right)^3\right]$$

$$\tau_{lcom} = \left[250 \cdot \frac{f_{lcm}}{100} - 460 \cdot \left(\frac{f_{lcm}}{100}\right)^2 + 310 \cdot \left(\frac{f_{lcm}}{100}\right)^3\right]$$

$$\tau_{cok} = \left(0.8 + \frac{f_{ck}}{1000}\right) \cdot \tau_{com} \text{ and } \tau_{lck} = \left(0.8 + \frac{f_{lck}}{1000}\right) \cdot \tau_{lcom}$$

for $\sigma_{com} = \sigma_{cok} = -240$ MPa and $\sigma_{lcom} = \sigma_{lck} = -60$ MPa

with f_{cm}, f_{ck}, f_{lcm} and f_{lck} in MPa.

The coefficients for normal weight concrete given in Figure 5.1-1 are the results of Eqs. (5.1-14) to (5.1-19b).

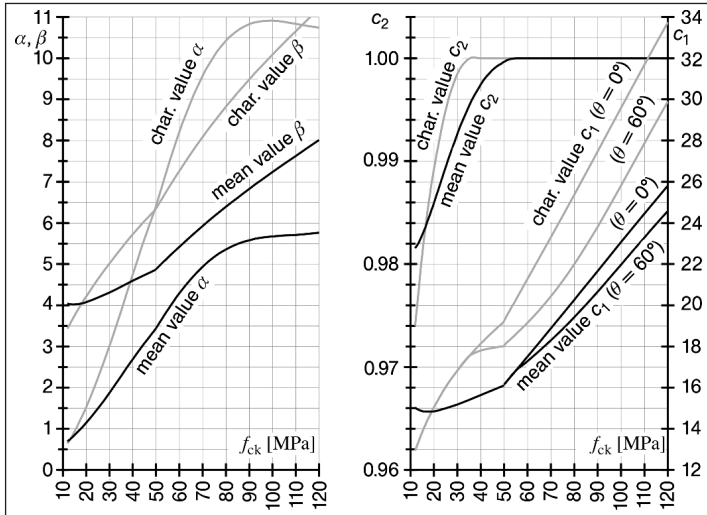


Figure 5.1-1: Coefficients for Eq. (5.1-11), normal weight concrete

$$\alpha = \frac{h \cdot \beta - \sqrt{2}}{y}; \quad \beta = \frac{\sqrt{2} - \frac{3 \cdot y}{k \cdot f_{2c}}}{h - \frac{9 \cdot y}{f_{2c} - k}} \quad (5.1-15)$$

$$\lambda_c = \lambda(\theta = 60^\circ) = \left(1 - \frac{h}{3 \cdot y}\right) \cdot \sqrt{3} \cdot \beta + \sqrt{3} + \frac{\sqrt{2}}{\sqrt{3} \cdot y} \quad (5.1-16)$$

$$\lambda_t = \lambda(\theta = 0^\circ) = \left(2 \cdot \sqrt{3} - \frac{f_{2c} \cdot h}{\sqrt{3} \cdot y}\right) \cdot \beta + \frac{\sqrt{3}}{f_{2c}} + \frac{\sqrt{2} \cdot f_{2c}}{\sqrt{3} \cdot y} \quad (5.1-17)$$

$$c_1 = [2 \cdot \cos \theta - 1] \cdot \lambda_t + 4 \cdot [1 - \cos \theta] \cdot \lambda_c \quad \text{for } \frac{\lambda_c}{\lambda_t} \leq \frac{1}{2} \quad (5.1-18a)$$

$$c_1 = \frac{\lambda_c}{\cos\left(\frac{\pi}{3} - \frac{1}{3} \cdot \arccos(c_2)\right)} \quad \text{for } \frac{\lambda_c}{\lambda_t} \geq \frac{1}{2} \quad (5.1-18b)$$

$$c_2 = 1 \quad \text{for } \frac{\lambda_c}{\lambda_t} \leq \frac{1}{2} \quad (5.1-19a)$$

$$c_2 = \cos\left\{3 \cdot \arctan\left[\frac{2 \cdot \frac{\lambda_c}{\lambda_t} - 1}{\sqrt{3}}\right]\right\} \quad \text{for } \frac{\lambda_c}{\lambda_t} \geq \frac{1}{2} \quad (5.1-19b)$$

To estimate a characteristic multiaxial strength, in Eqs. (5.1-11) and (5.1-14) the mean values of uniaxial compressive and tensile strength, biaxial and triaxial compressive strength have to be substituted by the characteristic values of these strengths.

The strength of concrete under biaxial states of stress ($\sigma_3 = 0$) may be estimated from the same criteria as given in Eqs. (5.1-11) to (5.1-19).

5.1.7 Modulus of elasticity and Poisson's ratio

5.1.7.1 Range of application

The information given in this section is valid for monotonically increasing compressive stresses or strains at a rate of $\dot{\sigma}_c \approx 0.6 \pm 0.4$ MPa/s or $\dot{\epsilon}_c \approx 15 \cdot 10^{-6} \text{ s}^{-1}$, respectively. For tensile stresses or strains, it is valid for $\dot{\sigma}_c \approx 0.06$ MPa/s or $\dot{\epsilon}_c \approx 1.5 \cdot 10^{-6} \text{ s}^{-1}$, respectively.

5.1.7.2 Modulus of elasticity

Values for the modulus of elasticity for normal weight concrete with natural sand and gravel can be estimated from the specified characteristic strength using:

$$E_{ci} = E_{c0} \cdot \alpha_E \cdot \left(\frac{f_{ck} + \Delta f}{10} \right)^{1/3} \quad (5.1-20)$$

where:

E_{ci} is the modulus of elasticity in MPa at the concrete age of 28 days;

f_{ck} is the characteristic strength in MPa according to subsection 5.1.4;

$\Delta f = 8$ MPa;

$E_{c0} = 21.5 \cdot 10^3$ MPa;

α_E is 1.0 for quartzite aggregates. For different types of aggregate qualitative values for α_E can be found in Table 5.1-6.

Where the actual compressive strength of concrete at an age of 28 days f_{cm} is known, E_{ci} may be estimated from:

$$E_{ci} = E_{c0} \cdot \alpha_E \cdot \left(\frac{f_{cm}}{10} \right)^{1/3} \quad (5.1-21)$$

The modulus of elasticity for lightweight aggregate concrete E_{lci} can be estimated from:

$$E_{lci} = \eta_E \cdot E_{ci} \quad (5.1-22)$$

where:

$$\eta_E = \left(\frac{\rho}{2200} \right)^2 = \text{reduction factor};$$

ρ is the oven-dry density of the lightweight aggregate concrete in kg/m^3 ;

E_{ci} is the modulus of elasticity in MPa according to Eq. (5.1-20) or Eq. (5.1-21); here $\alpha_E = 1.0$ for all types of lightweight aggregates.

Where only an elastic analysis of a concrete structure is carried out, a reduced modulus of elasticity E_c according to Eq. (5.1-23) should be used in order to account for initial plastic strain, causing some irreversible deformations.

$$E_c = \alpha_i \cdot E_{ci} \quad (5.1-23)$$

where:

$$\alpha_i = 0.8 + 0.2 \cdot \frac{f_{cm}}{88} \leq 1.0 \quad (5.1-24)$$

Values of the tangent modulus E_{ci} and the reduced modulus E_c for different concrete grades are given in Table 5.1-7.

The modulus of elasticity E_{ci} , as obtained from Eqs. (5.1-20) and (5.1-21), is defined as the tangent modulus of elasticity at the origin of the stress-strain diagram. It is approximately equal to the slope of the secant of the unloading branch for rapid unloading and does not include initial plastic deformations. It has to be used for the description of the stress-strain diagrams for uniaxial compression and uniaxial tension according to subsections 5.1.8.1 and 5.1.8.2, respectively, as well as for an estimate of creep according to Eq. (5.1-60), subsection 5.1.9.4.3. The reduced modulus of elasticity E_c according to Eq. (5.1-23) includes some irreversible strains.

The elastic deformations of concrete largely depend on its composition (especially the type of aggregate). The values given in this Model Code (Table 5.1-7) should be regarded as indicative for general applications. However, the modulus of elasticity should be specifically assessed or experimentally determined if the structure is likely to be sensitive to deviations from these general values. In this context, see RILEM CPC 8 (1975); a similar test procedure is published in ISO 1920-10.

Compared to the use of quartzite aggregates the modulus of elasticity can be increased by 20 % or decreased by 30 % only by changing the type of aggregate. Eqs. (5.1-20), (5.1-21) and Table 5.1-6 give the qualitative changes α_E in the modulus of elasticity for different types of aggregate.

Table 5.1-6: Effect of types of aggregate on the modulus of elasticity

Types of aggregate	α_E	$E_{c0} \cdot \alpha_E$ [MPa]
Basalt, dense limestone aggregates	1.2	25800
Quartzite aggregates	1.0	21500
Limestone aggregates	0.9	19400
Sandstone aggregates	0.7	15100

The modulus of elasticity E_{ci} does not include the initial plastic strain due to its definition. While the limit for the stress σ_c reached in the serviceability limit state (SLS) is set to $\sigma_c = -0.4 \cdot f_{cm}$ this stress level gives an upper limit for the reduction factor α_i (Figure 5.1-2, Eq. (5.1-23)). This factor $\alpha_i = E_c/E_{ci}$ is increasing with increasing concrete strength. For concrete grades higher than C80 the difference between first loading up to $\sigma_c = -0.4 \cdot f_{cm}$ and the unloading branch is smaller than 3% and could be neglected.

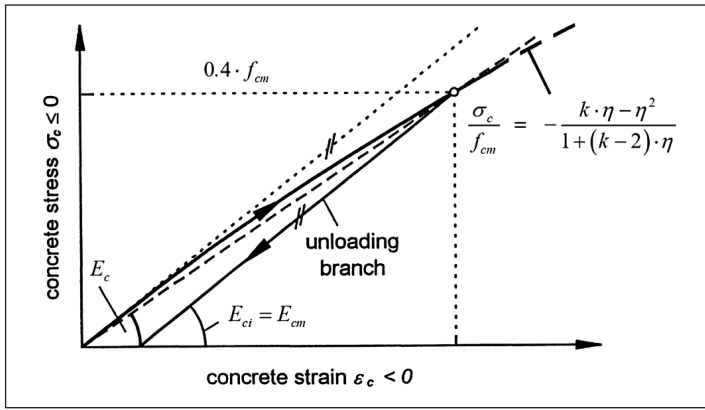


Figure 5.1-2: Definition of different moduli of elasticity (according to *fib* Bulletin 42 “Constitutive modelling for high strength/high performance concrete” (*fib*, 2008))

Note that E_{ci} is considered as the mean value of the tangent modulus of elasticity; hence $E_{ci} = E_{cm}$.

Self-compacting concrete (SCC), being produced with an increased binder content (powder type SCC), may have a reduced value of E_{ci} up to approximately 20 % at maximum compared to conventional concrete of equal strength. However, the E_{ci} values are within the scatter band for ordinary structural concrete.

E_c for normal weight concrete and E_{lc} for lightweight aggregate concrete are defined as the reduced or secant value of the modulus of elasticity.

Table 5.1-7: Tangent modulus and reduced modulus of elasticity

Concrete grade	C12	C16	C20	C25	C30	C35	C40	C45	C50
E_{ci} [GPa]	27.1	28.8	30.3	32.0	33.6	35.0	36.3	37.5	38.6
E_c [GPa]	22.9	24.6	26.2	28.0	29.7	31.4	33.0	34.5	36.0
α_i	0.845	0.855	0.864	0.875	0.886	0.898	0.909	0.920	0.932

Concrete grade	C55	C60	C70	C80	C90	C100	C110	C120
E_{ci} [GPa]	39.7	40.7	42.6	44.4	46.0	47.5	48.9	50.3
E_c [GPa]	37.5	38.9	41.7	44.4	46.0	47.5	48.9	50.3
α_i	0.943	0.955	0.977	1.0	1.0	1.0	1.0	1.0

The modulus of elasticity for lightweight aggregate concrete E_{lc} can be estimated by multiplying E_c from Eq. (5.1-23) with the reduction factor η_E given in Eq. (5.1-22):

$$E_{lc} = \eta_E \cdot E_c \quad (5.1-25)$$

5.1.7.3 Poisson's ratio

For a range of stresses $-0.6 \cdot f_{ck} < \sigma_c < 0.8 \cdot f_{ctk}$ the Poisson's ratio of concrete ν_c ranges between 0.14 and 0.26. Regarding the significance of ν_c , for the design of members, especially the influence of crack formation at the ultimate limit state (ULS), the estimation of $\nu_c = 0.20$ meets the required accuracy.

The value of $\nu_c = 0.20$ is also applicable for lightweight aggregate concrete.

5.1.8 Stress-strain relations for short term loading

5.1.8.1 Compression

The relation between σ_c and ε_c for short term uniaxial compression shown in Figure 5.1-3 is described by Eq. (5.1-26):

$$\frac{\sigma_c}{f_{cm}} = - \left(\frac{k \cdot \eta - \eta^2}{1 + (k - 2) \cdot \eta} \right) \quad \text{for } |\varepsilon_c| < |\varepsilon_{c,lim}| \quad (5.1-26)$$

where:

$$\eta = \varepsilon_c / \varepsilon_{c1} ;$$

$$k = E_{ci} / E_{c1} ;$$

ε_{c1} is the strain at maximum compressive stress (Table 5.1-8);

E_{c1} is the secant modulus from the origin to the peak compressive stress (Table 5.1-8);

k is the plasticity number (Table 5.1-8).

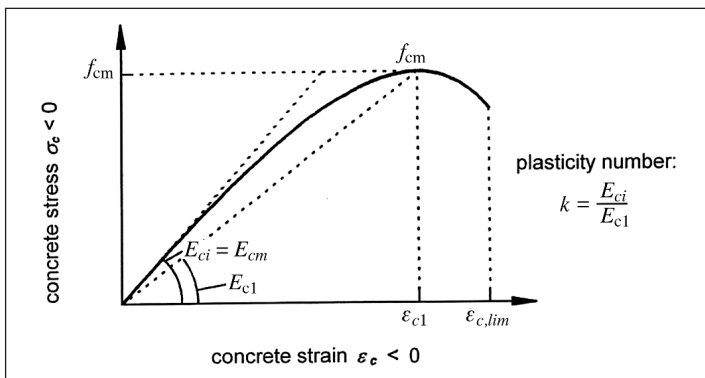


Figure 5.1-3: Schematic representation of the stress-strain relation for short term loading in uniaxial compression (according to *fib* Bulletin 42)

The stress–strain relations for concrete in compression generally comply with the schematic representation shown in Figure 5.1-3. The descending branch of the stress–strain relations should be considered as the envelope to all possible stress–strain relations of concrete, which tends to soften as a consequence of concrete micro-cracking.

The descending part of the stress–strain curve in compression is strongly dependent on the specimen or member geometry, the boundary conditions and the possibilities for load redistribution in the structure. In tests, a strong influence of the rigidity of the testing device used can be observed. During the softening process micro-cracking occurs in a fracture zone of a limited length and width. One single fracture zone is supposed to be decisive for the failure of a given member. The stress in the fracture zone drops down with a shear displacement in local shear bands of $w_c \approx 0.5$ mm.

The ultimate strain $\varepsilon_{c,lim}$ is caused by the displacement w_c related to a certain length which is given in Figure 4-4 in *fib* Bulletin 42 “Constitutive modelling for high strength/high performance concrete” (*fib*, 2008). The descending portion of the stress–strain relation is size-dependent and therefore not only a material property (see Figure 4-5 in *fib* Bulletin 42).

The stress–strain relation may be best determined by corresponding tests. If only the modulus of elasticity is available from experiments, this value may be used for estimating the stress–strain diagram. However, an accurate stress–strain diagram can only be found if the plasticity number k was investigated.

Eqs. (5.1-25) and (5.1-27) should be used with care when lightweight aggregate concretes with oven-dry densities < 1600 kg/m³ are considered.

Table 5.1-8: Moduli E_{ci} , E_{c1} , strains ε_{c1} , $\varepsilon_{c,lim}$ and plasticity number k for normal weight concrete

Concrete grade	C12	C16	C20	C25	C30	C35	C40	C45	C50
E_{ci} [GPa]	27.1	28.8	30.3	32.0	33.6	35.0	36.3	37.5	38.6
E_{c1} [GPa]	11.1	12.2	13.3	14.9	16.5	18.2	20.0	21.6	23.2
ε_{c1} [‰]	–1.9	–2.0	–2.1	–2.2	–2.3	–2.3	–2.4	–2.5	–2.6
$\varepsilon_{c,lim}$ [‰]	–3.5	–3.5	–3.5	–3.5	–3.5	–3.5	–3.5	–3.5	–3.4
k	2.44	2.36	2.28	2.15	2.04	1.92	1.82	1.74	1.66

Concrete grade	C55	C60	C70	C80	C90	C100	C110	C120
E_{ci} [GPa]	39.7	40.7	42.6	44.4	46.0	47.5	48.9	50.3
E_{c1} [GPa]	24.7	26.2	28.9	31.4	33.8	36.0	39.3	42.7
ε_{c1} [‰]	–2.6	–2.7	–2.7	–2.8	–2.9	–3.0	–3.0	–3.0
$\varepsilon_{c,lim}$ [‰]	–3.4	–3.3	–3.2	–3.1	–3.0	–3.0	–3.0	–3.0
k	1.61	1.55	1.47	1.41	1.36	1.32	1.24	1.18

For the calculation of ε_{lc1} for lightweight aggregate concrete a factor κ_{lc} is introduced, taking into account different types of sand:

$$\varepsilon_{lc1} = -\kappa_{lc} \cdot \frac{f_{lck} + 8}{E_{lc}} \quad (5.1-27)$$

where:

f_{lck} is the characteristic strength value for lightweight aggregate concrete in MPa according to Table 5.1-4;

E_{lc} is the modulus of elasticity in MPa for lightweight aggregate concrete according to Eq. (5.1-25);

κ_{lc} 1.1 for lightweight aggregate concrete with light sand;
1.3 for lightweight aggregate concrete with natural sand.

The stress–strain relation for unloading of the uncracked concrete may be described as:

$$\Delta\sigma_c = E_{ci} \cdot \Delta\varepsilon_c \quad (5.1-28)$$

where:

$\Delta\sigma_c$ is the stress reduction;

$\Delta\varepsilon_c$ is the strain reduction.

5.1.8.2 Tension

For uncracked normal weight concrete subjected to tension, the following bilinear stress–strain relation may be used (Figure 5.1-4):

$$\sigma_{ct} = E_{ci} \cdot \varepsilon_{ct} \quad \text{for } \sigma_{ct} \leq 0.9 \cdot f_{ctm} \quad (5.1-29)$$

$$\sigma_{ct} = f_{ctm} \cdot \left(1 - 0.1 \cdot \frac{0.00015 - \varepsilon_{ct}}{0.00015 - 0.9 \cdot f_{ctm} / E_{ci}} \right) \quad (5.1-30)$$

$$\text{for } 0.9 \cdot f_{ctm} < \sigma_{ct} \leq f_{ctm}$$

where:

E_{ci} is the tangent modulus of elasticity in MPa according to Eq. (5.1-20);

ε_{ct} is the tensile strain;

σ_{ct} is the tensile stress in MPa;

f_{ctm} is the tensile strength in MPa from Eq. (5.1-3).

Tensile failure of concrete is always a discrete phenomenon. Thus, to describe the tensile behaviour a stress–strain relation should be used for the uncracked concrete, and a stress–crack opening relation, as shown in Figure 5.1-4, should be used for the cracked section.

Since the post-cracking curve, as shown in Figure 5.1-4, is size-dependent it is recommended to carefully use this approach when constitutive relations for concrete need to be derived.

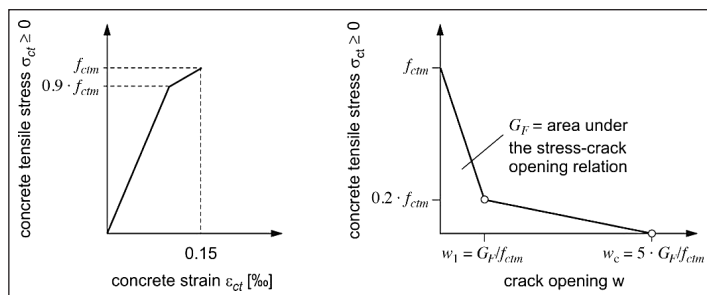


Figure 5.1-4: Schematic representation of the stress–strain and stress–crack opening relation for uniaxial tension (according to *fib* Bulletin 42)

At tensile stresses of about 90 % of the tensile strength f_{ct} , micro-cracking starts to reduce the stiffness in a small failure zone (Eqs. 5.1-29 and 5.1-30). The micro-cracks grow and form a discrete crack at stresses close to the tensile strength. All stresses and deformations in the fracture process zone can be related to a fictitious crack opening w (according to *fib* Bulletin 42).

Neglecting the small energy consumed by a complete loading cycle in the stress–strain relation, the maximum strain $\varepsilon_{ct,max}$ can be estimated as $\varepsilon_{ct,max} \approx f_{ctm}/E_{ci}$. For the analysis of the fracture zone a strain $\varepsilon_{ct,max} = 0.15$ ‰ can be assumed. Due to the localization of micro-cracking in the fracture zone and the large uncracked areas outside the damage zone, this strain is only valid inside the fracture zone.

Regarding the fracture energy in general, see subsection 5.1.5.2. To describe the stress–strain relation for uniaxial tension for lightweight aggregate concrete, see Faust, T., “Lightweight concrete in structural engineering” (Ernst & Sohn, Berlin, 2002; in German).

In the case of coinciding plastic potentials g and yield functions f the flow rule Eq. (5.1-34) is of the associated type, otherwise it is of the non-associated type. Non-associated flow rules should be used in concrete plasticity models to describe the inelastic volume change under compression, which is characteristic of frictional materials.

Basically, yield functions f and plastic potentials g can be chosen based on multiaxial failure criteria for concrete. These criteria should depend not only on shear stresses, but also on the first invariant I_1 of the stress tensor to consider the influence of the hydrostatic pressure on the ductility of the material. Thus, formulations such as:

- the Rankine criterion, where tensile failure occurs when the maximum principal stress reaches the uniaxial tensile strength f_{ct} ; refer to Rankine, W.J. M., “A Manual of Applied Mechanics” (London, 1868);
- the Drucker-Prager criterion, which is the modification of von Mises criterion including the influence of hydrostatic pressure on yielding; see Drucker, D.C. and Prager, W., “Soil mechanics and plastic analysis of limit design” (Quarterly of Applied Mechanics, Vol. 10, 1952);
- the Mohr–Coulomb criterion, where the maximum shear stress is the decisive measure of yielding, and the critical shear stress value depends on hydrostatic pressure; see Mohr, O., “Scientific paper on the area of technical mechanics” (Ernst & Sohn, Berlin, 1906; in German); and modifications or combinations of them can be used in concrete plasticity models.

For further criteria and detailed information, see:

- Chen, W.F. and Saleeb, A. F., “Constitutive Equations for Engineering Materials” (John Wiley & Sons, 1994);
- Jirásek, M. and Bažant, Z. P., “Inelastic Analysis of Structures” (John Wiley & Sons, 2002).

For a cracked section a bilinear approach for the stress–crack opening relation according to Figure 5.1-4 can be estimated by:

$$\sigma_{ct} = f_{ctm} \cdot \left(1.0 - 0.8 \cdot \frac{w}{w_1} \right) \quad \text{for } w \leq w_1 \quad (5.1-31)$$

$$\sigma_{ct} = f_{ctm} \cdot \left(0.25 - 0.05 \cdot \frac{w}{w_1} \right) \quad \text{for } w_1 < w \leq w_c \quad (5.1-32)$$

where:

- w is the crack opening in mm;
- $w_1 = G_F/f_{ctm}$ in mm when $\sigma_{ct} = 0.20 \cdot f_{ctm}$;
- $w_c = 5 \cdot G_F/f_{ctm}$ in mm when $\sigma_{ct} = 0$;
- G_F is the fracture energy in N/mm from Eq. (5.1-9);
- f_{ctm} is the tensile strength in MPa from Eq. (5.1-3).

5.1.8.3 Multiaxial states of stress

Constitutive relations of the elasto-plastic formulation, the damage formulation and combinations may be used to describe triaxial non-linear concrete behaviour at the macroscopic level in the short time range. Concrete is assumed to be an isotropic material in the initial unloaded state with an elasticity matrix $\mathbf{E}_0$, which is constant. Here the validity is restricted to small deformations.

The stress–strain relation of a general stress-based elasto-plastic formulation is given by:

$$\boldsymbol{\sigma} = \mathbf{E}_0 \cdot (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_p) \quad (5.1-33)$$

with the triaxial stress $\boldsymbol{\sigma}$, strain $\boldsymbol{\varepsilon}$ and plastic strain $\boldsymbol{\varepsilon}_p$. Occurrence of permanent plastic strain increments is determined by the flow rule:

$$\dot{\boldsymbol{\varepsilon}}_p = \lambda \frac{\partial g}{\partial \boldsymbol{\sigma}} \quad (5.1-34)$$

with the plastic potential g and the plastic multiplier λ .

The plastic potential g is a function of stress $\boldsymbol{\sigma}$ and state variables $\boldsymbol{\alpha}$, representing the load history. The multiplier λ is determined by the Kuhn-Tucker conditions:

$$\lambda \geq 0, \quad f\lambda = 0, \quad f \leq 0 \quad (5.1-35)$$

with a yield function f . The yield function f is also a function of stress $\boldsymbol{\sigma}$ and state variables $\boldsymbol{\alpha}$ and implies a limit condition for the material strength. The Kuhn-Tucker conditions distinguish unloading from loading and imply $\dot{\boldsymbol{\varepsilon}}_p = 0$ associated with $f < 0$ or $\dot{\boldsymbol{\varepsilon}}_p \neq 0$ in combination with:

$$\dot{f} = \frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \dot{\boldsymbol{\sigma}} + \frac{\partial f}{\partial \boldsymbol{\alpha}} \cdot \dot{\boldsymbol{\alpha}} = 0 \quad (5.1-36)$$

This consistency condition and an evolution law for the internal state variables

$$\dot{\boldsymbol{\alpha}} = \lambda \mathbf{h}(\boldsymbol{\sigma}, \boldsymbol{\alpha}) \quad (5.1-37)$$

result in an incremental constitutive law in case of loading:

$$\dot{\boldsymbol{\sigma}} = \left[\mathbf{E}_0 - \frac{\mathbf{E}_0 \cdot \frac{\partial g}{\partial \boldsymbol{\sigma}} \cdot \frac{\partial f^T}{\partial \boldsymbol{\sigma}} \cdot \mathbf{E}_0}{\frac{\partial f^T}{\partial \boldsymbol{\sigma}} \cdot \mathbf{E}_0 \cdot \frac{\partial g}{\partial \boldsymbol{\sigma}} - \frac{\partial f^T}{\partial \boldsymbol{\alpha}} \cdot \mathbf{h}} \right] \cdot \dot{\boldsymbol{\varepsilon}} \quad (5.1-38)$$

Examples for elaborated plasticity models are given in:

- Willam, K. and Warnke, E. P., “Constitutive model for the triaxial behaviour of concrete” (IABSE Report Vol. 19, Seminar on Concrete Structures Subjected to Triaxial Stresses, Bergamo, 1974);
- Oñate, E., Oller, S., Oliver, S. and Lubliner, J., “A constitutive model of concrete based on the incremental theory of plasticity” (Engineering Computations, Vol. 5, 1988);
- Etse, G. and Willam, K., “Fracture energy formulation for inelastic behaviour of plain concrete” (Journal of Engineering Mechanics, ASCE, Vol. 120, 1994);
- Grassl, P., Lundgren, K. and Gylltoft, K., “Concrete in compression: a plasticity theory with a novel hardening law” (International Journal of Solids and Structures, Vol. 39, 2002).

For a theoretical framework of damage models refer, for example, to:

- Carol, I., Rizzi, E. and Willam, K., “A unified theory of elastic degradation and damage based on a loading surface” (International Journal of Solids and Structures, Vol. 31, 1994).

The state variables β can be of scalar, vector and second or higher order tensor type. The use of scalar internal variables enables the description of isotropic damage, whereas tensor-valued state variables are needed for anisotropic damage formulations. Detailed information can be found for example in:

- Lemaitre, J., “A Course on Damage Mechanics” (Springer, 1992);
- Krajcinovic, D., “Damage Mechanics” (North-Holland, Elsevier, 1996);
- Skrzypek, J. and Ganczarski, A., “Modelling of Material Damage and Failure of Structures” (Springer, 1999).

The damage limit functions F can generally be chosen based on multiaxial limit criteria for concrete, which are defined in the stress space and can be transferred into the strain space. Relevant examples are given in:

- Ottosen, N. S., “A failure criterion for concrete” (Journal of Engineering Mechanics, ASCE, Vol. 103, 1977);
- Hsieh, S. S., Ting, E. and Chen, W. F., “A plasticity fracture model for concrete” (International Journal of Solids and Structures, Vol. 18, 1982);
- Willam, K. and Warnke, E. P., “Constitutive model for the triaxial behaviour of concrete” (IABSE Report Vol. 19, Seminar on Concrete Structures Subjected to Triaxial Stresses, Bergamo, 1974).

For more information see Chen, W.F.; Saleeb, A.F., “Constitutive Equations for Engineering Materials” (John Wiley & Sons, 1994).

Strain-based isotropic damage formulations with scalar internal variables which consider tensile as well as compressive damage can be found for example in:

- Mazars, J., “Application de la mécanique de l’endommagement au comportement nonlinéaire à la rupture du béton de structure” (Technical report, LMT, Université Paris, 1984);
- Tao, X. and Phillips, D.V., “A simplified isotropic damage model for concrete under bi-axial stress states” (Cement & Concrete Composites, Vol. 27, 2005).

An orthotropic damage approach based on the second-order integrity tensor as internal variable is described for example in:

The elastic law $\dot{\sigma} = \mathbf{E}_0 \cdot \dot{\epsilon}$ applies for unloading. The functions g, f and $\mathbf{h}$ are material functions, which have to be determined on the basis of experimental data. The elasto-plastic formulation may be extended by multiple yield surfaces and plastic potentials.

The stress–strain behaviour of a general strain-based damage formulation is given by:

$$\sigma = \mathbf{E} \cdot \epsilon \quad (5.1-39)$$

with the triaxial elasticity matrix $\mathbf{E}$, which is variable according to the damage formulation. Degradation of the elasticity or occurrence of damage is determined by:

$$\dot{\mathbf{E}} = -\lambda' \mathbf{G} \quad (5.1-40)$$

with a generalized damage direction $\mathbf{G}$ and a damage multiplier λ' . The generalized damage direction $\mathbf{G}$ depends on strain ϵ and state variables β representing the load history. The multiplier λ' is determined by the Kuhn-Tucker conditions:

$$\lambda' \geq 0, \quad F\lambda' = 0, \quad F \leq 0 \quad (5.1-41)$$

with a damage limit function F . The damage limit function F is also a function of strain ϵ and state variables β and again should imply a limit condition for the material strength. The Kuhn-Tucker conditions distinguish unloading from loading and imply $\dot{\mathbf{E}} = \mathbf{0}$ associated with $F < 0$ and $\dot{\mathbf{E}} \neq \mathbf{0}$ in combination with:

$$\dot{F} = \frac{\partial F}{\partial \epsilon} \cdot \dot{\epsilon} + \frac{\partial F}{\partial \beta} \cdot \dot{\beta} = 0 \quad (5.1-42)$$

This consistency condition and an evolution law for the internal state variables:

$$\dot{\beta} = \lambda' \mathbf{h}'(\epsilon, \beta) \quad (5.1-43)$$

result in an incremental constitutive law for loading:

$$\dot{\sigma} = \left[\mathbf{E} + \frac{1}{\frac{\partial F}{\partial \beta} \cdot \mathbf{h}'} \mathbf{G} \cdot \epsilon \cdot \frac{\partial F}{\partial \epsilon} \right] \cdot \dot{\epsilon} \quad (5.1-44)$$

The linear elastic law $\dot{\sigma} = \mathbf{E} \cdot \dot{\epsilon}$ with $\dot{\mathbf{E}} = \mathbf{0}$ applies for unloading. The functions $\mathbf{G}, F$ and $\mathbf{h}'$ are material functions, which have to be determined on the basis of experimental data.

The scalar isotropic damage is given as a special case:

$$\mathbf{E} = (1 - D) \mathbf{E}_0, \quad \dot{\mathbf{E}} = -\dot{D} \mathbf{E}_0, \quad \mathbf{G} = \mathbf{E}_0 \quad (5.1-45)$$

In Eq. (5.1-45) the restriction $0 \leq D \leq 1$ and the relation $\lambda' = \dot{D}$ hold. A scalar internal state variable is appropriate. The damage limit function F and the evolution function $\mathbf{h}'$ become scalar functions of strain invariants and of a scalar β . The value β is an equivalent strain measure with a restriction $\beta \geq 0$. Furthermore, for loading, simple relations like:

- Carol, I., Rizzi, E. and Willam, K., “On the formulation of anisotropic elastic degradation. I. Theory based on a pseudo-logarithmic damage tensor rate. II. Generalized pseudo-Rankine model for tensile damage” (International Journal of Solids and Structures, Vol. 38, 2001).

For an anisotropic formulation with a higher order tensorial damage variable refer for example to:

- Govindjee, S., Kay, G.J. and Simo, J. C., “Anisotropic modelling and numerical simulation of brittle damage in concrete” (International Journal for Numerical Methods in Engineering, Vol. 38, 1995).

Another approach for the material description of concrete is given with the microplane theory; see, for example:

- Bažant, Z. P., “Microplane model for progressive fracture of concrete and rock” (Journal of Engineering Mechanics, ASCE, Vol. 111, 1985);
- Ožbolt, J., Li, Y.-J. and Kožar, I., “Microplane model for concrete with relaxed kinematic constraint” (International Journal of Solids and Structures, Vol. 38, 2001).

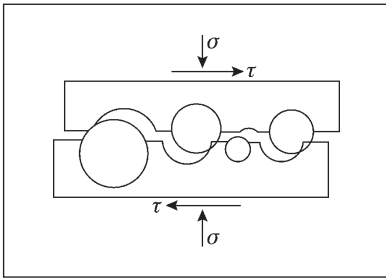


Figure 5.1-5: Principle of shear friction in concrete crack

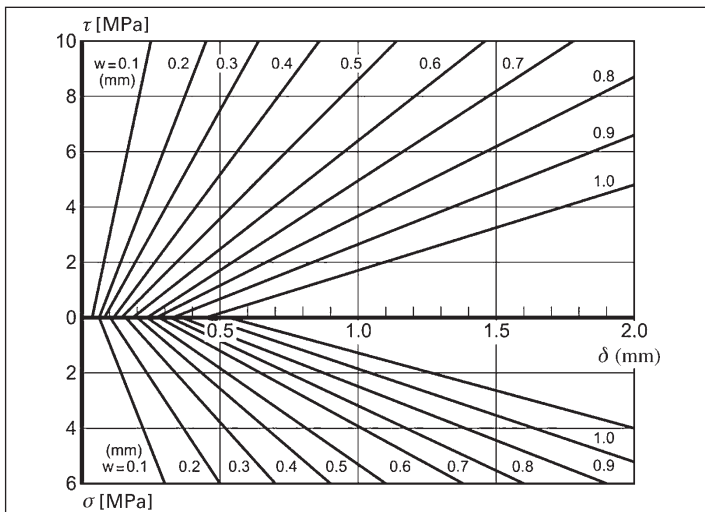


Figure 5.1-6: Relations according to Eqs. 5.1-48 and 5.1-49 for $f_{cm} = 30$ MPa

The compressive strength of concrete at an age t depends on the type and strength class of the cement, the type and amount of admixtures and additions, the water/cement ratio and environmental conditions, such as temperature and humidity.

$$D(\beta) = \begin{cases} 0 & \beta \leq \beta_0 \\ 1 - e^{-\left(\frac{\beta - \beta_0}{\beta_d}\right)^g} & \beta > \beta_0 \end{cases} \quad (5.1-46)$$

are appropriate where the material parameters β_0 , β_d and g may be derived from uniaxial material behaviour.

The description of anisotropic damage needs tensor notations, for example in case of orthotropic damage:

$$G_{ijpq} = \Lambda_0 [d_{ij} d_{pq} + d_{ij} \dot{d}_{pq}] + G_0 [d_{ip} d_{jq} + d_{ip} \dot{d}_{jq} + d_{iq} d_{jp} + d_{iq} \dot{d}_{jp}] \quad (5.1-47)$$

with the initial Lamé constant Λ_0 , the initial shear modulus G_0 and a second-order symmetric damage tensor $\mathbf{d}$ with components d_{ij} , whose principal values and directions describe damage in three orthogonal directions.

5.1.8.4 Shear friction behaviour in cracks

If in an open crack the crack faces are subjected to shear displacements with opposite signs, resisting shear stresses and normal (wedging) stresses develop as a result of the roughness of the crack faces.

The mean shear stress τ [MPa] and the mean normal stress σ [MPa] may be calculated from the following general relations:

$$\tau = C_f \{-0.04 f_{cm} + [1.8w^{-0.8} + (0.292w^{-0.7} - 0.25)f_{cm}] \delta\} \quad (5.1-48)$$

$$\sigma = C_f \{-0.06 f_{cm} + [1.35w^{-0.63} + (0.242w^{-0.55} - 0.19)f_{cm}] \delta\} \quad (5.1-49)$$

where:

δ is the shear displacement in mm;

w is the crack width in mm;

f_{cm} is the mean compressive strength in MPa at an age of 28 days.

C_f is an aggregate effectiveness factor, which is 1.0 if the aggregate does not fracture upon cracking of the concrete. For concrete with weak aggregates, or high strength concrete (with strong cement paste), in which most of the particles are broken, for C_f a value of about 0.35 applies. More accurate values for C_f can be found by carrying out a push-off test.

The crack opening path (development of shear displacement in relation to crack opening) can be constructed from diagrams as shown in Figure 5.1-6. If the relation between normal stress σ and crack opening w is given (analogous to spring stiffness), the corresponding values δ and τ can be read.

5.1.9 Time effects

5.1.9.1 Development of strength with time

For a mean temperature of 20°C and curing in accordance with ISO 1920-3 the relevant compressive strength of concrete at various ages $f_{cm}(t)$ may be estimated from:

The tensile strength of concrete primarily depends on those parameters which also influence the compressive strength of the concrete. However, tensile and compressive strength are not proportional to each other, and particularly for higher strength grades, an increase of the compressive strength leads only to a small increase of the tensile strength.

The development of tensile strength with time is strongly influenced by curing and drying conditions (internal stresses, surface cracking) as well as by the dimensions of the structural member. As a first approximation it may be assumed that for a duration of moist curing $t_s \leq 7$ days and a concrete age $t > 28$ days the development of the tensile strength is similar to that of the compressive strength. For a concrete age $t < 28$ days residual stresses may cause a temporary decrease of the tensile strength.

For high strength concrete the decrease of the tensile strength due to shrinkage cracks seems to be more pronounced than for normal strength concrete.

Where the development of tensile strength with time is important it is recommended to carry out experiments taking into account exposure conditions and dimensions of the structural member.

Eq. (5.1-50) was developed based on results obtained from experiments on structural concrete primarily made with CEM I and CEM III cements. If other cement types are used or if high amounts of pozzolans are used as partial replacement of CEM I and the development of the compressive strength with time has a major importance for the design, this effect should be determined experimentally.

Concretes with a high content of fly ash, natural pozzolans or fine granulated blast furnace slag (e.g. green concrete) show a reduced compressive strength at early age and a considerable further strength gain at higher ages. This effect may be more pronounced than considered in Eq. (5.1-51) for a low strength, normal hardening cement.

The compressive strength of lightweight aggregate concrete mainly depends on the strength of the cement paste. Lightweight aggregate concrete has a relatively rapid early strength development and a relatively slow long term strength development because of the low strength of the lightweight aggregates. Therefore hardly any strength gain may be observed after a certain concrete age when the cement paste approaches the strength of the lightweight aggregates with ongoing hydration. The age at which this situation is reached depends on the strength of the lightweight aggregates. An age in the range 1–4 weeks is a realistic estimate in most cases.

Due to the counteracting effects of the parameters influencing the strength under sustained loads, $f_{cm,sus}(t, t_0)$ passes through a minimum. The duration of loading for which this minimum occurs depends on the age at loading and is referred to as the critical period $(t-t_0)_{crit}$. For an age at loading of 28 days, a concrete made of normal cement, type N, $(t-t_0)_{crit} = 28$ days and the minimum value of the sustained loading strength is about $f_{c,sus,min} = 0.78f_{cm}$.

Research has shown a slight increase of the sustained load strength with increasing compressive strength of the concrete. However, due to the limited number of tests on high strength concrete the sustained load strength of normal strength concrete should be applied also for high strength concrete.

$$f_{cm}(t) = \beta_{cc}(t) \cdot f_{cm} \quad (5.1-50)$$

with:

$$\beta_{cc}(t) = \exp \left\{ s \cdot \left[1 - \left(\frac{28}{t} \right)^{0.5} \right] \right\} \quad (5.1-51)$$

where:

$f_{cm}(t)$ is the mean compressive strength in MPa at an age t in days;

f_{cm} is the mean compressive strength in MPa at an age of 28 days;

$\beta_{cc}(t)$ is a function to describe the strength development with time;

t is the concrete age in days adjusted according to Eq. (5.1-85) (taking into account the temperature during curing);

s is a coefficient which depends on the strength class of cement as given in the following Table 5.1-9.

Table 5.1-9: Coefficient s to be used in Eq. (5.1-51) for different types of cement

f_{cm} [MPa]	Strength class of cement	s
≤ 60	32.5 N	0.38
	32.5 R, 42.5 N	0.25
	42.5 R, 52.5 N, 52.5 R	0.20
> 60	all classes	0.20

For lightweight aggregate concrete the compressive strength in MPa at various ages may be estimated from:

$$f_{lcm}(t) = \beta_{lcc}(t) \cdot f_{lcm} \quad (5.1-52)$$

where:

$\beta_{lcc}(t)$ is a function describing the development with time;

$\beta_{lcc}(t) = \beta_{cc}(t)$ where s has to be replaced by s_{lc} ;

s_{lc} 0.05 for lightweight aggregates of high strength;

0.25 for lightweight aggregates of low strength;

f_{lcm} is the mean compressive strength in MPa at an age of 28 days.

5.1.9.2 Strength under sustained loads

5.1.9.2.1 Sustained compressive strength

When subjected to sustained high compressive stresses the compressive strength of concrete decreases with time under load due to the formation of micro-cracks. This strength reduction is counteracted by a strength increase due to continued hydration. The combined effect of sustained stresses and of continued hydration is given by:

$$f_{cm,sus}(t, t_0) = f_{cm} \cdot \beta_{cc}(t) \cdot \beta_{c,sus}(t, t_0) \quad (5.1-53)$$

with:

$$\beta_{c,sus}(t, t_0) = 0.96 - 0.12 \left\{ \ln \left[72(t - t_0) \right] \right\}^{1/4} \quad (5.1-54)$$

Resulting from the reduced strength gain of the lightweight aggregate concrete, as soon as the strength of the cement paste approaches the strength of the aggregates, the critical period is extended and the strength under sustained loads equals 70–75% of the short time strength at the age of 28 days.

Eq. (5.1-55) has been taken from *fib* Bulletin 42, 2008.

These relations have been developed based on experimental results primarily obtained with CEM I and CEM III cements. If other cement types are used or if high amounts of pozzolans are used in partial replacement of CEM I and the development of the modulus of elasticity with time has a high relevance for the design, this effect should be determined experimentally.

For lightweight aggregate concrete Eq. (5.1-57) has to be applied with caution. For structures sensitive to deformations, related tests have to be carried out.

The hydration of the cement in self-compacting concrete is basically controlled by the same mechanisms as that of vibrated concrete. Thus no particular difference in the time-development of properties is expected.

Concretes with a high content of fly ash, natural pozzolans or fine granulated blast furnace slag (e.g. green concrete) show a reduced modulus of elasticity at early age and a further gain of stiffness at higher ages. This effect may be more pronounced than suggested by Eq. (5.1-57) for low strength, normal hardening cement.

The distinction between creep and shrinkage is usually defined by convention. The delayed strains of loaded or unloaded concrete should be considered as two aspects of the same physical phenomena.

Also, separation of initial strain and creep strain is a matter of convention. In structural analysis, the total load-dependent strain as given by the creep function (subsection 5.1.9.4.3) is of importance. The initial and creep strain components are defined consistently, so that their sum results in the correct load-dependent strain.

where:

- $f_{cm,sus}(t, t_0)$ is the mean compressive strength of concrete in MPa at time t when subjected to a high sustained compressive stress at an age at loading $t_0 < t$;
- $\beta_{cc}(t)$ is the time development function according to Eq. (5.1-51);
- $\beta_{c,sus}(t, t_0)$ is a coefficient which depends on the time under high sustained loads $t - t_0$ in days. The coefficient describes the decrease of strength with time under load and is defined for $(t - t_0) > 0.015$ days (= 20 min);
- t_0 is the age of the concrete at loading in days;
- $t - t_0$ is the time under high sustained loads in days.

5.1.9.2.2 Sustained tensile strength

The tensile strength under sustained loading $f_{ctk,sus}$ in MPa can be estimated from:

$$f_{ctk,sus} = \alpha \cdot f_{ctk} \quad (5.1-55)$$

where:

- f_{ctk} is the short term strength in MPa;
- α = 0.60 for normal strength concrete and 0.75 for high strength concrete.

5.1.9.3 Development of modulus of elasticity with time

The modulus of elasticity of concrete at an age $t \neq 28$ days may be estimated from:

$$E_{ci}(t) = \beta_E(t) E_{ci} \quad (5.1-56)$$

with:

$$\beta_E(t) = [\beta_{cc}(t)]^{0.5} \quad (5.1-57)$$

where:

- $E_{ci}(t)$ is the modulus of elasticity in MPa at an age t in days;
- E_{ci} is the modulus of elasticity in MPa at an age of 28 days from Eq. (5.1-20);
- $\beta_E(t)$ is a coefficient which depends on the age of concrete, t in days;
- $\beta_{cc}(t)$ is the coefficient according to Eq. (5.1-51).

The coefficient s , to be inserted in $\beta_{cc}(t)$, depends on the type of cement (strength classes according to EN 197-1) and the compressive strength of the concrete and may be taken from Table 5.1-9.

5.1.9.4 Creep and shrinkage

5.1.9.4.1 Definitions

The total strain at time t , $\varepsilon_c(t)$, of a concrete member uniaxially loaded at time t_0 with a constant stress $\sigma_c(t_0)$ may be expressed as:

$$\varepsilon_c(t) = \varepsilon_{ci}(t_0) + \varepsilon_{cc}(t) + \varepsilon_{cs}(t) + \varepsilon_{cT}(t) \quad (5.1-58)$$

or

$$\varepsilon_c(t) = \varepsilon_{c\sigma}(t) + \varepsilon_{cn}(t) \quad (5.1-59)$$

For the prediction of the creep function, the initial strain $\varepsilon_{ci}(t_0)$ is based on the tangent modulus of elasticity as defined in Eqs. (5.1-20) and (5.1-56), that is

$$\varepsilon_{ci}(t_0) = \sigma_c(t_0) / E_{ci}(t_0)$$

The initial plastic strain occurring at first loading and being observed upon rapid unloading (see Figure 5.1-2) is considered to be part of the creep strain.

The formulation does not predict local rheological properties within the cross-section of a concrete member such as variations due to internal stresses, moisture states or the effects of local cracking.

The prediction formulation is not applicable to

- concrete subjected to extreme temperatures, high (e. g. nuclear reactors) or low (e. g. LNG tanks);
- very dry climatic conditions (average relative humidity RH < 40%).

The effect of temperature variations during hardening can be taken into account in accordance with Eq. (5.1-85). The effect of temperature in the range $0^\circ\text{C} < T < 80^\circ\text{C}$ is dealt with in subsection 5.1.10.

Here, concrete is considered as an ageing linear viscoelastic material. In reality, creep is a non-linear phenomenon. The non-linearity with respect to creep inducing stress may be observed in creep experiments at a constant stress, particularly if the stress exceeds $0.4f_{cm}(t_0)$, see subsection 5.1.9.4.3 (d), as well as in experiments with a variable stress history even below stresses of $0.4f_{cm}(t_0)$.

In this subsection, creep after a given duration of loading is described by means of the creep coefficient, which is a descriptive figure of the magnitude of creep effects. The creep coefficient is defined as the ratio of the creep strain to the elastic strain of concrete at an age of 28 days (reference elastic deformation) under the same stress.

In Müller, H. S., Anders, I., Breiner, R. and Vogel, M., “Concrete: treatment of types and properties in *fib* Model Code 2010” (Structure Concrete, Vol. 14, No. 4, December 2013) background information of the creep coefficient $\varphi(t, t_0)$ in combination with the modulus of elasticity E_{ci} is given.

The application of the principle of superposition is consistent with the assumption of linearity. However, due to the actual non-linear behaviour of concrete, some prediction errors are inevitable when linear superposition is applied to creep of concrete under variable stress, particularly for unloading or decreasing strains, respectively. These deviations are mainly caused by the neglect of hygral effects, interaction of stresses and ageing and material damage, including

where:

$\varepsilon_{ci}(t_0)$ is the initial strain at loading;

$\varepsilon_{cc}(t)$ is the creep strain at time $t > t_0$;

$\varepsilon_{cs}(t)$ is the shrinkage strain;

$\varepsilon_{cT}(t)$ is the thermal strain;

$\varepsilon_{c\sigma}(t)$ is the stress-dependent strain: $\varepsilon_{c\sigma}(t) = \varepsilon_{ci}(t_0) + \varepsilon_{cc}(t)$;

$\varepsilon_{cn}(t)$ is the stress-independent strain: $\varepsilon_{cn}(t) = \varepsilon_{cs}(t) + \varepsilon_{cT}(t)$.

5.1.9.4.2 Range of applicability

The relations for creep and shrinkage given below predict the time-dependent mean cross-section behaviour of a concrete member moist cured at normal temperatures for not longer than 14 days.

Unless special provisions are given, the relations are valid for ordinary structural concrete ($20 \text{ MPa} \leq f_{cm} \leq 130 \text{ MPa}$) subjected to a compressive stress $|\sigma_c| \leq 0.4f_{cm}(t_0)$ at an age at loading t_0 and exposed to a mean relative humidity in the range of 40 to 100% at a mean temperature in the range of 5°C to 30°C . The age at loading should be at least 1 day.

It is accepted that the relations apply as well to concrete in tension, though the relations given in the following are directed towards the prediction of creep of concrete subjected to compressive stresses.

5.1.9.4.3 Creep

(a) Assumptions and related basic equations

Within the range of service stresses $|\sigma_c| \leq 0.4 \cdot f_{cm}(t_0)$, creep is assumed to be linearly related to stress.

For a constant stress $\sigma_c(t_0)$ in MPa applied at time t_0 this leads to the subsequent creep strain $\varepsilon_{cc}(t, t_0)$:

$$\varepsilon_{cc}(t, t_0) = \frac{\sigma_c(t_0)}{E_{ci}} \varphi(t, t_0) \quad (5.1-60)$$

where:

$\varphi(t, t_0)$ is the creep coefficient;

E_{ci} is the modulus of elasticity at the age of 28 days according to Eqs. (5.1-20) or (5.1-21) in MPa.

The stress dependent strain $\varepsilon_{c\sigma}(t, t_0)$ at time t may be expressed as:

$$\varepsilon_{c\sigma}(t, t_0) = \sigma_c(t_0) \left[\frac{1}{E_{ci}(t_0)} + \frac{\varphi(t, t_0)}{E_{ci}} \right] = \sigma_c(t_0) J(t, t_0) \quad (5.1-61)$$

where:

$J(t, t_0)$ is the creep function or creep compliance, representing the total stress-dependent strain per unit stress;

$E_{ci}(t_0)$ is the modulus of elasticity at the time of loading t_0 according to Eq. (5.1-56); hence $1/E_{ci}(t_0)$ represents the initial strain per unit stress at loading.

For practical applications concrete may be considered as an ageing linear viscoelastic material, and for variable stresses and strains, the principle of superposition is assumed to be valid. On the basis of these assumptions and definitions given above, the constitutive equation for concrete may be written as:

$$\varepsilon_c(t) = \sigma_c(t_0) J(t, t_0) + \int_{t_0}^t J(t, \tau) \frac{\partial \sigma_c(\tau)}{\partial \tau} d\tau + \varepsilon_{cn}(t) \quad (5.1-62)$$

cracking and fracture. For linear creep prediction; models, the error depends also on the type of model underlying the creep prediction; see CEB Bulletin 177 “Summary and Analysis of Observations concerning the Revision of the CEB-FIP Model Code 1978 with discussion documents on new or revised clauses” (CEB, 1987).

The structural effects of time-dependent behaviour of concrete are dealt with in detail in subsection 7.2.4 of this Model Code, in CEB Bulletin 215 “Structural Effects of Time-Dependent Behaviour of Concrete” (CEB, 1993) and section 4.1.6 of *fib* Bulletin 52.

The relations to calculate the creep coefficient are empirical. They were calibrated on the basis of laboratory tests (creep in compression) on structural concretes. Total creep is separated into the components of basic creep and drying creep, reflecting the associated different physical mechanisms.

In this prediction model only those parameters are taken into account that are normally known to the designer, that is characteristic compressive strength, dimensions of the member, mean relative humidity to which the member is exposed, age at loading, duration of loading and type of cement. It should be pointed out, however, that creep of concrete does not depend on its compressive strength or age at loading per se, but rather on its composition and degree of hydration; creep of concrete decreases with decreasing water/cement ratio, decreasing cement paste content, increasing stiffness of the aggregates and increasing degree of hydration at the age of loading.

Due to the inherent scatter of creep and shrinkage deformations, the errors of the model and the general uncertainty caused by randomness of material properties and environment, a prediction of the deformation may result in a considerable error. After short durations of loading or drying, the prediction error is higher than after long durations of loading and drying.

Based on a computerized database of laboratory test results a mean coefficient of variation for the predicted creep function $V_c = 25\%$ has been found.

Assuming a normal distribution, this corresponds to a 10 and 5 per cent cut-off, respectively, on the lower and the upper side of the mean value of

$$\begin{aligned}\varphi_{0.10} &= 0.68\varphi; & \varphi_{0.05} &= 0.59\varphi \\ \varphi_{0.90} &= 1.32\varphi; & \varphi_{0.95} &= 1.41\varphi\end{aligned}$$

The prediction error should be taken into account in a probabilistic approach where appropriate.

It is not known whether creep approaches a finite value or not. However, in this constitutive approach the development of basic creep is predicted by a logarithmic function having no finite value, whereas a hyperbolic function was chosen for the description of drying creep which approaches an asymptotic value for $t \rightarrow \infty$.

Evaluations on the basis of test results indicate that these equations give a reasonably good approximation for the time development of creep up to 50 years of loading under the conditions indicated in Tables 5.1-10 and 5.1-11. From laboratory observations of creep up to 30 years, we can conclude that the increase of creep from 50 years up to 150 years of duration of loading will not exceed 10 % of the creep after 50 years. Recent observations in practice show after 30-50 years in service considerable higher deformations than expected from creep predictions. The reasons are not yet clarified. There might be excessive creep strains but also different constructional reasons.

(b) Creep coefficient

The creep coefficient $\varphi(t, t_0)$ may be calculated from:

$$\varphi(t, t_0) = \varphi_{bc}(t, t_0) + \varphi_{dc}(t, t_0) \quad (5.1-63)$$

where:

$\varphi_{bc}(t, t_0)$ is the basic creep coefficient according to Eq. (5.1-64);

$\varphi_{dc}(t, t_0)$ is the drying creep coefficient according to Eq. (5.1-67);

t is the age of concrete in days at the moment considered;

t_0 is the age of concrete at loading in days adjusted according to Eqs. (5.1-73) and (5.1-85).

The basic creep coefficient $\varphi_{bc}(t, t_0)$ may be estimated from:

$$\varphi_{bc}(t, t_0) = \beta_{bc}(f_{cm}) \cdot \beta_{bc}(t, t_0) \quad (5.1-64)$$

with:

$$\beta_{bc}(f_{cm}) = \frac{1.8}{(f_{cm})^{0.7}} \quad (5.1-65)$$

and the time development function

$$\beta_{bc}(t, t_0) = \ln \left[\left(\frac{30}{t_{0, adj}} + 0.035 \right)^2 \cdot (t - t_0) + 1 \right] \quad (5.1-66)$$

where:

f_{cm} is the mean compressive strength at an age of 28 days in MPa according to Eq. (5.1-1).

$t_{0, adj}$ is the adjusted age at loading in days according to Eq. (5.1-73)

The drying creep coefficient $\varphi_{dc}(t, t_0)$ may be estimated from:

$$\varphi_{dc}(t, t_0) = \beta_{dc}(f_{cm}) \cdot \beta(RH) \cdot \beta_{dc}(t_0) \cdot \beta_{dc}(t, t_0) \quad (5.1-67)$$

with:

$$\beta_{dc}(f_{cm}) = \frac{412}{(f_{cm})^{1.4}} \quad (5.1-68)$$

$$\beta(RH) = \frac{1 - \frac{RH}{100}}{\sqrt[3]{0.1 \cdot \frac{h}{100}}} \quad (5.1-69)$$

$$\beta_{dc}(t_0) = \frac{1}{0.1 + t_{0, adj}^{0.2}} \quad (5.1-70)$$

In cases where a lower level of accuracy is sufficient, the values given in Table 5.1-10 can be accepted as representative values for the creep coefficient after 50 years of loading of a normal weight ordinary structural concrete with a characteristic compressive strength between C20 and C50. For a normal weight high strength structural concrete with a characteristic compressive strength between C60 and C100 Table 5.1-11 is valid.

Table 5.1-10: Creep coefficient $\varphi(50y, t_0)$ of an ordinary structural concrete after 50 years of loading (service life according to Table 3.3-1)

Age at loading t_0 [days]	Dry atmospheric conditions ($RH = 50\%$, indoors)			Humid atmospheric conditions ($RH = 80\%$, outdoors)		
	Notional size $2A_c/u$ [mm]					
	50	150	600	50	150	600
1	4.8	4.0	3.3	3.2	2.9	2.6
7	3.5	2.9	2.4	2.4	2.2	2.0
28	2.7	2.3	1.9	1.9	1.7	1.5
90	2.1	1.8	1.5	1.5	1.3	1.2
365	1.6	1.3	1.1	1.1	1.0	0.9

Table 5.1-11: Creep coefficient $\varphi(50y, t_0)$ of a normal weight high strength concrete after 50 years of loading (service life according to Table 3.3-1)

Age at loading t_0 [days]	Dry atmospheric conditions ($RH = 50\%$, indoors)			Humid atmospheric conditions ($RH = 80\%$, outdoors)		
	Notional size $2A_c/u$ [mm]					
	50	150	600	50	150	600
1	2.3	2.0	1.7	1.7	1.6	1.5
7	1.7	1.5	1.3	1.3	1.2	1.1
28	1.3	1.1	1.0	1.0	0.9	0.9
90	1.0	0.9	0.8	0.8	0.7	0.7
365	0.7	0.7	0.6	0.6	0.5	0.5

The data given in Tables 5.1-10 and 5.1-11 apply for a mean temperature of the concrete between 10 °C and 20 °C. Seasonal variations of temperature between -20 °C and +40 °C can be accepted. The same holds true for variations in relative humidity around the mean values given in the same tables.

Creep of powder type SCC is affected by its high paste content. In general, the creep deformation is approximately 10–20 % higher than that of conventional concrete of equal strength. However, the deformations are within the scatter band for ordinary structural concrete, which is defined to be ± 30 %. If the structural response is sensitive to variations in creep behaviour tests are highly recommended.

The higher creep tendency of lightweight aggregate concrete due to the reduced stiffness of the aggregates is partially compensated by the lower creep capability of the stiffer cement paste matrix.

A more sophisticated and comprehensive model for calculating the creep deformations of normal and high strength lightweight aggregate concrete with expanded clay aggregates was published by Kvitsel, V. "Prediction of shrinkage and creep of normal strength and high-strength structural lightweight concrete made with expanded clay aggregates" (Dissertation, Institute of Concrete Structures and Building Materials, Karlsruhe Institute of Technology (KIT), 2011; in German).

The development of drying creep with time is described by:

$$\beta_{dc}(t, t_0) = \left[\frac{(t - t_0)}{\beta_h + (t - t_0)} \right]^{\gamma(t_0)} \quad (5.1-71a)$$

with:

$$\gamma(t_0) = \frac{1}{2.3 + \frac{3.5}{\sqrt{t_{0, adj}}}} \quad (5.1-71b)$$

$$\beta_h = 1.5 \cdot h + 250 \cdot \alpha_{f_{cm}} \leq 1500 \cdot \alpha_{f_{cm}} \quad (5.1-71c)$$

with:

$$\alpha_{f_{cm}} = \left(\frac{35}{f_{cm}} \right)^{0.5} \quad (5.1-71d)$$

where:

f_{cm} is the mean compressive strength at an age of 28 days in MPa according to Eq. (5.1-1);

RH is the relative humidity of the ambient environment in %;

$h = 2A_c/u$, is the notional size of the member in mm, where A_c is the cross-section in mm² and u is the perimeter of the member in contact with the atmosphere in mm;

$t_{0, adj}$ is the adjusted age at loading in days according to Eq. (5.1-73).

For lightweight aggregate concrete the relevant creep coefficient φ_l may be calculated according to Eq. (5.1-72):

$$\varphi_l = \eta_E \cdot \varphi(t, t_0) \quad (5.1-72)$$

where:

$\eta_E = (\rho/2200)^2$, with oven-dry density ρ in kg/m³;
 $\varphi(t, t_0)$ is the creep coefficient according to Eq. (5.1-63).

For concrete grades LC12 and LC16, the creep coefficient φ_l has to be additionally multiplied with a factor 1.3.

Different types of cement result in different degrees of hydration at the same age. Creep of concrete depends on the degree of hydration reached at a given age rather than on the age of the concrete. Therefore, the effect of the type of cement is taken into account by modifying the age at loading so that, for a given modified age, the degree of hydration is approximately independent of the type of cement. Note, that the duration of loading $(t - t_0)$ used in Eqs. (5.1-66) and (5.1-71a) is the actual time under load.

Eq. 5.1-73 was developed based on experimental results primarily with CEM I and CEM III cements. If other cement types are used or if large amounts of pozzolans are used in partial replacement of CEM I and the development of the creep deformations has high relevance for the design, this effect should be determined experimentally.

Green concretes may for example be produced by replacing a large part of the cement by the residual product fly ash. Mainly resulting from the reduced cement content a lower creep capability could be observed in corresponding creep experiments. However, when considering slowly hardening cement in Eq. (5.1-73) describing the delayed hydration of fly ash concretes the creep coefficient is increased due to the lower modified age at loading. The model may therefore overestimate the actual creep deformations of green concretes containing a large amount of fly ash.

The non-linear behaviour of concrete under high stresses mainly results from micro-cracking. Eq. (5.1-74) represents a simplification in so far as it just describes the increase of the magnitude of creep but does not take into account the observation that non-linearity decreases with increasing duration of loading. Further this approach neglects the differences in non-linear behaviour to be observed between basic creep and drying creep.

It should be noted that delayed elastic strains upon total unloading may be assumed as linear functions of stress up to stress levels of $|\sigma_c| = 0.6f_{cm}(t_0)$ though some experiments indicate some over-proportionality.

Due to microstructural mechanisms becoming dominant for high strength concrete, the total shrinkage has to be separated into basic shrinkage and drying shrinkage.

For curing periods of concrete members $t_s < 14$ days at normal ambient temperatures, the duration of moist curing does not significantly affect the total shrinkage. Hence, this parameter as well as the effect of curing temperature is not taken into account.

In Eqs. (5.1-77) and (5.1-82) the actual duration of drying $(t - t_s)$ has to be used.

Similar to creep, total shrinkage does not depend on concrete compressive strength per se. Drying shrinkage decreases with decreasing water/cement ratio and decreasing cement paste content, whereas basic shrinkage increases with decreasing water/cement ratio and decreases with decreasing cement paste content. The compressive strength serves as a convenient substitute parameter, always known at the design stage.

If the composition of concrete deviates considerably from ordinary structural concrete (e.g. green concrete) it is recommended to run tests. This also holds true for ordinary concrete in case the structural response is sensitive to shrinkage deformations or in case cement types are used other than CEM I, CEM II and CEM III, or if high amounts of pozzolans are used in partial replacement of CEM I.

(c) Effect of type of cement and curing temperature

The effect of the type of cement on the creep coefficient of concrete may be taken into account by modifying the age at loading t_0 to $t_{0, adj}$:

$$t_{0, adj} = t_{0, T} \cdot \left[\frac{9}{2 + t_{0, T}^{1.2}} + 1 \right]^\alpha \geq 0.5 \text{ days} \quad (5.1-73)$$

where:

$t_{0, T}$ is the age of concrete at loading in days adjusted according to Eq. (5.1-85);

α is a coefficient which depends on the type of cement:

$\alpha = -1$ for strength class 32.5 N;

$\alpha = 0$ for strength classes 32.5 R, 42.5 N;

$\alpha = 1$ for strength classes 42.5 R, 52.5 N, 52.5 R.

(d) Effect of high stresses

For stress levels in the range $0.4f_{cm}(t_0) < |\sigma_c| \leq 0.6f_{cm}(t_0)$ the non-linearity of creep may be taken into account using:

$$\varphi_\sigma(t, t_0) = \varphi(t, t_0) \cdot \exp[1.5(k_\sigma - 0.4)] \text{ for } 0.4 < k_\sigma \leq 0.6 \quad (5.1-74)$$

where:

$\varphi_\sigma(t, t_0)$ is the non-linear creep coefficient;

$\varphi(t, t_0)$ is the creep coefficient according to Eq. (5.1-63);

$k_\sigma = |\sigma_c| / f_{cm}(t_0)$, which is the stress-strength ratio.

5.1.9.4.4 Shrinkage

The total shrinkage or swelling strains $\varepsilon_{cs}(t, t_s)$ may be calculated as:

$$\varepsilon_{cs}(t, t_s) = \varepsilon_{cbs}(t) + \varepsilon_{cds}(t, t_s) \quad (5.1-75)$$

where shrinkage is subdivided into the basic shrinkage $\varepsilon_{cbs}(t)$ which occurs even if no moisture loss is possible:

$$\varepsilon_{cbs}(t) = \varepsilon_{cbs0}(f_{cm}) \cdot \beta_{bs}(t) \quad (5.1-76)$$

and the drying shrinkage $\varepsilon_{cds}(t, t_s)$ giving the additional shrinkage if moisture loss occurs:

$$\varepsilon_{cds}(t, t_s) = \varepsilon_{cds0}(f_{cm}) \cdot \beta_{RH}(RH) \cdot \beta_{ds}(t - t_s) \quad (5.1-77)$$

where:

t is the concrete age in days;

t_s is the concrete age at the beginning of drying in days;

$(t - t_s)$ is the duration of drying in days.

The basic shrinkage component $\varepsilon_{cbs}(t)$ may be estimated by means of the basic notional shrinkage coefficient $\varepsilon_{cbs0}(f_{cm})$ and the time function $\beta_{bs}(t)$:

$$\varepsilon_{cbs0}(f_{cm}) = -\alpha_{bs} \left(\frac{0.1 \cdot f_{cm}}{6 + 0.1 \cdot f_{cm}} \right)^{2.5} \cdot 10^{-6} \quad (5.1-78)$$

Tests should be performed according to:

- RILEM TC 107-CSP: “Creep and shrinkage prediction models: Principles of their formation. Recommendation for Measurement of time-dependent strains of concrete” (Materials and Structures, Vol. 31, October 1998, pp. 507–512);
- ISO 1920-8: “Testing of Concrete – Part 9: Determination of drying shrinkage for samples prepared in the field or in the laboratory”.

In some countries outside Europe, higher cement contents than recommended by EN 206 are applied for structural concrete. This leads to an increased shrinkage up to 20 % compared to the prediction of this model.

A mean coefficient of variation of predicted shrinkage has been estimated on the basis of a computerized database, resulting in $V_s = 35\%$. The corresponding 10 and 5% cut-off values are

$$\begin{aligned}\varepsilon_{cs0.10} &= 0.55\varepsilon_{cs}; & \varepsilon_{cs0.05} &= 0.42\varepsilon_{cs} \\ \varepsilon_{cs0.90} &= 1.45\varepsilon_{cs}; & \varepsilon_{cs0.95} &= 1.58\varepsilon_{cs}\end{aligned}$$

when a normal distribution is assumed.

In cases where a lower level of accuracy is sufficient, the values given in Table 5.1-13 can be accepted as representative values for total shrinkage after 50 years of drying of a normal weight ordinary structural concrete with a characteristic compressive strength between C20 and C50 produced with a cement of types 32.5 R or 42.5 N. Usually these values may be taken as final shrinkage values.

Though shrinkage reaches a final value, little information exists on the shrinkage strains of large members after long durations of drying. Therefore, the values calculated using Eq. (5.1-82) for $2A_c/u = 600$ mm, and the values given in Table 5.1-13 for shrinkage of members with a notional size of $2A_c/u = 600$ mm, respectively, are uncertain and may overestimate the actual shrinkage strains after 50 years of drying.

Table 5.1-13: Total shrinkage values $\varepsilon_{cs,50y} \cdot 10^3$ of an ordinary structural concrete after a duration of drying of 50 years (service life according to Table 3.3-1)

Dry atmospheric conditions ($RH = 50\%$, indoors)			Humid atmospheric conditions ($RH = 80\%$, outdoors)		
Notional size $2A_c/u$ [mm]					
50	150	600	50	150	600
-0.61	-0.60	-0.49	-0.38	-0.38	-0.31

The values given in Table 5.1-14 can be accepted as representative values for total shrinkage after 50 years of drying of a normal weight high strength structural concrete with a characteristic compressive strength between C60 and C100.

Table 5.1-14: Total shrinkage values $\varepsilon_{cs,50y} \cdot 10^3$ of a normal weight high strength concrete after a duration of drying of 50 years (service life according to Table 3.3-1)

Dry atmospheric conditions ($RH = 50\%$, indoors)			Humid atmospheric conditions ($RH = 80\%$, outdoors)		
Notional size $2A_c/u$ [mm]					
50	15	600	50	150	600
-0.51	-0.51	-0.44	-0.37	-0.36	-0.32

The shrinkage of powder type SCC is affected by its high paste content. The ultimate shrinkage deformation is approximately 20% higher than that of conventional concrete of equal strength. However, the deformations are within the scatter band, which is defined to be

$$\beta_{bs}(t) = 1 - \exp(-0.2 \cdot \sqrt{t}) \quad (5.1-79)$$

where:

- f_{cm} is the mean compressive strength at the age of 28 days in MPa according to Eq. (5.1-1);
- α_{bs} is a coefficient, dependent on the type of cement (see Table 5.1-12).

Table 5.1-12: Coefficients α_i used in Eqs. (5.1-78) and (5.1-80)

Strength class of cement	α_{bs}	α_{ds1}	α_{ds2}
32.5 N	800	3	0.013
32.5 R, 42.5 N	700	4	0.012
42.5 R, 52.5 N, 52.5 R	600	6	0.012

The drying shrinkage $\varepsilon_{cds}(t, t_s)$ is calculated by means of the notional drying shrinkage coefficient $\varepsilon_{cds0}(f_{cm})$, the coefficient $\beta_{RH}(RH)$, taking into account the effect of the ambient relative humidity, and the function $\beta_{ds}(t - t_s)$ describing the time-development:

$$\varepsilon_{cds0}(f_{cm}) = \left[(220 + 110 \cdot \alpha_{ds1}) \cdot \exp(-\alpha_{ds2} \cdot f_{cm}) \right] \cdot 10^{-6} \quad (5.1-80)$$

$$\beta_{RH} = \begin{cases} -1.55 \cdot \left[1 - \left(\frac{RH}{100} \right)^3 \right] & \text{for } 40 \leq RH < 99 \% \cdot \beta_{s1} \\ 0.25 & \text{for } RH \geq 99 \% \cdot \beta_{s1} \end{cases} \quad (5.1-81)$$

$$\beta_{ds}(t - t_s) = \left(\frac{(t - t_s)}{0.035 \cdot h^2 + (t - t_s)} \right)^{0.5} \quad (5.1-82)$$

$$\beta_{s1} = \left(\frac{35}{f_{cm}} \right)^{0.1} \leq 1.0 \quad (5.1-83)$$

where:

- $\alpha_{ds1}, \alpha_{ds2}$ are coefficients, dependent on the type of cement (see Table 5.1-12);
- f_{cm} is the mean compressive strength at the age of 28 days in MPa according to Eq. (5.1-1);
- RH is the relative humidity of the ambient atmosphere in %;
- h = $2A_c/u$, is the notional size of the member in mm, where A_c is the cross-section in mm^2 and u is the perimeter of the member in contact with the atmosphere in mm;
- t is the concrete age in days;
- t_s is the concrete age at the beginning of drying in days;
- $(t - t_s)$ is the duration of drying in days.

$\pm 30\%$. If the structural response is sensitive to variations in the shrinkage behaviour, tests are highly recommended.

Higher fly ash contents in concrete (e.g. green concrete) tend to decrease the total shrinkage deformations, which may result from the reduced cement content. As the given model considers slowly hardening cements which would correctly describe the delayed hydration of fly ash concretes but not the reduced cement content, shrinkage experiments are recommended when shrinkage deformations are decisive in the design of green concrete structures.

In contrast to normal weight concrete the shrinkage behaviour of lightweight aggregate concrete (LAWC) is characterized by swelling deformations in the young concrete age. This results from water stored in the porous aggregates which is only slowly released into the cement paste matrix. This shrinkage characteristic of LWAC is not taken into account in Eq. (5.1-84). The observed swelling deformations are turning into shrinkage deformations only after a longer duration of drying. The final value of drying shrinkage depends on the moisture content of the aggregates.

For details on shrinkage characteristics of LWAC and the related modelling, see information on left column referring to Eq. (5.1-72) and Müller, H. S., Anders, I., Breiner, R. and Vogel, M., "Concrete: treatment of types and properties in *fib* Model Code 2010" (Structural Concrete, Vol. 14, No. 4, December 2013).

For structures sensitive to shrinkage deformations tests are recommended.

The models were developed based on experimental results primarily on concretes with CEM I and CEM III cements as only those data have been available. If other cement types are used or if large amounts of pozzolans are used in partial replacement of CEM I, and temperature effects have a major importance for the design, they should be determined experimentally.

Eq. (5.1-85), originally developed for normal strength concrete, is based on an activation energy for cement hydration of 33 kJ/mol. Research has shown that the activation energy not only depends on the type and strength class of cement, but also on the water/cement ratio, additions and admixtures. Nevertheless, there is no data basis available which would enable a modification of Eq. (5.1-85) regarding the use of additions and admixtures in common normal strength and high strength concretes.

Note that it is useful to limit the temperature range to 65°C during the hydration process (very young concrete) in order to avoid the delayed formation of ettringite in hardened concrete.

The effect of elevated or reduced temperatures on maturity is prominent only until the compressive strength reaches about 50% of the 28-day value.

The coefficient of thermal expansion depends on the type of aggregates and on the moisture state of the concrete. Thus it may vary between approximately $6 \cdot 10^{-6} \text{ K}^{-1}$ and $15 \cdot 10^{-6} \text{ K}^{-1}$. The design value of $10 \cdot 10^{-6} \text{ K}^{-1}$ is valid for normal strength and high strength concrete, as well as for self-compacting concrete and concrete containing high amounts of fly ash (e.g. green concrete).

Dependent on the stiffness and the coefficient of thermal expansion of the aggregates, the coefficient of thermal expansion of lightweight aggregate concrete ranges between $5 \cdot 10^{-6} \text{ K}^{-1}$ and $11 \cdot 10^{-6} \text{ K}^{-1}$.

The shrinkage of lightweight aggregate concrete $\varepsilon_{lcs}(t, t_s)$ may be roughly estimated as:

$$\varepsilon_{lcs}(t, t_s) = \eta \cdot \varepsilon_{cs}(t, t_s) \quad (5.1-84)$$

where:

$\varepsilon_{cs}(t, t_s)$ is calculated according to Eq. (5.1-75);

$\eta = 1.5$ for LC8, LC12, LC16;

$= 1.2$ for LC20 and higher.

5.1.10 Temperature effects

5.1.10.1 Range of application

The information given in the preceding sections is valid for a mean temperature, taking into account seasonal variations between about -20°C and $+40^\circ \text{C}$. In the following section, the effect of substantial deviations from a mean concrete temperature of 20°C for the range of approximately 0°C to $+80^\circ \text{C}$ is dealt with.

5.1.10.2 Maturity

The effect of elevated or reduced temperatures on the maturity of concrete may be taken into account by adjusting the concrete age:

$$t_T = \sum_{i=1}^n \Delta t_i \exp \left[13.65 - \frac{4000}{273 + T(\Delta t_i)} \right] \quad (5.1-85)$$

where:

t_T is the temperature-adjusted concrete age which replaces t in the corresponding equations in days;

Δt_i is the number of days where a temperature T prevails;

$T(\Delta t_i)$ is the mean temperature in $^\circ \text{C}$ during the time period Δt_i .

5.1.10.3 Thermal expansion

Thermal expansion of concrete may be calculated as:

$$\varepsilon_{cT} = \alpha_T \Delta T \quad (5.1-86)$$

where:

ε_{cT} is the thermal strain;

ΔT is the change of temperature in K;

α_T is the coefficient of thermal expansion in K^{-1} .

Where the structural response is sensitive to thermal strains, tests should be performed according to RILEM TC 129-MHT: “Test methods for mechanical properties of concrete at high temperatures. Recommendations Part 6: Thermal strain” (Materials and Structures, Supplement March 1997, pp. 17–21).

Eq. (5.1-87) is valid for sealed and unsealed concrete tested in the hot state shortly after completion of the heating. Considering all experimental data, a large scatter of the compressive strength values can be observed; see Müller, H. S., Anders, I., Breiner, R. and Vogel, M., “Concrete: treatment of types and properties in *fib* Model Code 2010” (Structural Concrete, Vol. 14, No. 4, December 2013). If a higher accuracy is required tests must be performed, for example according to RILEM TC 129-MHT: “Test methods for mechanical properties of concrete at high temperatures. Recommendations: Compressive strength for service and accident conditions” (Materials and Structures, Vol. 28, 1995, pp. 410–414).

Sustained moderately elevated temperatures after sufficient curing may slightly increase the compressive strength compared to strength development at normal ambient environment if drying of the member is possible.

So far no information is available for self-compacting concrete and green concrete.

No information is available on high strength concrete, self-compacting concrete, lightweight aggregate concrete and green concrete.

If the tensile strength is a major input parameter in the design of a structure the values calculated by Eq. (5.1-88) may be reduced or increased by 20 %, due to the large scatter of data.

Tests are recommended and should be performed according to RILEM TC 129-MHT: “Test methods for mechanical properties of concrete at high temperatures. Recommendations Part 4: Tensile strength for service and accident conditions” (Materials and Structures, Vol. 33, May 2000, pp. 219–223).

No information is available on high strength concrete, self-compacting concrete, lightweight aggregate concrete and green concrete.

If moisture gradients may occur, the flexural tensile strength may be lower by up to 20 %.

No information is available on high strength concrete, self-compacting concrete, lightweight aggregate concrete and green concrete.

Eqs. (5.1-91a) and (5.1-91b) might give in some cases a somewhat more pronounced effect than observed. Available experimental data show a considerably broad scatter band.

For the purpose of structural analysis, the coefficient of thermal expansion may be taken as $\alpha_T = 10 \cdot 10^{-6} \text{ K}^{-1}$ for normal weight concrete, and $\alpha_T = 8 \cdot 10^{-6} \text{ K}^{-1}$ for lightweight aggregate concrete.

5.1.10.4 Compressive strength

The effect of temperature in the range $0^\circ\text{C} \leq T \leq 80^\circ\text{C}$ on the compressive strength of normal strength and high strength normal weight and lightweight aggregate concrete, $f_{cm}(T)$ and $f_{lcm}(T)$, respectively, may be calculated as:

$$f_{cm}(T) = f_{cm}(1.06 - 0.003 \cdot T) \quad (5.1-87a)$$

$$f_{lcm}(T) = f_{lcm}(1.04 - 0.002 \cdot T) \quad (5.1-87b)$$

where:

$f_{cm}(T), f_{lcm}(T)$ is the compressive strength in MPa at the temperature T in $^\circ\text{C}$;

f_{cm}, f_{lcm} is the compressive strength in MPa at $T = 20^\circ\text{C}$ from Eqs. (5.1-1) and (5.1-2);

T is the temperature in $^\circ\text{C}$.

5.1.10.5 Tensile strength and fracture properties

In the range $0^\circ\text{C} \leq T \leq 80^\circ\text{C}$, the uniaxial tensile strength f_{ctm} of normal strength concrete is significantly affected by temperature according to:

$$f_{ctm}(T) = f_{ctm}(1.16 - 0.008 \cdot T) \quad (5.1-88)$$

where:

$f_{ctm}(T)$ is the uniaxial tensile strength in MPa at the temperature T in $^\circ\text{C}$;

f_{ctm} is the uniaxial tensile strength in MPa at $T = 20^\circ\text{C}$ from Eq. (5.1-3);

T is the temperature in $^\circ\text{C}$.

In the range $0^\circ\text{C} \leq T \leq 80^\circ\text{C}$, the dependency of the splitting tensile strength $f_{ct,sp}$ on the temperature of normal strength normal weight concrete at the time of testing is described by:

$$f_{ct,sp}(T) = f_{ct,sp}(1.06 - 0.003 \cdot T) \quad (5.1-89)$$

where:

$f_{ct,sp}(T)$ is the splitting tensile strength in MPa at the temperature T in $^\circ\text{C}$;

$f_{ct,sp}$ is the splitting tensile strength in MPa at $T = 20^\circ\text{C}$;

T is the temperature in $^\circ\text{C}$.

To estimate the effect of elevated or reduced temperatures on flexural strength $f_{ct,fl}$ of normal strength normal weight concrete, we can use:

$$f_{ct,fl}(T) = f_{ct,fl}(1.1 - 0.005 \cdot T) \quad (5.1-90)$$

where:

$f_{ct,fl}(T)$ is the flexural strength in MPa at the temperature T in $^\circ\text{C}$;

$f_{ct,fl}$ is the flexural strength in MPa at $T = 20^\circ\text{C}$;

T is the temperature in $^\circ\text{C}$.

Fracture energy G_F is strongly affected by temperature and moisture content of concrete at the time of testing. The effect of temperature on G_F of normal strength normal weight concrete may be estimated as:

No information is available on high strength concrete, self-compacting concrete, lightweight aggregate concrete and green concrete.

Eq. (5.1-92) is valid for sealed and unsealed concrete. The observed scatter of data is indicated in Müller, H. S., Anders, I., Breiner, R. and Vogel, M., “Concrete: treatment of types and properties in *fib* Model Code 2010” (Structural Concrete, Vol. 14, No. 4, December 2013).

No information is available on self-compacting concrete.

If the structural response is sensitive to concrete stiffness, tests are recommended according to RILEM TC 129-MHT: “Test methods for mechanical properties of concrete at high temperatures. Recommendations: Modulus of elasticity for service and accident conditions” (Materials and Structures, Vol. 37, March 2004, pp. 139–144).

The relations to predict the effect of temperature up to 80 °C on creep given in this section are only rough estimates. For a more accurate prediction considerably more sophisticated models are required which take into account the moisture state of the concrete at the time of loading. Neglecting this parameter the relations given in this section are generally more accurate for thick concrete members with little change in moisture content than for thin members where significant changes in moisture content occur, particularly at elevated temperatures.

There is no information available on self-compacting concrete, lightweight aggregate concrete and green concrete.

If the structural response is sensitive to concrete creep, tests are urgently recommended, for example according to RILEM TC 129-MHT: “Test methods for mechanical properties of concrete at high temperatures. Recommendations Part 8: Steady-state creep and creep recovery for service and accident conditions” (Materials and Structures, Vol. 33, Jan.–Feb. 2000, pp. 6–13).

$$\text{dry concrete: } G_F(T) = G_F(1.06 - 0.003 \cdot T) \quad (5.1-91a)$$

$$\text{mass concrete: } G_F(T) = G_F(1.12 - 0.006 \cdot T) \quad (5.1-91b)$$

where:

$G_F(T)$ is the fracture energy in N/m at a temperature T in °C;

G_F is the fracture energy in N/m at $T = 20$ °C from Eq. (5.1-9);

T is the temperature in °C.

5.1.10.6 Modulus of elasticity

The effect of elevated or reduced temperatures at the time of testing on the modulus of elasticity of normal strength and high strength normal weight concrete and lightweight aggregate concrete at an age of 28 days may be estimated from:

$$E_{ci}(T) = E_{ci}(1.06 - 0.003 \cdot T) \quad (5.1-92a)$$

$$E_{lci}(T) = E_{lci}(1.04 - 0.002 \cdot T) \quad (5.1-92b)$$

where:

$E_{ci}(T), E_{lci}(T)$ is the modulus of elasticity in MPa at the temperature T in °C;

E_{ci}, E_{lci} is the modulus of elasticity in MPa at $T = 20$ °C from Eq. (5.1-20) and (5.1-22);

T is the temperature in °C.

5.1.10.7 Creep and shrinkage

5.1.10.7.1 Creep

The effect of temperature prior to loading may be taken into account using Eq. (5.1-85).

Eqs. (5.1-93) to (5.1-97) below describe the effect of a constant temperature differing from 20°C while a normal weight concrete is under load.

The effect of temperature on the time-development of creep is taken into account using $\beta_{h,T}$:

$$\beta_{h,T} = \beta_h \cdot \beta_T \quad (5.1-93)$$

with:

$$\beta_T = \exp[1500/(273 + T) - 5.12] \quad (5.1-94)$$

where:

$\beta_{h,T}$ is a temperature dependent coefficient replacing β_h in Eq. (5.1-71a);

β_h is the coefficient according to Eq. (5.1-71c);

T is the temperature in °C.

The effect of temperature on the creep coefficient is taken into account using:

$$\varphi_{bc,T} = \varphi_{bc} \cdot \varphi_T \quad (5.1-95)$$

$$\varphi_{dc,T} = \varphi_{dc} \cdot \varphi_T^{1.2} \quad (5.1-96)$$

with:

$$\varphi_T = \exp[0.015(T - 20)] \quad (5.1-97)$$

where:

$\varphi_{bc,T}$ is a temperature dependent coefficient which replaces φ_{bc} in Eq. (5.1-63);

$\varphi_{dc,T}$ is a temperature dependent coefficient which replaces φ_{dc} in Eq. (5.1-63);

φ_{bc} is the basic creep coefficient according to Eq. (5.1-64);

φ_{dc} is the drying creep coefficient according to Eq. (5.1-67);

T is the temperature in °C.

Here the additional creep (transient creep) is addressed which is observed when a temperature increase occurs while concrete is under constant load (temperature increase after loading).

If the structural response is sensitive to concrete creep, tests are recommended according to RILEM TC 129-MHT: “Test methods for mechanical properties of concrete at high temperatures. Recommendations Part 7: Transient creep for service and accident conditions” (Materials and Structures, Vol. 31, June 1998, pp. 290–295).

Some experiments indicate not only an acceleration of shrinkage but also an increased basic shrinkage deformation if the concrete is subjected to ongoing elevated curing temperatures. This effect decreases with increasing concrete strength and is neglected in this simplified approach.

If shrinkage at temperatures deviating from 20 °C is a major input parameter tests may be performed according to RILEM TC 129-MHT: “Test methods for mechanical properties of concrete at high temperatures. Recommendations Part 7: Shrinkage for service and accident conditions” (Materials and Structures, Vol. 33, May 2000, pp. 224–228).

The effect of elevated temperatures on shrinkage is influenced considerably by the moisture content of the concrete prior to heating and the moisture loss after an increase of temperature.

Whether a concrete specimen is shrinking or swelling under particular ambient climate conditions is determined by its internal relative humidity and the temperature-dependent water sorption capacity. The transition point at the relative humidity between shrinkage and swelling (RH_T) is therefore dependent on the concrete temperature and the concrete compressive strength (substitute parameter for the concrete composition).

For an increase of temperature while the structural member is under load, creep may be estimated from Eq. (5.1-98):

$$\varphi(t, t_0, T) = \varphi(t, t_0) + \Delta\varphi_{T,trans} \quad (5.1-98)$$

with:

$$\Delta\varphi_{T,trans} = 0.0004(T - 20)^2 \quad (5.1-99)$$

where:

$\varphi(t, t_0)$ is the creep coefficient according to Eq. (5.1-63) and temperature-adjusted according to Eqs. (5.1-95) to (5.1-97);

$\Delta\varphi_{T,trans}$ is the transient thermal creep coefficient which occurs at the time of the temperature increase;

T is the temperature in °C.

5.1.10.7.2 Shrinkage

Temperatures between 0 °C and 80 °C affect both components of shrinkage being defined by Eq. (5.1-75).

For basic shrinkage mainly the time-development is influenced. To take this effect into consideration, the concrete age t in Eq. (5.1-79) has to be replaced by the effective concrete age t_T according to Eq. (5.1-85).

The effect of a constant temperature differing from 20 °C while the concrete is drying is described by means of Eqs. (5.1-100) to (5.1-105) below.

The effect of temperature on the time-development of drying shrinkage is taken into account using $\alpha_{sT}(T)$:

$$\alpha_{sT}(T) = 0.035 \cdot h^2 \exp[-0.06(T - 20)] \quad (5.1-100)$$

where:

$\alpha_{sT}(T)$ is a temperature-dependent coefficient replacing the product $0.035 h^2$ in Eq. (5.1-82);

T is the temperature in °C.

The effect of temperature on the magnitude of drying shrinkage is taken into account using:

$$\beta_{RH,T} = \beta_{RH} \cdot \beta_{sT} \quad (5.1-101)$$

$\beta_{RH,T}$ is a temperature-dependent coefficient, which replaces β_{RH} in Eq. (5.1-77). It is calculated using:

$$\beta_{sT} = 1 + \left(\frac{4}{103 - RH} \right) \cdot \left(\frac{T - 20}{40} \right) \quad (5.1-102)$$

$$\beta_{RH} = \begin{cases} -1.55 \cdot \left[1 - \left(\frac{RH}{100} \right)^3 \right] & \text{for } 40 \leq RH < RH_T \\ 0.25 & \text{for } RH \geq RH_T \end{cases} \quad (5.1-103)$$

$$RH_T = 99 \cdot \beta_{s1} + \beta_{s1,T} \leq 100 \% \quad (5.1-104)$$

$$\beta_{s1} = \left(\frac{35}{f_{cm}} \right)^{0.1} \leq 1.0 \quad (\text{see Eq. (5.1-83)})$$

$$\beta_{s1,T} = \left(\frac{T - 20}{25} \right)^3 \quad (5.1-105)$$

For the material properties of concrete at high temperatures, see *fib* Bulletin 38 “Fire design of concrete structures – materials, structures and modelling” (*fib*, 2007) and *fib* Bulletin 46 “Fire design of concrete structures – structural behaviour and assessment” (*fib*, 2008) as well as chapter 3 “Material properties” in EN 1992-1-2:2004 and AC 208 “Eurocode 2: Design of concrete structures – Part 1-2: General rules – structural fire design”.

Approximate moisture content m of structural concrete [% by mass]:

- general indoor: 2 %;
- general outdoor: 4 %;
- exposed to rain: 6 %.

Eq. (5.1-106) is also valid for lightweight aggregate concrete.

No information is available for self-compacting concrete. However, it is supposed that the behaviour of self-compacting concrete does not deviate significantly from that of ordinary structural concrete; that is, data on self-compacting concrete are supposed to lie within the scatter band of ordinary concrete.

Fatigue tests exhibit a large scatter in the number of cycles to failure. Therefore, probabilistic procedures are often applied in evaluating fatigue behaviour of concrete. For further details, see CEB Bulletin 188 “Fatigue of concrete structures, State-of-the-Art Report” (CEB, 1988) and *fib* Bulletin 42 “Constitutive modelling of high strength/high performance concrete, State-of-the-Art Report” (*fib*, 2008) as well as Lohaus, L., Oneschkow, N., and Wefer, M., “Design model for the fatigue behaviour of normal-strength, high-strength and ultra-high-strength concrete” (Structural Concrete, Vol. 13, No. 3, September 2012).

The relations in Eqs. (5.1-107) to (5.1-109) are valid for concrete stored in a constant environment of approximately 20 °C, 65 % relative humidity. Figure 5.1-7 shows Eqs. (5.1-107) to (5.1-109) in a diagram. The curves were developed based on experiments with ultra high strength concrete (up to C200) and validated for high strength and normal strength concrete as well. The curves have been verified with experiments up to 10^7 load cycles to failure ($\log N = 7$). For $\log N > 8$, the curves asymptotically approach the minimum stress level of the respective curve.

Permeable concrete immersed in water may have a lower fatigue strength than expressed by these relations. If pores are filled with water, even lower fatigue strength may be obtained due to water pressure.

where:

- RH is the relative humidity of the ambient environment in %;
- T is the temperature in °C;
- f_{cm} is the mean compressive strength at the age of 28 days in MPa according to Eq. (5.1-1).

5.1.10.8 High temperatures

The material properties are negatively affected by the influence of high temperatures. Changes to the microstructure of concrete during a high temperature exposure result in corresponding changes in mechanical and physical properties which are mainly caused by the thermo-hydraulic and thermo-mechanical behaviour of the material.

For a structural fire design point of view see subsection 7.5.1 of this Model Code.

5.1.10.9 Low temperatures (cryogenic temperatures)

The compressive strength increases at low temperatures as a function of temperature and moisture content of the concrete. The strength gain can be estimated as:

$$\Delta f_{cm} = 12m \left[1 - \left(\frac{T + 170}{170} \right)^2 \right] \quad (5.1-106)$$

where:

- Δf_{cm} is the gain in compressive strength in MPa;
- m is the moisture content in % by mass;
- T is the temperature in °C.

Eq. (5.1-106) is valid for T between $0^\circ \geq T \geq -170^\circ\text{C}$, for a single temperature drop.

Tensile strength and modulus of elasticity increase at low temperature. They can be estimated by means of Eqs. (5.1-3) and (5.1-20) inserting the respective compressive strength.

5.1.11 Properties related to non-static loading

5.1.11.1 Fatigue

5.1.11.1.1 Fatigue strength

For constant stress amplitude, the number N of cycles causing fatigue failure of plain concrete may be estimated from Eqs. (5.1-107) to (5.1-112) below. They are valid for pure compression, compression-tension and pure tension, respectively.

(a) Pure compression

For $S_{c,min} > 0.8$, the S-N relations for $S_{c,min} = 0.8$ are valid. For $0 \leq S_{c,min} \leq 0.8$, we can use:

$$\log N_1 = \frac{8}{(Y-1)} \cdot (S_{c,max} - 1) \quad (5.1-107)$$

$$\log N_2 = 8 + \frac{8 \cdot \ln(10)}{(Y-1)} \cdot (Y - S_{c,min}) \cdot \log \left(\frac{S_{c,max} - S_{c,min}}{Y - S_{c,min}} \right) \quad (5.1-108)$$

with:

$$Y = \frac{0.45 + 1.8 \cdot S_{c,min}}{1 + 1.8 \cdot S_{c,min} - 0.3 \cdot S_{c,min}^2}$$

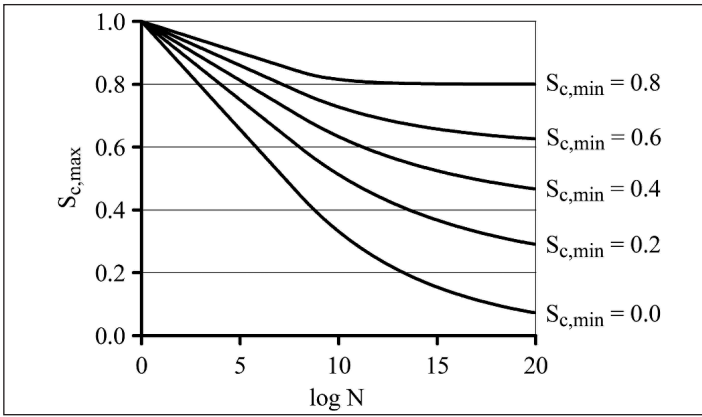


Figure 5.1-7: S-N relations according to Eqs. (5.1-107) to (5.1-109)

The fatigue reference compressive strength $f_{ck,fat}$ has been introduced to take into account the increasing fatigue sensitivity of concrete with increasing compressive strength.

Though experimental evidence has still to be given, the S-N relations may be assumed to apply also for self-compacting concrete due to material considerations.

If Eq. (5.1-111) is applied, it may be assumed that the concrete always fails in compression.

For concrete in tension, the crack propagation can be different for different types of concrete due to the differences in the internal material structure. For normal concrete the crack propagates in the cement paste and in the interface zone around the aggregates. However, for high strength concrete and concrete with lightweight aggregates the crack propagates in the cement paste and through the aggregates due to the relatively higher strength of the cement paste and the interface zone, respectively. Thus, for concrete types where the strength of the aggregates is of importance, the fatigue life of the aggregates should also be considered. However, test results have shown that the fatigue life seems relatively equal for the various concrete types, see *fib* Bulletin 42.

The S/N curves are mean curves of numbers of cycles to failure. The safety is taken care of by a further reduction of static strength.

Eqs. (5.1-107)–(5.1-112) are applicable for frequencies $f > 0.1$ Hz and for stress levels $S_{c,max}$ and $S_{ct,max} < 0.9$. For higher stress levels and lower frequencies, that is low cycle fatigue, lower values of $\log N$ than predicted by Eqs. (5.1-107) to (5.1-112) may be expected. For further details, see CEB Bulletin 188 “Fatigue of Concrete Structures – State-of-the-Art Report” (CEB, 1988) and *fib* Bulletin 42 “Constitutive modelling for high strength/high performance concrete” (*fib*, 2008).

A value of $\beta_{c,sus}(t, t_0) = 0.85$ has been chosen to take account of actual frequencies of loading which are in most practical cases significantly lower than those applied in experiments.

The value of the Palmgren-Miner sum indicating failure is varying in various codes from 0.2 to 1.0. Consequently, the Palmgren-Miner rule is only a very rough approximation of the actual concrete behaviour. It may over- or underestimate the actual fatigue strength of concrete subjected to varying repeated loads. Rest periods in the loading may increase the fatigue life.

Different parts of the concrete area are exposed to changing maximum and minimum stress levels. The different parts have to be treated using, for example, the Palmgren-Miner rule. Numerical simulations with, for example, the finite element method allow this to be treated effectively.

where:

$$(a) \text{ if } \log N_1 \leq 8, \text{ then } \log N = \log N_1 \quad (5.1-109a)$$

$$(b) \text{ if } \log N_1 > 8, \text{ then } \log N = \log N_2 \quad (5.1-109b)$$

with:

$$S_{c,max} = |\sigma_{c,max}| / f_{ck,fat};$$

$$S_{c,min} = |\sigma_{c,min}| / f_{ck,fat};$$

$$\Delta S_c = |S_{c,max}| - |S_{c,min}|.$$

The fatigue reference compressive strength $f_{ck,fat}$ may be estimated as:

$$f_{ck,fat} = \beta_{cc}(t) \beta_{c,sus}(t, t_0) f_{ck} (1 - f_{ck}/400) \quad (5.1-110)$$

(b) Compression-tension with $\sigma_{ct,max} \leq 0.026 |\sigma_{c,max}|$

$$\log N = 9(1 - S_{c,max}) \quad (5.1-111)$$

(c) Pure tension and tension-compression with $\sigma_{ct,max} > 0.026 |\sigma_{c,max}|$

$$\log N = 12(1 - S_{ct,max}) \quad (5.1-112)$$

with:

$$S_{ct,max} = \sigma_{ct,max} / f_{ctk,min}$$

where:

N is the number of cycles to failure;

$S_{c,max}$ is the maximum compressive stress level;

$S_{c,min}$ is the minimum compressive stress level;

$S_{ct,max}$ is the maximum tensile stress level;

ΔS_c is the stress level range;

$\sigma_{c,max}$ is the maximum compressive stress in MPa;

$\sigma_{c,min}$ is the minimum compressive stress in MPa;

$\sigma_{ct,max}$ is the maximum tensile stress in MPa;

f_{ck} is the characteristic compressive strength from Table 5.1-3;
 $f_{ck,fat}$ is the fatigue reference compressive strength from Eq. (5.1-110);

$f_{ctk,min}$ is the minimum characteristic tensile strength;

$\beta_{cc}(t)$ is a coefficient which depends on the age of concrete at the beginning of fatigue loading, to be taken from subsection 5.1.9.1, Eq. (5.1-51);

$\beta_{c,sus}(t, t_0)$ is a coefficient which takes into account the effect of high mean stresses during loading. For fatigue loading it may be taken as 0.85.

(d) Spectrum of load levels

To estimate the fatigue life for a spectrum of load levels the Palmgren-Miner summation may be applied. Fatigue failure occurs if $D = 1$.

$$D = \sum_i \frac{n_{Si}}{N_{Ri}} \quad (5.1-113)$$

where:

D is the fatigue damage;

n_{Si} is the number of acting stress cycles at a given stress level and stress range;

N_{Ri} is the number of cycles causing failure at the same stress level and stress range according to Eqs. (5.1-107) to (5.1-112).

In Eq. (5.1-114) it is assumed that creep due to repeated loading is equal to creep under a constant stress $(|\sigma_{c,max}| + |\sigma_{c,min}|)/2$ acting for a time $(t - t_0) = (n/f) = \text{duration of repeated loading [days]}$, where:

- n is the number of cycles applied at a frequency f ;
- f is the frequency of repeated loading [min^{-1}].

Therefore, Eq. (5.1-114) gives only a rough estimate of the creep strains due to repeated loads. It does not take into account variations of E_c due to repeated loads or tertiary creep which develops prior to fatigue failure. For further details, see CEB Bulletin 188 “Fatigue of Concrete Structures – State-of-the-Art Report” (CEB, 1988).

The given constitutive relations are valid also for lightweight aggregate concrete.

No information is available for self-compacting concrete. However, it is supposed that the behaviour of self-compacting concrete does not deviate significantly from that of ordinary structural concrete; that is, data on self-compacting concrete are supposed to lie within the scatter band of ordinary concrete.

5.1.11.2 Fatigue strains

For maximum compressive stresses $|\sigma_{c,max}| < 0.6f_{ck}$ and a mean stress $(|\sigma_{c,max}| + |\sigma_{c,min}|)/2 < 0.5f_{ck}$ the strain at maximum stress due to repeated loads of a given frequency f may be estimated as:

$$\varepsilon_{cf}(n) = \frac{\sigma_{c,max}}{E_{ci}(t_0)} + \frac{\sigma_{c,max} + \sigma_{c,min}}{2E_{ci}} \varphi(t, t_0) \quad (5.1-114)$$

where:

- ε_{cf} is the strain at maximum stress due to repeated loads;
- $\sigma_{c,max}$ is the maximum compressive stress in MPa;
- $\sigma_{c,min}$ is the minimum compressive stress in MPa;
- E_{ci} is the modulus of elasticity of concrete in MPa at a concrete age of 28 days according to Eq. (5.1-20);
- $E_{ci}(t_0)$ is the modulus of elasticity of concrete in MPa at a concrete age t_0 according to Eq. (5.1-56);
- $\varphi(t, t_0)$ is the creep coefficient according to Eq. (5.1-63);
- t_0 is the age of concrete at the beginning of repeated loading in days;
- t is the age of concrete at the moment considered in days.

5.1.11.2 Stress and strain rate effects – impact

5.1.11.2.1 Range of applicability

The information given below is valid for monotonically increasing compressive stresses or strains at a constant range of approximately $1 \text{ MPa/s} < |\dot{\sigma}_c| < 10^7 \text{ MPa/s}$ and $30 \cdot 10^{-6} \text{ s}^{-1} < |\dot{\varepsilon}_c| < 3 \cdot 10^2 \text{ s}^{-1}$, respectively. In the corresponding equations all strain and stress values have to be used as absolute values.

For tensile stresses or strains the information is valid approximately for $0.03 \text{ MPa/s} < \dot{\sigma}_{ct} < 10^7 \text{ MPa/s}$ and $1 \cdot 10^{-6} \text{ s}^{-1} < \dot{\varepsilon}_{ct} < 3 \cdot 10^2 \text{ s}^{-1}$, respectively.

5.1.11.2.2 Compressive strength

For a given strain and stress rate, respectively, the compressive strength under high rates of loading may be estimated as:

$$f_{c,imp,k}/f_{cm} = (\dot{\varepsilon}_c/\dot{\varepsilon}_{c0})^{0.014} \quad \text{for } \dot{\varepsilon}_c \leq 30 \text{ s}^{-1} \quad (5.1-115a)$$

$$f_{c,imp,k}/f_{cm} = 0.012(\dot{\varepsilon}_c/\dot{\varepsilon}_{c0})^{1/3} \quad \text{for } \dot{\varepsilon}_c > 30 \text{ s}^{-1} \quad (5.1-115b)$$

with $\dot{\varepsilon}_{c0} = 30 \cdot 10^{-6} \text{ s}^{-1}$

and

$$f_{c,imp,k}/f_{cm} = (\dot{\sigma}_c/\dot{\sigma}_{c0})^{0.014} \quad \text{for } \dot{\sigma}_c \leq 10^6 \text{ MPa s}^{-1} \quad (5.1-116a)$$

$$f_{c,imp,k}/f_{cm} = 0.012(\dot{\sigma}_c/\dot{\sigma}_{c0})^{1/3} \quad \text{for } \dot{\sigma}_c > 10^6 \text{ MPa s}^{-1} \quad (5.1-116b)$$

with $\dot{\sigma}_{c0} = 1 \text{ MPa s}^{-1}$,

where

$f_{c,imp,k}$ is the impact compressive strength;

$\dot{\varepsilon}_c$ is the strain rate in s^{-1} ;

f_{cm} is the mean compressive strength in MPa;

$\dot{\sigma}_c$ is the stress rate in MPa/s .

5.1.11.2.3 Tensile strength and fracture properties

(a) Tensile strength

For a given strain and stress rate, respectively, the tensile strength under high rates of loading $f_{ct,imp,k}$ may be estimated as:

$$f_{ct,imp,k}/f_{ctm} = (\dot{\varepsilon}_{ct}/\dot{\varepsilon}_{ct0})^{0.018} \quad \text{for } \dot{\varepsilon}_{ct} \leq 10 \text{ s}^{-1} \quad (5.1-117a)$$

$$f_{ct,imp,k}/f_{ctm} = 0.0062(\dot{\varepsilon}_{ct}/\dot{\varepsilon}_{ct0})^{1/3} \quad \text{for } \dot{\varepsilon}_{ct} > 10 \text{ s}^{-1} \quad (5.1-117b)$$

with $\dot{\epsilon}_{ct0} = 1 \cdot 10^{-6} \text{ s}^{-1}$

and

$$f_{ct,imp,k}/f_{ctm} = (\dot{\sigma}_{ct}/\dot{\sigma}_{ct0})^{0.018} \text{ for } \dot{\sigma}_{ct} \leq 0.3 \cdot 10^6 \text{ MPa s}^{-1} \quad (5.1-118a)$$

$$f_{ct,imp,k}/f_{ctm} = 0.0062(\dot{\sigma}_{ct}/\dot{\sigma}_{ct0})^{1/3} \text{ for } \dot{\sigma}_{ct} > 0.3 \cdot 10^6 \text{ MPa s}^{-1} \quad (5.1-118b)$$

with $\dot{\sigma}_{ct0} = 0.03 \text{ MPa s}^{-1}$.

(b) Fracture energy

The information available regarding the effect of stress or strain rate on the fracture energy is too incomplete to be included in this Model Code.

5.1.11.2.4 Modulus of elasticity

The effect of stress and strain rate on the modulus of elasticity $E_{c,imp}$ may be estimated as:

$$E_{c,imp}/E_{ci} = (\dot{\sigma}_c/\dot{\sigma}_{c0})^{0.025} \quad (5.1-119a)$$

$$E_{c,imp}/E_{ci} = (\dot{\epsilon}_c/\dot{\epsilon}_{c0})^{0.026} \quad (5.1-119b)$$

with $\dot{\sigma}_{c0} = 1 \text{ MPa s}^{-1}$ and $\dot{\epsilon}_{c0} = 30 \cdot 10^{-6} \text{ s}^{-1}$ for compression;
with $\dot{\sigma}_{ct0} = 0.03 \text{ MPa s}^{-1}$ and $\dot{\epsilon}_{ct0} = 1 \cdot 10^{-6} \text{ s}^{-1}$ for tension.

5.1.11.2.5 Stress–strain relations

There is little information regarding the effect of high stress or strain rates on the shape of the stress–strain diagram.

No information is available for the strain-softening region.

For monotonically increasing compressive stresses or strains up to the peak stress, as an approximation Eq. (5.1-26) may be used together with Eqs. (5.1-115) and (5.1-116) for the peak stress $f_{c,imp}$, Eq. (5.1-119) for the modulus of elasticity $E_{c,imp}$ and Eq. (5.1-120) for the strain at maximum stress $\epsilon_{c1,imp}$.

The effects of high stress and strain rates on the strains at maximum stress in tension and compression may be estimated as:

$$\epsilon_{c1,imp}/\epsilon_{c1} = (\dot{\sigma}_c/\dot{\sigma}_{c0})^{0.02} = (\dot{\epsilon}_c/\dot{\epsilon}_{c0})^{0.02} \quad (5.1-120)$$

with $\dot{\sigma}_{c0} = 1 \text{ MPa s}^{-1}$ and $\dot{\epsilon}_{c0} = 30 \cdot 10^{-6} \text{ s}^{-1}$ for compression;
with $\dot{\sigma}_{ct0} = 0.03 \text{ MPa s}^{-1}$ and $\dot{\epsilon}_{ct0} = 1 \cdot 10^{-6} \text{ s}^{-1}$ for tension;
where:

$\epsilon_{c1,imp}$ is the impact strain at maximum load for compression and tension, respectively;

ϵ_{c1} is the strain at maximum load for static loading from subsections 5.1.8.1 and 5.1.8.2 for compression and tension, respectively.

5.1.12 Transport of liquids and gases in hardened concrete

Transport characteristics are difficult to predict since they may vary by several orders of magnitude, depending on concrete composition (e.g. water/cement ratio), type of materials (e.g. cement, pozzolanic additives), age, curing and moisture content of the concrete (e.g. environmental conditions).

The relations presented in this chapter may be assumed to be reasonable approximations. However, all relations correlated with compressive strength have to be handled carefully, as the compressive strength represents first a substitute value for the microstructure and second a mean value over the whole concrete cross-section, whereas the transport characteristics in the concrete cover are decisive concerning concrete durability. Therefore, when a more accurate prediction of transport characteristics is required, they should be determined experimentally.

The subsequent relations are valid for ordinary normal strength and high strength normal weight concrete according to subsection 5.1.2, unless noted otherwise.

Liquids, gases or ions may be transported in hardened concrete by the transport mechanisms permeation, diffusion, capillary action and by mixed modes of transport mechanisms.

For further details concerning the transport properties of normal weight concrete, see RILEM TC 116 PCD: “State-of-the-Art Report: Performance Criteria for Concrete Durability” (1995) or to RILEM TC 146 TCF: “State-of-the-art Report: Penetration and Permeability of Concrete: Barriers to organic and contaminating liquids” (1997) as well as *fib* Bulletins 51 and 53, “Structural concrete” and *fib* Bulletin 70 “Code-type models for structural behaviour of concrete – Background of the constitutive relations and material models in MC2010”.

Self-compacting concrete (SCC) with a comparable strength usually exhibits a denser microstructure than normal weight concrete, so that the models presented in this chapter should be on the safe side for SCC. Nevertheless, for further details concerning self-compacting concrete, see RILEM TC 205 DSC: “State-of-the-art Report: Durability of Self-Compacting Concrete” (2007).

Regarding lightweight concrete, it appears that its transport coefficients are slightly lower compared to normal strength concrete of the same grade, mainly due to the usually higher quality of the inner contact zone. However, this difference becomes negligible for higher strength grades. Further details concerning lightweight aggregate concrete can be found in, for example, Faust, T., “Lightweight concrete in structural engineering” (Ernst & Sohn, Berlin, 2002; in German).

In normal strength concrete the flow of water does not only occur in the capillary pores of the hydrated cement paste but also through internal micro-cracks, as well as along the porous interfaces between the matrix and coarse aggregates. These effects increase the permeability of concrete, which therefore equals or exceeds the permeability of the hydrated cement paste matrix.

The flow of water in the hydrated cement paste depends on the presence of interconnected capillary pores which are mainly determined by the water/cement ratio of the mix and the degree of hydration of the cement. Despite a low water/cement ratio, insufficient curing – which may result in a low degree of hydration, especially in the near surface region – may lead to a high permeability.

The appropriate use of silica fume or fly ash (e.g. according to ISO 22965-1:2007 “Concrete – Part 1: Methods of specifying and guidance for the specifier”), as is often the case in high strength concrete, leads to a densification of the matrix and the porous interface because of the preceding pozzolanic reactions and the filler effect of those additives. Depending on age and composition of the concrete, this effect can be even more pronounced than is expressed by Eq. (5.1-122).

The experimental determination of the coefficient of water permeability is not standardized so far. However, the penetration of water into concrete can be measured according to EN 12390-8 “Testing hardened concrete – Depth of penetration of water under pressure” and converted into a coefficient of water permeability, but it has to be considered as an approximate value only.

Similar to the flow of water, gases may pass through the pore system and micro-cracks of concrete under the influence of an external pressure. The coefficient of permeability K_g [m²] in Eq. (5.1-123) represents a constant material parameter. Therefore, the viscosity η of the gas flowing, as well as the pressure level p , have to be considered in the calculation of the volume of gas V .

5.1.12.1 Permeation

Permeation is the flow of liquids, for example water, or of gases, for example air, caused by a pressure head.

5.1.12.1.1 Water permeability

The transport of water is generally described by Darcy’s law:

$$V = K_w \frac{A}{l} \Delta h_w t \quad (5.1-121)$$

where:

- V is the volume of water in m³ flowing during time t ;
- Δh_w is the hydraulic head in m;
- A is the penetrated area in m²;
- t is the time in s;
- l is the thickness in m;
- K_w is the coefficient of permeability for water flow in m/s.

For mature concrete the coefficient of water permeability may be estimated roughly from the mean compressive strength of concrete f_{cm} according to Eq. (5.1-122):

$$K_w = \frac{4 \cdot 10^{-3}}{f_{cm}^6} \quad (5.1-122)$$

where:

- K_w is the coefficient of water permeability in m/s;
- f_{cm} is the mean compressive strength in MPa.

5.1.12.1.2 Gas permeability

For a stratified laminar flow, the volume of gas flowing through a porous material is given by:

$$V = K_g \frac{A}{l} \frac{p_1 - p_2}{\eta} p_m \frac{1}{p} t \quad (5.1-123)$$

If only one type of gas is considered η is normally taken as unity. Then K_g represents the specific permeability of the gas considered, and is given in m/s.

If also the influence of the pressure level p_m is neglected, the volume of gas flowing can be calculated from:

$$V = \bar{K}_g \frac{A}{l} \frac{p_1 - p_2}{p} t \quad (5.1-124)$$

where:

$\bar{K}_g$ is the coefficient of gas permeability in m²/s.

As is the case for water permeability lower water/cement ratio may lead to a lower coefficient of gas permeability with higher compressive strength. The use of additives (e. g. according to ISO 22965-1:2007) may even result in a further densification, especially at very high strength grades.

Aside from the pore structure of the concrete, the moisture content exerts an essential influence on its gas permeability. Eq. (5.1-125) is valid for a relative pore humidity of the concrete of less than about 65%. With increasing relative humidity of the concrete, K_g may be reduced by a factor up to 10⁻³. In contrast for concrete specimens that have been oven-dried before testing, K_g should be assumed one order of magnitude higher (factor 10).

Considering all experimental data, a large scatter of the gas permeability values can be observed. Therefore, when a more accurate prediction is required, the gas permeability should be determined experimentally. This may be done according to the RILEM Technical Recommendation: "Measurement of the gas permeability by RILEM – CEMBUREAU method" (Materials and Structures, Vol. 32, pp. 176–178, 1999).

In most cases, transient diffusion phenomena occur, that is the amount of substance diffusing varies with location x and time t . In this case, Fick's second law of diffusion is valid, which describes the change in concentration for an element with time according to Eq. (5.1-126) considering one-dimensional flow and the diffusion coefficient D to be a constant:

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} \quad (5.1-126)$$

In cases where the diffusing substance becomes immobile, such as in the case of diffusion of chloride ions, Eq. (5.1-126) has to be expanded:

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} + s \quad (5.1-127)$$

where s = sink, that is the amount of transported substance that becomes immobile. Note that bounded chloride ions may also be released, for example by carbonation. In this case s is negative, that is s = source.

Frequently, the diffusion of ions is described by:

$$\frac{\partial c_{free}}{\partial t} = D_{eff} \frac{\partial^2 c_{free}}{\partial x^2} \quad (5.1-129)$$

where c_{free} = concentration of free ions, D_{eff} = effective diffusion coefficient. If some of the ions become immobile, this is taken into account by an adjustment of the diffusion coefficient. Therefore, D_{eff} in Eq. (5.1-129) is not a constant but varies with time of exposure.

The transport of water vapour in the pore system of concrete involves different transport mechanisms and driving forces,

where:

V is the volume of gas in m³ flowing during time t ;
 K_g is the coefficient of gas permeability in m²;
 A is the penetrated area in m²;
 l is the length in m of the penetrated concrete;
 $p_1 - p_2$ is the pressure difference in N/m²;
 p_m is the mean pressure = $(p_1 + p_2)/2$ in N/m²;
 η is the viscosity of gas in Ns/m²;
 p is the local pressure, at which V is observed in N/m²;
 t is the time in s.

As a rough estimate, K_g for air, oxygen and nitrogen may be determined from the mean compressive strength of concrete f_{cm} from Eq. (5.1-125):

$$K_g = \frac{2 \cdot 10^{-10}}{f_{cm}^4} \quad (5.1-125)$$

where:

K_g is the coefficient of gas permeability in m²;
 f_{cm} is the mean compressive strength in MPa.

5.1.12.2 Diffusion

Gases, liquids and dissolved substances are transported due to a constant concentration gradient according to Fick's first law of diffusion, as given in Eq. (5.1-128):

$$Q = D \frac{c_1 - c_2}{l} A \cdot t \quad (5.1-128)$$

where:

Q is the amount of substance transported in g;
 $c_1 - c_2$ is the difference in concentration in g/m³;
 l is the length of the penetrated concrete in m;
 A is the penetrated area in m²;
 t is the time in s;
 D is the diffusion coefficient in m²/s.

5.1.12.2.1 Diffusion of water vapour

The transport of water in the vapour phase can be described by Fick's first law of diffusion, introducing a gradient of the relative

therefore $D \neq \text{const}$. In most cases diffusion theory is applied to describe moisture migration. As the driving force, the local moisture concentration c [g/m³] may be considered (see Eq. (5.1-128)).

The diffusion coefficient D at local moisture concentration c may be determined experimentally according to EN 12086 "Determination of water vapour transmission properties". This test method has been widely used for concrete specimens, but it has to be kept in mind that it is intended originally for thermal insulating products.

A more convenient approach to describe the water vapour diffusion is achieved by the definition of a relative pore humidity $0 < H < 1$, which is correlated with the moisture concentration c by sorption isotherms.

For transient phenomena, such as drying of a concrete cross-section, the balance Eq. (5.1-126) is transformed to:

$$\frac{\partial H}{\partial t} = \frac{\partial}{\partial x} \left(D(H) \frac{\partial H}{\partial x} \right) \quad (5.1-130)$$

Eq. (5.1-132) is taken from Bazant, Z.P. and Najjar, L. J., "Drying of concrete as a non-linear diffusion problem" (Cement and Concrete Research, Vol. 1, pp. 461–473, 1971).

Eq. (5.1-132) is valid for normal strength concrete only. No test data covering high strength concrete are available.

pore humidity as the driving force. The diffusion coefficient D is a non-linear function of the local relative pore humidity H . The volume of water flowing is given by:

$$V = D(H) \frac{dH}{dx} A \cdot t \quad (5.1-131)$$

where:

V is the volume of transported water in m³;

$D(H)$ is the diffusion coefficient in m²/s at relative pore humidity H ;

dH/dx is the gradient in relative pore humidity in m⁻¹;

A is the penetrated area in m²;

t is the time in s.

For isothermal conditions, the diffusion coefficient can be expressed as a function of the relative pore humidity $0 < H < 1$:

$$D(H) = D_1 \left[\alpha + \frac{1 - \alpha}{1 + [(1 - H)/(1 - H_c)]^n} \right] \quad (5.1-132)$$

where:

D_1 is the maximum of $D(H)$ in m²/s for $H = 1$;

D_0 is the minimum of $D(H)$ in m²/s for $H = 0$;

$\alpha = D_0/D_1$;

H_c is the relative pore humidity at $D(H) = 0.5D_1$;

n is an exponent;

H is the relative pore humidity.

The following approximate values may be assumed:

$\alpha = 0.05$;

$H_c = 0.80$;

$n = 15$.

D_1 may be estimated from:

$$D_1 = \frac{D_{1,o}}{f_{cm} - 8} \quad (5.1-133)$$

where:

$D_{1,o} = 1 \cdot 10^{-8}$ [m²/s];

f_{cm} is the mean compressive strength in MPa.

5.1.12.2.2 Diffusion of gases

So far no international standards exist to determine the diffusion coefficients of gases such as oxygen or carbon dioxide through concrete.

Eqs. (5.1-135) and (5.1-136) are valid for normal strength concrete stored in a constant environment of approximately 20 °C, 65 % relative humidity. For concrete exposed to a natural environment, for instance to rain, the diffusion coefficients are substantially lower than estimated from Eq. (5.1-135) or Eq. (5.1-136).

Based on Eqs. (5.1-127), (5.1-129) and (5.1-136), the progress of carbonation of a concrete under controlled conditions may be estimated from:

$$d_c^2 = 2D_{CO_2} \frac{C_a}{C_c} t \quad (5.1-134)$$

The diffusion of gases such as air, oxygen (O₂) or carbon dioxide (CO₂) is primarily controlled by the moisture content of the concrete. For intermediate moisture contents the diffusion coefficient for carbon dioxide or oxygen is in the range of $10^{-7} < D < 10^{-10}$ m²/s.

The diffusion coefficient for oxygen D_{O_2} through non-carbonated concrete may be roughly estimated from:

$$\log D_{O_2} = -0.02f_{cm} + 6.5 \quad (5.1-135)$$

where:

D_{O_2} is the diffusion coefficient of O₂ in m²/s;

f_{cm} is the mean compressive strength in MPa.

The diffusion coefficient for carbon dioxide D_{CO_2} through carbonated concrete may be roughly estimated from:

$$\log D_{CO_2} = -0.05f_{cm} + 6.1 \quad (5.1-136)$$

where:

- d_c is the depth of carbonation at time t in m;
- D_{CO_2} is the diffusion coefficient of CO_2 through carbonated concrete in m^2/s (from Eq. (5.1-136));
- C_a is the concentration of CO_2 in the air in g/m^3 ;
- C_c is the amount of CO_2 required for complete carbonation of a unit volume of concrete in g/m^3 ;
- t is the time in s.

For normal weight concrete made of Portland cement and exposed to a standard environment, C_a/C_c may be taken as $8 \cdot 10^{-6}$.

However, one should keep in mind that, in particular, the relative humidity of the surrounding atmosphere as well as the properties and the composition (e.g. the use of blast furnace slag cements) of a particular concrete have a strong influence on D_{CO_2} , so that Eq. (5.1-134) cannot give a reliable estimate of the progress of carbonation of a structure in service.

A more sophisticated model concerning the progress of carbonation is presented in subsection 5.1.13.2. As its application requires well-founded statistical knowledge and the use of numerical programs, the simplified and generally accepted approach in subsection 5.1.12.2.2 may serve as an easily applicable tool for a first estimation of the carbonation progress.

The diffusion coefficients of dissolved substances increase with increasing moisture content of the concrete.

The prediction of the transport of chloride ions into concrete is very complex because chlorides penetrating into concrete may be transported not only by diffusion but also by capillary suction. In addition, the external chloride concentration is variable, and some of the intruding chloride ions become immobile due to chemical reaction or time dependent physical adsorption. The amount of bound chlorides depends on the type of cement used, and it must be in equilibrium with the concentration of chlorides dissolved in the pore water. Only the dissolved chlorides take part in the diffusion process. In carbonated concrete all chlorides are dissolved in the pore water.

Eq. (5.1-137) is valid for normal and high strength Portland cement concrete without additives and a mean compressive strength f_{cm} lower than 95 MPa.

Eq. (5.1-138) may be used for normal and high strength concrete with reasonable amounts of silica fume, fly ash (e.g. according to EN 206-1) or blast furnace slag cements up to a compressive strength f_{cm} of 130 MPa.

The determination of diffusion coefficients is standardized in CEN TS 12390-11. Furthermore, reasonable experimental results can be achieved with an electrically accelerated method according to Tang, L., "Electrically accelerated methods for determining chloride diffusivity in concrete" (Magazine of Concrete Research, Vol. 48, pp. 173–179, 1996), which is described in NT Build 492.

Further information and a sophisticated model concerning the penetration of chlorides into concrete can be found in subsection 5.1.13.3. As its application requires well-founded statistical knowledge and the use of numerical programs, the simplified approach based on the compressive strength in subsection 5.1.12.2.3 may serve as an easily applicable tool for a first estimation of the chloride diffusivity.

Similar to water permeability, capillary suction is strongly influenced by the moisture content of the concrete. As the pore humidity of the concrete increases, the rate of water absorption and thus M_w decrease.

where:

- D_{CO_2} is the diffusion coefficient of CO_2 in m^2/s ;
- f_{cm} is the mean compressive strength in MPa.

5.1.12.2.3 Diffusion of chloride ions

For chloride ions the effective diffusion coefficients in mature concrete as defined in Eq. (5.1-129) may be roughly estimated from the compressive strength of concrete f_{cm} :

$$D_{Cl^-} = \frac{5 \cdot 10^{-9}}{f_{cm}^{1.5}} \quad (5.1-137)$$

where:

- D_{Cl^-} is the effective diffusion coefficient in m^2/s ;
- f_{cm} is the mean compressive strength in MPa.

The use of additives or Portland blast furnace slag-cements may lead to lower diffusion coefficients which can be roughly expressed by:

$$D_{Cl^-,add} = \frac{5 \cdot 10^{-8}}{f_{cm}^{2.5}} \quad (5.1-138)$$

where:

- $D_{Cl^-,add}$ is the effective coefficient of diffusion in m^2/s related to the use of additives;
- f_{cm} is the mean compressive strength in MPa.

5.1.12.3 Capillary suction

Liquids, particularly water, may be transported into concrete by capillary suction or absorption. Water absorption may be expressed by Eq. (5.1-139):

For a uniform pore humidity and no substantial microstructural variations within a concrete section exposed to capillary suction, the exponent n in Eq. (5.1-139) may be taken as $n = 0.5$. If the moisture distribution is non-uniform, $n < 0.5$.

Eq. (5.1-140) is valid for a uniform pore humidity of the concrete of approximately 65 % and for moderately oven-dried concrete. The coefficient of water absorption depends not only on the moisture state of the concrete, but also on microstructural parameters which are linked with concrete composition and type of materials used (e.g. water/cement ratio, amount of cement, silica fume, fly ash etc.). Considering all experimental data, a large scatter of the capillary suction values has to be kept in mind, so that predictions solely based on the concrete strength are rather uncertain.

Therefore, when a more accurate prediction is required, the coefficient of water absorption may be determined experimentally according to EN ISO 15148 “Determination of water absorption coefficient by partial immersion” or alternatively according to RILEM Technical Recommendation: “Determination of the capillary absorption of water of hardened concrete” (Materials and Structures, Vol. 32, pp. 178–179, 1999).

The durability of structural concrete components in service is determined by the transport of aqueous and gaseous substances in the pore system of concrete and their interaction with the hydrated paste matrix, the aggregate or the steel reinforcement. The substances may cause degradation and loss of serviceability by their direct action on the concrete microstructure or, indirectly, enable other reactions leading to deterioration.

Some degradation models have found a relatively broad international acceptance. Such models usually contain parameters that need to be quantified for material and environmental effects on the deterioration process and transfer parameters which consider uncertainties resulting from experimental setups. However, operational standards are not available for the quantification of most parameters. Information must therefore be found by measurements with equivalent material or on existing structures and in the literature, for instance in *fib* Bulletin 34, “Model Code for Service Life Design” (*fib*, 2006) and Concrete Society, Technical Publication No. 61, “Enhancing reinforced concrete durability”.

The exposure of concrete structures to atmospheric CO_2 results in the carbonation of the hydration products accompanied by a reduction in pH value of the pore solution, which can induce corrosion of the steel reinforcement. The penetration of the carbonation front depends on the concentration of CO_2 in the atmosphere and the amount of hydration products able to react with CO_2 . If gas diffusion is assumed, the carbonation depth is proportional to the square root of time (see also subsection 5.1.12.2.2).

Eq. (5.1-141) has been developed in the European research project DuraCrete and slightly revised in the research project DARTS:

– The European Union – Brite EuRam III, “Modelling of Degradation. DuraCrete, Probabilistic Performance based Durability Design of Concrete Structures” (1998);

$$w = w_1 \left(t/t_1 \right)^n = M_w t^n \quad (5.1-139)$$

where:

w is the water absorbed per unit area at time t in m^3/m^2 ;

w_1 is the water absorbed at a given time t_1 in m^3/m^2 ;

t is the duration of water absorption in s;

$n = 0.5$;

M_w is the coefficient of water absorption in $\text{m/s}^{0.5}$.

For a rough estimate the coefficient of water absorption for a given concrete strength may be determined from Eq. (5.1-140):

$$M_w = \frac{0.2}{f_{cm}^{2.5}} \quad (5.1-140)$$

where:

M_w is the coefficient of water absorption in $\text{m/s}^{0.5}$;

f_{cm} is the mean compressive strength in MPa.

5.1.13 Properties related to durability

5.1.13.1 General

When considering concrete properties related to durability deterioration, models describing the time-dependent degradation of concrete are essential.

Indirect degradation of concrete may be caused by

- carbonation-induced corrosion of reinforcing steel;
- chloride-induced corrosion of reinforcing steel.

Direct degradation of concrete may be caused by

- freeze-thaw attack (internal damage, scaling);
- reactivity of aggregate and/or of the cement paste (internal damage);
- acid action (dissolving action);
- leaching processes.

Several models for indirect and direct deterioration are considered in the following sections.

5.1.13.2 Carbonation progress

The propagation of the carbonation front from the concrete surface may be described by:

$$x_c(t) = \sqrt{2 \cdot k_e \cdot k_c \cdot R_{NAC,0}^{-1} \cdot C_S \cdot \sqrt{t} \cdot W(t)} \quad (5.1-141)$$

where:

$x_c(t)$ is the carbonation depth at the time t in mm;

t is the time in years;

k_e is the environmental function [–];

k_c is the execution transfer parameter [–];

C_S is the CO_2 -concentration in the air in kg/m^3 ;

$W(t)$ weather function [–];

$R_{NAC,0}^{-1}$ is the inverse effective carbonation resistance of concrete in $(\text{mm}^2/\text{years})/(\text{kg}/\text{m}^3)$;

- DARTS, “Durable and Reliable Tunnel Structures. Deterioration Modelling” (DARTS R2.1, May 2004).

The inverse effective carbonation resistance $R_{ACC,0}^{-1}$ varies depending on the water/cement ratio and the type of binder.

Further details, including information on parameters and functions in Eqs. (5.1-141) and (5.1-142) which is not given here, may be found in *fib* Bulletin 34, “Model Code for Service Life Design” (*fib*, 2006).

The penetration of chlorides (e.g. de-icing salt) changes the chemical composition of the pore solution of concrete adjacent to the steel reinforcement causing corrosion to set in. If chloride penetration is diffusion-controlled (Fick’s second law being valid), an error function may be used to describe the penetration profiles.

Within the convection zone the chloride profile deviates from Fick’s second law.

In the European joint research projects DuraCrete and DARTS (see references above) a model for the prediction of time- and depth-dependent chloride content has been developed and validated (see Eq. (5.1-143)).

The apparent chloride diffusion coefficient $D_{app,C}$ may be calculated by means of an inverse analysis from a measured chloride profile. In this case it should be noted that the obtained value for $D_{app,C}$ depends on the value of Δx .

The chloride migration coefficient $D_{RCM,0}$ varies in dependence on the water/cement ratio and type of binder significantly. Further details may be found in NT Build 492.

The exponent a varies significantly according to cement type and type of exposure.

Further information, including definitions of variables etc. which are not given here, may be found in *fib* Bulletin 34 “Model Code for Service Life Design” (*fib*, 2006).

At present, no validated time-dependent model exists for the calculation of the resistance of a given concrete in a structural component to the action of frost or frost combined with de-icing agents. Current design aims at avoiding damage by the

and

$$R_{NAC,0}^{-1} = k_t \cdot R_{ACC,0}^{-1} + \varepsilon_t \quad (5.1-142)$$

where:

$R_{NAC,0}^{-1}$ is the inverse effective carbonation resistance of dry concrete (65 % RH) determined at a certain time t_0 using the normal carbonation test NAC in (mm²/years)/(kg/m³);

$R_{ACC,0}^{-1}$ is the inverse effective carbonation resistance of dry concrete, determined at a certain time t_0 using the accelerated carbonation test ACC in (mm²/years)/(kg/m³);

k_t is the regression parameter for the test effect of the ACC test [–];

ε_t is the error term for inaccuracies which occur conditionally when using the ACC test method in (mm²/years)/(kg/m³).

5.1.13.3 Ingress of chlorides

The change of the chloride content of concrete exposed to chloride ingress is given by:

$$C(x,t) = \left[C_0 + (C_{s,\Delta x} - C_0) \cdot \left[1 - \operatorname{erf} \left(\frac{x - \Delta x}{2 \cdot \sqrt{D_{app,C} \cdot t}} \right) \right] \right] \quad (5.1-143)$$

where:

$C(x,t)$ is the chloride content of concrete in % by mass of cement;

x is the depth in m;

t is the concrete age in s;

C_0 is the initial chloride content in % by mass of cement;

$C_{s,\Delta x}$ is the chloride content at a depth of Δx in % by mass of cement;

Δx is the depth of the convection zone in m;

$D_{app,C}$ is the apparent chloride diffusion coefficient in concrete in m²/s;

with

$$D_{app,C}(t) = k_e \cdot D_{RCM,0} \cdot k_t \cdot A(t) \quad (5.1-144)$$

where:

$D_{RCM,0}$ is the chloride migration coefficient in m²/s;

k_e is the environmental variable [–];

k_t is the test method variable [–];

$A(t)$ is the ageing function [–];

with

$$A(t) = \left(\frac{t_0}{t} \right)^a \quad (5.1-145)$$

where:

t is the concrete age in s;

t_0 is the reference concrete age in s;

a is the age exponent [–].

5.1.13.4 Freeze-thaw and freeze-thaw de-icing agent degradation

(a) Mechanisms

The degree of internal damage caused by freeze-thaw attack depends on:

- the material properties determined by concrete composition including porosity, pores size distribution and strength;

specification of concrete composition for a particular service environment and standard testing methods for resistance to freeze-thaw and freeze-thaw de-icing salt action.

The exposure of concrete structural components to subzero temperatures in service can result in internal cracking and thus in a loss of strength due to moisture transport and the expansion of water on freezing.

The deterioration of concrete caused by freeze-thaw attack with de-icing agents is related to complex processes associated with physical and chemical changes in the pore solution, binder paste matrix and aggregates. It results in scaling, that is external damage.

A service life model to describe the internal damage caused by freeze-thaw attack was developed by Fagerlund. The model is based on the observation that a critical water saturation degree S_{CR} exists, above which the material is damaged by frost. Below S_{CR} no severe damage occurs.

Further information can be found in *fib* Bulletin 70 “Code-type models for structural behaviour of concrete – Background of the constitutive relations and material models in MC2010”.

The formation of an expansive alkali silica gel leads to deformation and cracking when the internal pressure exceeds the tensile strength of the aggregate and/or the binder paste matrix including the transition zone. Ultimately, degradation and loss of serviceability of the concrete structure occur.

At present, no suitable predictive analytical or numerical method exists for durability modelling of concrete behaviour with respect to the alkali-aggregate reaction (AAR). Contemporary concrete design aims at the avoidance of AAR (also termed ASR = alkali-silica reaction) which is usually achieved by limiting the alkali content of the cement or the use of non-reactive aggregate. The third method, to guarantee a sufficiently low water content, is difficult to achieve in practice.

For further details see: CONTECVET, “A Validated Users Manual for Assessing the Residual Service Life of Concrete Structures – Manual for Assessing Structures Affected by ASR” (EC Innovation, Programme IN309021, 2001).

The service life of a structural component exposed to degradation by acids and leaching processes is defined by the time needed for the corrosion to reach a given depth.

On contact of an aggressive medium with the concrete surface, acid attack proceeds immediately without an initiation period. A corroded surface layer of low mechanical strength forms due to the dissolution of the binder matrix and, if dissolvable, the aggregate particles. The depth of corrosion increases as time passes. The attacking medium may be classified as:

- mineral acids;
- buffer solutions including organic acids, carbonic acid or ammonium salts.

- the actual service environment, that is the conditions at the concrete surface and their variation with time covering relative humidity, surface contact with water and temperature;
- the degree of saturation which varies with time and location in the concrete due to moisture transport by capillary suction, water vapour diffusion together with capillary condensation and water vapour sorption.

When combined with de-icing salt, freeze-thaw attack is also affected by material factors such as aggregate type and reactivity. Besides moisture content, factors such as the minimum freezing temperature, the rate of freezing and the cation types in the de-icing agent are important.

(b) Models

Models to be included here, though being rather crude, are still under discussion.

5.1.13.5 Alkali-aggregate reaction

(a) Mechanism

Alkalis in the pore solution of concrete react chemically with certain types of concrete aggregates forming an expansive alkali silica gel.

(b) Damage monitoring

The following methods may be used to predict the future expansion of structures affected by alkali-aggregate reaction (AAR):

- monitoring the expansion of cores taken from the structure;
- monitoring deformations of the structure;
- use of known expansion behaviour of similar concrete under similar exposure conditions.

The observed expansion behaviour has to be extrapolated after correcting the data for the effect of restraint.

(c) Models

Models to be included here, though being rather crude, are still under discussion.

5.1.13.6 Degradation by acids

The degree of degradation of concrete caused by acid attack is defined by a corrosion depth d with respect to the original surface. It comprises the depth of material removed by abrasion and/or crystallization pressure and the depth of corroded material remaining on the concrete surface.

If the loss of surface material is negligible and the strength of the acid is assumed to be constant, the corrosion depth d [m] may be estimated from:

$$d = k_c \sqrt{ct} \quad (5.1-146)$$

where:

c is the concentration of acid in mol/l, see Eqs. (5.1-147) or (5.1-148);

t is the contact time in s;

k_c is a constant.

So far, no prediction formula for the constant k_c has been given. This constant should be determined by appropriate experiments. For further details see Beddoe, R. E. and Schmidt, K., “Acid attack on concrete – effect of concrete composition” (Cement International, Part 1, Vol. 7, No. 3, pp. 88–94, 2009; Part 2, Vol. 7, No. 4, pp. 86–93, 2009).

The effect of concrete composition on the corrosion process is given by the constant k_c which includes the effect of cement content and type, additions, water/cement ratio and aggregate solubility. For mineral acids c [mol/l] is given by the cation concentration of the acid as calculated from its pH value by means of Eq. (5.1-147):

$$c = 10^{-pH} \quad (5.1-147)$$

In case of buffering media it is necessary to know the pH value and the total content c_{tot} of acid and acid anions (e.g. acetate and acetic acid), dissolved CO_2 or ammonium:

$$c = \frac{10^{-pH} c_{tot}}{(10^{-pH} + K_s)} \quad (5.1-148)$$

where:

K_s is the dissociation constant in mol/l;

c_{tot} is the total content of acid and conjugate base, dissolved CO_2 or ammonium in mol/l.

5.1.13.7 Leaching progress

The leaching of environmentally relevant substances such as Cr, V and Zn from concrete structural components commences on first contact of the concrete surface with water. Environmentally relevant substances on the concrete surface enter the water by the wash-off mechanism. The leaching rate is also determined by the solubility and dissolution kinetics of the environmentally relevant substances in the pore solution of concrete and the diffusion of the species through the pore solution to the concrete surface.

The leaching potential of the substance in question may be assessed for a particular concrete composition in terms of the cumulative leaching E_{56} [mol/m²] obtained after 56 days in a tank leach test, according to NEN 7345 (standard of the Netherlands).

Details on the leaching progress may be found in:

- Coté et al., “An evaluation of cement-based waste forms using the results of approximately two years of dynamic leaching” (Nuclear and Waste Management, Vol. 7, No. 2, pp. 129–139, 1987);
- Hohberg, I., “Characterization, modelling and evaluation of the leaching process in concrete related to environmentally relevant inorganic substances” (PhD thesis, RWTH Aachen, 2002; in German).

If leaching is purely diffusion-controlled, it may be described by a root-time law; for details see the references above.

Availability describes the total amount of a particular substance per cubic metre concrete which can be leached.

The leaching rate depends on the supply of water to the surface and dry periods. Leaching scenarios include the following:

- a) constant contact, for example ground water on foundations;
- b) intermittent contact, for example seepage water on foundations, rain on facades;
- c) flowing water, for example shotcrete on tunnel liners.

The size of a structural component limits the total amount of leachable substances. For small sizes, depletion progressively lowers the leaching rate. Further background information on the concrete properties, as treated in this section, is given by Müller, H. S., Anders, I., Breiner, R., Vogel, M. (2013), Concrete: treatment of types and properties in *fib* Model Code 2010. Structural Concrete, 14. doi: 10.1002/suco.201200048.

The cumulative leaching of a substance from a given concrete surface area in constant contact with water is given empirically by:

$$E = k_1(1 - e^{-k_2 t}) + k_3 \sqrt{t} + k_4 t \quad (5.1-149)$$

where:

E is the cumulative leaching in mol;

t is the total contact time in s;

k_1, k_2, k_3, k_4 are constants.

The constants k_i are essentially material constants determined by concrete composition (essentially content of cement and additions, water/cement ratio) and the availability of the substances in the concrete.

If wash-off and depletion effects are negligible and the dissolution kinetics of the substances in the pore solution is fast, leaching is controlled by diffusion, so Eq. (5.1-149) simplifies to:

$$E = k_3 \sqrt{t} \quad (5.1-150)$$

where:

$$k_3 = 2A c_{mo,0} \sqrt{\frac{D_{eff}}{\pi}} \text{ in mol/s}^{0.5} \quad (5.1-151)$$

with:

A is the area of concrete surface in m²;

$c_{mo,0}$ is the initial availability of substance in concrete in mol/m³ according to availability test NEN 7341;

D_{eff} is the effective diffusion coefficient of a substance in concrete in m²/s.

The effective diffusion coefficient is a material parameter depending on concrete composition and age. If diffusion-controlled leaching is assumed, D_{eff} can be calculated from the availability test and tank test results using Eqs. (5.1-150) and (5.1-151) according to NEN 7345.

Bars can be either directly produced in straight lengths or by straightening reinforcing steel from coils.

Surface characteristics include the characteristics of the ribs or indentations by means of which bond with the concrete is achieved, and the characteristics of the coating if applicable.

Bond aspects are covered in chapter 6 Interface characteristics.

Durability mainly covers the corrosion protection properties of “special” reinforcing steel such as (either metallic or organic) coated steel or corrosion resistant steel. Behaviour at extreme temperatures may include either the behaviour at low temperature (e.g. for cryogenic applications) or high temperature (e.g. behaviour in case of fire).

Examples of relevant international product standards are ISO 6935-1 to -3.

Types of reinforcement not covered by product standards may be used after it has been shown that they meet the specified minimum requirements.

Mechanical couplers for splicing are specified in subsection 7.13.2.6.

The properties of reinforcing steels are generally confirmed by certification programmes and certificates of compliance. Requirements to certification of conformity should follow relevant international or national standards.

Cutting and bending of steel reinforcement, welding and mechanical splicing may be controlled via a standard for execution of concrete structures such as EN13670 or ISO 22966.

The simultaneous use of steels of various types on the same site is allowed only on condition that no confusion between the types is possible during construction.

It should be possible to distinguish clearly between:

- plain bars of various grades and/or of various ductility classes;
- high bond bars of various grades and/or of various ductility classes;
- reinforcement that is weldable and that which is not.

5.2 Reinforcing steel

5.2.1 General

Products used as reinforcing steel may be:

- bars;
- wires;
- welded fabric.

Reinforcing steel is characterized by:

- geometrical properties:
 - size;
 - surface characteristics.
- mechanical properties:
 - yield strength and tensile strength;
 - ductility;
 - fatigue behaviour;
 - behaviour under extreme thermal conditions.
- technological properties:
 - bond characteristics;
 - bendability;
 - weldability;
 - thermal expansion;
 - durability;
 - behaviour at extreme temperature.

Reinforcing steels must comply with national or international product standards applicable at the location of the structure. The standards specify geometrical, mechanical and technological properties.

5.2.2 Quality control

The fabrication of reinforcing steel must be subject to factory production control by the manufacturer, and continuous external control by an independent qualified body, which includes certification and regular audits.

5.2.3 Designation

The designation of reinforcing steels normally includes:

- the relevant product standard;
- the nominal diameter or size;
- the steel grade related to the characteristic yield strength, the ductility properties and the weldability.

Each product must be clearly identifiable with respect to this designation.

The nominal diameter is a conventionally fixed value, for example in product standards which serves as a basis for the calculation of the nominal cross-sectional area taken as the area of a circle with a diameter equal to the nominal diameter.

The actual cross-sectional area is determined by weighing a given length of bar, assuming a steel density of 7850 kg/m^3 .

For welded fabric the following applies:

- twin bars are allowed in one direction only;
- adequate stiffness of the welded fabric should be ensured either by a limitation of the maximum spacing of the bars, or by introducing a minimum ratio between the diameter of the transverse bars and the diameter of the longitudinal bars.

Plain wires and bars should only be used for reinforced concrete in non-structural applications such as spacers, except in the form of welded fabric.

The rib parameters may be specified by either the relative rib area f_R , or by a combination of rib spacing, rib height and rib inclination of the transverse ribs.

The indentation parameters may be specified by either the relative indentation area f_P , or by a combination of indentation spacing, indentation depth and inclination of the indentations.

Poor straightening of ribbed or indented bars and wires from coils can significantly reduce the relative rib or indentation area and thus the bond properties of the straightened products.

The standardized requirements are, however, given to the straightened product.

The common coatings applied to reinforcing steel are either metallic (e.g. zinc or zinc alloy) or organic (e.g. epoxy).

Examples of relevant international product standards for coated reinforcing steel are ISO 14645 and ISO 14657.

For quality control purposes and design calculations, the mechanical properties of a product are referred to the nominal cross-sectional area.

Reference is made to the test methods for reinforcing steel given in ISO 15630 Part 1 (bars, wire rod and wire), Part 2 (welded fabric) and Part 3 (steel for prestressing).

The requirements apply to the product in the condition in which it is delivered. In the case of reinforcing steel delivered in coils (wire or rods), the requirements apply to the product after straightening.

The value of f_{yk} corresponds to the 0.2% offset in the characteristic stress-strain diagram.

For steels totally or partially cold-worked by means of axial tension, it will generally be the case that:

$$f_{yc} \neq f_{yt}$$

where f_{yc} and f_{yt} are the actual yield strengths, for compression and tension respectively. The value of f_{yc} to be used in a calculation should therefore be stipulated in the relevant standards.

5.2.4 Geometrical properties

5.2.4.1 Size

The size of reinforcing steel is defined by a nominal diameter for bars, wires and reinforcing steel in coils, and a set of nominal diameters for welded fabric.

The difference between actual and nominal cross-sectional area must not exceed the limiting values specified in relevant product standards.

5.2.4.2 Surface characteristics

The surface of reinforcing steel may be:

- ribbed;
- plain;
- indented.

Ribbed bars, wires and some indented products are considered as high bond reinforcements if they satisfy the conditions and requirements imposed by the relevant product standards.

Bars not satisfying these requirements should be treated as plain bars with respect to bond. Indented products, which cannot be considered as high bond reinforcement, must be treated according to relevant standards or technical specifications.

The characteristics of the coatings are in general:

- finish and appearance;
- adherence;
- mass of the coating deposited per unit area;
- continuity.

These characteristics must conform to the requirements specified in the relevant product standards.

5.2.5 Mechanical properties

The mechanical properties are defined on the basis of standard tests.

5.2.5.1 Tensile properties

The characteristic values of

- the tensile strength (f_t);
- the yield strength (f_y);
- the strain at maximum force (ϵ_u).

are respectively denoted as f_{tk} , f_{yk} and ϵ_{uk} .

There are a number of situations where the designer expects reinforcement to yield before failure of a member or structure. If in fact the reinforcement is appreciably stronger than assumed in the design, it is possible that, in a flexural situation, the concrete could reach its ultimate compression strain before the yielding of the tensile reinforcement. This could lead to brittle, rather than ductile failure. It is therefore important to ensure that the actual yield strength of the reinforcement is not significantly higher than that assumed in the design. An over-strength in the order of 30% may be appropriate.

See also CEB Bulletin 242 “Ductility of reinforced concrete structures” (CEB, 1998).

Due to the diversity and evolution of the manufacturing processes for bars and wires, various stress–strain relations may be encountered.

The Menegotto–Pinto formulation may be used as an idealization of the actual stress–strain relation of the reinforcing steel. The strength and strain values should correspond to the actual material values.

The Menegotto–Pinto function is:

$$\frac{\sigma}{\sigma_0} = b\left(\frac{\varepsilon}{\varepsilon_0}\right) + d = b\left(\frac{\varepsilon}{\varepsilon_0}\right) + \frac{(1-b)\left(\frac{\varepsilon}{\varepsilon_0}\right)}{\left[1 + \left(\frac{\varepsilon}{\varepsilon_0}\right)^n\right]^{1/n}} \quad (5.2-1)$$

where b is the ratio of the final to the initial stiffness and d is a value that is graphically defined in Figure 5.2-2. In the normalized space of stress and strain, the initial stiffness has a slope 1, the slope of the final tangent stiffness is b , and d varies from 0 to $(1 - b)$ as $\varepsilon/\varepsilon_0$ progressively increases from 0 to a maximum value at the last data point. The parameter n defines the transition between elastic and post-yield slopes. The transition tends to a pure bilinear curve with sharp yield value when $n = \infty$ (in practice $n > 15$), and to a smooth curve if n is low.

The Menegotto–Pinto idealization also applies well to special types of steel such as stainless steel and to prestressing steel – see section 5.3.

The Menegotto–Pinto idealization may also be applied to cyclic straining to varying amplitudes by restarting a branch at each strain reversal while adjusting the value of n (see reference below). In this way, it is commonly used in modelling seismic behaviour.

Relevant references are:

Menegotto M. and Pinto P.E. “Method of analysis for cyclically loaded RC plane frames including changes in geometry and non-elastic behaviour of elements under combined normal force and bending”, Preliminary Report, IABSE Symposium: Resistance and ultimate deformability of structures acted on by well defined repeated loads – IABSE Vol. 13, Lisboa, 1973.

Menegotto M. and Pinto P.E. “Strength of reinforced or prestressed concrete columns under biaxial load”, Preliminary Report, IABSE Symposium: Design and safety of reinforced concrete compression members – IABSE Vol. 16, Quebec, 1974.

5.2.5.2 Steel grades

The steel grade denotes the value of the specified characteristic yield strength in MPa and, possibly, the specified ductility and the weldability properties.

5.2.5.3 Stress–strain diagram

Indicative stress–strain diagrams of reinforcing steel in tension are shown in Figure 5.2-1.

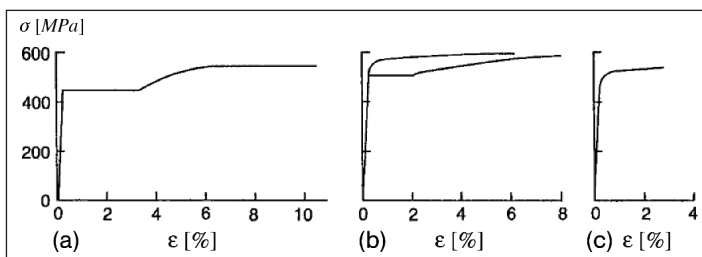


Figure 5.2-1: Stress–strain relationships of reinforcing steel: (a) hot-rolled bars; heat-treated bars; micro-alloyed bars; (b) low carbon, heat-treated bars (lower curve); cold-worked bars (upper curve); (c) cold-worked wires

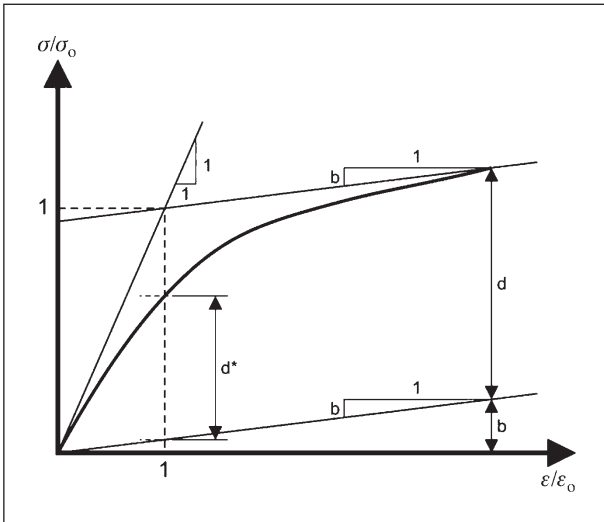


Figure 5.2-2: Menegotto–Pinto expression for the stress–strain relation of reinforcing steel

Adequate ductility is necessary, whether or not moment redistribution is taken into account in design.

The characteristic value of the ratio (f_t/f_y) , that is $(f_t/f_y)_k$, corresponds to the 5% fractile of the relation between actual tensile strength and actual yield strength.

Ductility class definitions A, B, C and D are only valid for steel grades with a characteristic yield strength ≤ 600 MPa.

Classes C and D should be used where high ductility of the structure is required (e.g. in seismic regions). In seismic design an additional requirement for $f_{y,act}/f_{yk}$ for classes C and D (e.g. $f_{y,act}/f_{yk} \leq 1.3$) can be introduced.

Should it be required to quantify a level of ductility in relation to the deformation capacity of a concrete member, it may be misleading to focus on isolated physical characteristics of the steel. In practice, quantification of ductility may be done by giving consideration to the “equivalent steel concept”, where the overall steel ductility parameter p may be regarded as equivalent to:

for cold-worked steel

$$p = \epsilon_u^{0.75} \left[\frac{f_t}{f_y} - 1 \right]^{0.8} \quad (5.2-2a)$$

for hot-rolled steel

$$p = \left((\epsilon_u - \epsilon_y) + 3 \cdot \epsilon_y \right)^{0.75} \left[\frac{f_t}{f_y} - 1 \right]^{0.8} \quad (5.2-2b)$$

Different types of reinforcing steels will show comparable ductility performance in a structure if they have the same p values.

For more information, see CEB Bulletin 218: “Ductility – Reinforcement – Progress Report” (CEB, 1993) and CEB Bulletin 242: “Ductility of Reinforced Concrete Structures” (CEB, 1998).

Provision of cross wires with properly welded joints will significantly reduce the bond length of longitudinal wires. This may under certain conditions induce strain localization in the longitudinal wires.

Fatigue behaviour depends on factors such as bar size, rib geometry, bending of bars and welded connections, thus making it difficult to

5.2.5.4 Ductility

Four ductility classes are defined for design purposes.

These classes are defined by minimum specified values for the characteristic value of the ratio f_t/f_y and the characteristic strain at maximum stress ϵ_{uk} as follows:

Class A: $(f_t/f_y)_k \geq 1.05$ and $\epsilon_{uk} \geq 2.5\%$;

Class B: $(f_t/f_y)_k \geq 1.08$ and $\epsilon_{uk} \geq 5\%$;

Class C: $(f_t/f_y)_k \geq 1.15$ and ≤ 1.35 and $\epsilon_{uk} \geq 7.5\%$;

Class D: $(f_t/f_y)_k \geq 1.25$ and ≤ 1.45 and $\epsilon_{uk} \geq 8\%$.

5.2.5.5 Shear of welded joints in welded fabric

Where welded joints are taken into account for the calculation of the anchorage length, each welded joint must be capable of withstanding a shear force not less than $0.3 A_s f_{yk}$, where A_s denotes the nominal cross-sectional area of the anchored wire.

5.2.5.6 Fatigue behaviour

The S–N fatigue behaviour of reinforcing steel is described in Table 7.4-1.

give generalized S–N curves. More information can be found in CEB Bulletin 188 “Fatigue of Concrete Structures – State-of-the-Art Report” (CEB, 1988).

When specifying the use of reinforcing steel that is expected to operate in concrete under extremely high or extremely low temperatures – that is, outside of what may be considered as the “normal” temperature range (–40°C to +40°C) – consideration should be given to using steel manufactured specifically for such purposes. This recognizes that factors such as the chemistry and processing conditions related to its manufacture will affect the steel’s performance.

Detailed information on the behaviour of steel reinforced concrete at extreme temperatures may be found in FIP Special Report SR 88/2, June 1988. Information on the behaviour of ribbed steels at very low temperatures may be found in G. Hartl, *Beton- und Stahlbetonbau* 10/1985, pp. 271–274, and in 11/1985, pp. 293–299.

For details on the strain rate effect on the reinforcing steel properties, see CEB Bulletin 187: “Concrete Structures under Impact and Impulsive Loading – Synthesis Report” (CEB, 1988).

Reinforcing bars should not be bent to a radius less than that used in the relevant bend and/or rebend test specified in the product standards by some margin. Margins of 1.5 against the bend test have been specified in some national standards.

Bending of steel at temperatures below –5°C is permitted only if allowed by the project specification and additional precautions are documented (ref: EN 13670, section 6.3).

Inappropriate welding procedures can adversely affect tensile or other properties of reinforcing steel – see subsection 8.3.4.

A factory production control system that complies with ISO 9001 and addresses the requirements of the relevant product standard is considered as appropriate. Processing may include cutting, cutting and bending, mechanical splicing and welding.

5.2.5.7 Behaviour under extreme thermal conditions

The properties of reinforcing steel as treated before are valid for “normal” temperatures between –40°C and +40°C.

Tensile strength and yield strength decrease and the strain at maximum stress increases if the temperature increases above the “normal” temperature range.

Tensile strength, yield strength and modulus of elasticity increase if the temperature decreases below the “normal” temperature range. The percentage total elongation at maximum stress increases initially but decreases rapidly below a temperature of about –150°C.

5.2.5.8 Effect of strain rate

The tensile properties of steel increase with the speed or the rate of strain application.

5.2.6 Technological properties

5.2.6.1 Bendability

The requirements concerning bendability are specified in the relevant standards.

5.2.6.2 Weldability

The requirements concerning weldability are generally given by requirements to the chemical composition of the reinforcing steel, specified in the relevant standards.

Depending on the type of reinforcement used, the methods for welding may be restricted.

5.2.6.3 Coefficient of thermal expansion

Within the temperature range between –20°C and 180°C, the coefficient of thermal expansion of steel may be taken as $10 \cdot 10^{-6} \text{ degC}^{-1}$.

5.2.6.4 Provisions for quality control

Reinforcing steel should be manufactured and processed under an appropriate permanent system of factory production control, which should include evaluation of the specified properties.

The evaluation of conformity of reinforcing steels should be based on the verification of their properties by batch sampling and testing as specified in the relevant product standards and may include determination of long term quality levels.

The conformity of reinforcing steel both as manufactured and subsequently processed may be attested by certification programmes operated by third party certification bodies.

Further information on these three special types of steels may be found in *fib* Bulletin 49: “Corrosion protection of reinforcing steels” (*fib*, 2009).

Some specific effects might have to be considered in design and/or detailing for special types of steels: for example, possible sensitivity to fretting fatigue of some stainless steels, see TR 51.

Other special types of steel that are not mentioned, such as low-carbon chromium steel bars, bars with stainless steel cladding or similar new developments are available.

Combination of non-coated and stainless steel does not cause problems – see TR 51.

Stainless steels may be classified according to their corrosion resistance.

The maximum diameter of reinforcing steel bar may be limited for certain design checks.

The actual diagram for a particular steel may be used if it is duly verified by the producer. See also commentary relating to subsection 5.2.5.1 on over-strength of reinforcement.

For high strength steels, the stress–strain diagram is non-symmetrical in compression and in tension.

Some cold-worked steels have a lower modulus of elasticity in compression than in tension. The difference is not important in practice.

The Menegotto–Pinto idealization (see subsection 5.2.5.3) may also be used for design purposes. In that case, the parameters of the idealization have to be adjusted to the relevant characteristic values of the stress–strain diagram.

5.2.7 Special types of steels

The following special types of steel that show enhanced corrosion protection properties can be used, subject to possible application provisions:

- galvanized steels;
- epoxy coated steels;
- stainless steels.

5.2.8 Assumptions used for design

The parameters of reinforcing steel to be used for design are as follows:

- modulus of elasticity, E_s ;
- characteristic yield strength, f_{yk} (or characteristic value of 0.2% proof strength, $f_{0.2k}$);
- ductility parameters, that is characteristic strain at maximum force ϵ_{uk} and characteristic ratio tensile strength/yield strength $(f_t/f_y)_k$.

As a simplification, actual stress–strain diagrams can in calculations be replaced by an idealized characteristic diagram according to Figure 5.2-3, assuming a modulus of elasticity E_s equal to 200 GPa.

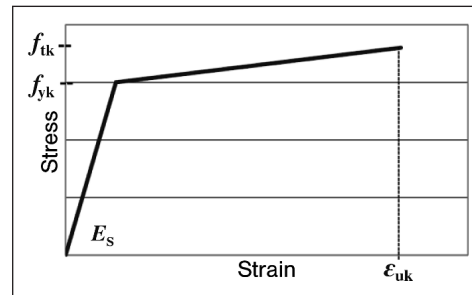


Figure 5.2-3: Idealized stress–strain diagram

The main parameters of reinforcing steel to be used for fire design are as follows:

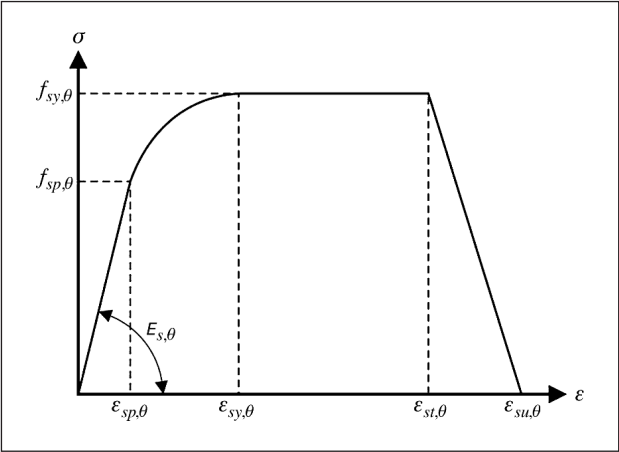
- modulus of elasticity at temperature θ , $E_{s,\theta}$,
- proportional limit at temperature θ , $f_{sp,\theta}$,
- maximum stress at temperature θ , $f_{sy,\theta}$.

As a simplification, the idealized characteristic diagram according to Figure 5.2-4 can be used for fire design.

The values for the parameters $E_{s,\theta}$, $f_{sp,\theta}$ and $f_{sy,\theta}$ given in Table 5.2-1 may be used.

Table 5.2-1: Values for the parameters $E_{s,\theta}$, $f_{sp,\theta}$ and $f_{sy,\theta}$

Steel temperature θ (°C)	$E_{s,\theta}/E_s$		$f_{sp,\theta}/f_{yk}$		$f_{sy,\theta}/f_{yk}$	
	Hot-rolled	Cold-worked	Hot-rolled	Cold-worked	Hot-rolled	Cold-worked
20	1.00	1.00	1.00	1.00	1.00	1.00
100	1.00	1.00	1.00	0.96	1.00	1.00
200	0.90	0.87	0.81	0.92	1.00	1.00
300	0.80	0.72	0.61	0.81	1.00	1.00
400	0.70	0.56	0.42	0.63	1.00	0.94
500	0.60	0.40	0.36	0.44	0.78	0.67
600	0.31	0.24	0.18	0.26	0.47	0.40
700	0.13	0.08	0.07	0.08	0.23	0.12
800	0.09	0.06	0.05	0.06	0.11	0.11
900	0.07	0.05	0.04	0.05	0.06	0.08
1000	0.04	0.03	0.02	0.03	0.04	0.05
1100	0.02	0.02	0.01	0.02	0.02	0.03
1200	0	0	0	0	0	0



Strain range	Stress σ_θ	Tangent modulus
$\varepsilon_{sp,\theta}$	$\varepsilon E_{s,\theta}$	$E_{s,\theta}$
$\varepsilon_{sp,\theta} \leq \varepsilon \leq \varepsilon_{sy,\theta}$	$f_{sp,\theta} - c + (b/a)[a^2 - (\varepsilon_{sy,\theta} - \varepsilon)^2]^{0.5}$	$\frac{b(\varepsilon_{sy,\theta} - \varepsilon)}{a[a^2 - (\varepsilon - \varepsilon_{sy,\theta})^2]^{0.5}}$
$\varepsilon_{sy,\theta} \leq \varepsilon \leq \varepsilon_{st,\theta}$	$f_{sy,\theta}$	0
$\varepsilon_{st,\theta} \leq \varepsilon \leq \varepsilon_{su,\theta}$	$f_{sy,\theta}[1 - (\varepsilon - \varepsilon_{st,\theta}) / (\varepsilon_{su,\theta} - \varepsilon_{st,\theta})]$	—
$\varepsilon = \varepsilon_{su,\theta}$	0	—
Parameters	$\varepsilon_{sp,\theta} = f_{sp,\theta} / E_{s,\theta}$ $\varepsilon_{sy,\theta} = 0.02$ $\varepsilon_{st,\theta} = 0.15$ $\varepsilon_{su,\theta} = 0.20$ except for Class A reinforcement for which: $\varepsilon_{st,\theta} = 0.05$ $\varepsilon_{su,\theta} = 0.10$	
Functions	$a^2 = (\varepsilon_{sy,\theta} - \varepsilon_{sp,\theta})(\varepsilon_{sy,\theta} - \varepsilon_{sp,\theta} + c / E_{s,\theta})$ $b^2 = c(\varepsilon_{sy,\theta} - \varepsilon_{sp,\theta})E_{s,\theta} + c^2$ $c = \frac{(f_{sy,\theta} - f_{sp,\theta})^2}{(\varepsilon_{sy,\theta} - \varepsilon_{sp,\theta})\varepsilon_{sp,\theta} - 2(f_{sy,\theta} - f_{sp,\theta})}$	

Figure 5.2-4: Idealized stress–strain diagram for fire design and corresponding mathematical model

5.3 Prestressing steel

5.3.1 General

Steels for prestressing are delivered as:

- wire;
- 2-wire strands, 3-wire strands, 7-wire strands, 19-wire strands;
- bars.

The standard tests are defined in ISO 15630-3.

Prestressing steel is characterized by:

- geometrical properties:
 - surface characteristics: plain, indented or ribbed;
 - nominal diameter;
 - nominal cross – sectional area;
 - nominal mass per metre;
 - nominal dimensions of indentations (wire and strand);
 - pitch length (strand);
 - left or right lay (strand);
 - nominal pitch and dimensions of ribs or threads (bars);
 - left or right thread (bars).
- mechanical properties:
 - tensile properties: ultimate tensile strength (UTS), 0.1% proof stress and total elongation at ultimate tensile strength;
 - modulus of elasticity;
 - fatigue behaviour;
 - behaviour under extreme thermal conditions;
 - bond characteristics;
 - ductility properties corresponding to the product type such as the percentage reduction of area, resistance to reverse bending and resistance to bending.

The 0.1% proof stress is sometimes called “yield stress”.

Behaviour at extreme temperature may cover either the behaviour at low temperature (e.g. for cryogenic applications) or high temperature (e.g. behaviour in case of fire).

Prestressing steel can be provided with the following coatings or sheathing:

- metallic coating;
- organic coating,
- exterior sheathing with a filling product.

Coatings mainly serve as supplementary corrosion protection. Some coatings modify the bond characteristics of prestressing steel.

Sheathings can either serve as corrosion protection and/or for containment of a lubricating filler inside the sheathing. For soft fillers inside the sheathing, like grease or wax, the prestressing steel remains permanently unbonded.

Resins have also been used as filler. Resins with delayed hardening can provide bond.

Examples of product standards for prestressing steels are: EN10138 –ASTM A416, A421 and A722 – JIS G3536 and G3109.

Types of prestressing steel not covered by product standards may be used after it has been shown that they meet the specified minimum requirements.

Prestressing steels must comply with national or international product standards that specify their geometrical, mechanical and technological properties. Testing of prestressing steel must be carried out in accordance with national or international standards such as ISO 15630-3.

5.3.2 Quality control

The properties of prestressing steels are generally ensured by certification schemes and certification compliance. The methods of testing and of certification of conformity are defined in the relevant national or international standards or recommendations by RILEM.

The fabrication of prestressing steels must be subject to factory production control by the manufacturer, and continuous external control by an independent qualified body, including certification and regular audits.

5.3.3 Designation

The designation of the product must consist of:

- reference of the relevant product standard;
- steel designation consisting of:
 - type of the prestressing steel (wire, strand, bar);
 - nominal ultimate tensile strength (UTS) in MPa;
 - nominal diameter;
 - relaxation class;

- fatigue class;
- stress corrosion resistance class.

Information on complementary characteristics specific to a product may include:

- for wire:
 - type of surface: plain or indented (with the type of indentation);
 - type of coating (if any);
- for strand:
 - number of wires (2, 3, 7 or 19);
 - type of coating (if any);
 - type of sheathing and filler (if any);
 - type of surface: plain, indented or compacted;
 - performance level of deflected tensile test behaviour;
- for bars:
 - type of surface: plain or ribbed;
 - type of ribs: hot-rolled or cold-rolled.

Each product must be clearly identifiable with respect to this designation.

5.3.4 Geometrical properties

Geometrical properties are mainly nominal diameter and nominal cross-sectional area.

Products must comply with the geometrical properties specified in national or international product standards.

The difference between actual and nominal geometrical properties must not exceed the tolerances specified in the relevant product standards.

5.3.5 Mechanical properties

The process of manufacture of prestressing steel may influence several properties. This is particularly true for bars and the following properties:

- modulus of elasticity;
- bending;
- ratio 0.1% proof stress to UTS;
- ductility.

The standard tests are defined in ISO 15630-3.

5.3.5.1 Tensile properties

The UTS value expressed in MPa is often denoted as the grade of prestressing steel.

The ratio of 0.1% proof stress to UTS:

- $\geq 88\%$ for wires;
- $\geq 86\%$ for strands.

The ratio for bars should be declared by the manufacturer.

Some standards specify an upper limit of the tensile strength (UTS) which is about 15% above the characteristic strength. Such an upper limit of the strength is given to ensure a certain homogeneity between lots of prestressing steel (small variations indicating good quality control during fabrication).

The characteristic values of

- UTS (f_{pt});
 - 0.1% proof stress ($f_{p0.1}$);
 - strain at maximum stress (ϵ_{pu});
- are respectively denoted as f_{ptk} , $f_{p0.1k}$ and ϵ_{puk} .

5.3.5.2 Stress–strain diagram

Due to specific details of the manufacturing process the stress–strain relation may differ between manufacturers.

As an idealization of the actual stress–strain relation the Menegotto–Pinto formulation presented in subsection 5.2.5.3 for reinforcing steel may be used for prestressing steel as well.

Indicative stress–strain relations for prestressing steel in tension are represented in Figure 5.3-1.

The nominal value of the modulus of elasticity of the prestressing steel E_p may be taken equal to:

- 205,000 MPa for wires;
- 195,000 MPa for strands (approximately).

The value for bars should be declared by the manufacturer.

The modulus of elasticity of the prestressing steel E_p must be declared and certified by the manufacturer.

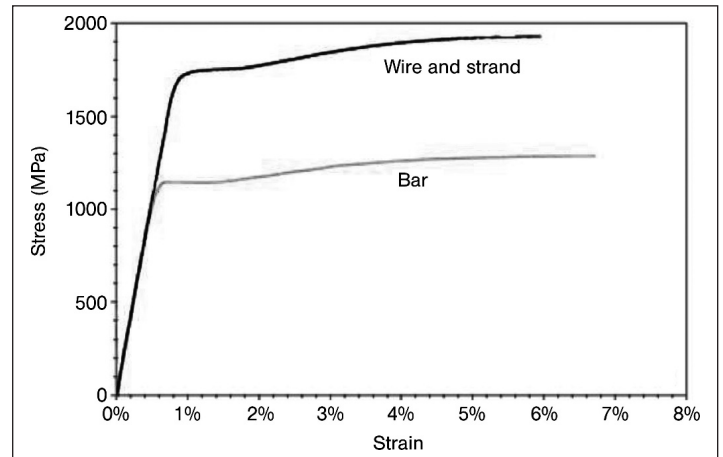


Figure 5.3-1: Typical stress–strain diagrams for prestressing steel

Other fatigue behaviour may be required if the steel is in an aggressive environment (e. g. near the sea) or at temperatures other than room temperature.

In Japan, the fatigue behaviour is defined by the standard of Japanese Society of Steel Construction.

Recommendations for fatigue testing:

The frequency of load cycles should not exceed:

- 120 Hz for wires and bars;
- 20 Hz for strands.

The temperature during the test should not exceed 40°C.

The minimum fatigue stress ranges indicated in Table 5.3-1 are valid for prestressing steels tested in air. Once installed as tendons, the prestressing steel is in contact with concrete/grout or metal surfaces, transverse stresses due to tendon curvature may occur, and anchorages may be provided. These conditions reduce the fatigue strength of prestressing steel (e. g. due to fretting fatigue). Therefore, this Model Code provides S–N curves only for tendons, see subsection 7.4.1.4, but not for the prestressing steel in air.

5.3.5.3 Fatigue behaviour

Prestressing steel must comply with fatigue stress range requirements determined at $2 \cdot 10^6$ load cycles performed at an upper stress level of 70% or 80% of the UTS. The minimum fatigue stress range must comply with Table 5.3-1.

Table 5.3-1: Minimum fatigue stress range of prestressing steels at upper stress level of 70% or 80% of the UTS

Type of steel		Stress range [MPa]
Plain wire		200
Indented wire		180
Plain strand		190
Indented strand		170
Plain bars	$d \leq 40$	200
	$d > 40$	150
Ribbed bars	$d \leq 40$	180
	$d > 40$	120

5.3.5.4 Behaviour under extreme thermal conditions

The above values are valid for “normal” temperatures between –40°C and +40°C.

Cryogenic conditions

Prestressing steels for cryogenic conditions must be specifically ordered for this purpose. The temperature for which they are tested must be specified.

The producer must provide test results of the prestressing steel for the specified cryogenic temperature.

It may be assumed that prestressing steels are typically suitable for use under cryogenic conditions showing a 0.1% proof stress and an ultimate tensile strength higher than at room temperature and a strain at maximum load over 2% when tested at –196°C.

Stress–strain relations are suitable for the assessment of the behaviour at cryogenic temperatures.

Reference is made to the FIP State of the Art Report “Cryogenic behaviour of materials for prestressed concrete” (FIP, 1982) and FIP Special Report SR88/2 “Appendix on ice load considerations to FIP recommendations: design and construction of concrete sea structures” (FIP, 1988).

The threshold value is in the temperature range 150°C to 250°C.

Figure 5.3-2 illustrates the actual ultimate tensile strength of prestressing steel when exposed to and tested at a given temperature. Figure 5.3-3 illustrates the residual ultimate tensile strength of prestressing steel tested at 20°C after previous exposure over 3 hours to a given temperature. Refer to Atienza, J.M. and Elices, M., “Behaviour of prestressing steel after simulated fire: Fire-induced damage”, Construction and Building Materials, Vol. 23, 2009, pp. 2923–2940. It should be noted that exposure to temperatures higher than 20°C increases the losses due to relaxation of prestressing steel – see subsection 5.3.6.1.

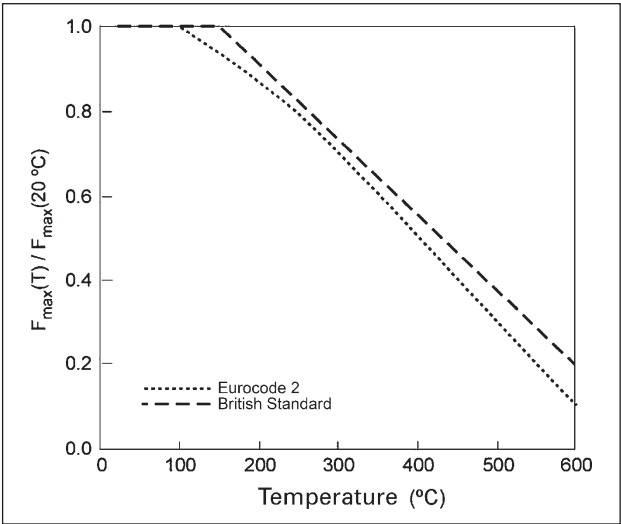


Figure 5.3-2: Effect of temperature on the ultimate tensile strength of prestressing steel when tested at a given temperature

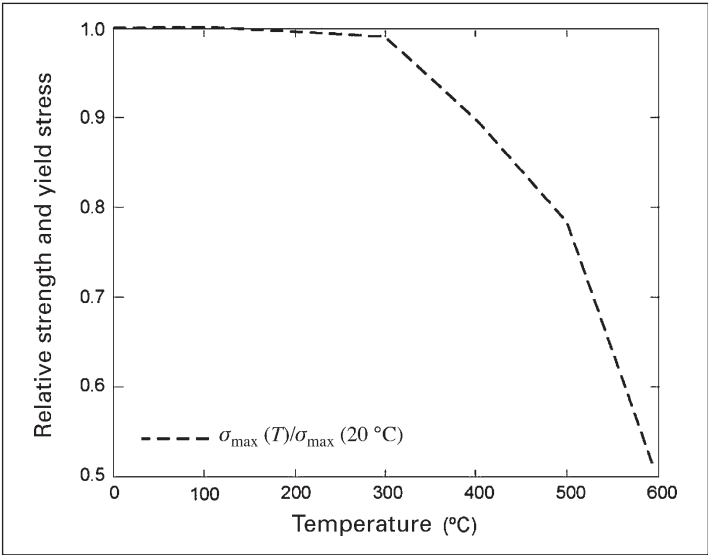


Figure 5.3-3: Effect of temperature on the ultimate tensile strength of prestressing steel when tested at 20°C after 3 hours exposure to a given temperature

Tests on prestressing steel show marginal effect of strain rate on the yield and ultimate tensile strength. Reference is made to: Galvez, F., Atienza, J.M. and Elices, M., “Behaviour of steel prestressing wires under extreme conditions of strain rate and temperature,” Structural Concrete 12 (2011), No. 4, pp. 255–261, and CEB Bulletin 187 “Concrete Structures under Impact and Impulsive Loading – Synthesis Report”, pp. 3.20–3.27 (CEB, 1988).

High temperature

The maximum stress and the 0.1% proof stress decrease and the strain increases if the temperature increases beyond a threshold value limiting the normal range.

5.3.5.5 Effect of strain rate

Any increase in yield and ultimate tensile strength of prestressing steel due to high strain rate may only be taken into account when tests on the particular prestressing steel confirm such improved values.

Quality control tests for the bond characteristics of prestressing steel should be performed in accordance with national or international standards. Alternatively, the North American Strand Producers (NASP) Bond Test protocol for strand bond test may be used. Minimum bond strength should comply with relevant standards or should be agreed with the customer.

Bond behaviour and strength for design purposes are covered in chapter 6.

The process of manufacture may influence several properties such as:

- stress relaxation;
- stress corrosion resistance (environmental cracking).

Relaxation tests at an initial stress of 80% of the actual ultimate tensile strength and at higher temperatures than 20°C over 1000 hours may be agreed between producers and purchasers.

In the past, mostly two classes of relaxation were used for wire and strand (very low and normal relaxation). At one stage even three classes were used (very low, low and normal relaxation).

The trend worldwide is to use mostly very low relaxation wire and strand. In Japan, wire and strand with a relaxation of 8% is also manufactured (ordinary products) which corresponds to the above-mentioned normal relaxation.

The CEB-FIP Model Code 1990 differentiated between three classes of relaxation:

- Class 1: normal relaxation characteristics for wire and strand;
- Class 2: improved relaxation characteristics for wire and strand;
- Class 3: relaxation characteristics for bars.

These three classes are also specified in EN 1992-1. This Model Code considers only wire and strand with very low relaxation (MC 90 Class 2), and bars (MC 90 Class 3). For information on MC 90 Class 1 (normal relaxation), see MC 90.

The loss by relaxation increases if the temperature of the prestressing steel increases above 20°C. Figure 5.3-4 illustrates the relaxation losses of very low relaxation wire and strand as a function of time when exposed to constant temperature.

Relaxation losses at 20°C may be conservatively assumed to be valid at lower temperatures than 20°C.

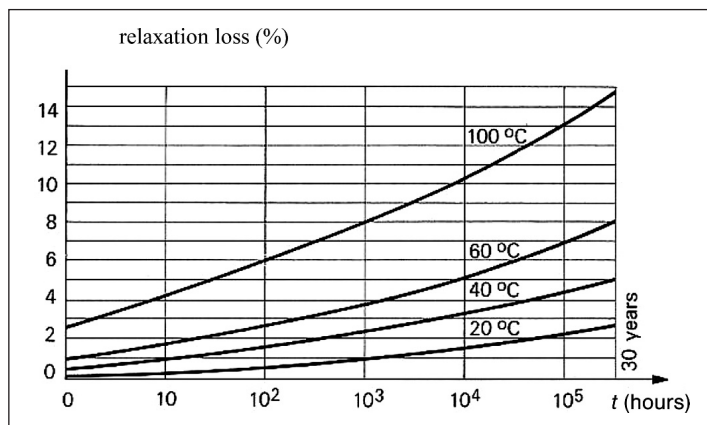


Figure 5.3-4: Relaxation losses of very low relaxation wire and strand as a function of time up to 30 years when exposed to constant temperature (information from prestressing steel manufacturer)

5.3.5.6 Bond characteristics

The bond characteristics of prestressing steel vary depending on the surface characteristics and the manufacturing process. For prestressing steel intended to be used in bonded conditions, the producer must document the bond characteristics through representative quality control testing performed on each production batch.

5.3.6 Technological properties

The standard tests are defined in ISO 15630-3.

5.3.6.1 Isothermal stress relaxation

The loss of stress by relaxation must be established by testing at a nominal temperature of 20°C for a period of 1000 h from an initial stress of 70% of the actual ultimate tensile strength of the prestressing steel.

Prestressing steels are available in different classes of relaxation. Table 5.3-2 gives values for very low relaxation wire and strand, and for prestressing bars.

Table 5.3-2: Maximum specified values of stress loss of prestressing wire and strand with very low relaxation and for prestressing bars after 1000 h.

Type of steel	Initial stress	Specified maximum loss
Wire/strand	70% UTS	2.5%
Wire/strand	80% UTS	4.5%
Bars ≤ 15mm	70% UTS	6%
Bars > 15mm	70% UTS	4%

Temperatures higher than 20°C accelerate the relaxation losses and increase the magnitude of relaxation loss of prestressing steel. When relevant, the relaxation losses of prestressing steel must be determined in relaxation tests performed at the relevant temperature to which the prestressing steel will be exposed for a significant period of time.

This test simulates the behaviour of strand in anchorages and curved ducts of post-tensioning tendons.

For details, see FIP recommendations “Deflected tensile test” (FIP, 1996).

For specific applications (stay cables or specific post-tensioning), a lower maximum reduction of 20% may be specified to the manufacturer.

The NH4SCN test (solution A of ISO 15630-3) is considered a suitable test for quality control of prestressing steels. It is a legitimate test when the dominant failure mechanism under service conditions is by hydrogen embrittlement.

Any new type of prestressing steel should be subject to stress corrosion testing using solution B of ISO 15630-3. In parallel, stress corrosion testing with solution A of ISO 15630-3 should be carried out to establish minimum individual and median values of lifetime to failure for such new types of prestressing steels.

New types of prestressing steel should include but not necessarily be limited to different chemical composition, different (in particular higher) ultimate tensile strength, different manufacturing process and so on.

Residual stresses, particularly the stresses created during cold-drawing of wires, have an influence on the shape of the stress-strain curve (i.e. the ratio of the 0.1% proof stress to UTS), on the stress relaxation losses and on cracking due to environmental influences.

Average values or an upper limit of residual stresses on the steel surface may be provided by the producer on request.

Usually it is assumed that prestressing wires of very low relaxation have surface residual tensile stresses lower than 50 MPa.

For more information see: Elices, M., “Influence of residual stresses in the performance of cold-drawn pearlitic wires”, *Journal of Materials Science*, Vol. 39, 2004, pp. 3889–3899.

Prestressing steels are available with coatings and sheathing either for improved corrosion protection and/or to maintain the prestressing steel permanently unbonded inside the concrete or grout.

Metallic coating may be applied to wire, strand and bar.

5.3.6.2 Deflected tensile behaviour (only for strands with nominal diameter ≥ 12.5 mm)

For normal prestressing applications the maximum permitted reduction of the tensile strength of prestressing steel in a deflected tensile test is 28%.

5.3.6.3 Stress corrosion resistance

Prestressing steels must be subject to ongoing quality control testing for stress corrosion. The minimum individual and median values of lifetime to failure must be determined using solution A of ammonium thiocyanate specified in ISO 15630-3. The values must be in accordance with the values given in the relevant product standards.

New types of prestressing steel must be subject to initial approval testing for stress corrosion. Such prestressing steels must pass 2000 hours in solution B specified in ISO 15630-3.

5.3.6.4 Coefficient of thermal expansion

Within the temperature range from -40°C to 180°C the coefficient of thermal expansion of prestressing steel may be taken as $10 \cdot 10^{-6} \text{degC}^{-1}$.

5.3.6.5 Residual stresses

Residual stresses influence the technological properties of prestressing steel.

5.3.7 Special types of prestressing steel

Prestressing steels are available with coatings and with sheathing containing different fillers.

5.3.7.1 Metallic coating

This coating process is made by hot-dip immersion in a zinc bath or zinc plus aluminium bath. It must be made before the final thermo-mechanical treatment.

Specific quality controls are specified in the relevant standards (or in approval documents) and include:

- mass per metre of coating;
- appearance of the coating;
- continuity of the coating;
- adherence of the coating on the steel.

Organic coating may be applied to wire, strand and bar.

The sheathing containing the filler is mostly applied to strand.

Properties of plastics and wax or grease are specified in NF A 35-037/NBNI10-008.

In Japan, see “Recommendation for design and construction of partially prestressed concrete (class III of prestressed concrete) structures” by the Architectural Institute of Japan.

The strand with low-friction sheathing is typically used for internal unbonded tendons (tendons cast inside concrete). This type of strand is filled with grease.

The strand with high-friction sheathing is typically used outside of concrete for non-injected tendons such as stay cables. This type of strand may be filled either with wax or grease depending on the application. If such strand is cast into concrete or inside grouted ducts, stressing is difficult or impossible due to the adherence.

All specific tests are defined in NF A 35-037 part 1/NBN I10-008 or in JIS K2220, 2246, 2265 and JIS K6922-2.

Typically, a minimum thickness of the sheathing is specified.

The sliding test measures the actual adherence of the sheathing on the strand:

- for post-tensioning (low-friction sheathing);
- for stay cable (high-friction sheathing).

Static and dynamic leak tightness tests have been specified.

For design purposes the Menegotto–Pinto formulation given in subsection 5.2.5.3 (Figure 5.2-2) may be used. The strength and strain values must be limited to the corresponding characteristic material values.

5.3.7.2 Organic coating

Organic coating is applied after the final thermo-mechanical treatment. Specific quality controls on the organic coating are specified in the relevant product standards.

5.3.7.3 Exterior sheathing with a filling product

Sheathing must be made of high-density polyethylene or polypropylene. Fillers may be either grease or wax.

Application of sheathing and filler is made after the final thermo-mechanical treatment of the prestressing steel.

Two basic types of sheathed products are available:

- strand with a low-friction sheathing (sliding strand) filled with grease;
- strand with high-friction sheathing (adherent strand) filled with either grease or wax.

Specific controls on the sheathing and on the filling products are specified in the relevant standards (or in approval documents) and include:

- mass of the sheathing;
- mass of the filling product;
- type of the sheathing and its geometrical characteristics;
- properties of the finished product based on the following tests:
 - sliding test;
 - bonding test under thermal variation;
 - splitting (cracking) test on the sheathing;
 - leak tightness test;
 - impact resistance test.

5.3.8 Assumptions used for design

The main parameters of prestressing steel to be used for design are as follows:

- characteristic UTS, f_{ptk} ;
- characteristic strain at maximum stress, ε_{puk} ;
- modulus of elasticity, E_p ;
- relaxation loss at 1000 hours at 70% of actual tensile strength and 20°C.

For design a stress–strain relation of prestressing steel similar to the real behaviour, see Figure 5.3-1, but limited to a maximum stress of f_{ptk} , may be used. However, an idealized bilinear stress–strain relation, as shown in Figure 5.3-5, may also be used assuming a nominal value for the modulus of elasticity of the particular type of prestressing steel up to the characteristic yield stress, and a second straight line up to characteristic ultimate tensile strength and characteristic strain at maximum stress.

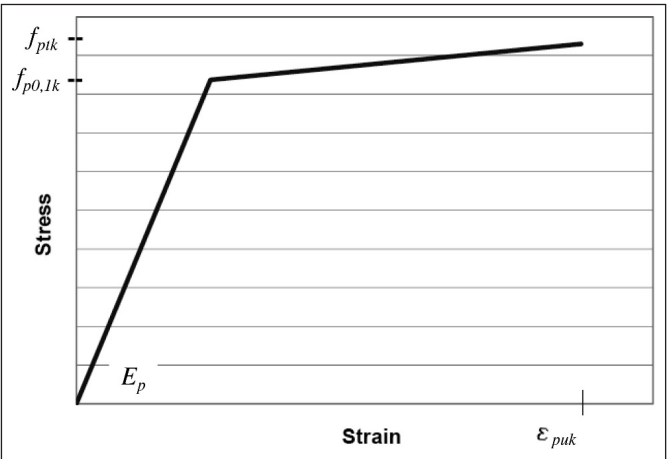


Figure 5.3-5: Idealized stress–strain relation for prestressing steel

The relaxation loss at 1000 hours must be taken either as the value specified in the relevant standard or as the average of three tests performed on the prestressing steel production batch intended to be used in the particular application. Tests must be done for a duration of 1000 hours minimum, at an initial stress of 70% of the actual tensile strength or the percentage relevant for the application, and at 20°C or at a temperature relevant for the particular application.

5.4 Prestressing systems

5.4.1 General

Prestressing steels are provided as wires, strands or bars.

FRP materials are provided as wires, strands, bars or plates.

Extradosed tendons and stay cables which are partially or totally outside the outline of the structure, and prestressed ground anchors are not covered by this Model Code. See: *fib* Bulletin 30, “SETRA Recommendations” and FIP Recommendations for Prestressed Ground Anchorages.

Optional features of prestressing tendons may be provided such as to obtain:

- re-stressable tendons;
- exchangeable tendons;
- tendons for cryogenic applications;
- fully encapsulated tendons;
- monitorable tendons.

In the absence of applicable national or international approval procedures, ETAG 013 is recommended to be used. It includes detailed test procedures and acceptance criteria for verification of the system performance.

The CEN Workshop Agreement CWA 14646 is recommended as reference for the qualification of specialist companies and for the training of supervisors and personnel.

Information relating to anchorage arrangements is given in the approval documents. When the assumptions or service conditions differ from those envisaged by these approval documents, additional experimental checks may be necessary.

Fixed anchorages can be mechanical devices or a tendon anchored by bond.

It may be necessary to place intermediate anchorages, functioning in both directions, or additional non-prestressed reinforcement, to reduce the risk of progressive collapse, when the strength of the structure is achieved by one set of tendons extending over many spans.

When tendons are bonded by cement grouting, the transfer of the prestressing force may be ensured by bond in sections adjacent to the failure.

The deviators have to be designed for both transverse (deviating) and longitudinal (friction) effects and the corresponding displacements of the tendons.

It is recommended to make special provisions for access and anchorage attachment in the diaphragms, and for deviation devices to permit future addition of external tendons. These provisions must be made for an additional prestressing force specified by the engineer. A minimum provision for 10% of the primary prestressing force or moment capacity is recommended.

The prestressing tendons considered in this Model Code are made either of prestressing steel in accordance with section 5.3, or of FRP materials in accordance with section 5.5. These tendons may be:

- internal to the concrete, and
 - pretensioned, or
 - post-tensioned – in this case they may be bonded by grouting, or temporarily or permanently unbonded;
- external to the concrete but totally within the external outline of the structure.

Prestressing may be used for any type of structure for:

- new construction;
- repair and strengthening of existing structures.

Post-tensioning systems must comply with national or international approval procedures.

Prestressing tendons with all necessary components must be installed by qualified specialist companies with sufficiently experienced supervisors and suitably trained personnel.

5.4.2 Post-tensioning system components and materials

5.4.2.1 Anchorages and coupling devices

General

After hardening of the concrete, the tensile elements of tendons are tensioned, and their extremities are fixed within anchorages, which transfer the prestressing forces to the concrete.

There are three basic types of anchorages:

- stressing anchorages which permit stressing of the tendon;
- fixed anchorages which do not permit stressing of the tendon;
- coupling devices to connect the end of a tendon that has been tensioned first, to a second tendon placed as an extension of the first, and which will be tensioned in a second stage.

With unbonded tendons (internal and external), special attention should be given to the potential consequences of an accidental tendon failure, because the tendon force is lost over the entire tendon length.

With external prestressing, deviating devices are placed between the tendons and the structure, to deflect the tendon as needed. These devices and their fixing zones, have to be designed to transfer the corresponding design actions, taking the permissible installation tolerances into account.

With external prestressing, provision must be made for the future replacement of the prestressing tendons.

The future addition of external tendons is intended for additional future load or deflection adjustment, if ever needed.

The producer should assess values of fracture toughness (K_{IC} or J_C) of new materials without prior experience for this application used for anchor heads and coupling devices. Susceptibility to environmentally assisted cracking for the particular exposure conditions of these materials should be ascertained.

Relevant test procedures are specified in the FIP “Recommendations for the acceptance of post-tensioning systems” (FIP, 1993) or in the ETAG 013.

The performance of tendons made of prestressing steel is based on either the actual ultimate tensile strength or the strength specified in standards. A minimum elongation of 2% is specified to ensure yielding and ductility of the tendon.

Since the material properties for FRP tendons are not standardized, the anchorage performance of tendons made of FRP materials is based on the tensile strength declared by the supplier, which takes into account:

- the short term tensile strength of the tendon for the given tendon/anchorage assembly;
- the level of strength retention during service life, whereas the residual strength of the tendon for the permissible permanent prestress and the given tendon/anchorage assembly is at least 95% of the short term tensile strength.

Since FRP materials do not yield, the minimum elongation of the tendon at maximum load must be declared by the supplier and be used as reference for acceptance.

Design of structures with FRP tendons must be based on a permissible permanent prestress, corresponding to the characteristic long term strength of the FRP tendon. Any applicable detrimental environment, like exhibition to temperature humidity and alkalinity, must be taken into consideration. In addition the permissible permanent prestress has to fulfil the strength retention criterion mentioned above.

When FRP tendons are used as external tendons, they are often applied without ducts.

Anchorage must be made of materials suitable for this purpose.

Anchorage and coupling device performance

Post-tensioning tendon anchorages and coupling devices must have the following minimum performance in the specified tests:

- (a) Tendons made of prestressing steel:
 - in the tensile test, achieve not less than 95% of the actual tensile strength of the prestressing steel with an elongation under maximum load of not less than 2%;
 - in the fatigue test, have not more than 5% of the tendon cross-section fail over $2 \cdot 10^6$ load cycles, for a stress range of 80 MPa at an upper stress of 65% of the tensile strength of the prestressing steel;
 - in the load transfer test, resist to not less than 110% of the specified tendon strength at the specified minimum concrete strength for stressing to the maximum force.
- (b) Tendons made of FRP materials:
 - in the tensile test, achieve not less than 95% of the declared tensile strength of the FRP tendon and a minimum elongation at maximum load which is equal to or exceeds the value declared in the system documentation;
 - in the fatigue test, have not more than 5% of the tendon cross-section fail over $2 \cdot 10^6$ load cycles, for a stress range of 80 MPa (or the value specified by the designer) at an upper stress equal to the permissible permanent prestress of the FRP tendon;
 - in the load transfer test, resist to not less than 110% of the declared tendon strength at the specified minimum concrete strength for stressing to the maximum force.

Technological aspects

Friction loss characteristics of the tendons inside the anchorages and coupling devices, and tendon seating loss characteristics in the anchorages and coupling devices, must be declared in the post-tensioning system documentation, and in technical approval documents where these exist.

5.4.2.2 Ducts

General

Depending on the intended use, ducts for prestressing systems have to provide one or several of the following features:

- forming a cavity in the structure for the installation of the tensile elements and defining the tendon path in the structure for internal tendons;
- forming a conduit for the installation of the tensile elements and deviating the tendon at specified locations in the structure for external tendons;
- providing an interface suitable for the transfer of bond stresses from the tensile elements to the structure for bonded tendons;

Other types of ducts may be considered if they provide the required features and if they offer equivalent performance to corrugated metal and plastic ducts or to smooth steel and plastic pipe, as may be applicable. The formation of deleterious galvanic cells between the tensile elements and the duct material should be avoided.

Performance requirements for ducts, testing procedures and corresponding acceptance criteria may be found for the different types of ducts in:

- corrugated metal ducts: EN 523 and 524;
- corrugated plastic ducts: *fib* Bulletin 7 “Corrugated plastic ducts for internal bonded post-tensioning” (*fib*, 2000);
- smooth steel pipe: ISO 4200 “Plain end steel tubes, welded and seamless; general tables of dimensions and masses per unit length”, (1991);
- smooth plastic pipe: ETAG 013.

The cross-sectional area of the ducts should normally be in the range of 2.0 to 2.5 times that of the actual area of the tensile elements, depending on the length and geometry of the tendon as well as on the installation method used (pushing strand by strand versus pulling the complete bundle). The above recommendations correspond to duct filling ratios between 0.4 and 0.5.

The diameter of ducts for external unbonded post-tensioning should be generally of the same size as for internal bonded since the requirement for complete filling of the duct is the same.

Relevant properties of ducts may include:

- range of recommended friction and wobble coefficients;
- bond characteristics;
- minimum duct wall thickness;
- permissible minimum radius of tendon curvature (for wear resistance);
- recommended support spacing and details;
- ageing characteristics/durability of plastic materials (e.g. minimum oxidation induction time).

Resins and other filling materials may be considered if they provide the required features and if they offer equivalent performance to cementitious grout, grease or wax.

FRP tendons often use resins for filling and/or bonding to the structure.

Performance requirements for filling materials, testing procedures and corresponding acceptance criteria may be found for the different types of materials in:

- providing a low friction interface suitable for unbonded tendons;
- providing corrosion protection to the tensile elements if made from a non-corrodible material and forming a leak tight envelope.

Ducts for prestressing tendons must be either:

- corrugated metal ducts;
- corrugated plastic ducts;
- smooth steel pipes;
- smooth plastic pipes.

Performance of ducts

Ducts must be designed, fabricated and installed such that they are fit for the intended purpose and that they have the expected durability.

Technological aspects

The relevant properties of ducts must be declared in the system documentation and in technical approval documents where these exist.

5.4.2.3 Filling materials

General

Depending on the intended use, filling materials for prestressing tendons have to provide one or several of the following features:

- permanent corrosion protection to the tensile elements;
- bond to the tensile elements and transferring bond stresses to the duct and structure for bonded tendons;
- lubrication between the tensile elements and the duct to permit stressing of permanently unbonded tendons.

Filling materials must be either:

- cementitious grout;
- grease;
- wax;
- resin.

Performance requirements

Filling materials must be designed, mixed and installed such that they are fit for the intended purpose and that they have the expected durability.

- cementitious grout: *fib* Bulletin 20 “Grouting of tendons in prestressed concrete” (*fib*, 2002), PTI Specification for grouting of post-tensioned structures (2003), EN 445-447 (2007)
- grease, wax: ETAG 013, FIP Recommendations for corrosion protection of prestressing steel.

Guidance for a suitable quality control of post-tensioning system components and materials may be found in ETAG 013.

Temporary corrosion protection may consist of suitable water-soluble oils. Attention should be paid that these water-soluble oils do not unacceptably reduce the bond characteristics of the tensile elements.

Guidance for the actual requirements, test procedures and acceptance criteria for different protection levels (PL1, PL2 and PL3) adapted to the protection provided by the structure may be found in *fib* Bulletin 33.

For internal and external post-tensioning tendons the following solutions may, for example, be considered to provide the required protection:

- PL1: tendon with a duct and a filling material providing durable corrosion protection;
- PL2: tendon with PL1 plus an envelope, enclosing the tensile element bundle over its full length (including the anchorages), and providing a permanent leak tight barrier;
- PL3: tendon with PL2 plus the integrity of tendon or encapsulation to be monitorable or inspectable at any time.

For pretensioning tendons the following solutions may, for example, be considered to provide the required protection:

- PL1: tendon with sufficient concrete cover adapted to the environmental conditions and exposure;

Technological aspects

Relevant technological aspects of the filling materials must be declared in the system documentation as required and in technical approval documents where these exist.

5.4.2.4 Quality control

The fabrication of post-tensioning system components and materials must be subjected to factory production control adapted to the importance and criticality of the particular component and material.

The factory production control must be subjected to continuous internal control by the manufacturer, and continuous external control by an independent qualified body, including regular audits.

5.4.3 Protection of tendons

5.4.3.1 Temporary corrosion protection

Tensile elements and anchorages in all pretensioned and post-tensioned applications must be given a suitable temporary corrosion protection adapted to the intended use, the expected environmental conditions and exposure, and the expected period until the permanent corrosion protection is applied.

5.4.3.2 Permanent corrosion protection

Tensile elements and anchorages for all pretensioning and post-tensioning tendons must be given a suitable permanent corrosion protection adapted to the intended use, the expected environmental conditions and exposure, and the specified design life of the structure in which the tendons are placed.

The protection provided by the structure and/or other protective systems on the structure may be considered for the required level of protection on the tendon itself.

Exposed surfaces of metal components must be given a suitable corrosion protection. The required maintenance procedures and intervals must be specified in the project specification.

5.4.3.3 Permanent corrosion protection of prestressing steel

Three protection levels for tendons are recommended, see Figure 5.4-1:

- PL1 for all tendons used in environments which have a relatively low aggressiveness and which are well protected by the structure;
- PL2 for all tendons used in all other combinations of environments and/or exposure, and protection not included in protection levels PL1 and PL3 provided by the structure;
- PL3 for all tendons used in aggressive environments and/or severe exposure and with low protection provided by the structure.

- PL2: tendon with PL1, and additionally protected with a suitable permanent coating and special protection of tendon ends;
- PL3: tendon with PL2, plus a suitable corrosion monitoring system.

		Structural protection layers		
		High	Medium	Low
Aggressivity / Exposure	High			
	Medium			
	Low			

Figure 5.4-1: Tendon protection levels

FRP materials are not susceptible to classical corrosion and are generally resistant against a wide range of aggressive environments. In particular cases, certain environmental effects such as UV radiation and alkalinity may be of concern. Typical influences such as temperature, humidity and alkalinity must be included in the declaration of the supplier, who must make reference to independent test data.

Fire protection is typically provided by sufficiently thick cover with a suitable material.

Resins in FRP tendons are particularly susceptible to the effect of fire.

Early application of prestress may prevent or reduce cracking of concrete due to shrinkage and temperature effects.

The minimum concrete strength required at the time when tensioning takes place depends mainly on the design of the anchorage, the provided local anchorage zone reinforcement, the edge distance of the anchorage and the spacing between adjacent anchorages.

The specified maximum force applies to the part of the tendon between end anchorages; that is, it does not apply to the part of the tendon in the stressing devices which do not form part of the permanent tendon.

In exceptional cases in post-tensioning where unforeseen deviation of frictional behaviour on the site occurs, it may be impossible to obtain the needed prestressing force under the limitations specified in this section. In such exceptional cases it is possible, if the actual tensile elements and prestressing technique allow it, to apply a higher stress at the end of the tendons. This stress should never exceed the value of $0.95 f_{p0.1k}$ – see chapter 8.

5.4.3.4 Permanent protection of FRP materials

FRP tendons must be given a suitable permanent protection adapted to the intended use, the expected environmental conditions and exposure, and the specified design life of the structure in which the tendons are placed.

5.4.3.5 Fire protection

Tensile elements and anchorages for all pretensioning and post-tensioning tendons must be given a suitable protection against the effect of fire, adapted to the intended use and exposure, and the specified fire rating of the structure or element in which the tendons are placed.

5.4.4 Stresses at tensioning, time of tensioning

5.4.4.1 Time of tensioning

The time when prestressing takes place should be fixed with due regard to:

- deformation conditions of the structure;
- safety with respect to local stresses and the compressive strength of the concrete;
- early application of a part of the prestress to reduce shrinkage effects.

The minimum concrete strength required at the time when tensioning takes place is given in the approval documents and/or system documentation for the prestressing system concerned, and must be specified in the project specifications.

5.4.4.2 Tendons made from prestressing steel

The maximum tensile stress in the tendons made from prestressing steel, both pretensioning and post-tensioning, should generally not exceed the lower of the following values before transfer of prestressing to the concrete:

$$\sigma_{p0,\max} = 0.80 f_{ptk} \quad (5.4-1)$$

$$\sigma_{p0,\max} = 0.90 f_{p0.1k} \quad (5.4-2)$$

The maximum tensile force in the tendons, both pretensioning and post-tensioning, should generally not exceed the lower of the following values after transfer of the prestressing to the concrete:

$$\sigma_{p0,\max} = 0.75 f_{ptk} \quad (5.4-3)$$

$$\sigma_{p0,\max} = 0.85 f_{p0.1k} \quad (5.4-4)$$

F_{pkT} is the characteristic long term tensile strength of the tendon made of FRP for the declared design life of the tendon. It considers stress rupture for the applicable tendon/anchorage assembly and detrimental environment. It considers the strength retention criterion of subsection 5.4.2.1(b). Indicative values for the permissible long term stress of FRP are given in subsection 5.5.5.6.

The coefficient α should be declared by the supplier of the FRP tendons; α is expected to be on the order of 0.75.

The initial prestress (at time $t = 0$) is calculated taking into account the prestressing force and the permanent actions present at tensioning.

Friction losses for deviated pretensioning tendons can be calculated similarly to post-tensioning tendons but neglecting the unintentional angular displacements, see subsection 5.4.5.3.

All values given in Table 5.4-1 below should be considered as indicative mean values.

The coefficient of friction μ is the product of the physical coefficient of friction μ_0 and the squeezing factor. This squeezing factor is dependent on the degree of filling of the duct. Where more exact investigations are not available, this squeezing factor can be assumed to be 1.3 to 1.35 for tendons filling the duct between 50% and 60%. The physical coefficient of friction μ_0 is influenced *inter alia* by the surfaces of tensile elements and ducts (micro- and macro-structures), rust, pressure, elongation of the tendon etc.

If more accurate values are not available and in the case of tensile elements and duct being both without rust, the values given below can be assumed, for μ with a 50% filling of ducts. These values which are indicative mean values can be multiplied by 0.9 if slight lubrication is present, for example by means of water-soluble oil.

Under site conditions, significant variations are possible. In the case of rust, variations of 50% or even higher may occur.

For the verification of the real values of prestressing losses at tensioning it is recommended to measure the transmission of prestressing force from one end of the tendon to the other (e. g. with lift-off testing at the dead-end anchorage) on some typical and/or critical tendons.

5.4.4.3 Tendons made from FRP materials

The maximum tensile force in the tendons made from FRP materials should generally, after transfer of prestressing to the concrete, not exceed the permissible permanent prestress declared by the supplier, with consideration of stress rupture:

$$F_{p0,\max} = \alpha \cdot F_{pkT} \quad (5.4-5)$$

5.4.5 Initial prestress

5.4.5.1 General

The value of the initial prestressing force (at time $t = 0$) at a given section of abscissa x along the tendon, is obtained by subtracting from the force at tensioning the different immediate losses described below.

5.4.5.2 Losses occurring in pretensioning beds

The following losses should be considered in design:

- loss due to friction at the bends (in the case of curved tensile elements);
- losses due to seating of the anchoring devices (at the abutments) when anchoring on a prestressing bed;
- loss due to relaxation of the tensile elements during the period which elapses between the tensioning of the tendons and prestressing of the concrete.

5.4.5.3 Immediate losses occurring during stressing

Losses due to the instantaneous deformation of concrete

Account should be taken of the loss in tendon force corresponding to the deformation of concrete:

- in the case of post-tensioning tendons, taking into account the order in which the tendons are stressed;
- in the case of pretensioning tendons, as a result of their action when they are released from the anchorages.

Losses due to friction (post-tensioning tendons)

In a cross-section which is at a distance x from a stressing anchorage, the stress $\sigma_{p0}(x)$ in the tendon being tensioned is lower than the stress at the anchorage $\sigma_{p0,\max}$. The difference between these two stresses corresponds to the losses due to friction:

$$\sigma_{p0}(x) = \sigma_{p0,\max} e^{-\mu(\alpha + kx)} \quad (5.4-6)$$

where:

- μ is the coefficient of friction between the prestressing steel and the duct;
- α is the sum of the angular displacements over a distance x , irrespective of direction or sign [radians];
- k is an unintentional angular displacement (per unit length) depending on the design layout (shape) of the tendon, stiffness of duct and spacing of duct supports [radians/m].

Values for μ and k are declared in the system documentation for the particular tendon details and deduced from previous experience or testing with the same type of materials and construction. These values μ and k must be declared in technical approval documents where these exist.

With external prestressing, the friction is concentrated at deviation devices.

For external tendons, the effect of unintentional angular displacement may be neglected.

(a) *Friction losses in case of bonded internal tendons made from prestressing steel*

Table 5.4-1: Typical friction coefficients μ for different types of prestressing steels and ducts

Type of prestressing steel	Corrugated metal duct	Corrugated plastic duct	Smooth steel pipe	Smooth plastic pipe
Cold-drawn wire	0.14–0.18	0.08–0.12	0.25–0.30	0.08–0.12
Strand	0.16–0.20	0.10–0.14	0.25–0.30	0.10–0.14
Deformed bar	0.63–0.68			
Smooth and round bar	0.30–0.35			

The coefficient k takes account of unintentional angular deviations. k is also called the wobble coefficient. Its value depends on the quality of workmanship and on the distance between supports of the tendon. Values for k are given in approval documents. The typical range may be assumed to $k = 0.005\text{--}0.01\text{ (m}^{-1}\text{)}$. For internal tendons in precast segmental construction higher values for k are recommended to be assumed in design.

(b) *Friction losses in the case of unbonded internal tendons made from prestressing steel*

Tests and practical experience have shown that the friction coefficients μ and k as listed below can be applied.

- For monostrands (individually greased and plastic sheathed strands; single or grouped): $\mu = 0.05\text{--}0.07$ $k = 0.006\text{--}0.01\text{ m}^{-1}$
- For multistrand or multiwire tendons inside plastic pipe and filled with grease: $\mu = 0.12\text{--}0.14$ $k = 0.004\text{--}0.008\text{ m}^{-1}$
- For dry multistrand or multiwire tendons (with dry air as subsequent corrosion protection) factors as for bonded internal tendons apply.

(c) *Friction losses in the case of external tendons made from prestressing steel*

- For bare dry strands or wires over steel saddle:
 $\mu = 0.25\text{--}0.30$ $k = 0$
- For lubricated strands or wires over steel saddle:
 $\mu = 0.20\text{--}0.25$ $k = 0$
- For dry strands or wires inside plastic pipe over saddle:
 $\mu = 0.12\text{--}0.15$ $k = 0$
- For bundle of monostrands (individually greased and plastic sheathed strands) over saddle: $\mu = 0.05\text{--}0.07$ $k = 0$

These values apply for saddle radii as given in subsection 5.4.9. For lower radii further test evidence may be needed.

(d) *Friction losses in the case of tendons made from FRP materials*

- Friction and wobble coefficients declared by the supplier of the FRP tendons should be considered for the calculation of immediate losses during stressing.

Seating of the prestressing steel in the anchorage causes a shortening (negative elongation) of the tendon with a corresponding loss of tendon force in the vicinity of the anchorage.

The values for seating to be taken into consideration are defined in the system documentation and/or approval documents for the prestressing system concerned.

Losses caused by seating of the tensile elements

Account must be taken of the loss which occurs during seating at the anchorages of post-tensioning tendons, that is during the operation of anchoring after tensioning.

Transfer of the prestressing force to the concrete in pretensioning tendons produces a loss in force in the tensile elements over the transfer length.

Due to this seating, the highest stress along the tendon is no longer at the anchorage.

Heat treatment is mostly used in precasting, and therefore, applies particularly to precast-pretensioned applications. The typical procedure exposes the prestressing steel during a well defined duration of some hours to elevated temperatures (typically less than 80–90°C).

For typical procedure the loss of prestress due to relaxation during the heat treatment can be equated to 75% of the total value of relaxation losses. The final loss of relaxation is slightly increased only. Figure 5.4-2 illustrates the effect of a typical heat treatment on very low relaxation and normal relaxation strand.

Effect of heat treatment curing

Two types of losses have to be taken into account:

- reduction of stress in the tensile elements due to an acceleration of relaxation during heat treatment;
- direct thermal effect.

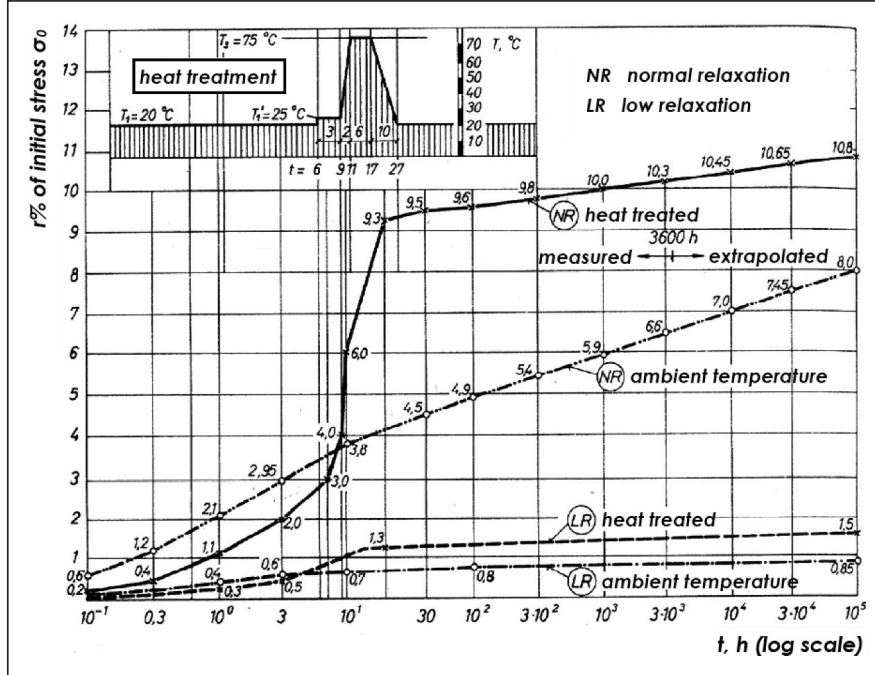


Figure 5.4-2: Effect of typical heat treatment on relaxation losses of strands stressed initially to 80% of actual tensile strength (Ref: Personal correspondence with A. Erdélyi)

(a) Relaxation losses

For non-typical heat treatment procedures, relaxation losses can be estimated by adding to the value of time a duration t_{eq} defined by:

$$t_{eq} = t_{p1} 1.14^{(T_{max} - 20)} \quad (5.4-7)$$

where:

T_{max} is the maximum temperature of the concrete during heat treatment in °C;

t_{p1} is the mean duration of the heating cycle, calculated by:

$$t_{p1} = (T_{max} - 20)^{-1} \int_0^{t_1} [T(t) - 20] dt \quad (5.4-8)$$

where:

t_1 is the age of the concrete when its temperature returns to ambient temperature;

$T(t)$ is the temperature of concrete, in °C, at time t .

An overall increase of final relaxation may have to be considered depending on the duration of exposure to elevated temperature.

(b) Losses of direct thermal origin

Direct thermal effect is caused by:

- the dilation of concrete, when it is not bonded to the tensile elements;
- the restraint to the dilation of concrete presented by the tensile elements when they are bonded.

This type of loss does not exist with moulds supporting the tension of tendons and heated together with concrete.

The losses of direct thermal origin can be calculated by:

$$\Delta\sigma = \kappa E_p \alpha_p (T_{\max} - T_0) \quad (5.4-9)$$

where:

- E_p is the elastic modulus of tensile elements;
- α_p is the coefficient of thermal expansion of tensile elements;
- T_0 is the temperature of tensile elements at tensioning;
- $T_{\max}$ is the maximum temperature of tensile elements during heat curing;
- κ is a coefficient which depends on effective bond of tendon to fresh concrete and the period of time before heat treatment starts. $\kappa = 0.9$ may be assumed conservatively. However, lower values have been measured down to $\kappa = 0.65$.

The time-dependent losses are usually determined under quasi-permanent loads on the structure. Other load combinations should be considered where and when relevant.

Data for calculation of the deformations of concrete under creep and shrinkage are given in section 5.1.

Ordinary reinforcement has an influence on the value of time-dependent shortening of concrete. The interaction can be estimated as described in CEB Bulletin 199 "Evaluation of the Time Dependent Behavior of Concrete" (CEB, 1999).

The reduction of strain in tensile elements due to time-dependent losses may be calculated by dividing the stress loss by the modulus of elasticity of tensile elements.

Basic data on the relaxation of tensile elements are given in subsection 5.3.6.1 for prestressing steel and subsection 5.5.5.7 for FRP materials.

For initial stress values below 70% of ultimate tensile strength, a straight line through the 70% point may be assumed for interpolation with zero relaxation at an initial stress of 50% of specified ultimate tensile strength.

Other immediate losses

Account should be taken of all possible causes of immediate loss of tendon force related to the tensioning process or the equipment used for tensioning.

5.4.6 Value of prestressing force during design life (time $t > 0$)

The initial prestressing force in a tendon is the force existing in the tendon at the end of the stressing operation. The initial prestressing force on a prestressed element is obtained by considering all the forces existing in the tendons, at the end of the last stressing operation.

The prestressing force at a given time t is obtained by subtracting from the initial prestressing force the value of the time-dependent losses at this time t .

These losses are due to creep and shrinkage of concrete and relaxation of tensile elements.

5.4.6.1 Calculation of time-dependent losses made of prestressing steel

The time-dependent losses are calculated by considering the following two reductions of stress:

- the reduction of stress, due to the reduction of strain, caused by the deformation of concrete due to creep and shrinkage, under quasi-permanent actions:
 - for bonded tendons, the local deformation at the level of the tendon has to be considered;
 - for unbonded tendons, the deformation of the whole structure between the anchorages of the tendons has to be taken into account;
- the reduction of stress within the tensile elements due to the relaxation of this material under tension.

5.4.6.1.1 Effect of initial stress on relaxation loss of prestressing steel

Relaxation losses at 20°C and for a duration of 1000 hours are specified in relevant standards. Values for prestressing steel at initial stresses of 70% and 80% of actual ultimate tensile strength are given in subsection 5.3.6.1. For design purposes, these values may be assumed conservatively to apply at 20°C for initial stresses of 70% and 80% of the specified ultimate tensile strength of prestressing steel.

For initial stress values between 70% and 80% of tensile strength, the relaxation loss at 1000 hours may be estimated based on linear interpolation.

Test results over extended periods of time have confirmed that a linear relationship in a bi-logarithmic presentation between relaxation loss and time represent the actual behaviour very closely. Figure 5.4-3 illustrates relaxation measurements made over a period of more than 40 years by: Müller, H.R. and Zetterholm, S. “Results of extreme long duration of a relaxation test (42 years) on prestressing steel”, Proceedings of 1st *fib* Congress, Osaka, pp. 385–390, 2002.

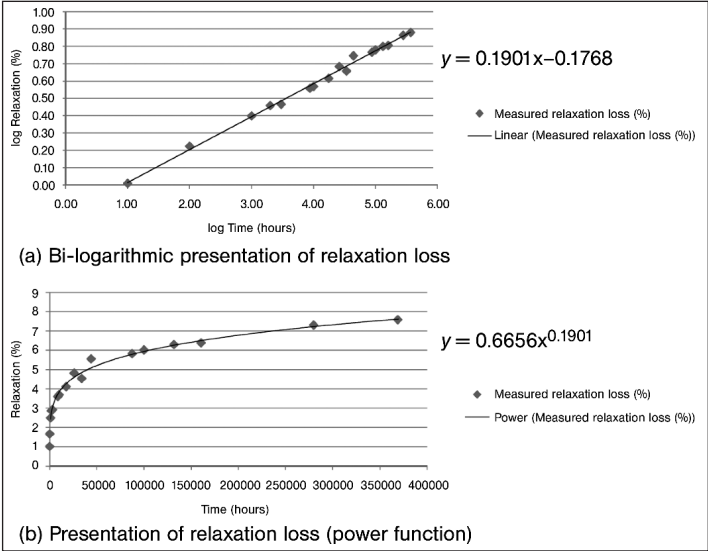


Figure 5.4-3: Results of long duration relaxation measurements (Müller, Zetterholm, 2002)

For level I approximation the following approximation may be used for the relationship of relaxation and time:

- straight line or power line curve fitting through the specified relaxation loss at 1000 hours and an estimated loss at 100 hours.

Table 5.4-2 may be used as an indication of how relaxation develops with time up to 1000 hours. Variations of ±5% may apply for given percentages for times of 100 hours and more, and variations of ±10% may apply for given percentages for times below 100 hours, respectively, for slow and rapid development of relaxation. For the above curve fitting, the value at 100 hours should be taken from the slow development of relaxation.

Table 5.4-2: Relationship between relaxation losses and time up to 1000 hours

Time in hours	1	5	20	100	200	500	1000
Slow development: relaxation losses as percentage of loss in 1000 hours	20	35	45	65	75	85	100
Mean development: relaxation losses as percentage of loss in 1000 hours	30	45	55	70	80	90	100
Rapid development: relaxation losses as percentage of loss in 1000 hours	40	55	65	75	85	95	100

5.4.6.1.2 Effect of time on relaxation loss of prestressing steel

For design purposes, a straight line relationship may be assumed in a bi-logarithmic presentation between relaxation loss and time.

The final value of relaxation loss may be assumed to be reached at the following times:

- 5·10⁵ hours for typical buildings (50 years design life);
- 10⁶ hours for bridges and engineering structures (100 years design life).

For design purposes the following level I and level II approximations for relaxation losses may be used:

- Level I approximation must apply when the assumed relaxation losses are not confirmed by test results for the actual prestressing steel;

For level I approximation of the loss due to relaxation, the following equation may be applied:

$$\rho_t = \rho_{1000} (t / 1000)^k \quad (5.4-10)$$

where:

ρ_t is the relaxation after t hours;

ρ_{100} is the relaxation after 100 hours (slow development);

ρ_{1000} is the specified relaxation after 1000 hours (Table 5.3-2);

$k \approx \log (\rho_{1000} / \rho_{100})$.

For level II approximation the following approximation may be used for the relationship of relaxation and time:

(a) straight line best fit of actual test results up to at least 1000 hours in bi-logarithmic presentation, or

(b) power line best fit of actual test results up to at least 1000 hours in presentation of relaxation loss versus time:

$$(a) \log \rho_t = a \log t + \log b$$

$$(b) \rho_t = b (t)^a \quad (5.4-11)$$

where a , b are coefficients from best fit curves.

See Figure 5.3-4 for the typical effect of temperature on relaxation losses of prestressing steel. As shown, relaxation losses are accelerated, and overall magnitude of relaxation loss is increased when the temperature is higher than 20°C over an extended period of time.

See de Halleux, B., “Explication physique de l’influence de la température sur le fluage et la relaxation des aciers de précontrainte”, FIP Symposium, Prague, 1970; and de Halleux, B., “Accelerated determination of the stress relaxation in prestressing reinforcement by an anisothermal relaxation test”, FIP VII Congress, New York, 1974.

In a first approximation, the slope of the straight line best fit curves remains approximately constant for different temperatures. However, the curves for temperatures higher than 20°C shift upwards along the ordinate (Figure 5.4-4). This can be considered by an increase of the relaxation loss with an amplification factor, AF (Figure 5.4-5).

- Level II approximation must apply when the assumed relaxation losses are confirmed by test results for the actual prestressing steel. Level II approximation has to be determined as the best fit curve of actual test results at a given initial stress and temperature with a duration of at least 1000 hours.

5.4.6.1.3 Effect of temperature on relaxation loss of prestressing steel

Relaxation losses for prestressing steel are specified in the relevant standards at a constant temperature of 20°C. For lower temperatures, these values may conservatively be assumed to apply.

In cases where tendons are exposed over a significant period of time to temperatures higher than 20°C, relaxation losses develop more rapidly and increase in magnitude when compared to losses at 20°C.

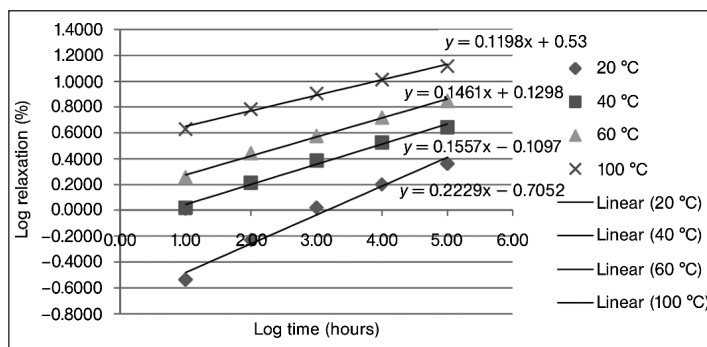


Figure 5.4-4: Effect of temperature on very low relaxation prestressing wire and strand (data from Figure 5.3-4 shown in bi-logarithmic presentation)

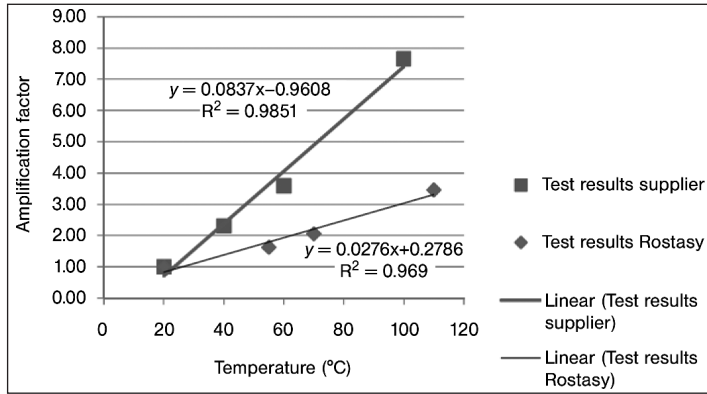


Figure 5.4-5: Amplification factor for $\rho_{1000}(T)$ for temperatures $T > 20^\circ\text{C}$ ("Test results supplier" = data from Figure 5.4-4; "Test results Rostasy" = data from Rostásy, F.S., Thienel, K.-Ch. and Schütt K., "On prediction of relaxation of cold-drawn prestressing wire under constant and variable elevated temperature", Nuclear Engineering and Design, Vol. 130 (1991), pp. 221–227)

For level I, the following approximation may be used for the relationship of relaxation and time:

$$\rho_t(T) = \text{AF} \cdot \rho_{1000}(t / 1000)^k \quad (5.4-12)$$

where:

$\rho_t(T)$ is the relaxation after t hours at temperature $T > 20^\circ\text{C}$

ρ_{1000} is the specified relaxation after 1000 hours (Table 5.3-2)

AF is the amplification factor to account for the effect of $T > 20^\circ\text{C}$

$$\text{AF} = T / 20^\circ\text{C} \quad (5.4-13)$$

$k \approx \log(\rho_{1000} / \rho_{100})$ as per subsection 5.4.6.1.2

It should be noted that the above suggested amplification factor, AF, is not necessarily conservative. Hence, whenever temperature is significantly above 20°C over extended periods of time, and relaxation losses are important for the performance of the structure or member, it is strongly recommended to perform relaxation tests at the particular expected temperature.

For level II the following approximation may be used for the relationship of relaxation and time:

(a) straight line best fit of actual test results at the relevant constant temperature up to at least 1000 hours in bi-logarithmic presentation, or

(b) power line best fit of actual test results at the relevant constant temperature up to at least 1000 hours in presentation of relaxation loss versus time:

$$(a) \log \rho_t(T) = a \log t + \log b$$

$$(b) \rho_t(T) = b(t)^a \quad (5.4-14)$$

where both a and b are coefficients from best fit curves as a function of the temperature T .

Alternatively, with at least one set of test results to at least 1000 hours at a constant temperature $T \geq 40^\circ\text{C}$ a straight line approximation for the amplification factor, AF, may be created passing through $\text{AF} = 1$ at $T = 20^\circ\text{C}$ and AF determined at $T \geq 40^\circ\text{C}$.

The effect of elevated constant and variable temperatures has been described for example by Rostásy, F.S., Thienel, K.-Ch. and Schütt K., "On prediction of relaxation of cold-drawn prestressing wire under constant and variable elevated temperature", Nuclear Engineering and Design, Vol. 130 (1991), pp. 221–227.

For design purposes the following level I and level II approximations for the effect of constant temperature on relaxation may be used:

– Level I approximation applies when no test results at the given constant temperature for the actual prestressing steel are available;

– Level II approximation applies when test results either at the given constant temperature T or at one temperature level $T \geq 40^\circ\text{C}$ for the actual prestressing steel are available.

Effects of variable temperature on relaxation losses should be suitably considered where relevant.

The level II approximation is generally recommended unless there is sufficient experience available for a particular FRP material which would justify use of a level I approximation.

In exceptional cases, several sets of prestressing tendons (practically never more than two) should be considered separately. These cases should be identified by judgement. The criteria, to be simultaneously satisfied, are that:

- the effect of the two sets are of contrary senses;
- these effects have the same order of magnitude;
- the dispersions are relatively high and there are qualitative reasons why they should not be considered as correlated.

In cases where the design value of the prestressing force influences the behaviour of the structure in a large over-proportional way, the designer may consider providing in the design the possibility for one or several of the following options:

- provide access at both tendon ends to allow stressing of the tendons from both ends;
- provide spare anchorages and ducts for additional internal tendons as and when required (if these spare ducts are not used at the time of construction, they should be filled after successful completion of all stressing and grouting operations);
- provide spare anchorages and deviators for additional external tendons as and when required. These anchorages and deviators may allow prestressing force to be added, not only at the time of construction but also at a later stage during the service life of the structure.

Elongation of multistrand tendons is measured on site before seating of the tendon in the stressing anchorage.

This procedure is generally applied for unbonded monostrand tendons and small bonded tendons used for post-tensioning of floors in buildings. Elongation of these tendons is measured on site only after anchorage seating.

5.4.6.2 Calculation of time-dependent losses made of FRP

Relaxation losses of tendons made of FRP must be considered similarly to the methods described in subsection 5.4.6.1. A level II approximation should generally be used.

5.4.7 Design values of forces in prestressing

5.4.7.1 General

Prestressing is usually exerted by a set of tendons. The total permanent force exerted at a given section (abscissa x), and at a time t , by the whole set is considered as the prestressing force.

5.4.7.2 Design values for SLS and fatigue verifications

For all verifications relating to cracking (decompression included) and deformations and for the analysis of the fatigue effect, the mean value of prestressing force is taken as design value.

In cases where the design value of the prestressing force or any variation on site would influence the behaviour of the structure in a large over-proportional way, the designer must take adequate precautions.

5.4.7.3 Design values for ULS verifications

For all verifications at ULS the prestrain corresponding to the mean value of prestressing force at SLS is taken and increased by the strain imposed onto the corresponding cross-section (bonded tendons) or on the overall tendon (unbonded tendons) at the relevant ULS load combination.

5.4.8 Design values of tendon elongations

For internal and external multistrand tendons, design values for tendon elongations must be calculated for the initial prestressing force – subsection 5.4.5.1 – before losses caused by seating of prestressing steel in the anchorage.

For internal unbonded monostrand tendons, design values for tendon elongations must be calculated by taking into account the losses caused by seating of the prestressing steel in the anchorage.

For pretensioning, the initial prestressing force – subsection 5.4.5.1 – before losses due to relaxation in the pretensioning bed and before heat treatment must be considered.

The assumed values for friction losses, μ and k , must be declared.

Minimum spacing may be assumed as twice the minimum cover required for bond transfer, and is a function of the concrete strength and strength of tensile elements.

Based on experience, the following minimum radii of curvature of tendons have provided satisfactory behaviour in practice for tendons made from prestressing steel:

(a) internal bonded tendons:

$$R_{\min} = 2.8 \sqrt{f_{ptk} A_p [\text{MN}]} \geq 2.5 \text{ m} \quad (5.4-15)$$

(b) internal unbonded monostrand tendons (greased and sheathed strands):

$$R_{\min} = 2.5 \text{ m for } 0.6'' \text{ and } R_{\min} = 2.0 \text{ m for } 0.5'' \quad (5.4-16)$$

(c) external tendons:

$$R_{\min} = 1.4 \sqrt{f_{ptk} A_p [\text{MN}]} \geq 2.0 \text{ m} \quad (5.4-17)$$

(d) loop tendons:

$$R_{\min} = 0.6 \sqrt{f_{ptk} A_p [\text{MN}]} \geq 0.6 \text{ m} \quad (5.4-18)$$

Loops are parts of tendons which are deviated by 180°, stressed simultaneously from both ends such that the prestressing steel inside the loop deviation basically does not move relative to the duct during stressing.

The minimum radius of curvature above recommended for internal bonded tendons limits the stresses on the concrete to values which do not require splitting or confinement reinforcement, in general.

Typically, the tendons are assumed to exit from the anchorage straight and perpendicular to the bearing plate. The minimum straight length depends on the specific type of anchorage and is specified in the post-tensioning system approval documents.

5.4.9 Detailing rules for prestressing tendons

5.4.9.1 Pretensioning tendons

Minimum radii of tendon curvature at deviators in the pretensioning bed must be such that the tendon capacity at deviators complies with the tensile strength requirement for anchorage and coupling devices – see subsection 5.4.2.1.

Individual pretensioned tensile elements must be spaced such that they comply with requirements for bond transfer.

5.4.9.2 Post-tensioning tendons

Minimum radii of tendon curvature for all types of tendons must be such that the tendon capacity in the curvature complies with the tensile strength requirement for anchorage and coupling devices – see subsection 5.4.2.1. The minimum radii of curvature given in the commentary can be deemed to comply with subsection 5.4.2.1.

The stresses on the concrete on the inside of the tendon curvature must be checked, and splitting or confinement reinforcement must be provided when required.

Minimum radii of tendon curvature must be declared by the tendon supplier in the system documentation and in technical approval documents where these exist.

The tendon must exit in a geometry from the anchorage or coupling devices similar to the set-up tested for anchorage and coupling device performance – see subsection 5.4.2.1.

Ducts of groups of curved tendons must be spaced such that the deviation forces from the curved tendons can be safely transferred around the adjacent ducts on the inside of the curve.

Ducts of groups of tendons must have minimum spacing which permits adequate placing and compacting of concrete.

Background information on designing with FRP, as treated in this Model Code, is given in Triantafillou, T., Matthys, S. (2013), *Fibre Reinforced Polymer Reinforcement Enters fib Model Code 2010. Structural Concrete*, 14. doi: 10.1002/suco.201300016. Fibres are commonly made of carbon, glass and aramid, but other materials are emerging, such as basalt and polyphenylene bezobisoxazole (PBO). As they occupy the largest volume fraction (40–70%) and have an elastic modulus much higher than that of the matrix, fibres are the principal stress-bearing constituent, while the resin transfers stresses among fibres and protects them.

FRP (non-metallic) reinforcement is available in various forms. For new structures bars, tendons and grids are used to reinforce and prestress concrete elements. In the repair sector, these elements are used to strengthen existing structures by means of external post-tensioning and near surface mounted reinforcement. Strips, laminates, sheets or fabrics are used for externally bonded reinforcement strengthening. Grids and fabrics may also be used in combination with shotcrete or mortar overlays.

The geometrical, mechanical and technological properties of FRP reinforcement basically depend on fibre and resin type and properties, constituent volume fractions, production parameters, shape and surface texture. In general, they are characterized by high axial strength, high ratio of axial to transverse strength, limited ultimate strain, low weight, excellent chemical resistance and non-susceptibility to a wide range of aggressive media, electromagnetic neutrality, excellent fatigue characteristics (depending on fibre type), limited ratio of long term to short term static strength for some fibre types.

Examples of relevant international product standards are ISO 10406 (Parts 1 and 2).

The methods of testing and certification of conformity are as defined in standards and recommendations, at the National, European or International level. For further information see *fib* Bulletin 14 “Externally bonded FRP reinforcement for RC structures” (*fib*, 2001) and *fib* Bulletin 40 “FRP reinforcement in RC structures” (*fib*, 2007).

5.5 Non-metallic reinforcement

5.5.1 General

Non-metallic reinforcing elements consist of a large number of continuous, directionalized, organic or inorganic fibres, typically embedded in a polymeric matrix. Both the terms “non-metallic reinforcement” and “FRP (fibre reinforced polymer) reinforcement” are used for this reinforcement type.

Non-metallic reinforcing products may be in the form of:

- pre-cured bars, tendons, strips, laminates, grids or profiles;
- sheets or fabrics applied by wet lay-up or pre-impregnated (prepreg).

Non-metallic reinforcement is characterized by:

- Geometrical properties:
 - configuration;
 - size;
 - surface characteristics.
- Mechanical properties:
 - tensile strength, modulus of elasticity and ultimate strain;
 - fatigue behaviour;
 - creep behaviour;
 - relaxation;
 - behaviour at elevated temperature and at extreme thermal conditions.
- Technological properties:
 - bond characteristics;
 - bendability;
 - thermal expansion;
 - durability.

Non-metallic reinforcement must comply with national or international product standards that specify their geometrical, mechanical and technological properties.

5.5.2 Quality control

The fabrication of non-metallic reinforcement must be subject to factory production control by the manufacturer, and continuous external control by an independent qualified body, including certification and regular audits.

5.5.3 Designation

The designation of non-metallic reinforcing elements normally includes:

- relevant product standard;
- fibre and matrix materials;
- nominal dimensions (e. g. diameter, width, thickness);
- characteristic tensile strength, modulus of elasticity and the ultimate strain in the direction of the fibres.

Each product must be clearly identifiable with respect to this designation.

Pre-cured systems are manufactured in various shapes, generally by pultrusion. Wet lay-up systems are manufactured with fibres lying in one (typically) or more directions and impregnated with the matrix at the job site.

Prepreg systems are manufactured with unidirectional or multidirectional fibre sheets or fabrics pre-impregnated at the manufacturing plant with partially polymerized resin. They may be bonded externally to concrete members with or without the use of additional resin.

The definition of the nominal cross-section is for pre-cured forms typically based on the global nominal dimensions (diameter, thickness, width). Alternatively or for wet lay-up/prepreg forms, an equivalent dry fibre cross-section may be used (referring to the continuous fibres as principal stress bearing component). The latter is obtained as the ratio of the fibre mass per length and the fibre density. For FRP with multiple fibre directions, where a different amount of fibres per unit length is applied in different fibre directions, the definition of the nominal dry fibre cross-section always relates to the specified fibre direction, and more than one nominal cross-section (or nominal thickness) may be given depending on the fibre direction.

As design verifications are based on equilibrium of forces, strength values should always be used with their corresponding nominal cross-section, as declared on the product data sheets. If data sheets of FRP products are compared, the possible difference in definition of the nominal cross-section should be taken into account.

Surface characteristics include the characteristics of the ribs or indentations or other surface deformations (e.g. sand-coating) by means of which bond with the concrete is achieved.

FRP to concrete bond quality is product specific, related to the surface characteristics. Further requirements are given in section 6.2.

For quality control purposes and design calculations, the mechanical properties of a product are referred to the nominal cross-sectional area.

The standard tests are defined in relevant standards and recommendations, at the national, European or international level. For further information, see *fib* Bulletin 14 “Externally bonded FRP reinforcement for RC structures” (*fib*, 2001) and *fib* Bulletin 40 “FRP reinforcement in RC structures” (*fib*, 2007).

The requirements apply to the standard product, as tested on straight test coupons. The design considers effective characteristic strength values where relevant, for example at bent regions.

5.5.4 Geometrical properties

5.5.4.1 Configuration

Non-metallic reinforcing elements may be:

- pre-cured;
- wet lay-up;
- pre-impregnated (prepreg).

The configuration is further characterized by the type of fibre(s) and matrix, fibre orientation(s) and constituent fractions. Constituent material fractions can either be given by mass (weight) or by volume.

5.5.4.2 Size

The size of non-metallic reinforcing elements is defined by a nominal diameter for circular bars or by the nominal cross-sectional dimensions for other products (e.g. thickness, width).

The difference between actual and nominal cross-sectional area must not exceed the limiting values specified in relevant product standards.

5.5.4.3 Surface characteristics

The surface of non-metallic reinforcement may be:

- plain;
- deformed.

5.5.5 Mechanical properties

The mechanical properties are defined on the basis of standard tests.

5.5.5.1 Tensile strength and ultimate strain

The characteristic values of

- tensile strength (f_f) and
- percentage total elongation at maximum force (ϵ_{fu}) are respectively denoted f_{fk} and ϵ_{fuk} .

Typical fibre material notation is C (carbon), A (aramid), G (glass), H (hybrid).

The stress–strain response is quasilinear elastic for most FRP elements. The slope of the stress–strain relation gives the modulus of elasticity and is typically defined as a secant modulus, following product standards.

The stress–strain diagram given in Figure 5.5-1 is indicative, as FRP reinforcement products are available in a range of strength and stiffness values (Table 5.5-1 gives overall ranges).

Table 5.5-1: Tensile properties of FRP reinforcement

Property	CFRP	GFRP	AFRP
Tensile strength f_f [MPa]	600–3000	400–1600	600–2500
Modulus of elasticity E_f [GPa]	80–500	30–60	30–125
Ultimate strain ϵ_{fu} [%]	0.5–1.8	1.2–3.7	1.8–4.0

Given the generally limited compressive modulus of elasticity and the risk of microbuckling or kinking of the fibre within the restraint of the matrix material, non-metallic reinforcement is generally not used to resist high compressive stresses.

The interlaminar transverse shear strength of non-metallic reinforcement is basically dominated by the matrix and the fibres in off-axis directions.

High modulus fibre composites have superior fatigue resistance. Cyclic tension fatigue strength of unidirectional CFRP and AFRP exceeds that of prestressing steel, while that of GFRP is lower. The fatigue strength of CFRP is higher than for AFRP. Indicative values are given in Table 5.5-2.

Table 5.5-2: Fatigue strength of reinforcement after $2 \cdot 10^6$ cycles

Type of reinforcement	$\sigma_{max}/f_{tk}^{(a)}$ [–]	$\Delta\sigma^{(b)}$ [MPa]
Prestressing steel	~ 0.60	~ 200
E-glass/polyester (rod)	~ 0.50	~ 60
E-glass/epoxy (rod)	~ 0.50	~ 75
Aramid/vinylester (rod)	~ 0.60	~ 235
Carbon/vinylester (rod)	~ 0.60	> 350
Carbon/epoxy (strand)	~ 0.60	~ 310

(a) Applied maximum stress as a function of the characteristic tensile strength of the reinforcement.

(b) Stress range yielding fatigue failure at $2 \cdot 10^6$ cycles.

5.5.5.2 Type

The non-metallic reinforcing element type denotes the fibre material, the value of the specified characteristic tensile strength in MPa, the mean secant modulus of elasticity in GPa and the specified characteristic ultimate strain in the principal direction of the fibres.

5.5.5.3 Stress–strain diagram and modulus of elasticity

An indicative stress–strain diagram of a non-metallic reinforcing element in tension is given in Figure 5.5-1.

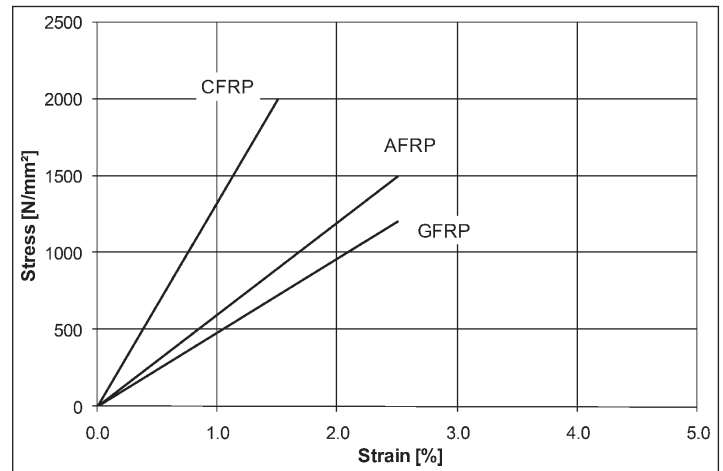


Figure 5.5-1: Stress–strain diagram of non-metallic reinforcement in the principal fibre direction

5.5.5.4 Compressive and shear strength

The compressive or transverse shear properties for a particular reinforcing element, if needed for a particular application, should be given by the manufacturer, who should also provide a description of the test method used to determine the properties.

5.5.5.5 Fatigue behaviour

If a non-metallic reinforcing element is subjected to a large number of load cycles, growth of internal or surface flaws may occur, resulting in a reduced mechanical strength compared to the short term static strength.

FRP reinforcement combines elastic fibres, which have excellent resistance to creep, with a viscoelastic polymer matrix, which may show significant creep deformations. As FRP tensile members normally have a high degree of fibre orientation, large fibre volume fractions and a high ratio of fibre over matrix stiffness, the tensile force shared by the matrix is extremely low, so that FRP creep deformations are negligible.

The permissible stress level against stress rupture depends on the fibre/resin system, the alignment of the fibres and the fibre volume fraction. Stress rupture is adversely influenced by the environmental conditions. Generally, CFRP can withstand stress levels up to at least 80% of its short term strength, while considerably lower stress levels apply for AFRP (about 50% on a 50 year basis) and GFRP (about 30% on a 50 year basis).

Relaxation of GFRP, CFRP and AFRP prestressing elements after 50 years of loading can be estimated as 4–14%, 2–10% and 11–25%, respectively. These values depend on the stress level and environmental influence.

Prestressing loss due to relaxation of FRP is compensated by a lower prestressing loss due to concrete shrinkage and creep (given the ratio of the modulus of elasticity E_f/E_c).

The glass transition temperature T_g is of particular importance, as it reflects the change of molecular mobility of polymer materials. For factory processed FRP elements, the matrix generally has a T_g in the range of 130–140°C. The T_g of cold-cured (ambient-cured) adhesives/saturating resins may be lower (typically in the range of about 50–80°C for epoxy).

FRP to concrete bond interaction is different from that of deformed steel rebars. Further requirements are given in section 6.2.

Thermoset resin based FRP elements are not bendable in situ. Bent FRP elements are factory made and pre-cured. Thermoplastic resin based FRP elements are bendable given proper application procedures.

Given the anisotropy of FRP elements, different CTE values are obtained in the longitudinal and the transverse direction. Indicative values are given in Table 5.5-3.

Table 5.5-3: Coefficient of thermal expansion (CTE)

FRP type	Longitudinal CTE [10 ⁻⁶ °C ⁻¹]	Transverse CTE [10 ⁻⁶ °C ⁻¹]
AFRP	≈ - 2	≈ 30
CFRP	≈ 0	≈ 25
GFRP	≈ 5	≈ 25

5.5.5.6 Creep behaviour

Provided that the glass transition temperature is well above the service temperature, FRP creep deformations are generally negligible.

The long term permissible stress of non-metallic reinforcement should be limited to avoid stress rupture.

5.5.5.7 Relaxation

Relaxation of non-metallic reinforcement is to be considered for prestressing loss calculations.

5.5.5.8 Behaviour under elevated temperature and under extreme thermal conditions

Although fibres exhibit relatively high thermal stability, polymer resins are strongly affected by temperature. As a result, the material and bond properties of FRP are influenced by temperature, and decrease drastically when reaching the glass transition temperature T_g . In the event of fire, sufficient concrete cover should be available so that the glass transition temperature is only reached after the required time span. For external reinforcement systems, fire protection systems may be required.

5.5.6 Technological properties

5.5.6.1 Bond characteristics

Bond characteristics of non-metallic reinforcement relate to the surface characteristics, as specified in subsection 5.5.4.3.

5.5.6.2 Bendability

The requirements concerning the bendability are specified in relevant standards.

5.5.6.3 Coefficient of thermal expansion

Values of the CTE for non-metallic reinforcement depend on the constituents and are product specific.

In general, FRP materials appear to have a good to excellent resistance against chemical attack. Glass fibre based FRP reinforcements have limited alkaline resistance, which is to be considered for internal reinforcement. FRP reinforcements may also be influenced by moisture, thermal stresses and UV radiation. For a detailed discussion on durability, see *fib* Bulletin 14 “Externally bonded FRP reinforcement for RC structures” (*fib*, 2001) and *fib* Bulletin 40 “FRP reinforcement in RC structures” (*fib*, 2007).

For serviceability limit state verifications a linear stress–strain response $\sigma_f = E_f \varepsilon_f$ is considered, referring to the mean value of the secant modulus of elasticity E_f .

For ultimate limit state verification, the design stress–strain curve is idealized by means of a linear response (Figure 5.5-2), given the characteristic tensile strength f_{fk} and ultimate strain ε_{fuk} . The slope of this design stress–strain curve refers to a modulus f_{fk}/ε_{fuk} .

For details corresponding to the serviceability and the ultimate limit states see *fib* Bulletin 14 “Externally bonded FRP reinforcement for RC structures” (*fib*, 2001), *fib* Bulletin 35 “Retrofitting of concrete structures by externally bonded FRPs, with emphasis on seismic applications” (*fib*, 2006) and *fib* Bulletin 40 “FRP reinforcement in RC structures” (*fib*, 2007).

The material factor γ_f for non-metallic reinforcement takes into account the scattering of strength values and the consistency in material failure mode. Sometimes the non-metallic reinforcement is designed explicitly for durability by considering the relative resistance of generic FRP types to aggressive environments and the desired service life of the structure. These allow to assume adapted values for the material factor and the allowable stress level, as outlined in *fib* Bulletin 14 “Externally bonded FRP reinforcement for RC structures” (*fib*, 2001) and *fib* Bulletin 40 “FRP reinforcement in RC structures” (*fib*, 2007).

5.5.6.4 Durability

Non-metallic reinforcement has excellent chemical resistance and non-susceptibility to a wide range of aggressive media.

5.5.7 Assumptions used for design

The parameters of non-metallic reinforcement to be used for design are as follows:

- modulus of elasticity, E_f ;
- characteristic tensile strength, f_{fk} ;
- ultimate strain, that is characteristic percentage of the total elongation at maximum force, ε_{fuk} .

For design purposes an idealized stress–strain diagram according to Figure 5.5-2 must be used.

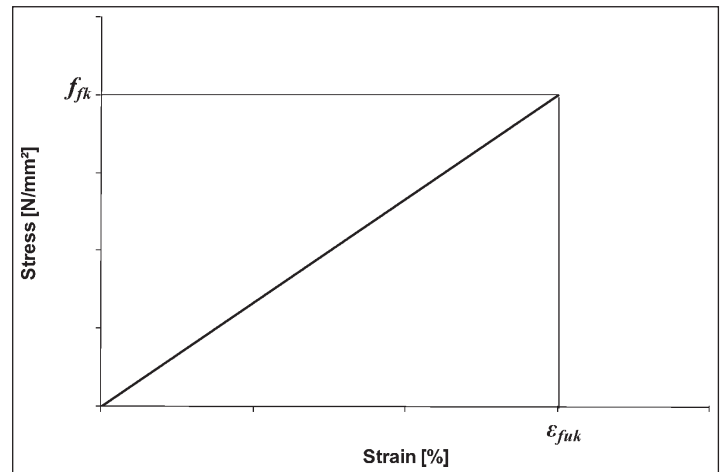


Figure 5.5-2: Idealized stress–strain diagram

The values of the material factor γ_f for non-metallic reinforcement are given in Table 5.5-4.

Table 5.5-4: Partial factors γ_f for non-metallic reinforcement

Fundamental basic variable	Design situation	
	Persistent/transient	Accidental
Non-metallic reinforcement Tensile strength (f_{fk}), γ_f	1.25	1.0

For stress limitation of non-metallic reinforcement, see subsection 5.5.5.6.

Unlike rebars or welded mesh, most fibres slip without failing in tension. Fibres are active as soon as micro-cracks are formed in the concrete. The main advantage of adding fibres to concrete or mortar is that they generate a post-cracking residual tensile strength in combination with a large tensile strain. As such, the material fibre reinforced concrete (FRC) is characterized by substantial ductility and toughness.

The properties of the composite depend on the characteristics of the constituting materials, as well as on their dosage. Other factors such as the geometry, the volume fraction and the mechanical properties of the fibres, the bond between fibre and concrete matrix, and the mechanical properties of the matrix, significantly affect the FRC properties.

Due to differences in casting and vibration procedures, FRC flowability and geometry of the moulds, anisotropic fibre distributions may occur, the effect of which should be taken into account since fibre orientation affects FRC properties after cracking.

The rules in this chapter are based most of all on experience with steel fibre reinforced concrete (SFRC).

For ultra high performance fibre reinforced concrete (UHPFRC), additional rules may apply. Further background information on the relations treated in this section is given by Di Prisco et al. (2013), Recommendations for fibre reinforced concrete in *fib* Model Code 2010. Structural Concrete, 14. doi: 10.1002/suco.201300021.

In the case of softening behaviour (a) the deformations localize in one crack. In the case of hardening behaviour (b) multiple cracking occurs before reaching the peak value.

The relation between strain softening and strain hardening behaviour is shown in Figure 5.6-1. Softening behaviour in tension can correspond to hardening behaviour in bending and a softening material in bending can result in a monotonically increasing load in the structure (Figure 5.6-1).

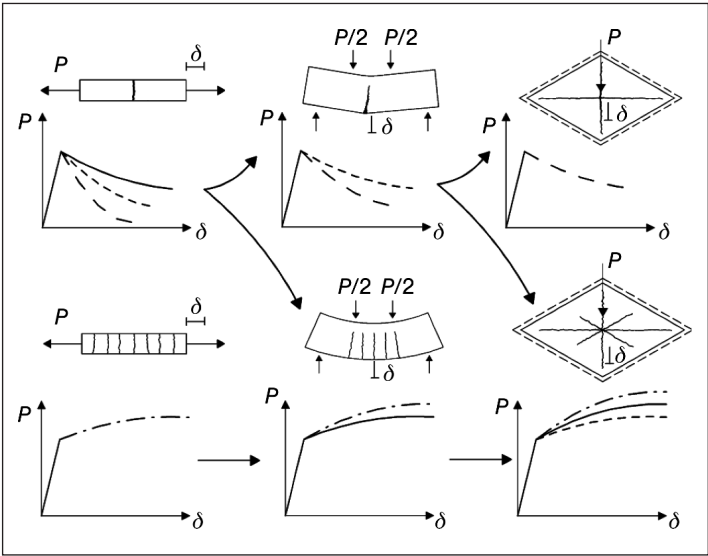


Figure 5.6-1: Different response of structures made of FRC having a softening or hardening behaviour under uniaxial tension or bending loads.

Fibres can reduce the brittleness of concrete in compression, especially in high or ultra high strength concrete (Figure 5.6-3).

5.6 Fibres/fibre reinforced concrete

5.6.1 Introduction

Fibre reinforced concrete (FRC) is a composite material characterized by a cement matrix and discrete fibres (discontinuous). The matrix is made of either concrete or mortar. Fibres can be made of steel, polymers, carbon, glass or natural materials.

Fibre materials with a Young's modulus which is significantly affected by time and/or thermo-hygrometrical phenomena are not covered by this Model Code.

Mixtures of different types and/or sizes of fibres can also be used (called hybrid fibre reinforced concrete).

Structural design of FRC elements is based on the post-cracking residual strength provided by fibre reinforcement. Other cases, such as early age crack-control or fire resistance, are considered non-structural use of FRC.

For structural use, a minimum mechanical performance of FRC must be guaranteed.

Fibres can be used to improve the behaviour at SLS since they can reduce crack spacing and crack width, thereby improving durability.

Fibres can be used to improve the behaviour at ULS where they can partially or totally substitute conventional reinforcement.

The mechanical properties of a cementitious matrix are modified when fibres are added. However, elastic properties and compressive strength are not significantly affected by fibres, unless a high percentage of fibres is used.

Depending on their composition, FRC can show hardening or softening behaviour under uniaxial tension (Figure 5.6-2).

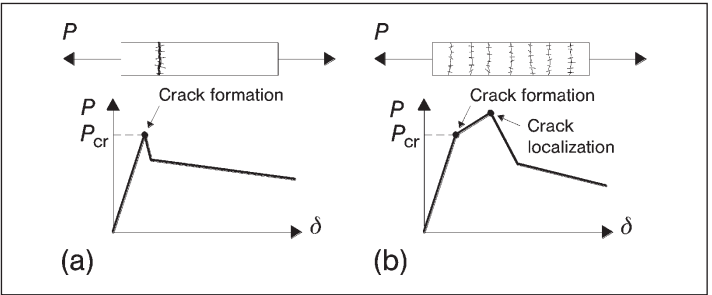


Figure 5.6-2: Softening (a) and hardening (b) behaviour in axial tension

5.6.2 Material properties

5.6.2.1 Behaviour in compression

Generally the compressive relations valid for plain concrete also apply to FRC.

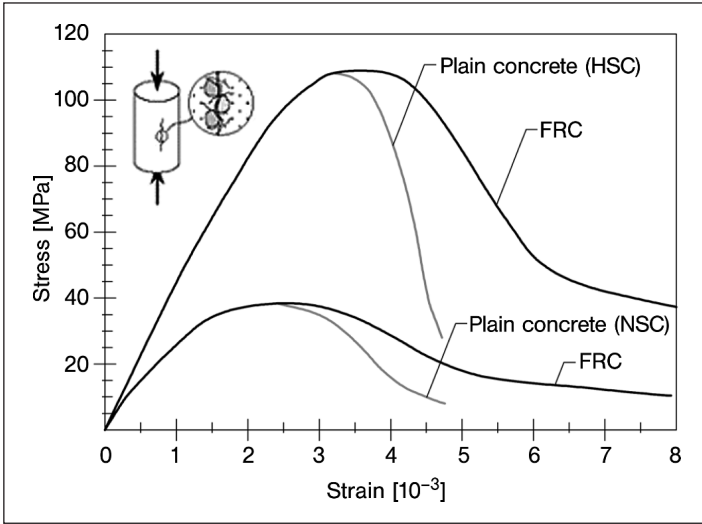


Figure 5.6-3: Main differences between plain and fibre reinforced concrete having both normal and high strength under uniaxial compression

Uniaxial tensile testing is not advised for standard testing of new mixtures, because tensile tests are difficult to carry out and interpret.

Since the specimens are normally small, the number of fibres in the governing plane will be small and it could present a fibre orientation effect due to the method of manufacturing.

5.6.2.2 Behaviour in tension

With regard to the behaviour in tension, which is the most important aspect of FRC, various test methods are possible.

Bending tests can be carried out aiming at determining the load-deflection relation. The results can be used for deriving the stress-crack width relations by inverse analysis, performing equilibrium calculations for numerous crack openings as shown in Figure 5.6-4. A simpler approach can be found in subsection 5.6.4.

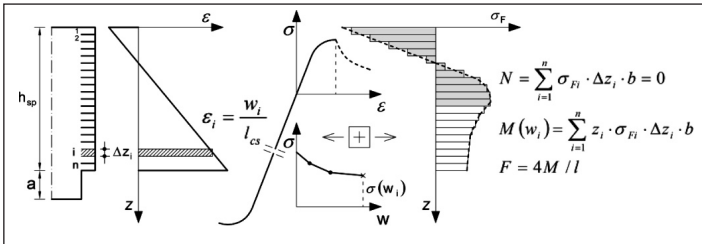


Figure 5.6-4: Inverse analysis of beam in bending performed to obtain stress-crack opening relation

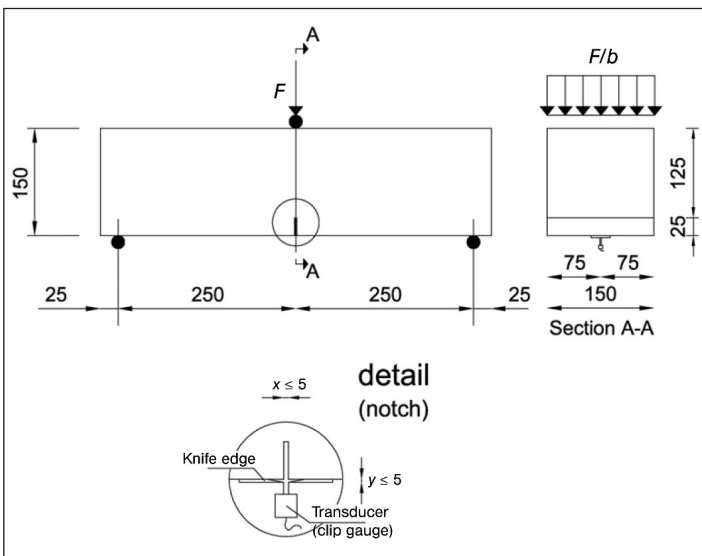


Figure 5.6-5: Test set-up required by EN 14651 (dimensions in [mm])

Nominal values of the material properties can be determined by performing a three-point bending test on a notched beam according to EN 14651 (Figure 5.6-5). The diagram of the applied force (F) versus the deformation must be produced (Figure 5.6-6). The deformation is generally expressed in terms of crack mouth opening displacement ($CMOD$), which is the opening of the notch at the bottom face of the beam (Fig. 5.6-5).

Parameters, $f_{R,j}$, representing the residual flexural tensile strength, are evaluated from the F - $CMOD$ relationship, as follows:

$$f_{R,j} = \frac{3F_j l}{2bh_{sp}^2}$$

where:

$f_{R,j}$ [MPa] is the residual flexural tensile strength corresponding to $CMOD = CMOD_j$;

F_j [N] is the load corresponding to $CMOD = CMOD_j$;

l [mm] is the span length;

b [mm] is the specimen width;

h_{sp} [mm] is the distance between the notch tip and the top of the specimen (125 mm).

Other tests can be accepted if correlation factors with the parameters of EN 14651 are proven.

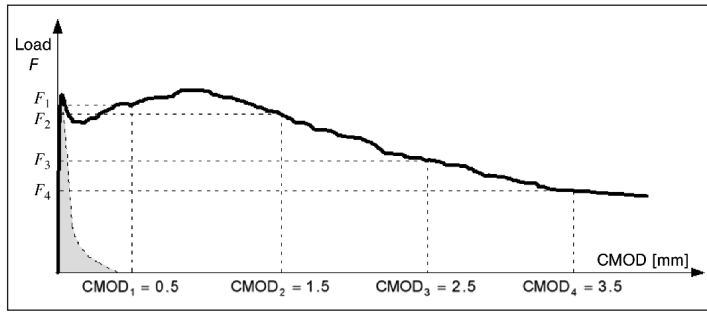


Figure 5.6-6: Typical load F – $CMOD$ curve for plain concrete and FRC

In case of organic and natural fibres, post-cracking long term behaviour can be affected by an additional creep of the fibres themselves.

For structural applications with normal and high-strength concrete, the material classification is based on the post-cracking residual strength.

For ultra-high strength fibre reinforced concrete, special design rules may be adopted.

For instance, a material denoted as “3b” has a strength f_{R1k} of 3–4 MPa and the f_{R3k}/f_{R1k} ratio of 0.7–0.9.

The limit of proportionality f_L , as defined in EN 14651, can be determined by applying the following equation:

$$f_L = \frac{3F_L l}{2bh_{sp}^2}$$

Besides these material requirements the requirements on a structural level indicated in subsection 7.7.2 should be met.

The rigid-plastic model takes the static equivalence into account as shown in Figure 5.6-8, that is f_{Ftu} results from the assumption that the whole compressive force is concentrated in the top fibre of the section.

For high fibre contents, strain hardening materials can be obtained. To guarantee the hardening in tension, the tensile behaviour must be identified by means of uniaxial tension tests carried out on unnotched specimens.

Long term behaviour of cracked FRC under tension has to be properly taken into account for those materials whose long term performance is affected by creep and/or creep rupture (see subsection 5.6.5.)

5.6.3 Classification

To classify the post-cracking strength of FRC, a linear elastic behaviour can be assumed, by considering the characteristic flexural residual strength values that are significant for serviceability (f_{R1k}) and ultimate (f_{R3k}) conditions, and, in particular, two parameters: f_{R1k} (representing the strength interval) and a letter a, b, c, d or e (representing the f_{R3k}/f_{R1k} ratio).

The strength interval is defined by two subsequent numbers in the series:

1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, ... [MPa]

while the letters a, b, c, d, e correspond to the residual strength ratios:

a if $0.5 < f_{R3k}/f_{R1k} < 0.7$

b if $0.7 \leq f_{R3k}/f_{R1k} < 0.9$

c if $0.9 \leq f_{R3k}/f_{R1k} < 1.1$

d if $1.1 \leq f_{R3k}/f_{R1k} < 1.3$

e if $1.3 \leq f_{R3k}/f_{R1k}$

(5.6-1)

The designer has to specify the residual strength class and the f_{R3k}/f_{R1k} ratio as well as the material of the fibre.

Fibre reinforcement can substitute (also partially) conventional reinforcement at ultimate limit state, if the following relationships are fulfilled:

$$f_{R1k}/f_{Lk} > 0.4 \quad (5.6-2)$$

$$f_{R3k}/f_{R1k} > 0.5 \quad (5.6-3)$$

5.6.4 Constitutive laws

A stress-crack opening law in uniaxial tension is defined for the post-cracking behaviour of FRC. Its identification can be obtained by following different procedures as shown in Figure 5.6-4.

Two simplified stress-crack opening constitutive laws may be deduced from the bending test results: a plastic rigid behaviour, or a linear post-cracking behaviour (hardening or softening) as schematically shown in Figure 5.6-7, where f_{Fts} represents the serviceability residual strength, defined as the post-cracking strength for serviceability crack openings, and f_{Ftu} represents the ultimate residual strength.

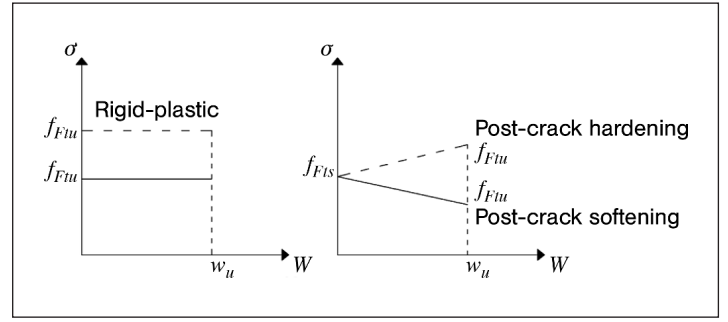


Figure 5.6-7: Simplified post-cracking constitutive laws: stress-crack opening (continuous and dashed lines refer to softening and hardening post-cracking behaviour, respectively)

$$M_u = \frac{f_{R3} b h_{sp}^2}{6} = \frac{f_{Ftu} b h_{sp}^2}{2}$$

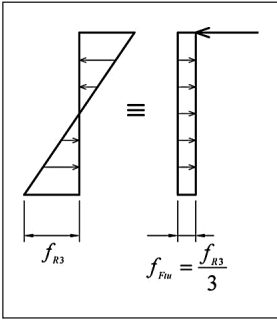


Figure 5.6-8: Simplified model adopted to compute the ultimate residual tensile strength in uniaxial tension f_{Ftu} by means of the residual nominal bending strength f_{R3}

The equation for f_{Ftu} and $w_u = CMOD_3$ is obtained, from the rotational equilibrium at ULS, when a stress block in tension along the section is taken into account, as shown in Figure 5.6-8.

Rigid-plastic model

The rigid-plastic model identifies an unique reference value, f_{Ftu} , based on the ultimate behaviour. Such a value is determined as:

$$f_{Ftu} = \frac{f_{R3}}{3} \quad (5.6-4)$$

Linear model

The linear model identifies two reference values, namely f_{Fts} and f_{Ftu} .

They have to be defined through residual values of flexural strength using the following equations:

$$f_{Fts} = 0.45 f_{R1} \quad (5.6-5)$$

$$f_{Ftu} = f_{Fts} - \frac{w_u}{CMOD_3} (f_{Fts} - 0.5 f_{R3} + 0.2 f_{R1}) \geq 0 \quad (5.6-6)$$

where w_u is the maximum crack opening accepted in structural design; its value depends on the ductility required.

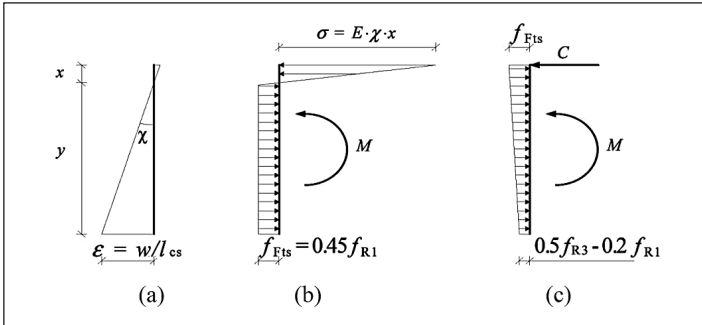


Figure 5.6-9: Stress diagrams for the determination of the residual tensile strength f_{Fts} (b) and f_{Ftu} (c) for the linear model, respectively

The limit value w_u applies particularly for design purposes.

The equation for f_{Ftu} and $w_u \neq CMOD_3$ is obtained by considering a linear constitutive law between points with abscissa $CMOD_1$ and $CMOD_3$, up to the point with abscissa w_u (Figure 5.6-10).

The stress value corresponding to the crack opening $CMOD_1$ is determined from equilibrium, with the assumption that the compressive stress distribution is linear (Figure 5.6-9b) and that the tensile behaviour is elasto-plastic until a crack opening displacement corresponding to the serviceability limit state ($CMOD_1$):

$$M(CMOD_1) = \frac{f_{R1} b h_{sp}^2}{6}$$

The variability introduced in the numerical coefficient introduced in Eq. (5.6-5) by the elastic modulus is here neglected and a common value is assumed.

The stress value corresponding to the crack opening $CMOD_3$ is determined from equilibrium, with the assumption that the compressive stress resultant is applied on the extrados chord (Figure 5.6-9c) and that the tensile behaviour is rigid-linear:

$$M(CMOD_3) = \frac{f_{R3} b h_{sp}^2}{6}$$

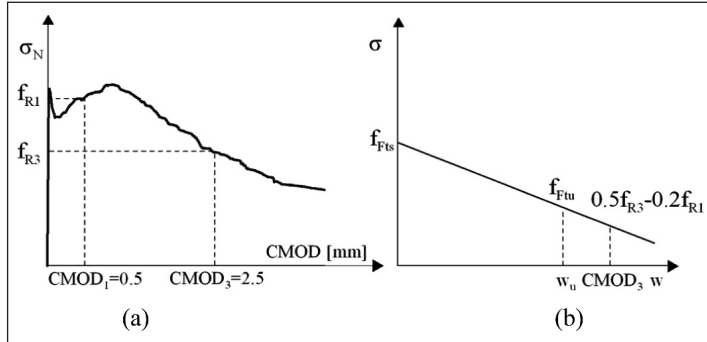


Figure 5.6-10: Typical results from a bending test on a softening material (a); linear post-cracking constitutive law (b)

For numerical analyses, more advanced constitutive laws are recommended, including first crack tensile strength.

When considering softening materials, the definition of the stress–strain law is based on the identification of the crack width and on the corresponding structural characteristic length, l_{cs} , of the structural element. Thus, the strain can be assumed equal to:

$$\varepsilon = w / l_{cs} \quad (5.6-7)$$

In elements with conventional reinforcement (rebars), l_{cs} , may be evaluated as:

$$l_{cs} = \min\{s_{rm}, y\} \quad (5.6-8)$$

where:

- s_{rm} is the mean distance between cracks;
- y is the distance between the neutral axis and the tensile side of the cross-section (Figure 5.6-9a), evaluated in the elastic cracked phase by neglecting the residual tensile strength of FRC, and for a load corresponding to the serviceability state of crack opening and crack spacing.

The ultimate tensile strength f_{Fu} in the linear model depends on the required ductility that is related to the allowed crack width. The ultimate crack width can be calculated as $w_u = l_{cs} \varepsilon_{Fu}$, by assuming ε_{Fu} is equal to 2‰ for variable strain distribution along the cross-section and 1‰ for constant tensile strain distribution along the cross-section. In any case, the maximum crack width may not exceed 2.5 mm.

In sections without traditional reinforcement under bending or under combined tensile-flexural and compressive-flexural forces with resulting force external to the section, $y = h$ is assumed. The same assumption can be taken for slabs.

When considering strain hardening materials, ε_{Fu} is equal to 2‰ for variable strain distribution along the cross-section and 1‰ for constant tensile strain distribution along the cross-section.

A material is considered as strain hardening when it shows a hardening behaviour in tension up to a $\varepsilon_{Fu} = 1\%$.

5.6.5 Stress–strain relationship

For the ULS, the constitutive laws as described in subsection 5.6.4 should be applied.

For softening materials at SLS (case (I)) the same constitutive relationship adopted for plain concrete in uniaxial tension is used

up to the peak strength f_{ct} . In the post-cracking stage, a bilinear relation applies (Figure 5.6-11a).

The post-peak propagation branch ($\overline{BC}$) is analytically described as:

$$\frac{\sigma - f_{ct}}{0,2f_{ct} - f_{ct}} = \frac{\varepsilon - \varepsilon_P}{\varepsilon_Q - \varepsilon_P}, \text{ for } \varepsilon_P \leq \varepsilon \leq \varepsilon_C \quad (5.6-9)$$

$$\text{with } \varepsilon_Q = \frac{G_F}{f_{ct} \cdot l_{cs}} + \left(\varepsilon_P - \frac{0,8f_{ct}}{E_c} \right) \quad (5.6-10)$$

where G_F represents the fracture energy of plain concrete, see Eq. (5.1-9).

Point A in the curves of Figure 5.6-11 (a), (b), (c) is defined in Figure 5.1-4.

For softening materials, the residual strength (fourth branch) is defined by two points corresponding to $(\varepsilon_{SLS}, f_{Fts})$ and $(\varepsilon_{ULS}, f_{Ftu})$ where:

$$\varepsilon_{SLS} = CMOD_1 / l_{cs} \quad (5.6-11)$$

$$\varepsilon_{ULS} = w_u / l_{cs} = \min(\varepsilon_{Fu}, 2.5 / l_{cs}) \quad (5.6-12)$$

The first and the second branch suggested in the pre-peak constitutive relationship and the post-peak crack propagation branch correspond to the behaviour of plain concrete until the intersection with the residual post-cracking behaviour which resumes fibre contribution.

When this condition does not apply, a new second branch is proposed, as shown in Figures 5.6-11b and 5.6-11c.

with $\varepsilon_{Fu} = 2\%$ for variable strain distribution along the cross-section and 1% for only tensile strain distribution along the cross-section – see subsection 5.6.4.

For materials characterized by a stable propagation up to ε_{SLS} with a tensile strength f_{Fts} larger than f_{ct} , two cases can be considered:

Case (II): the cracking process becomes stable up to the SLS strain and four branches again define the constitutive relationship. The first two branches remain those corresponding to plain concrete, while the third branch ($\overline{BD}$) is analytically described as:

$$\frac{\sigma - f_{ct}}{f_{Fts} - f_{ct}} = \frac{\varepsilon - \varepsilon_P}{\varepsilon_{SLS} - \varepsilon_P}, \text{ for } \varepsilon_P \leq \varepsilon \leq \varepsilon_{SLS}$$

Case (III): the cracking remains stable up to the SLS strain and three branches define the constitutive relationship. The second branch ($\overline{A'D}$) is defined as:

$$\frac{\sigma - \sigma_{A'}}{f_{Fts} - \sigma_{A'}} = \frac{\varepsilon - \varepsilon_{A'}}{\varepsilon_{SLS} - \varepsilon_{A'}}, \text{ for } \varepsilon_{A'} \leq \varepsilon \leq \varepsilon_{SLS}$$

where $\sigma_{A'}$ is on the elastic branch for a stress equal to $0.9 f_{Fts}$.

For both cases (II) and (III), the material can be softening ($\overline{DE}$) or hardening ($\overline{DE'}$) depending on the slope of the last branch.

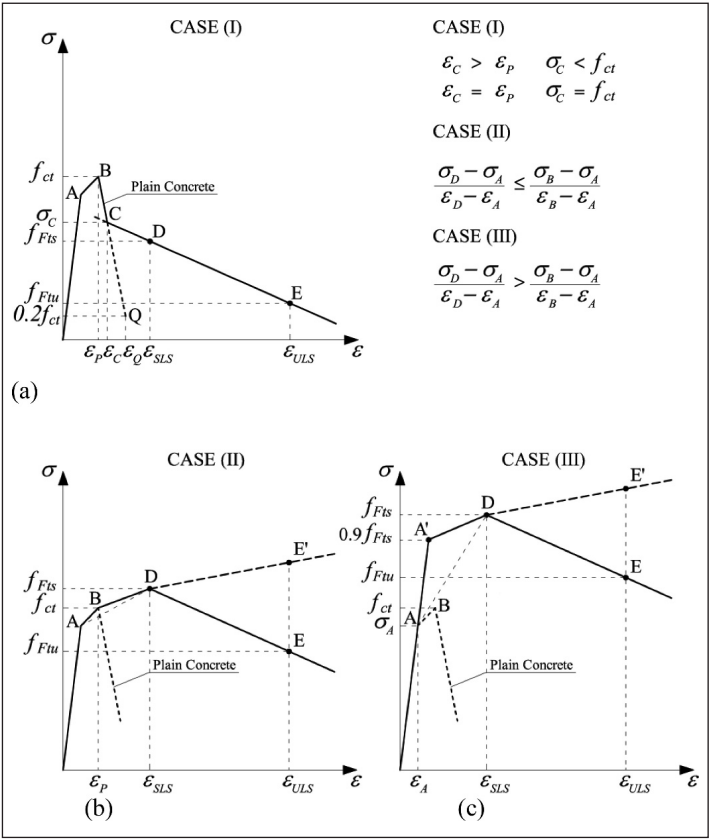


Figure 5.6-11: Stress–strain relations at SLS for softening (a) and softening or hardening (b, c) behaviour of FRC

5.6.6 Partial safety factors

A reduced safety factor $\gamma_F \geq 1.3$ may be adopted for improved control procedures.

Design values for the post-cracking strength parameter at ULS can be determined as (see Figure 5.6-7):

$$f_{Ftsd} = f_{Ftsk} / \gamma_F$$

and

$$f_{Ftud} = f_{Ftuk} / \gamma_F$$

The recommended values for the partial safety factors are given in Table 5.6-1:

Table 5.6-1: Partial safety factor

Material	Partial safety factors
FRC in compression	As plain concrete
FRC in tension (limit of linearity)	As plain concrete
FRC in tension (residual strength)	$\gamma_F = 1.5$

For serviceability limit states (SLS), the partial factors should be taken as 1.0

5.6.7 Orientation factor

The behaviour observed in the standard tests can deviate substantially (beneficial and non-beneficial) from the behaviour of the corresponding FRC in the structural element or structure. Thus the manufacturing method and the concrete consistency should be taken into account by the designer.

In general, an isotropic fibre distribution is assumed, so that the fibre orientation factor $K = 1.0$.

For favourable effects, an orientation factor $K < 1.0$ may be applied if experimentally verified.

For unfavourable effects, an orientation factor $K > 1.0$ must be experimentally determined and applied.

The values f_{Ftsd} and f_{Ftud} should then be modified to:

$$f_{Ftsd,mod} = f_{Ftsd}/K; f_{Ftud,mod} = f_{Ftud}/K$$

6 Interface characteristics

6.1 Bond of embedded steel reinforcement

Bond is the term used to denote the interaction and transfer of force between reinforcement and concrete. Bond influences performance of concrete structures in several ways. At the serviceability limit state, bond influences width and spacing of transverse cracks, tension stiffening and curvature. At the ultimate limit state, bond is responsible for strength of end anchorages and lapped joints of reinforcement, and influences rotation capacity of plastic hinge regions.

6.1.1 Local bond-slip relationship

6.1.1.1 Local bond stress-slip model, ribbed bars

Under well-defined conditions, it is possible to consider that there is an average “local bond” versus “local slip” relationship, for short anchorage lengths, statistically acceptable.

This section covers anchorage of ribbed reinforcing bars which satisfy the requirements for classification as “high bond” in accordance with requirements of section 5.2.

For monotonic loading the reference value τ_b of the bond stresses between concrete and reinforcing bar for pull-out and splitting failure can be calculated as a function of the relative displacement s parallel to the bar axis as follows (Figure 6.1-1):

$$\tau_b = \tau_{bmax} (s/s_1)^\alpha \quad \text{for } 0 \leq s \leq s_1 \quad (6.1-1)$$

$$\tau_b = \tau_{bmax} \quad \text{for } s_1 \leq s \leq s_2 \quad (6.1-2)$$

$$\tau_b = \tau_{bmax} - (\tau_{bmax} - \tau_{bf}) (s-s_2)/(s_3-s_2) \quad \text{for } s_2 \leq s \leq s_3 \quad (6.1-3)$$

$$\tau_b = \tau_{bf} \quad \text{for } s_3 < s \quad (6.1-4)$$

where the parameters are given in Table 6.1-1

Explicit consideration of the influence of bond on performance at the serviceability limit state and on rotation capacity is not normally required for members reinforced with conventional steel bars. The influence is instead represented by a modification of the tension stiffening from the surrounding concrete.

The bond stress-slip relationship depends on a considerable number of influencing factors including rib geometry (relative rib area), concrete strength, position and orientation of the bar during casting, state of stress, boundary conditions and concrete cover. The parameters given in Table 6.1-1 are valid for ribbed reinforcing steel with a relative rib area $f_r \geq f_{r,min}$ according to relevant international standards – see section 5.2.

The bond stress-slip curves for confined and unconfined concrete presented in Figure 6.1-1 can be considered applicable as an average formulation for a broad range of cases. Further reliability handling would be needed to derive design bond stress-slip curves.

The ascending part refers to the stage in which the ribs penetrate into the mortar matrix, characterized by local crushing and micro-cracking. A sustained plateau occurs only for confined concrete, during which advanced crushing and shearing off of the concrete between the ribs takes place. This level represents a residual bond capacity which is maintained only where a large concrete cover, dense transverse reinforcement or transverse compression is present to keep a certain degree of integrity intact. The descending branch refers to the reduction of bond resistance as concrete corbels between the ribs are sheared off. In the case of unconfined concrete splitting, failure occurs which is reflected by a sudden drop in bond stress before a constant residual level is reached. The peak value of bond strength in a splitting failure mode is denoted $\tau_{bu,split}$ – see Figure 6.1-1 and Eq. (6.1-5).

With regard to the development of bond stresses, the following considerations apply:

Reinforcement and concrete have the same strain ($\varepsilon_s = \varepsilon_c$) in those areas of the structure where the steel is in compression (outside eventual load introduction areas) and in those areas where the steel is in tension in uncracked parts of the structure.

In cracked cross-sections tension forces are transferred across the crack by the reinforcing steel. In general, the absolute displacements of the steel u_s and of the concrete u_c adjacent to a crack are different. Differences in displacements are similarly found along the transmission length l_{bpt} of pretensioned prestressed concrete members.

Due to the relative displacement $s = u_s - u_c$ bond stresses are generated between concrete and reinforcing steel or prestressing tendons. For $s < s_1$, the magnitude of these bond stresses depends predominantly on the slip s , but is influenced by the surface of the steel, concrete strength f_{cm} , the position of the reinforcing steel during concreting and by confinement from concrete cover, secondary reinforcement and the state of stress in the concrete around the bar. Between cracks or along the transmission length

l_{bpt} a part of the tension force of the reinforcing steel at the cracked section is transferred into the concrete by bond (tension stiffening effect). The local change in relative displacement is characterized by the strain difference $ds/dx = \varepsilon_s - \varepsilon_c$.

Depending on the selection of the coefficient α ($0 \leq \alpha \leq 1$) in Eq. (6.1-1) all usual forms of a bond stress–slip relationship can be modelled, starting from a bond characteristic with a constant stress ($\alpha = 0$) up to a bond stress–slip relationship with linear increasing bond stress ($\alpha = 1$).

The parameters from Table 6.1-1 to be applied are dependent primarily on the failure mode, pull-out or splitting. Further distinctions are made to account for differences in bond conditions within each failure mode.

The rationale underpinning the following rules is given in *fib* Bulletin “Bond and Anchorage of Reinforcement: Background to *fib* Model Code 2010”.

Table 6.1-1: Parameters defining the mean bond stress–slip relationship of ribbed bars (according to Eqs. (6.1-1)–(6.1-4))

	1	2	3	4	5	6
	Pull-out (PO)		Splitting (SP)			
	$\varepsilon_s < \varepsilon_{s,y}$		$\varepsilon_s < \varepsilon_{s,y}$			
	Good bond cond.	All other bond cond.	Good bond cond.		All other bond cond.	
			Unconfined	Stirrups	Unconfined	Stirrups
τ_{bmax}	$2.5\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$	$2.5\sqrt{f_{cm}}$	$2.5\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$
$\tau_{bu,split}$	—	—	$7.0 \cdot \left(\frac{f_{cm}}{25}\right)^{0.25}$	$8.0 \cdot \left(\frac{f_{cm}}{25}\right)^{0.25}$	$5.0 \cdot \left(\frac{f_{cm}}{25}\right)^{0.25}$	$5.5 \cdot \left(\frac{f_{cm}}{25}\right)^{0.25}$
s_1	1.0 mm	1.8 mm	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$
s_2	2.0 mm	3.6 mm	s_1	s_1	s_1	s_1
s_3	$c_{clear}^1)$	$c_{clear}^1)$	$1.2s_1$	$0.5c_{clear}^1)$	$1.2s_1$	$0.5c_{clear}^1)$
α	0.4	0.4	0.4	0.4	0.4	0.4
τ_{bf}	$0.40\tau_{max}$	$0.40\tau_{max}$	0	$0.4\tau_{bu,split}$	0	$0.4\tau_{bu,split}$

1) c_{clear} is the clear distance between ribs

The values in Table 6.1-1, columns 1 and 2 (pull-out failure) are valid for well-confined concrete (concrete cover $\geq 5\emptyset$, clear spacing between bars $\geq 10\emptyset$), or suitable confining reinforcement.

The values in Table 6.1-1, columns 3 to 6 (splitting failure) are derived from Eq. (6.1-5), which is derived from Eq. (6.1-19) by setting $l_b/\emptyset = 5$, assuming a uniform bond stress over this length and evaluating $\tau_{bu,split}$ for $\emptyset = 25$ mm, $c_{max}/c_{min} = 2.0$, $c_{min} = \emptyset$ and $K_{tr} = 0.02$ in the case where bars are confined by stirrups or $K_{tr} = 0$ where no stirrups are provided.

$$\tau_{bu,split} = \eta_2 6.5 \left(\frac{f_{cm}}{25}\right)^{0.25} \left(\frac{25}{\emptyset}\right)^{0.2} \left[\left(\frac{c_{min}}{\emptyset}\right)^{0.33} \left(\frac{c_{max}}{c_{min}}\right)^{0.1} + k_m K_{tr} \right] \quad (6.1-5)$$

where:

- η_2 = 1.0 for good bond conditions;
= 0.7 for all other bond conditions (see subsection 6.1.3.2 for definition of bond conditions);
- f_{cm} is the mean cylinder concrete compressive strength [N/mm²];
- $\emptyset$ is the diameter of the anchored bar considered [mm];
- $c_{min} = \min\{c_s/2, c_x, c_y\}$ (Figure 6.1-2);
- $c_{max} = \max\{c_s/2, c_x\}$ (Figure 6.1-2).

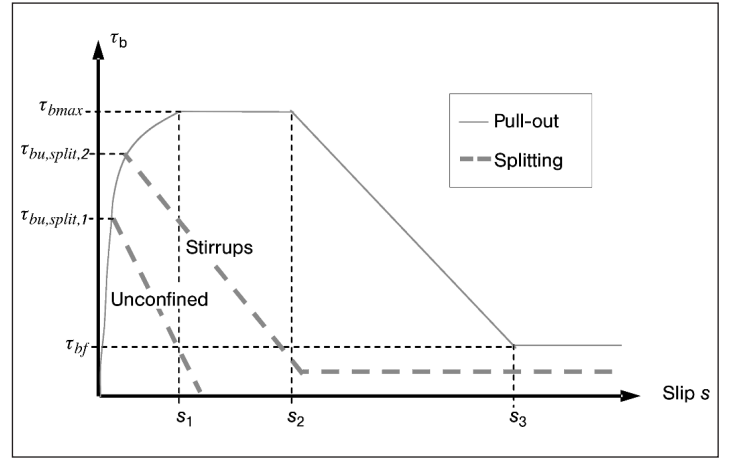


Figure 6.1-1: Analytical bond stress–slip relationship (monotonic loading). $\tau_{bu,split,1}$ and $\tau_{bu,split,2}$ denote peak local bond resistance in the absence ($K_{tr} = 0$) and presence, respectively, of confining stirrups

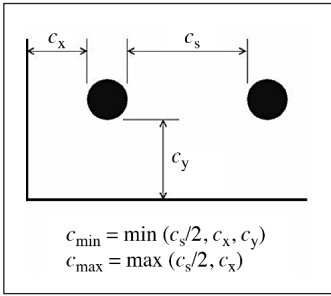


Figure 6.1-2: Notation for bar spacing and cover: straight bars

k_m represents the efficiency of confinement from transverse reinforcement, and has a value of 12 where bars are confined inside a bend of links passing round the bar of at least 90°. Where no confining reinforcement is provided between bars and the nearest face, $k_m = 0$, as shown in Figure 6.1-3.

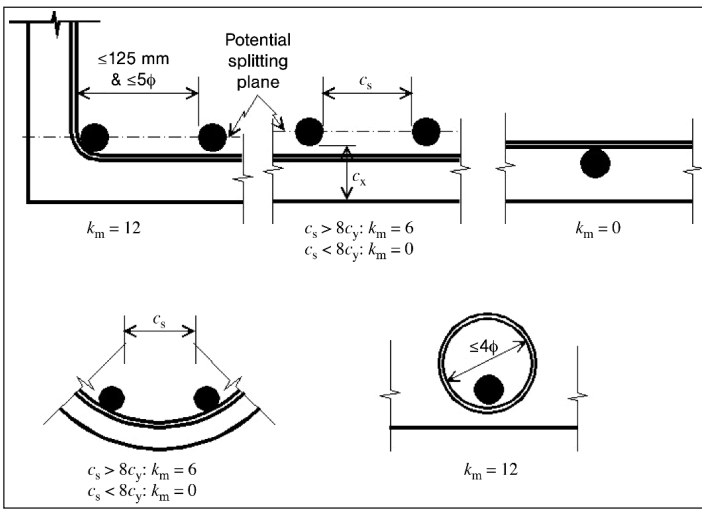


Figure 6.1-3: Confinement coefficients for transverse reinforcement

$$K_{tr} = n_t A_{st} / (n_b \bar{\sigma} s_t) \leq 0.05 \quad (6.1-6)$$

where:

n_t is the number of legs of confining reinforcement crossing a potential splitting failure surface at a section;

A_{st} is the cross-sectional area of one leg of a confining bar [mm²];

s_t is the longitudinal spacing of confining reinforcement [mm];

n_b is the number of anchored bars or pairs of lapped bars in the potential splitting surface.

6.1.1.2 Influence of transverse cracking

For those parts of the reinforcing bar which are at a distance $x \leq 2 \bar{\sigma}$ from a transverse crack, the bond stress τ is to be reduced by the factor λ where:

$$\lambda = 0.5 x / \bar{\sigma} \leq 1 \quad (6.1-7)$$

6.1.1.3 Influence of yielding, transverse stress and longitudinal cracking and cyclic loading

The bond stress–slip curve is considerably influenced by reinforcement yielding, by transverse pressure and cracking along the bar, and by cyclic, repeated or sustained loading.

The bond stress according to Eqs. (6.1-1)–(6.1-4) should be modified by the factors Ω_y , $\Omega_{p,tr}$ and Ω_{cr} and Ω_{cyc} in case of bar yielding, transverse pressure and cracking parallel to the bar axis and cyclic loading respectively:

$$\tau_{b,m} = \tau_{b0} \cdot \Omega_y \cdot \Omega_{p,tr} \cdot \Omega_{cr} \cdot \Omega_{cyc} \quad (6.1-8)$$

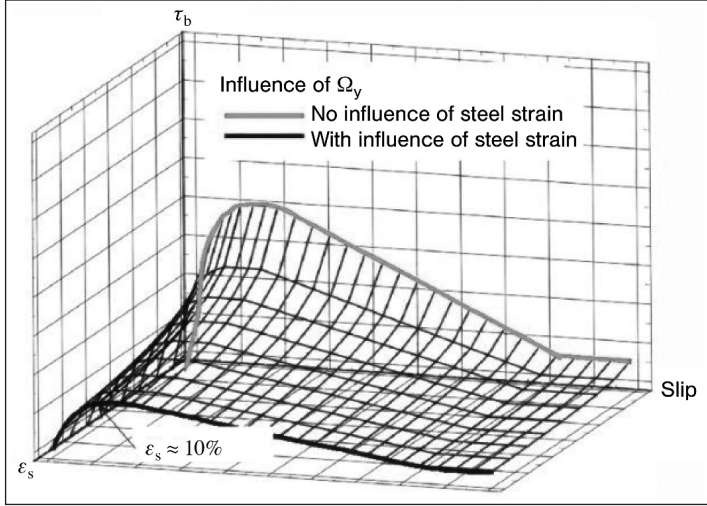


Figure 6.1-4: Influence of steel strains on local bond stress-slip relationship in case of pull-out failure
Bond resistance is decreased over the portion of an embedment length once reinforcement yield is reached, Figure 6.1-4.

Bond resistance is decreased in the presence of transverse tension and increased in the presence of compressive stresses perpendicular to the bar axis – Fig. 6.1-5 and Eq. (6.1-10a) and Eq. (6.1-10b).

A negative quantity for p_{tr} in Eq. (6.1-10) denotes a compressive stress.

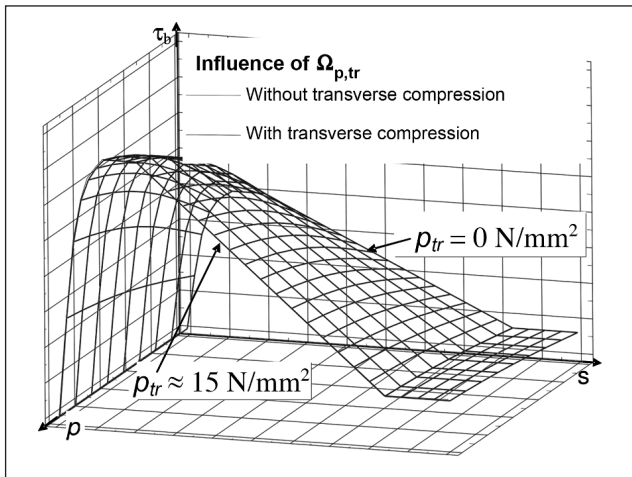


Figure 6.1-5: Influence of transverse pressure on local bond-stress slip relationship in case of pull-out failure

where:

$\tau_{b,m}$ is bond stress according to the modified bond stress-slip curve;

τ_b is bond stress according to the bond stress-slip curve defined by Eq. (6.1-1)–(6.1-4).

The modified bond stress τ_{bm} then replaces τ_b in Eqs. (6.1-1)–(6.1-4).

6.1.1.3.1 Influence of yielding of reinforcement

If yielding of reinforcement occurs along the embedment length, the corresponding reduction of the bond stress is given by the factor Ω_y as:

$$\Omega_y = 1.0 \quad \text{for } \varepsilon_s \leq \varepsilon_{sy} \quad (6.1-9a)$$

$$\Omega_y = 1.0 - 0.85 \cdot (1 - e^{-5a^b}) \quad \text{for } \varepsilon_{sy} < \varepsilon_s \leq \varepsilon_{su} \quad (6.1-9b)$$

with

$$a = \frac{\varepsilon_s - \varepsilon_{sy}}{\varepsilon_{su} - \varepsilon_{sy}}; \quad b = \left[2 - \frac{f_{tm}}{f_{ym}} \right]^2 \quad (6.1-9c)$$

f_{tm} and f_{ym} are mean values of tensile and yield strength respectively.

6.1.1.3.2 Influence of transverse pressure

If transverse pressure perpendicular to the bar axis is present, the bond stress slip curve for pull-out failure should be modified by the factor $\Omega_{p,tr}$ as:

$$\Omega_{p,tr} = 1.0 - \frac{0.3p_{tr}}{f_{cm}} \quad \text{for } f_{ctm} \geq p_{tr} \geq 0 \text{ MPa (tension)} \quad (6.1-10a)$$

$$\Omega_{p,tr} = 1.0 - \tanh \left[0.2 \frac{p_{tr}}{0.1f_{cm}} \right] \quad \text{for } p_{tr} \leq 0 \text{ MPa (compr.)} \quad (6.1-10b)$$

where:

p_{tr} is the mean stress in the concrete (orthogonal to the bar axis) averaged over a volume around the bar with a diameter of $3 \varnothing$.

A transverse tensile stress equal to the tensile strength of the concrete could split the concrete cover and leave zero bond resistance were there is no reinforcement crossing the splitting crack. Eq. (6.1-11) represents behaviour where reinforcement crosses potential splitting cracks and is not stressed above its yield strength.

Eq. (6.1-11) may be considered valid for cracks up to 0.5 mm wide.

Reversed cyclic loading reduces the bond strength compared to monotonic loading. This is shown in Figure 6.1-6. This figure explains also the calculation of the dissipated energy during monotonic and cyclic loading (area under the bond stress–slip curve). In Figure 6.1-7 the factor Ω_{cyc} is plotted as a function of the ratio Λ_{cyc}/Λ_0 .

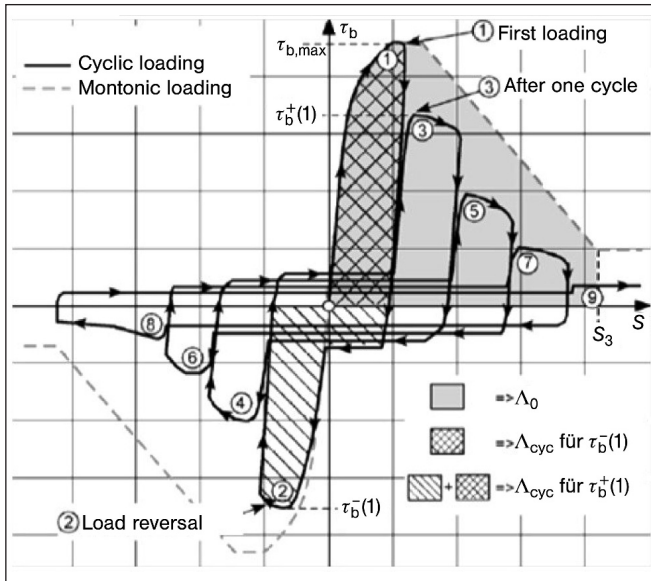


Figure 6.1-6: Bond stress–slip relationship and definition of the dissipated energy under monotonic and cyclic loading

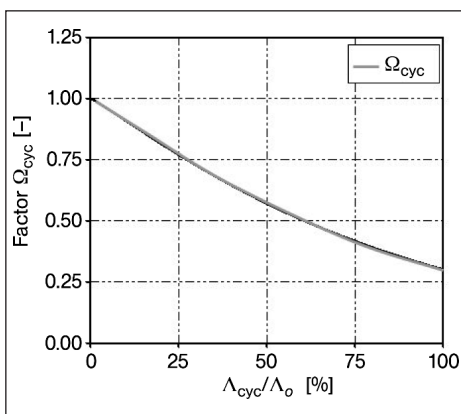


Figure 6.1-7: Factor Ω_{cyc} as a function of the ratio Λ_{cyc}/Λ_0

Bond slip increases with time under a constant sustained load.

The terms “repeated” or “fatigue loading” are used here to imply loading in which the bar stress does not change sign and the bond stress does not exceed $\tau_{b,max}$ under short term monotonic loading. Repeated loading produces a progressive increase in slip between bar and concrete which may lead to failure at cyclic bond stress

6.1.1.3.3 Influence of longitudinal cracking

If cracks parallel to the bar axis are present, the bond stress slip curve for pull-out failure should be modified by the factor Ω_{cr} :

$\Omega_{cr} = 1.0$ where concrete is uncracked parallel to the bar axis

$\Omega_{cr} = 1 - 1.2 \cdot w_{cr}$ where the concrete is cracked parallel to the bar axis and w_{cr} is the crack width in mm. (6.1-11)

6.1.1.3.4 Influence of reversed cyclic loading

After the bond stress has reached $\tau_{b,max}$ the ordinates (i.e. the stress values) of the bond stress slip curve for pull-out failure should be modified by the factor Ω_{cyc} on subsequent load cycles as:

$$\Omega_{cyc} = e^{\left(-1.2 \left[\Lambda_{cyc}/\Lambda_0\right]^{1.1}\right)} \quad (6.1-12)$$

Λ_{cyc} = dissipated energy during cyclic loading (Figure 6.1-7);

Λ_0 = dissipated energy during monotonic loading (Figure 6.1-7).

6.1.1.4 Influence of creep and fatigue loading

Creep reduces the slope of the ascending part of the bond stress–slip relationship of Figure 6.1-1.

The creep displacements may be described by isochrone curves, as shown in Figure 6.1-8. The slip $s_{n,t}$ due to permanent or repeated loading can be calculated as:

$$s_{n,t} = s(1 + k_{n,t}) \quad (6.1-13)$$

levels lower than the ultimate bond strength under monotonic loading. The rate of increase is influenced principally by the number and frequency of load cycles and the load level.

Provided that bond failure does not occur under repeated loading, then preceding repeated loads do not adversely influence either bond–slip behaviour on subsequent monotonic loading to failure or bond strength at ultimate load.

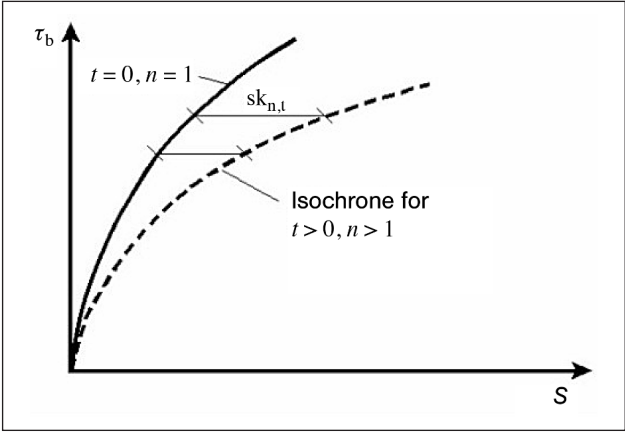


Figure 6.1-8: Creep effects on the bond stress–slip curve

For further information on bond under cyclic loading, see chapter 3, “Bond under repeated loading”, of *fib* Bulletin 10 “Bond Models”.

The slope S_s of the unloading branch is taken as the secant modulus at a slip of 0.01 mm.

Unloading from a maximum slip of less than 0.01 mm is taken back to the origin, that is to zero slip.

where the displacement factor $k_{nt} = k_t$ for a permanent load may be calculated as:

$$k_t = (1 + 10t)^{0.080} - 1 \tag{6.1-14}$$

where t is the load duration (hours)

For repeated constant amplitude loading the displacement factor $k_{n,t} = k_n$ may be determined as:

$$k_n = (1 + n)^{0.1070} - 1 \tag{6.1-15}$$

where n is the number of load cycles.

The validity of this relationship is restricted to the ascending branch of the bond–slip relationship.

6.1.1.5 Unloading branch

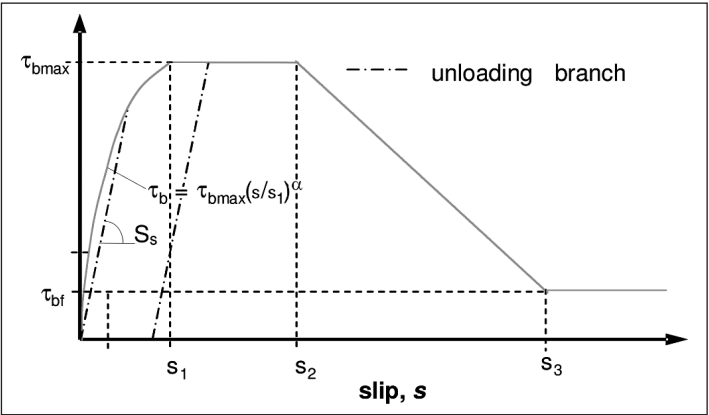


Figure 6.1-9: Unloading branch of the τ_b – s relationship

The unloading branch of the bond stress–slip relationship is linear and valid for the increasing and horizontal part of the diagram. The slope S_s (see Figure 6.1-9) is independent of the slip value s , and is given by Eq. (6.1-16).

$$S_s = 6.0 \tau_{bmax} \text{ [N/mm}^3\text{]} \tag{6.1-16}$$

6.1.1.6 Plain (non-ribbed) surface bars

The parameters given in Table 6.1-2 are valid for plain (i.e. non-ribbed) reinforcing steel, depending on the main influencing factors: roughness of the bar surface, bond conditions and concrete strength. They are valid for confined and unconfined concrete. They are applicable only in loading states for which the concrete is not subjected to lateral tension, in the elastic range of the reinforcement and for those parts of the reinforcing bar with the distance $x > 2\varnothing$ from a transverse crack. For those parts of the reinforcing bars with a distance $x \leq 2\varnothing$ from a transverse crack, the

Table 6.1-2: Parameters defining the bond stress–slip relationship of plain surface bars (according to Eqs. (6.1-1)–(6.1-4))

	Cold-drawn wire		Hot-rolled bars	
	Good bond conditions	All other bond cond.	Good bond conditions	All other bond cond.
$s_1 = s_2 = s_3$	0.01 mm	0.01 mm	0.1 mm	0.1 mm
α	0.5	0.5	0.5	0.5
$\tau_{bmax} = \tau_{bf}$	$0.1\sqrt{f_{cm}}$	$0.05\sqrt{f_{cm}}$	$0.3\sqrt{f_{cm}}$	$0.15\sqrt{f_{cm}}$

The parameters given in Table 6.1-2 are mean values.

The scatter in measured slip is considerable, especially for small values of slip. For a given value of the slip, the coefficient of variation of the bond stresses may amount to around 30%. The scatter is due to the use of different test specimens and the resulting differences in the state of stress in the concrete surrounding the reinforcing bar, to the different measuring techniques and to the different loading and deformation velocities. The heterogeneity of the concrete and the geometry of the reinforcing bars (relative rib area, diameters) also have a significant influence on the τ - s relationship.

In the absence of test data, the β coefficient for epoxy coated ribbed bars should be taken as 50% of that for a “normal” ribbed bar.

The influence of bond on crack control and tension stiffening may be estimated from the secant modulus of local bond-slip behaviour measured in accordance with RILEM pull-out test procedures at a free end slip of 0.01 mm.

Only one of the additional components may be considered to contribute to anchorage at a particular location.

Note that the resistance of two components acting in combination will generally be less than the sum of their resistances determined individually. The rules provided here make allowance for interaction effects.

Anchorage at support reactions and under concentrated loads may be considered to be subject to transverse compression.

As straight bars are less effective than links in restraining splitting, a higher bar spacing is required.

Minimum transverse reinforcement at laps and anchorages is to be provided to restrain a brittle mode of failure. Reinforcement provided for other purposes (e.g. for shear resistance) may be included in computation of ΣA_{st} .

bond stress τ_b and the slip s are to be reduced by the factor λ according to Eq. (6.1-7).

6.1.2 Influence on serviceability

Rules for serviceability behaviour have been derived for ribbed bars with a relative rib area f_r within the range 0.05–0.07. More highly ribbed bars may be capable of developing higher bond stiffness, while coatings may cause a reduction. Coefficients β representing the influence of bond in Eqs. (7.6-5) and (7.6-16) in section 7.6 may be modified in proportion to measured bond stiffness.

6.1.3 Anchorage and lapped joints of reinforcement

Reinforcement may be anchored by bond alone or by a combination of bond along the straight portion of a bar together with a resistance provided by one of:

- a hook or bend (but only for bars in tension);
- welded transverse bar(s);
- a head welded to the end of the bar;
- bearing of the end of the bar on the concrete (only for bars in compression).

Lapped joints may alternatively be made by welding or by mechanical couplers.

6.1.3.1 Minimum detailing requirements

Minimum anchorage lengths are specified in subsection 6.1.3.4, in Eq. (6.1-26). Minimum lap lengths are specified in subsection 6.1.3.7, in Eq. (6.1-29).

Minimum cover is equal to one bar diameter. For anchorage of a bundle, minimum cover is the equivalent diameter of the bundle, Eq. (6.1-32).

Clear spacing $c_{s,min}$ between anchored bars or bars belonging to different lapped joints confined by links must be at least two times bar diameter except in zones subject to transverse compression, where the minimum clear spacing may be reduced to one bar diameter. In walls and slabs where confining reinforcement comprises straight bars, clear spacing $c_{s,min}$ must be at least three times bar diameter, except in zones subject to transverse compression, where the minimum clear spacing may be reduced to one bar diameter.

Unless the bond zone is subjected to transverse compression, a minimum quantity of transverse reinforcement is to be provided within the anchorage or lap length.

In beams and columns, no lapped or anchored bar should be further than the lesser of 5ϕ or 125 mm from a leg of a link which lies perpendicular to the plane passing through the bar axes, Figure 6.1-10.

Where the diameter ϕ of the anchored bars is less than 20 mm and concrete is of Grade C60 or below, transverse reinforcement or links provided for other reasons may be assumed sufficient to

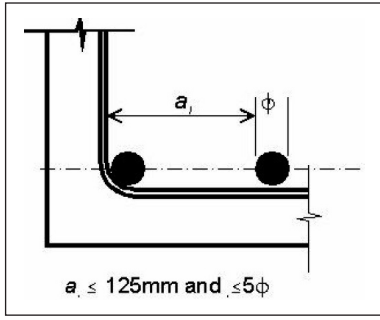


Figure 6.1-10: Confinement of beam and column bars by links

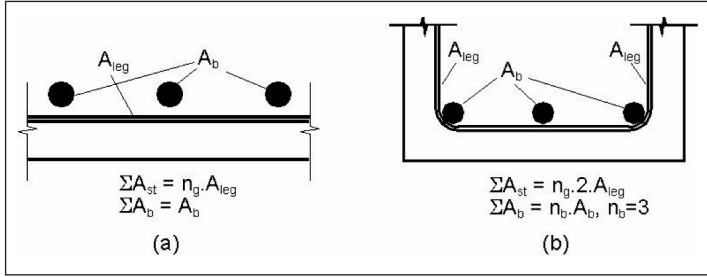


Figure 6.1-11: Calculation of minimum reinforcement: examples

For ribbed bars in a “good” casting position, reinforcement stress f_{stm} is derived from the semi-empirical expression of Eq. (6.1-19), which has been calibrated using results from over 800 tests.

$$f_{stm} = 54 \left(\frac{f_{cm}}{25} \right)^{0.25} \left(\frac{25}{\emptyset} \right)^{0.2} \left(\frac{l_b}{\emptyset} \right)^{0.55} \left[\left(\frac{c_{min}}{\emptyset} \right)^{0.25} \left(\frac{c_{max}}{c_{min}} \right)^{0.1} + k_m K_{tr} \right] \quad (6.1-19)$$

with $f_{stm} \leq f_y$, and $f_{stm} \leq 2.5\sqrt{f_c} 4(l_b / \emptyset)$ or $f_{stm} \leq 1.25\sqrt{f_c} 4(l_b / \emptyset)$

in good and poor bond conditions respectively.

Eq. (6.1-19) is valid for $15 \text{ MPa} < f_{cm} < 110 \text{ MPa}$, $0.5 < c_{min}/\emptyset < 3.5$, $1.0 < c_{max}/c_{min} < 5.0$ and $K_{tr} \leq 0.05$. The parameters are defined in Eq. (6.1-5).

f_{stm} is a mean value, and may not be directly used in design.

Eq. (6.1-19) has been derived from tests on bars with a relative rib area f_r within the range 0.05–0.14. Other rib patterns may be capable of developing higher bond strengths. The appropriate η_1 value should be calibrated from tests.

The basic design bond strength expression Eq. (6.1-20) has been derived from Eq. (6.1-19) as follows:

- A characteristic strength expression is obtained by altering the lead coefficient of 54 in the mean strength expression of Eq. (6.1-19) to 41 through analysis of the statistical accuracy of the expression.
- Eq. (6.1-19) is rearranged to allow bond length l_b to develop design strength of reinforcement $f_{yd} = f_{yk}/\gamma_c$ to be determined, with f_{yk} taken as 500 MPa.
- The basic bond strength $f_{bd,0}$ is then obtained by setting the part of Eq. (6.1-19) in square brackets to a value of 1.0 and dividing bar force $f_{yd} \cdot A_s$ by $\pi \emptyset l_b$, the nominal bar surface over which f_{yd} is developed.

satisfy minimum requirements for confining reinforcement without further justification.

In other circumstances, Eq. (6.1-17) should be satisfied.

$$\sum A_{st} \geq \alpha_t \cdot \alpha_1 \cdot \sum A_s \quad (6.1-17)$$

where for confinement by straight bars, $\sum A_s$ is the area of a single bar, (Figure 6.1-11a), or $\sum A_s$ is the total cross-sectional area of all bars lapped or anchored at the section where confinement is provided by links, (Figure 6.1-11b).

$$\sum A_{st} = n_g \cdot n_t \cdot A_{st} \quad (6.1-18)$$

n_g is the number of items of confining reinforcement within the bond length;

$n_t = 1$ for straight transverse reinforcement (Figure 6.1-11a), or n_t = the number of legs of links crossing a potential splitting failure surface at a section (Figure 6.1-11b);

A_{st} is the cross-sectional area of one leg of a confining bar;

$\alpha_t = 0.5$ for bars up to and including size 25;

$\alpha_t = 1.0$ for bars of size 50;

$\alpha_t = 0$ for distribution reinforcement in walls and slabs.

Linear interpolation may be used to determine α_t for intermediate sizes.

$\alpha_1 = A_{s,cal}/A_{s,ef}$ for anchorage or lap-splice zones subject to transverse compression, $A_{s,cal}$ is the calculated area of reinforcement required by the design and $A_{s,ef}$ is the area of reinforcement provided;

$\alpha_1 = 1.0$ in other circumstances.

6.1.3.2 Basic bond strength

Bond strength $f_{bd,0}$ is considered as an average stress on the nominal surface of a straight length of bar over the bond length l_b . The basic bond strength $f_{bd,0}$ is:

$$f_{bd,0} = \eta_1 \eta_2 \eta_3 \eta_4 (f_{ck}/25)^{0.5}/\gamma_c \quad (6.1-20)$$

where :

η_1 is a coefficient taken as 1.75 for ribbed bars (including galvanized and stainless reinforcement), 1.4 for fusion bonded epoxy coated ribbed bars;

η_2 represents the casting position of the bar during concreting:

$\eta_2 = 1.0$ when good bond conditions are obtained, as for:

- all bars with an inclination of 45–90° to the horizontal during concreting, and
- all bars with an inclination less than 45° to the horizontal which are up to 250 mm from the bottom or at least 300 mm from the top of the concrete layer during concreting (but see also “special circumstances” section later);

$\eta_2 = 0.7$ for all other cases where ribbed bars are used;

η_3 represents the bar diameter:

$\eta_3 = 1.0$ for $\emptyset \leq 25 \text{ mm}$;

$\eta_3 = (25/\emptyset)^{0.3}$ for $\emptyset > 25 \text{ mm}$ ($\emptyset$ in mm);

η_4 represents the characteristic strength of steel reinforcement being anchored or lapped;

$\eta_4 = 1.0$ for $f_{yk} = 500 \text{ MPa}$;

$\eta_4 = 1.2$ for $f_{yk} = 400 \text{ MPa}$;

$\eta_4 = 0.85$ for $f_{yk} = 600 \text{ MPa}$;

$\eta_4 = 0.75$ for $f_{yk} = 700 \text{ MPa}$;

$\eta_4 = 0.68$ for $f_{yk} = 800 \text{ MPa}$.

Intermediate values may be obtained by interpolation.

The partial safety coefficient for bond γ_c is taken as 1.5

- d) Values for cover and confining reinforcement corresponding to minimum detailing requirements are inserted, and indices and coefficients rounded to more convenient values.

More detailed background information may be found in a forthcoming *fib* Bulletin, “Bond and Anchorage of Reinforcement: Background to *fib* Model Code 2010”.

Bond of bars that are not in a “good” casting position may be impaired by consolidation of the fluid concrete under the bar. Greater reductions in bond strength tend to be measured in deeper pours. Measures to minimize plastic settlement cracking will also be effective in minimizing the reduction in bond where the casting position is not classified as “good”. The coefficient proposed for η_2 lies towards the conservative end of the range of values measured in laboratory tests.

Confinement from cover, transverse reinforcement and transverse pressure in excess of that specified in 6.1.3.1 has a beneficial influence on bond strength. Conversely, lower cover, less confinement from secondary reinforcement and transverse tension reduce bond strength.

Confinement from transverse pressure initially has a strongly beneficial influence on bond strength where it restrains a splitting failure mode. Once confinement is sufficient to restrain splitting, the rate of increase reduces. It will usually be beneficial to include the influence of transverse pressure when considering anchorage of reinforcement in deep beams and corbels.

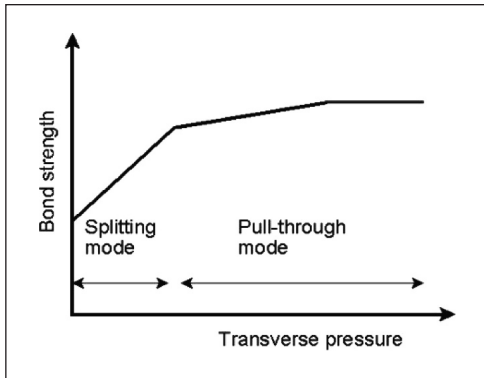


Figure 6.1-12: Influence of transverse pressure

Tension generated by the anchorage of the bar itself should not be included as “transverse tension” here.

Cover parameters c_{max} and c_{min} are shown in Figure 6.1-2. Parameter $c_{s,min}$ is specified in subsection 6.1.3.1.

6.1.3.3 Design bond strength

The design ultimate bond strength f_{bd} of ribbed bars may be modified from the basic value where concrete cover, bar spacing or transverse reinforcement differ from their respective minima, as stated in 6.1.3.1, or where the bar is subjected to transverse compression.

$$f_{bd} = (\alpha_2 + \alpha_3) f_{bd,0} - 2p_{tr}/\gamma_c < 2.5 f_{bd,0} - 0.4p_{tr}/\gamma_c < 1.5(\sqrt{f_{ck}})/\gamma_c \quad (6.1-21)$$

where:

α_2 and α_3 represent the influence of passive confinement from cover (α_2) and from transverse reinforcement (α_3). Provided minimum detailing provisions in 6.1.3.1 are satisfied, α_2 may conservatively be taken as 1.0 and α_3 may be conservatively taken as 0.

p_{tr} is the mean compression stress perpendicular to the potential splitting failure surface at the ultimate limit state; where transverse compression perpendicular to the bar axis acts over a portion of the bond length, bond strength may be increased over that portion. p_{tr} is negative when transverse stress is compressive.

Transverse tension

Transverse tensile stress reduces bond strength. Good detailing practice will provide transverse reinforcement to resist tension perpendicular to potential splitting failure planes. No reduction needs to be considered, however, provided that additional transverse reinforcement is applied to resist the applied transverse force. Otherwise the reduction may conservatively be taken as linearly proportional to p_{tr}/f_{ct} .

Passive confinement from cover: straight reinforcing bars

$$\text{Ribbed bars: } \alpha_2 = (c_{min}/\emptyset)^{0.5} \cdot (c_{max}/c_{min})^{0.15} \quad (6.1-22a)$$

$$\text{Epoxy coated bars: } \alpha_2 = (c_{min}/\emptyset)^{0.8} \cdot (c_{max}/c_{min})^{0.15} \quad (6.1-22b)$$

$$0.5 \leq c_{min}/\emptyset \leq 3.5, 1 \leq c_{max}/c_{min} \leq 5$$

Passive confinement from transverse reinforcement

$$\alpha_3 = k_d \cdot (K_{tr} - \alpha_t/50) \geq 0.0, K_{tr} \leq 0.05 \quad (6.1-23)$$

Reinforcement provided for other purposes (e.g. for shear resistance) may be included in computation of K_{tr} .

The factor $\alpha_t/50$ in Eq. (6.1-23) approximates the minimum transverse reinforcement according to 6.1.3.1, and results in a conservative value for α_3 . α_t is defined in 6.1.3.1.

At least one item of transverse reinforcement should be positioned within a lap length no further than 50 mm from the end of the bar – Figure 6.1-13. This requirement should be observed even if α_3 is conservatively taken as 1.0. Other items of transverse reinforcement should be spaced evenly throughout the lap length.

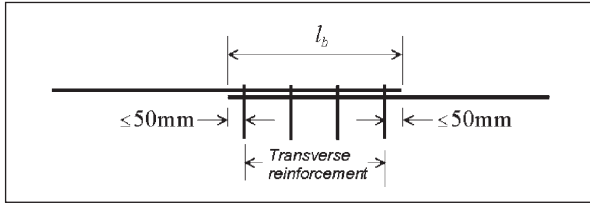


Figure 6.1-13: Location of links near ends of lap

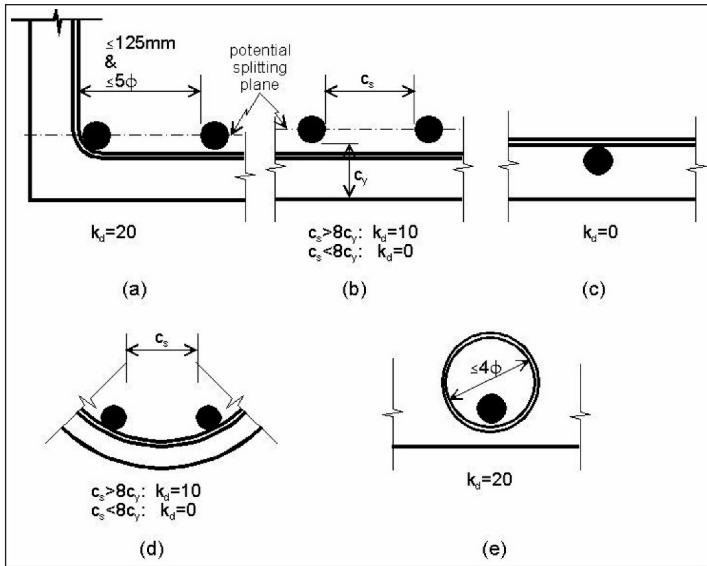


Figure 6.1-14: Transverse reinforcement factor k_d

Bond length l_b is measured to the end of a straight bar or to the outside of a hook or bend.

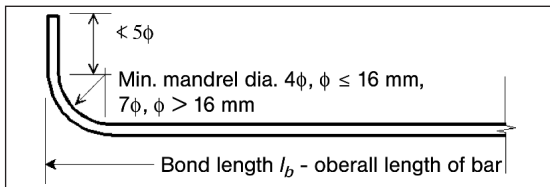


Figure 6.1-15: Bond length, bar terminating in hook or bend

where:

- $K_{tr} = n_t A_{st} / (n_b \emptyset s_t)$ is the density of transverse reinforcement, relative to the anchored or lapped bars;
- n_t is the number of legs of confining reinforcement crossing a potential splitting failure surface at a section;
- A_{st} is the cross-sectional area of one leg of a confining bar [mm²];
- s_t is the longitudinal spacing of confining reinforcement [mm];
- n_b is the number of anchored bars or pairs of lapped bars in the potential splitting failure surface;
- $\emptyset$ is the diameter of the anchored bar or of the smaller of a pair of lapped bars (mm).

k_d is an effectiveness factor dependent on the reinforcement detail, Figure 6.1-14. k_d factors in Figure 6.1-14 are derived from k_m factors in Figure 6.1-3, taking account of the non-linear relationship between anchorage or lap length and the stress developed in the bar.

$k_d = 20$ where the legs of a link are perpendicular to the splitting plane, provided no anchored bar or pair of lapped bars are further than either $5\emptyset$ or 125 mm from where the leg crosses the splitting plane, Figure 6.1-14a;

$k_d = 20$ where an individual bar or pair of lapped bars are wholly confined within a helix of internal diameter not exceeding 4 times the diameter of the lapped or anchored bar – Figure 6.1-14e;

$k_d = 10$ where bars are confined by straight bars or helix within the cover thickness, provided that the clear spacing between the main bars is at least 8 times the cover – Figures 6.1-14b and 6.1-14d;

$k_d = 0$ in other circumstances.

6.1.3.4 Design anchorage length

The stress in the reinforcement to be anchored by bond over the distance l_b , Figure 6.1-15, is:

$$\sigma_{sd} = \alpha_1 f_{yd} - (F_h / A_s) \quad (6.1-24)$$

where:

F_h is the force developed by the other measures listed in 6.1.3.

$F_h = 0$ in the case of straight tension bars;

A_s is the cross-sectional area of the bar considered;

α_1 is defined in 6.1.3.1.

The design anchorage length l_b may be calculated from:

$$l_b = \frac{\emptyset \sigma_{sd}}{4 f_{bd}} \geq l_{b,min} \quad (6.1-25)$$

Minimum anchorage length

$$l_{b,min} > \max\{0.3 \emptyset f_{yd} / (4 f_{bd}); 10 \emptyset; 100 \text{ mm}\} \quad (6.1-26)$$

With the exception of compression bars of columns or walls in footings or equivalent locations where cover perpendicular to the bar axis is very high, bends and hooks are not permitted for compression reinforcement. The exceptions are in footings and in exterior beam/column joints where the bend is toward the inside of the joint, the end cover parallel to the bar axis is at least 3.5ϕ and dense transverse reinforcement is provided in that cover over the entire length of the bend or hook.

Eq. (6.1-27) is based on the standard bend radius. For larger bend radii, see subsection 7.13.2.

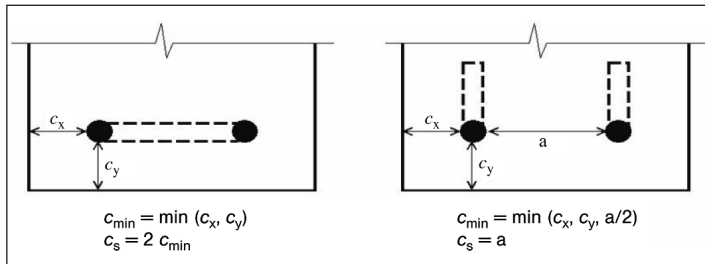


Figure 6.1-16: Notation for bar spacing and cover: hooks and bends

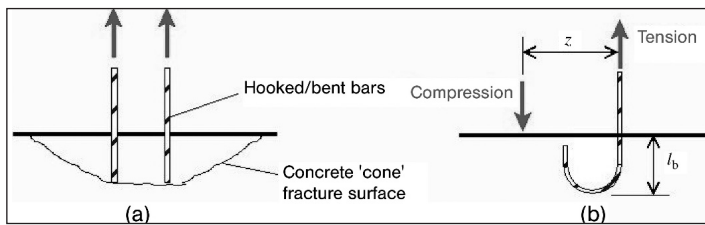


Figure 6.1-17: Cone failure hooked bars

The concrete cone failure mode may be modified when the tension force in the hooked or bent bar forms part of a moment couple (Figure 6.1-17b).

For more details reference should be made to information on cast-in anchors in the *fib* No. 58. “Design of anchorages in concrete. Guide to good practice.”

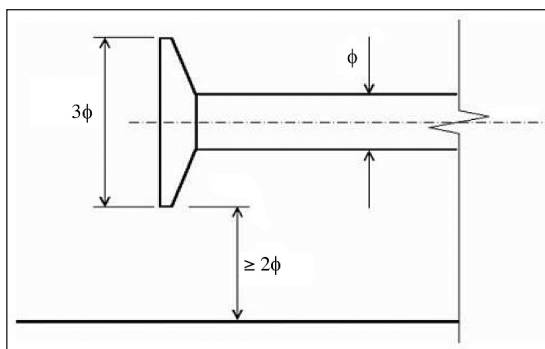


Figure 6.1-18: Condition (a), where the full yield capacity of the bar must be developed at the head

While it is evident that the strength of an anchorage comprising a straight portion of bar plus a welded head is less than the sum of the strengths of the two components acting independently, as yet no consensus model for condition (b) is available.

6.1.3.5 Contribution of hooks and bends

End bends or hooks contribute to transfer of force for bars in tension. The value of F_h in Eq. (6.1-24) for a standard bend or hook may be taken as:

$$F_h = 60 f_{bd} A_s \quad (6.1-27)$$

In the calculation of f_{bd} in Eq. (6.1-21):

α_2 is to be calculated using dimensions as shown in Figure 6.1-16;

α_3 is to be determined from transverse reinforcement perpendicular to the plane of the hook or bend, Eq. (6.1-21);

$p_{tr} = 0$ where the transverse compression acts parallel to the plane of the hook – Eq. (6.1-21).

Where hooked bars are closely spaced and anchor a predominantly tensile force, the possibility of a concrete cone type failure mode must be considered (Figure 6.1-17a).

6.1.3.6 Headed reinforcement

Two conditions may be considered:

- where the full yield capacity of the bar must be developed at the head, for example in double headed studs for shear or as wall ties, and concrete is uncracked perpendicular to the axis of the bar.
- other circumstances in which the reinforcement is anchored by a combination of anchorage by the head and bond along the length of the bar.

Condition (a) may be satisfied by a stud with head diameter 3 times that of the bar, where the minimum cover to the side of the head is not less than 2 bar diameters, spacing between bar centres is not less than 6 bar diameters, $f_{cd} > f_{yd}/24$, and concrete is uncracked perpendicular to the axis of the bar.

Option (ii) for condition (b) represents a conservative approach.

Lapped joints should be located away from regions of high stress whenever possible. If this is not possible, particular attention should be paid to ensuring robustness of the joint by confining reinforcement.

Recent research demonstrates the α_6 factor provided in MC90 for the proportion of bars lapped at a section to be invalid.

Where bars are lapped in regions of low stress – for example in the vicinity of points of contraflexure in continuous beams – the lap will not be required to develop yield under normal loading conditions, although it might be required to do so under accidental loadings for which a lower partial safety factor would be appropriate. The factor α_4 is intended to allow for lower partial safety factors in such circumstances.

End bearing should only be considered to contribute to strength of laps and anchorages of bars in compression where the end of the bar is no closer than 3.5ϕ (measured parallel to the bar axis) from an unsupported face.

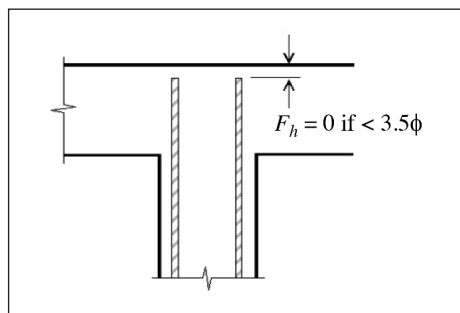


Figure 6.1-19: Minimum end cover to compression bars

Note: The minimum end cover is derived from source expressions and not from the design expressions given in this document. See forthcoming *fib* Bulletin “Bond and Anchorage of Reinforcement: Background to *fib* Model Code 2010”.

For condition (b), anchorage capacity may be determined in one of three ways:

- i) the headed bar may be treated as if it were a bar terminated by a hook or bend (subsection 6.1.3.5) provided the net projected area of the head is equal to that of a standard bend and the weld between head and bar is able to develop the full yield strength of the bar;
- ii) as the capacity of the head alone with no contribution from bond along the straight portion of the bar according to subsection 7.2.3.1.7;
- iii) anchorage capacity may be determined by test.

In all cases the head must have sufficient embedment beyond the most highly stressed point of the bar to preclude a premature concrete cone type failure.

6.1.3.7 Laps of bars in tension

The design lap length may be calculated from:

$$l_b = \alpha_4 \frac{\sigma_{yd}}{4 f_{bd}} \geq l_{b,min} \quad (6.1-28)$$

where:

α_4 may be taken as 0.7 where the calculated stress in reinforcement at the ultimate limit state throughout the lap length does not exceed 50% of the characteristic strength of the reinforcement, or no more than 34% of bars are lapped at the section, otherwise $\alpha_4 = 1.0$.

Minimum lap length

$$l_{b,min} > \max\left\{0.7 \frac{\sigma_{yd}}{4 f_{bd}}; 15\phi; 200 \text{ mm}\right\} \quad (6.1-29)$$

6.1.3.8 Laps of bars in compression

Bearing of ends of bars in compression supplements bond in the transfer of force between bar and concrete. Provided that the bar terminates at a distance of at least 3.5ϕ from an unsupported face, Figure 6.1-19, or is bent into a footing, Figure 6.1-20, the value of F_h may be taken as:

$$F_h = 60 f_{bd} A_s \quad (6.1-30)$$

and lap length then calculated as:

$$l_b = \frac{\sigma_{yd}}{4 f_{bd}} \left(f_{yd} - F_h / A_s \right) \geq l_{b,min} \quad (6.1-31)$$

and $l_{b,min}$ is given by Eq. (6.1-29).

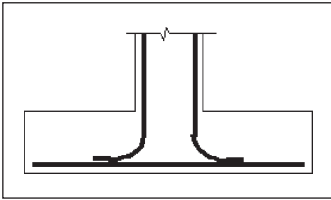


Figure 6.1-20: Column bars anchored in footing

6.1.3.9 Anchorage of bundled bars

Unless otherwise stated, the rules for individual bars also apply for bundles of bars. In a bundle, all the bars should be of the same characteristics (type and grade). Bars of different sizes may be bundled provided that the ratio of larger to smaller diameter does not exceed 1.7.

In design of anchorages, the bundle is replaced by a notional bar having the same sectional area as the bundle. The equivalent diameter, $\emptyset_n$ of this notional bar is such that:

$$\emptyset_n = \emptyset \sqrt{n_b} \quad (6.1-32)$$

where n_b is the number of bars in the bundle.

Where individual bars are anchored with a staggered distance greater than $1.3 l_b$ (where l_b is based on the bar diameter), the diameter of the individual bar may be used in assessing l_b . Otherwise the equivalent diameter of the bundle, $\emptyset_n$, should be used throughout all bond length calculations.

6.1.3.10 Lapped joints of bundled bars

Where the bundle comprises two bars with an equivalent diameter $\emptyset_n$ not exceeding 32 mm, the bundle may be lapped at the same section with the equivalent diameter $\emptyset_n = 1.41\emptyset$ used throughout all bond length calculations.

For bundles which consist of two bars with an equivalent diameter ≥ 32 mm or of three bars, laps of individual bars within a bundle should be staggered in the longitudinal direction by at least $1.3 l_b$ as shown in Figure 6.1-21, where l_b is based on a single bar, Eq. (6.1-28). There are to be no more than four bars in any individual lap cross-section. The individual diameter $\emptyset$ is used in bond length calculations.

Minimum cover and clear spacing dimensions for the bundle are to be used to determine α_2 .

6.1.4 Anchorage and lapped joints of welded fabric

6.1.4.1 Design anchorage length of welded fabric

The design anchorage length calculated in accordance with Eq. (6.1-21) may be reduced by 30%.

6.1.4.2 Design lap length of welded fabric in tension

6.1.4.2.1 Lap length for main reinforcement

The minimum number of welded cross wires over the lap length is:

$n = 1$ for fabric made of ribbed wires

$n = 5 (A_{s,calc}/A_{s,ef})$ for fabric made of plain indented wires (n to be rounded up to the next whole number).

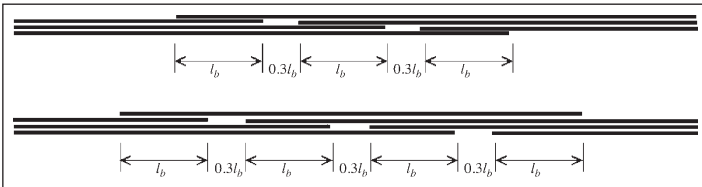


Figure 6.1-21: Laps of bars in a bundle (schematic)

Each welded joint should be capable of withstanding the shearing force given in subsection 5.2.5.5.

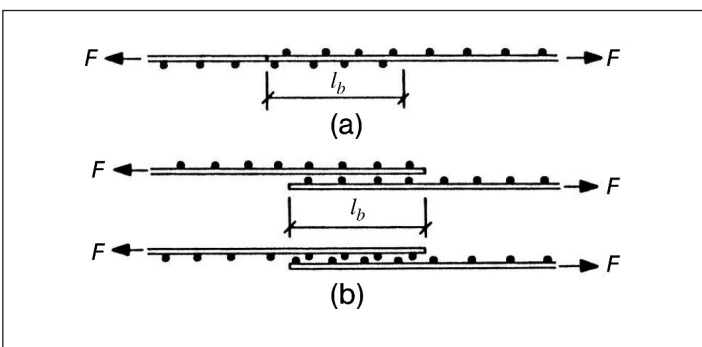


Figure 6.1-22: Lapping of welded fabric: (a) intermeshed fabric (longitudinal section); (b) layered fabric (longitudinal section)

Splicing of welded fabric in structures assessed for fatigue loads should be done with intermeshed fabrics.

For welded fabric placed in more than one layer, the values of l_b from Eq. (6.1-33) may be reduced by 20% for the fabric further from a surface.

The design lap length is given by:

- with intermeshed fabrics (Figure 6.1-22a), according to 6.1.3.7;
- with layered fabric (Figure 6.1-22b):

$$l_b \geq \alpha_4 \alpha_5 \frac{\emptyset f_{yd}}{4 f_{bd}} \geq l_{b,min} \quad (6.1-33)$$

where:

$$\alpha_5 = 0.75 + 50 / s_w \text{ and } 1.0 \leq \alpha_5 \leq 2.0 \quad (6.1-34)$$

and

$$l_{b,min} > \max \left\{ 0.7 \frac{\emptyset f_{yd}}{4 f_{bd}} ; 15\emptyset ; s_w ; 200 \text{ mm} \right\} \quad (6.1-35)$$

where s_w is the spacing of the longitudinal wires, in mm.

6.1.4.2.2 Laps in the transverse direction: secondary reinforcement

For intermeshed fabrics provisions as for the main welded fabric reinforcement apply (see subsection 6.1.4.2.1).

For layered fabrics, the length of lap is chosen from Table 6.1-3:

Table 6.1-3: Required lap lengths for splices of the secondary reinforcement (layered fabrics)

Diameter of wires	Lap lengths	Other requirements
$\emptyset \leq 6 \text{ mm}$	$\geq 150 \text{ mm}$	at least 1 wire pitch within the lap
$6 \text{ mm} < \emptyset \leq 8.5 \text{ mm}$	$\geq 250 \text{ mm}$	at least 2 wire pitch
$8.5 \text{ mm} < \emptyset \leq 12 \text{ mm}$	$\geq 350 \text{ mm}$	at least 2 wire pitch

6.1.4.3 Design lap length of welded fabric in compression

For the main reinforcement the design lap length should comply with Eq. (6.1-36).

$$l_b \geq \frac{\emptyset f_{yd}}{4 f_{bd}} \quad (6.1-36)$$

For the secondary reinforcement, subsection 6.1.4.2.1 applies.

6.1.5 Special circumstances

6.1.5.1 Slipform construction

The value of coefficient η_2 , Eq. (6.1-20), should be taken as 0.7 for bars in structural parts built using slipform construction.

6.1.5.2 Bentonite walling

The value of coefficient η_2 , Eq. (6.1-20), should be taken as 0.7 for bars in concrete cast under bentonite or polymer drilling fluids.

6.1.5.3 Post-installed reinforcement

Rebar connections using post-installed rebars are permissible for all applications where straight cast-in-place rebars are allowed. The design can be performed in a simplified way using the provisions for cast-in-place rebars in this code. However, the following restrictions must be considered:

- the system for inspecting the hole must be suitable for the application in question;
- larger minimum concrete cover;

The suitability of the systems has to be proven by an independent approval process. For more details, see EOTA Technical Report 023 “Assessment of post-installed rebar connections”, Brussels, Nov. 2006.

Considerations must be given to the need for temporary support or load restrictions.

Both bond strength and stiffness increase at temperatures below -80°C . The increase is influenced by a number of factors, and hence is too complex for a Code type formulation. The approach given in subsection 6.1.6.1 is conservative.

Changes in bond strength and stiffness with increasing temperature are influenced by a number of factors, and hence are too complex for a Code type formulation. The approach given in subsection 6.1.6.2 represents a reasonable simplification.

Most data on bond resistance of corroded reinforcement are obtained from tests in which corrosion activity has been accelerated, and corrosion rates are in excess, or well in excess, of those measured in field exposure. Consequently, experimental data must be interpreted with caution.

Table 6.1-4: The magnitude of the reduction in residual bond strength for corroded reinforcement

Corrosion penetration (mm)	Equivalent surface crack (mm)	Confinement	Residual capacity (as % of f_{bd}) Bar type	
			Ribbed	Plain
0.05	0.2–0.4	No links	50–70	70–90
0.10	0.4–0.8		40–50	50–60
0.25	1.0–2.0		25–40	30–40
0.05	0.2–0.4	Links	95–100	95–100
0.10	0.4–0.8		70–80	95–100
0.25	1.0–2.0		60–75	90–100

- larger minimum clear bar spacing;
- limited design compression strength;
- special requirements for fire safety.

6.1.5.4 Electrochemical extraction of chlorides (ECE)

The correct application of electrochemical techniques for chloride extraction from or re-alkalization of concrete does not cause any significant long term reduction in bond strength, except where:

- alkali aggregate reactions (AAR) are induced in concrete containing susceptible aggregates by the electrochemical process;
- an appreciable amount of corrosion has occurred prior to treatment. Plain (unribbed) round bars are likely to be particularly susceptible to loss of bond in these circumstances – see subsection 6.1.7.1.

There may be a temporary reduction in bond strength while treatment is underway. The reduction does not exceed 50%, and strength is restored within a few days of cessation of treatment.

6.1.6 Conditions of service

6.1.6.1 Cryogenic conditions

The basic bond strength given by Eq. (6.1-20) may safely be used in conditions of low temperature.

6.1.6.2 Elevated temperatures

The reduction in bond strength of ribbed bars at elevated temperatures may be taken as similar to that for tensile strength of concrete. Note that explosive spalling of HSC may reduce concrete cover.

Bond strength of plain round bars at 300°C and 500°C may be taken as 50% and 10% respectively of bond strength at normal temperatures.

6.1.7 Degradation

6.1.7.1 Corrosion

Corrosion of reinforcement embedded in concrete may affect residual capacity of reinforced concrete structures. The effects of corrosion in hardened concrete differ from those associated with corrosion prior to concreting.

Small amounts of corrosion, up to the level required to induce longitudinal cracking, do not cause loss of bond capacity, and may even augment bond strength to a modest degree, particularly where the bar is in a “poor” casting position.

At greater levels of corrosion, residual bond strength is strongly influenced by the degree of confinement provided by secondary reinforcement in the form of links and by the surrounding structure. Transverse pressure from support reactions increases bond strength.

The residual capacity of anchorages and lapped splices should be checked at the ultimate limit state at locations of high reinforcement stress where longitudinal cracking develops. Away from anchorages and laps, a substantial loss of bond may be tolerated without ultimate strength being affected. Rotation capacity at plastic hinges is likely to increase with corrosion, as the length of bar over which plastic strains develop will increase as longitudinal cracking develops and bond is reduced.

For the purpose of assessing performance at the serviceability limit state, it is likely that (a) the influence of bond stiffness on

The magnitude of the reduction in residual bond strength is highly dependent on the confinement to the bar, and is also affected by concrete quality and environment. The values in Table 6.1-4 may be taken as indicative; however, detailed guidance should be sought in cases where residual strength of a corroding structure is of concern.

The equivalent surface crack indicates the width of corrosion induced longitudinal crack which correlates with the residual strength indicated in typical conditions.

It should be appreciated that residual strength of concrete structures is also affected by cross-section loss of both steel and concrete.

Alkali aggregate reactions (AAR) occur in concrete when hydroxyl ions in the pore water react with certain components of the aggregate. The reaction product is a highly expandable gel which imbibes water and swells. If sufficient expansion takes place the result is cracking and expansion of the surrounding concrete. The surface of the concrete does not expand to the same extent as the interior and this causes tensile stresses to be set up in the surface of the concrete which can in turn lead to surface macro-cracks. The most common form of AAR is alkali silica reaction (ASR) with other less common forms being alkali silicate reaction and alkali carbonate reaction. The guidance in this section is appropriate for ASR. Further information is contained in: Structural Effects of alkali silica reaction, The Institution of Structural Engineers, July 1992.

Surface scaling is generally found where the surface of the concrete is subjected to weak solutions of salt or urea, typically for de-icing purposes.

Internal freeze-thaw damage results from expansive stresses generated by water on freezing when the pore structure of the concrete is saturated above a critical value, and leads to internal micro-cracking. Internal damage is likely only in concrete subjected to long term wet/saturated conditions. Care needs to be taken in diagnosis for internal frost damage as visual observation is not conclusive.

Tensile strength undergoes greater reductions than compressive strength under frost attack, and it is not acceptable to use estimated tensile strengths based on compressive strength measurements.

For more details, see CONTECVET (2001) A validated Users Manual for assessing the residual service life of concrete structures. Manual for assessing structures affected by frost. Geocisa, Madrid. Available on CD from BCA, Crowthorne, Berkshire, UK

deflections will be small compared to that of loss of reinforcement and of concrete cross-section, (b) the width of corrosion induced longitudinal cracks is likely to exceed that of flexural cracks by the time change in flexural crack widths would be observed, and (c) by this stage of deterioration the serviceability limit state of durability will in any case be the more critical.

6.1.7.2 Alkali silica reaction (ASR)

When laps and anchorages are restrained by links, expansion within the limits 0.45‰ (restrained) and 4.50‰ (unrestrained) induced by ASR has been shown to have no significant effect on bond strength.

Where transverse reinforcement is not present then bond strength may be reduced by up to 50%.

Care is necessary in the assessment of residual resistance of anchorage zones of prestressed elements, particularly of pretensioned members in which the tendons are not contained by links.

6.1.7.3 Frost

Two types of damage to concrete may occur as a result of freezing and thawing:

- surface scaling;
- internal damage.

Surface scaling leads to a reduction in concrete cover to reinforcement. The effect on bond strength may be accounted for by use of a reduced cover in Eq. (6.1-22).

Residual bond capacity of ribbed bars not confined by links in freeze-thaw damaged concrete may be assessed using splitting tensile strength measurements on cores taken from the affected structure. Concrete compressive strength f_{ck} used in Eq. (6.1-20) may be substituted by residual concrete compressive strength after freeze-thaw attack $f_{ck,ft}$ given by:

$$f_{ck,ft} = 3.3 f_{ctk,is}^{1.5} \quad (6.1-37)$$

where:

$f_{ctk,is}$ [N/mm²] is the characteristic measured in-situ tensile strength.

Bond strength of ribbed bars is not degraded as severely where bars are confined by secondary reinforcement detailed in compliance with the requirements of this Code. Residual bond strength is expected to be at least 70% of that “as constructed”.

6.1.7.4 Fire

The reduction in residual bond strength of ribbed bars in fire-damaged concrete structures may be taken as similar to that for tensile strength of concrete. Where surface spalling has occurred, the reduced concrete cover should be used.

Residual bond strength of plain round bars after heating to 300°C and 500°C may be taken as 50% and 10% respectively of bond strength at normal temperatures.

6.1.8 Anchorage of pretensioned prestressing tendons

6.1.8.1 General

Two different bond situations should be considered due to the transverse deformations of the tendon. “Push-in” along the transmission length, where the tendons become thicker at release and “pull-out”, which refers to the anchorage length where the opposite occurs when the steel stress is increased due to loading.

The bond strength of pretensioned prestressing tendons depends on the bond situation. The highest value applies to the transmission length, the length required to introduce the prestressing force. Beyond that length a lower bond strength has to be taken into account, which results in a bilinear diagram for the embedment length that is required to develop the design steel stress (Figure 6.1-23).

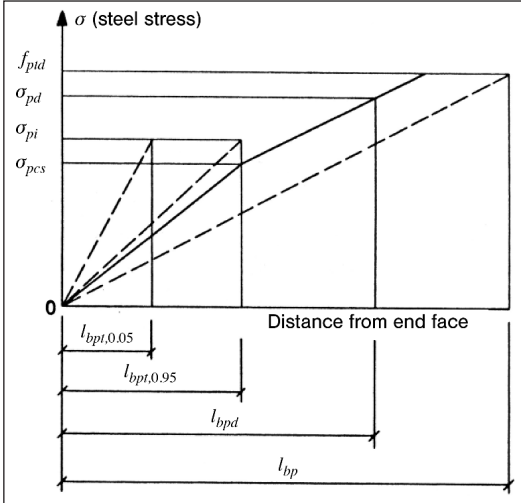


Figure 6.1-23: Variation in steel stress along the anchorage zone of a pretensioned member

6.1.8.2 Design bond strength

The design value of the bond strength for prestressing tendons is:

$$f_{bpd} = \eta_{p1} \eta_{p2} f_{ctd} \quad (6.1-38)$$

where:

$f_{ctd} = f_{ctk,min}(t)/\gamma_c$ is the lower design concrete tensile strength:
for the transmission length the strength at the time of release, for the anchorage length the strength at 28 days;

$\eta_{p.1}$ takes into account the type of prestressing tendon:

$\eta_{p.1} = 1.4$ for indented and crimped wires;

$\eta_{p.1} = 1.2$ for 7-wire strands;

$\eta_{p.2}$ takes into account the position of the tendon:

$\eta_{p.2} = 1.0$ for all tendons with an inclination of 45–90° with respect to the horizontal during concreting;

$\eta_{p.2} = 1.0$ for all horizontal tendons which are up to 250 mm from the bottom or at least 300 mm below the top of the concrete section during concreting;

$\eta_{p.2} = 0.7$ for all other cases.

6.1.8.3 Basic anchorage length

The basic anchorage length defines the length that is required to develop the full strength in an untensioned tendon.

The factor $A_{sp} / (\emptyset\pi)$ depends on the type of tendon:

$$\frac{A_{sp}}{\emptyset\pi} = \frac{\emptyset}{4} \text{ for tendons with a circular cross-section;}$$

$$\frac{A_{sp}}{\emptyset\pi} = \frac{7}{36} \emptyset \text{ for 7-wire strands.}$$

The basic anchorage length of an individual pretensioned tendon is:

$$l_{bp} = \frac{A_{sp} f_{ptd}}{\emptyset\pi f_{bpd}} \quad (6.1-39)$$

where:

A_{sp} is the cross-sectional area of the tendon;

$f_{ptd} = f_{ptk}/\gamma_s$ is the design tendon strength;

f_{ptk} is defined in subsection 5.3.5.1;

$\emptyset$ is the nominal diameter of the tendon.

The use of narrow spaced stirrups or helices around the tendons and transverse prestressing may result in shorter transmission length. This is not considered due to lack of experimental data.

Tendon release that is obtained by sawing through the concrete and the steel should be considered as gradual release.

The transmission length can be estimated from the draw-in value (δ_e) of the tendons at the end face of the concrete member. However, when the concrete member is sawn from a longer production unit, the draw-in cannot be estimated properly. Assuming a linear steel stress along the transmission length, this draw-in must be:

$$\delta_e < 0.5 \frac{\sigma_{pi}}{E_p} l_{bpt}$$

with $\alpha_{p2} = 1.0$ in Eq. (6.1-40) for l_{bpt} ; E_p is the modulus of elasticity of the prestressing steel. Note that $\alpha_{p2} = 0.5$ is associated with $l_{bpt,0.05}$ and $\alpha_{p2} = 1.0$ with $l_{bpt,0.95}$ in Figure 6.1-23.

See commentary to subsection 6.1.8.1 for different bond situations. The basic anchorage length is related to “pull-out”. The transmission length is connected to “push-in”.

If necessary, the required anchorage capacity may be obtained by additional end anchorages or non-prestressed reinforcement.

For non-rectangular sections, the development length can be found in a similar way as assumed for post-tensioning.

6.1.8.4 Transmission length

The transmission length of a pretensioned tendon is:

$$l_{bpt} = \alpha_{p1} \alpha_{p2} \alpha_{p3} l_{bp} \frac{\sigma_{pi}}{f_{bpd}} \quad (6.1-40)$$

where:

σ_{pi} is the steel stress just after release;

α_{p1} considers the type of release:

$\alpha_{p1} = 1.0$ for gradual release;

$\alpha_{p1} = 1.25$ for sudden release;

α_{p2} considers the action effect to be verified:

$\alpha_{p2} = 1.0$ for calculation of anchorage length when moment and shear capacity is considered;

$\alpha_{p2} = 0.5$ for verification of transverse stress due to development and distribution of prestress in the anchorage zone;

α_{p3} considers the influence of bond situation:

$\alpha_{p3} = 0.5$ for strands;

$\alpha_{p3} = 0.7$ for indented or crimped wires.

6.1.8.5 Design anchorage length

The design anchorage length of a pretensioned prestressing tendon is:

$$l_{bpd} = l_{bpt} + l_{bp} \frac{\sigma_{pd} - \sigma_{pcs}}{f_{ptd}} \quad (6.1-41)$$

where:

σ_{pd} is the tendon stress under design load ($\sigma_{pd} \leq f_{ptd}$);

σ_{pcs} is the tendon stress due to prestress including all losses.

6.1.8.6 Development length

The development length is the distance from the end face to the concrete cross-section beyond which the distribution of the longitudinal stresses over the sections follow the plane-sections hypothesis.

For a rectangular cross-section and straight tendons situated near the bottom edge of the concrete section the development length is:

$$l_p = \sqrt{h^2 + (0.6l_{bpt})^2} > l_{bpt} \quad (6.1-42)$$

where:

h is the total depth of the concrete section.

The bond behaviour of FRP reinforcement to concrete depends mainly on the reinforcement geometry, application type (e.g. internal or externally bonded) and surface characteristics. It varies from that of conventional steel reinforcement, given for example the following aspects:

- the modulus of elasticity of FRP is generally lower than that of steel, especially in the transverse direction;
- the shear stiffness of FRP is significantly lower than that of steel;
- the surface deformations relate to the resin matrix, which has a lower shear strength than steel.

It is generally possible to obtain bond strengths for non-metallic reinforcement of similar or greater magnitude than for steel reinforcement.

The bond stress–slip relationship (monotonic loading) for deformed steel rebars, given in subsection 6.1.1, is applicable for FRP reinforcement, provided that the use of model parameters is calibrated on the basis of experimental results. Generally, a modified bond stress–slip relationship is assumed with an ascending and descending branch and whereas the constant shear stress branches are not applicable in the case of FRP. Further details are given in subsections 6.2.1.1 and 6.2.1.2

The constitutive model of Figure 6.2-1 has been proposed by Cosenza, Manfredi and Realfonzo (1995) and is also discussed in *fib* Bulletin 10 “Bond of reinforcement in concrete” (*fib*, 2000) and *fib* Bulletin 40 “FRP reinforcement in RC structures” (*fib*, 2007). The parameters have to be calibrated on the basis of experimental results.

6.2 Bond of non-metallic reinforcement

Bond of non-metallic reinforcement is the term used to denote the interaction and transfer of forces between fibre reinforced polymer (FRP) reinforcement and concrete. At the serviceability limit state, bond influences width and spacing of cracks, tension stiffening and curvature. At the ultimate limit state, bond is responsible for strength at end anchorages or at intermediate regions (the latter in the case of externally bonded reinforcement).

6.2.1 Local bond stress–slip model

The bond stresses between concrete and non-metallic reinforcement can be calculated as a function of the relative displacement, s .

6.2.1.1 Local bond stress–slip model for FRP rebars

The bond stress τ_b can be calculated in terms of the slip s according to the following equations (see also Figure 6.2-1):

$$\tau_b = \tau_{bm}(s/s_m)^\alpha \quad \text{for } 0 \leq s \leq s_m \quad (6.2-1a)$$

$$\tau_b = \tau_{bm} - \tau_{bm} p(s-s_m)/s_m \quad \text{for } s_m \leq s \leq s_u \quad (6.2-1b)$$

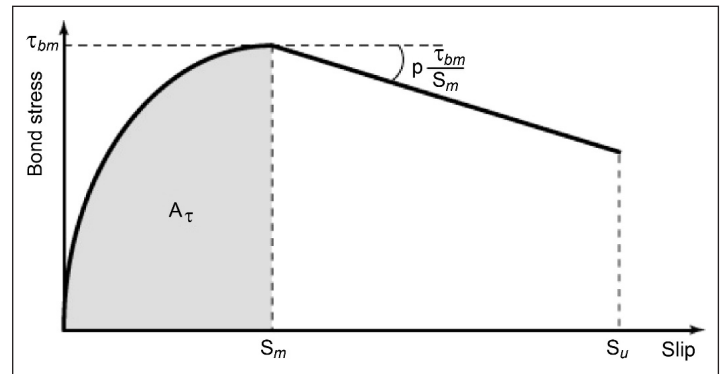


Figure 6.2-1: Analytical bond stress–slip relationship (embedded FRP reinforcement)

6.2.1.2 Local bond stress–slip model for externally bonded FRP

The bond stress can be calculated according to the following equations (see also Figure 6.2-2):

$$\tau_b = \tau_{bm}(s/s_m) \quad \text{for } 0 \leq s \leq s_m \quad (6.2-2a)$$

$$\tau_b = \tau_{bm} - \tau_{bm}(s-s_m)/(s_u-s_m) \quad \text{for } s_m \leq s \leq s_u \quad (6.2-2b)$$

The constitutive bond model for externally bonded FRP reinforcement is assumed bilinear – see *fib* Bulletin 14 “Externally bonded FRP reinforcement for RC structures” (*fib*, 2001). The model parameters basically relate to the fracture energy G_F of the glued joint connection between the external reinforcement and the concrete and have to be calibrated on the basis of experimental results.

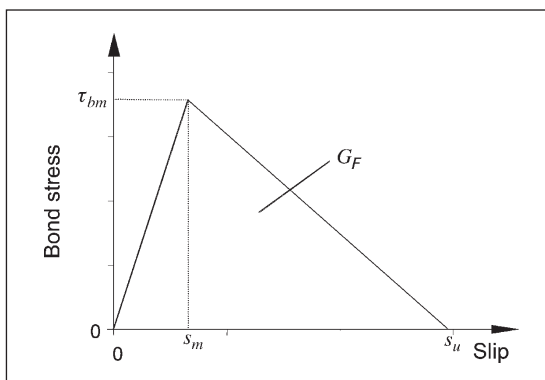


Figure 6.2-2: Analytical bond stress–slip relationship (externally bonded FRP reinforcement)

6.2.2 Bond and anchorage of internal FRP reinforcement

For a discussion on available models for the basic development length and lap splices of non-prestressed FRP reinforcement and the transfer length of prestressed FRP reinforcement, see *fib* Bulletin 10 “Bond of reinforcement in concrete” (*fib*, 2000) and *fib* Bulletin 40 “FRP reinforcement in RC structures” (*fib*, 2007). Bulletin 40 also offers models for tension stiffening, deflections and crack opening, taking into account bond interaction between FRP reinforcement and concrete.

Models are currently being developed by *fib* Task Group 9.3.

Bond of plain (smooth) bars is governed by the adhesion between the bar surface and the concrete, provided that the interlaminar shear strength between the fibres is higher. The bond strength of plain bars is generally low and splitting bond forces can be neglected. Their use is limited as they need to be combined with other anchoring devices such as bends and transverse bars.

Bond of deformed (surface treated) bars is often governed by the shear strength of the deformations, provided that the interlaminar shear strength between the fibres is higher. In this case, the influence of concrete strength is limited compared to the bond of steel bars. For high strength deformations, concrete shear failure similar to deformed steel bars is more predominant.

The bond strength of deformed FRP bars is similar or superior to that of steel bars. Splitting bond forces can govern in case insufficient confinement by the surrounding concrete is provided. Depending on the surface texture, the splitting tendency of FRP bars is lower or higher compared to deformed steel bars. Analytical modelling of bond splitting is provided in Appendix A of *fib* Bulletin 40.

6.2.3 Bond and anchorage of externally bonded FRP reinforcement

A further discussion on bond modelling of externally bonded reinforcement (EBR) is given in *fib* Bulletin 14 “Externally bonded FRP reinforcement for RC structures” (*fib*, 2001).

In the case of externally bonded reinforcement (EBR) the bond behaviour relates to the glued joint between the external FRP reinforcement and the concrete. As part of any flexural or shear strengthening design, the evaluation of the maximum force that may be transferred from the concrete to the external reinforcement, as well as the evaluation of shear and normal stresses at the concrete–FRP interface is required. The former is necessary when designing for the ULS; the latter when designing for the SLS.

Debonding of EBR can be predicted by considering the different bond failure modes which can occur, as outlined in the following subsections.

6.2.3.1 Bond-critical failure modes

Depending on the cohesive and adhesive strengths of the concrete, adhesive (glue) and FRP, the bond failure may occur at different interfaces, as shown in Figure 6.2-3. Given the high shear strength of the adhesive and the FRP, the failure plane is generally located in the concrete layer near the adhesive.

Depending on the starting point of the debonding process, the bond-critical failure modes in flexural and shear strengthening with EBR can be classified into two main categories:

- end debonding;
- intermediate crack debonding.

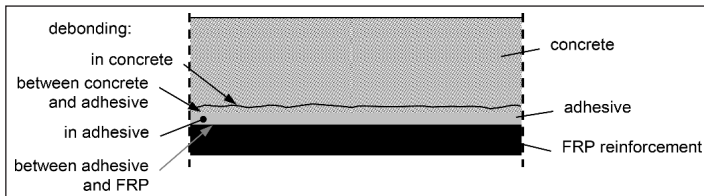


Figure 6.2-3: Bond interface

End debonding

If insufficient anchorage capacity is provided, interfacial end debonding occurs as shown in Figure 6.2-4 (top). A specific form of end debonding is concrete cover separation or concrete rip-off – see Figure 6.2-4 (bottom). The latter failure mode is obtained when a shear crack in the end region of the FRP reinforcement propagates into a debonding mode at the level of the internal reinforcement.

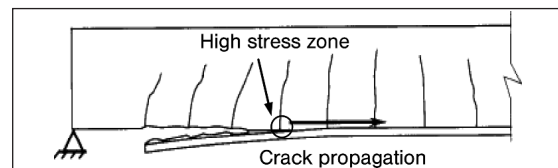


Figure 6.2-4: Anchorage (top) and concrete rip-off (bottom) failure

Intermediate crack debonding

Debonding of FRP can also be caused by bridging of intermediate cracks as illustrated in Figure 6.2-5.

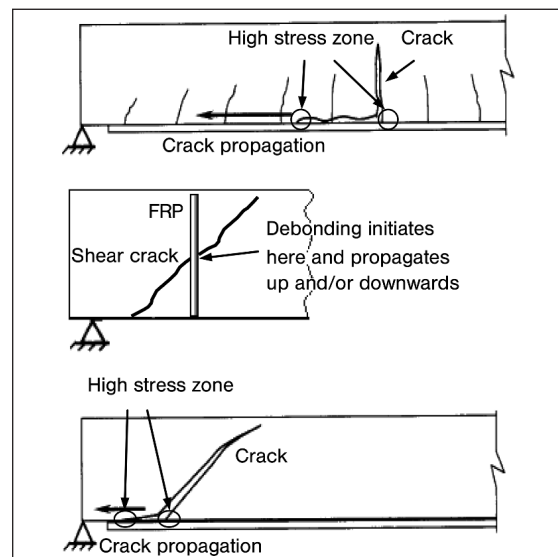


Figure 6.2-5: Intermediate crack debonding failure aspects

6.2.3.2 Maximum bond length

With reference to a typical bond test, as represented in Figure 6.2-6, the ultimate value of the force transferred to the FRP system prior to debonding depends on the length, l_b , of the (uncracked) bonded area. The maximum bond length, $l_{b,max}$, is defined as the length that, if exceeded, no further increase in the force transferred between the concrete and the EBR would be possible.

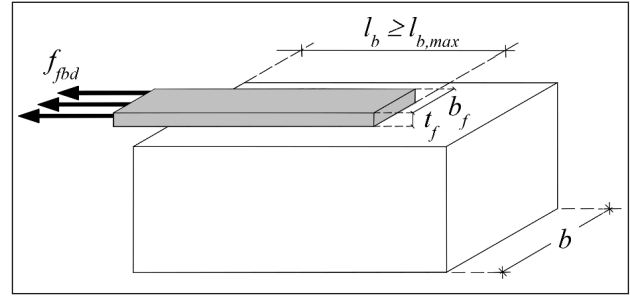


Figure 6.2-6: FRP pure shear bond test configuration

The maximum bond length may be estimated as follows:

$$l_{b,max} = \sqrt{\frac{E_f t_f}{k_{bl} f_{ctm}}} \quad (6.2-3)$$

(length s in mm, stresses in MPa)

where:

- E_f is the modulus of elasticity of the FRP in the direction of the stress;
- t_f is the thickness of the FRP;
- f_{ctm} is the mean tensile strength of the concrete substrate;
- k_{bl} is the bond length calibration factor obtained from test results; for FRP this can be taken equal to 2.

6.2.3.3 Ultimate strength for end debonding – anchorage capacity

The maximum bond anchorage capacity is valid for bond lengths equal to or higher than the maximum bond length. If smaller bond lengths are provided, the bond anchorage capacity is reduced, assuming a parabolic relationship between the anchorage capacity and the bond length, as expressed by the factor β_l .

The mean and the design ultimate bond strengths, that is the maximum tensile stress in the EBR limited by bond to concrete in a single (uncracked) anchorage zone, are:

$$f_{fbm} = k_m k_b \beta_l \sqrt{\frac{2E_f}{t_f} f_{cm}^{2/3}} \quad (6.2-4a)$$

$$f_{fbd} = \frac{k_k}{\gamma_{f,b}} k_b \beta_l \sqrt{\frac{2E_f}{t_f} f_{cm}^{2/3}} \quad (6.2-4b)$$

(lengths in mm, stresses in MPa)

where:

- f_{cm} is the mean compressive strength of concrete;
- $\gamma_{f,b}$ is the FRP partial safety factor for debonding (equal to 1.5);
- β_l is the length factor, defined as:

$$\beta_l = \frac{l_b}{l_{b,max}} \cdot \left(2 - \frac{l_b}{l_{b,max}} \right) \text{ if } l_b \leq l_{b,max}, \beta_l = 1 \text{ otherwise} \quad (6.2-5)$$

k_b is the shape factor, equal to:

$$k_b = \sqrt{\frac{2 - b_f / b}{1 + b_f / b}} \geq 1 \quad (6.2-6)$$

b_f and b are the FRP and concrete section widths, respectively.

On the basis of calibration with experimental results, for epoxy bonded CFRP systems $k_m = 0.25$ and, under the hypothesis of normal distribution of the bond strength, the 5% percentile can be calculated assuming $k_k = 0.17$.

If a shear crack occurs at the end of the FRP, this shear crack can propagate into concrete cover separation, related to the missing tension link between the internal and external reinforcement when considering the classical truss analogy for shear capacity of beams. Prediction of occurrence of a shear crack at the end of the FRP is provided in *fib* Bulletin 14 “Externally bonded FRP reinforcement for RC structures” (*fib*, 2001) and acts as a lower bound prediction model for concrete rip-off.

For more detailed modelling of intermediate debonding, see for example *fib* Bulletin 14 “Externally bonded FRP reinforcement for RC structures” (*fib*, 2001).

Bond stresses (shear and normal) at serviceability limit state can be calculated on the basis of linear elastic analysis.

6.2.3.4 Ultimate strength for end debonding – concrete rip-off

This debonding mode can be avoided by providing shear strengthening at the end of the FRP. The design of the shear strengthening aims at extending the existing shear links up to the level of the EBR.

6.2.3.5 Ultimate strength for intermediate debonding

According to a simplified procedure, the ultimate bond strength for intermediate debonding is obtained by multiplying f_{fbm} and f_{fbd} by a factor k_c . If specific experimental data are not available, k_c may be taken equal to 2.0 and 1.5, for the mean and the design ultimate bond strength, respectively. Alternative and more detailed approaches to prevent the debonding failure at intermediate cracks can be adopted, based on the envelope line of tensile stress and on the force transfer between the concrete and the EBR.

6.2.3.6 Interfacial stresses for the serviceability limit state

It is assumed that bond interface crack initiation will not occur under service load, provided proper detailing and limitation of deflections and crack widths has been carried out.

6.2.4 Mechanical anchorages for externally bonded FRP reinforcement

Debonding at the ends of the EBR can be avoided, or an enhancement of the debonding load can be achieved, using anchorage systems. Various solutions are available and can be designed for the specific case, employing the fibres themselves with suitable configurations (e.g. transverse wrapping, spike anchors) or additional devices such as bolts or plates.

The ultimate value of the force transferred to the EBR system prior to debonding depends on the efficiency of the anchorage system. In the presence of mechanical anchorage, the design bond strength f_{fad} may be taken equal to:

$$f_{fad} = k_a f_{fd} \quad (6.2-7)$$

where:

- f_{fd} is the design tensile strength of the FRP, equal to the characteristic tensile strength f_{fk} divided by the partial safety factor γ_f (see subsection 5.5.7);
- k_a is the effectiveness coefficient of the specific anchorage system ($k_a \leq 1.0$), provided on the basis of experimental results.

Background information on this subject is given by Randl, N. (2013), Design recommendations for interface shear transfer in *fib* Model Code 2010. Structural Concrete, 14. doi: 10.1002/suco.201300003.

In practice, the following aspects have to be considered:

- contamination of the concrete surface just before casting the overlay may reduce bonding significantly; likewise unfavourable climatic conditions (such as strong dry winds or solar radiation) can dry up the concrete surface and reduce bond strength;
- inappropriate roughening methods may harm the top surface of the concrete (e.g. leading to micro-cracking);
- insufficient quality of the overlay concrete may cause larger constraint forces (e.g. due to shrinkage) and reduce bonding;
- edge zones (where significant internal tensile and shear stresses may occur due to forces of constraint) have to be sufficiently secured.

6.3 Concrete to concrete

6.3.1 Definitions and scope

Concrete-to-concrete load transfer across interfaces has to be considered when two concretes are cast against each other at different times, that is when the hardening process of the older concrete is already finished. While tensile loads have to be transferred via reinforcement when designing for the ULS, shear forces with their load direction parallel to the interface, normal forces perpendicular to the interface or a combination of both are to some extent transferred directly from concrete to concrete.

The topic is relevant in practice in the following situations:

- repair and strengthening of existing RC-members by means of new concrete layers;
- supplement of precast elements with additional concrete cast at the site;
- all situations at the site where, due to interruptions in the erection process, new concrete is cast against already completely hardened concrete;
- post-installed concrete elements (e.g. corbels) attached to existing members for the introduction of loads.

6.3.2 Interface roughness characteristics

There are several indicators to describe and quantify the roughness of a concrete surface. The most commonly used parameter is the mean roughness R_a (Fig. 6.3-1) which represents the average deviation of the profile from a mean line ($\bar{y}$). It is determined as the mean value of profile heights along an assessment length l :

$$R_a = \frac{1}{l} \cdot \int_0^l |y(x) - \bar{y}| \cdot dx \approx \frac{1}{n} \sum_{i=1}^n |y_i - \bar{y}| \quad (6.3-1)$$

$$\bar{y} = \frac{1}{l} \cdot \int_0^l y(x) \cdot dx \approx \frac{1}{n} \sum_{i=1}^n y_i \quad (6.3-2)$$

where:

l is the assessment length;

$y(x)$ is the profile height at position x .

Another frequently used parameter is the mean peak-to-valley height R_z (Fig. 6.3-2), representing the average of maximum valley-to-peak-deviations R_{zi} within a certain number of assessment lengths, for example for 5:

$$R_z = \frac{1}{5} \cdot \sum_{i=1}^5 R_{zi} \quad (6.3-3)$$

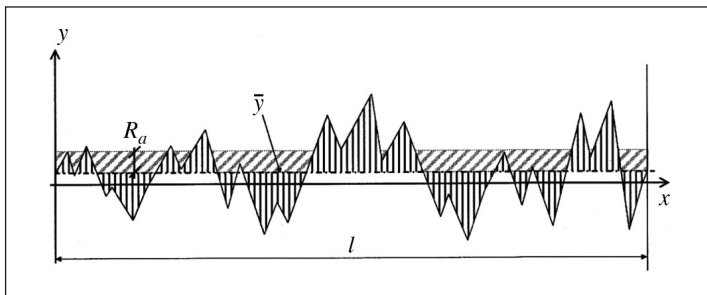


Figure 6.3-1: Average roughness R_a

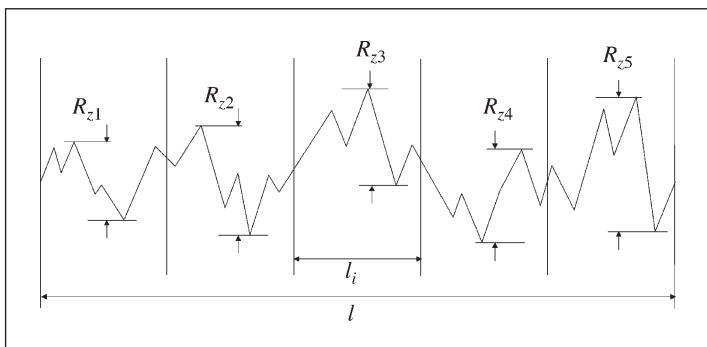


Figure 6.3-2: Mean peak-to-valley height R_z ($l_i = l/5$)

There are several methods to measure and specify the roughness of a concrete surface.

A simple method widely used on the site is the *sand patch method*: A defined volume of fine sand is spread on the surface; depending on the diameter of the circle the average roughness can be estimated as the mean height of the sand cylinder, that is, the sand volume divided by the circle's area. The advantage of this method is its simplicity and quickness. The disadvantages are that it is not very exact, only possible on horizontal or slightly inclined surfaces and that only the "peak-to-mean" roughness R_t ($\approx R_z/2$) can be determined.

Other advanced methods are due to their increased complexity preferably used for laboratory investigations. The advantage of these methods is that they allow for exact measurements of roughness profiles and therefore give a more detailed specification of the surface topography.

Examples are as follows:

- contact methods: mechanical contact profilometer
- non-contact methods: laser triangulation
photogrammetry – digital imaging

The simplified classification into the four categories very smooth, smooth, rough and very rough on the basis of the roughness R_t is not an exact method of categorization, but useful for practical design. However, the designer should be aware that identical values of the roughness R_t might still lead to different shear resistances due to differences in the actual surface topography and therefore engineering judgement is always required when determining the appropriate roughness category.

Substantial experimental investigations have been performed since about 1960. The results show a very large scatter for the following reasons:

- The test setup affects the flow of forces within the specimen and a clear determination of the shear force distribution along the interface may be difficult even with small scale specimens.
- Depending on whether or not reinforcement or connectors cross the interface, the loadbearing behaviour may be completely different.
- Bond–slip and ultimate load are strongly influenced by the bond strength which, under laboratory conditions, can on the one hand be very good or, on the other hand, be intentionally eliminated by pre-cracking the specimen or reduced by the use of debonding agents in order to account for example for possible contamination at the site.
- In the case of an uncracked joint without reinforcement the shear stresses are transferred primarily into the edge zones so that the size and shape of the shear interface affect the ultimate shear strength τ_u .

For design purposes, the surface roughness has to be classified into different categories. Depending on the roughness R_t derived from the sand patch method and the applied roughening method the categories in Table 6.3-1 can be defined:

Table 6.3-1: Classification of surface roughness

Category:	R_t [mm]
Very smooth (e. g. cast against steel formwork)	not measureable
Smooth (e. g. untreated, slightly roughened)	< 1.5 mm
Rough (e. g. sand blasted, high pressure water blasted etc.)	≥ 1.5 mm
Very rough (e. g. high pressure water blasted, indented)	≥ 3 mm

6.3.3 Mechanisms of shear transfer

The shear strength of interfaces between concretes cast at different times can be investigated experimentally on large scale members such as beams or slabs or on small scale specimens. There is a variety of small scale test setups such as slant shear test, pull-off and push-off tests. The main parameters decisive for the actual loadbearing capacity observed in tests are:

- interface roughness;
- cleanliness of surface;
- concrete strength and concrete quality;
- eccentricity/inclination of shear force;
- strong bond/pre-cracking/debonding before testing;
- ratio of reinforcement crossing the interface.

The main contributions to the overall shear resistance result from the following mechanisms:

- mechanical interlocking and adhesive bonding;
- friction due to:
 - external compression forces perpendicular to the interface;
 - clamping forces due to reinforcement and/or connectors;
- dowel action of reinforcement and/or connectors crossing the interface.

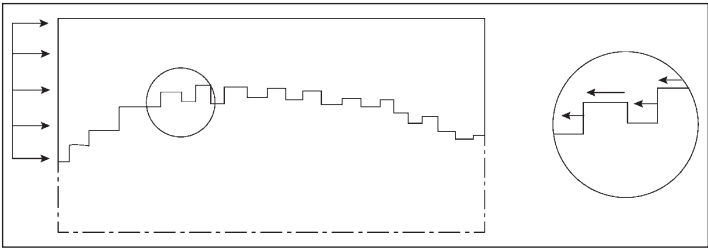


Figure 6.3-3: Simplified representation of the effects of adhesive bonding and mechanical interlocking

Parameters influencing adhesive bonding and mechanical interlocking are, among others:

- preparation (roughening etc.) and cleanliness of interface;
- strength class of old and new concrete;
- quality of top layer of old concrete;
- porosity and moisture content of old concrete;
- quality, composition and properties of fresh new concrete;
- degree of shrinkage of new concrete;
- age of concretes.

The *shear-slip characteristics* of adhesive bonding and mechanical interlocking are characterized by a very stiff behaviour, only effective for shear slip values at about 0.05 mm or less when adhesive bonding predominates or somewhat more when mechanical interlocking contributes.

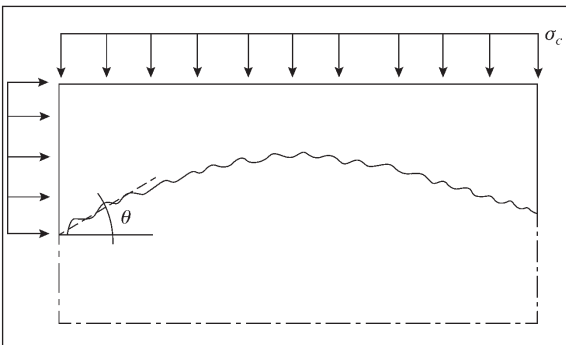


Figure 6.3-4: Simplified representation of shear friction principle

According to a simplified shear friction theory (Figure 6.3-4), for a smooth contact area (with only the general undulation) the shear resistance would be:

$$\tau = \sigma_c \tan \theta \quad (6.3-4a)$$

where $\tan \theta$ is often replaced by the friction coefficient μ . Including the micro-roughness at the contact area (and mechanical interlocking effects respectively) results in the general basic expression:

$$\tau = \tau_a + \mu \sigma_c \quad (6.3-4 b)$$

The confining stress σ_c can as well be generated by reinforcing steel or connectors crossing the interface.

Shear-slip characteristics for friction at constant σ_c show a slight decrease of the shear resistance with increasing slip due to interface deterioration. If the confining action is obtained by reinforcement or connectors crossing the interface, the shear-slip relation may increase as long as the axial force in the reinforcement or connectors increases due to joint opening.

Shear-slip characteristics of dowel action show an increase of the shear resistance with increasing slip.

Adhesive bonding and mechanical interlocking

Adhesive bonding and mechanical interlocking (Fig. 6.3-3) can significantly contribute to the overall shear resistance provided the adhesion and interlocking effects are not reduced by contaminants along the interface. While adhesive bonding – that is adhesive forces due to chemical and physical bonding – can develop along smooth interfaces as well, mechanical interlocking requires an appropriate surface roughness. Taking into account the effect of adhesive bonding and mechanical interlocking for the ultimate bearing capacity implies that the related slip at failure is very small; for larger shear displacements the effects of adhesion and mechanical interlocking are considerably reduced due to loss of bond and fracture of protruding parts of the interface.

Compared to the other mechanisms contributing to interface shear transfer, adhesive bonding and to some extent also mechanical interlocking are sensitive to any contamination of the surface.

Provided good bonding and mechanical interlocking are achieved by appropriate measures (clean surface, appropriate roughening, good concrete quality etc.) representative values for the mean shear resistance are for concrete grades $\leq C50$:

Rough interface (e. g. sand blasted):	$\sim 1.5\text{--}2.5 \text{ N/mm}^2$
Very rough interface (e. g. high pressure water jetted):	$\sim 2.5\text{--}3.5 \text{ N/mm}^2$

Shear friction

In the case of compression forces perpendicular to the interface, a so-called shear friction mechanism can develop depending on the roughness of the interface. Reinforcement and connectors can generate those compressive forces indirectly, since shear sliding normally goes along with joint opening, which leads to stretching of the reinforcement or connectors.

For a constant confining stress σ_c representative mean values for the coefficient of friction μ are for concrete grades $\leq C50$ in the following ranges:

Smooth interface:	0.5–0.7
Rough interface:	0.7–1.0
Very rough interface:	1.0–1.4

Dowel action

Dowel action refers to the bending resistance of connectors (reinforcing bars or dowels) crossing the interface: a shear slip

Dowel action of the connectors crossing the interface means first of all the bending resistance (Fig. 6.3-5) which develops to its maximum with shear slips of up to approximately 0.1–0.2 times the bar diameter. With large slips the so-called kinking effect can be observed (Fig. 6.3-5): in the interface intersection zone the inclined bar with large slips increasingly provides a contribution to the shear resistance due to the horizontal component of the tensile force in the bar.

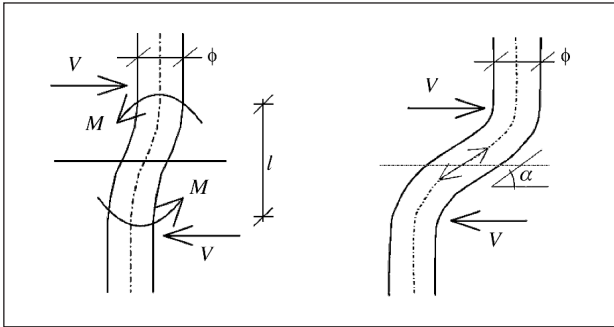


Figure 6.3-5 Dowel action: bending and kinking effect with large slips

along an interface leads to a lateral displacement between upper and lower connector ends, thereby inducing bending stresses in these bars which are superimposed by axial tensile forces due to the opening of the joint.

These tensile forces, on the other hand, do not allow for the full bending resistance resulting from the plastic moment of the bar to develop.

Effects of interaction between the shear resisting components

In a real structure subject to shear loading the various mechanisms (i. e. adhesive bonding, mechanical interlocking, shear-friction and dowel action) interact, thereby affecting each other as a function of the shear slip. After failure of adhesive bond, with increasing shear slip also the mechanical interlock effect decreases quickly (reduction of micro-roughness and fracture of protruding parts of the interface, Figure 6.3-4). While also the shear-frictional resistance declines somewhat, the bending resistance of the connectors increases with larger slips.

If there is no interface reinforcement present, the behaviour of an unreinforced joint is quite brittle; typically failure occurs with the loss of adhesion and mechanical interlocking with slips at around 0.05 mm or even less.

Reinforced joints with $\rho \geq 0.05\%$ show a more ductile behaviour, depending on the reinforcement ratio and the interface roughness failure occurs typically with much larger slips (~0.5–1.5 mm).

6.3.4 Modelling and design

The ultimate resistance of an interface subject to shear forces can be approached by superposition of the single mechanisms of adhesion and mechanical interlocking, shear-friction and dowel action. As these mechanisms interact with each other and reach their maxima at different shear slips it is not possible to add them all together with their maximum values.

Depending on bond strength, roughness of the interface and degree of reinforcement either adhesive bonding (or mechanical interlocking respectively) or the other mechanisms of shear-friction and dowel action will dominate the overall loadbearing behaviour. Therefore two situations can be distinguished in general:

- strong adhesive bonding → “rigid” bond-slip behaviour:
adhesive bonding is the main contributing mechanism to shear resistance
- weak adhesive bonding → “non-rigid” bond-slip behaviour:
shear-friction and dowel action are the main contributing mechanisms to shear resistance

A “rigid” bond-slip behaviour is to be expected when, in the case of smooth interfaces, no reinforcement or, in the case of rough interfaces, no or only small amounts of reinforcement, cross the interface. A noticeable amount of reinforcement ($\geq 0.05\%$) on the other hand allows for larger deformations until the ultimate failure load is reached, that is a “non-rigid” bond-slip behaviour where friction forces and dowel action are activated due to the slip.

The single mechanisms can be summarized and approached with the formulae given below.

Besides other influencing parameters (see subsection 6.3.3) adhesive bond between concretes cast at different times is linked to the hydrophobicity of the existing concrete surface. Experimental investigations indicate that the adhesive bond strength can therefore be related among others to the roughness factor of *Wenzel* (ratio of the true contact area to the apparent contact area), whose identification in turn requires advanced measurement devices and depends on the density of the measuring grid.

Strong adhesive bond (“rigid” bond–slip) means that τ_a becomes predominant and any effect of the reinforcement becomes rather small.

For post-installed reinforcement or connectors with circular cross-sections, the interaction factor κ_I may be assumed equal to 0.5 for rough and very rough interfaces (for background information, see Randl, N. and Wicke, M., Schubübertragung zwischen Alt- und Neubeton. (Beton- und Stahlbetonbau, Heft 8, 2000)).

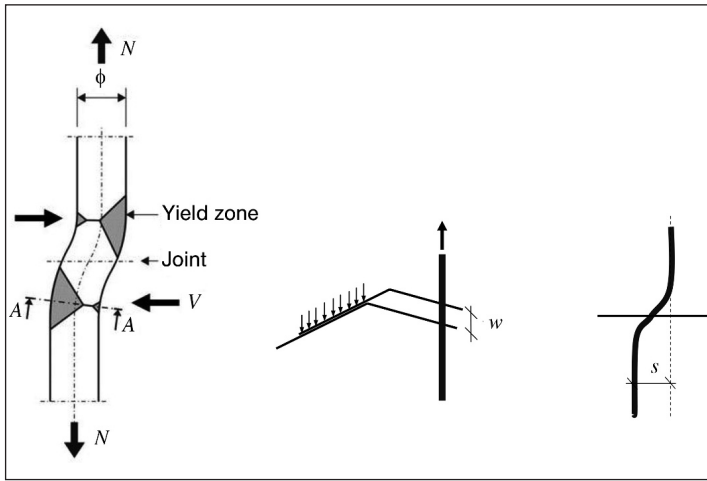


Figure 6.3-6: Clamping force effect and dowel action

The upper limit of the bending resistance in Eq. (6.3-6) is derived on the basis of the “von Mises”-criterion (steel shear failure), usually becoming relevant only in high strength concrete.

Depending on the strength of adhesive bonding, the degree of reinforcement and the kind and intensity of roughening, either τ_a is decisive (rigid bond–slip characteristics) or the other mechanisms (shear friction, dowel action) may deliver the main contribution.

Adhesion/interlocking + friction:

$$\tau_u = \tau_a + \mu \cdot (\sigma_n + \kappa_I \cdot \rho \cdot f_y) \quad (6.3-5)$$

where:

- τ_a is the shear resistance due to adhesive bond/interlocking;
- κ_I is interaction (“effectiveness”) factor;
- σ_n is (lowest) compressive stress resulting from a normal force acting on the interface;
- ρ is ratio of reinforcement crossing the interface ($\rho = A_s/A_c$)

The tensile force in the reinforcement/connectors may be limited due to simultaneous bending (Fig. 6.3-6) and/or reduced anchorage of the bars and, moreover, due to the fact that interface shear failure may occur already at low slip values: $\kappa_I = \sigma_s/f_y \leq 1.0$.

Dowel action

The resistance V_F of the reinforcement or connectors to an acting shear force can be approximated as follows:

$$V_F(s) \approx V_{F,\max} \cdot \left(\frac{s}{s_{\max}} \right)^{0.5} = \kappa_{2,\max} \cdot A_s \cdot \sqrt{f_{cc} \cdot f_y} \cdot \left(\frac{s}{s_{\max}} \right)^{0.5} \leq \frac{A_s \cdot f_y}{\sqrt{3}} \quad (6.3-6)$$

where:

- $V_{F,\max}$ is the maximum value of dowel action as defined in Eq. (6.3-6): $V_{F,\max} = \kappa_{2,\max} \cdot A_s \cdot (f_{cc} \cdot f_y)^{0.5}$;
- $s_{\max}$ is the slip when $V_{F,\max}$ is reached: $s \leq s_{\max} \approx 0.10\phi - 0.20\phi$;
- $\kappa_{2,\max}$ is the interaction coefficient for flexural resistance at slip $s_{\max}$, where $\kappa_{2,\max} \leq 1.6$ for circular cross-sections and C20–C50.

Interaction of tensile forces and bending leads to a reduction of the maximum possible dowel action, especially when the surface is rough so that substantial tensile forces are generated in the reinforcement/connectors. The reduction of the shear strength can be approached by multiplying $V_{F,\max}$ with the interaction factor κ_I taking into account the degree of utilization:

$$V_F(s) = V_{F,\max} \cdot \left(\frac{s}{s_{\max}} \right)^{0.5} \cdot \sqrt{1 - \left(\frac{\sigma_s}{f_y} \right)^2} = V_{F,\max} \cdot \left(\frac{s}{s_{\max}} \right)^{0.5} \cdot \sqrt{1 - \kappa_I^2} \quad (6.3-7)$$

Superposition

When superposing the different mechanisms, the following aspects have to be taken into account:

- the different mechanisms interrelate and thereby influence each other;
- interaction of tension and bending in the connectors leads to an interrelationship between clamping force and dowel action;
- maximum contributions of different mechanisms occur at different slips.

The ultimate shear stress at the interface resulting from the different single mechanisms can be described in an overall simplified approach as follows:

For design, this in consequence implies that either τ_a is the main contributor in the interaction formula, or τ_a becomes rather small (possibly representing some interlocking effects at very rough interfaces) or even disappears compared to the other mechanisms of shear friction and dowel action.

$$\tau_u = \underbrace{\tau_a}_{\text{Adhesion/interlock}} + \underbrace{\mu \cdot (\rho \cdot \kappa_1 \cdot f_y + \sigma_n)}_{\text{Shear friction}} + \underbrace{\kappa_2 \cdot \rho \cdot \sqrt{f_y \cdot f_{cc}}}_{\text{Dowel action}} \leq \beta_c \cdot v \cdot f_{cc} \quad (6.3-8)$$

where:

- ρ is ratio of reinforcement crossing the interface ($\rho = A_s/A_c$);
- β_c is a coefficient for the strength of the compression strut – see also Table 7.3-2;
- v is the effectiveness factor for the concrete – see also Eq. (7.3-51).

The interaction factors κ_1 and κ_2 in Eq. (6.3-8) take into account that

- the reinforcement or connectors are subject to bending and axial forces simultaneously;
- the maximum values of the different contributions occur with different slips.

Design values are given in subsection 7.3.3.6, Table 7.3-2.

Depending on the roughness, expected deformation characteristics (rigid/non-rigid bond–slip) and the anchorage of the connectors, the following scenarios can be differentiated:

Strong adhesive bond in combination with rather low amounts of reinforcement intersecting the interface ($\rho < 0.05\%$) implies $\tau_u \approx \tau_a$, effects of the reinforcement should then not be taken into account.

When adhesive bond is lost due to large shear stresses or contamination at the site, a certain number of connectors are required. With reinforcement or dowels intersecting the interface ($\rho \geq 0.05\%$), a factor $\kappa_1 = 0.5$ should be taken when the interface has been roughened. For smooth interfaces dowel action is the main resistance mechanism resulting from connectors, that is $\kappa_1 = 0$ and $\kappa_2 \leq 1.6$ (corresponding design values are given in subsection 7.3.3.6).

Additional design recommendations for interfaces subject to fatigue loading:

The main question concerning interface shear behaviour under fatigue loading is whether or not cracks along the interface are to be expected. In the case of monolithic behaviour (good bonding provided) the overall behaviour of the member can be judged according to subsection 7.4.1. However, as a simple approximation, a reduction of τ_a to 50% under fatigue loading may be assumed.

As soon as cracks appear, they tend to cause more significant deteriorations along the interface under dynamic loading than observed with monolithic concrete members. When cyclic shear loading is expected, an overall reduction of all contributing mechanisms to about 40% of the static resistance according to Eq. (6.3-8) is recommended, if no further evidence is available.

6.3.5 Detailing

Arrangement of connectors

Reinforcement or dowels intersecting the interface are required if the shear load in the interface under consideration cannot be resisted by the adhesive/interlocking effect of the roughened joint surface alone. A stepped distribution of the connectors with respect to the interface shear load distribution may be used. In the case of smooth surfaces, the flexibility of the connectors even allows for a redistribution of forces and thus a uniform distribution of the connectors.

For background information to Eqs. (6.3-7), (6.3-8) and appropriate parameters depending on different interface roughness characteristics, see Randl, N., Design recommendations for interface shear transfer in MC2010 (Structural Concrete, Vol. 14, No. 3, 2013).

Tests performed with large scale specimen (e.g. beam tests) indicate that with rough interfaces (good bonding provided) under dynamic loading usually no significant strength reduction occurs compared to monolithic beams subject to fatigue loading.

However, if the adhesive bond resistance is exceeded under characteristic load (or likewise ~50% under fatigue loading), cracks are likely to occur. In that case, due to the relative displacements in the crack, more significant deteriorations have to be considered under fatigue loading:

- reduction of mechanical interlocking;
- reduction of shear-friction due to an increasing amount of fine grains;
- local plasticizing and deterioration of concrete around reinforcing bars/connectors crossing the interface.

The recommended reinforcement ratio ρ_{min} has been derived based on the model that the shear force at loss of adhesion is taken up by shear-friction, dowel action and (only rough interfaces) partly by mechanical interlocking effects.

In the case of slabs, a total overall collapse of adhesion and interlocking effects does not have to be taken into account so that the minimum reinforcement may be reduced to ~50%.

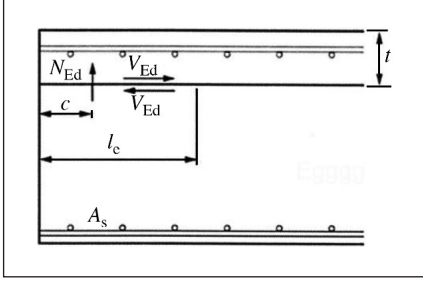


Figure 6.3-7: Edge reinforcement

At the perimeter of a new concrete layer, the concrete dries out and tends to contract, thereby producing tensile forces and delaminating perpendicular to the interface, leading eventually to cracking of the concrete. On the load side usually in practical design restraint forces along the perimeter are not taken into account because the realistic determination of the effects of differential shrinkage and temperature gradients is difficult without knowing all the influencing factors. Therefore appropriate edge reinforcement is highly recommended.

The thickness of the additional concrete overlay is usually limited and lower than the height of the existing concrete member. Moreover, especially in bridge design, the requirements concerning the rather large concrete covers for bridges have to be fulfilled. Therefore the connectors usually need a special kind of end anchorage at the upper end. This end anchorage has to be designed in such a way that premature pull-out or concrete cone breakout can be excluded.

The minimum anchorage length of a connector with a diameter $\varnothing$ in the existing concrete should not be less than $6\varnothing$ in the case of smooth and $10\varnothing$ in the case of rough interfaces.

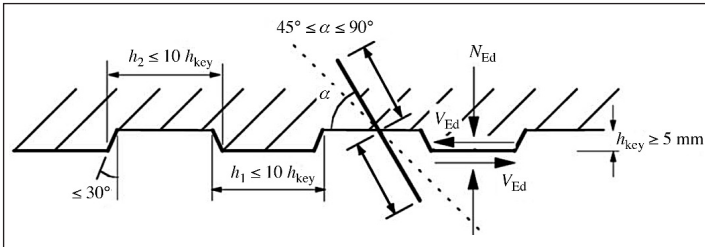


Figure 6.3-8: Shear key geometry

Minimum interface reinforcement

If interface reinforcement or dowels are required because the shear load in the interface under consideration cannot be resisted by adhesive bonding and interlocking effects alone, a minimum amount of reinforcement $\rho_{min} = A_{s,min}/A_c$ should be foreseen in order to prevent brittle failure at loss of adhesion:

$$\begin{aligned} \text{beam members:} \quad & \rho_{min} = 0.20 f_{ctm}/f_{yk} \geq 0.001 \\ \text{slabs:} \quad & \rho_{min} = 0.12 f_{ctm}/f_{yk} \geq 0.0005 \end{aligned}$$

Interface edge reinforcement

In order to counteract possible delaminating of the concrete overlay, reinforcement or dowels should be installed all along the perimeter if other adequate structural provisions are not taken. In the absence of more exact calculations, as an upper limit, the tensile cracking force of the new concrete overlay may be introduced as the maximum interface shear force to be expected along the edges (Fig. 6.3-7), and the reinforcement/connectors may be designed accordingly:

$$V_{Ed} = t \cdot b \cdot f_{ctd} \quad (6.3-9)$$

where:

- t is thickness of the new concrete layer;
- b is the width of the interface;
- f_{ctd} is the tensile strength of the new concrete layer (reduction due to early age effects might be taken into account).

Anchorage of interface connectors

The connectors have to be anchored appropriately in the old as well as in the new concrete in order to avoid premature pull-out failure.

If post-installed reinforcement or dowels are used in combination with the design approach based on the provisions given in subsection 6.3.4 and subsection 7.3.3.6, the tensile force to be anchored may be assumed as:

$$N_{Ed} = \kappa_1 \cdot A_s \cdot f_{yd} \quad (6.3-10)$$

For detailed design of the anchorage loaded by N_{Ed} the possible failure modes known for bonded anchors have to be checked (see also section 6.4).

Shear keys

A shear key should typically have proportions as recommended in Figure 6.3-8 in order to use the design values for very rough interfaces (Table 7.3-2). In addition, the base length h_1 of a key should be at least three times its height h_{key} ($h_1 \geq 3 \cdot h_{key}$).

Concrete to steel interfaces play a governing role with regard to the design of many hybrid structures. This section is intended to give general guidance.

Interaction between concrete and steel components can be classified as follows:

- adhesion (pure bond) between two materials;
- frictional interlock provided by peculiar shapes of the interface profile;
- mechanical interlock provided by specific treatments and deformations of the steel interface (i.e. indentations and embossments);
- dowel action provided by anchor devices and systems.

The choice of interaction mechanisms depends on the type of members and/or structure and on the type of loading.

Relevant standards can be referred to for design under gravity and horizontal forces.

Interaction between the interfaces aims at limiting interface shear slip, so that elastic and/or plastic analysis of cross-sections under both flexural and axial forces is allowed; thus this interaction strongly influences the overall structural response of members either in the load introduction zones or in the critical regions, where high internal forces develop.

Relevant standards apply to manufacture and design of steel sheeting as formwork; relevant rules are also available for the determination of the minimum nominal thickness of the steel sheeting.

An alternative solution that fits design requirements for composite members is represented by a sheeting with a re-entrant cross-section, so that frictional interface stresses can take advantage of lateral confinement actions due to shrinkage and flexural deformation.

Relevant standards can be used to make sure that the requirements concerning the aspect ratio of steel components and/or flanges depending on the steel grade are satisfied.

The requirements for the structural behaviour of interfaces between steel and concrete depend on the type of cross-section, the design level of composite interaction, and on the load introduction mechanisms and details adopted.

6.4 Concrete to steel

6.4.1 Classification of interaction mechanisms

Design of members with components made of concrete and steel, both structural and cold-formed, needs proper consideration of interaction mechanisms.

A variety of details exist, depending on the type of members to be connected, the actions to be transmitted and the design performance requirements.

A fundamental classification can be made according to the nature of interaction that is needed:

- mutual restraint between steel and concrete members and/or substructures;
- interconnection between components which allows steel and concrete components to behave as single structural members.

In the first case, steel members are connected to concrete via mechanical devices and generate localized actions in the concrete.

In the second case, stiffness and strength of the interface lead to the activation of the composite behaviour of the member and/or of the structure; the interaction can be both local and distributed.

6.4.2 Bond of metal sheeting and profiles

Combination of concrete and steel members requires the development of composite action that involves extended interface surfaces.

This applies to composite encased members and slabs made of metal sheeting and reinforced concrete slabs.

6.4.2.1 Metal sheeting

The use of metal sheeting is a common solution to produce composite slabs and meanwhile takes advantage of the loadbearing capacity of the cold-formed steel as temporary formwork supporting the hardening concrete in the construction stage.

Pure bond is not suitable to transmit the shear forces across the contact interface in order to develop the composite action to be realized, thus a plain (smooth) sheeting is basically inappropriate for composite applications.

Conversely, mechanical interlock provided by deformations in the steel profile, that is indentations and embossments, is an appropriate solution.

Mechanical connecting devices can be used in order to prevent end relative slip between the two components; deformation of end steel webs can be used as an alternative to additional steel devices when re-entrant profiles are used.

6.4.2.2 Steel profiles

A variety of solutions can be used to increase the performance of steel profiles under generalized (axial force, moment and shear) external actions.

The structural performance of members can be increased either by fully encasing the steel profile or by filling hollow steel sections.

In both cases, the effectiveness of the steel–concrete interface is critical in view of the development of composite action. Adequate structural performance is needed in order to ensure a sufficient composite behaviour of the members and prevent local buckling of steel components.

Pure bond is not suitable in the case of partially encased members subjected to flexure. Compressive members and composite columns

Relevant codes give detailed data concerning characteristic strength values to be used in specific design contexts.

can take advantage of such an interaction mechanism, so only segments inside the load transfer regions are critical.

6.4.2.3 Interface strength

The interface strength is dependent on the type of shear transmission mechanism, such as chemical adhesion, interface interlock and active friction.

Chemical adhesive bond between concrete and steel has a strength of around 0.1 MPa and occurs over the total area between the steel profile and the adjacent concrete. This contact area is higher in the case of trapezoidal ribs and low deformed ribs. Adhesion significantly drops as slippage starts and cannot be recovered. Until reaching the adhesion capacity full interaction exists, so that the theory of elasticity can be used to determine the interface shear stresses.

Interface interlock is related to local passive restraint given by embossments and indentations of the steel profile. Strength is then strongly influenced by the shape of the ribs, the thickness of the sheeting, the size and frequency of the embossments. The strength generated by this mechanism can range from zero for smooth trapezoidal ribs without embossment to about 0.8 MPa for deformed ribs with embossments. The strength levels are related to the area of the steel rib that is in contact with the concrete.

Active friction provides a contribution to the shear strength depending on the normal forces applied across the interface and by the surface treatment that govern the value of the friction coefficient. The coefficient of friction between steel and concrete can range from very low values, virtually zero, to about 0.6 when the interface is not greased.

Common values of active frictional strength are about 0.003 MPa, much lower than values provided by adhesion and interface interlock. However, its contribution generally is not negligible, since it develops all over the interface surface.

6.4.2.4 Shear stress–slip relationships

The shear stress–slip relation for the interface depends on the surface properties of the steel–concrete interface.

If indentations and surface treatments are not applied, the constitutive relationship is of a rigid plastic type: as soon as the chemical bond is exceeded, large shear-slip starts with a residual stress related to development of friction (curve A in Figure 6.4-1).

When indented interfaces are concerned, typical shear bond interaction exhibits a strength increase until the ultimate (peak) shear stress value and a consequent descending curve until a stabilized shear stress value has been reached due to friction interaction (curve B in Figure 6.4-1). During this phase the steel deck stiffness (related to thickness and rib height) plays an important role.

For design purposes, in analogy to similar interaction phenomena (i.e. bond of smooth reinforcing bars), in the theoretical stress–strain relations the peaking of the stress can be neglected and a linear elastic–plastic response, with a plastic plateau at the residual stress level, can be assumed.

6.4.2.5 Influence of the type of loading

The structural behaviour of the interface is influenced by the type of loading. A satisfactory assessment of the influence of repeated loading and sustained loading has not been fully established yet.

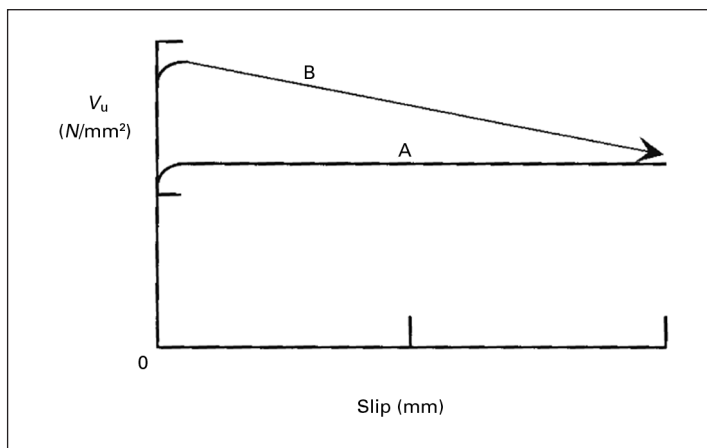


Figure 6.4-1: Shear stress–slip relationship for different types of sheeting

Requirements for flexural tests on full scale members (type 1) – for example assessing the interface behaviour between steel and concrete components – can be found in relevant codes.

The experimental setup of shear bond tests on small scale elements (type 2) is similar to those used for tests on connection devices. Average forces and slip measured at the steel–concrete interface enable the determination of local constitutive relationships. The latter depend on the interface normal stresses depending on the type of the composite member: thus specific force controlled devices have to be used in order to reproduce realistic interface stress conditions.

Relevant codes of practice have to be used for design and structural detailing of fixtures.

Reliable fastening applications require qualified and experienced designers. Installations should be carried out by experienced and skilled personnel, and proper maintenance of the structure of the fixtures and of anchorages should be provided. Furthermore, the specified use of fastenings should not be changed during their intended service life without recalculation.

Long term phenomena that develop in the concrete components should be taken into consideration, since they can influence the performance of the interface, especially when concrete filled members are used.

Repeated loading – that is fatigue or seismic actions – needs stress and slip limitations at the interface level and should be taken into careful consideration.

Damage tolerance at the interface level can become critical for specific applications.

6.4.2.6 Determination of properties by testing

Shear stress–slip constitutive relationships can be obtained from two main types of experiments:

- flexural tests on full scale members (type 1);
- shear bond tests on small scale elements (type 2).

6.4.3 Mechanical interlock

Mechanical interlock is generally used for load transfer into concrete components and structures or to connect elements.

Anchors can act either as single components or as groups when common fixtures are used. Load transfer mechanisms can be either statically determinate or statically indeterminate.

Construction drawings should clearly give at least the following information:

- location of the anchorage in the structure including tolerances;
- number and type of anchors, including embedment length;
- spacing and edge distance of the anchors, including (positive) tolerances;
- position of the attachments on the fixture including tolerances;
- maximum thickness of any non-bearing layers below the fixture;
- any special installation instruction.

6.4.3.1 Classification of devices

Fastening of steel to concrete may be classified as follows (Figure 6.4-2):

- cast-in-place devices;
- post-installed devices.

Cast-in-place devices are positioned in the formwork before the concrete is cast and thus can be used in members with dense reinforcement.

Post-installed systems may either be installed into drilled holes (drill installation) or be driven into the base material with impact energy (direct installation).

Different mechanisms can be activated to transfer tensile loads from the steel anchor to the surrounding concrete matrix:

- friction;
- mechanical interlock;
- dowel action;
- bond.

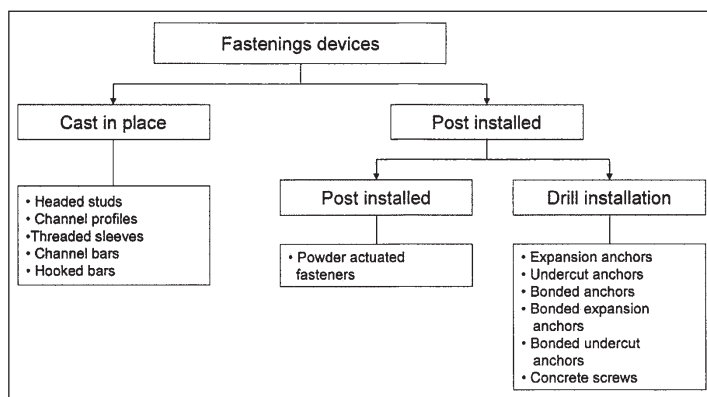


Figure 6.4-2: Summary of fastening techniques in concrete

Cast-in-place systems such as headed studs, hooked bars and channel bars transfer loads mainly by mechanical interlock. Similar mechanisms are activated in the case of undercut anchors, while in the case of expansion anchors, tensile forces are transferred to concrete basically by friction.

6.4.3.2 Strength evaluation

Strength of anchors depends basically on the type of loading. Axial loads are induced by fixtures subjected to axial forces and bending moments.

A summary of the most relevant failure modes of anchors is reported in Figure 6.4-3.

There are several types of failure modes which are exhibited by anchors when they are loaded beyond their capacity:

- steel fracture;
- concrete cone failure;
- concrete splitting;
- edge failure;
- bond failure, or pull-out (including slip or pull-through).

Strength and failure mode of anchors are related to the concrete strength, depth of embedment, loading type, loading direction, edge distance or anchor spacing.

Due to several causes tensile stresses in concrete can reach the concrete strength, so fasteners and connectors are often installed in cracked concrete regions. The influence of concrete cracking on the anchor performances depends on the type of anchor.

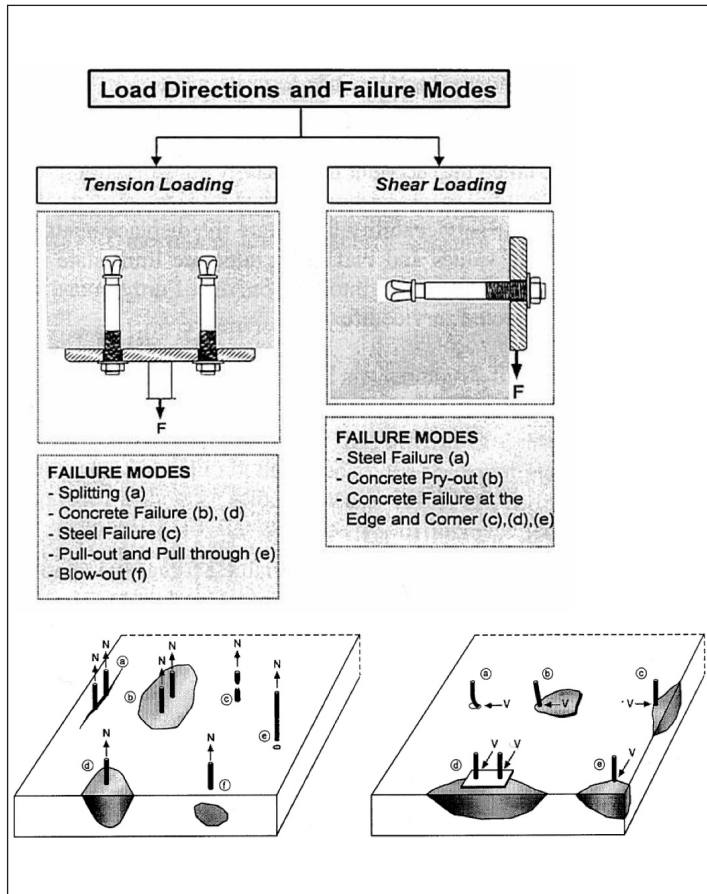


Figure 6.4-3: The most relevant failure modes of anchors

Design equations given by relevant codes have to be properly used depending on the types of anchors and the nature of the reinforced concrete components or members to be connected.

The force components perpendicular to the interface arise during the loading process of the composite members, but can play a secondary role if specific detailing of the devices and the connection system is provided.

Only fasteners with a predictable behaviour in cracked concrete should be used; the anchor suitability has to be proven on an experimental basis by means of pre-qualification tests carried out according to relevant codes.

Characteristic resistance of anchors can be based on the computation or test evaluation of the steel tensile and shear resistance, the concrete breakout tensile and shear resistance, the concrete splitting resistance and the tensile pull-out resistance of the anchors.

The models should take into account factors affecting strength such as embedment depth, spacing and edge distance, depth of structural members as well as the presence or the absence of concrete cracking. Limits on edge distance and anchor spacing in the design model should be consistent with the set of reference tests.

Interaction of tensile and shear loads should be considered in design, using interaction curves that result in a prediction of the strength in good agreement with results of comprehensive tests.

Connection devices are widely used in composite structures, where structural combination of concrete and steel is achieved basically through a shear type (parallel to the interface) force transmission. As a result, commonly fastenings in composite structures are defined as “shear connectors”.

Shear strength of headed stud connectors, which represent the most common device for composite structures, is basically derived from empirical correlations between relevant parameters and results of experimental programmes:

$$P = P(A, f_t, f_c, E_s, E_c) \quad (6.4-1)$$

It is worth noting that the strength of headed studs is influenced by:

- ultimate axial resistance of studs ($A f_t$);
- interaction between steel and concrete via the ratio between elastic moduli and resistances;
- the number of connectors; the factor $(5.3 - 1.3/\sqrt{n})$ depends on the number n of the studs in the group. Assuming a very large number of connectors (n tends to infinity) the factor 5.3 gives the mean shear strength of the connectors.

where:

- A is the connector equivalent cross-section;
- f_c is the compressive strength of the concrete;
- E_c is the modulus of elasticity of the concrete;
- f_t is the ultimate tensile stress in steel;
- E_s is the modulus of elasticity of steel.

Due to the very limited variation of E_s , the connector shear strength depends basically on four parameters.

A statistical analysis of data gives the following characteristic shear strength of headed stud connectors in solid slabs:

$$P_k = \left(5.3 - \frac{1.3}{\sqrt{n}} \right) \cdot A \cdot f_c^{0.35} \cdot f_t^{0.65} \cdot \left(\frac{E_c}{E_s} \right)^{0.4} \quad (6.4-2)$$

where f_c and f_t are in N/mm^2 and P_k is in N.

Eq. 6.4-2 was derived for the following ranges of parameters:

$10,000 \text{ MPa} \leq E_c \leq 33,000 \text{ MPa}$

$430 \text{ MPa} \leq f_t \leq 640 \text{ MPa}$

$24 \text{ MPa} \leq f_c \leq 81 \text{ MPa}$

Headed studs are furthermore characterized by an aspect ratio h_c/d and a height of the weld equal to $0.31 d_c$:

h_c = stud nominal height;

d = diameter of stud;

d_c = stud nominal diameter.

Collapse of concrete due to cracking and pull-out are not covered.

If composite decks are used, the studs are placed within a rib, and their performance is fairly different from that described for the previous case of a solid concrete slab. Completely different aspects concern the concrete stiffness, degree of confinement and the resistance mechanism of the studs, which are loaded eccentrically.

The prime parameters affecting the stud behaviour are:

- the orientation of the ribs relative to the beam span;
- the rib geometry as characterized by the b_r/h_r ratio;
- the stud height h_{sc} relative to the rib height h_r .

Available data are not suitable for the development of a comprehensive design method.

In codes, the effects of the main factors are accounted for via a suitable reduction factor that ranges between 0.4 and 1.0 depending on the geometry of sheeting and studs dimensions and location.

Relevant codes provide limits for the use of the relationship mentioned previously in terms of the ratio b_r/h_r and the height of the headed stud compared to height of the profiled sheeting (Figure 6.4-4).

6.4.3.3 Force-shear slip constitutive relationships

The strength of the connection between concrete and steel members may be influenced by the capacity of the anchors to redistribute loads among the devices connected by fixtures.

This emphasizes the relevance of the force–slip relationships on the overall performances of the composite substructure.

Structurally relevant aspects related to the force–shear slip relationships are:

- stiffness: a shear connection realizes either full interaction (the connection is “rigid” and no slip occurs under stress at the steel–concrete interface) or partial interaction (the connection is flexible and interface slip occurs);
- resistance: strength is required to the single connector or to the group of connectors in order to enable in a plastic approach the full transfer of forces related to the assumed structural mechanism;

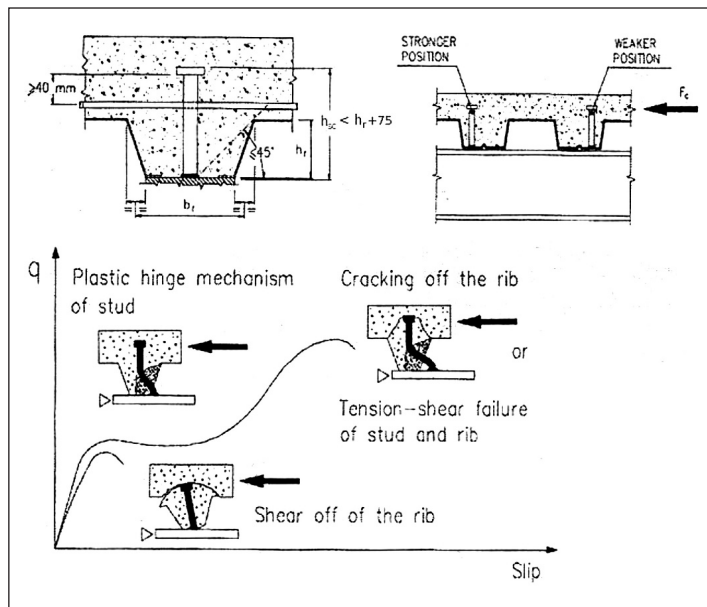


Figure 6.4-4: Sheeting geometry vs. headed stud performance.

The so-called full connection condition is obtained if the design ensures that the ultimate flexural capacity of the composite member (beam or slab) is reached before interface failure.

A partially composite structural element is characterized by the property that the resistance of the shear connection is reached before flexural failure of the composite member. The ratio F_c/F between the resistance of the shear connection and the minimum resistance required by the full connection condition defines the degree of shear connection.

Figure 6.4-5 shows a number of force–slip relationships for shear connectors commonly used in composite construction.

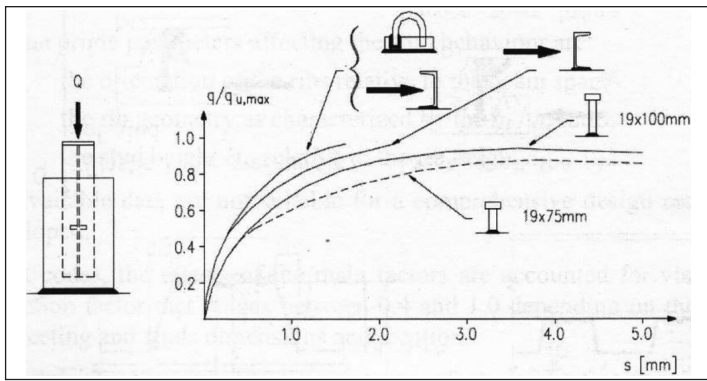


Figure 6.4-5: Non-dimensional force slip relationships for shear connectors

Generally, all types of mechanical shear connectors possess a limited deformation capacity. In general, the associated shear slip is sufficient to develop the design flexural resistance and rotation capacity of the composite section.

Relevant codes give provisions about ductility requirements related to span and the degree of shear connection and to the characteristic values of slip capacity for design.

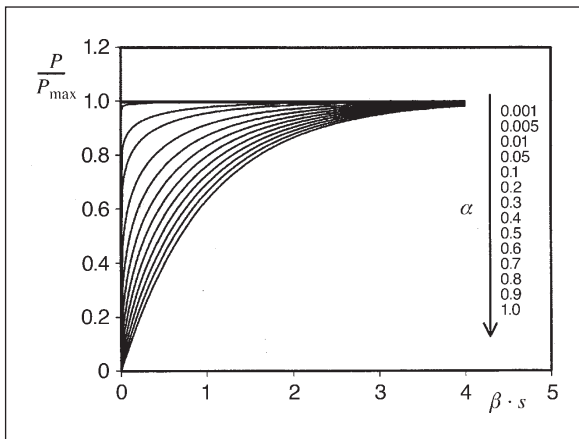


Figure 6.4-6: Force–slip relationship depending on parameters α and β

- ductility: a connection is ductile if its deformation (shear and/or pull-out) capacity is adequate for a complete redistribution of the forces acting on the individual connector. In composite structures, the ductility demand depends on the member length and the degree of shear connection.

Modelling the force–slip relationship can be carried out by evaluating specific tests on connectors that can be influenced by interface greasing of the steel profiles. In fact, due to chemical adhesion, force–slip relationships can show an initial rigid (zero slip) branch and then a very steep curve. Conversely, the presence of grease at the interface enables the generation of a response completely depending on the dowel action.

If the deformation of headed connectors in solid slabs is concerned, four relevant parameters can be mentioned:

- strength of the stud, given in the previous section;
- ultimate slip s_u , corresponding to the maximum load P of the stud;
- service slip s_s , corresponding to a half of the maximum load;
- slip capacity s_c evaluated on the post-peak descending branch of the relationship at 95% of the peak shear load.

Statistical analyses, in analogy with the results discussed for the stud resistance, give the following relationship for the ultimate slip and the slip development respectively:

$$s_u = (0.389 - 0.0023 \cdot f_c) \cdot d_c \quad (6.4-3)$$

$$s_c = (0.453 - 0.0018 \cdot f_c) \cdot d_c \quad (6.4-4)$$

The results are relevant for a range of concrete strengths between 20 MPa and 70 MPa, d_c being the stud nominal diameter.

For refined modelling of composite construction advantage can be taken of theoretical curves that represent the force–slip relationships of studs (Figure 6.4-6):

$$P(s) = P_{\max} \cdot (1 - e^{-\beta \cdot s})^\alpha \quad (6.4-5)$$

where:

- s is the generic slip (mm);
- $P_{\max}$ is the stud resistance;
- α and β are constants that enable a characterization of the curve.

In particular, parameter α governs the initial stiffness, and parameter β influences the shape of the curve by a proper scaling of the slip.

Theoretically α can range between zero and unity – Figure 6.4-6.

Based on experimental results, α ranges between 0.5 and 1, while β ranges between 0.7 and 1.5.

An alternative formulation can be derived from the well-known bond–slip relationship of reinforcing bars in concrete, that fits properly the basic requirements of experimental force–slip relationships:

$$P(s) = P_{\max} \cdot \left(\frac{s}{s_u} \right)^\gamma \quad (6.4-6)$$

The shape of the curve is governed by exponent γ , that plays a role similar to parameter α used in Eq. 6.4-5 for the slip at peak load, that is conceptually analogous to the product $\beta \cdot s$, which represents an equivalent slip s_{eq} , but enables an easier interpretation.

With regard to serviceability and deflection analysis of composite members, a linear approximation of the force–shear-slip relationship can be used. Different definitions of linear stiffness of connectors exist; a possible evaluation can be based on the secant stiffness evaluated in correspondence with the slip and the shear force under serviceability conditions, at about 50–60% of the resistance.

6.4.3.4 Influence of the type of loading

Fatigue loading of the structural members serving as base material or of the anchorage may be allowed for certain anchors, if this is stated in the relevant approval certificate or if it has been shown in the pre-qualification procedure that fatigue loads can be sustained by the anchor. In both cases the corresponding conditions (e.g. a permanent prestressing force of sufficient magnitude) and the allowable load should be met in the design.

6.4.3.5 Determination of properties by testing

Relevant standards can cover the need for design equations that relate material and geometrical parameters to the strength of mechanical devices.

Since the equations do not always cover the selected anchors, experiments can supply design data.

The variables to be investigated include the geometry and the mechanical properties of the concrete slab, the connectors and the reinforcement.

The resistance to types of loading, other than fatigue, may be determined by specific tests in accordance with relevant standards.

In general, the test rigs should allow the formation of an unrestricted rupture cone. For this reason the clear distance between the support reaction and an anchor (single anchor) or an outer anchor (anchor group) must be at least twice the effective depth of the anchor or twice the distance between the anchor and the edge in the direction of the load.

During all tests, the load must be applied to the anchor by a fixture representing the conditions found in practice.

In tests on single anchors where there are no influences of edge and spacing, the centre-to-centre distance and the distances from free edges must be large enough to allow the formation of an unrestricted rupture cone of vertex angle 120° in the concrete.

In the case of shear connectors for composite structures, specific push-off tests should be carried out such that the slabs and the reinforcement are suitably dimensioned in comparison with the beams for which the test is developed.

Thus, the length l of each slab should be related to the longitudinal spacing of the connectors in the composite structure; the width of each slab should not exceed the effective width of the beam's slab; the thickness of each slab should not exceed the minimum thickness of the slab in the member.

The applied load and the relative displacements between the single anchor and the base concrete component should be measured. The slip should be measured at least until the load has dropped down to 20% of the maximum load. Average displacements can be used as reference values for the force–slip relationships.

The slip capacity of a specimen should be taken as the maximum slip measured at the characteristic load level. The characteristic slip capacity should be taken as the minimum test value of the slip capacity reduced by 5% or determined by a statistical evaluation of all the test results.

7 Design

7.1 Conceptual design

7.1.1 General

The conceptual design stage is the most important phase of a project. Without an idea, without a proper solution to the problem under study there is no established safety concept, no adequately defined behaviour and essentially no solution to the defined problem, without which a successful construction project cannot be realized.

Many iterations of the design process are commonly required to refine the design concepts to accord with the functional requirements and associated financial/other constraints. The analytic tools applied at this stage to the investigation of the problem and evaluation of potential options may be relatively crude.

Further background information and illustrations on this topic can be found in Corres-Peiretti, H. (2013), Sound engineering through conceptual design according to the *fib* Model Code 2010. Structural Concrete, 14: 89–98. doi: 10.1002/suco.201200042.

The conceptual design stage is the period during which identified needs are examined, requirements for potential solutions are defined, potential solutions are evaluated and a suitable structural concept for further design is developed.

The basic approach to design relies on decomposition and integration. Since design problems are large and complex, they have to be decomposed into sub-problems that are small enough to solve. There are numerous alternative ways to decompose design problems, such as decomposition by functions of the facility, by spatial locations of its parts, or by links among various functions or parts. Solutions to sub-problems must be integrated into an overall solution. The integration and rationalization process often creates conceptual conflicts which must be identified and resolved.

Various ideas for solving the problem under study, taking into account the owners programme and the stakeholders' expectations, are produced during the conceptual design stage, with one that complies in an optimal manner with the specified requirements. These ideas, even though lacking in detail, must describe the solution from the points of view of functionality, environmental integration (physical, social and historical), structural adequacy, sustainability, construction, economy and so on. This phase should identify the more critical aspects which need to be more thoroughly developed in the following stages of the design process.

7.1.2 Methodology

Conceptual design is a creative act for which it is not easy to establish a methodology. Figure 7.1-1 illustrates a process which may provide some insight and be of assistance for carrying out this activity.

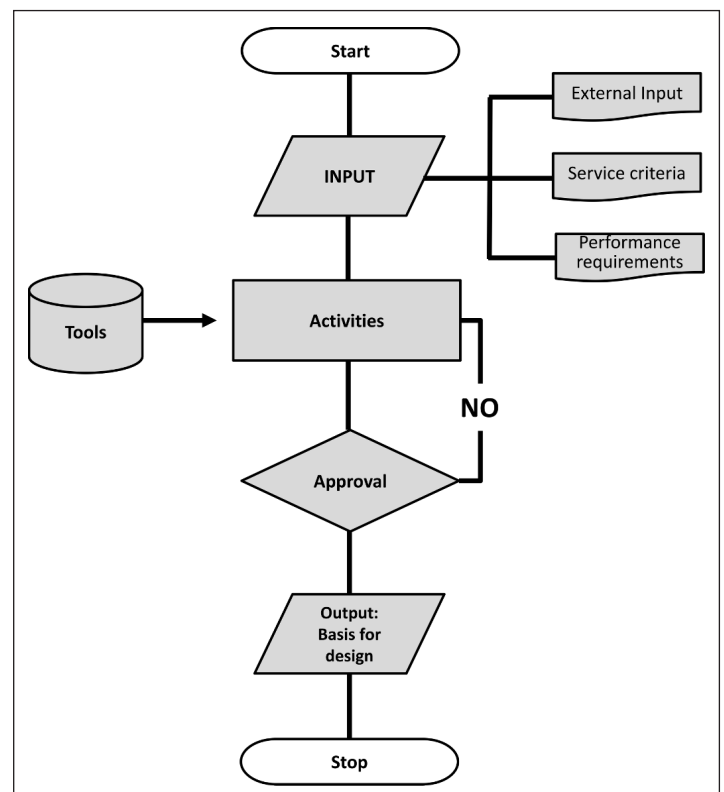


Figure 7.1-1: Methodological flowchart for conceptual design

7.1.2.1 Input

Initial information must be established with regard to:

- general criteria;
- basic external input data;
- service criteria;
- performance requirements.

General criteria

- owner and stakeholder requirements and expectations;
- functionality;
- aesthetics;
- environmental integration (physical, social and historical);
- sustainability;
- structural adequacy;
- feasibility;
- economy;
- etc.

Basic external input data

- basic data applicable, including third party interactions (geotechnical data, metocean data, topographical and bathymetrical data, climatological data, environmental data (earthquake, hurricanes), material properties, accessibility and transport facilities, local construction rules, in the case of buildings architectural requirements, such as form, mass/volume, surface texture and colour.)

Service criteria

- general aims for the use of the construction works (efficiency, comfort, safety etc.);
- operational and maintenance requirements (efficiency, economy etc.);
- special requirements of the stakeholders (upgrading, replacement etc.);
- objectives of protection and special risks;
- loadings and loading combinations;
- environmental conditions;
- codes and regulatory requirements.

Performance requirements

- performance criteria for serviceability and safety (including durability and robustness);
- service life constraints (temporary, replaceable, evolutive, long term);
- reliability constraints;
- performance requirements for sustainability.

7.1.2.2 Activities

If the basic external input data is not available to the designer, a process will need to be established so that it can be obtained either from the owner, the architect, the authorities or some other source, or via an appropriate process instigated by the designer. Basic data must be clearly specified in the Service Criteria Agreement, see subsection 3.5.3.5.

The service criteria have to be discussed and established with the owner or the architect; it must be approved by all and be clearly specified in the Service Criteria Agreement, see subsection 3.5.3.5.

The performance requirements have to be established, proposed and explained by the designer, in conjunction with the owner, and must be clearly specified in the Service Criteria Agreement, see subsection 3.5.3.5.

In general, activities performed during the stage of conceptual design of construction works are related to:

- constraint analysis and classification;
- environment analysis (including local politics and local traditions);
- general conception;
- choice of materials (considering economy and energy consumption for production and elimination);
- structural concept (structural logic, dimensions);
- integration and aesthetics (legibility, simplicity, proportions, equilibrium, shapes, detail philosophy);
- construction method (sequences);

The conceptual design process can be characterized by a series of interactive activities:

- formulation, which refers to the definition or description of a design problem in broad terms, through the synthesis of ideas describing alternative concepts;
- analysis, which refines the problem definition or description by separating important from peripheral information and by pulling together the essential detail; interpretation and prediction are usually required as part of the analysis;
- search, which involves gathering a set of potential solutions for performing the specified functions and satisfying the user requirements;

- rough cost estimate;
- comparison of alternatives;
- successive presentation, explanation and discussions with the owner (architect);
- after acceptance by the owner – preparation of the basis for design (drawings, notes, reports).

Attributes and tools such as the following may be employed during the conceptual design stage:

- experience, plus insight from background, feedback, database sources;
- intuition, feeling, sensitivity for the circumstances;
- creativity, imagination, capacity of simultaneous analysis and integration of diverse criteria and constraints taking into account their relative weights;
- quick pre-design methods;
- development of ideas, concepts and design details by sketching (ranging from rough freehand sketches to accurate drawings);
- visualization tools.

It is always interesting to involve experts from other relevant fields, depending on the type of structures (architects, urban planners, landscape artists, archaeologists, historians, etc.).

The extent and content of the basis for design must be adapted to the importance of the construction works and the associated hazards and environmental risks, but it must always exist no matter how minor the project might be considered to be.

- decision, which means that each of the potential solutions is evaluated and compared to the alternatives until the best solution is obtained;
- specification, which is to describe the chosen solution in a form which is detailed enough for implementation;
- modification, which refers to the change in the solution or re-design if the solution is found to be insufficient or if new information is discovered in the process of design.

7.1.2.3 The role of expertise, insight and tools

The conceptual design process aims to find acceptable solutions for the defined requirements, constraints and the associated opportunities provided by the circumstances. The process is guided by the experience gathered in comparable construction works, along with insight and intuition obtained in other relevant circumstances. A variety of tools and aids may be used to assist the process, including those for visualization of candidate schemes and alternative options, basic dimensioning of elements, preliminary evaluation of economic outcomes, etc.

7.1.3 Structural concept and basis for design

The structural concept derived from the conceptual design includes:

- the chosen structural system;
- information on the most important dimensions, construction material properties and construction details;
- comments on the envisaged methods of construction.

The structural concept derived from the conceptual design must be described in the basis for design, including the bases and requirements for the subsequent design, execution, use and conservation.

The basis for design describes:

- the design working life;
- the service situations considered;
- the hazard scenarios considered;
- the requirements of structural safety, serviceability and durability together with the measures needed to guarantee them, including division of responsibilities, processes, controls and corrective mechanisms;
- the assumed ground conditions;
- the important assumptions in the structural and analytical models;
- the accepted risks;
- other conditions relevant to the design.

In particular cases, structural models may be based on experiments carried out in relation to the particular design or on a combination of testing and analytical calculations.

Typical D-regions are regions with static and/or geometric discontinuities, where plane sections do not remain plane after deformation, such as deep beams, beams/columns/slabs with geometrical discontinuities, bridge diaphragms, foundations and pile caps. Other examples of typical D-regions are the areas where structural components are connected (e.g. beam column, load introduction areas and supports).

Imposed deformations can result from differential settlements, temperature gradients or differences in humidity or from seismic actions.

With regard to the theory of plasticity, both the upper bound and the lower bound theorems of plasticity can be applied.

The application of the lower bound theorem of plasticity implies that a safe bearing mode is found, if a statically admissible bearing system applies in which, under the actions defined, the admissible stresses are nowhere exceeded. Examples of such systems are strut and tie models and the strip method, used for the design of slabs. The solutions found can be more or less economic, but represent a lower bound for the bearing capacity.

The application of the upper bound theorem of plasticity requires the adoption of a pattern of yield lines, generating a kinematic mechanism. The pattern that fails at the lowest load represents the bearing capacity. This method is particularly valuable for finding the bearing capacity of existing structures.

An example of a case where a detailed investigation is necessary in the SLS is crack width control for leakage in a statically indeterminate structure. Examples of cases for the ULS are the control of rotation capacity and the determination of the magnification factor for verifying the stability of a structure.

7.2 Structural analysis and dimensioning

7.2.1 General

Structural analysis starts with a general determination of the actions in the structure, tracing the flow of forces and moments through the structure as a whole.

Structural analysis, both on a general and a more detailed level, comprises the determination of action effects such as internal stress fields, forces and moments, support reactions and deformations carried out on the basis of a structural model. To that aim, the structure can be subdivided into components, such as beams, slabs, walls and shells and connecting areas, like B- and D- regions. In B-regions the forces and moments vary gradually. The assumption that the sections remain plane after deformation (hypothesis of Bernouilli) is valid. In D-regions the forces and moments vary discontinuously, so the hypothesis of Bernouilli is no longer valid.

Analyses have to be carried out using idealizations of both the geometry and the behaviour of the structure. Idealizations must be appropriate to the case considered.

The effect of geometry and the properties of the structure and its behaviour at each stage of construction and service has to be considered in the design. Second order effects have to be taken into account where they are likely to affect the overall stability of a structure significantly and for the attainment of the ultimate limit state at critical sections.

The internal forces and moments in a structure follow from a system of loads or from imposed deformations or from a combination of both.

Internal forces, moments and deformations in statically indeterminate structures may be determined based on:

- theory of linear elasticity;
- theory of linear elasticity with limited redistribution;
- theory of plasticity;
- non-linear methods.

The effect of time dependent behaviour of concrete (creep and shrinkage) must be accounted for according to the guidelines of subsection 7.2.4. Proper consideration must be given to the effect of relaxation of steel.

Except for seismic action, the effects of imposed deformations may be neglected in verifying structural safety if an adequate deformation capacity is ensured for all parts of the structure.

If detailed investigations are necessary for the determination of forces and moments in the serviceability limit state or the ultimate limit state (ULS), an analysis can be carried out with adequately reduced stiffness of structural areas due to cracking.

7.2.2 Structural modelling

7.2.2.1 General

The static and geometrical boundary conditions as well as the transmission of support reactions must be taken into account when idealizing and delimiting the system.

Soil structure interaction must be considered appropriately.

Deviations in cross-sectional dimensions are normally taken into account in the material safety factors. Therefore these need not be included in structural analysis.

In the case of bridge piers or highly stressed building columns, the inclination resulting from an unintended base inclination can be of importance for the dimensioning of the bracing structural members (e. g. floor slabs, bracings of buildings, bridge bearings). The effect of unintended inclination must be estimated and if necessary taken into consideration in the calculations.

The definitions of l and m in Eqs. (7.2-1) and (7.2-2) depend on the effect considered, for which three principal cases can be considered:

- effect on the single element: l = real length of structural member, $m = 1$;
- effect on stabilizing structure: l = height of building, m = number of vertical structural members that contribute to the horizontal force on the stabilizing structure;
- effect on floors or roofs which transmit horizontal loads: l = storey height, m = number of vertical elements in the storeys that contribute to the total horizontal force on the floor considered.

7.2.2.2 Geometric imperfections

The unfavourable effects of possible deviations in the geometry of the structure and the position of the loads have to be taken into account in the analysis of members and structures.

Imperfections must be taken into account for the verification of the ultimate limit state for persistent and accidental design situations. In the case of slender compression members, the second order effects and the influences of creep of concrete must be taken into account (subsection 7.3.7).

Imperfections need not be considered for the verification of the serviceability limit state.

Unless specified otherwise in the basis of design, the unintended inclination α_i of vertical compression members amounts to:

$$\frac{1}{200} \geq \alpha_i = \frac{0.01}{\sqrt{l}} \geq \frac{1}{300} \quad (l \text{ in m}) \quad (7.2-1)$$

where:

l denotes the height of the compression member or compression members standing on top of one another.

In buildings, the average unintended inclination α_{im} of a group of vertical compression members can be estimated from:

$$\alpha_{im} = \alpha_i \sqrt{0.5(1.0 + \frac{1.0}{m})} \quad (7.2-2)$$

where:

m denotes the number of compression members which have to be included in determining the effect of unintended inclination – see Figure 7.2-1.

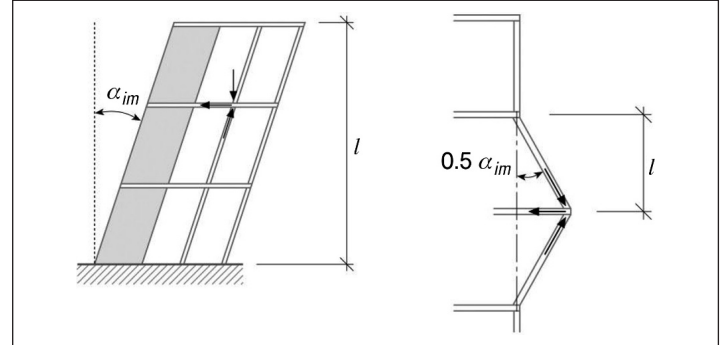


Figure 7.2-1: Geometrical imperfections

7.2.2.3 Structural geometry

For structural analysis, the structure has to be idealized using suitable models; examples are plane or space frames and B- and D-regions of structural components.

In the case of T-beams, the effective slab width depends on the web and the flange dimensions, the type of action, the span and the support conditions. The effective slab width may be estimated (Figure 7.2-2) from:

$$b_{eff} = \Sigma b_{eff,i} + b_w \leq b \quad (7.2-3)$$

where:

$$b_{eff,i} = 0.2b_i + 0.1l_0 \leq 0.2l_0 \quad (7.2-4)$$

The distance l_0 between the points of zero moment may be determined for usual cases according to Figure 7.2-3, based on the following assumptions:

- the cantilever length is less than half the adjacent span;
- the ratio between adjacent spans is between 2/3 and 1.5.

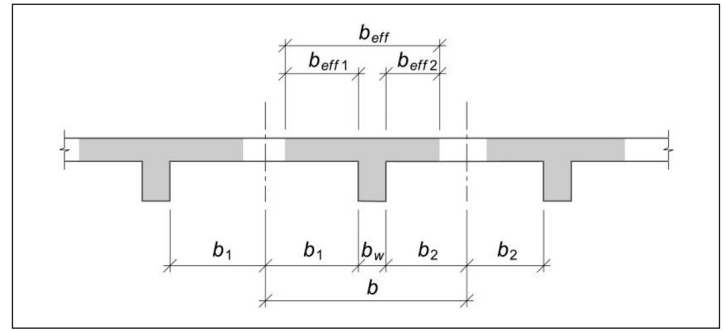
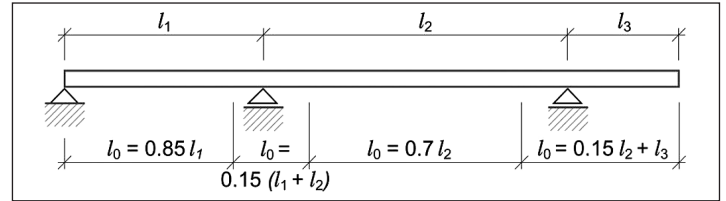


Figure 7.2-2: Effective slab width

Figure 7.2-3: Relevant distances l_0 for the determination of the effective slab width

7.2.2.4 Calculation methods

7.2.2.4.1 Analysis based on linear elasticity

This approach implies that the “response relationship” is linear, and the assumption of reversible deformations is retained. The results are realistic only under the assumption that actions are low and members are uncracked.

For ULS verifications, existing practice allows the use of linear elastic analysis without direct verification of sufficient ductility. This is based on the assumption that there is enough ductility to balance the lack of compatibility. The method is normally used in combination with the uncracked cross-section of concrete members; therefore it requires a definition of the geometry of the structure, but not necessarily of the reinforcement.

Cracked cross-sections may, however, be used if, in the limit state under consideration, a fully developed crack pattern can be expected. The results of a linear analysis are also used in the verification for the serviceability limit state.

Analysis of elements based on the theory of linear elasticity may be used for both the serviceability and the ultimate limit states.

For the determination of the action effects, linear elastic analysis may be carried out assuming:

- uncracked cross-sections;
- linear stress–strain relationships;
- the mean value of the modulus of elasticity.

For determining the effect of imposed deformations at the ultimate limit state a reduced stiffness corresponding to cracked sections may be assumed. For the serviceability limit state, a gradual evolution of cracking should be considered.

7.2.2.4.2 Analysis according to linear elasticity with limited redistribution

If redistribution of moments is applied in determining the reinforcement this may have an influence on deflection and crack width.

Linear analysis with limited redistribution may be applied to the analysis of structural members for the verification at the ULS.

The influence of any redistribution of moments on other aspects of design has to be considered.

The moments at the ULS calculated using a linear elastic analysis may be redistributed, provided that the resulting distribution of moments remains in equilibrium with the applied loads.

Redistribution of bending moments without explicit check on the rotation capacity is allowed for continuous beams or slabs which are predominantly subjected to flexure and have a ratio of the lengths of adjacent spans in the range of 0.5 to 2. In this case the following relations should apply:

$$\delta \geq k_1 + k_2 x_u / d \quad \text{for } f_{ck} \leq 50 \text{ MPa} \quad (7.2-5)$$

$$\delta \geq k_3 + k_4 x_u / d \quad \text{for } f_{ck} > 50 \text{ MPa} \quad (7.2-6)$$

and

$\delta \geq k_5$ where class B, class C or class D reinforcement is used, see subsection 5.2.5.4;

$\delta \geq k_6$ where class A reinforcement is used, see subsection 5.2.5.4;

where:

δ is the ratio of the redistributed moment to the elastic bending moment

x_u is the depth of the neutral axis at the ULS after redistribution;

d is the effective depth of the section;

$k_1 = 0.44$;

$k_2 = 1.25(0.6 + 0.0014/\varepsilon_{cu2})$;

$k_3 = 0.54$;

$k_4 = 1.25(0.6 + 0.0014/\varepsilon_{cu2})$;

$k_5 = 0.7$;

$k_6 = 0.8$;

ε_{cu2} is ultimate strain according to subsection 7.2.3.1.1.

Redistribution should not be carried out in circumstances where the rotation capacity cannot be defined with confidence (e.g. in corners of frames with opening moments).

For the design of columns it should be checked whether the moment before redistribution is governing for the design.

7.2.2.4.3 Theory of plasticity

General

Plastic analysis should be based either on the lower bound (static) or the upper bound (kinematic) theorem.

When applying methods based on the theory of plasticity it should be ensured that the deformation capacity of critical areas is sufficient for the envisaged mechanism to be developed.

The effects of previous applications of loading may generally be ignored and a monotonic increase of the intensity of the actions may be assumed.

Plastic analysis of beams, frames and slabs with the kinematic theorem

Plastic analysis without any check of the rotation capacity may be used for the ultimate limit state if all the following conditions are met:

- the area of tensile reinforcement is limited to such a value that at any section
 - $x_u/d \leq 0.25$ for concrete strength classes $\leq C50$;
 - $x_u/d \leq 0.15$ for concrete strength classes $\geq C55$;
- reinforcing steel is either Class B or C;
- the ratio of the moments at intermediate supports to the moments in the span is between 0.5 and 2.

Columns should be checked for the maximum plastic moments which can be transmitted by connecting members. For connections to flat slabs this moment should be included in the punching shear calculation.

When plastic analysis of slabs is carried out, account should be taken of any non-uniform reinforcement, corner tie down forces and torsion at free edges.

Rotation capacity

If continuous beams or continuous one-way slabs do not meet the conditions for which no check of rotation capacity is required, as formulated previously, a simplified procedure can be used. This procedure is based on a control of the rotation capacity. The rotation capacity is determined over a length of approximately 1.2 times the depth of the section. It is assumed that these zones undergo a plastic deformation (formation of yield hinges) under the relevant combination of actions. The verification of the plastic

For the application of the simplified procedure, it is supposed that the plastic hinge considered occurs as the first of those resulting in a kinematic system.

If the demand of rotation θ_s is calculated by integrating the curvatures between the plastic hinges, in general the application of a trilinear moment–curvature relation is appropriate. The moments caused by prestressing should be considered as a part of the effect of the load on the structure.

rotation in the ultimate limit state is considered to be fulfilled, if it is shown that under the relevant combination of actions the demand in rotation of the plastic hinge, calculated θ_s is smaller than or equal to the allowable plastic rotation $\theta_{pl,d}$ (Figure 7.2-4).

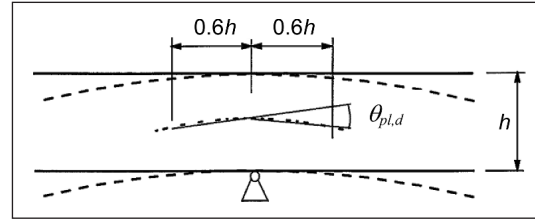


Figure 7.2-4: Rotation capacity $\theta_{pl,d}$ of reinforced concrete sections for continuous beams and continuous one-way spanning slabs

In regions of yield hinges, x_u/d should not exceed the value 0.45 for concrete strength classes less or equal to C50/60, and 0.35 for classes higher than or equal to C55/67.

The rotation $\theta_{pl,d}$ should be determined on the basis of design values for action and materials and on the basis of mean values for prestressing at the relevant time.

According to the simplified procedure, the allowable plastic rotation may be determined by multiplying the basic value of allowable rotation $\theta_{pl,d}$ by a correction factor k_λ that depends on the shear slenderness. The values for the allowable rotation for the steel classes B and C (the use of steel class A is not recommended for plastic analysis) and concrete strength classes below or equal to C50/60 and C90/105 are given in Figure 7.2-5.

In Figure 7.2-5 the x_u values are calculated considering design values of material properties, bilinear elasto-plastic stress-strain response of steel with a strain limit of $\epsilon_s = \epsilon_{ud}$ and the behaviour of concrete under compression simplified with a parabola-rectangle stress-strain diagram with nominal strain limits according to Figure 7.2-9.

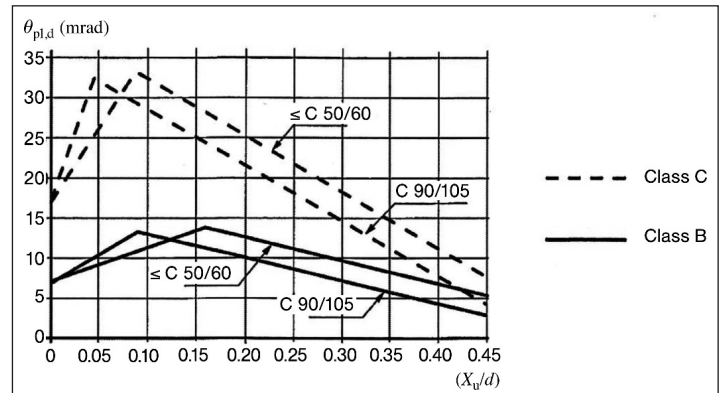


Figure 7.2-5: Basic values for allowable rotation $\theta_{pl,d}$ of reinforced concrete sections for class B and C reinforcement. The values apply for a shear slenderness $\lambda = 3.0$

The values for the strength classes C50/60 to C90/105 may be interpolated accordingly.

The values in Figure 7.2-5 apply for a shear slenderness $\lambda = 3.0$. For different values of the shear slenderness $\theta_{pl,d}$ should be multiplied by:

$$k_\lambda = \sqrt{\lambda / 3} \quad (7.2-7)$$

where:

λ is the ratio of the distance between the point of zero and maximum moment after redistribution and the effective depth d .

Analysis with strut and tie models

Analysis with a strut and tie model is a method according to the lower bound theorem of plasticity (static method). According to this method a state of equilibrium between external and internal forces

Orienting the struts to the compressive stress trajectories in the uncracked state, assuming linear elastic behaviour, aims at minimizing redistribution of forces after cracking, which could result in violation of service requirements (crack width and deformation) or even premature failure. If the redistribution of forces is minimized, for SLS and ULS the same strut and tie model is appropriate.

has to be found, which fulfils the static boundary conditions and nowhere violates the yield conditions.

Strut and tie models may be used for design of the reinforcement in continuity regions (B-regions) in the ULS (cracked state of beams and slabs) and for the design and detailing of discontinuity regions (D-regions) in the ULS. In general, D-regions extend up to a distance h from the discontinuity, where h is the largest cross-sectional dimension.

Verifications in the SLS may also be carried out using strut and tie models, for example for verification of steel stresses and crack width control, if approximate compatibility for strut and tie models is ensured; in particular the position and direction of important struts should be oriented according to the compression trajectories in the linear elastic stage.

Strut and tie models consist of struts representing compressive stress fields, of ties representing the reinforcement and of connecting nodes. Both for ULS and SLS calculations, the struts should be oriented to the compressive stress trajectories in the uncracked stage based on linear elastic behaviour. The forces in the elements of a strut and tie model should be determined by maintaining the equilibrium with the applied loads in the ultimate limit state. The elements of strut and tie models should be dimensioned according to the rules given in subsection 7.3.6.

The ties in a strut and tie model should coincide in position and direction with the corresponding reinforcement.

7.2.2.4.4 Non-linear analysis

Non-linear methods of analysis may be used for both ULS and SLS, provided that equilibrium and compatibility are satisfied and adequate non-linear behaviour for materials is assumed. The analysis may be first or second order.

Non-linear analysis should be carried out on the basis of the principles given in section 7.11. For predicting the mean behaviour mean values of the material characteristics should be used as defined in section 5.1.

7.2.3 Dimensioning values

7.2.3.1 Concrete

7.2.3.1.1 Strength and strain characteristics

The compressive strength of concrete is denoted by concrete strength classes which are related to the characteristic (5%) cylinder strength f_{ck} or the cube strength $f_{ck,cube}$ at an age of 28 days.

The characteristic strengths for f_{ck} and the corresponding mechanical characteristics necessary for design are given in Table 7.2-1.

Concrete can be tailored for particular applications. In that case the relation between the compressive strength of the concrete and relevant mechanical properties may deviate from the relations found in Table 7.2-1. In such case, different relations can be obtained by testing.

In certain situations (e.g. prestressing) it may be appropriate to assess the compressive strength of the concrete before or after 28 days. General relations between the compressive and tensile strengths of the concrete at other ages than 28 days are given in section 5.1.

The tensile strength of concrete can be determined directly by a uniaxial tensile test or by a splitting tensile test. Where the tensile strength is determined as the splitting tensile strength $f_{ct,sp}$ an approximate value of the axial tensile strength f_{ct} is found from:

$$f_{ct} = 1.0 f_{ct,sp} \quad (7.2-8)$$

The mechanical characteristics given in Table 7.2-1 represent average values on the basis of the concrete compressive strength for a wide range of compositions.

General rules for design by testing are given in section 7.13.

Axial tensile tests are very sensitive to the way the test is carried out. For further information on axial tensile testing of concrete, see van Mier, J. G. M., van Vliet, M.R. "Uniaxial tension test for determination of fracture parameters of concrete: state of the art", Engineering Fracture Mechanics, Vol. 69, Issue 2, Jan. 2002, pp. 235–247.

See also subsection 5.1.5.

Table 7.2-1 Strength classes for concrete

Concrete grade	C12	C16	C20	C25	C30	C35	C40	C45	C50	C55	C60	C70	C80	C90	C100
f_{ck} (MPa)	12	16	20	25	30	35	40	45	50	55	60	70	80	90	100
$f_{ck,cube}$ (MPa)	15	20	25	30	37	45	50	55	60	67	75	85	95	105	115
f_{cm} (MPa)	20	24	28	33	38	43	48	53	58	63	68	78	88	98	108
f_{ctm} (MPa)	1.6	1.9	2.2	2.6	2.9	3.2	3.5	3.8	4.1	4.2	4.4	4.6	4.8	5.0	5.2
$f_{ctk,0.05}$ (MPa)	1.1	1.3	1.5	1.8	2.0	2.2	2.5	2.7	2.9	3.0	3.1	3.2	3.4	3.5	3.7
$f_{ctk,0.95}$ (MPa)	2.0	2.5	2.9	3.3	3.8	4.2	4.6	4.9	5.3	5.5	5.7	6.0	6.3	6.6	6.8
E_{cm} (GPa)	27.1	28.8	30.3	32.0	33.6	35.0	36.3	37.5	38.6	39.7	40.7	42.6	44.4	46.0	47.5
ϵ_{c1} (‰)	1.8	1.9	2.1	2.2	2.3	2.3	2.4	2.4	2.5	2.5	2.6	2.7	2.8	2.9	3.0
ϵ_{cu1} (‰)	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.4	3.4	3.3	3.2	3.1	3.0	3.0
ϵ_{c2} (‰)	2.0									2.2	2.3	2.4	2.5	2.6	2.7
ϵ_{cu2} (‰)	3.5									3.1	2.9	2.7	2.6	2.6	2.7
n	2.0									1.75	1.6	1.45	1.4	1.4	1.3
ϵ_{c3} (‰)	1.75									1.8	1.9	2.0	2.2	2.3	2.4
ϵ_{cu3} (‰)	3.5									3.1	.9	2.7	2.6	2.6	2.4

The values in Table 7.2-1 are based on the following analytical relations (in all formulas the strengths f_{cm} , f_{ck} , f_{ctm} and f_{ctk} are in MPa and the strains ϵ in ‰).

$f_{cm} = f_{ck} + 8$
 $f_{ctm} = 0.30 f_{ck}^{2/3}$
 $f_{ctm} = 2.12 \ln (1+ f_{cm}/10)$
 $f_{ctk;0.05} = 0.7 f_{ctm}$
 $f_{ctk;0.95} = 1.3 f_{ctm}$
 $E_{cm} = 21.5(f_{cm}/10)^{1/3}$
 ϵ_{c1} (‰)
 ϵ_{cu1} (‰) = $\epsilon_{c,lim}$
 ϵ_{c2} (‰) = $2.0 + 0.085 (f_{ck} - 50)^{0.53}$
 ϵ_{cu2} (‰) = $2.6 + 35 [(90 - f_{ck}) / 100]^4$
 $n = 1.4 + 23.4 [(90 - f_{ck}) / 100]^4$
 ϵ_{c3} (‰) = $1.75 + 0.55 [(f_{ck} - 50) / 40]$
 ϵ_{cu3} (‰) = $2.6 + 35 [(90 - f_{ck}) / 100]^4$

with f_{cm} and with f_{ck} in MPa
for concrete classes $\leq$ C50
for concrete classes $>$ C50

with f_{cm} in MPa
see Table 5.1-8
see Table 5.1-8
for $>$ C50
for $>$ C50
for $>$ C50
for $>$ C50
for $>$ C50

Further see subsection 5.1.4 (compressive strength), subsection 5.1.5.1 (tensile strength), subsection 5.1.7.2 (modulus of elasticity) and subsection 5.1.8.1 (stress–strain relations in compression for short term loading).

The constant 0.06 in Eq. (7.2-9) is appropriate for normal strength concrete. With increasing brittleness of the concrete the coefficient decreases. This means that for high strength concrete and lightweight concrete lower values for the constant in Eq. (7.2-9) apply – see subsection 5.1.5.1 and the comment about Eq. (5.1-8a,b).

Table 7.2-1 shows a survey of design values for concrete strength classes up to C100. Although in section 5.1 mechanical properties of concrete are given up to a strength class C120, in this table the strength classes are limited to C100. The reason is that for strength classes $>$ C100 not all areas of application have sufficient data yet for all aspects of behaviour.

The flexural tensile strength can be formulated as a function of the axial tensile strength. A general relation is:

$f_{ctm,fl} = f_{ctm} \frac{1 + 0.06 h_b^{0.7}}{0.06 h_b^{0.7}}$

(7.2-9)

where:
 f_{ctm} is the mean axial tensile strength [MPa];
 $f_{ctm,fl}$ is the mean flexural strength [MPa];
 h_b is the overall depth of beam [mm].

This relation applies as well to the characteristic values of the concrete strength.

7.2.3.1.2 Elastic deformation

The elastic deformations of concrete largely depend on its composition (especially the aggregates). The values given in Table 7.2-1 should be

regarded as indicative for general applications. The values given in Table 7.2-1 are approximate values for the E modulus E_{cm} , being the secant value between $\sigma_c = 0$ and $0.4f_{cm}$ for concrete with quartzite aggregate, subjected to short term loading. More detailed information for concrete with other aggregates is given in section 5.1. The elastic deformations should be specially assessed if the structure is likely to be sensitive to deviations from the indicative values. More detailed information on the E modulus is given in section 5.1.

The Poisson's ratio may be taken equal to 0.2 for uncracked concrete and 0 for cracked concrete.

Unless more accurate information is available, the linear coefficient of thermal expansion may be taken equal to $10 \cdot 10^{-6} \text{ K}^{-1}$

7.2.3.1.3 Stress-strain relation for non-linear structural analysis

The relation between σ_c and ε_c shown in Figure 7.2-6 (compressive stress and shortening strain shown as absolute values) for short term uniaxial loading is described by the expression:

$$\frac{\sigma_c}{f_{cm}} = \frac{k\eta - \eta^2}{1 + (k-2)\eta} \quad (7.2-10)$$

where:

$$\eta = \varepsilon_c / \varepsilon_{c1};$$

ε_{c1} is the strain at peak stress according to Table 7.2-1;

$$k = 1.05 E_{cm} \cdot |\varepsilon_{c1}| / f_{cm} \quad (f_{cm} \text{ according to Table 7.2-1}).$$

Eq. (7.2-10) is valid for $0 < |\varepsilon_c| < |\varepsilon_{cu1}|$ where ε_{cu1} is the nominal ultimate strain.

Other idealized stress-strain relations may be applied, if they adequately represent the behaviour of the concrete considered.

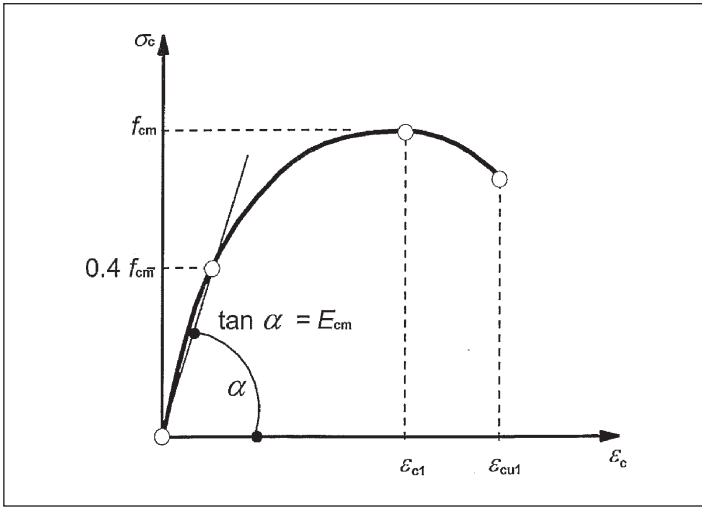


Figure 7.2-6: Schematic representation of the stress-strain relation for structural concrete

Generally the first variable loads on a structure are applied months after the determination of the 28-days strength. Since then the strength of the concrete has increased by continued hydration of the cement. This increase approximately balances the unfavourable effect of sustained loading, so that $\alpha_{cc} = \alpha_{ct} = 1.0$ is appropriate. For concrete strength determined at an age greater than 28 days, the effect of hydration may not be able any more to compensate the effect of sustained loading, so that $\alpha_{cc} = \alpha_{ct} = 0.85$ is more suitable, unless tests show otherwise.

7.2.3.1.4 Design compressive and tensile strengths

The value of the design compressive strength is defined as:

$$f_{cd} = \alpha_{cc} f_{ck} / \gamma_c \quad (7.2-11)$$

where:

γ_c is the partial safety factor for concrete, being 1.5 for transient and persistent situations and 1.2 for incidental situations;

α_{cc} is a coefficient taking account of long term effects on the compressive strength and of unfavourable effects from the way the load is applied. For normal design situations it may be assumed that the increase of the compressive strength after 28 days compensates the effect of sustained loading, so that $\alpha_{cc} = 1.0$ for new structures.

The value of the tensile strength f_{ctd} is defined as:

$$f_{ctd} = \alpha_{ct} f_{ctk} / \gamma_c \quad (7.2-12)$$

For the same reason as mentioned for α_{cc} , for new structures $\alpha_{ct} = 1.0$

7.2.3.1.5 Stress strain relations for the design of cross-sections

For the design of cross-sections a choice can be made between two types of stress-strain relations. A parabola-rectangle relation (Fig. 7.2-7 and 7.2-8) is defined according to:

$$\sigma_c = f_{cd} \left[1 - \left(1 - \frac{\varepsilon_c}{\varepsilon_{c2}} \right)^n \right] \quad \text{for } 0 \leq \varepsilon_c \leq \varepsilon_{c2} \quad (7.2-13)$$

$$\sigma_c = f_{cd} \quad \text{for } \varepsilon_{c2} \leq \varepsilon_c \leq \varepsilon_{cu2} \quad (7.2-14)$$

For concrete strength classes > C50 the stress–strain relation is characterized by a reduced length of the horizontal plateau (Figure 7.2-7).

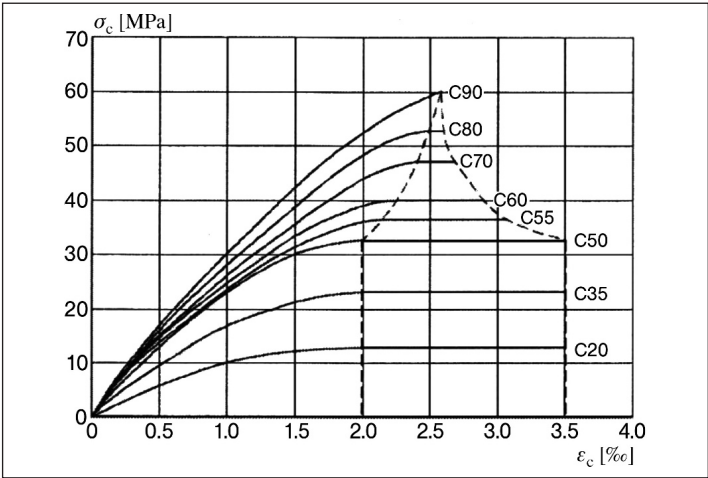


Figure 7.2-7: Design stress strain relations for various concrete strength classes (parabola–rectangle) for $\gamma_c = 1.5$

where:

- n is the exponent according to Table 7.2-1;
- ϵ_{c2} is the strain at reaching the maximum strength according to Table 7.2-1;
- ϵ_{cu2} is the ultimate strain according to Table 7.2-1.

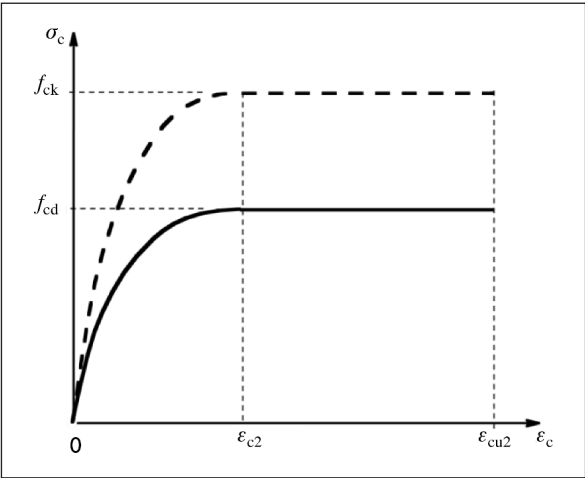


Figure 7.2-8: Parabola–rectangle diagram for concrete in compression (basic curve used in Fig. 7.2-7)

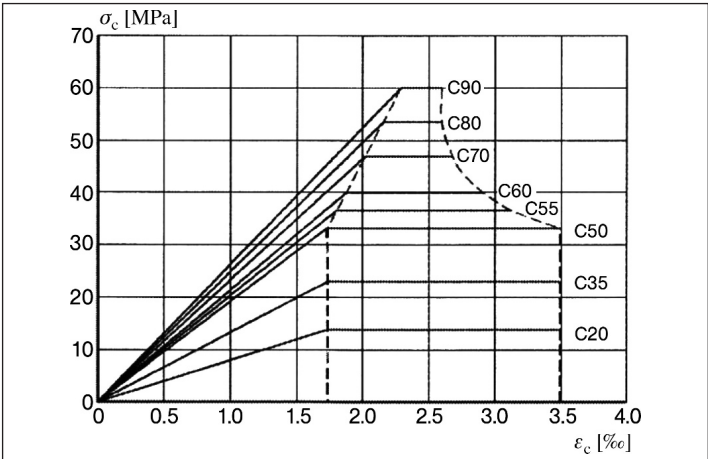


Figure 7.2-9: Design stress–strain relation for various concrete strength classes (bilinear relation) for $\gamma_c = 1.5$

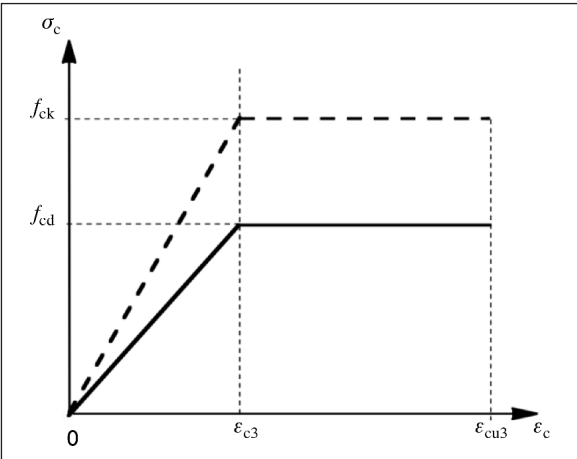


Figure 7.2-10: Bilinear stress–strain relation for concrete in compression (basic curve used in Fig. 7.2-9)

The values for η and λ have been derived in such a way that they give the same results as the parabola–rectangle stress distribution, see Figure 7.2-11.

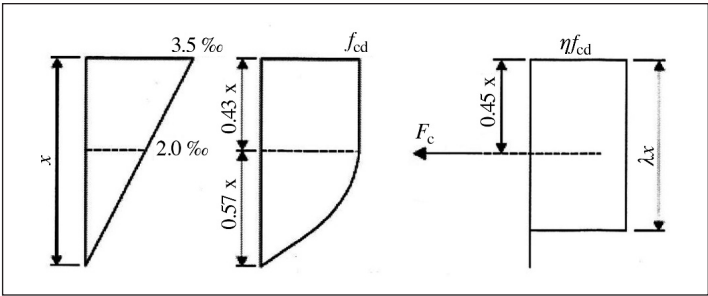


Figure 7.2-11: Derivation of rectangular stress block from parabolic rectangle stress distribution for concrete strength class $\leq C50$

As an alternative, a bilinear relation can be chosen, based on the values ϵ_{c3} and ϵ_{cu3} (Fig. 7.2-9 and 7.2-10).

Finally a rectangular stress distribution can be used, as shown in Figure 7.2-12.

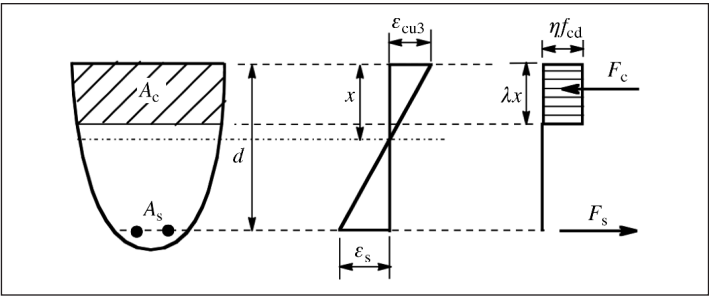


Figure 7.2-12: Rectangular stress distribution

The factor λ , defining the height of the compression zone and the factor η , defining the effective strength, follow from:

The equations for the rectangular stress block have been derived for the basic case that the neutral axis is inside the cross-section. For concrete strength classes $\leq C50$ they are also correct if the neutral axis is outside the cross-section. However, for higher concrete strength classes the results underestimate the strength if the cross-section is completely in compression.

By appropriate confining of concrete the axial strength and the ductility are increased. If these properties are to be exploited in terms of calculations, it must be checked whether the ultimate resistance is still sufficient after spalling of the cover concrete, and it must be ensured that premature buckling of the longitudinal reinforcement is prevented by means of closely spaced stirrups.

The confining pressure σ_2 (with positive sign) can be calculated with the expressions:

$$\sigma_2 = \omega_c f_{cd} \left(1 - \frac{s_c}{d_c}\right) \quad (\text{circular cross-section confined by spiral reinforcement})$$

$$\sigma_2 = \omega_c f_{cd} \left(1 - \frac{s_c}{d_c}\right)^2 \quad (\text{circular cross-section confined by circular hoops})$$

$$\text{where } \omega_c = \frac{A_{sc} f_{yd}}{s_c d_c f_{cd}}$$

$$\sigma_2 = \omega_c f_{cd} \left(1 - \frac{s_c}{a_c}\right) \left(1 - \frac{s_c}{b_c}\right) \left(1 - \frac{\sum b_i^2 / 6}{a_c b_c}\right) \quad (\text{rectangular cross-sections})$$

$$\text{where } \omega_c = \min \left\{ \omega_y = \frac{A_{sy} f_{yd}}{a_c s_c f_{cd}}, \omega_z = \frac{A_{sz} f_{yd}}{b_c s_c f_{cd}} \right\}$$

and b_i is the centre line spacing along the section parameter of longitudinal bars (indexed by i) engaged by a stirrup corner or a cross-tie.

The dispersion of the concentrated forces causes biaxial or triaxial compression immediately under the load, whereas it produces transverse tension further away. Reinforcement should be provided for those tensile forces.

The increased stress according to Eq. (7.2-22) can only be combined with the increased stress due to confining action (7.2.3.1.6) if confirmed by appropriate test results.

For very large confining stresses or very small loading area, failure occurs due to wedging action under the loaded area, at a stress f_{cd}^* which can be significantly higher than $3f_{cd}$.

$$\lambda = 0.8 \quad \text{for } f_{ck} \leq 50 \text{ MPa} \quad (7.2-15)$$

$$\lambda = 0.8 - (f_{ck} - 50) / 400 \quad \text{for } 50 < f_{ck} \leq 100 \text{ MPa} \quad (7.2-16)$$

and

$$\eta = 1.0 \quad \text{for } f_{ck} \leq 50 \text{ MPa} \quad (7.2-17)$$

$$\eta = 1.0 - (f_{ck} - 50) / 200 \quad \text{for } 50 < f_{ck} \leq 100 \text{ MPa} \quad (7.2-18)$$

If the width of the compression zone decreases in the direction of the extreme compression fibre, the value ηf_{cd} should be reduced by 10%.

7.2.3.1.6 Confined concrete

Confinement of concrete results in a modification of the effective stress-strain relationship: higher strength and higher critical strains are achieved. The other basic material characteristics may be considered as unaffected by design.

In the absence of more precise data, the stress-strain relation given by Eqs. (7.2-13)–(7.2-14) may be used, with increased characteristic strength and strains according to:

$$\frac{f_{ck,c}}{f_{ck}} = 1 + 3.5 \left(\frac{\sigma_2}{f_{ck}} \right)^{\frac{3}{4}} \quad (7.2-19)$$

$$\varepsilon_{c2,c} = \varepsilon_{c2} \left[1 + 5 \left(\frac{f_{ck,c}}{f_{ck}} - 1 \right) \right] \quad (7.2-20)$$

$$\varepsilon_{cu2,c} = \varepsilon_{cu2} + 0.2 \sigma_2 / f_{ck} \quad (7.2-21)$$

where σ_2 ($= \sigma_3$) is the effective lateral compressive stress at the ULS due to confinement, and ε_{c2} and ε_{cu2} follow from Table 7.2-1.

Confinement can be generated by adequately closed links or cross-ties, which reach the plastic condition due to lateral extension of the concrete – see Figure 7.2-13.

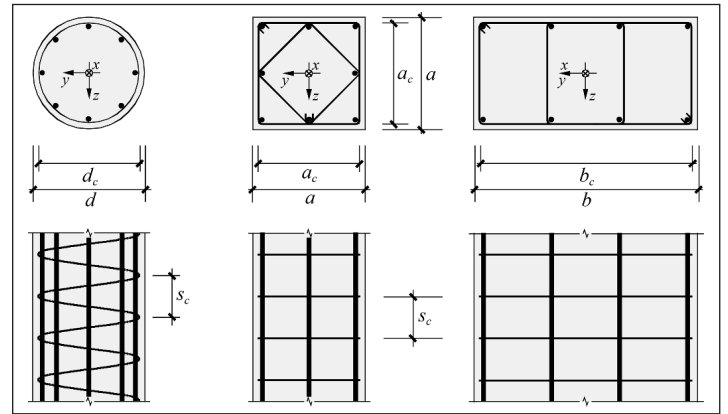


Figure 7.2-13: Compression members with confining reinforcement

7.2.3.1.7 Partially loaded areas

For a uniform distribution of load on an area A_{c0} (Figure 7.2-14) the concentrated resistance force may be determined as follows:

$$F_{Rdu} = A_{c0} f_{cd} \sqrt{A_{c1} / A_{c0}} \leq 3.0 f_{cd} A_{c0} \quad (7.2-22)$$

where:

A_{c0} is the loaded area;

A_{c1} is the maximum design distribution area with a similar shape to A_{c0} .

The design distribution area A_{c1} required for the resistance force F_{Rdu} should correspond to the following conditions:

– the height for the load distribution in the load direction should correspond to the conditions given in Figure 7.2-14;

If no further data are available the average bearing capacity can be calculated with the expression:

$$\frac{f_{cc}^*}{f_{cc}} = 12.5 \sqrt{(40 / f_{cc})} \quad (7.2-23)$$

(f_{cc} in MPa)

However, if limited penetration is considered, f_{cc}^* values not higher than $4f_{cc}$ should be taken.

In Figure 7.2-14 the following limits to h apply:

$$h \geq a_2 - a_1$$

$$h \geq b_2 - b_1$$

- the centre of the design distribution area A_{c1} should be on the line of action passing through the centre of the load area A_{c0} ;
- if there is more than one compression force acting on the concrete cross-section, the designed distribution areas should not overlap.

The value F_{Rdu} should be reduced if the load is not uniformly distributed on the area A_{c0} or if high shear forces exist.

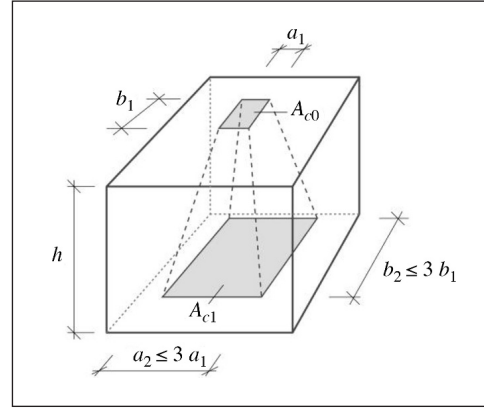


Figure 7.2-14: Load distribution for partially loaded areas

7.2.3.2 Reinforcing steel

The design tensile strength of reinforcing steel f_{yd} follows from:

$$f_{yd} = f_{yk} / \gamma_s \quad (7.2-24)$$

where:

f_{yk} is the characteristic yield stress of the steel;

$\gamma_s = 1.15$ for persistent and transient situations and 1.0 for accidental situations.

The behaviour of reinforcing steel in tension and compression is idealized in accordance with the stress–strain relations given in Figure 7.2-15.

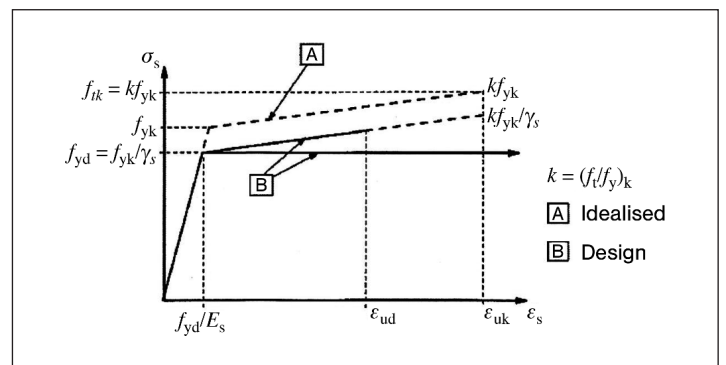


Figure 7.2-15: Idealized and design stress–strain relations for reinforcing steel in tension and compression

For situations in which the plastic structural deformations are of importance, it is suitable to assume a linear strain-hardening behaviour of the steel. In this case, the maximum stress $k f_{yd} / \gamma_s$ is reached at a strain of $\epsilon_{ud} = 0.9 \epsilon_{uk}$. The value k follows from $k = f_t / f_y$. For the values of the ultimate strength f_t and the yield strength f_y – see section 5.2.

In general, the structural analysis and dimensioning may be based on the mean value of the modulus of elasticity E_s given in section 5.2.

7.2.3.3 Prestressing steel

The dimensioning values of the yield strength of the prestressing steel are determined on the basis of section 5.3.

The behaviour of prestressing steel under tensile or compressive stresses is idealized in accordance with the stress–strain diagram given in Figure 7.2-16.

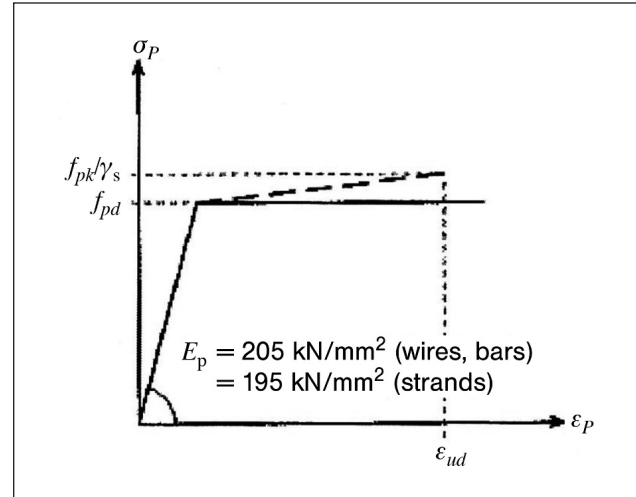


Figure 7.2-16: Idealized stress strain diagram for prestressing steel

The dimensioning is based on the nominal cross-sectional areas of the prestressing steel.

In general, a perfectly plastic behaviour may be assumed.

For situations in which the plastic structural deformations are of importance, it is practical to assume a linear strain hardening behaviour of the prestressing steel. The ultimate strain must be limited to $\epsilon_{ud} = \epsilon_{uk}$ as specified in the relevant product standard.

In general, the structural analysis and dimensioning may be based on the mean value of the modulus of elasticity E_p according to section 5.3.

7.2.4 Analysis of structural effects of time-dependent behaviour of concrete

7.2.4.1 General

Shrinkage strains influence the state of deformation and induce stresses when they are restrained; they also cause stress redistributions in non-homogeneous and composite structures and sections and stress losses in prestressed structures.

The consequences of creep can be either beneficial or detrimental. On the one hand, creep exercises a beneficial effect by relieving undesirable stresses due to unintentional imposed strains such as shrinkage, extreme initial temperatures, settlement of supports and yielding of restraints. On the other hand, the long term reliability of structures may be adversely affected, as creep:

- increases by an important factor their initial deformations;
- reduces the beneficial effects of stresses artificially imposed to improve the performance of the structure with regard to serviceability, either causing prestress losses in structures prestressed with tendons or strands, or significantly jeopardizing stress corrections enforced by jacking;
- activates the delayed restraints in case of changes in the structural system after the application of sustained loads, inducing in some cases significant redistributions of internal actions and stresses that may lead to unfavourable increases of their values in some regions of the structures;

The inelastic strains due to creep and shrinkage of concrete may cause non-negligible variations of deformations and/or of internal actions and stresses in structures and structural elements.

The overall dimensions and the slenderness of structures and structural elements, in particular if combined with the adoption of thin sections, magnify their sensitivity to the time-dependent behaviour of concrete. In fact, while the initial deformations are large, the creep amplification factor and the shrinkage strains are augmented, as a consequence of the drying in thin elements.

The time sequence of casting, loading and application of restraints, as well as the presence of important non-homogeneities exert a significant influence on the time-dependent response.

Creep and shrinkage affect primarily the long term serviceability and durability of structures. In particular, if the effects of time-dependent strains are mainly in the sense of an increment of the deformations, the limit state of deformations should be checked. If the state of stress is mainly influenced by time-dependent effects, the limit states of stresses and of cracking should be checked.

- reintroduces, for the same reason, a significant part of the internal actions and stresses due to self-weight that were provisionally eliminated in statically indeterminate structures by temporary reductions, during the construction stages, of their degree of restraint (e.g. bending moments in fixed end arches due to axial shortening, provisionally eliminated through the adoption of temporary hinges).

In non-homogeneous structures, creep-induced stress redistributions transfer stresses from the parts of the structure creeping more to the parts creeping less, or from concrete to steel elements. Non-homogeneities may be due to differences in casting and loading ages, mixture proportions and components, size and shapes of structural elements and cross-section components, environmental conditions and so on, in the various concrete parts of the structure, and to the association of concrete and steel elements.

For example, in high-rise buildings, the combined effects of creep and shrinkage, of their non-uniform development and of sequential construction, besides generating several serviceability concerns regarding both structural and non-structural components, can also cause reductions of the safety margins with respect to the ultimate limit state. This second concern can depend on the possible increase with time of action effects on some structural elements, and may become relevant especially when a limited ductility is available because of high axial loads, as in vertical elements, and of the use of high-strength concrete, or when buckling of slender steel elements is a concern. The influence of non-symmetrical time-dependent vertical shortenings on gravity-induced side-sway and its effects on structural reliability must also be investigated.

When checking the stability of compressed members, the long term deformations of concrete may approximately be taken into account as indicated in subsection 7.3.7.1 for level II approximation. When more refined analyses are required, the indications given in the same subsection for level IV approximation must be respected.

For more information see the CEB Design Manual “Structural Effects of Time-dependent Behaviour of Concrete”, CEB Bulletin 142, 1984; CEB Bulletin 215, 1993; section 4.1.6 of “Structural Concrete Textbook” *fib* Bulletin 52, 2010; “Analysis of Creep and Shrinkage Effects in Concrete Structures” to be published as ACI document 209.3R.

Structures may be conveniently classified according to their levels of sensitivity to time-dependent effects. The lower levels are represented by small and simple structures. The higher levels refer to important, large and technically complex structures. Typical examples of highly sensitive structures are segmentally built large span cantilever and arch bridges joined at a later stage, cable-stayed bridges and structures, steel-tied arches, concrete or steel and concrete structures for high-rise and super-tall buildings, steel-concrete composite beams or framed structures, structures with high ratios of prestressing and reinforcing steel, large span slender concrete arches and shells, structures stressed by jacking, off-shore, marine and nuclear structures etc.

In general, a high sensitivity of the structure to the time-dependent behaviour of concrete may be responsible of final uneconomies in its service life costs. A reduction of this sensitivity by proper design and construction provisions, which may require higher initial costs, besides contributing to the reduction of the margins of uncertainty in the assessment of long term reliability, may also result in service life economies.

Influence on the safety margins with respect to the ultimate limit state of strength depends on the ductile behaviour of the structure or structural element and can become a concern in the presence of significant time-dependent amplifications of action effects such as internal forces and moments in cases where these effects cannot be redistributed, especially when the ultimate limit state is governed by non-plastic failure of concrete.

In slender or thin structures or structural elements and whenever second order effects are of importance, the increase of deflections due to creep reduces the safety margins with respect to instability and may lead to creep buckling; the unfavourable influence of shrinkage should be considered as well.

The guidelines given in the following apply essentially to the verifications with respect to serviceability limit states.

7.2.4.2 Levels of refinement of the analysis

When choosing the level of refinement for the analysis, the following aspects should be considered:

- sensitivity of the structure to the time-dependent behaviour of concrete;

Excessive refinement is not warranted for low to moderate sensitivity structures, and in the preliminary and conceptual design stages of all types of structures. Sophisticated and laborious analyses should be reserved to important, sensitive and very sensitive structures in their final stages of design.

A refined creep and shrinkage prediction model ought always to be used for structures analysed by sophisticated computational methods, while excessive refinement in the analysis should be avoided if the prediction of material properties is poor. The error caused by replacing an accurate prediction of creep and shrinkage values on the basis of the most realistic available models with a simple but crude estimation is often larger than the gain from replacing simplified analysis approaches with sophisticated approaches.

Important uncertainties affect both, and to large extent, the prediction of the material response and the evaluation of the consequent structural response.

Appropriate confidence limits should be better considered also in these cases on the basis of adequate estimations.

As evidenced in subsection 5.1.9.4, the deformation prediction suffers from important uncertainties, due to inherent scatter of creep and shrinkage strains, errors of the model and randomness of material properties and the environment, and may result in a considerable prediction error.

In ACI 209.2R-08 “Guide for Modeling and Calculating Shrinkage and Creep in Hardened Concrete”, ACI 2008, besides the 1999 edition of the model of subsection 5.1.9.4, three other alternative mean-cross-section prediction models are presented: ACI 209R-92, last updated in 1992, and the two more recent models termed GL2000 and B3.

In cases of highly sensitive structures for which cracking is a dominant consideration for serviceability and safety (e.g. marine and nuclear structures), a realistic analysis of the time evolution of internal stresses and strains within cross-sections and throughout the structure by appropriate discretization techniques should be based on the description of local rheological properties (constitutive laws for a material point of the concrete), taking into account their intrinsic non-linearities, coupled with moisture and temperature distributions and non-linear effects of cracking. See the specialized literature such as Bažant, Z. P. et al. “Prediction of creep and shrinkage and their effects in concrete structures: Critical appraisal”, in: Tanabe et al. (eds.), Proc. Eighth Int. Conf. Creep, Shrinkage and Durability Mechanics of Concrete and Concrete Structures, Taylor and Francis, 2009, pp. 1275–89, and referenced literature. See also subsection 7.2.4.11(b).

- importance of the limit state under consideration;
- design stage;
- reliability of the information on material properties (prediction on the basis of the prediction models such as the model given in subsection 5.1.9.4, or prediction accompanied by tests at early ages, or test extrapolation; mean cross-section behaviour or local rheological properties within the cross-section etc.).

Balanced attention should be given in any case to both the material properties problem and the structural analysis problem.

7.2.4.3 Probabilistic and deterministic approach

The problem of evaluation of time-dependent effects in concrete structures in the serviceability domain is statistical in nature, since most of the contributing factors are inherently random with significant coefficients of variation.

A deterministic approach based on mean prediction of time-dependent effects may be adequate for low to moderate sensitivity structures and whenever a refined analysis is not required.

A probabilistic approach is highly recommendable for sensitive structures, and becomes mandatory for very sensitive ones. The design of these structures should be based on predictions of extreme values of time-dependent effects that are exceeded with a certain specified small probability, such as 5%. The probabilistic approach should take into account the various factors of uncertainty.

Concerning the prediction of creep and shrinkage properties, reference should be made to the statistical indicators of the prediction model being considered. Updating the model parameters by testing is recommended in any case.

As for the evaluation of the consequent structural effects, the use of refined analyses contributes to the reduction of uncertainties.

7.2.4.4 Prediction models for concrete and significance of the analysis

When adopting a prediction model for creep and shrinkage of concrete of the type given in subsection 5.1.9.4, attention must be paid to its range of applicability as specified in subsection 5.1.9.4.2.

While models of this kind that predict the mean cross-section behaviour of a concrete member are generally suitable for the analysis of time-dependent effects on the overall response of structures (with the exception of highly sensitive structures) in terms of internal actions, restraint reactions and deformations, they should be applied with caution in the analysis of creep and shrinkage effects on local stress distributions within cross-sections, where larger errors may be introduced.

Rather, the results of such cross-sectional analyses should be considered to have essentially a nominal value, and not as realistic descriptions of the actual stress values in the sections. However, for ordinary structures, in which fine cracking of concrete is not of much concern, or ample prestress is provided, more refined analyses are not normally needed.

In the case of cross-sections composed of elements with different thickness or environmental exposure conditions (e.g. in box girders), the non-uniformity of drying can be captured, in a simple but rather approximate way, applying the mean cross-section prediction model for creep and shrinkage to each component separately.

For the range of validity of the linearity assumption, see subsection 5.1.9.4.3. Hygrothermal effects and cracking introduce major deviations from the principle of superposition. Non-linearities due to high stresses may be taken into account on the basis of subsection 5.1.9.4.3 (d).

Shrinkage strains may be considered as imposed stress-independent strains.

In constitutive relations (5.1-62) or (7.2-25) strain or, respectively, stress may be considered as a functional of the previous stress (or strain) history alone.

If, on the contrary, the histories of the strain or, conversely, of the stress are prescribed in Eq. (5.1-62) or in Eq. (7.2-25), respectively, these equations represent linear Volterra hereditary integral equations for the determination of the corresponding stress or, respectively, strain histories.

Generalization to multiaxial stress of constitutive relation (5.1-62) requires the additional assumption of constancy of creep Poisson's ratio of concrete ν_c . For further information see Bažant, Z. P. "Theory of Creep and Shrinkage in Concrete Structures: a Précis of Recent Developments", *Mechanics Today*, Vol. 2, Pergamon Press, 1975, pp. 1–93; Salençon, J. "Viscoélasticité pour le Calcul des Structures", Presses des Ponts et Chaussées, 2009.

Different values of t_0 must be considered and repeated solutions of Eq. (7.2-26) must be performed to obtain $R(t, t')$ from $J(t, t')$. The procedure for numerical solution indicated in subsection 7.2.4.11(a) must be adopted.

The ranges of applicability of these simplifications are given in the corresponding subsections.

The application of this assumption is based on the following considerations:

7.2.4.5 Time-dependent analysis based on ageing linear viscoelasticity

Within the limits specified in subsections 5.1.9.4. and 7.2.4.4, the time-dependent analysis of concrete structures, for what concerns the evaluation of the effects of creep, may be based on the theory of ageing linear viscoelasticity.

Ageing linear viscoelasticity implies the validity of the principle of superposition of the responses in terms of strains or stresses to all imposed stress or strain increments applied during the time.

7.2.4.6 Constitutive laws in ageing linear viscoelasticity

The application of ageing linear viscoelasticity gives the stress–strain constitutive relation in the form of the hereditary integral relation (5.1-62) representing the strain response to a sustained variable imposed stress history with initial value $\sigma_c(t_0)$. In this case, the time-dependent behaviour of concrete is fully characterized by the compliance function $J(t, t')$.

Alternatively, with the same assumptions and range of validity, the stress response to a sustained variable imposed strain history with initial value $\varepsilon_{c\sigma}(t_0)$ may be written as:

$$\sigma(t) = \varepsilon_{c\sigma}(t_0)R(t, t_0) + \int_{t_0}^t R(t, t')d\varepsilon_{c\sigma}(t') \quad (7.2-25)$$

where:

$R(t, t')$ is the relaxation function, representing the stress response at time t to a sustained constant unit imposed stress-dependent strain applied at time t' ;

$R(t, t')$ can be obtained from the compliance function $J(t, t')$, specified by the creep prediction model being considered, as the stress response in Eq. (5.1-62) for a constant unit imposed stress-dependent strain $\varepsilon_{c\sigma} = 1$. Therefore, the compliance and the relaxation functions are reciprocally related by the integral equation:

$$1 = R(t_0, t_0)J(t, t_0) + \int_{t_0}^t J(t, t')dR(t', t_0) \quad (7.2-26)$$

7.2.4.7 Simplified approaches for time-dependent analysis

When a highly refined analysis is not required, some convenient simplifications may be introduced:

- at the level of the structural model, by the introduction of the assumption of an effective rheological homogeneity for the concrete structure (subsection 7.2.4.8), or for the concrete part of a structure that includes steel structural elements (subsection 7.2.4.9), when performing the analysis of the overall behaviour of the structure;
- at the constitutive level, through the adoption of the approximate algebraic formulation of the AAEM method for the constitutive relation (5.1-62) (subsection 7.2.4.10).

7.2.4.8 Effective homogeneous concrete structures with rigid or stress-independent yielding of restraints

A simplification of the structural model consists in neglecting the non-homogeneities in the rheological properties of the material and

- the differences in the rheological properties of the concrete along the structure, in terms of long term values of creep and shrinkage strains, are usually rather contained if compared to the large magnitude of these values obtained from prediction models;
- the influence of the non-homogeneities due to the presence of reinforcement is generally small in most prestressed structures, due to the small geometrical percentage of reinforcement and to the lack of cracking;
- although this influence may be larger in ordinary reinforced concrete structures, as a consequence also of the interaction between creep and cracking, it may equally be disregarded in most cases, in particular if the analysis does not concern specifically local effects and the structure does not pertain to the high ranks of sensitivity;
- the stress-independent yielding of the restraints may be considered as a set of imposed deformations at the points of applications of the restraints of the structure;
- the case of restraints characterized by an elastic behaviour, or equivalently of the presence of steel structural elements considered as elastic elements, is discussed in subsection 7.2.4.9.

The formulations obtained from the assumption of effective rheological homogeneity are particularly apt for the serviceability assessment of low to moderate sensitivity structures, and especially in the preliminary and conceptual design stages for a wider class of structures. In fact, their sound theoretical fundamentals guarantee their reliability from the conceptual point of view, while the reference to averaged rheological properties allows to capture the basic orders of magnitude. Therefore, if extreme and very special cases characterized by significant non-homogeneities in the concrete properties are excluded, the general trends of the time-dependent phenomena under consideration are unquestionably seized.

Assuming the principle of superposition to be valid, the solutions presented here in sections (a) to (d) for each separate problem in terms of stress and strain histories may be superimposed.

referring to an effective homogeneous concrete structure of averaged creep and shrinkage properties with rigid restraints.

Accordingly, a set of simple formulations for the determination of the overall time-dependent response of the structure is obtained from the fundamental theorems of the theory of ageing linear viscoelasticity. The system of the stresses $S(t)$ and of the deformations $D(t)$ of the structure, under sustained imposed loads or deformations or consequent to a modification of the restraint conditions, may then be evaluated on the basis of the stresses $S^{el}(t)$ and deformations $D^{el}(t)$ for an elastic structure of constant reference elastic modulus E_c .

(a) Imposed loads

Under sustained variable imposed loads the elastic stresses are not modified by creep. The deformations at time t may be evaluated summing up the increments of the elastic deformations factored by the non-dimensional creep factor $E_c J(t, t')$ (first theorem of ageing linear viscoelasticity):

$$S(t) = S^{el}(t) \quad (7.2-27)$$

$$D(t) = E_c \int_0^t J(t, t') dD^{el}(t') \quad (7.2-28)$$

where:

$S(t)$ is the system of the stresses (internal stresses, internal forces, restraint reactions) at time t ;

$S^{el}(t)$ is the elastic solution for the system of the stresses in the associated elastic problem at time t ;

$D(t)$ is the system of the deformations (internal strains, internal deformations, external displacements) at time t ;

$D^{el}(t)$ is the elastic solution for the system of the deformations in the associated elastic problem at time t ;

E_c is the reference elastic modulus for the associated elastic problem.

Under sustained constant loads, creep does not alter the initial stresses in the structure. The initial deformations are followed by their creep-induced gradual increase, the deformation history being related by an affinity to the compliance function (*creep problem* at the structural level).

The response to a system of sustained constant loads imposed at $t = t_0$ is thus given by:

$$S(t) = S^{el, t_0} \quad (7.2-29)$$

$$D(t) = E_{ci}(t_0) J(t, t_0) D^{el, t_0} \quad (7.2-30)$$

For a normal weight ordinary structural concrete, the magnification factor $E_{ci}(t_0)J(t, t_0)$ at the end of a service life of 100 years can easily reach values in the range of 3 to 5 for typical ages t_0 at loading, depending on the creep prediction model, on the values of the various influencing parameters and on the reference value of the elastic modulus $E_{ci}(t_0)$ considered in the evaluation of the initial deformations (CEB Bulletin 215 and *fib* Bulletin 52).

If reference is made to the prediction model of subsection 5.1.9.4.3, the magnification factor is $E_{ci}(t_0)J(t, t_0) = [1 + \varphi_{28}(t, t_0)E_{ci}(t_0)/E_{ci}]$ (see subsection 7.2.4.10 for notation φ_{28}).

When applying Eq. (7.2-32) for the evaluation of the effects of cyclic time histories of the imposed deformations – for example when considering the effects of restrained annual thermal variations – reference to the comments of subsection 7.2.4.11(a) is appropriate.

Under sustained constant imposed deformations, creep does not alter the initial deformations. The initial stresses are followed by their creep-induced gradual decrease, the stress history being related by an affinity to the relaxation function (*relaxation problem* at the structural level).

For a normal weight ordinary structural concrete, the creep-induced mitigation factor $R(t, t_0)/E_{ci}(t_0)$ lies in the range 0.1 to 0.3 at the end of a service life of 100 years for typical ages t_0 at application of imposed deformations, depending on the creep prediction model, on the values of the various influencing parameters and on the reference value of the elastic modulus $E_{ci}(t_0)$ considered in the evaluation of the initial stresses (CEB Bulletin 215 and *fib* Bulletin 52). Therefore, as a result of relaxation, creep exerts a beneficial action in significantly relieving the stresses due to unintentional sudden imposed deformations. On the other hand, it drastically reduces the beneficial effects of stress corrections artificially induced, for example, by jacking.

As a general indication, the formulations of Eqs. (7.2-35) to (7.2-37) show the importance of stress redistributions due to creep in case of changes in the structural system. In fact, the inspection of the diagrams of the redistribution function $\xi(t, t_0, t_1)$ shows that for average creep properties and for typical ages t_0 at loading and t_1 at change in structural system, characterizing usual construction sequences, the long term values of ξ are contained in a range between 0.5 and 0.9, if reference is made to the prediction model of subsection 5.1.9.4.3. Higher values, in the range of 0.7÷0.9, are obtained for GL2000 and B3 models (CEB Bulletin 215 and *fib* Bulletin 52).

As a result, the long term distribution of stresses, internal actions and restraint reactions tends to approach those of the structure in its final structural system. Therefore, the creep-induced stress redistribution must be accounted for in the serviceability analyses of the structure and may require additional reinforcement and/or the introduction of additional prestressing after the change in the structural system. When assessing the reliability with respect to the ultimate limit state, this redistribution must be properly considered in case of structures in which the internal actions cannot be redistributed due to limited available ductility, the collapse being governed by the brittle failure of concrete, or when buckling is an issue.

where:

S^{el, t_0}, D^{el, t_0} are the elastic solutions for the system of the stresses and deformations calculated with the initial value $E_{ci}(t_0)$ of the elastic modulus.

(b) Imposed deformations

Under sustained variable imposed deformations the elastic deformations are not modified by creep. The stresses at time t may be evaluated by summing up the increments of the elastic stresses factored by the non-dimensional relaxation factor $R(t, t')/E_c$ (second theorem of ageing linear viscoelasticity):

$$D(t) = D^{el}(t) \quad (7.2-31)$$

$$S(t) = 1/E_c \int_0^t R(t, t') dS^{el}(t') \quad (7.2-32)$$

The response to a system of sustained constant imposed deformations imposed at $t = t_0$ is thus given by:

$$D(t) = D^{el, t_0} \quad (7.2-33)$$

$$S(t) = \frac{R(t, t_0)}{E_{ci}(t_0)} S^{el, t_0} \quad (7.2-34)$$

(c) Modification of restraint conditions after loading

In a structure subjected to sustained constant loads imposed at time t_0 , whose initial structural system 1 is modified into a final structural system 2 by the introduction of additional restraints at time $t_1 \geq t_0^+$, the stress distribution evolves for $t > t_1$ according to Eq. (7.2-35) (third theorem of ageing linear viscoelasticity):

$$S^2(t) = S^{el, 1} + \Delta S^1(t) = S^{el, 1} + \Delta S^{el, 1} \xi(t, t_0, t_1) \quad (7.2-35)$$

where:

$S^2(t)$ is the system of the stresses at $t > t_1$ in the structure in the final structural system 2;

$S^{el, 1}$ is the elastic solution for the system of the stresses in the structure in the initial structural system 1;

$\Delta S^{el, 1}$ is the correction to be applied to the elastic solution $S^{el, 1}$ in order to comply with the elastic solution $S^{el, 2}$ corresponding to an assumption of application of the loads in the final structural system 2, that is:

$$S^{el, 2} = S^{el, 1} + \Delta S^{el, 1}$$

$\xi(t, t_0, t_1)$ is the redistribution function.

One possible strategy of reducing the creep-induced stress redistribution consists of reducing the difference in the stress distributions corresponding to the two reference elastic configurations of the original and final structural systems, respectively. For instance, in prestressed structures under bending, in cases where the additional restraints prevent the delayed flexural deformations, this may be done by balancing the permanent loads in the original structural system by prestressing, thus reducing the corresponding flexural deformations. Therefore, the additional restraints have a limited effect in altering the original elastic stress distribution.

The delayed introduction of additional restraints is sometimes conceived as an artifice to improve the behaviour of the structure under permanent loads, when the corresponding stress distribution in the original structural system is more favourable, if compared to the one that would be obtained, for the same loads, in the final system. In such cases, the introduction of the additional restraints, intended to improve the response to live loads and the final robustness of the structure by an increase of its statical redundancy, substantially reduces the original benefits, because of the significant creep-induced stress redistributions altering the original response to permanent loads (e.g. fixed concrete arches provisionally built as three-hinged arches with a centre line corresponding to the funicular curve for dead loads).

Different values of t_0 and t_1 must be considered, and repeated solutions of Eq. (7.2-36) must be performed to obtain $\xi(t, t_0, t_i)$ from $J(t, t')$. The procedure for numerical solution indicated in subsection 7.2.4.11(a) must be adopted.

For further information, see Chiorino, M. A. "A Rational Approach to the Analysis of Creep Structural Effects", ACI SP-227, 2005, pp. 107–141.

Design aids. For a given creep prediction model, and for the corresponding compliance function $J(t, t')$, design aids can be provided for the evaluation of the related functions $R(t, t')$ and $\xi(t, t_0, t_i)$ in terms of sets of graphs of these three functions and of the ageing coefficient $\chi(t, t')$ introduced in subsection 7.2.4.10 (refer to CEB Bulletin 215 for the creep prediction model of CEB Model Code 1990; a few examples for the models indicated in subsection 7.2.4.4 are given in *fib* Bulletin 52), or in terms of computational programs to be downloaded (see for example www.polito.it/creepanalysis) or inserted in computational software. For further information see Sassone, M. and Chiorino, M. A. "Design Aids for the Evaluation of Creep Induced Structural Effects", ACI SP-227, 2005, pp. 239–259.

The redistribution function ξ is a non-dimensional factor whose values lie in the interval (0,1) (with $\xi = 0$ for $t = t_1$), and is related to the compliance function through the integral equation:

$$J(t, t_0) - J(t_1, t_0) = \int_{t_1}^1 J(t, t') d\xi(t', t_0, t_1) \quad (7.2-36)$$

(d) Multiple changes in the structural system

In cases where the transition from the initial to the final structural system is obtained by means of several different restraint modifications applied at times $t_i \geq t_0^+$ ($i = 1, \dots, j$), the redistribution effects consequent to every single change in the structural system may be superimposed in time. Therefore, the system of the stresses evolves for $t > t_1$ according to Eq. (7.2-37) (fourth theorem of ageing linear viscoelasticity):

$$S^{j+1}(t) = S^{el,1} + \sum_{i=1}^j \Delta S^{el,i} \xi(t, t_0, t_i) \quad (7.2-37)$$

where:

$\Delta S^{el,i}$ is the correction to be applied, in the associated elastic problem, to the elastic solution $S^{el,i}$, in order to respect the geometrical conditions imposed by the additional restraints of structural system $i+1$, imagined as introduced before the loads.

7.2.4.9 Effective homogeneous concrete structures with additional steel structural elements

If the main cause of heterogeneity is represented by the presence of steel structural elements that may be considered equivalent to redundant elastic restraints, while the concrete part of the structure may still be approximately regarded as an effective homogeneous structure of averaged rheological properties, a set of simple formulations can be provided for the determination of the overall time-dependent response of the structure for the cases of constant imposed loads or of constant deformations η_i imposed at the points of applications of the restraints, or of a change in structural system (Figure 7.2-17).

Reference should be made to the specialized literature indicated in *fib* Bulletin 52, p. 53, and to ACI 2090.3R Guide.

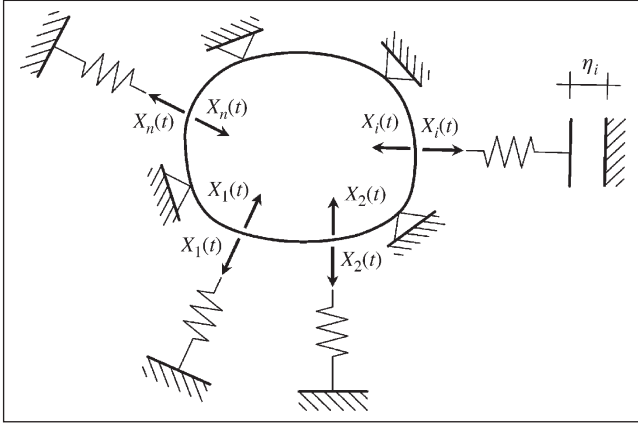


Figure 7.2-17: Schematic representation of a homogeneous structure with n redundant elastic restraints

Although the theoretical condition of invariance can hardly be obtained in practical cases, due to construction sequences and differences in the creep properties along the structure, an appropriate stressing of the steel restraints can substantially reduce the variation in time of the stress state. This artifice, which may be responsible of higher initial costs – as the steel restraints must be designed for higher forces – contributes to the reduction of uncertainties in the long term reliability assessments, and may result in service life economies.

In cable-stayed bridges, stressing of the stays is normally performed in more than one step to allow for adjustments, as required by the usual cantilever sequential construction techniques. One other reason for stressing the stays is the need of eliminating the non-linear mechanical behaviour of the stays due to cable sag. For more information, see Casalegno, C. et al. “Time dependent effects in cable-stayed bridges built by segmental construction”, Proceedings 3rd *fib* International Congress, Washington 2010.

Eq. (7.2-39), with the expression (7.2-40) for the ageing coefficient χ , corresponds exactly to Eq. (5.1-62) for all one-step imposed action histories resulting from linear combinations of a creep and a relaxation problem, that is for all the strain histories of the type:

$$\begin{aligned} \varepsilon_{c\sigma}(t) &= c_1 + c_2 J(t, t_0) = c_1 + c_2 \left[\frac{1}{E_{ci}(t_0)} + \frac{\phi_{28}(t, t_0)}{E_{ci}} \right] \\ &= a + b \phi_{28}(t, t_0) \end{aligned} \quad (7.2-38)$$

These formulations, which are expressed in matrix form, represent the extension to the case of elastic restraints of the ageing linear viscoelastic formulations presented in subsection 7.2.4.8.

Typical structures that can be schematically modelled as effective homogeneous concrete structures with redundant elastic restraints are tied arches and frames, and cable-stayed bridges and structures. Although the final assessment of these types of structures – being inherently complex and usually characterized by intricate construction procedures and phases – must normally be performed by proper sequential approaches adopting the numerical techniques indicated in subsection 7.2.4.11 and accounting for possible causes of non-linearity (e.g. due to cable sag in stayed structures), the reference to this schematic model and the inherent formulations allows setting some general guidelines listed in the following, which are particularly useful in the preliminary and conceptual design stages.

(a) Constant imposed loads

Differently from the case of rigid restraints, for which the invariance of the state of stress is stated by Eq. (7.2-27), the initial elastic state of stress in the restraints and in the structure is significantly altered by creep. The higher the deformability of the restraints, the higher is the difference between the initial and long term values. In the long term, the system of stresses tends to approach the solution corresponding to the case of rigid restraints.

A reduction of the time dependence of the state of stress under permanent loads when highly deformable restraints are adopted (e.g. high strength steel stays of small section as in cable-stayed bridges), can be obtained through a convenient stressing of these restraints. In fact, in the theoretical case of an effective homogeneous structure with elastic restraints introduced all at the same time, the invariance of the stress state is obtained forcing the restraints up to the values of the rigid restraints reactions.

(b) Constant imposed deformations

The stress losses are less pronounced with respect to the case of rigid restraints represented by Eq. (7.2-34). This is a consequence of the elastic energy stored in the system of the restraints.

(c) Modification of restraint conditions after loading

The theoretical solutions show that the system of elastic restraints contributes to a lower degree to the variation of the original system of the stresses in the structure, attracting lower values of restraint reactions, with respect to the case of delayed additional rigid restraints discussed in subsection 7.2.4.8(c).

7.2.4.10 Approximate algebraic formulation for the constitutive relation: age-adjusted effective modulus (AAEM) method

For a compliance function expressed in the form of Eq. (5.1-61), the hereditary integral constitutive relation of Eq. (5.1-62) may be written in the following equivalent algebraic form:

$$\begin{aligned} \varepsilon_c(t) &= \sigma_c(t_0) J(t, t_0) + [\sigma_c(t) - \sigma_c(t_0)] \left[\frac{1}{E_{ci}(t_0)} + \chi(t, t_0) \frac{\phi_{28}(t, t_0)}{E_{ci}} \right] \\ &+ \varepsilon_{cn}(t) = \frac{\sigma_c(t_0)}{E_{c,ef}(t, t_0)} + \frac{\sigma_c(t) - \sigma_c(t_0)}{E_{c,adj}(t, t_0)} + \varepsilon_{cn}(t) \end{aligned} \quad (7.2-39)$$

where the time-dependent part is related by an affinity to the compliance function $J(t, t_0)$ or, equivalently, to the creep coefficient $\phi_{28}(t, t_0)$. This includes a broad range of strain (and corresponding stress) histories. With sufficient accuracy, its use may be extended to cover a large number of actual action histories in structures showing an initial finite or zero value at $t = t_0$ and a time-dependent part varying at a gradually decreasing rate over wide time intervals. In current use of the AAEM method, Eq. (7.2-39) is given a quasi-elastic incremental formulation relating the variations of the total strain $\Delta \epsilon_c(t)$ and of the stress $\Delta \sigma(t)$ occurring over the interval (t_0^+, t) after the initial stress state at time t_0^+ . In both alternative procedures, the responses to multistep load histories can be obtained by superimposing the solutions for several one-step histories. For further information, see Bažant, Z. P. “Numerical determination of long-range stress history from strain history in concrete”, *Material and Structures*, Vol. 5, 1972, pp. 135–141; Jirásek, M. and Bažant, Z. P. “Inelastic Analysis of Structures”, Wiley and Sons, 2002.

Under these conditions, the AAEM method may be applied firstly to the analysis of the overall time-dependent response of concrete structures that may be considered as homogeneous on the basis of the same assumption of effective rheological homogeneity discussed in subsection 7.2.4.8. The same ranges of applicability of the solution have to be considered.

The accuracy of the results normally remains satisfactory if the application is extended to cover the cases of heterogeneous structures consisting of concrete portions with different creep properties and/or containing steel elements. However, in case of very complex structures and construction sequences, preference has to be given to the numerical approaches of the general method illustrated in subsection 7.2.4.11.

The AAEM algebraic formulation of the constitutive relation in association with the assumption of plane sections can normally be adopted for the estimation of stress redistributions due to creep and shrinkage in non-homogeneous and composite cross-sections of one-dimensional elements, such as prestressed concrete sections with prestressing and reinforcing steel in one or multiple layers, concrete–concrete and steel–concrete composite sections. In fact, while on the one hand the AAEM solutions are generally very accurate with respect to the use of the integral formulation (5.1-62) of the constitutive law, on the other hand reference to the observations of subsection 7.2.4.4 on the nominal character of these estimations is appropriate.

For any given compliance function $J(t, t')$, the ageing coefficient $\chi(t, t')$ can be determined from Eq. (7.2-40) for different values t_0 of t' introducing the values of the relaxation function $R(t, t')$ calculated on the basis of the numerical procedure indicated in subsection 7.2.4.11(a). For design aids, refer to subsection 7.2.4.8.

Notation. In the referenced literature, and in the ACI 209R-92 and B3 prediction models, the creep coefficient ϕ (noted also as ϕ) represents the ratio between the creep strain and the initial elastic strain $I/E_{ci}(t_0)$ at the age t_0 at loading. By contrast, the notation ϕ_{28} adopted here is intended to correspond to the definition of the creep coefficient in Eq. (5.1-60).

Reference is made to constitutive relation (5.1-62) as the majority of creep prediction models specify the compliance function $J(t, t')$. No real advantage would be obtained from a computational point of view if reference were made to the equivalent constitutive relation (7.2-25), even in the frame of the equilibrium method. In fact, the

having introduced the ageing coefficient:

$$\begin{aligned} \chi(t, t_0) &= \frac{1}{1 - R(t, t_0) / E_{ci}(t_0)} - \frac{1}{E_{ci}(t_0) J(t, t_0) - 1} \\ &= \frac{E_{ci}(t_0)}{E_{ci}(t_0) - R(t, t_0)} - \frac{E_{ci}}{E_{ci}(t_0) \phi_{28}(t, t_0)} \end{aligned} \quad (7.2-40)$$

the effective modulus:

$$E_{c,ef}(t, t_0) = \frac{1}{J(t, t_0)} = \frac{E_{ci}(t_0)}{1 + [E_{ci}(t_0) / E_{ci}] \phi_{28}(t, t_0)} \quad (7.2-41)$$

and the age-adjusted effective modulus:

$$E_{c,adj}(t, t_0) = \frac{E_{ci}(t_0)}{1 + \chi(t, t_0) [E_{ci}(t_0) / E_{ci}] \phi_{28}(t, t_0)} \quad (7.2-42)$$

$\chi(t, t_0)$ varies relatively little with the age t_0 for sufficiently long elapsed times. For typical values of t_0 and other influencing parameters, its long term values are in the range 0.6–0.9 for the prediction model of subsection 5.1.9.4 and in a narrower range for other models like GL2000 and B3.

The adoption of a fixed long term value within this narrower range, independently of the age at loading and of the creep properties of the structural element being considered, often leads to satisfactory accuracies in the evaluation of the long term structural responses, particularly in the conceptual and preliminary design stages and in the assessment of structures of low sensitivity to time-dependent effects.

In such situations, it is often adequate to use the value $\chi = 0.8$.

7.2.4.11 General method

The most general and refined approach for the evaluation of creep and shrinkage structural effects in the frame of ageing linear viscoelasticity consists in the incorporation of the constitutive relation (5.1-62) for concrete into the computational algorithms of continuum mechanics or of structures composed of beams.

relaxation function $R(t, t')$ needs to be numerically calculated through Eq. (7.2-26), from the given compliance function

The linear Volterra integral equations of structural creep problems can be solved analytically only for some simple forms of the compliance function $J(t, t')$. For the compliance function of the creep prediction model given in subsection 5.1.9.4.3, and for the other modern creep prediction models referenced in subsection 7.2.4.4, an incremental numerical solution of the type indicated here is required.

In consideration of the particular shape of the creep curves and of the typical imposed strain histories in most structural problems, both being characterized by increments at decreasing rate (if fluctuating strains such as for example cyclic thermal strains are excluded), it is possible to gradually increase the time steps Δt_k in order to reach the long term response with an acceptable number of steps, considering that the first intervals should be of the order of fractions of a day (due to the high initial slope of the creep curves).

The same incremental numerical procedure has to be adopted for the solution of Eqs. (7.2-26) and (7.2-36) to obtain the relaxation function $R(t, t')$ and the redistribution function $\xi(t, t_0, t_i)$ from the given compliance function $J(t, t')$. The ageing coefficient $\chi(t, t')$ can then be determined from Eq. (7.2-40). For further information and reference to the computer programs, see *fib* Bulletin 52.

This storage requirement used to be a computational burden, but for modern computers it is only a problem for structural systems with a huge number of unknowns.

The rate-type form is also advantageous for dealing with variable humidity and temperature, as it allows to separate the effects of variable pore humidity or temperature on creep viscosity from those on the ageing rate, and with material damage (fracture and distributed cracking) as they are rate-dependent processes. Only with the rate-type form is it possible to meet the thermodynamic restrictions and introduce physical concepts related to these processes. See the specialized literature (Bažant 2009-2012; ACI document 209.3R).

The use of the ageing linear viscoelastic model for the concrete portions of the structure and of the elastic model for steel leads to a system of linear Volterra integral compatibility or equilibrium equations, when the force or, respectively, the deformation method is adopted for the structural analysis.

A completely general, accurate and most effective computational approach is to obtain first the incremental form of Eq. (5.1-62) for a small time step, by one of the procedures indicated in the following. The incremental form represents a linear elastic stress-strain relation with initial strains, in which the elastic moduli and initial strains vary through the body, and from step to step. Thus the problem of ageing linear viscoelasticity gets converted, already at the constitutive level, to a sequence of elasticity problems.

Any elastic finite element program, used repeatedly in a loop, can thus be generalized for creep, allowing the formulation of the solution in terms of a time-history.

(a) Incremental numerical solution based on the hereditary integral

An incremental form may be obtained by replacing the integral over the past stress or strain history with a sum.

The procedure is based on the approximation of the superposition integral of Eq. (5.1-62) with finite sums using the second order trapezoidal integration rule. The time t is subdivided by discrete times $t_0, t_1, t_2, \dots, t_i, \dots, t_k$ into sub-intervals $\Delta t_i = t_i - t_{i-1}$ ($i = 1, 2, \dots, k$) and at each step the average value of the compliance function $\frac{1}{2}[J(t_k, t_i) + J(t_k, t_{i-1})]$ is adopted in the calculation. In order to allow strain histories with an initial finite step, times t_0 and t_1 are assumed to be coincident, so that $\Delta t_1 = t_1 - t_0 = 0$, and consequently $\Delta \epsilon_{c\sigma}(t_1) = \epsilon_{c\sigma}(t_0)$.

At each step the calculations of the previous steps must be stored, so that the entire history of stress and strain can be stored in computer memory.

(b) Incremental numerical solution based on rate-type creep laws

The computation can be made more efficient, approximating the integral-type constitutive law with a rate-type relation based on Kelvin chains of spring-dashpot rheological models for ageing materials. In that case, the history does not need to be stored because it is implied by the current values of a few hidden variables. The resulting differential equations are of second order. Rate-type laws are particularly helpful for the solution of structural problems by means of the finite elements method, because they are immediately compatible with this computational approach.

The effective shear depth z may be taken as $0.9d$, where d denotes the distance between the extreme compression fibre and the mean planes of the reinforcement layers at the opposite face. The effective shear depth needs not be taken as less than $0.72h$.

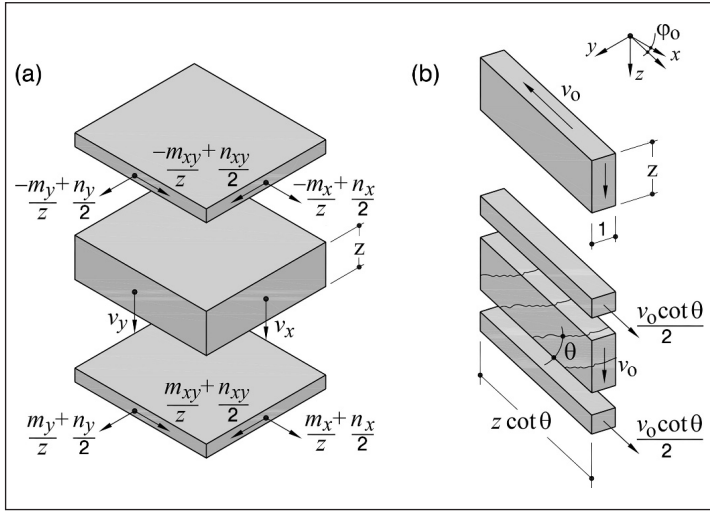


Figure 7.3-3: (a) Layer forces in sandwich model and (b) transfer of transverse shear force in uncracked and cracked core

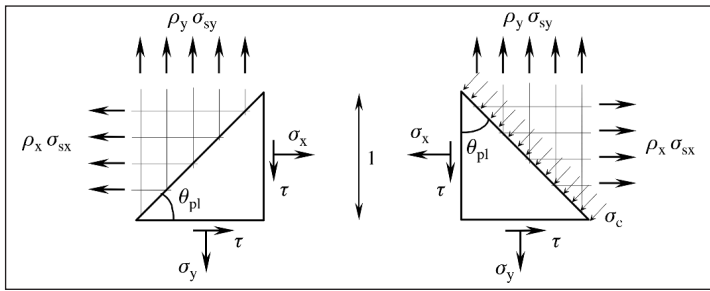


Figure 7.3-4: Stresses acting on and within a reinforced concrete element

To ensure that the ductility demand is met, the term $|\theta_{pl} - \theta_{el}|$ in Eq. (7.3-9) should not be greater than 15° , unless refined calculations are undertaken to justify a higher value.

$$n_{y\inf,sup} = \frac{n_y}{2} \pm \frac{m_y}{z} + \frac{v_y^2}{2v_o} \cot \theta \quad (7.3-2)$$

$$n_{xy\inf,sup} = \frac{n_{xy}}{2} \pm \frac{m_{xy}}{z} + \frac{v_x v_y}{2v_o} \cot \theta \quad (7.3-3)$$

where:

θ is the inclination of the compression stresses in the core layer;

z is the average lever arm between the forces in the x and y directions in the top and bottom layers and the effective shear depth, respectively;

v_o is the principal transverse shear force per unit length and follows from:

$$v_o = \sqrt{v_x^2 + v_y^2} \quad (7.3-4)$$

For members with shear reinforcement, the angle θ is to be selected in accordance with subsection 7.3.3.3. For members without shear reinforcement, a value of $\cot \theta = 2$ may be used as is (implicitly) suggested in subsection 7.3.3.2.

Design of outer membrane layers

If at least one principal stress is in tension, the outer layers may be designed as membrane elements according to plasticity theory such that (Figure 7.3-4):

$$\sigma_{sx} = \frac{1}{\rho_x} (\sigma_x + \tau \cot \theta_{pl}) \leq f_{yd} \quad (7.3-5)$$

$$\sigma_{sy} = \frac{1}{\rho_y} (\sigma_y + \tau \cot \theta_{pl}) \leq f_{yd} \quad (7.3-6)$$

$$\sigma_c = \frac{\tau}{\sin \theta_{pl} \cos \theta_{pl}} \leq \frac{v f_{ck}}{\gamma_c} \quad (7.3-7)$$

If no reinforcement is yielded and at least one principal stress is in tension:

$$v = \frac{1.18}{1.14 + 0.00166 \sigma_{si}} \leq 1.0 \quad (7.3-8)$$

where σ_{si} is the maximum tensile stress (in MPa) in any layer of reinforcing steel ($i = x, y$).

If one or more layers of reinforcement yield:

$$v = \left(1 - 0.032 |\theta_{pl} - \theta_{el}| \right) \cdot \frac{1.18}{1.14 + 0.00166 f_{yd}} \quad (7.3-9)$$

where:

θ_{pl} is the compression field angle with respect to x-axis at the ULS

θ_{el} is the first cracking angle with respect to the x-axis.

If both principal stresses are compressive:

$$|\sigma_2| \leq v f_{cd} \quad (7.3-10)$$

where σ_2 is the minor principal (compressive) stress and v may be taken as 1.0 or determined in accordance with subsection 5.1.6.

Design of inner core layer

The shear core should be designed in accordance with subsection 7.3.3.

7.3.3 Shear

7.3.3.1 General

Design shear force and shear resistance

The models presented in this section represent an advance in philosophy to more physical based models. The change is made at this time, in recognition of the maturity of the new methods and of their capacity for further development, using a consistent framework, over the future years.

For the past 30 years, empirical approaches have formed the basis of models for design for shear and are widely adopted in national design standards. Such models have been validated for a wide range of structural applications and may continue to be used in design, including as models for beam shear and for punching shear.

The following equations are provided for the shear resistance of the webs of beams and the core layers of slabs and do not include the effects of flanges. Figures 7.3-3 and 7.3-5 show the regions of members being designed both for slabs and beams, respectively.

In beams, a minimum quantity of shear reinforcement in accordance with subsection 7.13.5.2 must be provided.

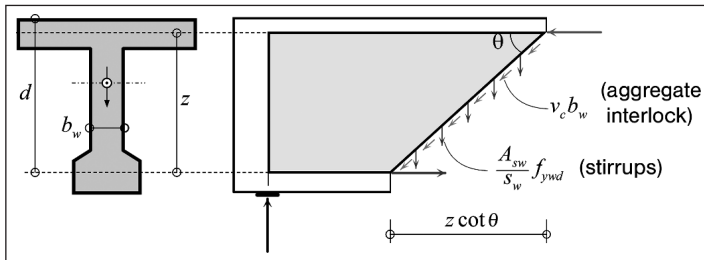


Figure 7.3-5: Forces in the web of a beam

Further background information on shear provisions treated in this section is given by Sigrist, V., Bentz, E., Fernández Ruiz, M., Foster, S., Muttoni, A. (2013), Background to the *fib* Model Code 2010 Shear Provisions – Part I: Beams and Slabs. Structural Concrete, 14. doi: 10.1002/suco.201200066.

The depth d denotes the effective depth in flexure which is defined as the distance from the extreme compressive fibre of the concrete to the resultant tensile force in the tensile reinforcing steel and tendons.

The dimension z may also be taken as the distance between the centrelines of the top and the bottom chord, where the depth of the compression chord may be calculated for the location of maximum

The shear resistance of a web or slab is determined according to:

$$V_{Rd} = V_{Rd,c} + V_{Rd,s} \geq V_{Ed} \quad (7.3-11)$$

where:

V_{Rd} is the design shear resistance;

$V_{Rd,c}$ is the design shear resistance attributed to the concrete;

$V_{Rd,s}$ is the design shear resistance provided by shear reinforcement;

V_{Ed} is the design shear force.

The design can be based on a stress field analysis or a strut-and-tie model, as outlined in subsection 7.3.6. Such models are especially suitable for the design of discontinuity regions (D-regions) at supports or transverse applied forces.

Alternatively, a cross-sectional design procedure may be applied. The corresponding rules are given in the following subsections.

Cross-sectional design

For a cross-sectional design, the design shear force must in general be determined for control sections at a location d from the face of supports (see Figure 7.3-6) and from discontinuities of geometry or applied loads.

For the effective shear depth z a value of $0.9d$ can be assumed.

Other control sections may be required, for example in case of varying web widths along a span, for non-uniform or

bending and a stress block according to Figure 7.2-12. For non-prestressed members z must not be less than $0.9d$.

For members containing mild steel reinforcement as well as prestressed tendons, the effective shear depth z can be taken as:

$$z = \frac{z_s^2 A_s + z_p^2 A_p}{z_s A_s + z_p A_p} \quad (7.3-12)$$

where z_s and z_p denote the distances between the centreline of the compressive chord and the reinforcement and tendon axes, respectively.

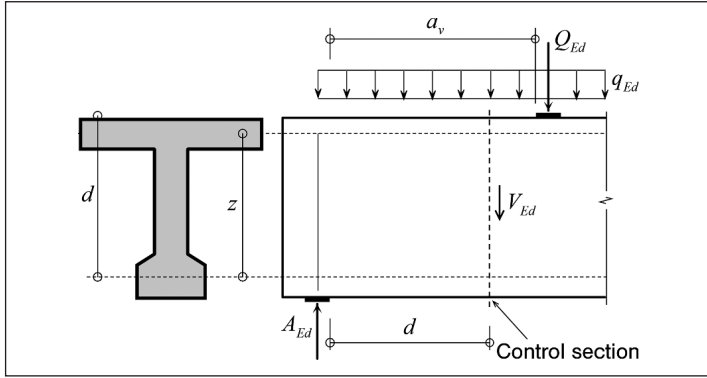


Figure 7.3-6: Definition of control section for sectional design

The effect of redistribution of internal forces in slabs with concentrated loads can result in higher shear capacities when compared to one-way slabs or beams subjected to uniformly distributed loading. This effect may be accounted for by assuming a uniform distribution of the shear force along a control width b_w , as shown in Figure 7.3-7.

significant concentrated loads, or at sections near points of curtailment of reinforcement.

Sections closer to supports than the distance d may be designed for the same shear force as at the control section provided that the member is directly supported.

Unless more refined modelling techniques are used to consider loads taken directly to a support through strut or arch action (see subsection 7.3.6), the following rules apply:

- the contribution of point loads applied within a distance of $d < a_v \leq 2d$ from the face of the support to the design shear force V_{Ed} may be reduced by the factor:

$$\beta = a_v / (2d) \quad (7.3-13)$$

- in the case of point loads applied as close as $a_v < d$ from the face of the support, the design shear force V_{Ed} must be calculated with $\beta = 0.5$ as if the load was applied at $a_v = d$.

Where a concentrated load is applied to a slab near a support line, its capacity must be checked for punching at the control perimeter around the loaded area, as described in subsection 7.3.5, and for shear at a control section taken parallel to the line of the support, as defined in Figure 7.3-7.

The control section is taken at the lesser of the distances equal to d and $a_v/2$ from the face of the support. The load distribution angle must be taken as $\alpha = 45^\circ$ for the case of clamped edges and $\alpha = 60^\circ$ for simply supported edges.

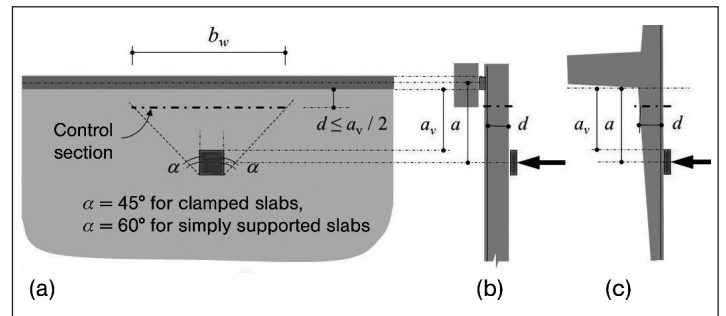


Figure 7.3-7: Location and length of the control section, b_w , for the determination of the shear resistance of slabs with point loads located near a support-line; (b) simple edge support; (c) clamped edge support

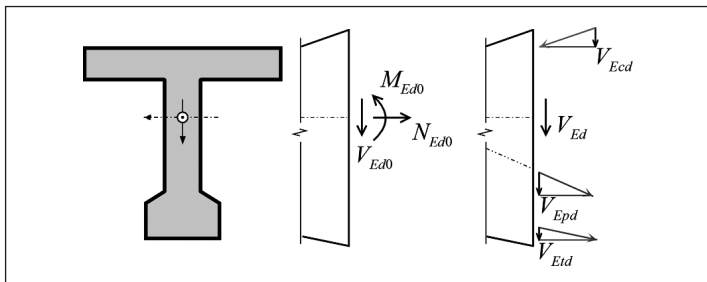


Figure 7.3-8: Contributions of inclined chord forces to design shear force (M_{Ed0} , V_{Ed0} and N_{Ed0} denote bending moment, shear and normal forces resulting from sectional analysis)

For determining V_{Ed} , the shear force from the sectional analysis V_{Ed0} may be reduced by favourable contributions resulting from any inclined tension chords (V_{Etd}), compression chords (V_{Ecd}) and prestressing tendons (V_{Epd}) – see Figure 7.3-8. In determining V_{Epd} , an eventual reduction in prestress due to the development length must be considered.

Any unfavourable contributions from inclined chord and prestressing tendon forces must be added to V_{Ed0} .

The level I equation is derived from the more general level II approximation with the assumption that the mid-depth strain at the control section can be taken as $\epsilon_x = 0.00125$, which corresponds to half the yield strain for a reinforcing bar with $f_{yk} = 500$ MPa ($\epsilon_x \approx f_{yk} / (2E_s)$).

In higher strength concrete and lightweight aggregate concretes, the fracture surface may go through the aggregate particles, rather than around, reducing the crack roughness.

There is evidence that the shear resistance of members without shear reinforcement is influenced by the maximum size of the aggregate d_g . If concrete with a maximum size of the aggregate different from $d_g = 16$ mm is used, the value k_{dg} may be calculated with:

$$k_{dg} = \frac{32}{16 + d_g} \geq 0.75 \quad (7.3-20)$$

For concrete strengths in excess of 70 MPa and for lightweight concrete, d_g in Eq. (7.3-20) should be taken as zero, in order to account for the loss of aggregate interlock in the cracks due to fracture of aggregate particles.

The web reinforcement ratio given by Eq. (7.3-22) corresponds to the minimum reinforcement ratio as defined in subsection 7.13.5. Members containing a lower reinforcement ratio are to be treated according to section 7.3.3.2.

Size effects are limited in members with web reinforcement greater than that required by Eq. (7.3-22).

The strength reduction factor k_c consists of two parts: the state of strain in the webs of beams or the core layers of slabs is taken into account by k_ϵ ; the effect of more brittle failure behaviour of concrete of strengths greater than 30 MPa is considered in η_{fc} .

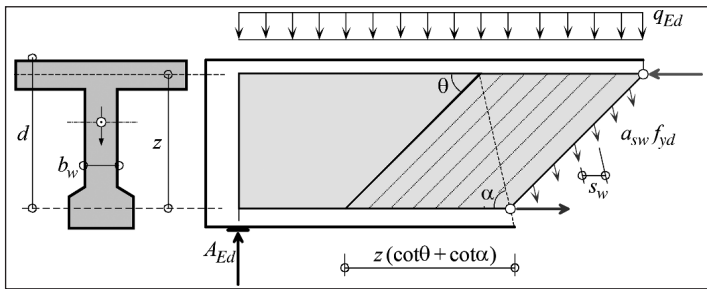


Figure 7.3-10: Geometry and definitions

In the case of stirrups that are inclined relative to the beam axis the Eqs. (7.3-26) and (7.3-29) are replaced by:

$$V_{Rd,max} = k_c \frac{f_{ck}}{\gamma_c} b_w z \frac{\cot \theta + \cot \alpha}{1 + \cot^2 \theta} \quad (7.3-24)$$

and

$$V_{Rd,s} = \frac{A_{sw}}{s_w} z f_{ywd} (\cot \theta + \cot \alpha) \sin \alpha \quad (7.3-25)$$

where α is the inclination of the stirrups as shown in Figure 7.3-10.

Level I approximation

For members with no significant axial load, with $f_{yk} \leq 600$ MPa, $f_{ck} \leq 70$ MPa and with a maximum aggregate size of not less than 10 mm:

$$k_v = \frac{180}{1000 + 1.25z} \quad (z \text{ in mm}) \quad (7.3-19)$$

Level II approximation

For the level II approximation, the design shear resistance is determined with:

$$k_v = \frac{0.4}{1 + 1500\epsilon_x} \cdot \frac{1300}{1000 + k_{dg}z} \quad (z \text{ in mm}) \quad (7.3-21)$$

Provided that the size of the maximum aggregate particles, d_g , is not less than 16 mm, k_{dg} in Eq. (7.3-21) can be taken as $k_{dg} = 1.0$.

7.3.3.3 Members with shear reinforcement

General

This subsection applies to members that meet the demand for minimum shear reinforcement according to:

$$\rho_w \geq 0.08 \frac{\sqrt{f_{ck}}}{f_{yk}} \quad (f_{ck} \text{ and } f_{yk} \text{ in MPa}) \quad (7.3-22)$$

The design shear resistance is then determined from:

$$V_{Rd} = V_{Rd,c} + V_{Rd,s} \quad (7.3-23)$$

but must not be taken as greater than:

$$V_{Rd,max} = k_c \frac{f_{ck}}{\gamma_c} b_w z \sin \theta \cos \theta \quad (7.3-26)$$

where θ denotes the inclination of the compressive stress field.

The strength reduction factor is defined as:

$$k_c = k_\epsilon \eta_{fc} \quad (7.3-27)$$

with k_ϵ as given in the following and:

$$\eta_{fc} = \left(\frac{30}{f_{ck}} \right)^{1/3} \leq 1.0 \quad (f_{ck} \text{ in MPa}) \quad (7.3-28)$$

The limitation on $\sqrt{f_{ck}}$ is provided due to the larger observed variability in shear strength of members with higher strength concrete, particularly for members without stirrups such as slabs.

Values of k_D depend on the material of the duct and whether it is grouted or not. Suggested values for design are:

- grouted steel duct: $k_D = 0.5$;
- grouted plastic duct: $k_D = 0.8$;
- ungrouted duct: $k_D = 1.2$.

Factor k_D may be reduced in presence of reinforcement transverse to the plane of the web.

In the case of stirrups that are inclined relative to the beam axis the Eq. (7.3-34) is replaced by:

$$\Delta F_{td} = \frac{V_{Ed}}{2} (\cot \theta - \cot \alpha) \quad (7.3-32)$$

In the level III approach, where a concrete contribution $V_{Rd,c} \neq 0$ is considered, in the Eqs. (7.3-32) and (7.3-34) the design shear force V_{Ed} is replaced by V_{Ed}^* where:

$$V_{Ed}^* = V_{Ed} + V_{Rd,c} \quad (7.3-33)$$

The value of $\theta_{\min}$ is determined by the level of approximation.

The level I approximation represents a variable angle truss model approach.

The design shear resistance provided by stirrups is:

$$V_{Rd,s} = \frac{A_{sw}}{s_w} z f_{ywd} \cot \theta \quad (7.3-29)$$

where f_{ywd} denotes the design yield strength of the shear reinforcement.

The design shear resistance attributed to the concrete can be taken as:

$$V_{Rd,c} = k_v \frac{\sqrt{f_{ck}}}{\gamma_c} b_w z \quad (7.3-30)$$

where the value of $\sqrt{f_{ck}}$ must not be taken as greater than 8 MPa.

In Eq. (7.3-30), the effective web width b_w must be taken as the minimum concrete web width within the effective shear depth z .

In the case of prestressing tendons with duct diameters $\varnothing_D \geq b_w/8$, the ultimate resistance of the compression struts must be calculated on the basis of the nominal value of the web width:

$$b_{w,nom} = b_w - k_D \sum \varnothing_D \quad (7.3-31)$$

where $\sum \varnothing_D$ is to be determined for the most unfavourable prestressing tendon configuration.

The longitudinal reinforcement in the flexural tensile chord at every section of interest is to be designed to resist the additional force due to shear of:

$$\Delta F_{td} = \frac{V_{Ed}}{2} \cot \theta \quad (7.3-34)$$

However, the total demand on longitudinal reinforcement need not exceed the demand at the maximum moment location due to moment alone.

The limits of the compressive stress field inclination θ , relative to the longitudinal axis of the member (Figure 7.3-10), are:

$$\theta_{\min} \leq \theta \leq 45^\circ \quad (7.3-35)$$

where θ may be chosen freely between these limits for design.

Level I approximation

In the level I approach, the design shear resistance is given by:

$$V_{Rd} = V_{Rd,s} \leq V_{Rd,max} \quad (7.3-36)$$

but must not to be taken less than the resistance of the same member without shear reinforcement.

The minimum inclination of the compressive stress field is:

$$\theta_{\min} = 25^\circ \quad \text{for members with significant axial compression or prestress;}$$

$$\theta_{\min} = 30^\circ \quad \text{for reinforced concrete members;}$$

$$\theta_{\min} = 40^\circ \quad \text{for members with significant axial tension.}$$

The width of the beam or web must be checked for the selected inclinations of the compression stresses where k_ϵ is taken as:

$$k_\epsilon = 0.55 \quad (7.3-37)$$

Eqs. (7.3-36) and (7.3-37) apply to cross-sections where the longitudinal strain ϵ_x remains below a value of 0.001.

Level II approximation

In the level II approach, the design shear resistance is given by:

$$V_{Rd} = V_{Rd,s} \leq V_{Rd,max} \quad (7.3-38)$$

but must not to be taken less than the resistance of the same member without shear reinforcement.

The level II approximation is based on a generalized stress field approach.

The stress field approach allows the strut inclination θ to be selected within certain limits and is confirmed by experimental observations.

For a preliminary design or analysis, ε_x maybe taken as 0.001.

The variable value k_ε considers the influence of the state of strain in the web. This influence is important for prestressed members or members in compression, but less significant for reinforced elements and/or sections with higher θ and ε_x values.

Within the limits of Eq. (7.3-35) the inclination of the compressive stress field can be freely selected for design, or analytically determined for assessment.

The minimum inclination of the compressive stress field is:

$$\theta_{\min} = 20^\circ + 10000\varepsilon_x \quad (7.3-39)$$

where ε_x represents the longitudinal strain at the mid-depth of the effective shear depth as shown in Figure 7.3-9.

The design shear resistance attributed to the concrete is neglected, that is $k_v = 0$. The width of the beam or web should be checked for the respective inclination of the compression stresses where k_ε is taken as:

$$k_\varepsilon = \frac{1}{1.2 + 55\varepsilon_1} \leq 0.65 \quad (7.3-40)$$

where:

$$\varepsilon_1 = \varepsilon_x + (\varepsilon_x + 0.002) \cot^2 \theta \quad (7.3-41)$$

The longitudinal strain ε_x at the mid-depth of the effective shear depth is calculated on the basis of Eq. (7.3-14) or (7.3-16).

Level III approximation

Level III approximation represents a general form of sectional shear equations and is based on the simplified modified compression field theory.

A comparison of the relative predictions for modelling levels I to III is shown in Figure 7.3-11.

In the level III approach, the design shear resistance in the range of $V_{Rd} < V_{Rd,max}(\theta_{\min})$ is given by:

$$V_{Rd} = V_{Rd,s} + V_{Rd,c} \quad (7.3-42)$$

where $V_{Rd,max}(\theta_{\min})$ is calculated from Eq. (7.3-26) for $\theta = \theta_{\min}$.

In the range of $V_{Rd} \geq V_{Rd,max}(\theta_{\min})$ the resistance is determined as in the level II approximation.

The inclination $\theta_{\min}$ is taken as given by Eq. (7.3-39).

For determining the design shear resistance $V_{Rd,c}$ attributed to the concrete the following expression should be used:

$$k_v = \frac{0.4}{1 + 1500\varepsilon_x} \left(1 - \frac{V_{Ed}}{V_{Rd,max}(\theta_{\min})} \right) \geq 0 \quad (7.3-43)$$

The strain ε_x at the mid-depth of the effective shear depth is calculated with Eq. (7.3-14) or (7.3-16).

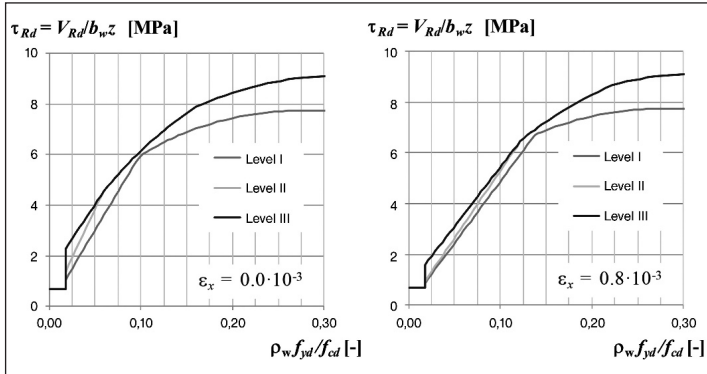


Figure 7.3-11: Comparison of levels I, II and III results for members with $f_{ck} = 50$ MPa (Note: for the curves shown in the figure the value f_{cd} is defined as $\eta_{fc} f_{ck} / \gamma_c$)

The use of tools based on advanced methods of analysis often requires extensive experience to ensure that safe and consistent results are obtained.

For hollow core slabs and similar structural members the design shear resistance may be calculated on the basis of this subsection or alternatively of subsection 7.3.3.3; the higher of the results may be adopted as the capacity.

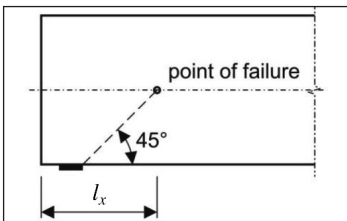


Figure 7.3-12: Basis for derivation Eq. (7.3-44)

Level IV approximation

The resistance of members in shear, or shear combined with torsion, may be determined by satisfying the applicable conditions of equilibrium and compatibility of strains and by using appropriate stress-strain models for the steel and for diagonally cracked concrete.

7.3.3.4 Hollow core slabs

In single span prestressed hollow core slabs without shear reinforcement, shear failure occurs when the principal tensile stress in the web exceeds the tensile strength of the concrete.

Level I approximation

The design shear resistance can be determined by:

$$V_{Rd,ct} = 0.8 \frac{I_c \cdot b_w}{S_c} \sqrt{f_{ctd}^2 + \alpha_l \cdot \sigma_{cp} \cdot f_{ctd}} \quad (7.3-44)$$

where:

I_c is second moment of area;

S_c is first moment of area above and about the centroidal axis;

b_w is width of the cross-section at the centroidal axis;

σ_{cp} is concrete compressive stress at the centroidal axis due to prestressing, in the area where the prestressing force is fully introduced;

$\alpha_l = l_x / l_{bpt,95\%}$;

l_x follows from Figure 7.3-12;

$l_{bpt,95\%}$ follows from Eq. (7.13-5a).

Level II approximation

In a level II approximation, the design shear resistance is determined by:

$$V_{Rd,ct} = \frac{I_c \cdot b_w(y)}{S_c(y)} [\sqrt{f_{ctd}^2 + \alpha_l \cdot \sigma_{cp}(y) \cdot f_{ctd}} - \tau_{cp}(y)] \quad (7.3-45)$$

where:

I_c is second moment of area;

$S_c(y)$ is first moment of area above height y and about the centroidal axis;

$b_w(y)$ is width of the cross-section at the height y ;

y is the height of the critical point at the line of failure;

$\sigma_{cp}(y)$ is concrete compressive stress at height y and distance l_x ;

$\tau_{cp}(y)$ is the shear stress in the concrete due to transmission of prestress at height y and distance l_x .

The concrete compressive stress at height y and distance l_x is determined from:

$$\sigma_{cp}(y) = \left[\frac{1}{A_c} + \frac{y_c - y}{I} \right] \cdot F_p(l_x) \quad (7.3-46)$$

and the shear stress in the concrete due to transmission of prestress:

$$\tau_{cp}(y) = \frac{1}{b_w(y)} \cdot \left[\frac{A_c(y)}{A_c} - \frac{S_c(y) \cdot (y_c - y_{pt})}{I} \right] \cdot \frac{dF_p(l_x)}{dx} \quad (7.3-47)$$

where:

y_c is height of concrete centroidal axis;

A_c is area of concrete cross-section;

$A_c(y)$ is concrete area above height y ;

y_{pt} is height of centroidal axis of prestressing steel;

$F_p(l_x)$ is the prestressing force at distance l_x .

By varying y in the calculation, the lowest value of $V_{Rd,ct}$ in Eq. (7.3-45) is found.

7.3.3.5 Shear between web and flanges of T-sections

The introduction of tensile or compressive forces into the flanges (Figure 7.3-13) creates shear forces at the transition of web and flanges, inducing corresponding transverse tensile and compressive forces in the flanges.

The spread of the compressive forces in the flanges must be examined with the aid of stress fields. Recommended values for the angle of spread are:

$25^\circ \leq \theta_f \leq 45^\circ$ for compressive flanges; and

$35^\circ \leq \theta_f \leq 50^\circ$ for tensile flanges.

Unless a more detailed analysis is undertaken, reinforcement for the introduction of forces into the flanges is to be superimposed on that required for transverse bending.

In addition, the minimum transverse reinforcement must not be less than that required by subsection 7.13.5.

The longitudinal flange reinforcement must be anchored in accordance with the assumed stress field requirements.

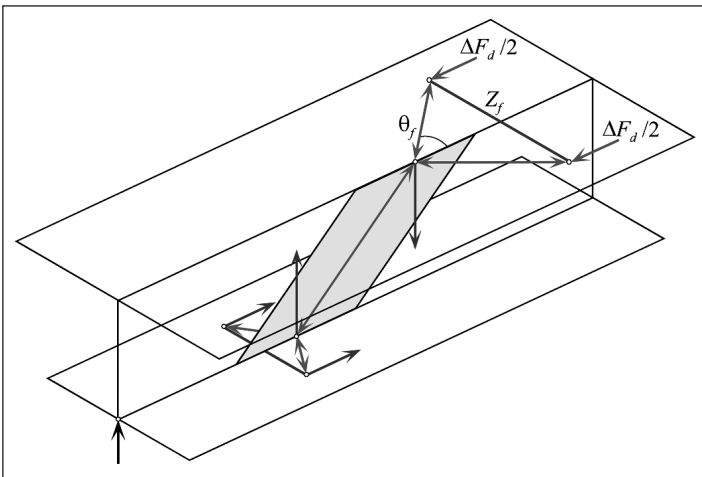


Figure 7.3-13: Strut-and-tie model for force introduction into the flanges

Background information on this subject is given by Randl, N. (2013), Design recommendations for interface shear transfer in *fib* Model Code 2010. Structural Concrete, 14. doi: 10.1002/suco.201300003.

If a “rigid” bond-slip behaviour is expected and very good adhesive bonding guaranteed on the site, the adhesive bond effect should be taken into account without superimposing effects of interface reinforcement. The most important precondition for the assumption of good adhesive bond is a well prepared and very clean concrete surface at the time of casting.

Adhesive bond resistance should only be applied for design if no tensile loading perpendicular to the interface is expected.

The adhesion coefficients c_a actually depend on a variety of influencing parameters (see subsection 6.3.3); nevertheless the c_a factors given in Table 7.3-1 represent reasonable values on the safe side for the given roughness categories.

Special attention must be given to edge zones – see detailing rules in section 6.3.

For the definition of roughness of the classes distinguished in Table 7.3-1, see subsection 6.3.2.

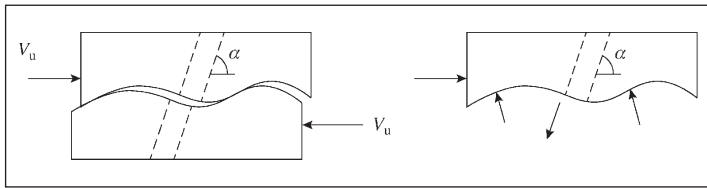


Figure 7.3-14: Transmission of shear forces across an interface intersected by reinforcing bars

Eq. (7.3-51) relates to interfaces intersected by dowels or reinforcement and characterized by a rather non-rigid bond-slip behaviour.

Connectors may be omitted in interface regions where the design shear stress does not exceed the resistance given in Eq. (7.3-50).

7.3.3.6 Shear at the interface between concrete cast at different times

In addition to the requirements formulated in subsections 7.3.3.1–7.3.3.5 the shear stress at the interface between concrete cast at different times should also satisfy the following condition:

$$\tau_{Edi} \leq \tau_{Rdi} \quad (7.3-48)$$

where τ_{Edi} is the design value of the shear stress in the interface, given by

$$\tau_{Edi} = \beta \cdot V_{Ed} / (z b_i) \quad (7.3-49)$$

where:

- β is the ratio of the longitudinal force in the new concrete and the total longitudinal force either in the compression or tension zone, both calculated for the section considered;
- z is the inner lever arm of the composed section;
- b_i is the width of the interface and V_{Ed} is the shear force on the composed section.

Interface without reinforcement (rigid bond-slip behaviour)

The design limit value τ_{Rdi} for the interface shear in Eq. (7.3-31) follows from:

$$\tau_{Rdi} = c_a \cdot f_{ctd} + \mu \cdot \sigma_n \leq 0.5 \cdot v \cdot f_{cd} \quad (7.3-50)$$

where:

- c_a is the coefficient for the adhesive bond;
- μ is the friction coefficient from Table 7.3-2;
- σ_n is the (lowest expected) compressive stress resulting from an eventual normal force acting on the interface.

The adhesion factor c_a depends on the roughness of the interface (see Table 7.3-1; R_t is derived from the sand patch method).

Table 7.3-1: Coefficients for the adhesive bond resistance

Surface characteristics of interface		c_a
Very rough (including shear keys)	$R_t \geq 3.0$ mm	0.5
Rough (strongly roughened surface)	$R_t \geq 1.5$ mm	0.40
Smooth (concrete surface without treatment after vibration or slightly roughened when cast against formwork)		0.20
Very smooth (steel, plastic, timber formwork)		0.025

Under fatigue or dynamic loads the values for c_a as found in Table 7.3-1 have to be reduced to 50%.

Interface intersected by dowels or reinforcement

If strong adhesive bond cannot be guaranteed on the site or the design shear resistance provided by adhesive bond from Eq. (7.3-50) is lower than the design shear stress, interface connectors are required and the design limit value τ_{Rdi} follows from:

$$\tau_{Rdi} = c_r \cdot f_{ck}^{1/3} + \mu \cdot \sigma_n + \kappa_1 \cdot \rho \cdot f_{yd} \cdot (\mu \cdot \sin \alpha + \cos \alpha) + \kappa_2 \cdot \rho \cdot \sqrt{f_{yd} \cdot f_{cd}} \leq \beta_c \cdot v \cdot f_{cd} \quad (7.3-51)$$

where strength values are in N/mm² and:

- c_r is the coefficient for aggregate interlock effects at rough interfaces;
- κ_1 is the interaction coefficient for tensile force activated in the reinforcement or the dowels;

Note that for a bar in tension, as explained in section 6.3, the tensile strength of the bar is reduced when the bar is also subjected to dowel action, as shown in Figure 7.3-15.

The detailing rules given in section 6.3 concerning embedment depth of connectors and minimum amount of steel cross-section must be obeyed.

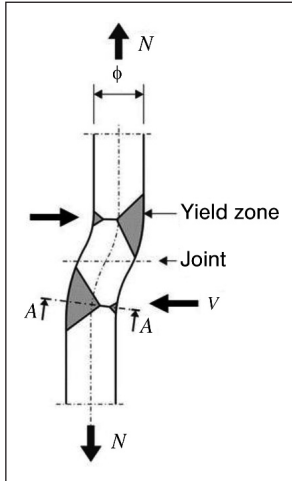


Figure 7.3-15: Dowel action under simultaneous tension and shear

For the background of the values in Table 7.3-2, see Randl, N., Design recommendation for interface shear transfer in MC2010 (Structural Concrete, Vol. 14, No. 3, 2013).

The roughness of a concrete surface can be measured in various ways (see subsection 6.3.2). An appropriate way is the *sand patch method*, as depicted in Figure 7.3-16 (Kaufmann, N., “Das Sandflächenverfahren”, Strassenbautechnik (1971), Nr. 3). A volume of sand V is spread on the rough surface in a circular area with diameter D . The roughness parameter R_t follows from:

$$R_t [mm] = \frac{40 \cdot V}{\pi D^2} \quad (7.3-52)$$

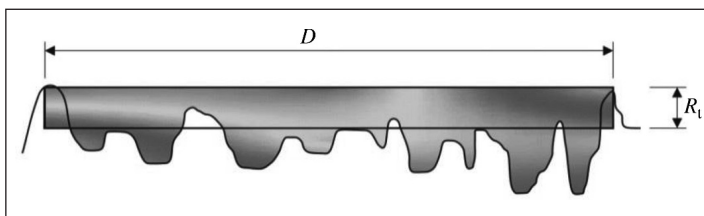


Figure 7.3-16: Principle of sand patch method for the qualification of the roughness of an interface

- κ_2 is the interaction coefficient for flexural resistance;
- μ is the friction coefficient;
- ρ is the reinforcement ratio of the reinforcing steel crossing the interface;
- σ_n is the (lowest expected) compressive stress resulting from an eventual normal force acting on the interface;
- α is the inclination of the reinforcement crossing the interface (see Figure 7.3-14);
- β_c is the coefficient for the strength of the compression strut;

$$\nu = 0.55 \left(\frac{30}{f_{ck}} \right)^{1/3} < 0.55.$$

The coefficients for different surface roughness in interfaces reinforced with dowels or rebars are given in Table 7.3-2.

Table 7.3-2: Coefficients for different surface roughness

Surface Roughness	c_r	κ_1	κ_2	β_c	μ	
					$f_{ck} \geq 20$	$f_{ck} \geq 35$
Very rough* $R_t \geq 3.0$ mm	0.2	0.5	0.9	0.5	0.8	1.0
Rough $R_t \geq 1.5$ mm	0.1	0.5	0.9	0.5	0.7	
Smooth	0	0.5	1.1	0.4	0.6	
Very smooth	0	0	1.5	0.3	0.5	

* valid also for shear keys

Under fatigue or dynamic loads, the values for τ_{Rdi} according to Eq (7.3-51) have to be reduced to 40%.

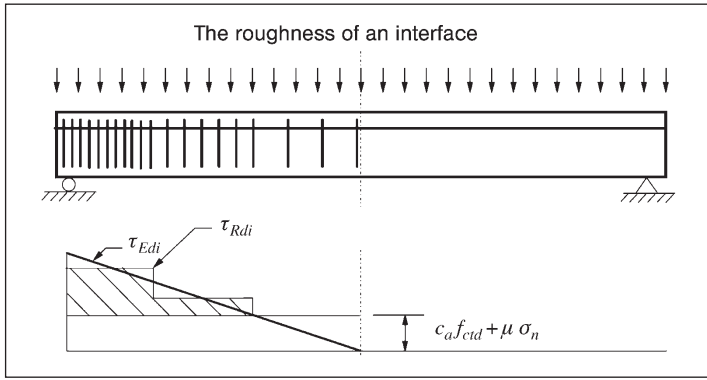


Figure 7.3-17: Shear diagram representing the required interface reinforcement

A stepped distribution of the transverse reinforcement may be used, as indicated in Figure 7.3-17.

7.3.4 Torsion

Where static equilibrium depends on the torsional resistance of elements of the structure, a full torsional design for both the ultimate and serviceability limit states must be undertaken.

Where in structures torsion arises from consideration of compatibility only, and the structure is not dependent on torsional resistance for its stability, it will normally not be necessary to consider torsion at the ultimate limit state. In such cases minimum torsional reinforcement must be provided in the form of stirrups, and longitudinal bars should be provided to prevent excessive cracking, as per the requirements of subsection 7.13.5.2.

The determination of the torsional resistance of box-girders and beams of solid cross-section is based on an ideal hollow cross-section as shown in Figure 7.3-18.

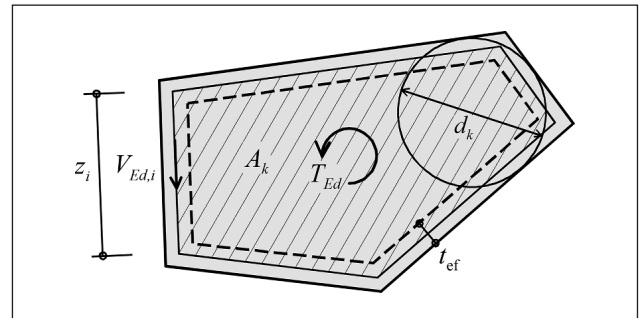


Figure 7.3-18: Definition of the ideal hollow cross-section

If the internal forces and moments, the cross-sectional dimensions and the reinforcement do not change abruptly in the longitudinal direction, it may be assumed that the shear flow due to torsion is constant over the circumference of the effective cross-sectional area. The torsional moment T_{Ed} may then be resolved into equivalent panel forces such that:

$$V_{Ed,Ti} = \frac{T_{Ed}}{2A_k} z_i \quad (7.3-53)$$

where A_k is the area within the centre line of the thin-walled effective cross-section, including inner hollow areas.

The provisions of subsection 7.3.3 apply analogously for the dimensioning of the reinforcement and checking of the panel dimensions. The effective panel thickness of solid cross-sections (Figure 7.3-19) can thereby be taken into account as:

$$t_{ef} \leq \frac{d_k}{8} \quad (7.3-54)$$

where d_k denotes the diameter of the circle that might be inscribed at the most narrow part of the cross-section.

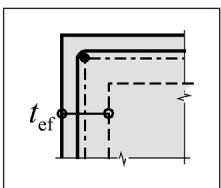


Figure 7.3-19: Minimum effective panel thickness

In the case of combined action of torsion, bending and shear force in a solid section, the core within the idealized hollow cross-section may be used for the transmission of the shear forces.

A minimum effective panel thickness $t_{ef,min}$ of twice the distance between the concrete surface and the centre of the longitudinal reinforcement may be considered.

In the case of box-girders, the effective panel thickness corresponds to the wall thickness, if the wall is reinforced on all sides.

The longitudinal reinforcement due to torsion must either be distributed evenly over the length of the panels or concentrated at the corners.

In the case of combined action of torsion, bending and shear force, the internal forces and moments are replaced by a statically equivalent set of normal and shear forces. The reinforcement is then determined according to the provisions of subsections 7.3.2 and 7.3.3.

For box-girder sections the maximum resistance of a panel is given by $V_{Rd,c}$ for members without and $V_{Rd,max}$ for members with shear reinforcement, respectively.

For other sections (such as rectangular cross-sections) the maximum resistance has to be checked by

$$\left(\frac{T_{Ed}}{T_{Rd,max}} \right)^2 + \left(\frac{V_{Ed}}{V_{Rd,max}} \right)^2 \leq 1 \quad (7.3-55)$$

where $T_{Rd,max}$ is calculated as:

$$T_{Rd,max} = k_c \frac{f_{ck}}{\gamma_c} t_{ef} 2A_k \sin \theta \cos \theta \quad (7.3-56)$$

and the definitions of the parameters involved are given in subsection 7.3.3.3. Moreover $V_{Rd,max}$ follows from Eq. (7.3-27).

7.3.5 Punching

7.3.5.1 General

Punching failures may develop with limited deformations (brittle behaviour). Therefore, the effects of imposed deformations (temperature, creep and shrinkage, settlements etc.) should be taken into account in design. The influence of imposed deformations can, however, be neglected if sufficient deformation capacity is provided. Strategies for increasing the deformation capacity are:

- choice of a sufficiently large supported area and depth of slab in combination with low bending reinforcement ratios (rules are given in subsection 7.3.5.3);
- use of punching shear reinforcement (rules are given in subsection 7.3.5.3).

With flat slabs, safety against punching is particularly significant as the failure of one column can propagate to adjacent columns leading to a complete collapse of a structure. To avoid such progressive collapses, one (or both) of the following strategies should be adopted:

- increase of the deformation capacity of the potential failure zones (see above) to allow internal forces to redistribute;
- arrange integrity reinforcement for slabs with limited deformation capacity (rules are given in subsection 7.3.5.6).

Further background information on the punching shear models treated in this section is given by Muttoni et al. (2013), Background to the *fib* Model Code 2010 Shear Provisions – Part II Punching Shear. Structural Concrete, 14. doi: 10.1002/suco.201200064

Punching can result from a concentrated load applied on a relatively small area of the structure. In flat slabs, punching shear failures normally develop around supported areas (columns, capitals, walls). In other cases (e.g. foundation slabs, transfer slabs, deck slabs of bridges) punching failures can also develop around loaded areas. The rules presented hereafter for flat slabs apply by analogy to loaded areas.

7.3.5.2 Design shear force, shear-resisting effective depth and control perimeter

(1) Design shear force

The design shear force with respect to punching (V_{Ed}) is calculated as the sum of design shear forces acting on a basic control perimeter (b_1).

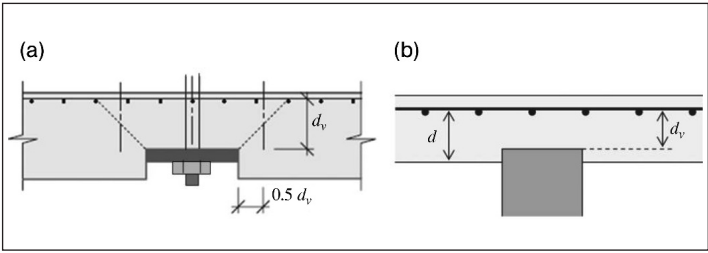


Figure 7.3-20: Effective depth of the slab considering support penetration (d_v) and effective depth for bending calculations (d)

For flat slabs and footings, the design shear force is equal to the value of the support reaction reduced by the actions applied inside the basic control perimeter (such as gravity loads, soil pressure at footings and deviation forces of prestressing tendons).

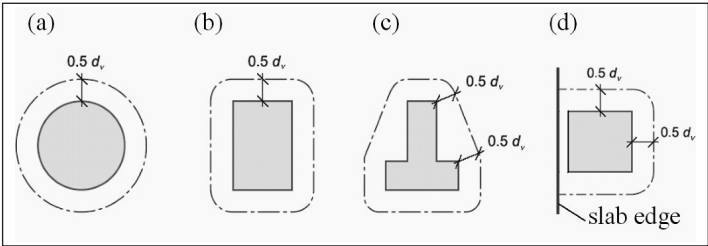


Figure 7.3-21: Basic control perimeters around supported areas

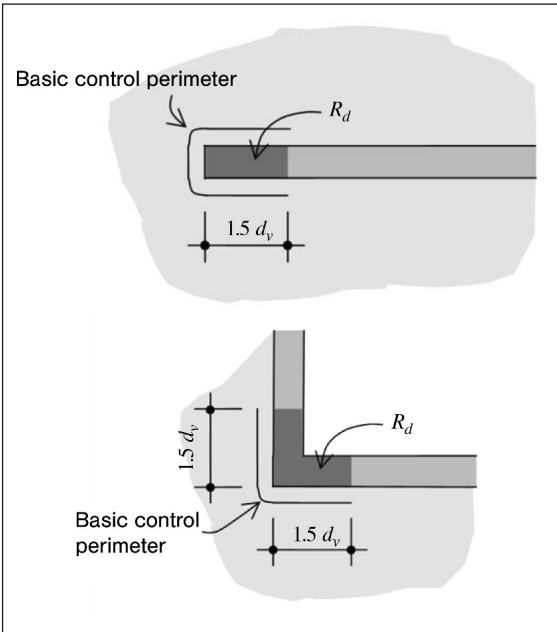


Figure 7.3-22: Basic control perimeter around walls

(2) *Shear-resisting effective depth*

The shear-resisting effective depth of the slab (d_v) is the distance from the centroid of the reinforcement layers to the supported area (Figure 7.3-20).

(3) *Basic control perimeter (b_1)*

The basic control perimeter b_1 may normally be taken at a distance $0.5d_v$ from the supported area (Figure 7.3-21 and Figure 7.3-22) and should be determined in order to minimize its length (Figure 7.3-21c). The length of the control perimeter is limited by slab edges (Figure 7.3-21d).

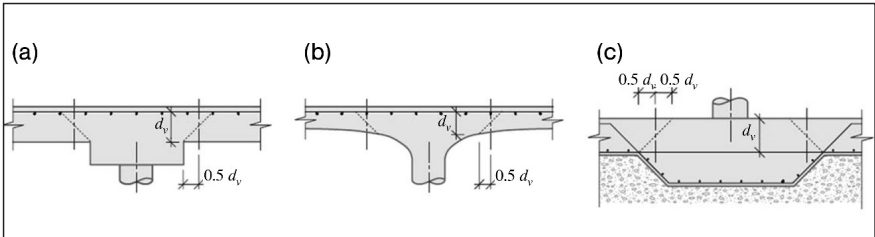


Figure 7.3-23: Choice of the potentially governing control perimeter

In the case of slabs with variable depth, control sections at a greater distance from the supported area may be governing (refer to Figure 7.3-23).

The shear-resisting control perimeter b_0 can be obtained on the basis of a detailed shear field analysis as:

$$b_0 = \frac{V_{Ed}}{v_{perp,d,max}} \quad (7.3-57)$$

where $v_{perp,d,max}$ is the maximum shear force per unit length perpendicular to the basic control perimeter (Figure 7.3-24).

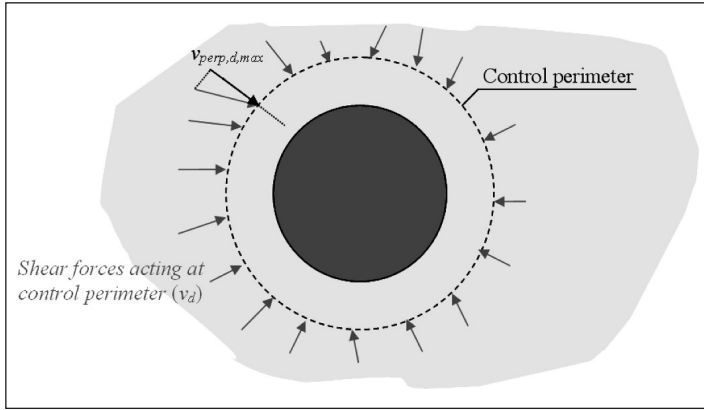


Figure 7.3-24: Shear force per unit length (v_d) and maximum value perpendicular to the basic control perimeter

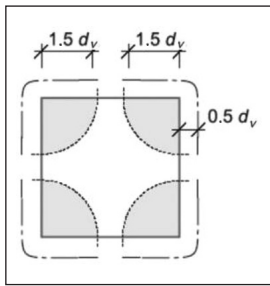


Figure 7.3-25: Reduction of basic control perimeter for large supported areas ($b_{1,red}$)

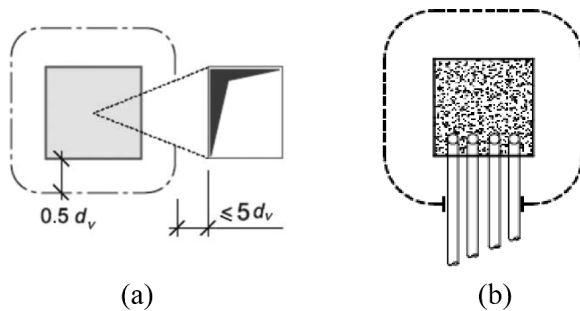


Figure 7.3-26: Reduction of basic control perimeter ($b_{1,red}$) in presence of: (a) openings; and (b) pipes or inserts

Cast-in pipes, pipe bundles or slab inserts, where the distance from the supported area is less than $5d_v$ must be arranged perpendicular to the control perimeter. In these cases, the control perimeter should be reduced in accordance to Figure 7.3-26.

The coefficient of eccentricity can be determined as a function of the moment transferred from the column to the slab as:

$$k_e = \frac{1}{1 + e_u/b_u} \quad (7.3-59)$$

where e_u is the eccentricity of the resultant of shear forces with respect to the centroid of the basic control perimeter – see Figure 7.3-27b – and b_u is the diameter of a circle with the same surface as the region inside the basic control perimeter. For design purposes,

(4) Shear-resisting control perimeter (b_0)

For calculating the punching shear resistance, a shear-resisting control perimeter (b_0) is used. The shear-resisting control perimeter accounts for the non-uniform distribution of shear forces along the basic control perimeter.

A non-uniform distribution of the shear forces may result due to:

1. *Concentrations of the shear forces at the corners of large supported areas.* This effect can approximately be taken into account by reducing the basic control perimeter ($b_{1,red}$) assuming that the length of its straight segments does not exceed $3d_v$ for each edge (Figure 7.3-25).
2. *Geometrical and statical discontinuities of the slab.* In the presence of openings and inserts, the basic control perimeter ($b_{1,red}$) is to be reduced according to the rules of Figure 7.3-26.
3. *Concentrations of the shear forces due to moment transfer between the slab and the supported area.* This effect can approximately be taken into account by multiplying the length of the reduced basic control perimeter ($b_{1,red}$) by the coefficient of eccentricity (k_e):

$$b_0 = k_e \cdot b_{1,red} \quad (7.3-58)$$

4. *Presence of significant loads near the supported area.* In cases where significant concentrated loads ($\geq 0.2V_{Ed}$) are applied near the supported area (closer than $3d_v$ from the edge of the supported area) the general procedure for calculating b_0 should be used, refer to Eq. (7.3-57).

In cases where the lateral stability does not depend on frame action of slabs and columns and where the adjacent spans do not differ in length by more than 25%, the following approximated values may be adopted for the coefficient k_e :

- 0.90 for inner columns;
- 0.70 for edge columns;
- 0.65 for corner columns;
- 0.75 for corners of walls (horizontal shear resisting members where the rules of Figure 7.3-22 apply).

the location of the centroid of the basic control perimeter can be calculated by approximating its shape with straight lines, see Figure 7.3-27b.

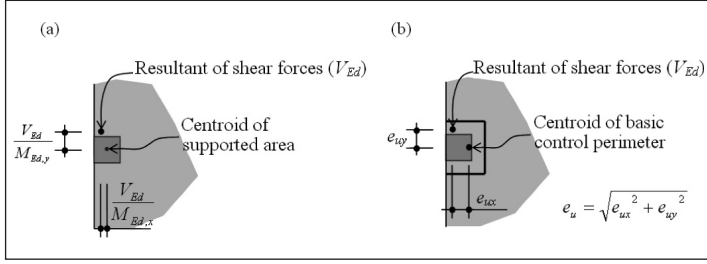


Figure 7.3-27: Resultant of shear forces: (a) location with respect to the centroid of the supported area; and (b) approximated basic control perimeter for calculation of the position of its centroid and eccentricity between the resultant of shear forces and the centroid of the basic control perimeter

The calculation of the punching shear strength is based on the critical shear crack theory.

The parameter ψ refers to the rotation of the slab around the supported area (Figure 7.3-28).

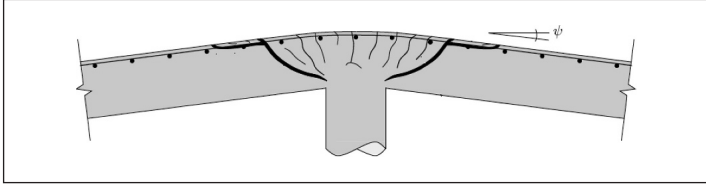


Figure 7.3-28: Rotation (ψ) of a slab

There is evidence that the punching shear resistance is influenced by the maximum size of the aggregate (d_g). If concrete with a maximum aggregate size smaller than $d_g = 16$ mm is used, the value of k_{dg} in Eq. (7.3-63) is assessed as:

$$k_{dg} = \frac{32}{16 + d_g} \geq 0.75 \quad (7.3-62)$$

where d_g is in mm. For aggregate sizes larger than 16 mm, Eq. (7.3-63) may also be used. For high strength and lightweight concrete, the aggregate particles may break, resulting in a reduced aggregate interlock contribution. In that case, the value d_g should be assumed to be 0.

For inclined shear reinforcement or bent-up bars (Figure 7.3-29), Eq. (7.3-66) is replaced by:

$$V_{Rd,s} = \sum A_{sw} k_e \sigma_{swd} \sin \alpha \quad (7.3-64)$$

and Eq. (7.3-67) is replaced by:

$$\sigma_{swd} = \frac{E_s \psi}{6} (\sin \alpha + \cos \alpha) \cdot \left(\sin \alpha + \frac{f_{bd}}{f_{ywd}} \frac{d}{\phi_w} \right) \leq f_{ywd} \quad (7.3-65)$$

7.3.5.3 Punching shear strength

The punching shear resistance is calculated as:

$$V_{Rd} = V_{Rd,c} + V_{Rd,s} \geq V_{Ed} \quad (7.3-60)$$

The design shear resistance attributed to the concrete may be taken as:

$$V_{Rd,c} = k_{\psi} \frac{\sqrt{f_{ck}}}{\gamma_c} b_0 d_v \quad (7.3-61)$$

with f_{ck} in [MPa].

The parameter k_{ψ} depends on the deformations (rotations) of the slab and follows from:

$$k_{\psi} = \frac{1}{1.5 + 0.9 k_{dg} \psi d} \leq 0.6 \quad (7.3-63)$$

where d is the mean value [in mm] of the (flexural) effective depth for the x and y directions.

Provided that the size of the maximum aggregate particles, d_g , is not less than 16 mm, k_{dg} in Eq. (7.3-63) can be taken as $k_{dg} = 1.0$.

The design shear resistance provided by the stirrups may be calculated as

$$V_{Rd,s} = \sum A_{sw} k_e \sigma_{swd} \quad (7.3-66)$$

where ΣA_{sw} is the sum of the cross-sectional area of all shear reinforcement suitably anchored, or developed, and intersected by the potential failure surface (conical surface with angle 45°) within the zone bounded by $0.35d_v$ and d_v from the edge of the supported area (Figure 7.3-29).

The term σ_{swd} refers to the stress that is activated in the shear reinforcement and can be calculated as:

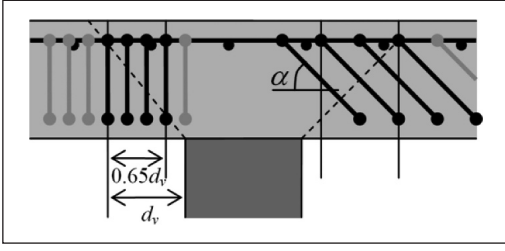


Figure 7.3-29: Shear reinforcement activated at failure

If more restrictive detailing rules are adopted ($s_0 \leq 0.5d_v$ and $s_1 \leq 0.6d_v$, with s_0 and s_1 according to Figure 7.13-10 and if the placing of the transverse reinforcement is checked at the construction site (distance between transverse reinforcements, top and bottom cover), the value k_{sys} can be increased as follows:

- $k_{sys} = 2.4$ for stirrups with sufficient development length at the compression face of the slab and bent (no anchorages or development length) at the tension face;
- $k_{sys} = 2.8$ for studs (diameter of heads larger or equal than three times the bar diameter).

Other values may be used for the coefficient k_{sys} provided that they are experimentally verified.

Slabs calculated under this assumption comply with deformation capacity requirements stated in subsection 7.3.5.1.

The value of r_s can be approximated as $0.22 L_x$ or $0.22 L_y$ for the x and y directions, respectively, for regular flat slabs where the ratio of the spans (L_x/L_y) is between 0.5 and 2.0. In level I approximation, the maximum value of r_s has to be considered in Eq. (7.3-70).

The average bending moment acting in the support strip (m_{Ed}) can be approximated for each reinforcement direction and support type as:

- for inner columns (top reinforcement in each direction):

$$m_{Ed} = V_{Ed} \cdot \left(\frac{1}{8} + \frac{|e_{u,i}|}{2 \cdot b_s} \right) \quad (7.3-71)$$

- for edge columns:

when calculations are made considering the tension reinforcement parallel to the edge:

$$m_{Ed} = V_{Ed} \cdot \left(\frac{1}{8} + \frac{|e_{u,i}|}{2 \cdot b_s} \right) \geq \frac{V_{Ed}}{4} \quad (7.3-72)$$

when calculations are made considering the tension reinforcement perpendicular to the edge:

$$m_{Ed} = V_{Ed} \cdot \left(\frac{1}{8} + \frac{|e_{u,i}|}{b_s} \right) \quad (7.3-73)$$

$$\sigma_{swd} = \frac{E_s \psi}{6} \left(1 + \frac{f_{bd}}{f_{ywd}} \cdot \frac{d}{\phi_w} \right) \leq f_{ywd} \quad (7.3-67)$$

where ϕ_w denotes the diameter of the shear reinforcement and f_{ywd} is its yield strength. The bond strength (f_{bd}) can be calculated according to subsection 6.1.3.2. Alternatively, a value $f_{bd} = 3$ MPa for corrugated bars may be used for design.

In order to ensure sufficient deformation capacity, in slabs with punching shear reinforcement a minimum amount of punching shear reinforcement is required such that:

$$\sum A_{sw} k_e f_{ywd} \geq 0.5 V_{Ed} \quad (7.3-68)$$

The maximum punching shear resistance is limited by crushing of the concrete struts in the supported area:

$$V_{Rd,max} = k_{sys} k_\psi \frac{\sqrt{f_{ck}}}{\gamma_c} b_0 d_v \leq \frac{\sqrt{f_{ck}}}{\gamma_c} b_0 d_v \quad (7.3-69)$$

The coefficient k_{sys} accounts for the performance of punching shear reinforcing systems to control shear cracking and to suitably confine compression struts at the soffit of the slab. In the absence of other data, and provided that reinforcement is detailed as per the provisions of subsection 7.13.5.3, a value $k_{sys} = 2.0$ can be adopted.

7.3.5.4 Calculation of rotations around the supported area

Level I approximation

For a regular flat slab designed according to an elastic analysis without significant redistribution of internal forces, a safe estimate of the rotation at failure is:

$$\psi = 1.5 \cdot \frac{r_s}{d} \cdot \frac{f_{yd}}{E_s} \quad (7.3-70)$$

where r_s denotes the position where the radial bending moment is zero with respect to the support axis.

Level II approximation

In cases where significant bending moment redistribution is considered in the design, the slab rotation can be calculated as:

$$\psi = 1.5 \cdot \frac{r_s}{d} \cdot \frac{f_{yd}}{E_s} \cdot \left(\frac{m_{Ed}}{m_{Rd}} \right)^{1.5} \quad (7.3-75)$$

where:

m_{Ed} is the average moment per unit length for calculation of the flexural reinforcement in the support strip (for the considered direction);

m_{Rd} is the design average flexural strength per unit length in the support strip (for the considered direction).

The rotation has to be calculated along the two main directions of the reinforcement.

The width of the support strip for calculating m_{Ed} is:

$$b_s = 1.5 \cdot \sqrt{r_{s,x} \cdot r_{s,y}} \leq L_{min} \quad (7.3-76)$$

where close to slab edges, the width of the strip is limited to b_{sr} according to Figure 7.3-30. The same value for r_s as that for Level I approximation can be adopted.

- for corner columns (tension reinforcement in each direction):

$$m_{Ed} = V_{Ed} \cdot \left(\frac{1}{8} + \frac{|e_{u,i}|}{b_s} \right) \geq \frac{V_{Ed}}{2} \quad (7.3-74)$$

In these equations, the term e_{ui} refers to the eccentricity of the resultant of shear forces with respect to the centroid of the basic control perimeter in the direction investigated ($i = x$ and y for x and y directions respectively, see Figure 7.3-27).

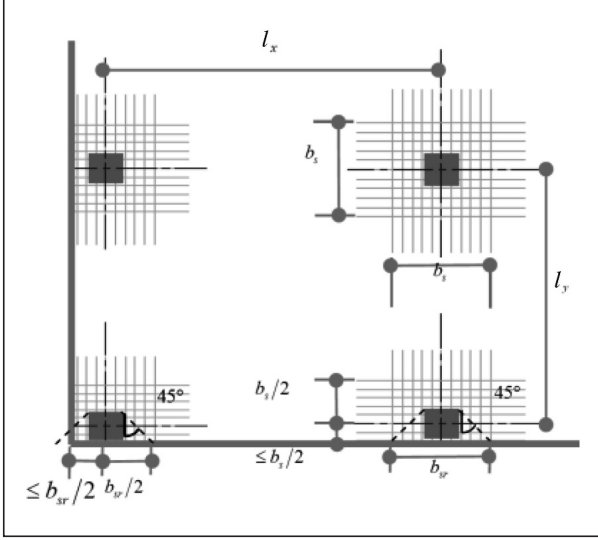


Figure 7.3-30: Support strip dimensions

Slabs calculated under this assumption do not comply with deformation capacity requirements stated in subsection 7.3.5.1. Therefore, they need to be provided with integrity reinforcement.

The design average flexural strength per unit length in the support strip is to be calculated accounting for both ordinary and prestressing steel at yielding.

Level III approximation is recommended for irregular slabs or for flat slabs where the ratio of the span lengths (l_x/l_y) is not between 0.5 and 2.0.

Parameter m_{Ed} has to be calculated consistently with the method used for determining the flexural reinforcement and is to be determined at the edge of the supported area maximizing m_{Ed} , see Figure 7.3-31.

Eq. (7.3-75) also applies to slabs with a flexural reinforcement that is increased over the supported areas in order to increase their punching shear strength.

For prestressed slabs, Eq. (7.3-75) can be replaced by:

$$\psi = 1.5 \cdot \frac{r_s}{d} \cdot \frac{f_{yd}}{E_s} \cdot \left(\frac{m_{Ed} - m_{Pd}}{m_{Rd} - m_{Pd}} \right)^{1.5} \quad (7.3-77)$$

where m_{Pd} denotes the average decompression moment over the width of the support strip (b_s) due to prestressing. Constrained forces and moments and losses due to shrinkage, creep and relaxation must be taken into account.

Level III approximation

The coefficient 1.5 in Eqs. (7.3-75) and (7.3-77) can be replaced by 1.2 if:

- r_s is calculated using a linear elastic (uncracked) model;
- m_{Ed} is calculated from a linear elastic (uncracked) model as the average value of the moment for design of the flexural reinforcement over the width of the support strip (b_s).

The width of the support strip can be calculated as in level II approximation taking $r_{s,x}$ and $r_{s,y}$ as the maximum value in the direction investigated. For edge or corner columns, the following minimum value of r_s has to be considered:

$$r_s \geq 0.67 b_{sr} \quad (7.3-78)$$

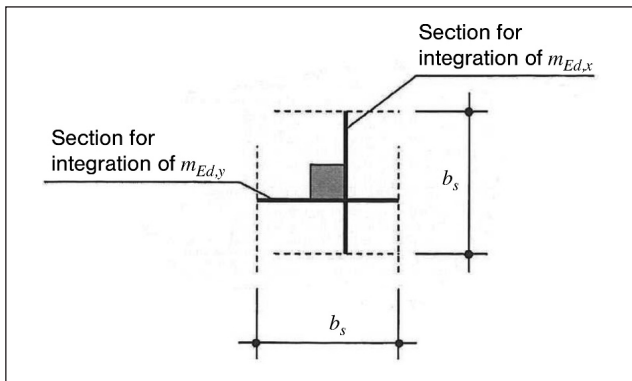


Figure 7.3-31: Example of sections for integration of support strip moments

Analytical or numerical techniques (e.g. finite elements, finite differences etc.) may be used for level IV approximation.

Level IV approximation

The rotation ψ can be calculated on the basis of a non-linear analysis of the structure and accounting for cracking, tension-stiffening effects, yielding of the reinforcement and any other non-linear effects relevant for providing an accurate assessment of the structure.

7.3.5.5 Punching shear resistance outside the zones with shear reinforcement or shearheads

The extent of the slab with shear-reinforcement can be determined by checking the resistance of the slab outside this region. Subsection 7.3.5.3 applies by accounting for a control perimeter with a maximum effective distance between two shear reinforcing elements of $3d_v$ (Figure 7.3-32).

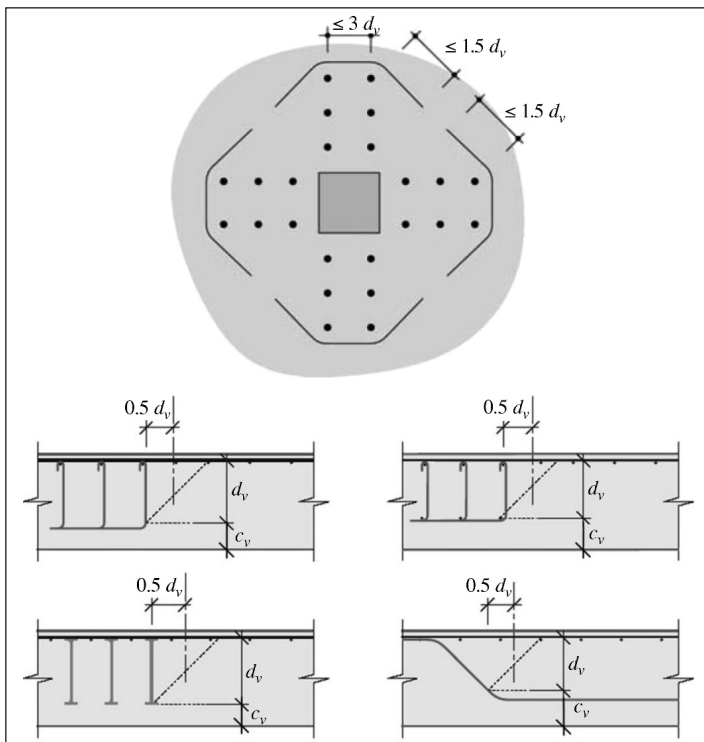


Figure 7.3-32: Reduced control perimeter and shear-resisting effective depth

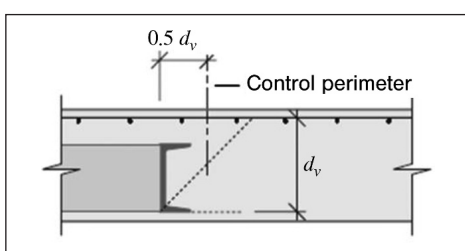


Figure 7.3-33: Shear-resisting effective depth and control perimeter accounting for shearhead penetration

The punching shear resistance of a slab outside of the shearhead is calculated on the basis of subsection 7.3.5.3 considering the shearhead as a rigidly supported area. The shear-resisting effective depth must account for the position of the shearhead in the slab as shown in Figure 7.3-33.

The design shear for calculation of the integrity reinforcement can be calculated on the basis of an accidental situation where progressive collapse has to be avoided.

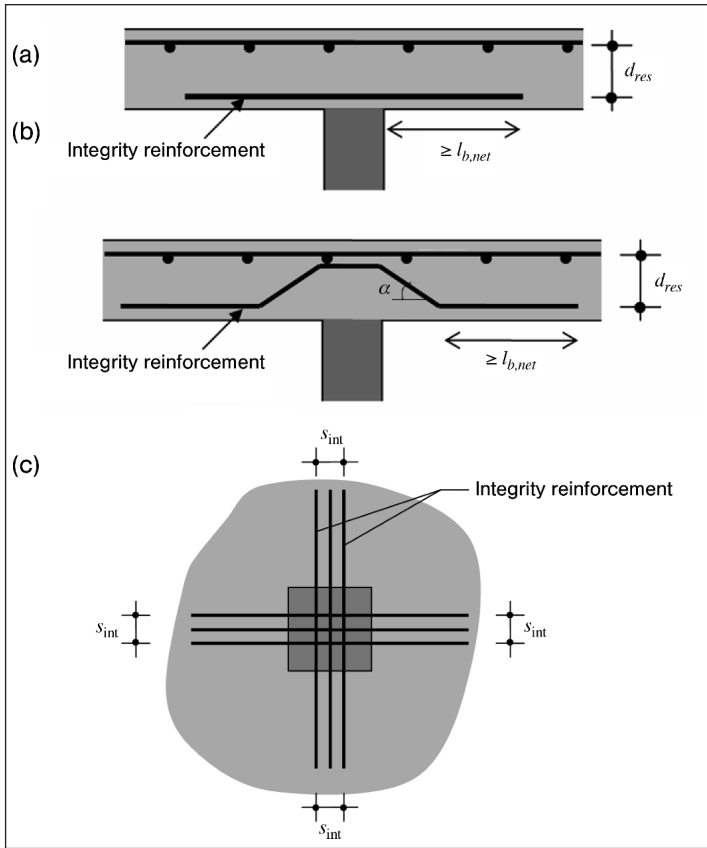


Figure 7.3-34: Integrity reinforcement: (a) straight bars; (b) bent-up

The integrity reinforcement should at least be composed of four bars placed over the supported area and correctly developed on the compression side of the slab. Post-tensioning tendons can also be considered as integrity reinforcement.

In order to allow full activation of the integrity bars, the diameter of the integrity bars (ϕ_{int}) has to be chosen such that: $\phi_{int} \leq 0.12 d_{res}$.

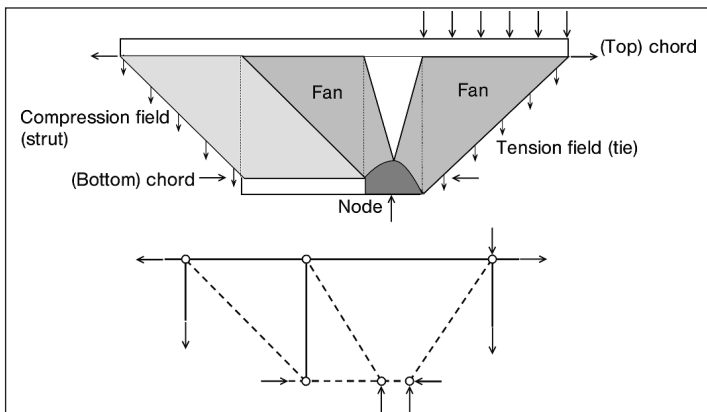


Figure 7.3-35: Basic elements for stress field analysis as well as for strut-and-tie modelling and statically equivalent truss models

7.3.5.6 Integrity reinforcement

Slabs without shear reinforcement, or with insufficient deformation capacity, must be provided with integrity reinforcement (Figure 7.3-34) to avoid progressive collapse.

The resistance provided after punching by the integrity reinforcement can be calculated as:

$$V_{Rd,int} = \sum A_s f_{yd} \left(\frac{f_t}{f_y} \right)_k \sin \alpha_{ult} \leq \frac{0.5 \sqrt{f_{ck}}}{\gamma_c} d_{res} b_{int} \quad (7.3-79)$$

where:

A_s refers to the sum of the cross-sections of all reinforcement suitably developed beyond the supported area on the compression side of the slab or to well-anchored bent-up bars;

f_{yd} is the design yield strength of the integrity bars; the ratio $(f_t/f_y)_k$ and parameter ϵ_{uk} are defined in subsection 5.2.5.4 and depend on the ductility class of the reinforcement;

α_{ult} is the angle of the integrity bar with respect to the slab plane at failure (after development of plastic deformations in the post-punching regime):

α_{ult}	Type of integrity reinforcement
0°	Straight bars, class of ductility: A
20°	Straight bars, class of ductility: B
25°	Straight bars, class of ductility: C or D
$\alpha \leq 40^\circ$	Inclined or bent-up bars, class of ductility: B, C or D

where α is the angle of the integrity bars with respect to the slab plane (before punching occurs), Figure 7.3-34;

d_{res} is the distance between the centroid of the flexural reinforcement ratio and the centroid of the integrity reinforcement, see Figure 7.3-34(a) and (b);

b_{int} is the control perimeter activated by the integrity reinforcement after punching. It can be calculated as:

$$b_{int} = \sum \left(s_{int} + \frac{\pi}{2} d_{res} \right) \quad (7.3-80)$$

where the summation refers to the groups of bars activated at the edge of the supported area and s_{int} is equal to the width of the group of bars (refer to Figure 7.3-34).

7.3.6 Design with stress fields and strut-and-tie models

7.3.6.1 General

Structures can be subdivided into B-regions, where the assumption of a plane section may be used (B for Bernoulli) and D-regions, where a non-linear strain distribution exists (D for discontinuity); D-regions typically are located at supports or at places of concentrated loads.

Stress fields and strut-and-tie models represent the force flow within a structural member or structure and consist of nodes, compressive fields, struts, fans, tensile fields, ties and chords (Figure 7.3-35). These elements may be characterized as follows:

- node: highly bi- or triaxially stressed zone within a stress field;
- compressive field, strut: parallel stresses of uniaxial compression with uniform stress intensity;
- fan: fan-shaped stress field of uniaxial compression with variable stress intensity;

- tensile field, tie: parallel stresses of uniaxial tension with uniform stress intensity; and
- chord (tension, compression): tensile or compressive elements with a stress intensity that varies along their axis.

The forces within a stress field or strut-and-tie model can be calculated with help of equilibrium conditions.

When developing a model, it is advisable to roughly take compatibility of deformations into consideration. In a first approximation, directions and magnitudes of the forces of the model may be orientated at the corresponding linear elastic stress state.

When applying stress fields or strut-and-tie models (Figure 7.3-36), the following steps may be considered:

- the geometry of the D-region may be assumed and have a minimum length equal to the maximum width of spread;
- a free-body with a (first) strut-and-tie model may be sketched. In order to minimize the effects of redistribution of forces (with consequences for crack width in the SLS) the struts should as much as possible be oriented to the compressive stress trajectories in the uncracked state;
- forces of the model have to be calculated such that they represent an equilibrium system of internal forces and external loads;
- the cross-section of the struts (compressive fields) and ties (tensile fields) shall be determined and checked for strength;
- geometry of nodes must be checked and detailing of reinforcement developed;
- the model has to be refined if necessary;
- nodes, struts and ties of the final model must comply with the detailing of the reinforcement.

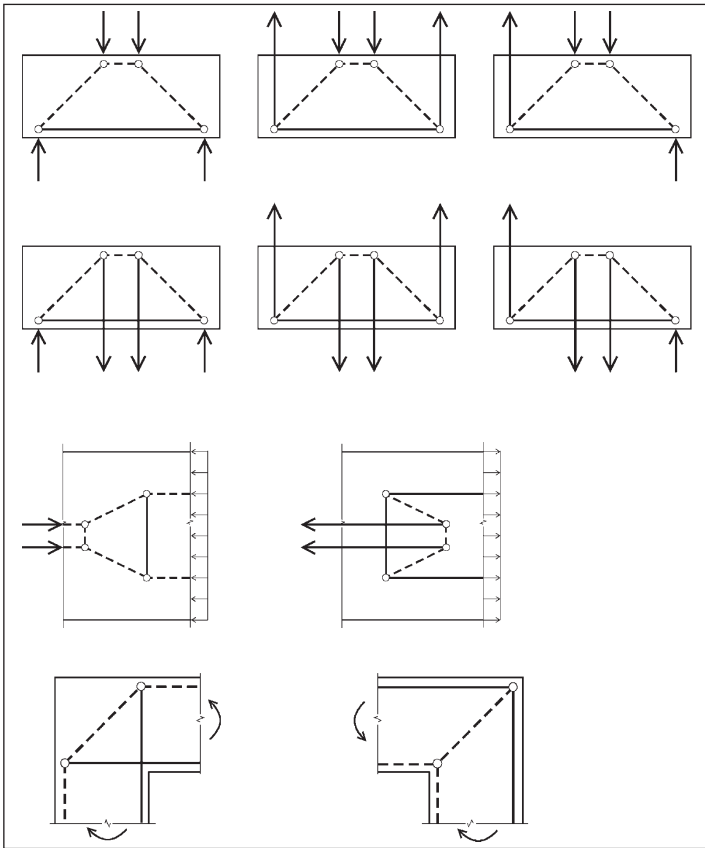


Figure 7.3-36: Exemplary basic elements for strut-and-tie modelling

7.3.6.2 Struts

For the dimensioning of the struts or stress fields the reduced concrete compressive strength, $k_c f_{ck}/\gamma_c$ must be used.

For undisturbed uniaxial compression states (Figure 7.3-37a) and for regions with transverse compression, the reduction factor is:

$$k_c = 1.0\eta_{fc} \quad (7.3-81)$$

For struts or stress fields with cracks parallel to the direction of compression and tension reinforcement perpendicular to this (see Figure 7.3-37b), the reduction factor is:

$$k_c = 0.75\eta_{fc} \quad (7.3-82)$$

For struts or stress fields with reinforcement running obliquely (with angles smaller than 65°) to the direction of compression, for example webs of beams (see Figure 7.3-37c), the reduction factor is:

$$k_c = 0.55\eta_{fc} \quad (7.3-83)$$

$$\eta_{fc} = \left(\frac{30}{f_{ck}} \right)^{1/3} \leq 1.0 \quad (f_{ck} \text{ in MPa}) \quad (7.3-84)$$

The design compressive stress values given above may be increased by 10%, where a biaxial state of compression is assured or all angles between struts and ties are $\geq 45^\circ$ and where the reinforcement is arranged in multiple layers.

7.3.6.3 Ties

The design strengths of tensile fields or ties are defined as:

$$f_{yd} = \frac{f_{yk}}{\gamma_s} \quad \text{for normal strength steel} \quad (7.3-85)$$

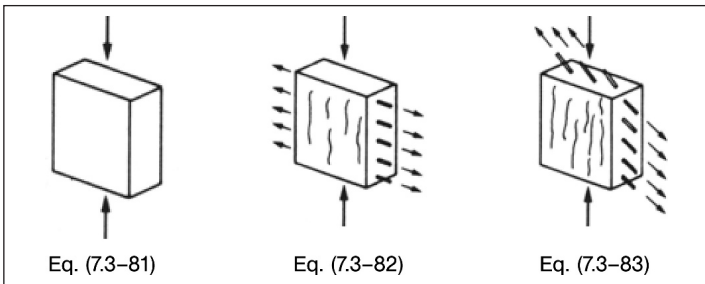


Figure 7.3-37: Various states of stress: (a) undisturbed uniaxial compression; (b) tension normal to the direction of compression; (c) tension oblique to the direction of compression

Refer to subsection 7.3.3.3 for the background of the parameter η_{fc} .

$$f_{pd} = \frac{f_{p0.1k}}{\gamma_s} \text{ for prestressing steel} \quad (7.3-86)$$

In the case of post-tensioned members, the initial prestressing state (defined by σ_{p0}) may be considered with the relevant anchorage and deviation forces. When applying this procedure the additional stress in the prestressing steel is limited to:

$$\Delta\sigma_p = f_{pd} - \sigma_{p0} \quad (7.3-87)$$

If appropriate, the reinforcement required to resist the forces may be distributed over the relevant length (Figure 7.3-35).

7.3.6.4 Nodes

The maximum stress to be applied at the edge of a node is limited to:

$$\sigma_{Rd,max} = k_c f_{ck} / \gamma_c \quad (7.3-88)$$

For compression nodes where no ties are anchored at the node, Figure 7.3-38, the reduction factor is the same as for Eq. (7.3-81).

A value of 1.1 η_{fc} may be assumed in regions where significant biaxial compression exists. This increase also applies if the stresses at supports or point loads are uniform and the node is confined by stirrups and if the node is reliably confined by means of bearing arrangement or friction. In cases with triaxial compression, subsections 7.2.3.1.6 and 7.2.3.1.7 apply.

For compression-tension nodes with anchored ties provided in one or two directions, Figure 7.3-39, the reduction factor is the same as for Eq. (7.3-82), where η_{fc} is defined in subsection 7.3.6.2.

The anchorage of the reinforcing bars in compression-tension nodes starts at the beginning of the node, for example in the case of a support anchorage starting at its inner face (Figure 7.3-39). The anchorage lengths should extend over the entire node length.

The anchorage of the reinforcing bars behind the node is strongly recommended in cases where the member dimensions are large enough. This type of anchorage is beneficial as it creates a state of pure compression in the node.

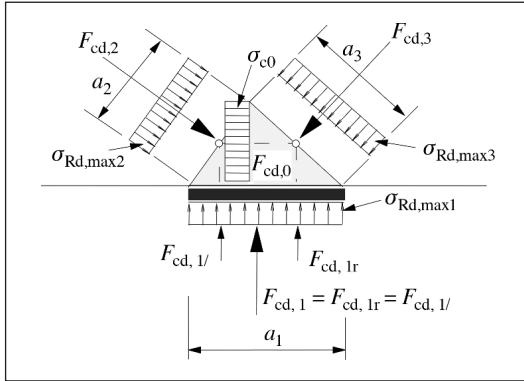


Figure 7.3-38: Compression nodes without ties

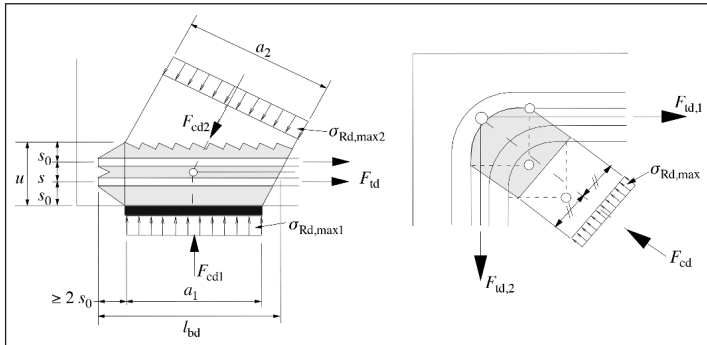


Figure 7.3-39: Compression-tension node with reinforcement provided in one and two directions

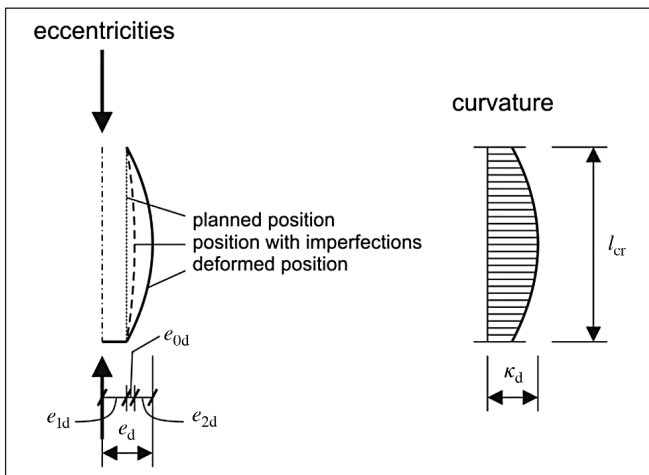


Figure 7.3-40: Compression resultant with eccentricities and curvature variation

7.3.7 Compression members

7.3.7.1 Stability of compressed members in general

Where the behaviour of a member is significantly influenced by second order effects (Figure 7.3-40), the verification is carried out for the deformed system and the dimensioning values of the actions. The geometrical imperfections according to subsection 7.2.2.2 have to be taken into account. Furthermore the influence of cracking, non-linear deformation of the structural materials and time-dependent deformations should be considered.

The dimensioning value of the bending moment is:

$$M_d = -N_d e_d \quad (7.3-89)$$

The maximum eccentricity e_d , that is the maximum distance between the compression resultant and the deformed axis of the compression member, may be determined as:

$$e_d = e_{0d} + e_{1d} + e_{2d} \quad (7.3-90)$$

where e_{0d} is eccentricity due to imperfections, being the greater value of:

$$e_{0d} = \alpha_i l_0 / 2 \text{ and } e_{0d} = d / 30 \quad (7.3-91)$$

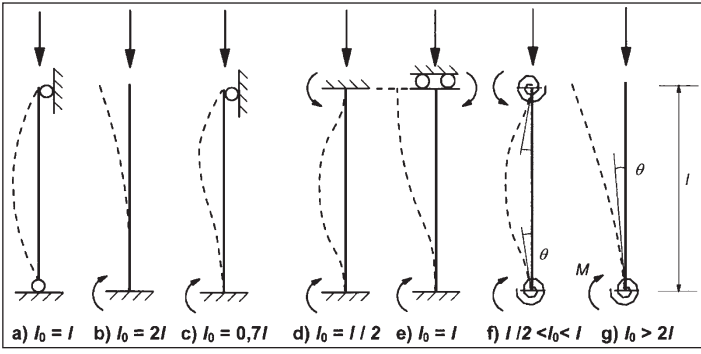


Figure 7.3-41: Examples of different buckling modes and corresponding effective lengths for isolated members

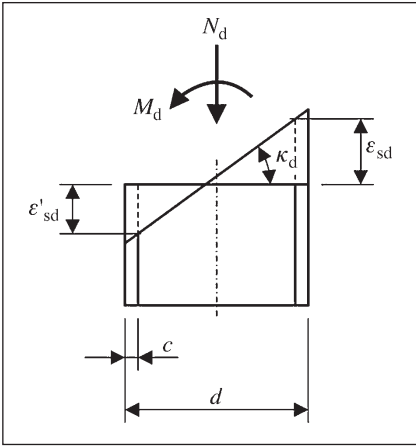


Figure 7.3-42: Strain plane corresponding to maximum curvature

Eq. (7.3-98) is based on interpolation using the interaction diagram, which is normally used as a design aid to determine the reinforcement in cross-sections subjected to a normal force and a bending moment. Figure 7.3-43 shows a simplified representation of such a diagram. At point B, the reinforcement yields at both sides of the column, so that the curvature is $\kappa = \epsilon_{yd} / (0.45d)$. At point A, the curvature $\kappa = 0$. So, the curvature in point C can be obtained by interpolation from Eq. (7.3-98).

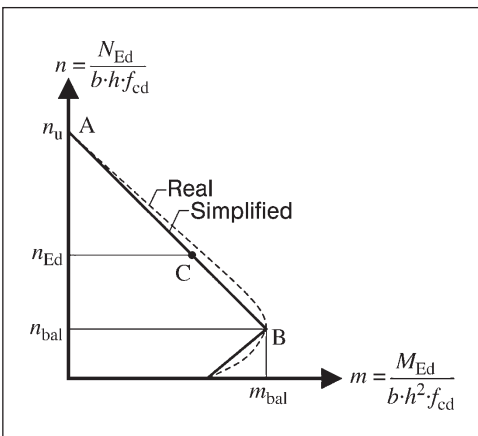


Figure 7.3-43: Simplified representation of interaction diagram

where l_0 is the effective length of the compressed member, see for example Figure 7.3-41. The value of α_i can be estimated as:

$$\frac{1}{200} \geq \alpha_i = \frac{0.01}{\sqrt{l_0}} \geq \frac{1}{300} \quad (l_0 \text{ in m}) \quad (7.3-92)$$

The first order eccentricity e_{1d} can be estimated according to:

$$e_{1d} = \frac{M_{1d}}{-N_d} \quad (7.3-93)$$

The eccentricity due to the deformation of the compression member e_{2d} may be calculated by:

$$e_{2d} = \kappa_d \frac{l_0^2}{c_0} \quad (7.3-94)$$

where κ_d is the maximum design curvature (Figure 7.3-42):

$$\kappa_d = \frac{\epsilon_{sd} - \epsilon'_{sd}}{d - 2c} \quad (7.3-95)$$

and c_0 is the integration factor accounting for the curvature distribution along the member.

Level I approximation

The value of the integration factor can be assumed as:

$$c_0 = \pi^2 \quad (7.3-96)$$

The maximum design curvature may be obtained from Eq. (7.3-95) with:

$$\epsilon_{sd} = \frac{f_{yd}}{E_s} \quad \text{and} \quad \epsilon'_{sd} = -\frac{f_{yd}}{E_s} \quad (7.3-97)$$

Level II approximation

A more accurate value of the maximum design curvature can be obtained from the equation:

$$\kappa_d = \left(\frac{n_u - n_d}{n_u - n_{bal}} \right) \cdot \frac{\epsilon_{yd}}{0.45d} \quad (7.3-98)$$

where:

$$n_u = 1 + \omega;$$

$$n = N_{Ed} / (A_c f_{cd});$$

$$n_{bal} = \text{value of } n \text{ at maximum moment resistance} \approx 0.4 \quad (\text{point B in Figure 7.3-43});$$

$$\omega = A_s f_{yd} / (A_c f_{cd}).$$

The long term deformations due to creep and shrinkage of the concrete may approximately be taken into account as pre-curvature of the cross-section. The maximum curvature according to Eq. (7.3-98) must be increased by adding the irreversible portion:

$$\kappa_{d,\infty} = \frac{|\epsilon_{c\infty}|}{d} \quad (7.3-99)$$

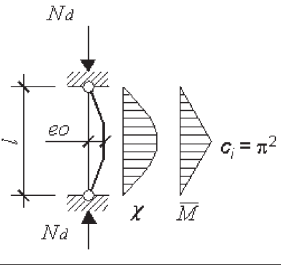
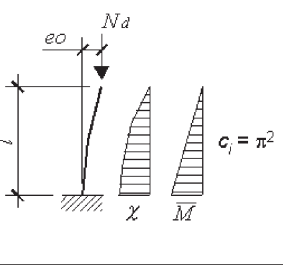
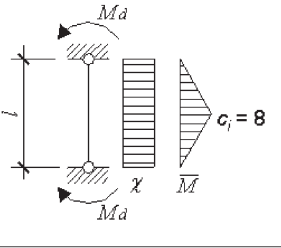
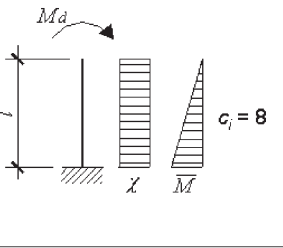
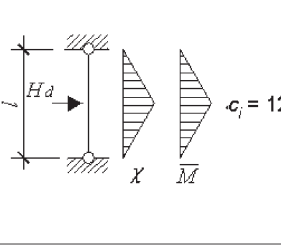
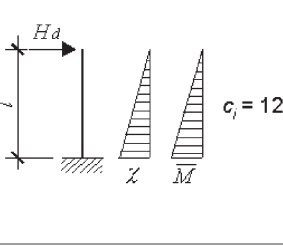
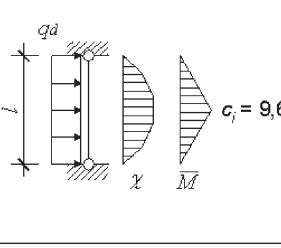
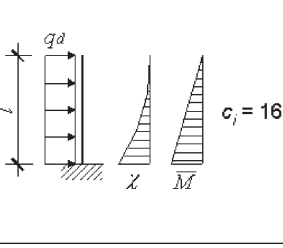
	Double-hinged column	Free cantilever column
Significant second order effects	 $c_i = \pi^2$	 $c_i = \pi^2$
Moment at column ends	 $c_i = 8$	 $c_i = 8$
Horizontal point loading	 $c_i = 12$	 $c_i = 12$
Horizontal distributed loading	 $c_i = 9.6$	 $c_i = 16$

Figure 7.3-44: Values of integration factors c_i as a function of the load type and the boundary conditions

Level III approximation

A more refined value of the integration factor c_0 can be calculated on the basis of the values of the various integration factors c_i for each action according to:

$$c_0 = \pi^2 \cdot \frac{N}{N_{cr}} + \frac{\sum_{i=1}^n M_i}{\sum_{i=1}^n \frac{M_i}{c_i}} \cdot \left(1 - \frac{N}{N_{cr}}\right) \tag{7.3-100}$$

where the factors c_i are given in Figure 7.3-44.

Level IV approximation

A refined calculation of second order effects can be determined using an analysis that accounts for non-linear behaviour of concrete in compression, cracking, creep and shrinkage, reinforcement yielding and other non-linear effects important to the change in behaviour over time and loading state.

7.3.7.2 Biaxial eccentricities and out-of-plane buckling

For members with rectangular cross-sections, separate verifications in the two principal planes y and z are permissible, if the point of application of N_{Ed} is located close to one principal axis, for example within the hatched zones in Figure 7.3-45. The ratios of the corresponding eccentricities e_{y1}/b and e_{z1}/h have to satisfy one of the following conditions:

$$(e_{z1}/h)/(e_{y1}/b) \leq 1/4; \text{ or} \tag{7.3-101}$$

$$(e_{y1}/h)/(e_{z1}/b) \leq 1/4; \tag{7.3-102}$$

where the eccentricities e_{y1} and e_{z1} are those in the directions of the section dimensions b and h , respectively, and include an imperfection allowance e_{0d} , as defined in Eq. (7.3-91).

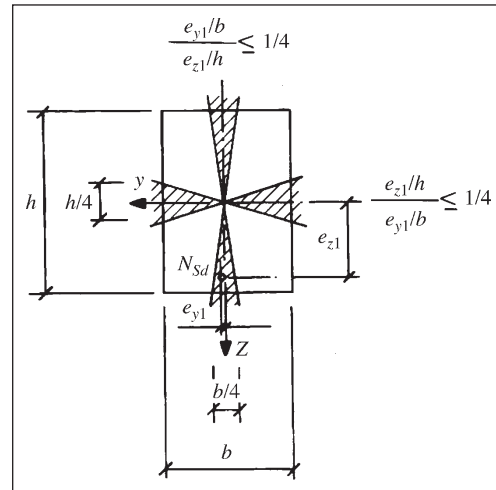


Figure 7.3-45: Condition for separate verification in the two principal planes

If the criteria expressed by Eqs. (7.3-101) and (7.3-102) are not satisfied, the cross-section should be designed for biaxial bending, including the second order effects in each direction. In the absence of a refined cross-sectional analysis for biaxial bending, the following simplified criterion may be used:

$$\left(\frac{M_{Edx}}{M_{Rdx}} \right)^a + \left(\frac{M_{Edy}}{M_{Rdy}} \right)^a \leq 1.0 \quad (7.3-103)$$

where:

$M_{Edx/y}$ is the design moment around the respective axis, including nominal second order moments;

$M_{Rdx/y}$ is the moment resistance of the cross-section in the respective direction;

a is an exponent which is 2 for circular and elliptical cross-sections and for rectangular cross-sections follows from:

N_{Ed}/N_{Rd}	0.1	0.7	1.0
a	1.0	1.5	2.0

N_{Ed} is the design value of axial force;

N_{Rd} is the design axial resistance of section = $A_c f_{cd} + A_s f_{yd}$.

7.3.8 Lateral instability of beams

Situations where a check is necessary are, for example, slender precast beams during transport and erection, and beams with insufficient lateral bracing in the finished structure. In such cases, geometric imperfections should be taken into account.

A check of the lateral instability of beams is relevant in long beams where lateral bracing is lacking.

A lateral deflection of $l/300$ should be assumed as a geometric imperfection in the verification of beams in unbraced conditions, where l is the total length of the beam. In finished structures, bracing from connected members may be taken into account.

Second order effects with regard to lateral instability may be ignored if the following condition is fulfilled:

$$\frac{l_{of}}{b} \leq \frac{50}{(h/b)^{1/3}} \quad (7.3-104)$$

where:

l_{of} is the unbraced length of the compression flange;

h is the total depth of the cross-section of the beam in the central part;

b is the width of the compression flange.

For background see Foster, S., Marti, P. and Mojsilović, N. “Design of Reinforced Concrete Solids Using Stress Analysis” (ACI Structural Journal, V100, N6, Nov–Dec. 2003, pp. 758–764).

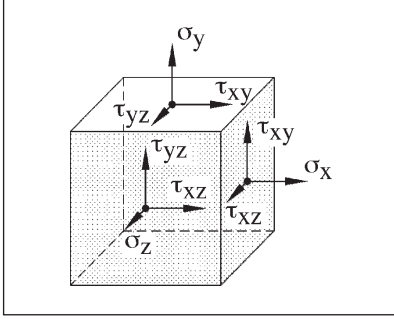


Figure 7.3-46: 3D stresses at a point defined in the orthogonal xyz axis system

In the application of this design method, the xyz axes are taken to correspond with reinforcing directions. The normal stresses applied at a point in a reinforced concrete solid element are carried by reinforcing steel and/or the concrete while shear stresses are carried by the concrete alone. Given that the applied stress tensor has been determined, for example, by 3D finite element solid modelling, the Mohr's circles of applied stress may be plotted, as shown in Figure 7.3-47. Within the circles the stress points (σ_i, S_i) are also plotted where $i = x, y, z$. As the reinforcing steel cannot carry shear stress it follows that the points denoting the concrete stresses (σ_{ci}, S_{ci}) must fall within the hatched region of the concrete stress circles where $\sigma_{ci} = \sigma_i - \rho_{s,j}\sigma_{s,j}$ and $S_{ci} = S_i$.

In the xyz space σ_x , σ_y and σ_z are, by definition, normal to the yz, xz and xy planes, respectively. The magnitudes of the shear stresses on these planes are given by

$$S_x = \sqrt{\tau_{xy}^2 + \tau_{xz}^2}; \quad S_y = \sqrt{\tau_{xy}^2 + \tau_{yz}^2}; \quad S_z = \sqrt{\tau_{xz}^2 + \tau_{yz}^2}$$

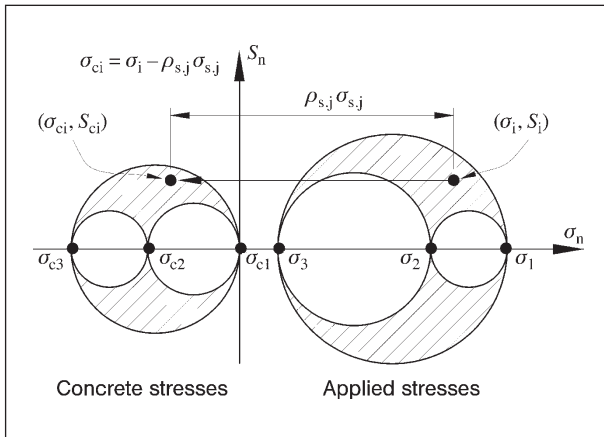


Figure 7.3-47: Compression field for 3D stress at a point

In developing solutions, the designer must “respect” the limitations of the concrete material. In a solid subject to a constant ratio of normal and shear stresses (with at least one tensile principal stress), before cracking the stress field in the concrete remains relatively elastic and the stresses in the reinforcement are negligible. After cracking, the tensile stresses in the concrete reduce, while those in the reinforcing steel increase. If the concrete does not fail in compression then the crack directions will remain relatively stable

7.3.9 3D solids

7.3.9.1 Stress limit requirements

Considering a stress tensor for a reinforced concrete 3D element (Figure 7.3-46), the applied stresses on an element can be replaced by equivalent stresses in the concrete (subscript *c*) and in the reinforcement (subscript *s*), according to:

$$\begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} = \begin{bmatrix} (\sigma_{cx} + \sigma_{sx}) & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & (\sigma_{cy} + \sigma_{sy}) & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & (\sigma_{cz} + \sigma_{sz}) \end{bmatrix} \quad (7.3-105)$$

where *x*, *y* and *z* are the axis directions of the orthogonal reinforcing steel.

The equivalent reinforcement stresses are limited by:

$$|\sigma_{s,j}| \leq \rho_{s,j} f_{yd,j} \quad (7.3-106)$$

where $\rho_{s,j}$ ($j = x, y, z$) are the reinforcement ratios in the *x*, *y* and *z* directions, respectively.

The concrete stresses (ordered as $\sigma_{c3} \leq \sigma_{c2} \leq \sigma_{c1}$ as shown in Figure 7.3-47) are required to satisfy:

$$-\sigma_{c3} \leq v f_{cd} \quad (7.3-107)$$

If no reinforcement has yielded and at least one principal stress is in tension then:

$$v = \frac{1.18}{1.14 + 0.00166 \sigma_{si}} \leq 1.0 \quad (7.3-108)$$

where σ_{si} is the maximum tensile stress (in MPa) in any layer of reinforcing steel ($i = x, y, z$).

If one or more layers of reinforcement yield:

$$v = (1 - 0.032 |\delta_i|) \cdot \frac{1.18}{1.14 + 0.00166 f_{yd}} \quad (7.3-109)$$

where δ_i is given by Eq. (7.3-116).

If all principal stresses are compressive, *v* may be taken as 1.0 or determined in accordance with subsection 5.1.6.

7.3.9.2 Ductility requirements

The ductility demand can be assessed by comparing the principal stress directions (Figure 7.3-48) of the resulting concrete stress tensor with that of the applied, factored loads.

The direction cosines of the principal stresses of the loading tensor $\mathbf{n}_i = \{n_{ix}, n_{iy}, n_{iz}\}$ ($i = 1, 2, 3$) are:

$$n_{ix} = \frac{-c_{iy} c_{iz}}{C} \quad (7.3-110)$$

until yielding of the steel in one direction. After yielding in one direction, the forces are continuously redistributed to balance the applied tractions until yielding in all directions has occurred. Concrete elements have a limit on the amount of redistribution that can be achieved. As a rule, concrete elements should not be pushed far beyond that which is “natural”. Designers should critically examine the load path being assumed, to ensure that a sufficient level of ductility is available to meet the demands of the imposed tractions. It is suggested to limit $|\delta_i| \leq 15^\circ$ (Figure 7.3-48).

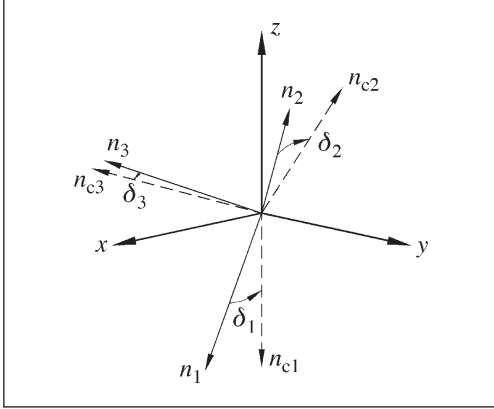


Figure 7.3-48: Comparison of concrete principal stress directions and the principal stress directions due to the applied tractions for the case of optimum reinforcement

$$n_{iy} = \frac{-c_{ix} c_{iz}}{C} \quad (7.3-111)$$

$$n_{iz} = \frac{-c_{ix} c_{iy}}{C} \quad (7.3-112)$$

where $C = \sqrt{c_{ix}^2 c_{iy}^2 + c_{ix}^2 c_{iz}^2 + c_{iy}^2 c_{iz}^2}$ and where:

$$c_{ix} = (\sigma_x - \sigma_i) \tau_{yz} - \tau_{xy} \tau_{xz} \quad (7.3-113)$$

$$c_{iy} = (\sigma_y - \sigma_i) \tau_{xz} - \tau_{xy} \tau_{yz} \quad (7.3-114)$$

$$c_{iz} = (\sigma_z - \sigma_i) \tau_{xy} - \tau_{xz} \tau_{yz} \quad (7.3-115)$$

The direction cosines of the principal concrete stresses $\mathbf{n}_{ci}$ are calculated from Eqs. (7.3-110) to (7.3-112) and Eqs. (7.3-113) to (7.3-115) with σ_{cx} , σ_{cy} and σ_{cz} substituted for σ_x , σ_y and σ_z , respectively.

The enclosed angles between the concrete stresses and those from the applied loading δ_i ($i = 1, 2, 3$) are given by:

$$\delta_i = \cos^{-1} |n_{ix} n_{cix} + n_{iy} n_{ciy} + n_{iz} n_{ciz}| \quad (7.3-116)$$

The rotational demands as indicated by the differences in the stresses resulting from the applied loading and those in the concrete should be limited to ensure sufficient ductility capacity to meet the demands.

The relations given in section 7.4 are valid for concrete stored in a constant environment of approximately 20°C, 65% relative humidity (see subsection 5.1.11). The fatigue strength of steel is given both for a normal environment and for a marine environment.

7.4 Verification of structural safety (ULS) for non-static loading

7.4.1 Fatigue design

7.4.1.1 Scope

The following design rules apply for the entire service life of concrete structures. The rules for reinforcing and prestressing steel should be applied if more than 10^4 load repetitions are expected; low-cycle fatigue is not covered.

The verification of the design principle (see subsection 4.5.2.3) can be performed according to four methods with an increasing refinement. The methods according to level II, level III and level IV approximation are given in subsections 7.4.1.3, 7.4.1.4 and 7.4.1.5. The models for the analysis of stresses in reinforced and prestressed concrete members under fatigue loading are treated in subsection 7.4.1.2 as well as concrete stress gradients.

Subsection 7.4.1.6 deals with shear design and in 7.4.1.7 a method for calculating the increased deflections under fatigue loading is given. The relevant combination of loads is treated in subsection 4.5.2.3.

7.4.1.2 Analysis of stresses in reinforced and prestressed members under fatigue loading

Linear elastic models may generally be used, and reinforced concrete in tension is considered to be cracked. The ratio of moduli of elasticity for steel and concrete may be taken as $\alpha = 10$.

In the case of prestressed members it should be verified if the relevant section is sensitive to cracking. This holds true if any combination of loads (see subsection 4.5.2.5) causes tensile stresses at the concrete surface. In that case the stress ranges for reinforcing steel and prestressing steel should be calculated assuming the cracked state.

The effect of differences in bond behaviour of prestressing and reinforcing steel has to be taken into account. Unless a more refined method is used, this can be done using a linear elastic model for stress calculation which fulfills the compatibility in strains and multiplying the stress in the reinforcing steel by the following factor:

$$\eta_s = \frac{1 + (A_p / A_s)}{1 + (A_p / A_s) \sqrt{\xi (\phi_s / \phi_p)}} \quad (7.4-1)$$

where:

η_s is the factor which increases the stress in the reinforcing steel due to differences in bond behaviour between prestressing and reinforcing steel;

A_s is the area of reinforcing steel;

A_p is the area of prestressing steel;

ϕ_s is the smallest diameter of the reinforcing steel in the relevant cross-section;

ϕ_p is the diameter of the prestressing steel (for bundles an equivalent diameter $1.6\sqrt{A_p}$ is chosen, where A_p is the cross-section area of the bundle);

ξ is the ratio of bond strength of prestressing steel and high-bond reinforcing steel.

For post-tensioned members the following values may be used:

$\xi = 0.2$ for smooth prestressing steel;

$\xi = 0.4$ for strands;

$\xi = 0.6$ for ribbed prestressing wires;

$\xi = 1.0$ for ribbed prestressing bars.

For pretensioned members the following values may be used:

$\xi = 0.6$ for strands;

$\xi = 0.8$ for ribbed prestressing steels.

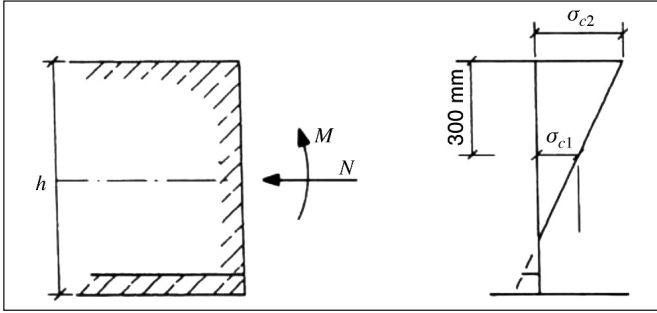


Figure 7.4-1: Definition of stress σ_{c1} , σ_{c2}

For level I approximation see subsection 4.5.2.3. This is just a qualitative verification that no variable action is able to produce fatigue.

Values for $\gamma_{s,fat}$ and $\gamma_{c,fat}$ are given in subsection 4.5.2.3.

The fatigue reference strength is defined as follows (see also subsection 5.1.11).

Compression

$$f_{cd,fat} = 0.85 \cdot \beta_{cc}(t) \cdot f_{ck} \cdot \left(1 - \frac{f_{ck}}{400}\right) / \gamma_{c,fat}$$

where:

- $\beta_{cc}(t)$ is the coefficient which depends on the age t of the concrete in days when fatigue loading starts (see subsection 5.1.9.1).

Tension

$$f_{ctd,fat} = f_{ctk,0.05} / \gamma_{c,fat}$$

For the value of $\gamma_{c,fat}$, see subsection 4.5.2.3.

For $\sigma_{c,max}$, $\sigma_{ct,max}$, see subsection 4.5.2.3.

When the unique value Q can be chosen satisfactorily (e.g. as fatigue equivalent), this method is a more precise assessment than the simplified procedure.

The stress gradient for concrete in the compression zone of a cracked section may be taken into account by multiplying the maximum stress in the compression zone by a factor η_c , equal to:

$$\eta_c = \frac{1}{1.5 - 0.5|\sigma_{c1}|/|\sigma_{c2}|} \quad (7.4-2)$$

where:

- η_c is the averaging factor of concrete stresses in the compression zone considering the stress gradient;
- $|\sigma_{c1}|$ is the minimum absolute value of the compressive stress within a distance of 300 mm from the surface under the relevant load combination of actions (Figure 7.4-1);
- $|\sigma_{c2}|$ is the maximum absolute value of the compressive stress within a distance of 300 mm from the surface under the same load combination as that for which $|\sigma_{c1}|$ was determined (Figure 7.4-1).

7.4.1.3 Level II approximation: the simplified procedure

This procedure is only applicable to structures subjected to a limited number ($\leq 10^8$) of low stress cycles.

Steel

The fatigue requirements will be met, if the maximum calculated stress range under the frequent combination of loads, $\max \Delta\sigma_{Es}$, satisfies the condition:

$$\gamma_{Ed} \max \Delta\sigma_{Es} \leq \Delta\sigma_{Rsk} / \gamma_{s,fat} \quad (7.4-3)$$

where:

$\Delta\sigma_{Rsk}$ is the characteristic fatigue strength at 10^8 cycles.

Values for $\Delta\sigma_{Rsk}$ are given in Tables 7.4-1 and 7.4-2.

Concrete

Detailed fatigue design needs not be carried out if the maximum calculated stresses under the frequent combination of loads, $\sigma_{c,max}$ (compression), $\sigma_{ct,max}$ (tension), respectively, satisfy the following criteria:

Compression

$$\gamma_{Ed} \sigma_{c,max} \eta_c \leq 0.45 f_{cd,fat} \quad (7.4-4)$$

where:

- $\sigma_{c,max}$ is the maximum compressive stress;
- η_c is an averaging factor considering the stress gradient Eq. (7.4-2);
- $f_{cd,fat}$ is the design fatigue reference strength for concrete in compression.

Tension

$$\gamma_{Ed} \sigma_{ct,max} \leq 0.33 f_{ctd,fat} \quad (7.4-5)$$

where:

- $\sigma_{ct,max}$ is the maximum tensile stress in the concrete;
- $f_{ctd,fat}$ is the design fatigue reference tensile strength of the concrete.

7.4.1.4 Level III approximation: verification by means of a single load level

This method takes into account the required service life with a foreseen number of cycles, n . This number intervenes in the verification with the maximum fatigue effects of the action, Q , as

When it is considered necessary to carry out fatigue tests to determine the performance of reinforcing steel, the tests should be made according to standardized procedures, as described in ISO 15630-1, -2 and -3.

The characteristic fatigue strength function for steel consists of segments (see Figure 7.4-2) of the form $(\Delta\sigma_{Rsk})^m \cdot N = \text{const.}$ Values for the S–N curves are given in Table 7.4-1 and Table 7.4-2.

The values given in Tables 7.4-1 and 7.4-2 are characteristic and do not incorporate partial safety factors. These values or higher values must be validated by appropriate approval documents.

The code does not cover coiled and re-straightened bars.

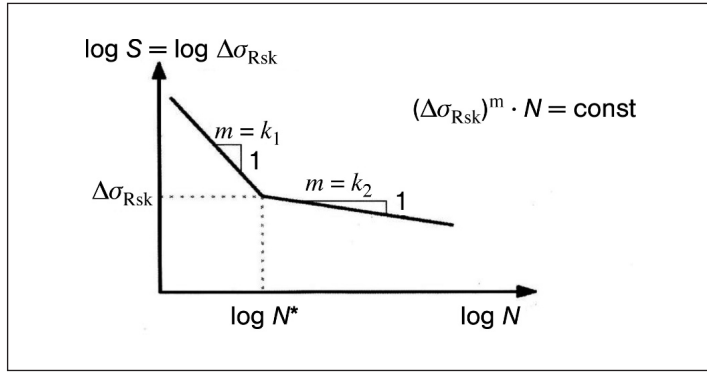


Figure 7.4-2: Shape of the characteristic fatigue strength curves (S–N curves) for steel

Where appropriate information is provided by specific approval documents for the steel to be used, higher fatigue strength values may be used accordingly.

Data on the fatigue behaviour of bars with diameters larger than 40 mm are scarce. Therefore no data for these bars are given here.

The values given in Table 7.4-2 are on the safe side compared to the strength values for the basic material given in subsection 5.1.11.

The reduction of the $\Delta\sigma_{Rsk}$ values for curved tendons compared with the values of straight tendons is due to fretting corrosion, which results from the lateral pressure and slip between prestressing strands and/or ribs of the steel sheaths.

Characteristic S–N curves for concrete can be used without any restriction for frequencies higher than 0.1 Hz. For lower frequencies, the fatigue life should be reduced – see chapter 3 in CEB Bulletin 188 “Fatigue of Concrete Structures” (CEB, 1988) for guidance.

defined in subsection 4.5.2.3 (part on level III approximation), subsection 7.4.1.2 and the paragraphs below.

Steel

The fatigue requirement will be met if the calculated maximum acting stress range, $\max \Delta\sigma_{Es}$, satisfies the condition:

$$\gamma_{Ed} \max \Delta\sigma_{Es} \leq \Delta\sigma_{Rsk}(n) / \gamma_{s, fat} \quad (7.4-6)$$

where:

$\Delta\sigma_{Es}$ is the steel stress range under the acting loads;

$\Delta\sigma_{Rsk}(n)$ is the stress range relevant to n cycles obtained from a characteristic fatigue strength function.

Table 7.4-1: Parameters of S–N curves for reinforcing steel (embedded in concrete)

	N^*	Stress exponent		$\Delta\sigma_{Rsk}$ (MPa) ^(e)	
		k_1	k_2	at N^* cycles	at 10^8 cycles
Straight and bent bars $D \geq 25\phi$	10^6	5	9	210	125
	$\phi \leq 16$ mm	5	9	160	95
	$\phi > 16$ mm ^(a)	5	9	— ^(c)	— ^(c)
Bent bars $D < 25\phi$ ^(b)	10^7	3	5	50	30
Welded bars ^(b) including tack	10^7	3	5	65	40
Welding and butt joints					
Mechanical connectors					
Marine environment ^{(b),(d)}					

(a) The values given in this line represent the S–N curve of a 40 mm bar; for diameters between 16 and 40 mm, interpolation between the values of this line and those of the line above is permitted.

(b) Most of these S–N curves intersect the curve of the corresponding straight bar. In such cases the fatigue strength of the straight bar is valid for cycle numbers lower than that of the intersection point.

(c) Values are those of the according straight bar multiplied by a reduction factor ξ depending on the ratio of the diameter of mandrel D and bar diameter ϕ : $\xi = 0.35 + 0.026D/\phi$.

(d) Valid for all ratios D/ϕ and all diameters ϕ .

(e) In cases where $\Delta\sigma_{Rsk}$ values calculated from the S–N curve exceed the stress range $f_{yd} - \sigma_{min}$, the value $f_{yd} - \sigma_{min}$ is valid.

Table 7.4-2: Parameters of S–N curves for prestressing steel (embedded in concrete)

	N^*	Stress exponent		$\Delta\sigma_{Rsk}$ (MPa) ¹
		k_1	k_2	at N^* cycles
<i>Pretensioned steel</i>				
Straight strands and wires	10^6	5	9	185
<i>Post-tensioned steel</i>				
Single strands in plastic ducts	10^6	5	9	185
Straight tendons or curved tendons in plastic ducts	10^6	5	10	150
Curved tendons in steel ducts	10^6	5	7	120
Splicing devices	10^6	5	5	80

¹ In cases where the S–N curve intersects that of the straight tendon, the fatigue strength of the straight tendon is valid.

Concrete

The fatigue requirements under cyclic loading will be met if the required lifetime (number of cycles) is lower than or equal to the number of cycles to failure:

$$n \leq N$$

where:

- n is the foreseen number of cycles during the required design service life;
 N is the number of resisting stress cycles, to be calculated from the fatigue strength functions given below.

Compression

For $S_{cd,min} > 0.8$, the S–N relations for $S_{cd,min} = 0.8$ are valid.

For $0 \leq S_{cd,min} \leq 0.8$, the following equations apply:

$$\log N_1 = \frac{8}{Y-1} \cdot (S_{cd,max} - 1) \quad (7.4-7a)$$

$$\log N_2 = 8 + \frac{8 \cdot \ln(10)}{Y-1} \cdot (Y - S_{cd,min}) \cdot \log \left(\frac{S_{cd,max} - S_{cd,min}}{Y - S_{cd,min}} \right) \quad (7.4-7b)$$

with

$$Y = \frac{0.45 + 1.8 \cdot S_{cd,min}}{1 + 1.8 \cdot S_{cd,min} - 0.3 \cdot S_{cd,min}^2}$$

where:

- (a) if $\log N_1 \leq 8$, then $\log N = \log N_1$;
 (b) if $\log N_1 > 8$, then $\log N = \log N_2$;

where:

$\sigma_{c,min}$ is the minimum compressive stress;
 $S_{cd,min} = \gamma_{Ed} \sigma_{c,min} \eta_c / f_{cd,fat}$ is the minimum compressive stress level

$S_{cd,max} = \gamma_{Ed} \sigma_{c,max} \eta_c / f_{cd,fat}$ is the maximum compressive stress level

where η_c follows from Eq. (7.4-2).

Compression – tension with $\sigma_{ct,max} \leq 0.026 |\sigma_{ct,max}|$

$$\log N = 9(1 - S_{cd,max}) \quad (7.4-8)$$

Pure tension and tension-compression with $\sigma_{ct,max} > 0.026 |\sigma_{ct,max}|$

$$\log N = 12(1 - S_{td,max}) \quad (7.4-9)$$

where:

$\sigma_{ct,max}$ is the maximum tensile stress;

$S_{td,max} = \gamma_{Ed} \sigma_{ct,max} / f_{ctd,fat}$ is the maximum tensile stress level.

7.4.1.5 Level IV approximation: verification by means of a spectrum of load levels

The partial coefficients are applied in this procedure as follows:

For steel, values N_{Ri} are calculated from the S–N curves given in Tables 7.4-1 and 7.4-2 using an increased stress range $\gamma_{Ed} \gamma_{s,fat} \Delta \sigma_{Esi}$.

For concrete, values N_{Ri} are calculated directly from the fatigue strength functions given in subsection 7.4.1.4.

This method takes account of the required service life, the load spectrum (which is divided into j blocks) and the characteristic fatigue strength functions.

Fatigue damage D is calculated using the Palmgren–Miner summation

$$D = \sum_{i=1}^j \frac{n_{Ei}}{N_{Ri}} \quad (7.4-10)$$

where:

D is fatigue damage;

n_{Si} denotes the number of acting stress cycles associated with the stress range for steel and the actual stress levels for concrete;

N_{Ri} denotes the number of resisting stress cycles at a given stress level.

The fatigue requirement will be satisfied if $D \leq D_{lim}$.

Appropriate values for D_{lim} should be adopted for concrete. Under increasing stress levels, D_{lim} can be safely used.

For concrete D_{lim} depends on the stress history (Zang et al., Mag. Of Concrete Research, 49 (180), 1997, pp. 241–252). Under decreasing stress levels D_{lim} can be significantly smaller than 1.0.

If it can be demonstrated that in the structural member no shear cracks will occur under the relevant combination of loads, fatigue in shear need not be verified.

Eq. (7.4-11) is also valid for beams and slabs in shear and punching.

According to Eq. (7.4-12), the strut inclination θ_{fat} is steeper than the value θ used for design under static loading. This takes account of the higher value of θ in the SLS.

The fatigue reference strength is to be reduced in the same way as the compressive strength of the concrete subjected to simultaneously acting compressive and transverse tensile forces.

7.4.1.6 Shear design

Members without shear reinforcement

The fatigue requirements will be met, if under cyclic loading the number of cycles corresponding to the required service life is smaller than or equal to the numbers of cycles to failure:

$$n \leq N.$$

N should be calculated from the fatigue strength functions given below.

$$\log N = 10(1 - V_{\max} / V_{\text{ref}}) \quad (7.4-11)$$

where:

$V_{\max}$ is the maximum shear force under the relevant representative values of permanent loads including prestress and maximum cyclic loading;

$V_{\text{ref}} = V_{Rd,c}$ (see subsection 7.3.3).

Members with shear reinforcement

The stress in the shear reinforcement should be calculated according to chapter 6, assuming the following inclination of the compression struts under fatigue loading:

$$\tan \theta_{fat} = \sqrt{\tan \theta} \quad (7.4-12)$$

where:

θ_{fat} is the angle between the web compression and the chords valid for verification of the reinforcement.

For assessment of the θ value, see subsection 7.3.3.

The resistance of compressive struts can be verified using Eq. (7.4-4) or Eqs. (7.4-7a) and (7.4-7b) reducing the fatigue reference strength given in subsection 7.4.1.3 by a factor of k_c according to Eq. (7.3-27). The compression of web concrete subjected to fatigue loading should be calculated using the angle θ (see subsection 7.3.3).

7.4.1.7 Increased deflections under fatigue loading in the SLS

Under cyclic loading, progressive deflection can occur in reinforced concrete members in addition to the deflection produced by creep. The cyclic effect can be calculated from

$$a_n = a_1[1.5 - 0.5 \exp(-0.03n^{0.25})] \quad (7.4-13)$$

where:

a_n final deflection after n cycles;

a_1 deflection in the first cycle due to the maximum load including effects of shear strains;

n is the number of cycles.

7.4.2 Impact and explosion

7.4.2.1 General remarks

Loads caused by impact and explosions are characterized as high dynamic loads with strain rates which are significantly higher than those for example of traffic loads or earthquake loads (Figure 7.4-3).

With increasing strain rates caused by loadings with high velocities, special effects occur which must be taken into account in the design process:

- mass effects;
- resonance effects;
- spalling and scabbing effects;
- punching.

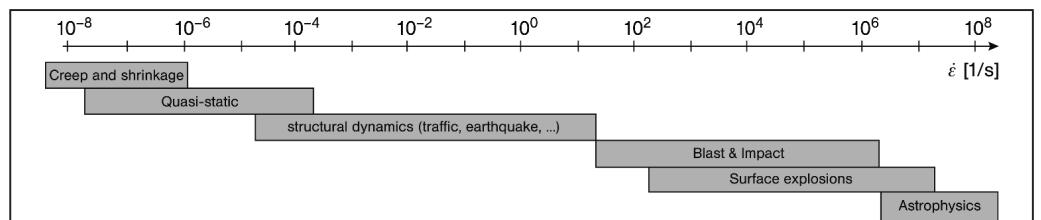


Figure 7.4-3: Strain rates for different loading velocities

Structures that are designed to resist high dynamic loads are for example:

- nuclear power plants;
- shelters for planes and ammunition;
- office buildings with a high degree of threat, such as embassies and military buildings;
- chemical factories;
- piers of bridges which can be hit by trains or vehicles.

For strain rates lower than 10^{-1} , caused by traffic or earthquake, quasi-static loads can be used to model the additional load effects (level I).

For higher strain rates, a dynamic calculation is required. For known phenomena such as free-air burst explosions or plane impact, idealized load–time curves or pressure–time curves are available in codes like UFC 3-340-02 or DIN 25449, which can be adapted to the respective situation.

High dynamic loads can be caused by:

- impact of vehicles, trains or airplanes;
- impact of projectiles and missiles;
- impact of debris;
- surface burst explosion;
- blast caused by nearby explosions;
- blast caused by distant explosions;
- explosions in rooms, including multiple reflections.

7.4.2.2 Determination of design loads

Concerning safety factors, high dynamic loads usually belong to extraordinary loads. For values of the safety factors, see subsection 4.5.2.4. The characteristic loads strongly depend on the type and intensity of the explosion or impact so that a general definition of the load is not possible. The following different approaches are possible:

- quasi-static equivalent loads (level I);
- standardized pressure–time curves (level II);
- individual load calculation (level III).

For some configurations, loads can be obtained from literature or special codes (such as UFC 3-340-02). In other cases numerical calculations with the use of hydrocodes, FE-codes or tests are necessary to get load data (level III).

Figures 7.4-4 and 7.4-5 give examples for pressure–time curves for loads caused by free-air burst explosions.

For the determination of internal forces and stresses in the structure, three levels of analysis are distinguished:

Level I

For quasi-static equivalent loads a linear or non-linear static calculation can be done using the static model which is used for dead and live loads.

Level II

For time–pressure curves a linear or non-linear dynamic calculation using a time-step method is required.

Level III

For a detailed modelling of impact or explosion by hydrocode or FE-calculation, the internal stresses can be obtained directly from this calculation. As an alternative, pressure–time curves can be derived from the results of these calculations which are the basis for a calculation on level II.

The effect of a free air burst explosion can be modelled using a standard curve as in Figure 7.4-4. The load value and the time of duration depend on several factors such as the distance between the explosive and the concrete structure and the type and quantity of the explosive material.

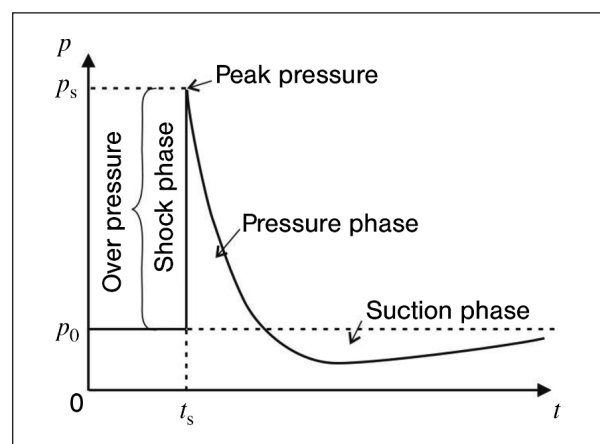


Figure 7.4-4: Principle pressure-time-curve caused by an air burst explosion

Figure 7.4-5 gives a more simplified pressure-time-function without taking account the suction phase which can be used for structures with a high weight and a high stiffness which are loaded by low pressures.

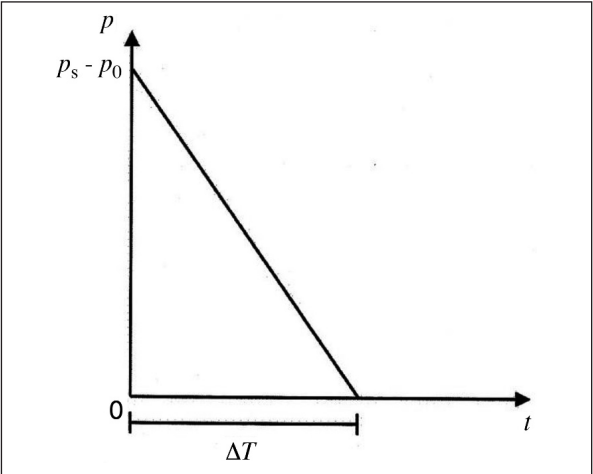


Figure 7.4-5: Free air burst explosion

The calculation methods for level I and II analysis are similar to those for seismic design, see subsection 7.4.3

7.4.2.3 Dimensioning for overall stresses

For the dimensioning of RC structures for overall stresses such as bending moments, normal and shear forces, principally the same procedures can be used as for seismic loads. Three different approaches are available:

- static dimensioning with regard to internal forces (subsection 7.4.2.3.1);
- dynamic dimensioning with regard to internal forces (7.4.2.3.2);
- dynamic dimensioning with regard to ductility (7.4.2.3.3).

7.4.2.3.1 Static dimensioning with regard to internal forces

For static dimensioning with regard to internal forces the following options are available:

- linear or non-linear calculation of the static internal forces calculated at level I;
- dimensioning of the structure with the usual methods for static loads;
- dimensioning with normal values for the strength of the concrete and reinforcement without dynamic increase factors.

Loading at high strain rates usually results in very high stresses. However, even at the side of the structural resistance observable special effects should be taken into account. The material strength of concrete and of steel under compression and under tension increases with increasing strain rates – see Figures 7.4-6 and 7.4-7. A simplified approach for the increase of material strength is given in Tables 7.4-3 and 7.4-4.

Table 7.4-3 includes different values for near and far detonations. One possibility to distinguish between the nearby and distant detonations is given by Mayrhofer Chr., “Grundlagen zu den Methoden der dynamischen Grenztragfähigkeitsberechnung bei terroristischen Ereignissen”, 2. Workshop Bau-Protect, Berichte aus dem konstruktiven Ingenieurbau 06/4, München, 2006.

Table 7.4-3: Dynamic increase factors (DIFs) f_{dy}/f_y and f_{du}/f_u for the design of reinforced concrete elements

Type of stress	Distant design range			Nearby design range		
	Reinforcing bars		Concrete	Reinforcing bars		Concrete
	f_{dy}/f_y	f_{du}/f_u	f'_{dc}/f'_c	f_{dy}/f_y	f_{du}/f_u	f'_{dc}/f'_c
Bending	1.17	1.05	1.19	1.23	1.05	1.25
Diagonal tension	1.00	—	1.00	1.10	1.00	1.00
Direct shear	1.10	1.00	1.10	1.10	1.00	1.10
Bond	1.17	1.05	1.00	1.23	1.05	1.00
Compression	1.10	—	1.12	1.13	—	1.16

$z = r/W^{1/3}$
where:
 r is the distance between explosion epicentre and structure [m];
 W is the weight of explosives as TNT equivalent [kg];
with
 $z \leq 0.5$ for nearby explosion;
 $z > 0.5$ for distant explosion.

For prestressing steel, strands or wires, a dynamic increase factor f_{dy}/f_y should always be taken as 1.

The calculation of large structures such as nuclear power plants is usually done at level II. For level II calculations the DIFs in Tables 7.4-3 and 7.4-4 should be used because local strain rates are not known.

Tables 7.4-3 and 7.4-4 are taken from UFC 3-340-02 – Structures to resist the effects of accidental explosion, Dept. of Defence, USA, 5/12/2008.

Table 7.4-4: Dynamic design stresses for the design of reinforced concrete elements

Type of stress	Type of reinforcement	Maximum support rotation, Θ_m (degrees)	Dynamic design stress	
			Reinforcement f_{ds}	Concrete, f'_{dc}
Bending	Tension and compression	$0 < \Theta_m \leq 2$	$f_{dy}^{(1)}$	$f'_{dc} \text{ ---}^{(2)}$
		$2 < \Theta_m \leq 6$	$f_{dy} + (f_{du} - f_{dy})/4$	$\text{---}^{(2)}$
		$6 < \Theta_m \leq 12$	$(f_{dy} + f_{du})/2$	$\text{---}^{(2)}$
Diagonal tension	Stirrups	$0 < \Theta_m \leq 2$	f_{dy}	f'_{dc}
		$2 < \Theta_m \leq 6$	f_{dy}	f'_{dc}
		$6 < \Theta_m \leq 12$	f_{dy}	f'_{dc}
Diagonal tension	Lacing	$0 < \Theta_m \leq 2$	f_{dy}	f'_{dc}
		$2 < \Theta_m \leq 6$	$f_{dy} + (f_{du} - f_{dy})/4$	f'_{dc}
		$6 < \Theta_m \leq 12$	$(f_{dy} + f_{du})/2$	f'_{dc}
Direct shear	Diagonal bars	$0 < \Theta_m \leq 2$	f_{dy}	$f'_{dc} \text{ ---}^{(3)}$
		$2 < \Theta_m \leq 6$	$f_{dy} + (f_{du} - f_{dy})/4$	$\text{---}^{(3)}$
		$6 < \Theta_m \leq 12$	$(f_{dy} + f_{du})/2$	$\text{---}^{(3)}$
Compression	Column	— ⁽⁴⁾	f_{dy}	f'_{dc}

(1) Tension reinforcement only.

(2) Concrete crushed and not effective in resisting moment.

(3) Concrete is considered not effective, and shear is resisted by the reinforcement only.

(4) Capacity is not a function of support rotation.

7.4.2.3.2 Dynamic dimensioning with regard to internal forces

With regard to dimensioning concrete structures for internal forces caused by dynamic loads, the following considerations apply:

- the internal forces caused by dynamic loads (level II, III) have to be calculated using of a dynamic numerical calculations. The numerical material behaviour of reinforced concrete has to be considered in compression and tension;
- the reinforcement and the concrete in any cross-section can be dimensioned for bending with the same methods as used for static loading, considering the increase of strength of concrete and reinforcement steel according to Figures 7.4-6 and 7.4-7, under high dynamic strain rates;
- the use of the dynamic increase factors for reinforcing bars is only possible if the steel has sufficient ductility, as for seismic requirements – see Table 7.4-3;
- the structure has to be dimensioned for bending with normal forces, shear and torsion for the global internal forces.

In the numerical calculation of the internal forces on level III the impact or the explosion or the impact process itself is part of the numerical calculation. So, special codes are necessary, by which the relevant phenomena such as wave propagation, large deformations and fracture can be modelled adequately. This can be done for example by hydrocodes or special finite element codes. On level III the structure is modelled in a very detailed way, local strain rates are available generally.

In Figures 7.4-6 and 7.4-7 curves for the increase of the strength of concrete and of reinforcing steel are given in relation to the strain rates.

The behaviour of concrete in compression and tension is based on equations 5.1-115a and 5.1-115b, 5.1-117a and 5.1-117b respectively in subsection 5.1.11.2.

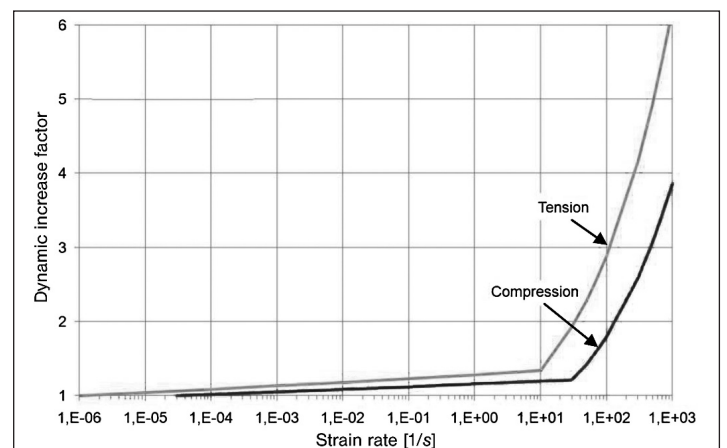


Figure 7.4-6: Increase of concrete strength under high strain rates for compression and tension in a semi-log format

The equations in that describe the yield and ultimate stress of reinforcing steel used Figure 7.4-7 are listed below and can be found in Malvar L.J., Crawford J.E., “*Dynamic Increase Factor for Steel Reinforcing Bars*”, 28. DDESB Seminar, Orlando, Florida, 1998.

$$\text{DIF} = (\dot{\epsilon} \cdot 10^{-4})^{\alpha}$$

where $\alpha_{fy} = 0.074 - 0.040 (f_y/414)$;

$\alpha_{fu} = 0.019 - 0.009 (f_y/414)$;

with f_y as static yield strength of reinforcing bar [MPa].

All of these data are derived from tests and can be used in calculations on level 3. More detailed information concerning stress and strain rate effects for concrete are given in subsection 5.1.11.2.

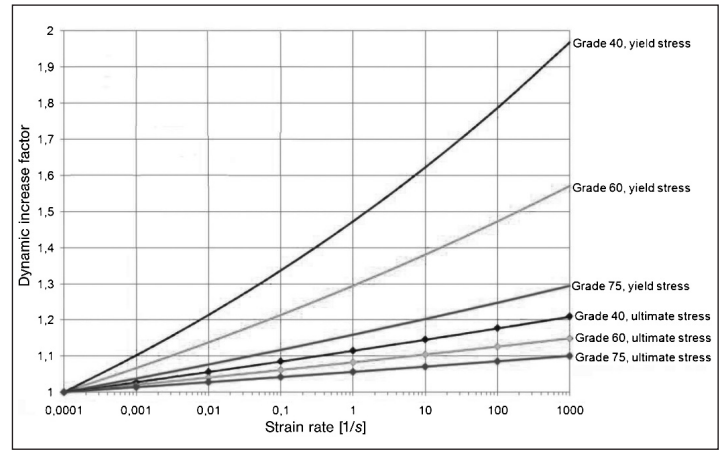


Figure 7.4-7: Increase of reinforcing steel strength under high strain rates

7.4.2.3.3 Dynamic dimensioning with regard to ductility

For dynamic dimensioning with regard to ductility the following considerations apply:

- The deflections caused by dynamic loads (level II, III) have to be calculated by use of a dynamic numerical calculation. The material behaviour of reinforced concrete has to be considered in compression and tension.
- As for earthquake design, the plastic rotations are limited (Table 7.4-4).
- The dynamic design stresses are limited to the values given in Table 7.4-4. With these parameters the load–deflection behaviour can be determined for each cross-section, for example to establish moment–curvature diagrams.

7.4.2.4 Structural detailing and other measures

7.4.2.4.1 Possibilities for a reduction of the loads in the design phase

In the design phase, the engineer has several options to reduce the high dynamic loads. If there is sufficient space, this can be done by increasing the distance between the structure and the threat, or by barriers. If this is not possible, the materials have to be selected and the structure has to be designed for the dynamic loads concerned.

In the design stage other measures could also be considered to reduce the danger of structural failure due to dynamic loads, such as:

- measures to secure sufficient distance between structure and threat;
- security walls for a reduction of pressure;
- sandwich cross-sections with hard and soft layers to get a sufficient dissipation of energy, adapted to the special case of loading;
- symmetric design of the lateral load-resisting elements in plane, as for seismic design to avoid high local stresses.

7.4.2.4.2 General recommendations for detailing

There are two general options for design concerning loads with high strain rates when failure should be taken into account. One possibility is to define alternative load paths in the structure. This leads to a safe design only if the quantity of the loads can be calculated with sufficient accuracy. The alternative solution is the use of defined breaking points or areas in the structure to make sure that no progressive collapse can occur. In this case, local damage is accepted but the collapse of the total structure has to be avoided.

Generally it is important that the cross-sections, as well as the complete structure, have a high ductility. This involves the choice of the material, the layout of the connections and of the cross-sections as well.

With regard to detailing the following possibilities may be considered:

- use concrete with reduced Young’s modulus;
- use concrete with high ductility;
- generate alternative load paths and redundant structural systems;
- provide minimum dimensions;
- ensure high ductility of cross-sections and connections;
- use steel quality as required for seismic design;
- use adequate confinement to secure the cross-sections.

Strengthening of existing structures can be done by the same methods as used in seismic design.

Background information on the seismic design models is also given by Fardis, M. N. (2013), Performance- and Displacement-Based Seismic Design and Assessment of Concrete Structures in the *fib* Model Code 2010. Structural Concrete, 14. doi: 10.1002/suco.201300001.

Under certain conditions linear elastic analysis may be used for the determination of the deformations due to the seismic action, but not of the internal forces (see subsection 7.4.3.2.2).

As the seismic action induces large inelastic rotation demands to beam ends, the effective slab width on either side of the web at the ends of T-beams in Figure 7.2-2, $b_{eff,i}$, is taken as the lesser of 25% of the beam span and of the mid-distance to the adjacent parallel beam, b_i . All slab bars which are parallel to such a beam and fall within this width are considered as fully effective as longitudinal reinforcement of the beam's end section.

Seismic deformation demands are underestimated, if sources of flexibility are neglected and stiffness is overestimated. To this end, tension stiffening is neglected, as it diminishes with load cycling. Slippage of the tension bars from their anchorage in joints or foundation elements increases the member's apparent flexibility, owing to the fixed-end rotation of the end section it produces:

$$\theta_{slip} = \frac{\phi d_b L \sigma_s}{8 \tau_b} \quad (7.4-13)$$

7.4.2.4.3 Strengthening of existing reinforced concrete structures

The following methods for strengthening existing structures may be considered:

- increase of strength by implicating external reinforcement, for example carbon retrofitting;
- application of sandwich elements for energy dissipation;
- limitation of debris by, for example, an additional layer of textile reinforced concrete.

7.4.3 Seismic design

7.4.3.1 Format of the verifications

Seismic actions impose deformations on structures. These deformations in turn produce internal forces and moments.

Action effects, S_{Ed} , in seismic situations and the corresponding resistances, R_d , for the verification of the ULS of resistance according to subsection 4.5.2.2 are normally deformations for ductile modes of behaviour and failure (e.g. for flexure with limited axial load), or internal forces for brittle ones (notably, for shear):

- deformation action effects, S_{Ed} , are determined through non-linear analysis according to 7.4.3.2.4, or – under certain conditions – via linear elastic analysis according to 7.4.3.2.2.
- deformation resistances are normally plastic chord rotations at member ends, established according to 7.4.3.4.
- shear force action effects, S_{Ed} , are determined through non-linear analysis according to 7.4.3.2.4, or by capacity design calculations according to 7.4.3.2.3.
- shear force resistances, R_d , are determined according to section 7.3.3, with the special rules of subsection 7.4.3.5 applying in flexural plastic hinges.

For the verification of the SLS of deformations in seismic situations according to subsection 4.5.2.5, a is normally the chord rotation at a member end and is verified according to subsection 7.4.3.7.

7.4.3.2 Determination of seismic action effects through analysis

Non-linear methods of analysis should normally be used for seismic actions, taking into account the dynamic response of the structure and representing the action through the system of structural deformations or inertia loads it produces.

7.4.3.2.1 Modelling

The model of the structure should represent the distribution of its stiffness, mass and resistance, so that all significant deformation shapes and inertia forces are properly accounted for.

For both the ULS and the SLS the elastic stiffness should be based on fully cracked sections of those members which are expected to yield in the limit state of interest, without tension stiffening. The fixed-end-rotation of the member's end section due to slippage of longitudinal bars from their anchorage zone outside the member length should also be taken into account.

where:

- φ is the curvature at the end section;
- σ_s is the stress of the tension bars at the end section;
- d_{bL} is the mean diameter of the tension bars;
- τ_b is the mean bond stress along the straight anchorage length of the tension bars outside the member length.

At yielding of the end section, φ and σ_s may be taken equal to their yield values, φ_y and f_y , while for ribbed bars τ_b (in MPa) may be taken equal to $\sqrt{f_c}$ (MPa), giving a value of θ_{slip} denoted as $\theta_{slip,y}$. The increase in the member's apparent flexibility due to this fixed-end rotation is equivalent to adding a rotational spring at the end section, with stiffness $M_y/\theta_{slip,y}$, where M_y is the yield moment there (proportional to φ_y).

In the background of seismic design based on controlled inelastic response stands a bilinear force–deformation law of the lateral load resisting system as a whole and of its individual components in primary (monotonic) loading. The effective stiffness to the yield point should therefore be used as element elastic stiffness. The default damping of 5% of critical used in linear elastic seismic analysis of concrete structures reflects hysteretic energy dissipation in load cycles up to element yielding.

The chord rotation at the end of a one-dimensional member, θ , is the angle between the normal to the end section and the chord connecting the member ends at the member's displaced position. It includes flexural and shear deformations along the member, as well as the fixed-end rotation of the end section due to slippage of longitudinal bars from their anchorage zone outside the member length. In the linear elastic regime the member deformations that determine – together with any loading between the ends – the bending moments and shears along the member are the chord rotations at the two member ends.

Estimation of EI_{eff} before the analysis for the seismic action may be based on presumed values of L_s equal to 50% of the clear length of beams between columns and of columns between beams in the plane of bending, as well as of bridge piers fixed against rotation by the deck in the plane of bending. For the strong direction of walls in buildings, the value of L_s within a storey may be taken equal to 50% of the height from the wall's base section in that storey to the top of the wall in the building. In members cantilevering within the plane of bending, L_s is the member clear length.

An average value of EI_{eff} at: (a) the two ends where it frames into other members in the plane of bending and (b) for positive and negative moments may be used for a member.

Application of Eq. (7.4-14) presumes that the longitudinal reinforcement of the end sections is known. In new structures, this may entail first dimensioning this reinforcement from the non-seismic situations and minimum reinforcement and then estimating how much it may increase for the verifications in the seismic situation. As the value of EI_{eff} depends weakly on the amount of longitudinal reinforcement, use may also be made of empirical expressions giving the ratio of EI_{eff} of different types of members to the uncracked gross section stiffness as a function of known parameters (e.g. the shear span to depth ratio at the end of the member, the mean axial stress, N/A_c , the ratio of mean bar diameter to the section depth etc.).

45° cracking near the member end precedes flexural yielding if the value of $V_{Rd,c}$ of Eq. (7.3-17) is less than the shear force at flexural yielding, M_y/L_s .

The shear deformation of the member at the time it yields in flexural mode is fairly small, in the order of:

- 0.0014(1 + 1.5h/L_s) for beams or rectangular columns;
- 0.0013 for walls and members with hollow rectangular section;
- 0.0027 max[0; 1 – 2L_s/15D] for circular piers or columns.

The secant stiffness to the yield-point should be used as effective elastic stiffness of members which yield at the limit state of interest.

For one-dimensional concrete members (including slender walls) the secant stiffness to the yield point may be estimated as:

$$EI_{eff} = \frac{M_y L_s}{3\theta_y} \quad (7.4-14)$$

where:

- M_y is the yield moment;
- θ_y is the chord rotation at the yielding member end;
- $L_s = M/V$ is the shear span (moment to shear ratio) at that end section in the seismic situation.

θ_y is the sum of:

- a flexural component, equal to $\varphi_y(L_s+z)/3$ if 45° cracking of the member precedes flexural yielding of its end section, or to $\varphi_y L_s/3$ if it doesn't;
- shear deformation, which normally increases with decreasing member shear span to depth ratio, L_s/h ; and

Eq. (7.4-14) applied with computed values of θ_y and with the prestress taken as part of the actions, generally underestimates the effective stiffness of prestressed components with bonded tendons. Hence, its use is safe-sided in the context of displacement-based seismic design or assessment.

If a rotational spring with stiffness $M_y/\theta_{slip,y}$ is added at the member end, to account for the apparent increase in member flexibility due to this fixed-end rotation, then the term $\theta_{slip,y}$ is not included in θ_y .

This is the “equal displacement rule” at the level of member deformations. For a single-degree-of-freedom oscillator with elasto-plastic force-deformation law in monotonic loading, the rule states that the maximum displacement in the inelastic seismic response is about the same as that of an elastic oscillator with the same mass, damping and elastic stiffness.

Concrete structures have fundamental periods in the range where the equal displacement rule gives fairly accurate results for an equivalent single-degree-of-freedom oscillator and applies well on average for member deformations (albeit with considerable deviations for individual members).

The end section of a strong column framing into weaker beams, or of a strong beam framing into weaker columns, does not yield in the seismic situation, even when their M_E/M_R ratio exceeds 1.0. Except at such sections, the ratio M_E/M_R is about equal to the corresponding ductility ratio demand for the chord rotation at the end of the member (peak inelastic chord rotation demand divided by the corresponding value at yielding of the end section).

The directions of the two orthogonal horizontal components should be chosen along two main directions of the structural layout in plan:

- in bridges: along the longitudinal direction (connecting the two points on the deck axis at the abutments) and the orthogonal, transverse one;
- in buildings or similar structures: along directions of near-symmetry, or to which a large number of plane frames and/or walls are aligned etc.

The response history analyses may be carried out either simultaneously for all seismic action components of interest, or separately for each one and the results superposed.

The resultant of the equivalent forces in the direction of the seismic action component, V_b , is determined as:

$$V_b = m_{eff,1} S_a(T_1)$$

where:

$S_a(T_1)$ is the value of the elastic response spectrum at period T_1 ;

$m_{eff,1}$ is an estimate of the effective modal mass of the dominant mode.

It is safe-sided to take $m_{eff,1}$ as equal to the total vibrating mass, m . In the horizontal direction of a building type structure, m is the total mass above the foundation or the top of a rigid basement. In the horizontal direction of a bridge, it is the total mass of the deck

- the fixed-end rotation due to slippage of longitudinal bars from their anchorage zone outside the member length (the value at yielding of the end section, $\theta_{slip,y}$).

7.4.3.2.2 Linear elastic analysis for the calculation of seismic deformation demands (action effects)

Member inelastic flexural deformations due to the seismic action may be determined through linear elastic analysis with damping 5% of critical, under the condition that they are not concentrated at particular parts of the structure (notably at one side of a building in plan, or in one or a few building storeys or bridge piers) but are distributed fairly uniformly throughout the structure.

Applicability of linear elastic analysis for the estimation of inelastic flexural deformations may be checked by inspection of the distribution over the structure of the ratio of the moment from the linear analysis at member end sections, M_E , to the corresponding moment resistance, M_R .

The pertinent moment-deformation relations should be used to calculate moments at member ends from inelastic flexural deformations estimated via linear elastic analysis.

The maximum values of member inelastic flexural deformations due to the relevant components of the seismic action may be estimated using one of following methods of linear elastic analysis:

1. Response history dynamic analysis with damping 5% of critical.
2. Modal response spectrum analysis, using the 5% damped elastic response spectrum:
The number of normal modes should be sufficient to capture with their effective modal masses at least 90% of the total vibrating mass in each direction that a seismic action component is applied. Peak modal responses for any deformation measure of interest should be combined via the complete quadratic combination (CQC) rule.

Modal response spectrum analysis may be simplified into separate linear static analyses under equivalent forces in the direction of each one of the relevant seismic action components, with the structure considered as an equivalent single-degree-of-freedom system having the period of the normal mode, T_1 , which has the largest modal mass in that direction.

and of the upper half of the pier height in the relevant part of the bridge. In the vertical direction of a long component on relatively rigid supports, having significant mass distributed along its span, m is the total mass of the component.

If T_1 is shorter than one-half of the period that marks the applicability limit of the simplification of modal analysis as equivalent static (see condition (iii) below), it is permissible to take:

$m_{eff,1} = 0.85m$ for:

- the horizontal direction of buildings with more than two storeys above the foundation or the top of a rigid basement;
- the transverse direction of bridges with continuous deck laterally restrained at the abutments (except when the transverse stiffness of piers is large, or it has large differences between adjacent piers or decreases from the abutments to deck mid-length).

The simplification cannot be applied to only one of the two horizontal seismic action components, but it may be applied to the vertical alone.

Condition (i) may be considered to be met if that mode's effective modal mass in the direction of interest accounts for at least 75% of the total.

Condition (ii) may be considered to be met:

- in building type structures, in the horizontal direction, if:
 - (a) the storey mass and lateral stiffness are either constant in all storeys, or decrease gradually and smoothly from the base to the top; and
 - (b) no lateral-load-resisting subsystem (frame, wall etc.) is vertically interrupted before the top of the corresponding part of the structure; and
 - (c) any setbacks of each side are limited (e.g. at each storey to 10% of the parallel dimension of the one below and in total to one-third of the building's parallel dimension at the base).
 - The shape of the single mode considered may then be taken proportional to elevation above the top of the foundation or of a rigid basement.
- in bridges with piers having total mass much less (e.g. $\leq 20\%$) than the deck:
 - (a) in any horizontal direction, if lateral stiffness is provided only by piers that are not coupled through the deck (e.g. if spans are simply supported) and may be considered as structurally independent;
 - (b) in the longitudinal direction, if the deck is continuous, almost straight and not restrained in that direction at the abutments;
 - (c) in the transverse direction, if the deck is continuous and approximately straight, unless the transverse stiffness of piers is of the same order as that of the abutments, or exhibits large differences between adjacent piers, or decreases from the abutments to deck mid-length.
 - In cases (a) and (b) the relevant part of the deck is considered to have the same translation in the horizontal direction of interest. In case (c) the mode shape is taken proportional to the elastic displacements due to the gravity loads applied in the horizontal direction of interest.
- in the vertical direction of long components supported on relatively rigid supports and having significant mass distributed along the span; in that case the vertical mode is taken proportional to the component's elastic deflection due to gravity loads.

T_1 is usually the longest of all the normal modes with significant effective modal mass in the direction of the seismic action component in question (the fundamental period). As spectral displacements increase until a value T_D of the period, above which

The simplification of modal response spectrum analysis into equivalent static analyses in the direction of the seismic action component is allowed, if all of the following conditions are fulfilled:

- i. The response to the relevant component is dominated by the normal mode of vibration having the largest effective modal mass in that direction.
- ii. It is possible to identify, to a good approximation, the shape of the single normal mode taken into account.

- iii. Response spectral displacements at T_1 are much larger than at the period of any other mode with significant effective modal mass in the direction of the seismic action component.

they stay about constant or fall with increasing period T , condition (iii) may be considered as met, if:

$$T_1 \leq \min(T_D; 4T_C)$$

where:

T_C is the period at which spectral accelerations attain their largest value (normally spectral velocities are about constant between T_C and T_D).

The Rayleigh quotient estimates T_1 quite accurately, using the displaced shape of the structure from the very equivalent static analysis under the force distribution used there (forces proportional to mass times the assumed modal shape).

Regardless of whether a full modal response spectrum analysis or its equivalent static simplification is applied, peak absolute values of seismic action effects from (concurrent) seismic action components in the x, y and z directions are determined via the SRSS rule from the peak values computed for separate application of these components. These peak absolute values should be combined with other action effects with a plus or minus sign.

If modal response spectrum analysis is used, the combination of modal contributions through the CQC rule and of E_X , E_Y (and E_Z) via Eq. (7.4-15) can be done in a single modal response spectrum analysis covering all relevant seismic action components. Such an analysis gives the expected value of peak seismic action effects under concurrent statistically independent seismic action components along x, y (and z). Moreover, this value of E is independent of the choice of the horizontal directions x and y. Values of E_Z from equivalent static analysis for the vertical component are still combined via Eq. (7.4-15) with the outcome of the combination of E_X , E_Y within a single modal response spectrum analysis for the two horizontal components x and y.

Probabilistic models give the values and signs of other action effects (e.g. the column deformation in the orthogonal direction) expected to take place simultaneously with the maximum value of the action effect obtained via the SRSS rule.

If M_R denotes a moment resistance component with vector normal to the shear force of interest, index b is used for beams, c for columns, w for walls and l_{cl} is the clear length of a beam, column or bridge pier (Figure 7.4-8).

The equivalent static analysis is carried out under a force distribution over the structure (or its relevant part) proportional to the product of mass and the shape of the single normal mode considered. The value of T_1 is estimated on the basis of mechanics, possibly using results of such an analysis.

Peak values of seismic action effects (i.e. of deformations in this case), E_X , E_Y , E_Z , due to separate application of the seismic action components in the x, y and z directions should be combined through the square root of the sum of squares (SRSS) rule:

$$E = \pm \sqrt{E_X^2 + E_Y^2 + E_Z^2} \quad (7.4-15)$$

7.4.3.2.3 Capacity design for shear forces when linear elastic analysis is used for the estimation of deformation seismic action effects

In capacity design, the maximum force demands in brittle components or modes of behaviour or failure are estimated from equilibrium and from the moments delivered to the component in question at its connections to the others. Each of these moments is taken equal to the minimum of:

- (a) the moment component with vector normal to the shear force of interest from the analysis, M_E ;
- (b) the moment resistances at plastic hinge formation – multiplied by $\gamma_{Rd} \geq 1.0$ for over-strengths not explicitly modelled (e.g. due to steel strain hardening) – with plastic hinges assumed to form at those two opposite faces of a joint between the weaker elements.

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at those two opposite faces of a joint between the weaker elements.

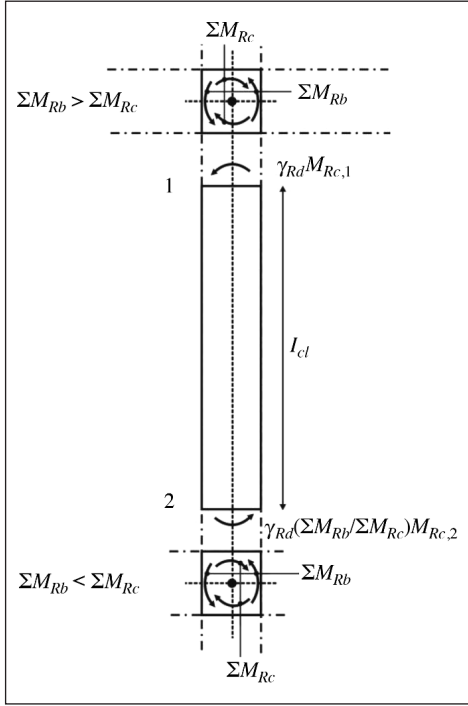


Figure 7.4-8 End moments for capacity design shear in a frame column

The capacity design shear forces can be calculated in the following way:

- (1) Between ends 1 and 2 of a frame column, bridge pier or similar member:

$$V_{CD} = \frac{\min \left[M_{E,c1}; \gamma_{Rd} M_{R,c1} \min \left(1; \frac{\sum M_{R,b}}{\sum M_{R,c}} \right)_1 \right] + \min \left[M_{E,c2}; \gamma_{Rd} M_{R,c2} \min \left(1; \frac{\sum M_{R,b}}{\sum M_{R,c}} \right)_2 \right]}{l_{cl}} \quad (7.4-16)$$

where:

M_E and M_R enter with positive values;

$\sum M_{Rc}$ or $\sum M_{Rb}$ are the sums of moment resistances of the columns or the beams framing into the joint, respectively;

At the connections with the foundation or of a pier with a deck integral with it, it is always: $\sum M_{Rb} / \sum M_{Rc} > 1$;

For convenience, the moment resistances of columns or piers, M_{Rc} , are computed for the axial force due to gravity loads alone.

- (2) At distance x from end section i of a frame beam (Figure 7.4-9) (the other end denoted by j):

$$V_{CD}(x) = V_{g+\psi 2q,o}(x) + \frac{\min \left[M_{E,bi}^-; \gamma_{Rd} M_{R,bi}^- \min \left(1; \frac{\sum M_{R,c}}{\sum M_{R,b}} \right)_i \right] + \min \left[M_{E,bj}^+; \gamma_{Rd} M_{R,bj}^+ \min \left(1; \frac{\sum M_{R,c}}{\sum M_{R,b}} \right)_j \right]}{l_{cl}} \quad (7.4-17)$$

where:

all moments enter as positive, but index $(-)$ is used for moments inducing tension at the top flange of a beam (hogging ones) and $(+)$ for those inducing tension at the bottom (sagging);

$V_{g+\psi 2q,o}(x)$ is the shear force at section x due to the quasi-permanent gravity loads $g+\psi 2q$, with the beam considered as simply supported (index: o).

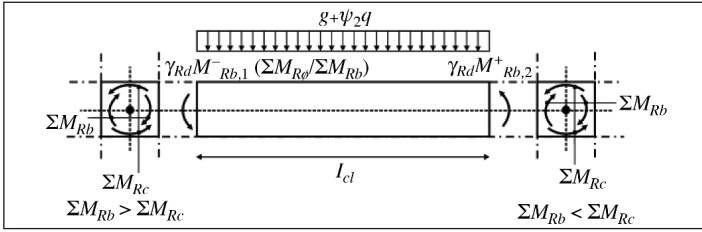


Figure 7.4-9: End moments for capacity design shear in a frame beam

- (3) In a joint where a horizontal element (a beam – including a foundation beam – a bridge deck etc.) and a vertical one (frame column, structural wall, bridge pier etc.) frame into each other:

The nominal shear stress in the joint core, v_j , is the same, regardless of whether it is derived:

from the horizontal shear force in the joint, V_{jh} , as:
$$v_j = \frac{V_{jh}}{b_j h_{jc}} \quad (7.4-18)$$

or

from the vertical shear force in the joint, V_{jv} , as:
$$v_j = \frac{V_{jv}}{b_j h_{bj}} \quad (7.4-19)$$

where:

b_j is the effective joint width in the orthogonal horizontal direction;

h_{jc} is the horizontal distance between the outermost layers of vertical reinforcement in the joint in the direction of V_{jh} ;

h_{bj} is the clear depth between the top and bottom reinforcement in the joint.

If $\Sigma M_{Rb} < \Sigma M_{Rc}$, v_j is controlled by the horizontal shear force in the joint:

$$V_{jh} = \Sigma M_{Rb} \left(\frac{1}{h_{bj}} - \frac{1}{h_{st}} \frac{L_b}{L_{bn}} \right) \quad (7.4-20)$$

Otherwise (i.e. if $\Sigma M_{Rb} > \Sigma M_{Rc}$), it is governed by the vertical shear force:

$$V_{jv} \approx \Sigma M_{Rc} \left(\frac{1}{h_{cj}} - \frac{1}{L_b} \frac{h_{st}}{h_{st,n}} \right) + \frac{1}{2} \left| [V_{g+\psi_2 q, b}]_l - [V_{g+\psi_2 q, b}]_r \right| \quad (7.4-21)$$

where:

h_{st} , $h_{st,n}$ are the theoretical and clear height of the vertical element, respectively;

L_b , L_{bn} are the theoretical and clear span, respectively, of the horizontal element (beam or deck);

$V_{g+\psi_2 q, b}$ is the shear force due to quasi-permanent actions at the end of the horizontal element on the left (index: l) or on the right (index: r) side of the joint.

If the moments from the analysis, M_{Eb} , M_{Ec} , are such that $\Sigma M_{Eb} < \Sigma M_{Rb}$ and $\Sigma M_{Ec} < \Sigma M_{Rc}$, v_j may be estimated either from:

- the horizontal shear force in the joint, V_{jh} , from Eq. (7.4-20) using there ΣM_{Eb} instead of ΣM_{Rb} ;

or

- the vertical shear force, V_{jv} , from Eq. (7.4-21) using there ΣM_{Ec} instead of ΣM_{Rc} .

- (4) At elevation z above the base (top of foundation or of a rigid basement) of a structural wall in a building:

Capacity-design shears along the height of a multistorey wall cannot be established only from equilibrium and the moment capacities of the wall at the base and of beams framing into it

Whenever capacity design effects cannot be determined solely from the moment capacities at plastic hinge formation on the basis of equilibrium, they may be found assuming that the seismic action effects at the instant that the moment capacities at plastic hinge are reached are proportional to the corresponding outcomes of the linear elastic analysis for the seismic situation.

at floor levels. They may be conveniently estimated, instead, as follows:

For “squat” walls (with height-to-length ratio $h_w/l_w \leq 2$):

$$V_{CD,w}(z) = \min \left(1; \frac{M_{R,w}(z=0)}{M_{E,w}(z=0)} \right) V_{Ed,w}(z) \quad (7.4-22)$$

For “slender” walls ($h_w/l_w > 2$), also taking into account higher mode effects on the shears of slender walls after yielding at the base:

$$V_{CD,w}(z) = \sqrt{\min \left(1; \frac{M_{R,w}(z=0)}{M_{E,w}(z=0)} \right)^2 + 0.1 \left(\frac{\max S_a}{S_a(T_1)} \right)^2} V_{Ed,w}(z) \quad (7.4-23)$$

where:

$M_{E,w}(z)$, $V_{E,w}(z)$ are the moment and shear from the linear analysis for the seismic action of interest and the concurrent quasi-permanent actions;

$M_{R,w}(z=0)$ is the moment resistance of the base section for the value of axial load due to the quasi-permanent gravity loads alone;

$\max S_a$ is the maximum value of the elastic acceleration response spectrum of the seismic action;

$S_a(T_1)$ is the value of the elastic acceleration response spectrum at the fundamental period T_1 in the horizontal direction (closest to that) of the wall shear force.

(5) At the interface of foundation elements and the ground:

All linear analysis results for the seismic action effects transferred to the ground are multiplied by a factor a_{CD} , as follows:

(a) For individual footings:

$$a_{CD} = \min \left[\left(1; \frac{M_{R,y}}{M_{E,y}} \right); \left(1; \frac{M_{R,z}}{M_{E,z}} \right) \right] \quad (7.4-24)$$

where:

$M_{E,y}$, $M_{E,z}$ are the two orthogonal moment components at the base of the vertical element from linear elastic analysis for the seismic action of interest and the quasi-permanent gravity loads;

$M_{R,y}$, $M_{R,z}$ are the uniaxial moment resistances at the base of the vertical element for the value of axial load due to the quasi-permanent actions alone.

(b) For $N > 1$ vertical elements with a common foundation (a foundation beam, a pile cap, a box-type foundation, a raft etc.):

$$a_{CD} = \frac{\sum_{i=1}^N a_{CD,i} M_{E,i}}{\sum_{i=1}^N M_{E,i}} \quad (7.4-25)$$

where:

$a_{CD,i}$ is the value of a_{CD} for individual vertical element i ;

$M_{E,i}$ is that value among the two moment components $M_{E,y}$, $M_{E,z}$ from linear analysis for the seismic action of interest and the quasi-permanent actions which gives the minimum ratio $M_{R,y}/M_{E,y}$ or $M_{R,z}/M_{E,z}$ at the base of vertical element i and governs plastic hinging there.

(6) In brittle or sensitive components designed to remain elastic after plastic hinging of vertical supports (notably, for the deck, any fixed bearings, shock transmission units, abutments flexibly connected to the deck, seismic links consisting of shear keys,

buffers and/or linkage bolts or cables and so on of bridges with plastic hinging in piers).

Linear analysis results for the seismic action effects in brittle or sensitive components, are multiplied by a factor applying for the entire structure under the seismic action direction of interest:

$$a_{CD} = \frac{\sum V_{CD,i}}{\sum V_{E,i}} \quad (7.4-26)$$

where:

i is the index that refers to the vertical supporting elements that yield in the seismic situation considered and summations extend over all of them;

$V_{E,i}$ is the seismic shear force from linear analysis;

$V_{CD,i}$ is the capacity design shear in the direction of $V_{E,i}$ calculated as in case 1 above (see Eq. (7.4-16)).

Unlike linear elastic analysis, which may be relied upon – under certain conditions – to estimate seismic deformations but not internal forces, non-linear analysis may be used to determine both types of seismic action effects.

When bending is mainly in a single plane, it may be sufficient to use for each member end a moment-(chord) rotation model within that plane. The effect of significant variation of the axial load during the response (e.g. in the exterior columns of tall frames) on the moment-rotation behaviour should be taken into account. Simplified treatment of the coupling between the two directions of bending in vertical elements of three-dimensional models is normally possible.

In the end, all components (including those of an existing structure being retrofitted for earthquake resistance) are normally verified to remain elastic if they are brittle, or to have some margin against their ultimate deformation – after which the drop in resistance is significant – if ductile. So, the reduction in resistance after ultimate strength may be neglected and the force–deformation relation in primary loading taken bilinear, especially if the initial post-yield strain hardening is also neglected, with the full post-yield behaviour taken perfectly plastic. It is on such a straight post-yield branch that the limit deformation is verified, with corrective measures taken if this limit is exceeded.

Bilinear unloading-reloading parallel to the elastic and post-yield branches in monotonic loading is characteristic of steel but not of structural concrete, producing unrealistically large hysteretic damping.

When part of the deformation is due to bond-slip (e.g. from a joint) or to the effects of shear (as for example in members with low shear-span-to-depth ratio), hysteresis loops are “pinched” in the form of an inverted S and hysteretic energy dissipation is reduced.

Hysteresis rules also play an important role for the estimation of residual deformations of members (for local damage) or of the structure as a whole (e.g. the permanent tilt) after the earthquake.

7.4.3.2.4 Non-linear analysis for seismic action

Non-linear dynamic analysis, with solution of the equations of motion in the time-domain, is the reference analysis method for seismic actions. The action should be specified as a suite of independent seismic events in terms of time histories of the ground motion components (see subsection 4.5.1.4).

Non-linear models should be employed, at least for those components expected to enter the inelastic regime in the seismic situation of interest.

One-dimensional members (including slender walls) designed or retrofitted for earthquake resistance will have all their inelasticity concentrated in flexural plastic hinges at their ends, where they are connected with other components. As a minimum, the non-linear model of such a member should employ a force–deformation relation for these end regions – notably, one between the end moment and the corresponding (chord) rotation.

As a minimum, non-linear models for components should use a bilinear force-deformation law in primary (monotonic) loading. Positive post-yield stiffness (due to strain hardening) may be neglected and elastic perfectly plastic behaviour assumed instead. Strong post-yield softening should be taken into account through negative post-yield stiffness.

The force–deformation law in primary loading should be supplemented with unloading-reloading (hysteresis) rules, realistically reflecting the amount of post-yield hysteretic energy dissipation and the reduction of unloading and reloading stiffness with increasing peak deformation of a cycle (stiffness degradation) which characterizes concrete components. The degradation of resistance with load cycling should be included if it is significant (e.g. in brittle or poorly detailed components).

The elastic stiffness of components and the viscous damping associated with the elastic regime should be the same as in linear elastic analysis (i.e. from subsection 7.4.3.2.1 and 5%, respectively).

Under seismic loading, material failure at the local level (even rupture of a tension bar) does not constitute by itself member failure. The plastic hinge fails owing to local material failures gradually accumulating during cycling of the imposed deformations.

The dependence of the plastic rotation capacity on each basic variable (material property or geometric dimensions) is not always monotonic and does not lend itself to application of partial factors on these variables.

The plastic part of the chord rotation at a member end is essentially the same as the plastic hinge rotation there, plus the post-yield part of the fixed-end-rotation, θ_{slip} , due to slippage of longitudinal bars from their anchorage zone outside the member length.

Under cyclic loading θ_u^p is the value of θ_{pl} beyond which an increase in imposed deformation cannot increase the moment resistance above 80% of its maximum ever value.

7.4.3.3 ULS verifications of inelastic flexural deformations

In seismic situations the verification at the ULS of one-dimensional members (including slender walls) takes place in terms of inelastic flexural deformations in plastic hinges forming at member ends.

Deformation measures used in the verifications should represent the behaviour of the plastic hinge region as a whole.

The global safety format in section 4.6 is appropriate for the ULS verification of a plastic hinge region, considered as an element.

An appropriate deformation measure for the plastic hinge region is the plastic part of the chord rotation at the member end, θ_{pl} .

The ultimate value of θ_{pl} , denoted as θ_u^p , is conventionally identified with a 20% drop in moment resistance.

The characteristic value (lower 5% fractile) of θ_u^p is obtained from its mean value, $\theta_{u,m}^p$, as:

$$\theta_{u,k}^p = \theta_{u,m}^p / \gamma_{Rd} \quad (7.4-27)$$

where the safety factor γ_{Rd} accounts for model uncertainty.

Resistance values that enter the verifications are obtained by dividing the characteristic value of the resistance, $\theta_{u,k}^p$, by a global safety factor γ_R^* , which depends on the ULS being verified:

- for the life safety limit state: $\gamma_R^* = 1.35$;
- for the near collapse (NC) limit state: $\gamma_R^* = 1.0$.

7.4.3.4 Cyclic plastic chord rotation capacity

The expected (mean) value of the plastic part of the ultimate chord rotation at a member end may be estimated as:

$$\theta_{u,m}^p = (\phi_u - \phi_y) L_{pl} \left(1 - \frac{L_{pl}}{2L_s} \right) + \Delta\theta_{slip,u-y} \quad (7.4-28)$$

where:

ϕ_u and ϕ_y are the ultimate and the yield curvature, respectively, of the end section;

L_s is the shear span (M/V ratio) at the member end;

L_{pl} is the plastic hinge length;

$\Delta\theta_{slip,u-y}$ is the post-yield part of the fixed-end-rotation due to slippage of longitudinal bars from their anchorage zone outside the member length.

ϕ_u and ϕ_y are determined from plane section analysis.

- for ϕ_y , linear elastic stress–strain relations may be assumed until yielding of the tension or the compression chord;
- for ϕ_u , the parabola–rectangle diagram of subsection 7.2.3.1.5 and Figure 7.2-9 should be used for concrete in compression and the idealized one of subsection 7.2.3.2 and Figure 7.2-15 (with linear strain hardening) for the reinforcing steel. Calculation of ϕ_u should take into account all possible failure modes:
 - (a) rupture of tension reinforcement in the full, unspalled section;
 - (b) exceedance of the concrete ultimate strain ε_{cu2} at the extreme compression fibres of the unspalled section;
 - (c) rupture of tension reinforcement in the confined core after spalling of the cover;
 - (d) exceedance of the ultimate strain $\varepsilon_{cu2,c}$ of the confined core after spalling.

Owing to the large local inelastic strains and surface cracking that develop in a bar when it buckles in one half cycle of loading, the bar may rupture during the subsequent tensile half cycle under a strain much lower than its nominal elongation at maximum force (which in practical applications may be taken equal to its characteristic value, $\epsilon_{u,k}$). Moreover, being erratic and unpredictable under cyclic loading, bar buckling of one out of several bars in a section may take place early, leading then to bar rupture.

Bonded tendons of prestressed components do not buckle under cyclic loading. It is safe-sided to take their strain at rupture as equal to that applying in monotonic loading, namely:

$$\epsilon_{su,mon} = \left(1 - \frac{1}{3} \sqrt{\ln N_{t,tension}}\right) \epsilon_{su,no\ min\ al} \quad (7.4-30)$$

where:

$N_{t,tension}$ is the number of prestressing tendons in the tension zone.

There is a tendency for the ultimate strain of the extreme fibres of the confined core after spalling of the cover under cyclic loading to increase with decreasing neutral axis depth, x_o , or depth, h_o , of the confined core of the section as:

$$\epsilon_{cu2,c} = 0.0035 + \left(\frac{1}{x_o(mm)}\right)^{3/2} + 0.4 \frac{\alpha \rho_w f_{yw}}{f_{cc}} \quad (7.4-32)$$

or:

$$\epsilon_{cu2,c} = 0.0035 + \left(\frac{10}{h_o(mm)}\right)^2 + 0.4 \frac{\alpha \rho_w f_{yw}}{f_{cc}} \quad (7.4-33)$$

For the full, unspalled section, the confinement term does not apply, and the dimensions x and h of the full section are used, in lieu of those of the confined core.

Failure mode (b) governs over (c) or (d) if the moment resistance of the confined core is less than 80% of that of the full unspalled and unconfined section.

For the purposes of the determination of ϕ_u under cyclic loading, the strain at rupture of ribbed tension bars should be taken equal to:

$$\epsilon_{su,cyc} = (3/8) \epsilon_{u,k} \quad (7.4-29)$$

while the ultimate strain of the extreme compression fibres of a concrete core confined by closed ties with 135° hooks may be taken as:

$$\epsilon_{cu2,c} = 0.0035 + 0.4 \frac{\alpha \rho_w f_{yw}}{f_{cc}} \quad (7.4-31)$$

where:

ρ_w is the ratio of transverse reinforcement in the direction of bending (or the minimum in the two transverse directions for biaxial bending)

f_{yw} is the yield stress of the transverse reinforcement;

α is the confinement effectiveness factor, which may be estimated as:

– for rectangular sections:

$$\alpha = \left(1 - \frac{s_h}{2b_o}\right) \left(1 - \frac{s_h}{2h_o}\right) \left(1 - \frac{\sum b_i^2 / 6}{b_o h_o}\right) \quad (7.4-34)$$

– for circular sections with circular hoops:

$$\alpha = \left(1 - \frac{s_h}{2D_o}\right)^2 \quad (7.4-35)$$

– for circular sections with spiral reinforcement:

$$\alpha = \left(1 - \frac{s_h}{2D_o}\right) \quad (7.4-36)$$

where:

s_h denotes the centreline spacing of stirrups;

D_o , b_o , h_o are the confined core dimensions to the centreline of the hoop;

b_i is the centreline spacing along the section perimeter of longitudinal bars (indexed by i) engaged by a stirrup corner or a cross-tie.

Between yielding of the end section and the ultimate curvature under cyclic loading, the yielding of the tension bars penetrates into their anchorage zone, increasing the fixed-end-rotation of the end due to slippage of longitudinal bars from their anchorage by:

$$\Delta\theta_{slip,u-y} = 5.5 d_{bL} \phi_u \quad (7.4-37)$$

For ϕ_u , ϕ_y and $\Delta\theta_{slip,u-y}$ determined as above, the plastic hinge length, L_{pl} , for cyclic loading may be estimated as:

– for beams, rectangular columns or walls and members of T-, H-, U- or hollow rectangular section:

$$L_{pl} = 0.2h \left[1 + \frac{1}{3} \min \left(9; \frac{L_s}{h} \right) \right] \quad (7.4-38)$$

– for circular columns or piers with diameter D :

$$L_{pl} = 0.6D \left[1 + \frac{1}{6} \min \left(9; \frac{L_s}{D} \right) \right] \quad (7.4-39)$$

Two practically equivalent purely empirical expressions for beams, rectangular columns or walls and members of T-, H-, U- or hollow rectangular section, are

$$\theta_{u,m}^{pl} = \alpha_{st}^{pl} (1 - 0.44a_{w,r}) \left(1 - \frac{a_{w,nr}}{4} \right) (0.25)^v \left(\frac{\max(0.01; \omega_2)}{\max(0.01; \omega_1)} \right)^{0.3} f_c^{0.2} \left(\frac{L_s}{h} \right)^{0.35} 25^{\left(\frac{\alpha \rho_w f_{yw}}{f_c} \right)} 1.275^{100\rho_d} \quad (7.4-40)$$

$$\theta_{u,m}^{pl} = \alpha_{st}^{hb} \left(1 - 0.052 \max \left(1.5; \min \left(10; \frac{h}{b_w} \right) \right) \right) (0.2)^v \left(\frac{\max(0.01; \omega_2)}{\max(0.01; \omega_1)} \frac{L_s}{h} \right)^{\frac{1}{3}} f_c^{0.2} 25^{\left(\frac{\alpha \rho_w f_{yw}}{f_c} \right)} 1.225^{100\rho_d} \quad (7.4-41)$$

where:

$\theta_{u,m}^{pl}$ is in rads, f_c in MPa, and:

α_{st}^{pl} , α_{st}^{hb} are coefficients for the type of steel:

- for Class B, C or D steel: $\alpha_{st}^{pl} = 0.0143$ and $\alpha_{st}^{hb} = 0.017$;
- for Class A: $\alpha_{st}^{pl} = 0.0069$, $\alpha_{st}^{hb} = 0.0073$.

$a_{w,r}$ is a zero-one variable for rectangular walls:

- $a_{w,r} = 1$ for rectangular walls;
- $a_{w,r} = 0$ otherwise;

$a_{w,nr}$ is a zero-one variable for non-rectangular sections:

- $a_{w,nr} = 1$ for T-, H-, U- or hollow rectangular section;
- $a_{w,nr} = 0$ for rectangular ones;

$v = N/bhf_c$

where:

b is the width of the compression zone;

N is the axial force, positive for compression;

$\omega_1 = (\rho_l f_{yl} + \rho_w f_{yw})/f_c$ is the mechanical reinforcement ratio for the entire tension zone, including the tension chord (index 1) and the web longitudinal bars (index v);

$\omega_2 = \rho_2 f_{y2}/f_c$ is the mechanical reinforcement ratio for the compression zone;

$L_s/h = M/Vh$ is the shear-span-to-depth ratio at the section of maximum moment;

α is the confinement effectiveness factor from Eq. (7.4-34);

$\rho_w = A_{sh}/b_w s_h$ is the ratio of transverse steel parallel to the plane of bending;

ρ_d is the steel ratio of diagonal bars (if any) in each diagonal direction of the member;

b_w is the width of *one* web, even in cross-sections with two or more parallel webs.

With $\theta_{u,m}^{pl}$ from these expressions, the safety factor for its conversion to characteristic value via Eq. (7.4-27) is $\gamma_{Rd} = 1.75$.

For prestressed components with bonded tendons Eqs. (7.4-40) and (7.4-41) may be applied (as a matter of fact, they are safe-sided), if the prestressing tendons are included in the calculation of ω_1 and ω_2 and the prestressing force in N .

7.4.3.5 Cyclic shear resistance at the ULS in members with shear reinforcement

The design value of shear resistance within flexural plastic hinges forming at member ends in the seismic situation is determined according to subsection 7.3.3.3, but using the following values for the minimum inclination of the compression struts, ϑ :

- $\cot \vartheta = 1$, wherever the plastic part of the chord rotation demand at the yielding member end is more than twice the elastic part, θ_y ;
- $\cot \vartheta = 2.5$, for elastic flexural response (zero plastic chord rotation);

with interpolation in between these values.

The shear resistance of walls for web crushing, $V_{Rd,max}$ is computed with a reduction coefficient k_c applied on the concrete compressive strength equal to:

$$k_c = 0.25 \left(\frac{30}{f_{ck}} \right)^{\frac{1}{3}} \leq 0.25 \quad (7.4-42)$$

7.4.3.6 ULS verification of joints between horizontal and vertical elements

Joints between horizontal and vertical elements (beams and columns in frames, piers and the deck in a bridge) may be verified at the ULS as plates under in-plane loading consisting of:

- n_x, n_y : mean axial stresses in the joint core from the vertical and the horizontal element;
- v_j : nominal shear stress in the joint core.

In the ULS verifications a reduction coefficient k_c is applied on the concrete compressive strength to account for the reinforcement running obliquely to the direction of compression.

$$k_c = 0.55 \left(\frac{30}{f_{ck}} \right)^{\frac{1}{3}} \leq 0.55 \quad (7.4-43)$$

Vertical reinforcement between the extreme bars of the vertical element in the plane of loading the frame counts as joint vertical reinforcement. Hoops should be placed as joint horizontal reinforcement.

7.4.3.7 SLS verifications of flexural deformations

For the verification of the SLS of deformations in seismic situations according to subsection 4.5.2.5, a is normally the chord rotation at a member end. Its value is determined through non-linear analysis according to subsection 7.4.3.2.4 or via linear elastic analysis according to subsection 7.4.3.2.2, if applicable.

- If the operational limit state in subsection 3.3.1.1 is verified, it is appropriate to use as C_d the value of the chord rotation at yielding, θ_y , of the member end of interest.
- If the immediate use limit state in subsection 3.3.1.1 is verified, an appropriate value for C_d is equal to $2\theta_y$.

Several nominal fire curves are proposed in the codes to be used in the design process for representing the action of the fire. The most often used are the ISO 834 fire curve, the ASTM E119 curve, the hydrocarbon fire and the external fire curve. All are formed of a simple relationship giving one temperature (the temperature of the gases in the compartment) as a function of time. They are thus representing a fully developed fire. For a large compartment, such a situation is not encountered before a significant amount of time has elapsed since the very beginning of the fire. This initial period of time is thus not taken into account in the calculated fire resistance whereas, as far as safety of people is concerned, this is the most important period; in fact, the only period during which evacuation from the fire compartment is possible.

The cooling down phase of the fire is generally not taken into account in the nominal fire curves. In fact, when a particular fire resistance time is required, no consideration is given to the period beyond this duration.

These relationships hardly depend on the particular characteristics of the situation for which the design is performed. The quantity of combustible material, the dimensions of the compartment and the conditions of ventilation, for example, are not taken into account.

If these characteristics have to be taken into account, it is necessary to apply fire safety engineering, where the fire risk is considered and the exposure is determined based on the actual conditions and utilization. If so, a fire curve is estimated for the actual room/structure. It should be noted that the fire safety in this case is dependent on the assumptions for this determination of the fire exposure.

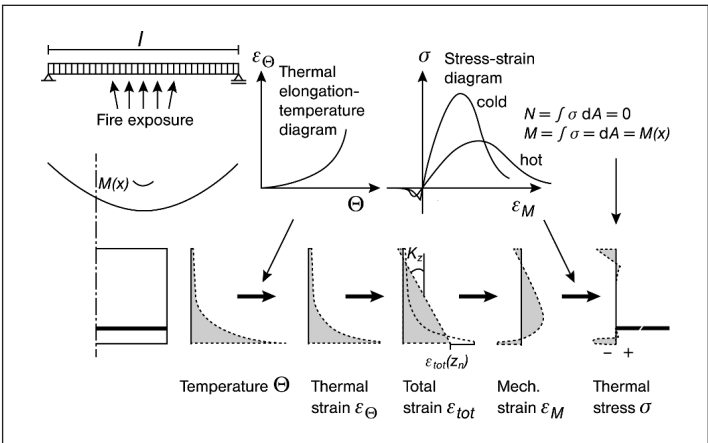


Figure 7.5-1: Graphical representation of strains and stresses in a simply supported beam exposed to the fire along the bottom face

7.5 Verification of structural safety (ULS) for extreme thermal conditions

7.5.1 Fire design

7.5.1.1 Introduction

Fire design includes consideration of a transiently changing material in interaction with the exposure: the fire, the temperatures, the thermal characteristics and the impact on the structural behaviour.

Fire design is necessary in order to achieve a suitable performance in the case of fire and to minimize the damage from a fire event.

The present chapter on fire contains the fire design principles, and the calculation methods and it outlines the design of selected important structural elements. Finally the topic of compartmentation is briefly discussed.

In the present chapter the fire risk and the fire exposures have not been specified. Even though these topics are very important, it is out of the scope of this Model Code to specify loads and exposures.

Normal structural design is based on design of components. The safety format takes the component considerations as a basis.

In the fire design of concrete structures, it is strongly recommended to consider the behaviour of the entire structure and the interaction between deformations and lost strength in one part of the structure. This may in some case lead to critical load situations in other parts of the structure not directly exposed to the fire.

7.5.1.1.1 Direct and indirect effects of fire

The most direct effect of a fire on a structure is that the temperature in the structure will increase, in the first phase, then decrease progressively as the fire decreases until extinction. During the heating phase, heat is introduced into the structure by a combination of convection from the surrounding gas and radiation (Figures 7.5-1 and 7.5-2).

During a fire the temperature is far from uniform in the concrete elements. The main reason for this is the massive dimensions of the structural elements. Because the material characteristics change as a function of temperature, the material properties at various locations in the cross-section will transiently change. This influences the cross-sectional behaviour and may result in restraint forces, internal stresses and deformations.

Indirect actions are those effects of actions that arise from restrained thermal expansion. For example, a beam that cannot freely expand longitudinally will be subjected to an increase in the axial force especially during the first minutes of the fire.

In fact, even in a single member that is completely free to expand, indirect effects do appear at the local level.

Thermal gradients in concrete slabs generate the large deflections that are required for the membrane tension effect to develop. In this case, thermal expansion has a positive effect on the stability.

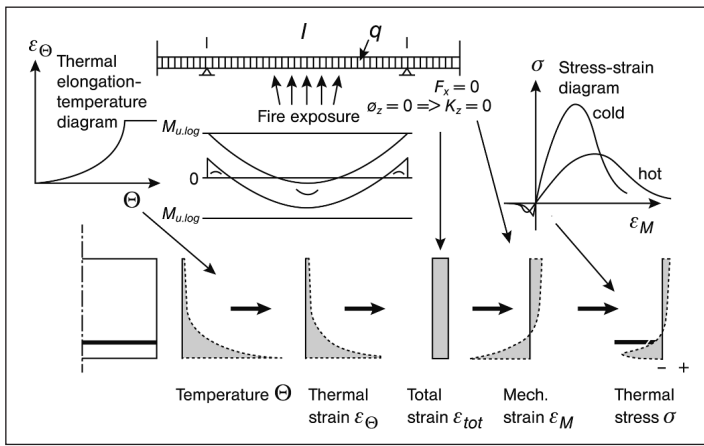


Figure 7.5-2: Graphical representation of strains and stresses in a continuously supported beam exposed to the fire along the bottom face. The indication of zero rotation and zero curvature is valid for a point in the adjacent spans

The symbols in Figure 7.5-2 are defined as:

- $M_{u,hog}$ is the moment at support;
- $M_{u,sag}$ is the mid-span moment;
- F_x is the axial force in x direction;
- κ_z is the curvature in z direction;
- φ_z is the angular rotation in z direction.

Restraint to axial expansion induces axial compression forces in beams which, depending on the position of the restraint, can have a favourable effect (restraints close to the bottom of the end sections), or an unfavourable effect (restraints close to the top of the end sections), in both simply supported and continuous beams (favourable effect = reduction of the tensile stresses and of the bending moments).

Restraint to thermal expansion increases the compressive force in columns but the effect may not be as detrimental as expected if the behaviour of the building as a whole is taken into account.

Restraint to thermal bowing dramatically modifies the bending moment diagram in continuous beams or slabs with a clear tendency to have more negative bending than under ambient conditions.

7.5.1.2 Fire design principles

7.5.1.2.1 Ultimate limit state

The fire situation is an accidental situation that requires only verifications against the ultimate limit state (as opposed to the serviceability limit state). Ultimate limit states are these states associated with structural collapse or other similar forms of structural failure such as loss of equilibrium, failure by excessive deformation, formation of a kinematic mechanism, rupture or loss of stability.

In the semiprobabilistic approach, the design against the ultimate limit state is based on the comparison between the resistance of the structure calculated with the design values of the material properties, on the one hand, and the effects of the design values of the actions on the other hand, as in Eq. (7.5-1).

$$R_{d,fi}(X_{d,fi}) \geq E_{d,fi}(F_{d,fi}) \quad (7.5-1)$$

where:

- $R_{d,fi}$ is the design value of the resistance in the case of fire;
- $X_{d,fi}$ is the design value of the material properties in the case of fire;
- $E_{d,fi}$ is the design value of effects of actions in the case of fire;
- $F_{d,fi}$ is the design value of the actions in the case of fire.

The resistance and the effects of actions are both based on characteristic values of geometrical data, usually the dimensions specified in the design, for example cross-sections. Geometrical imperfections such as a bar out of straightness or frame initial inclinations are represented by design values.

Different actions generally occur simultaneously on the structure. In an accidental situation, they have to be combined as follows:

- design values of permanent actions;
- design value of the accidental action;
- frequent or quasi-permanent value of the dominant variable action;
- quasi-permanent values of other variable actions.

The design value of the accidental action that has been mentioned previously does not appear in Eq. 7.5-1 because, in the case of fire, the fire action is not of the same form as the other actions. It does not consist of some N or some N/m^2 that could be added to the dead weight or to the wind load. The fire action consists of indirect effects of actions induced in the structure by differential and/or restrained thermal expansion.

Table 7.5-1 given here (Table A1.1 of Eurocode 0 – EN 1990, 2003, Table A1.1 in Annex A1) gives the relevant ψ factors for the fire situation in buildings.

Table 7.5-1: Coefficients for combination ψ for buildings

Action	ψ_1	ψ_2
Imposed load in buildings		
category A: domestic, residential	0.5	0.3
category B: offices	0.5	0.3
category C: congregation areas	0.7	0.6
category D: shopping	0.7	0.6
category E: storage	0.9	0.8
Traffic loads in buildings		
category F: vehicle weight ≤ 30 kN	0.7	0.6
category G: 30 kN $<$ vehicle weight < 160 kN	0.5	0.3
category H: roofs	0.0	0.0
Snow loads for $H < 1000$ m amsl	0.2	0.0
Wind loads	0.5	0.0

The rationale for using 1.0 as a partial safety factor for material properties and for the actions lies in the theory of conditional probabilities. Assuming that the probability of failure at ambient condition meets a particular target value, the following possibilities exist:

$P(\text{failure at ambient conditions}) \leq \text{Target value}$

The probability that the structure ever fails in a fire is the product of two probabilities: the probability that a severe fire occurs and the probability that this fire causes failure.

$P(\text{failure in fire condition}) = P(\text{there is a fire}) \cdot P(\text{failure caused by this fire})$

The coupled probability of having a fire and having a failure under that fire should be the same as the probability of having a failure at ambient temperature. This is why more favourable values of the partial safety factors are used in the fire situation, as well as in any accidental situation.

$P(\text{failure in fire condition}) \leq \text{Target value}$

From the above equations it follows:

$P(\text{failure caused by the fire}) \leq \text{Target value} / P(\text{there is a fire})$

An explanation of why the material factors are taken equal to 1.0 in fire, instead of 1.2 in accidental conditions, is that in many accidental conditions (e.g. earthquakes) the uncertainty is taken care of partly in the material ($\gamma_m = 1.2$) and partly in the actions. In fire, however, the uncertainty is totally taken care of in the actions, by means of very severe temperature–time curves (standard fire), or rather refined and realistic temperature–time

When it is not obvious how to determine which one among the variable actions is the dominant one, each variable action should be considered in turn as the dominant action, which leads to as many different combinations to be considered.

In the case of fire, and if the variability of the permanent action is small (i.e. in most cases), the following symbolic equations hold:

$$\sum_{j \geq 1} G_{k,j} + P + \psi_{1,1} Q_{k,1} + \sum_{i > 1} \psi_{2,i} Q_{k,i} \quad (7.5-2a)$$

$$\sum_{j \geq 1} G_{k,j} + P + \psi_{2,1} Q_{k,1} + \sum_{i > 1} \psi_{2,i} Q_{k,i} \quad (7.5-2b)$$

where:

- G_k, Q_k, P_k are the characteristic values of the permanent, variable and prestressing action, respectively;
- $G_{d,fi}, Q_{d,fi}, P_{d,fi}$ are the design values of these actions in the case of fire;
- ψ_1 is the coefficient for frequent value of a variable action;
- ψ_2 is the coefficient for quasi-permanent value of a variable action.

Eq. 7.5-2a is based on the frequent value and Eq. 7.5-2b on the quasi-permanent value for the dominant variable action.

The design values of the material properties, $X_{d,fi}$, are described by the general equation:

$$X_{d,fi} = \frac{X_k(\Theta)}{\gamma_{M,fi}} \quad (7.5-3)$$

where:

- $\gamma_{M,fi}$ is the partial safety factor for material property in fire design, normally taken as 1.0;
- Θ is the temperature.

curves (natural fires). In the specific case of concrete, it is well known that high temperature reduces the random scattering of the material properties, and so the uncertainties.

7.5.1.2.2 Concrete and steel

The present chapter concerns design of concrete structures, and the detailed description of concrete as a material is beyond the scope of this section. Therefore only some principal features are mentioned with respect to concrete material and fire design.

Concrete is in itself a versatile material, which, based on the mix-design including the admixtures, can exhibit many different characteristics, depending on the actual requirements. In the case of a fire the material properties change because of the progressive thermal damage, so the structure or the cross-section actually consists of a multitude of different concrete layers with different material properties. These properties constantly change during and after the fire.

The strength as well as the stiffness of steel and concrete are reduced by a temperature increase. The evolution of the strength and stiffness characteristics with temperature is not yet sufficiently known to describe the modification of the material characteristics, because the whole stress–strain relationships are modified. See Figures 7.5-3 and 7.5-4.

The modifications of the material properties comprise the following:

Compressive strength is the most extensively analysed property of concrete. The strength at room temperature, the water/cement ratio, the type of cement, the maximum aggregate size and the rate of heating appear to have little influence on the relative reduction, in terms of the percentage of the original strength. The type of aggregate has an influence, the decrease being less important with calcareous or lightweight aggregates compared to siliceous aggregates. The aggregate/cement ratio also has an effect, with the reduction being proportionally smaller for lean mixes than for rich mixes. Finally, the reduction highly depends on the test procedure, with more favourable results obtained when a certain compression stress level is maintained during heating, as is generally the case in reinforced concrete columns.

The modulus of elasticity of concrete is influenced in the same way by the factors previously mentioned for the compressive strength. The reduction as a function of the temperature is larger than for the compressive strength because of the simultaneous decrease of the compressive strength and increase of the strain at the peak stress.

Steady state creep is of importance essentially for service conditions, that is at temperatures below 150°C applied for very long periods, in concrete reactor vessels for example. In a fire situation, the creep rates observed under steady state conditions are very much less important than the creep values observed under transient temperature conditions.

Load-induced thermal strain is the particular deformation that occurs in concrete during first heating under load. It is influenced mainly by the aggregate type, the aggregate/cement ratio and the curing conditions; air-cured and oven-dried specimens exhibit a significantly lower transient strain than water-cured specimens.

The tensile strength of concrete has a tendency to decrease faster with the temperature than the compressive strength, but less than the elastic modulus.

The fracture energy of concrete has a tendency to keep constant or to slightly increase with temperature, but how to define and measure the fracture energy is still a highly debated issue, in spite of the many test results.

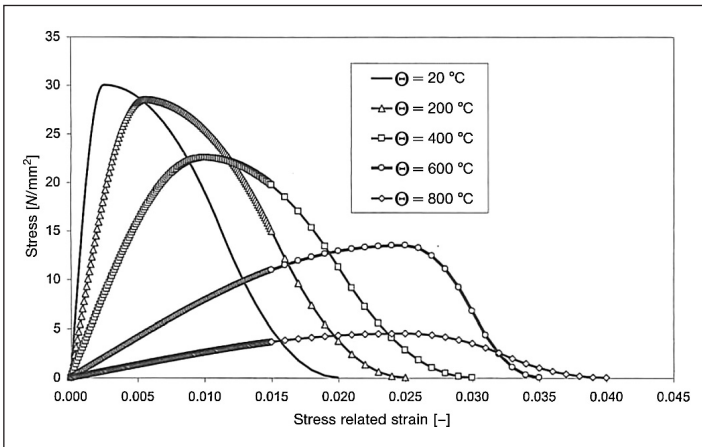


Figure 7.5-3: Example of stress–strain relationships of concrete under compression at elevated temperatures

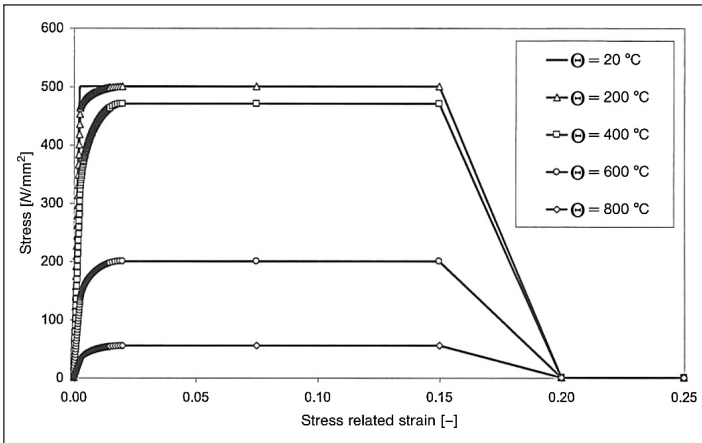


Figure 7.5-4: Stress–strain relationships of reinforcing steel at elevated temperatures

The stress–strain relation of concrete in compression reflects the modifications due to elevated temperatures of the compressive strength, of the modulus of elasticity and of the strain at the peak stress. As for the ultimate strain at 20°C, its value is defined in such a way that the M–N envelopes at the ultimate limit state obtained by means of an incremental-iterative procedure be in agreement with those obtained with the traditional approach based on strain limitations, and on the usual parabola–rectangle (or bilinear) stress–strain curve.

The thermal properties of concrete are also modified by a temperature increase. The thermal conductivity is normally reduced, whereas the specific heat is increased by an elevation of the temperature. As a result, the thermal diffusivity decreases with increasing temperatures.

The yield strength of steel is reduced by an elevation of the temperature. The relative reduction does not depend on the value of the yield strength at room temperature but it varies with the type of steel, hot-rolled (less temperature sensitive) or cold-worked (more temperature sensitive) reinforcing steel, quenched and tempered or cold-worked prestressing steel (very temperature sensitive).

The elastic modulus of steel is also reduced at elevated temperatures, somewhat faster than the yield strength.

The thermal properties of steel are also modified by a temperature elevation, with a decrease of the thermal conductivity and a slight increase of the specific heat. This is not particularly relevant in concrete structures because the amount of steel is generally so low that it hardly influences the temperature distribution.

The geometrical size of concrete and steel is modified by a temperature increase. This is the well known *thermal expansion*. The expansion is not a linearly increasing function of the temperature. The order of magnitude of the thermal expansion can reach 1% at very high temperatures in the range of 800°C. This phenomenon plays an important role in the behaviour of structures, because it induces either large displacements that may generate geometrical second order effects or indirect effects of actions if the expansion is restrained.

The bond strength between concrete and steel has been shown to decrease with temperature similar to the tensile strength rather than to the compressive strength. Experience has yet very rarely produced evidence of failures by debonding in reinforced structures. The problem is more critical in prestressed pretensioned members.

Spalling in concrete structures is a very important characteristic linked to high temperatures. Different types of behaviour are often called spalling, from the very progressive sloughing off at the surface that progressively exposes the inner part of the section and the reinforcing bars to elevated temperatures, to the explosive spalling that suddenly completely destroys the material. Extensive research activity is still going on in order to understand and mitigate this phenomenon, the problem becoming more crucial with the introduction of high strength concretes because they have a more closed and dispersed porosity, which favours the increase of vapour pressure in the pores. The factors most often mentioned as playing a negative role in spalling are: fast temperature increase, high moisture content, high compression stress level, young age, low porosity, thin members and geometrical effects (corner spalling). Some of these are related to the material itself, but it seems that structural effects also play a role in this phenomenon.

7.5.1.2.3 Heating of materials

After having defined or determined the fire exposure, and in order to be able to evaluate the modifications of the material and the structural effects, it is necessary to investigate the thermal field, which depends on the fire exposure and on the thermal properties of the material, as well as on the structural context.

The thermal properties of the material will not be discussed here.

The direct consequence of the heating of the concrete is a modification of the material characteristics such as a decrease of the strength and of the stiffness, as well as the generation of additional deformations associated with the stress level during first heating. These deformations are usually called load-induced thermal strain (LITS) or transient creep in the literature. Furthermore, thermal elongation is a direct result of the temperature increase, which is not uniformly distributed on the section. These strains and their non-uniform distribution have several effects on the behaviour of concrete members and the effect may be different depending on section type. The most important effects have to do with: spalling, elongation and deflections.

7.5.1.3 Calculation method

7.5.1.3.1 Sectional analysis

Sectional analysis is a common approach to verify the fire resistance. The bearing capacity of reinforced concrete sections subjected to a fire is usually evaluated by means of different approaches:

- by using tabulated data (first-level method);
- by using the reference-isotherm method or the zone method (second-level methods);
- by using temperature-dependent stress–strain relationships within the framework of an incremental-iterative procedure (third-level method).

In the present section focus is given to non-linear analysis. Four issues are addressed:

- the use of non-linear analysis implemented with simplified constitutive laws, as an alternative to realistic (but more complex) laws, at room temperature;
- the use of incremental-iterative procedures (“exact” method) and non-linear analysis in fire conditions;
- the validity of the well-known 500°C isotherm method in fire conditions, under an eccentric axial force;
- the relevance of the eigenstresses generated by the thermal gradients.

Tabulated data

The approach of tabulated data allows the designer to give a quick response in many practical cases with well-defined boundary conditions. On the other hand, this approach does not allow the designer to refer to materials’ properties and fire scenarios other than ordinary concrete and the standard ISO834 Fire Curve. Neither the mechanical nor the thermal aspects of the problem are explicitly addressed by this approach.

For the approach based on the use of tabulated data, see EN 1992-1-2, section 5.

For the approach based on the use of the 500°C isotherm method, see EN 1992-1-2, Annex B1. For the approach based on the zone method, see EN 1992-1-2, Annex B2.

For the approach based on the use of stress–strain, temperature-dependent laws, see EN 1992-1-2, section 3.2.2.1.

See EN 1992-1-1.

See EN 1992-1-2.

The tabulated data are based on past experience and on the theoretical evaluation of tests (Naranayan R.S. and Beeby A., “Designer’s Guide to EN1992-1-1 and EN1992-1-2”, Thomas Telford, 2005). These data provide a set of admissible values for the main geometric parameters of a section, including the cover of the reinforcement, as a function of the fire duration that the element is required to withstand.

The reference isotherm method is based on reasonable assumptions, and was originally devised for reinforced concrete sections subjected to pure bending, where the failure is generally controlled by the yielding of the tensile reinforcement. The possible extension to sections subjected to an eccentric axial force is still under discussion.

Investigations show that the larger is the fire duration, the worse is the agreement between the 500°C isotherm method and the exact method, particularly under large axial forces. Of course the two methods give almost the same results in the case of less temperature-sensitive large sections. The bearing capacity in pure compression at low values of the fire duration is slightly overestimated by the isotherm method, which on the whole is rather conservative.

For the temperature-dependent stress–strain curves, see EN 1992-1-2, section 3.2.2.1.

Reference isotherm method (500°C isotherm)

The reference isotherm method (or effective section method) is based on the assumption that concrete is fully damaged above the temperature of 500°C, while it is fully effective (fully undamaged) for temperatures below 500°C. On the contrary, the mechanical decay of the reinforcing steel is explicitly introduced.

This method can be applied within the context of non-linear analysis, by assuming the parabola–rectangle stress–strain curve at ambient temperature for the concrete, with the usual strain limitations, and by considering only the undamaged part of the concrete section.

Zone method

The zone method retains the philosophy of the 500°C isotherm method, but considers a more complex and realistic effective section, whose size depends on the temperature distribution. Also the characteristics of the concrete in the effective section (compressive strength and Young's modulus) depend on the temperature distribution. In order to perform the calculation, the section is divided into a finite number of zones. The temperature is determined in the centroid of each zone, on the basis of the thermal analysis. The method is more complex than the 500°C isotherm method, but yields better results, especially in the case of pure compression and thin webs exposed to the fire on both sides. Moreover, the method allows to consider second-order effects, by introducing proper correction coefficients.

Exact method – incremental-iterative procedure

The incremental-iterative procedure is based on the temperature-dependent stress–strain curves. At first, a thermal analysis is performed in order to determine the temperature distribution in the section, and thus the level of the thermal damage at each point, for any given fire duration. The mechanical properties of concrete and steel at each point can then be related to the maximum temperature reached locally by means of the temperature-dependent stress–strain curves. In this way, the section is considered as a composite section, consisting of many different materials, whose properties and spatial distribution are related to the thermal field.

The next step is to determine the ultimate value of the bending moment M_u for a suitable set of values N_u of the axial force. This is done by working out, for each value N_u , the corresponding moment–curvature diagram of the section, by means of an incremental iterative procedure (see also 7.5.1.3.2). Once the moment–curvature diagram is known, the maximum value of the bending moment is the ultimate bending moment M_u , corresponding to the assigned value of the axial force N_u . The couples of values (N_u, M_u) identify as many points in an M–N domain, and the interaction envelope is obtained by connecting these points.

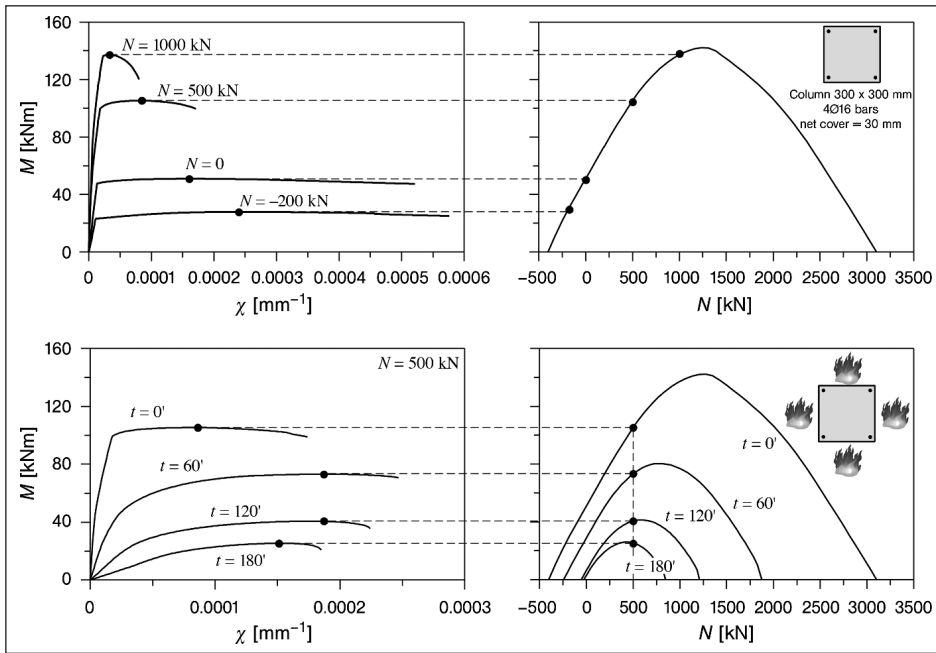


Figure 7.5-5: Examples of the application of the incremental-iterative procedure at room temperature (a); and for different values of the fire duration (b)

The application of the incremental-iterative procedure is illustrated in Figure 7.5-5.

This procedure is rather time consuming when compared with the previous methods (500°C isotherm and zone methods). Nevertheless, non-linear analysis based on strain limitations cannot be used with the stress-strain curve proposed in EN 1992-1-2, section 3.2.2.1, because the attainment of the ultimate strains in one of the two materials does not correspond – in general – to the attainment of the ultimate bearing capacity.

Note that the strain limitation method, that is commonly adopted in ambient conditions, can be extended to high temperature, only if suitable monotonic stress-strain curves are introduced (see the strain limitation method below).

The sectional capacity is underestimated if the fire curve is not used properly, that is by adopting the aforementioned incremental iterative procedure. It seems that in order to achieve a reliable result with non-linear analyses, the incremental iterative procedure will have to be followed.

Strain limitation model

The strain limitation method is based on simple monotonic stress-strain curves, that are the extension of the well-known parabola-rectangle – or similar – curve used in the design at room temperature. Each curve is valid for a given value of the temperature. After (a) the thermal analysis of the section has been performed for a given value of the fire duration; (b) the “coldest” chord of the section has been identified (coldest chord = farthest chord from the heated surface, according to the inflexion plane; the coldest chord is characterized by the least deformation capacity); (c) the distribution of the ultimate strains along this chord has been identified; and (d) a number of linear strain profiles have been plotted considered along the same coldest chord, so as to respect the ultimate strain in each point of the section, the stresses can then be worked out for each strain profile and, by integration, M_u and N_u can be evaluated. For each strain profile, there is a single couple of values for M_u and N_u . The couples M_u – N_u make it possible to draw the envelope for the given value of the fire duration.

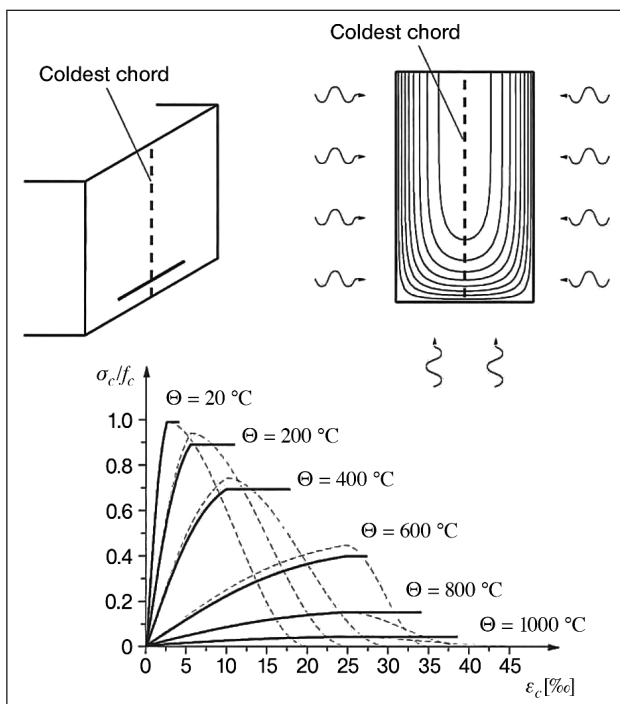


Figure 7.5-6: Strain-limitation method and qualitative parabola-rectangle diagrams

These curves can be easily worked out from the softening curves proposed by EN 1992-1-2.

Note that once the deformation capacity of the coldest chord is respected, this is also true of any other point.

For the constitutive laws, see EN 1992-1-2.

It is stated in Eurocode 2 (EN 1992-1-2, 2004) that an important question is whether load redistributions between different sections of a member in bending can be accepted in the case of fire, these redistributions being allowed by the plastic behaviour of both the reinforcement and the concrete not only at room temperature but (even more) at high temperature. One of the key conditions for this plastic behaviour is the ductility of the section, that is the capacity of the section to keep on developing the plastic bending moment, when the curvature increases to very high values. This seems to be the case according to some numerical examples that show how the ductility of a section tends to increase during a fire.

The main difference between the hot and cold situations is the ratio between the ultimate plastic moment and the first-yielding moment. This ratio is much higher in fire, which means that much higher rotations have to take place before the full plastic moment is reached. This is in no way a contradiction to what is generally observed during laboratory tests, where the failure of RC structures is often accompanied by very large displacements.

This method does not require iterative procedures and is as reliable as the exact method, provided that a suitable set of monotonic, temperature-dependent stress–strain curves is available for the concrete.

7.5.1.3.2 The role of thermal strain

Including the effect of the thermal strains requires a clear understanding of the various assumptions concerning the different strain components acting on the section, that is the total strain ε_{tot} , the thermal strain ε_{Θ} and the stress-induced strain ε_{σ} :

- the hypothesis that plane sections remain plane is still valid, but with reference to the total strain ε_{tot} ;
- the usual non-linear analysis with strain limitations can still be applied, provided that the limitations are referred to the stress-induced strain ε_{σ} .

To work out the moment–curvature diagrams for any given value N of the axial force, the following incremental iterative procedure is applied, for a number of values of the curvature χ :

- for each value of χ , a tentative deformation ε_0 is assumed in the centroid of the section, and the corresponding total deformation ε_{tot} is determined at each point of the section, assuming that plane sections remain plane;
- the corresponding stress-related strain ε_{σ} is determined at each point of the section:

$$\varepsilon_{\sigma} = \varepsilon_{tot} - \varepsilon_{\Theta}$$
- the value of the stress σ is determined at each point, by using the constitutive laws;
- the axial force N' is evaluated: if $N' = N$ (within a prefixed tolerance), the bending moment M is calculated, and a next value of χ is considered;
- if $N' \neq N$, the procedure is repeated until $N' = N$, by varying ε_0 ; then, the bending moment M is calculated, and a next value of χ is considered.

7.5.1.3.3 Plastic analyses

The theory of plasticity gives a theoretical validation to the fact that several effects leading to self-equilibrated stress distributions can be neglected in non-linear numerical analysis. Among these effects, it is worth mentioning (a) those occurring either in the construction phases or during the service life at room temperature, before the fire starts (due for instance to shrinkage, creep and thermal strains), and (b) those occurring during the fire (due to creep and thermal expansion). A consequence of neglecting these effects is that the strains, stresses and tangent moduli that are computed in any given point of a structure are only approximate – or “mean” – values compared to the “true” values that would be computed if all these effects were taken into account. The computed values are indeed based on the hypothesis of a “virgin” initial stress distribution, which is far from reality.

Neglecting these self-equilibrated stress distributions has some limitations, since it is justified only as long as the ensuing displacements are small. Hence, the effects of thermal expansion during the fire must be taken into account.

Another strain component that may affect deformations – second order effects included – in concrete structures submitted to fire is the transient-creep strain.

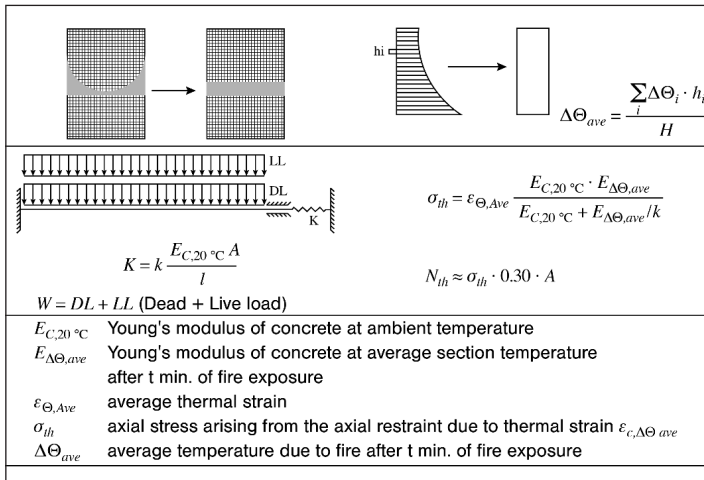


Figure 7.5-7: Evaluation of the axial force in axially restrained beams

Comments to verification of the method:

In the case of no axial restraint ($K = 0$), the ultimate bending capacity of the critical sections at any given fire duration is generally underestimated. As a result, plastic analysis underestimates the ultimate load-carrying capacity of a beam, thus leading to conservative results.

The assumed axial force in axially restrained beams results in a lower-bound estimate of the actual force (as given by non-linear analysis). Hence, the proposed method underestimates the effects of the axial restraint.

In most cases, the ultimate bending moment of fully axially restrained beams is overestimated, particularly close to the end sections. As a result, plastic analysis leads – in most cases – to a non-conservative estimate of the ultimate load-carrying capacity.

For partially restrained beams, plastic analysis leads in most cases to acceptable results, that is conservative or slightly non-conservative results.

It is observed that in plastic analysis, by completely neglecting the effects of the axial restraint, the estimate of the ultimate loadbearing capacity of a beam is always on the safe side.

Use of plastic analysis

Within the scope of fire design applications, the objective of plastic analysis is generally to evaluate the load-carrying capacity of a beam.

According to plastic analysis the verification is positive if the ultimate load at the requested fire duration is larger than the applied load, at the onset of beam collapse because of the formation of a suitable number of plastic hinges.

The plastic (e.g. ultimate) moments at critical sections may be determined according to various sectional analysis methods, such as the 500°C isotherm method or the zone method.

In axially unrestrained beams, the application of plastic analysis is straightforward and generally leads to a conservative estimate of the ultimate load.

For axially restrained beams, the ultimate bending moment depends on the axial force developed during the fire (Figure 7.5-7). Its value may be determined by performing the analysis of the entire structure. Since this procedure involves sophisticated models and heavy computing, the axial force has to be evaluated in a simpler way, based on fire effects (i.e. thermal field of the section), and on the actual axial restraints of the member in question, before performing the mechanical analysis of the section. An approximate estimate of the axial force, to be adopted in plastic analysis, can be performed via the simplified procedure outlined in the following:

- based on the results of the thermal analysis after a time t of exposure to the fire, the average temperature distribution at each level along the section is determined and the average temperature in the section is found;
- the axial force ensuing from the restrained thermal elongation, corresponding to the mean temperature, is evaluated (a) by computing the longitudinal stress σ_{th} arising in a beam of stiffness $K_{beam} = (E_{\Theta,ave} A)/L$, axially restrained by a spring of stiffness $K = k (E_{20} A)/L$, as a consequence of the average thermal elongation $\epsilon_{\Theta,ave}$; and (b) by multiplying such a stress by a suitably reduced cross-section A' . The following definitions/values should be adopted: $A' = 0.30A$, where A is the cross-section area of the beam; $E_{\Theta,ave}$ (for the evaluation of σ_{th}) is the elastic modulus corresponding to the average temperature of the section; k has to do with the number of equal spans connected to the heated beam (e.g. $k = 1.00, 0.50, 0.33$ for a two-span, three-span and four-span continuous beam, with one end span in fire); $N_{th} = \sigma_{th} A'$ applied in the geometric centroid of the section. At any temperature T , excluding creep and transient strain, the elastic modulus may be calculated by multiplying the elastic modulus in virgin conditions (E_{20}) by the normalized decay inferred from the s – e curves given in the EN 1992-1-2: $E_{\Theta} = E_{20} \cdot (E_{\Theta}/E_{20})_{EC2-Fire}$;
- the plastic (i.e. ultimate) moments of the critical sections are determined by means of either the 500°C isotherm method or the zone method considering also the axial force.

Plastic analysis is a simple and straightforward method that is very sensitive to the evaluation (a) of the plastic moments at the inner intermediate supports, and (b) of the effects of the axial restraints.

Neglecting the effects of the axial restraints always leads to a conservative estimate of the ultimate load-carrying capacity in statically redundant beams.

7.5.1.4 Structural elements

In the following sections some important conclusions concerning the fire design of structural elements are given.

For further reference and explanation, see *fib* Bulletin 46: “Fire design of concrete structures – structural behaviour and assessment” (*fib*, 2008). M–N envelopes for different cross sections and fire exposures are exemplified in Figure 7.5-8.

For the integrity requirements, see EN1991-1-2.

Not only the structural resistance (R) will have to be estimated and documented according to the fire design principles (see subsection 7.5.1.2) but also the insulation requirements (I) and the integrity requirements will have to be documented. This may require a detailed investigation of the deformations.

7.5.1.4.1 Beams and slabs

The following requirements and recommendations are the result of recent studies:

It is recommended to thoroughly consider all the sectional forces when designing beams and frames. It has been observed that axial forces in the design of reinforced concrete beams (predominantly exposed to bending moment) under fire conditions may contribute to an increased fire resistance.

Hence, neglecting the axial force in the design of reinforced concrete beams under fire conditions leads to a conservative estimate of the fire resistance.

The collapse of continuous reinforced concrete beams is generally controlled by the intermediate support sections, where concrete damage may lead to anticipated section failures.

During the fire, the bending moment can shift upwards due to the negative bending moment induced by thermal gradients. This effect will occur at an early stage in fire before the stiffness degradation takes place.

In such a situation, top reinforcement is required in the mid-span sections to avoid a premature collapse during a fire, especially in (one-way) slabs.

The shear capacity in continuous beams may become critical even though it would not be critical at ambient temperature. Especially for (one-way) slabs the shear forces may become critical as result of the fire.

For beams which are partly restrained in the axial direction, axial displacements increase rapidly in earliest phases of the fire, whereas they keep constant with rather low values after the fire-induced stiffness degradation has taken place. Also the axial force increases rapidly at the beginning of a fire and moderately later on.

Second order effects have a marginal influence on rather massive RC members subjected to bending; on the contrary, in rather thin members like one-way slabs, second-order effects may induce shear forces larger than those at ambient temperature.

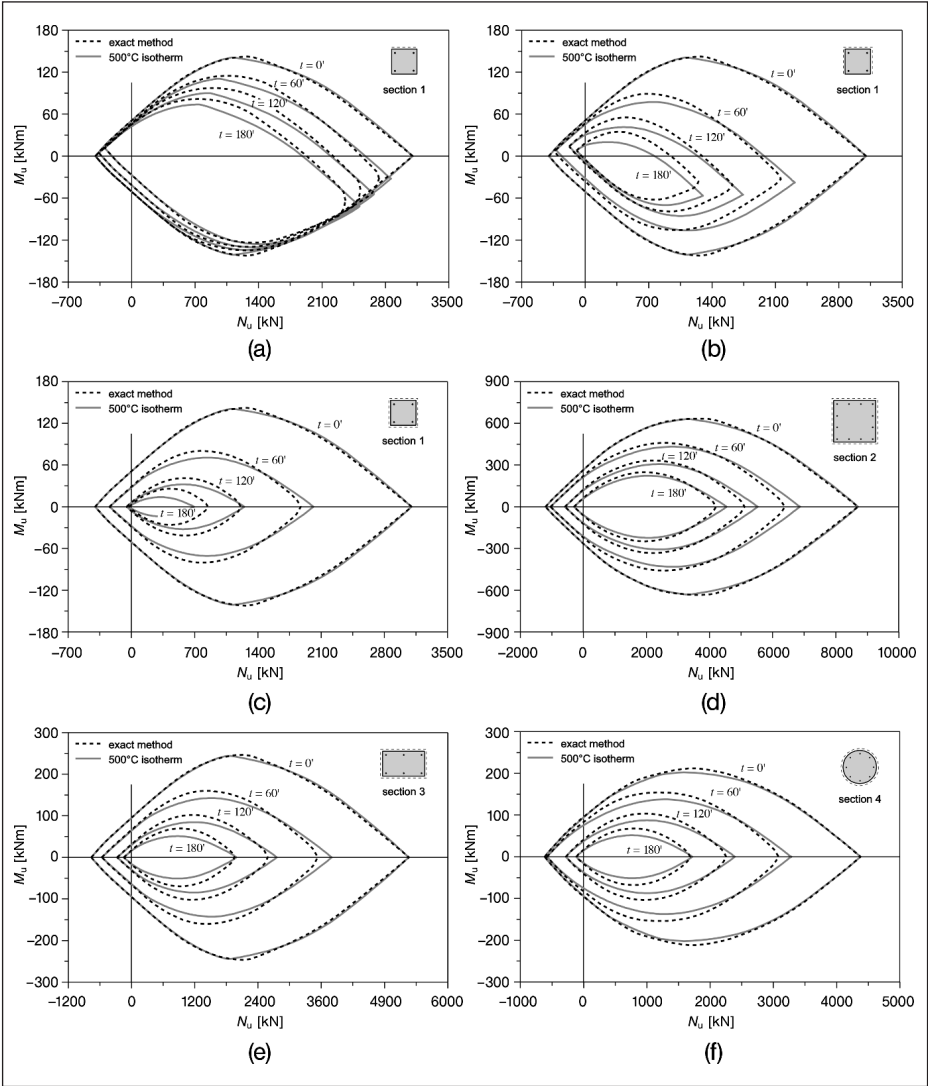


Figure 7.5-8: Example of M – N envelopes for different fire exposures (a,b,c) and for different sectional geometries (d,e,f). The heated sides of the sections are indicated with dashed lines. The M – N curves illustrate a comparison between the reference isotherm method (using 500°C) and the exact method.

7.5.1.4.2 Frames

The following requirements and recommendations are the result of recent studies.

All the sectional forces should be thoroughly considered in the design of frames.

Neglecting the effects of beam thrust in the design of RC frames may lead to highly non-conservative results, because of the increasing bending and shear in the columns during the first 30 minutes of fire exposure.

It is required to thoroughly consider and model the support and foundation conditions of the frames in the estimation of the fire resistance of the structure. The thermal effects on bending and shear depend on the type of the foundation adopted in column design, since – for instance – isolated footings provide less rotational stiffness than continuous foundation beams and even less than 2D foundations mats. Consequently, bending and shear in fire increase less in the first case.

Second-order effects have a marginal influence on rather massive reinforced concrete frames subjected to bending.

Detailing rules for the columns similar to those generally adopted in seismic design seem to be suitable also in fire design. In fact, the adoption of densely spaced closed stirrups (hoops) is instrumental in improving section strength and ductility under combined bending and axial force, and helps in controlling concrete spalling.

7.5.1.4.3 Columns

It should be observed that even if concrete above 500°C has little residual strength, it still contributes markedly to the inertia of the section, because of its peripheral location.

Reduced M–N envelopes taking into account second order effects are still hardly available in the literature, and only qualitative sketches can be found.

It may be possible – as a simplified and rough model – to evaluate the slenderness on the basis of the effective section using the reference isotherm method; in the case of normal strength concrete (NSC) the 600°C isotherm may be adopted, whereas a lower value should be chosen in the case of high performance concrete (HPC) – see also subsection 7.5.1.3.1.

For a more detailed consideration, the reduction of elasticity should also be taken into account.

Except for the slenderness considerations, the columns may be considered in accordance with the M–N envelopes discussed above.

7.5.1.5 Compartmentation

For horizontal members, failure in a standard test is assumed to have occurred when the deflection reaches a value of $L/20$ (L = clear span of the specimen) or the rate of deflection (mm/min) exceeds a prefixed value (for instance $L^2/9000d$ where d is the effective depth in mm). The rate-of-deflection criterion only applies once the deflection has reached a value of $L/30$.

The origin of the deflection limits are unclear and tests have demonstrated that the loadbearing capacity can be maintained for deflection values much larger than this limit. For concrete floor members, compartmentation failure is generally related to the insulation capacity rather than to the loadbearing capacity.

The origins of the definitions and of the values concerning the limits of deformability, integrity and insulation are unclear and may deserve to be worked over in order to reflect the situation in real fires.

However, the basic idea is that, for any given fire duration, structural members should behave properly according to the

Compartmentation has traditionally been defined according to the concept of fire resistance, with reference to collapse (R criterion), fire penetration (E criterion), and excessive heat transfer (I criterion).

The purpose of subdividing spaces into separate fire compartments is twofold. Firstly, it prevents any rapid fire spread, which could trap occupants of the building and secondly it restricts the overall size of the fire.

Continuity at the junctions of the fire-resisting members enclosing a compartment must be guaranteed. Typically, this would be the junction between a wall (either loadbearing or non-loadbearing) and a floor.

Loss of integrity is deemed to occur when gaps and fissures allow the ignition of a cotton pad on the unexposed side, or when gaps are wider than 25 mm.

Insulation failure is deemed to occur when the mean temperature exceeds by more than 140°C its initial value, or the local temperature on the unexposed face exceeds by 180°C its initial value.

original design philosophy, which means, for instance, that slabs and walls should remain almost flat during a fire.

In the design of structures for storing refrigerated liquefied gases (RLG) generally a distinction is made between “double containment” tank systems and “full containment tank systems”. Double containment tank systems consist of a closed primary and an open top secondary container. The primary container is a single containment RLG structure. Under normal operating conditions it contains both the refrigerated liquid and the associated vapours. The secondary containment is often an open top concrete wall serving two basic functions:

- protection of the primary container from external loads under normal operating conditions;
- containing the leakage from the primary container, but not the vapour generated from such leakage under accidental spill conditions.

A full containment tank systems consists of an open top primary and a closed top secondary container. Under normal operating conditions the primary tank contains the refrigerated liquid and vapour, while the secondary tank contains vapour and provides protection to the primary container. Under accidental spill conditions the secondary container contains the leakage from the primary container and contains or controls the vapour generated from such leakage.

Storage structures for refrigerated liquefied gases require taking into account different scenarios for coping with the increased risk of such structures.

A primary container must at least remain liquid tight under hydrotesting, operational loads and operational loads in combination with a specified basic seismic load. The primary container should also retain its containment capability under specified seismic loading (SSE = safe shutdown earthquake) and aftershock events (SSE_{aft} – safe shutdown aftershock event). For the secondary container, under all conditions, including but not limited to spill, SSE and SSE_{aft} plus spill events, the structural integrity of the wall should be maintained. The secondary containment must be designed for SSE_{aft} while containing the total volume of spilled product with sufficient prestressing force and reinforcement provided to prevent through-thickness cracking of concrete during the SSE_{aft} .

Details about these scenarios can be found, for example, in ACI 376 Code “Requirements for Design and Construction of Concrete

7.5.2 Cryogenic design

7.5.2.1 General

Design of structures under cryogenic circumstances is most relevant for the design of containment structures for refrigerated liquefied natural gas (LNG). Concrete is particularly well suited for the storage of cryogenic liquids because most of its properties and its behaviour improve substantially as temperature is lowered into the cryogenic range. In this section the most important properties of concrete in cryogenic conditions are treated.

7.5.2.2 Design loads to be considered in the design of structures for refrigerated liquefied gases

Design loads to be considered for the design of structures containing refrigerated liquefied gases are:

- (a) *regular loads*: dead load, product pressure and weight, prestressing forces, installation loads, construction specific loads, wind;
- (b) *incidental loads*: testing, commissioning and decommissioning loads;
- (c) *accidental loads*: seismic loads, explosion and impact, fire, environmental loads;
- (d) *imposed deformation*: loads due to thermal and moisture gradients, loads induced by shrinkage and creep, differential settlement, thermal and moisture gradient loading under spill conditions.

7.5.2.3 Failure mechanisms to be regarded in the design of structures for storing refrigerated liquefied gases

The following ULS criteria apply:

- loss of overall equilibrium;
- failure of critical regions;
- instability resulting from large deformations;
- instability resulting from plastic or creep deformation.

The following SLS criteria apply:

- cracking and spalling;
- deformations;
- corrosion of reinforcement or deterioration of concrete;
- vibrations;
- tightness.

Structures for the Containment of Refrigerated Liquefied Gases”, American Concrete Institute, Jan. 2010

Thermal shock loading will not cause any appreciable loss in strength if the relative humidity (RH) is less than a critical value of about 85%. For higher values of RH, a thermal shock can cause a significant loss of strength only if: (a) the concrete is water-saturated and/or (b) a high w/c ratio has been used.

The influence of the moisture content is demonstrated in Figure 7.5-9.

7.5.2.4 Concrete material properties under cryogenic conditions

7.5.2.4.1 Concrete compressive strength

The concrete cylinder compressive strength under a temperature Θ can be formulated as:

$$\text{for } \Theta > -120^\circ\text{C } f_c(\Theta, mc) = f_c + 12 \cdot mc \cdot \left[1 - \left(\frac{\Theta + 170}{170} \right)^2 \right] \quad (7.5-4a)$$

$$\text{for } \Theta \leq -120^\circ\text{C } f_c(mc) = f_c + 10.7 \cdot mc \quad (7.5-4b)$$

where:

Θ is the temperature in $^\circ\text{C}$;

mc is the percentage of moisture content by weight.

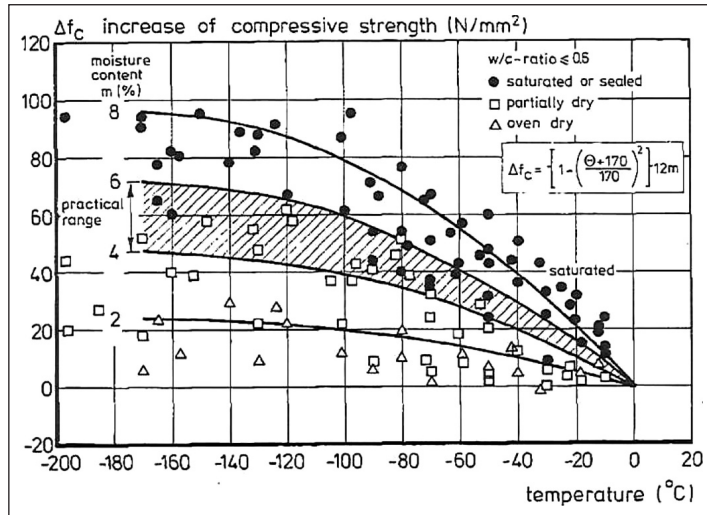


Figure 7.5-9: Increase of compressive strength as a function of temperature and moisture content (Van der Veen, “Cryogenic bond stress – strain relationship” PhD thesis TU Delft, 1990)

7.5.2.4.2 Compressive stress–strain relation

The compressive stress–strain relation is a function of the temperature and the moisture content. The strain $\epsilon_{fc}(\Theta)$ at the ultimate strength at a temperature Θ can be expressed by:

$$\epsilon_{fc}(\Theta) = \epsilon_{fc}(\Theta = 20^\circ\text{C}) + \kappa \cdot \Delta\epsilon_{fc}^{\max}(\Theta = -60^\circ\text{C}) \quad (7.5-5)$$

where:

$\Delta\epsilon_{fc}^{\max}(\Theta = -60^\circ\text{C})$ is the maximum increase in strain experienced at -60°C and the coefficient κ is defined as:

$$\text{for } \Theta > -60^\circ\text{C} \quad \kappa = 1 - \left(\frac{\Theta - 60}{60} \right)^2 \quad (7.5-6a)$$

$$\text{for } -170^\circ\text{C} < \Theta < -60^\circ\text{C} \quad \kappa = \frac{\Theta - 170}{110} \quad (7.5-6b)$$

For an overview of the mechanical properties of concrete under cryogenic conditions see Rostásy, F. S., “Assessment of mechanical properties of structural materials for cryogenic application”

In Eq. (7.5-5) it can be assumed that $\epsilon_{fc}(\Theta = 20^\circ\text{C}) = 0.2\%$ and $\Delta\epsilon_{fc}^{\max}(\Theta = -60^\circ\text{C}) = 0.1\%$ for Portland cement-based concretes and 0.15% for blast furnace cement-based concretes.

(Braunschweig, iBMB, Institut für Baustoffe, Massivbau und Brandschutz, Technical University of Braunschweig, Report H 216, ISBN 978-3-89288-200-8).

The following relationship can be used for predicting the effect of temperature on the stress–strain relation:

$$\sigma_c(\Theta) = f_c(\Theta, mc) \cdot \left[1 - \left(1 - \frac{\varepsilon_c(\Theta)}{\varepsilon_{fc}(\Theta)} \right)^n \right] \quad (7.5-7)$$

where:

$\sigma_c(\Theta)$ and $\varepsilon_c(\Theta)$ are stress and strain at a given point of the stress–strain curve;

$\varepsilon_{fc}(\Theta)$ is the strain at ultimate strength defined in Eq. (7.5-5);

n is an exponent defined in Eq. (7.5-8a, b, c):

$$\text{at } \Theta = 20^\circ\text{C} \quad n = 2 \quad (7.5-8a)$$

$$\text{for } -170^\circ\text{C} < \Theta < 20^\circ\text{C} \quad n = 1 + \frac{\Theta + 170}{170} \quad (7.5-8b)$$

$$\text{at } \Theta = -170^\circ\text{C} \quad n = 1 \quad (7.5-8c)$$

7.5.2.4.3 Splitting tensile strength

The splitting tensile strength $f_{c, split}(\Theta)$ can be derived from the cylinder compressive strength $f_c(\Theta, mc)$ according to the relation:

$$f_{c, split}(\Theta, mc) = C \cdot f_c(\Theta, mc)^{0.67} \quad (7.5-9)$$

The coefficient C varies between 0.3 and 0.56 in case of air-dried and water-saturated concrete.

7.5.2.4.4 Modulus of elasticity

Neville (1995) reported on the E modulus at a temperature of -190°C which was about 1.7 times as high as when measured at room temperature. Van der Veen (1990) reported an increase of 1.15, 1.5 and 1.8 for air-dried, sealed and water-saturated samples, respectively, exposed to -165°C .

The modulus of elasticity increases with decreasing temperature, but not as significant as the concrete strength. However, there is significant scatter.

7.5.2.4.5 Coefficient of thermal expansion

The coefficient of thermal expansion decreases with decreasing temperature. For dry concrete this decrease is about 10%. The response of concrete stored at an RH above 86% will show a sudden decrease in the coefficient of thermal expansion. This decrease is dominated by the content of free moisture. For moist concrete, at a temperature of -100°C even an expansion can occur.

7.5.2.4.6 Creep

The creep of concrete decreases with decreasing temperature. At a temperature of -30°C creep is about 50% of that measured at room temperature. However, because the elastic strains are also reduced with a reduction of temperature, the influence of temperature on the ratio between elastic and creep strains almost cancel each other out. Hence the general relationship between instantaneous (elastic) and long term strain values still remains the same as at room temperature.

7.5.2.4.7 Bond and crack width control

For background information about bond and crack width control, see Van der Veen, “Cryogenic bond stress–strain relationship” (PhD thesis TU Delft, 1990).

The bond properties improve at decreasing temperature. Crack width control, carried out for room temperature conditions, gives a reasonable, slightly conservative, approach for the determination of crack widths in structures at cryogenic conditions.

Extended background information on verification of serviceability can be found in Balázs et al. (2013), Design for SLS according to *fib* Model Code 2010. Structural Concrete, 14: 99–123. doi: 10.1002/suco.201200060. The serviceability limit states are listed in subsection 3.3.1.

The verification of SLS is performed under service load conditions and the operational failure probability to exceed the limit state is about a thousand times higher than that of ULS (see Table 3.3-5). However, if the SLS criteria are exceeded, this does not yet mean that the structure should be rejected.

Exceeding the limit state of stresses or limit state of cracking may lead to limited local structural damage mainly affecting the durability of the structure, its tightness or its appearance.

Excessive deformations may produce damage in non-structural elements or loadbearing walls and can affect the efficient use or appearance of structural or non-structural elements.

Vibrations may cause discomfort, alarm or loss of functionality.

For special problems, other suitable combinations may be agreed by client and designer.

In general for the limit state of deformations, the mean value of the prestressing force at the time considered is appropriate, while for the limit state of cracking the upper or lower value (P_{sup} or P_{inf}) according to subsection 4.5.1.4.2 is suitable.

If the value of the prestressing force is known (e.g. from site measurement) the mean value of the prestressing force can be also used for the crack width analysis.

Linear or non-linear methods may be used. For most SLS problems linear analysis is sufficient. If, however, a non-linear analysis is carried out for ULS, the action effects under service loads may be calculated by the same model. Plastic analysis is unsuitable for SLS calculations.

In selecting appropriate stress limits, the effect of the absolute dimensions of the member should be taken into account. Lower limits will be appropriate for larger members due to size-effects.

Further information is given in subsection 7.6.4.

7.6 Verification of serviceability (SLS) of RC and PC structures

7.6.1 Requirements

It should be demonstrated that the structure and the structural elements perform adequately in normal use. To meet this requirement the serviceability limit states (SLS) should be verified.

Depending on the type and function of a structure or a structural element the verification of different serviceability limit states may be relevant, such as the limitation of:

- stresses (see subsection 7.6.3);
- crack widths (see subsection 7.6.4);
- deformations (see subsection 7.6.5);
- vibrations (see subsection 7.6.6).

7.6.2 Design criteria

For SLS verifications the partial safety coefficients are generally taken equal to 1.0.

The combination of loads to be considered depends on the type of SLS and on the specific problem. It is suitable to utilize one of the combinations given in subsection 4.5.2.5 under representative values of variable actions, being:

- characteristic value;
- combination value;
- frequent value;
- quasi-permanent value.

The relevant values of the prestressing force depend on the type of SLS and the problem considered. Prestressing force values to be considered are suggested in subsection 4.5.2.5.

For structural analysis any appropriate method may be used, which takes account of the material behaviour under service loads.

7.6.3 Stress limitation

Under service load conditions, limitation of stresses may be required for

- tensile stresses in the concrete;
- compressive stresses in the concrete;
- tensile stresses in the steel.

The limitation of tensile stresses in the concrete is an adequate measure to reduce the probability of cracking.

The limitation of compressive stresses in concrete aims at avoiding excessive compression, producing irreversible strains and longitudinal cracks (parallel to the compressive strains).

Stresses are calculated using section properties corresponding to either the uncracked or the fully cracked condition, whichever is appropriate.

Where the maximum tensile stress in the concrete calculated on the basis of an uncracked section under the characteristic combination of loads exceeds f_{ctm} (see Table 7.2-1), the cracked state should be assumed.

Where the section is assumed to be uncracked, the whole concrete section is considered to be active and both concrete and steel are assumed to be elastic, both in tension and compression.

In specific cases, for example in incremental launching with precast elements, a minimum compressive stress may be required.

As a rule, the limit state of decompression should be required, if cracking or reopening of cracks has to be avoided under a given load combination.

In a beam, the state of decompression is reached when the section under consideration is compressed and the extreme fibre concrete stress is equal to zero.

The occurrence of longitudinal cracks may lead to a reduction in durability. In the absence of other measures (such as an increase of concrete cover) it is recommended to limit the compressive stress for exposure classes XD, XS and XF (subsection 4.7.2). However, no limitation in serviceability conditions is necessary for stresses under bearings and anchorages through mechanical devices (e.g. anchor plate of prestressing tendons).

The limit of $0.6f_{ck}(t)$ is not a sharply defined value. Consequently, in the corresponding verification, the prestressing force may be represented by its mean value, and in transient situations where the magnitude of variable actions is small (especially at transfer of prestressing forces in beams) the quasi-permanent combination of actions may be replaced by the characteristic combination. On the other hand, the prestressing force and concrete strength should be considered in the verification with their values at the time at which the maximum stresses are reached.

Measures should be envisaged for deformations if the span/effective depth ratio exceeds 85% of the value given in Table 7.6-6.

If creep is likely to significantly affect the behaviour of the member considered (e.g. with regard to loss of prestress, deformation, validity of the structural analysis) an alternative measure would be to limit the stress to $0.4f_{ck}(t)$. However, the limitation may be taken as a value between $0.4f_{ck}(t)$ and $0.6f_{ck}(t)$ for verifications relating to a transient situation (e.g. during construction) depending on the duration of the loading.

Creep effects in a cracked cross-section may be taken into account by assuming a modular ratio of 15 for situations where more than 50% of the stress arises from quasi-permanent actions. Otherwise, they may be ignored.

Tensile stresses in reinforcement should be limited with an appropriate safety margin below the yield strength, preventing uncontrolled cracking.

In calculating the stress, account must be taken of whether the section is expected to crack under service loads. Moreover, the effects of creep, shrinkage, relaxation of prestressing steel and differential temperatures should be considered.

7.6.3.1 Tensile stresses in the concrete

Depending on the limit state considered, various stress limitations may apply. The limit state of decompression is the most relevant. Stresses may be calculated on the basis of a homogeneous uncracked concrete section. The contribution of reinforcement to the area and the section modulus of the cross-section may be taken into account.

7.6.3.2 Limit state of decompression

The limit state of decompression is defined as the state where concrete stresses are below or equal to zero in all principal directions.

7.6.3.3 Compressive stresses in the concrete

Excessive compressive stress in the concrete under service load may lead to longitudinal cracks and high and hardly predictable creep, with serious consequences to prestress losses. When such effects are likely to occur, measures should be taken to limit the stresses to an appropriate level.

If the stress does not exceed $0.6f_{ck}(t)$:

- under the characteristic combination, longitudinal cracking is unlikely to occur;
- under the quasi-permanent combination of actions, creep and the corresponding prestress losses can be predicted with adequate accuracy.

If under the quasi-permanent combination of actions the stress exceeds $0.4f_{ck}(t)$, non-linearity should be considered for the assessment of creep deformation (see subsection 5.1.9.4.3 (d)).

7.6.3.4 Steel stresses

Tensile stresses in the steel under serviceability conditions, which could lead to inelastic deformations of the steel should be avoided as this will lead to large, permanently open cracks. The allowed stress in steel is in accordance with subsection 5.4.4.2

This requirement will be met provided that, under the characteristic combination of loads, the tensile stress in the ordinary reinforcement does not exceed $0.8f_{yk}$. Where the stress is due entirely to imposed deformations, a stress of $1.0f_{yk}$ is acceptable.

Stress verifications should be carried out for fatigue effects in partially prestressed members.

Cracks may be due to other causes such as plastic shrinkage or chemical reactions accompanied by expansion of the hardened concrete. The avoidance and the control of the width of such cracks is not covered by this chapter.

It should be noted for the determination of the crack width in practice that the phenomenon of cracking is of highly probabilistic nature. Calculated crack widths (w_d) are nominal values for comparison with nominal limit values (see definition of w_d and w_{lim} in 7.6.4.3). The actual crack widths observed on the real structure may differ from these nominal calculated values. Therefore the comparison of calculated crack widths with nominal crack widths limits may only serve as an approximate means of satisfying the design criteria. High accuracy may not be expected.

Loss of functionality occurs, for instance, in liquid retaining or containment structures, where leakage can occur if the crack widths are too large.

For containment structures a concrete compression zone with a particular minimum height can be specified, or a lining can be provided in more demanding cases.

The durability of a structure is predominantly governed by the thickness of the concrete cover and the quality of the concrete if the crack widths are limited to the maximum characteristic crack widths given in subsection 7.6.4.3.

Limit values of crack widths should be agreed with the owner.

If the cover is larger than required for the environmental conditions concerned, those practical limit values for the crack width can even be enlarged proportionally.

In some particular cases more severe demands should be formulated in agreement with the owner, such as if de-icing agents are frequently used on top of tension zones in reinforced or partially prestressed structures.

The concrete may have a higher strength than was ordered. A higher tensile strength of the concrete may lead to larger crack widths. Another uncertainty is the development of the tensile strength in time due to the continued hydration of the cement, in combination with the loading scenario.

7.6.4 Limit state of cracking

7.6.4.1 Requirements

It should be ensured that, with adequate probability, cracks will not impair the serviceability and durability of the structure.

Cracks do not, per se, indicate a lack of serviceability or durability; in reinforced concrete structures, cracking due to tension, bending, shear, torsion (resulting from either direct loading or restraint of imposed deformations) is often inevitable and does not necessarily impair serviceability or durability.

Design crack widths can be specified to satisfy requirements concerning functionality (subsection 7.6.4.1.1), durability (subsection 7.6.4.1.2) or appearance (subsection 7.6.4.1.3).

7.6.4.1.1 Functional requirements

The functionality of the structure should not be impaired by the cracks formed.

In relevant cases, nominal crack widths (w_{lim}) may be agreed with the client.

7.6.4.1.2 Durability

The durability of the structure during its intended lifetime should not be impaired by the cracks formed.

7.6.4.1.3 Appearance of the structure

The appearance of the structure should not be unacceptable because of cracking.

7.6.4.1.4 Uncertainties

Uncertainties related to the actual local concrete tensile strength, as well as to unforeseen tensile stresses, should be appropriately considered in design and construction.

At all sections which are expected to be subjected to significant tension (due to imposed or restrained deformation, combined or not combined with direct loading), a minimum amount of reinforcement should be placed, to ensure that the reinforcement does not yield immediately after cracking in the SLS.

This applies also to prestressed members in regions where tension is expected to develop in the concrete in the SLS.

Different combinations of actions may be considered under particular conditions.

- (a) For reinforced concrete members, for exposure classes XC, XD, XF, XS (as specified in subsection 4.7.2), a $w_{lim} = 0.30$ mm may be assumed satisfactory under the quasi-permanent combination of actions with respect to both appearance and durability, in the absence of specific requirements (e.g. water tightness). For exposure class X0, this limit may be relaxed provided that it is not necessary for reasons other than durability. When de-icing agents are expected to be used on top of tensioned zones of reinforced concrete elements, appropriate w_{lim} values should be specified in accordance with the client, depending on the thickness and quality of the concrete, and of additional protective layers.
- (b) For prestressed members, if more detailed data are not available, the crack width limit values presented in Table 7.6-1 may be used.

Table 7.6-1: Crack width limits (mm) for reinforced members and for prestressed members with bonded prestressing steel

	RC	PL1	PL2	PL3
X0	0.3	0.2	0.3	0.3
XC	0.3	0.2	0.3	0.3
XD	0.2	$\sigma < 0$ *	0.2	0.2
XS	0.2	$\sigma < 0$ *	0.2	0.2
XF	0.2	$\sigma < 0$ *	0.2	0.2

For PL1, PL2 and PL3 see subsection 5.4.3.3.

* Stress in concrete at the level of prestressed reinforcement

For structural members subjected to exposure classes XD, XF, XS and bending, no tension is allowed in a distance of 100 mm from the duct or the perimeter of the prestressed reinforcement towards the tension face. This value may be reduced to 25 mm in cases where the overall height of the member is below 400 mm.

With regard to crack width limitation for fluid-tightness, different requirements may apply for various cases (0.15–0.25 mm). Special recommendations should be used. The requirement generally depends on the consequences of leakage and the pressure of the fluid:

- if leakage should be limited to a small amount, and some surface staining is acceptable, $w_{lim} = 0.20$ mm may be used as a limit for self-healing cracks. Otherwise 0.1 mm may be more appropriate;

7.6.4.2 Design criteria versus cracking

- (a) The specific requirements of subsections 7.6.4.1.1 to 7.6.4.1.4 may be met by an appropriate limitation of crack widths. This may be achieved either by means of analytical procedures (subsection 7.6.4.4 and 7.6.4.5) or by appropriate practical rules (subsection 7.6.4.6).
- (b) The rules given in this chapter only apply if a minimum reinforcement is provided according to section 7.13 if stress limitation requirements according to subsection 7.6.3 are fulfilled.

7.6.4.3 Limitation of crack width

In order to meet the demands with regard to functionality, durability and/or appearance, the crack width has to satisfy the following condition:

$$w_d \leq w_{lim} \quad (7.6-1)$$

where:

w_d denotes the design crack width calculated as in subsection 7.6.4.4.1 under the appropriate combination of actions (subsection 4.5.2.5) considered at the concrete surface;

w_{lim} denotes the nominal limit value of crack width considered at the concrete surface, which is specified for cases of expected functional, appearance-related or in some particular cases durability-related consequences of cracking. The nominal limit value of crack width corresponds to the nominal concrete cover c_{nom} according to subsection 7.13.2.2.

increase. When enough cracks have been formed to ensure that no undisturbed areas (white areas in Figure 7.6-1) are left, the tensile strength of the concrete cannot be reached anymore between the cracks, so that no new cracks can appear. This is the start of the stabilized cracking stage (3). In this stage no new cracks are formed but existing cracks widen. Finally, the steel will start yielding at stage (5).

For the model, as a simplification line (2) has been assumed to be horizontal, corresponding to a constant tensile cracking force:

$$N_r = A_{c,ef} \cdot f_{ctm}(1 + \alpha_e \rho_{s,ef}).$$

In reality this line will not be horizontal but inclined, ranging from the first crack for $f_{ctk,0.05}$ to an upper value of $f_{ctk,0.95}$ (sawtooth line in Figure 7.6-1 right).

The approach with the horizontal line (2) is regarded as accurate enough, considering the influence of a number of uncertainties, such as the accuracy of placement of reinforcement, the real effective tensile strength and the influence of construction quality.

The transmission of forces in a disturbed area adjacent to a crack is shown in Figure 7.6-3. Again the relations are simplified by linearization.

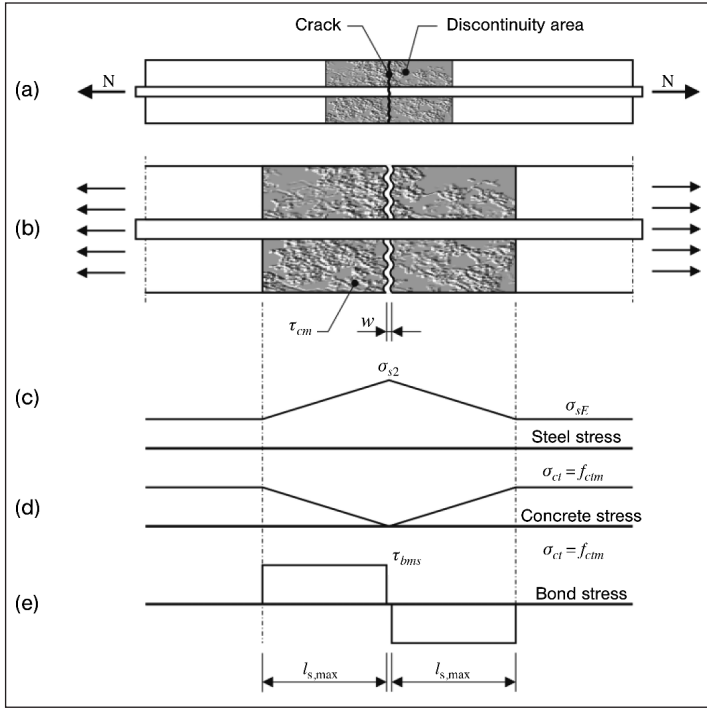


Figure 7.6-3: Steel, concrete and bond stresses in disturbed area in the crack formation stage (simplified representation)

- (a) centrally reinforced tensile member with crack;
 (b) discontinuity area;
 (c) steel stress development in discontinuity area;
 (d) concrete stress development in discontinuity area;
 (e) development of bond stress in discontinuity area

The formulation given for the value of the crack width (Eqs. (7.6-3)–(7.6-5)) provides an estimate of the surface crack width for members subjected to pure tension. For members subjected to bending, the values represent the crack width at the level of the reinforcement. In the latter case crack spacing and crack width will generally be larger at the extreme tensile fibre. In order to estimate the value of the crack width at the extreme tensile fibre, the crack width may be multiplied with a factor $(h - x)/(d - x)$.

Equation 7.6-4 is valid for structures, where the concrete cover is not larger than 75 mm. For a larger concrete cover a more detailed analysis is required. Procedures based on the fracture mechanics

For all stages of cracking, the design crack width w_d may be calculated by:

$$w_d = 2l_{s,max}(\varepsilon_{sm} - \varepsilon_{cm} - \varepsilon_{cs}) \quad (7.6-3)$$

where:

$l_{s,max}$ denotes the length over which slip between concrete and steel occurs. The steel and concrete strains, which occur within this length, contribute to the width of the crack;

$l_{s,max}$ is calculated with Eq. (7.6-4);

ε_{sm} is the average steel strain over the length $l_{s,max}$;

ε_{cm} is the average concrete strain over the length $l_{s,max}$;

ε_{cs} is the strain of the concrete due to (free) shrinkage.

For the length $l_{s,max}$ the following expression applies:

$$l_{s,max} = k \cdot c + \frac{1}{4} \cdot \frac{f_{ctm}}{\tau_{bms}} \cdot \frac{\varphi_s}{\rho_{s,ef}} \quad (7.6-4)$$

where:

k is an empirical parameter to take the influence of the concrete cover into consideration; as a simplification, $k = 1.0$ can be assumed;

c is the concrete cover;

τ_{bm} is mean bond strength between steel and concrete (Table 7.6-2).

The relative mean strain in Eq. (7.6-3) follows from:

$$\varepsilon_{sm} - \varepsilon_{cm} - \varepsilon_{cs} = \frac{\sigma_s - \beta \cdot \sigma_{sr}}{E_s} - \eta_r \cdot \varepsilon_{sh} \quad (7.6-5)$$

where:

σ_s is the steel stress in a crack;

σ_{sr} is the maximum steel stress in a crack in the crack formation stage, which, for pure tension, is:

$$\sigma_{sr} = \frac{f_{ctm}}{\rho_{s,ef}} (1 + \alpha_e \rho_{s,ef}) \quad (7.6-6)$$

approach would be appropriate. According to the contemporary knowledge $k = 1$ may be used in Eq. 7.6-4.

In the absence of a more refined model Figure 7.6-4 may be used to assess the effective concrete area in tension.

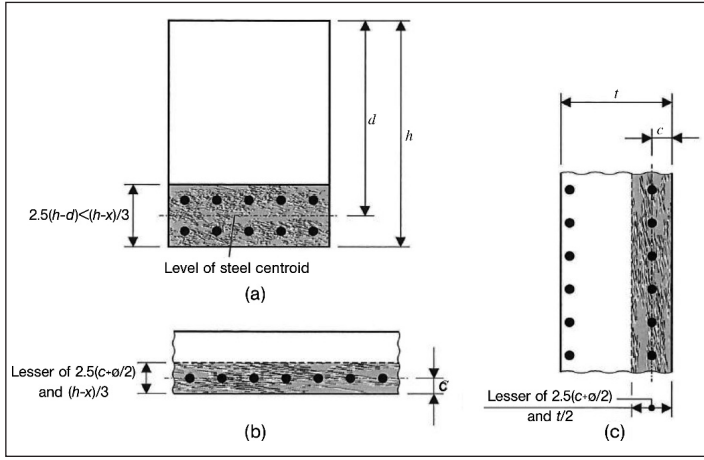


Figure 7.6-4: Effective tension area of concrete $A_{c,ef}$ for: (a) beam; (b) slab; (c) wall in tension (shaded areas)

where:

$$\rho_{s,ef} = \frac{A_s}{A_{c,ef}}$$

with $A_{c,ef}$ = effective area of concrete in tension (Figure 7.6-4);

α_e is the modular ratio = $\frac{E_s}{E_c}$;

β is an empirical coefficient to assess the mean strain over $l_{s,max}$ depending on the type of loading (Table 7.6-2);

η_r is a coefficient for considering the shrinkage contribution;

ϵ_{sh} is the shrinkage strain.

The value for τ_{bms} and the coefficients β and η_r can be taken from Table 7.6-2.

Table 7.6-2: Values for τ_{bms} , β and η_r for deformed reinforcing bars

	Crack formation stage	Stabilized cracking stage
Short term, instantaneous loading	$\tau_{bms} = 1.8 \cdot f_{ctm}(t)$ $\beta = 0.6$ $\eta_r = 0$	$\tau_{bms} = 1.8 \cdot f_{ctm}(t)$ $\beta = 0.6$ $\eta_r = 0$
Long term, repeated loading	$\tau_{bms} = 1.35 \cdot f_{ctm}(t)$ $\beta = 0.6$ $\eta_r = 0$	$\tau_{bms} = 1.8 \cdot f_{ctm}(t)$ $\beta = 0.4$ $\eta_r = 1$

For the stabilized cracking stage, if different bar diameters are used in the tensile area, the value $\bar{\sigma}_s$ in Eq. 7.6-4 may be replaced by $\bar{\sigma}_{eq}$ according to:

$$\bar{\sigma}_{eq} = \frac{\sum n_{s,i} \bar{\sigma}_{s,i}^2}{\sum n_{s,i} \bar{\sigma}_{s,i}} \quad (7.6-7)$$

where:

$n_{s,i}$ is the number of bars with diameter $\bar{\sigma}_{s,i}$.

The effective area of concrete in tension $A_{c,ef}$ accounts for the non-uniform normal stress distribution by bond forces in the concrete cross-section at the end of the transmission length.

With the method given in this section, the design crack width within the effective area of concrete in tension is controlled. Outside this area, larger cracks may occur; this can be avoided by providing crack distribution reinforcement (Figure 7.6-5).

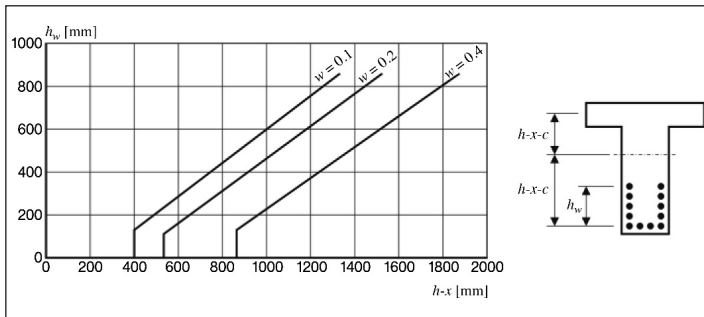


Figure 7.6-5: The height of the web that should be provided with crack distribution reinforcement in order to prevent wide cracks outside the effective tension area (illustrative diagram)

When a more refined model is not available, the following expression for $l_{s,max}$ may be used:

7.6.4.4.2 Combined effects of load and imposed deformations

The effect of cracking may be considered for the analysis of combined effects of loads and imposed deformations. Hence, where cracking is due to imposed deformations and loads, the steel stresses at the cracks due to loads as well as imposed deformations should be taken into account.

7.6.4.4.3 Orthogonal reinforcement directions

If the cracks in a member reinforced in two orthogonal directions are expected to form at an angle which differs substantially ($>15^\circ$)

$$l_{s,max,\theta} = \left(\frac{\cos \theta}{l_{sx,k}} + \frac{\sin \theta}{l_{sy,k}} \right)^{-1} \quad (7.6-8)$$

where:

θ denotes the angle between the reinforcement in the x direction and the direction of the principal tensile stress;
 $l_{sx,k}$, $l_{sy,k}$ denote the slip lengths in the two orthogonal directions, calculated according to Eq. (7.6-4).

The design crack width can then be calculated from:

$$w_d = 2 \cdot l_{s,max,\theta} (\varepsilon_{\perp} - \varepsilon_{c,\perp}) \quad (7.6-9)$$

where:

$\varepsilon_{\perp}$ and $\varepsilon_{c,\perp}$ represent the mean strain and the mean concrete strain, evaluated in the direction orthogonal to the crack (Figure 7.6-6).

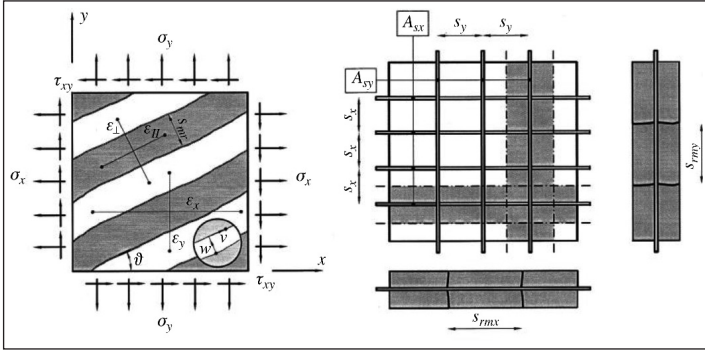


Figure 7.6-6: Basis for calculation of crack width for reinforcement deviating from the direction orthogonal to the crack

When prestressed and non-prestressed types of steel are simultaneously used, since the bond behaviour of prestressing tendons is different from the bond behaviour of deformed reinforcing bars, different steel stresses will be developed in each type of steel. For the calculation of the equivalent effective reinforcing ratio $\rho_{s+p,ef}$ both equilibrium and compatibility are respected, Figures 7.6-7 and 7.6-8.

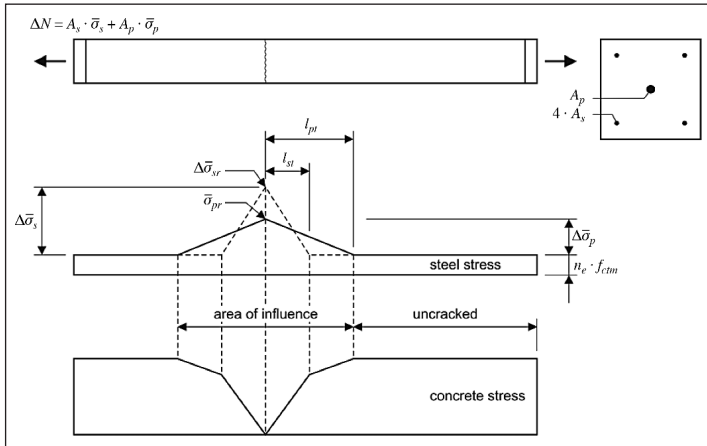


Figure 7.6-7: Crack formation stage: development of the steel and concrete stresses beyond decompression for a combination of reinforcing steel and prestressing steel

from the direction of the reinforcement, the approximation by Eqs. (7.6-8) and (7.6-9) may be used to calculate $l_{s,max}$ and w_d .

7.6.4.5 Calculation of crack width in prestressed concrete members

The calculation of crack width concerns structural members with bonded prestressing reinforcement.

In general, the calculation of crack widths for prestressed structural members follows the procedure and the formulas given for reinforced concrete members in subsection 7.6.4.4.

For the general case of combined reinforcement consisting of reinforcing steel and prestressing steel, the prestressing steel is replaced by an equivalent cross-sectional area of reinforcing steel, which takes the lower bond quality of the prestressing steel into account. So, Eq. (7.6-3) is extended to the more general case of reinforcing and prestressing steel by replacing $\rho_{s,ef}$ by $\rho_{s+p,ef}$, where:

$$\rho_{s+p,ef} = \frac{A_s + \xi_1^2 \cdot A_p}{A_{c,ef}} = \rho_s + \xi_1^2 \cdot \rho_p \quad (7.6-10)$$

where ξ is a bond factor according to:

$$\xi_1 = \sqrt{\frac{\tau_{bmp}}{\tau_{bms}} \cdot \frac{\phi_s}{\phi_{p,eq}}} \quad (7.6-11)$$

and the equivalent diameter $\phi_{p,eq}$ of the prestressing steel is given as:

$$\phi_{p,ef} = \frac{4 \cdot A_p}{u_p}$$

where:

$A_p = \sum A_{p,i}$ is the total actual area of the prestressing steel;
 $u_p = \sum u_{p,i}$ is the total equivalent perimeter of the prestressing steel;
 where:

$$\begin{aligned} u_{p,i} &= 1.6 \cdot \pi \cdot \sqrt{A_{p,i}} \quad \text{for bundles;} \\ u_{p,i} &= 1.75 \cdot \pi \cdot \phi_{wire} \quad \text{for 7-wire strands;} \\ u_{p,i} &= 1.20 \cdot \pi \cdot \phi_{wire} \quad \text{for 3-wire strands;} \end{aligned}$$

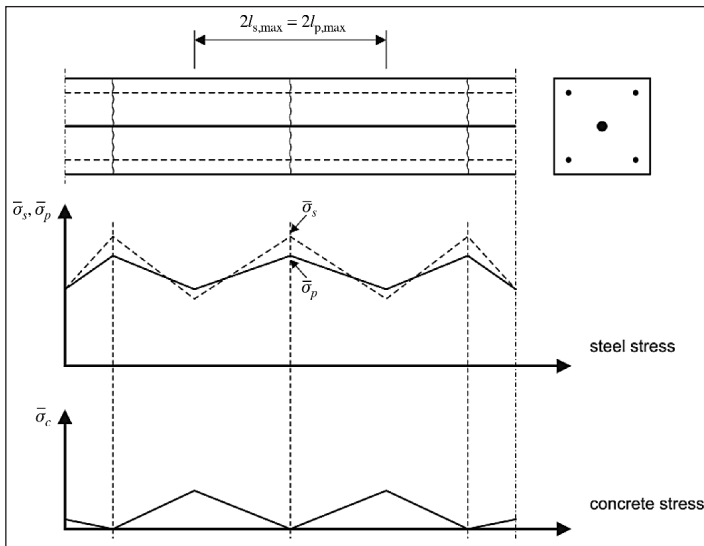


Figure 7.6-8: Stabilized cracking stage: development of the steel and concrete stresses beyond decompression for a combination of reinforcing steel and prestressing steel

For reinforced or prestressed slabs subjected to bending without significant axial tension, no special measures to control cracking are needed, provided that the overall depth of the slab does not exceed 200 mm.

If the values given in Tables 7.6-4 and 7.6-5 are respected, crack widths do generally not exceed 0.30 mm for reinforced elements and 0.20 mm for prestressed elements.

- For cracking caused mainly by restraint (crack formation stage), crack widths will generally not exceed the values stated above, provided the bar sizes given in Table 7.6-4 are not exceeded; the σ_s values referred to in Table 7.6-4 have to be calculated at cracking of the element (σ_{sr}).
- For cracks caused mainly by imposed loads (stabilized cracking stage), crack widths will generally not exceed the values stated above, provided either the provisions of Table 7.6-4 or those of Table 7.6-5 are satisfied.

Table 7.6-4: Maximum bar diameters (deformed bars) for crack width control without calculation for reinforced sections.

Steel stress σ_s (MPa)	Maximum bar diameter (mm)
160	40
200	30
240	20
280	14
320	10
360	8
400	6
450	4

Note: The steel stresses are calculated under the quasi-permanent loads (reinforced concrete). The values in Table 7.6-4 have been obtained under the following assumptions:

- The reinforcement provided is the minimum amount of reinforcement in pure bending. Larger reinforcement values will provide larger values for the maximum bar diameter.
- (h – d) is taken as approximately 0.10 h.
- In order to evaluate the reduction in tensile strength due to self-equilibrating stresses, a height of 400 mm has been assumed

where ϕ_{wire} is the single wire diameter in the strands.

The maximum transmission length is given by Eq. (7.6-4) replacing $\rho_{s,ef}$ by $\rho_{s+p,ef}$.

The relative mean strain in the cracked section beyond decompression is given by Eq. (7.6-5) replacing $\rho_{s,ef}$ by $\rho_{s+p,ef}$.

In the absence of more detailed data, approximate values for τ_{bmp}/τ_{bms} can be taken from Table 7.6-3.

Table 7.6-3: Bond factors τ_{bmp}/τ_{bms} for different types of prestressing steel

Surface condition	Pretensioned steel	Post-tensioned steel	No bond
Smooth wire	0.40	0.20	0
Strand	0.60	0.40	0
Indented wire	0.80	0.60	0
Ribbed bar	—	1.00	0

7.6.4.6 Control of cracking without calculation

Under well specified conditions, the fulfilment of the requirements in subsections 7.6.4.1, 7.6.4.2 and 7.6.4.3 may also be achieved by means of appropriate practical rules:

- when small depth elements subjected mainly to bending are considered, no special measures are needed for crack control;
- under the condition that the minimum reinforcement specified in section 7.13 (minimum reinforcement) is provided, the design crack width may be kept to acceptably low values, if appropriately chosen bar diameters and bar spacings are used.

Further guidance concerning the choice of the bar diameter is given in section 7.13.

As a simplification for prestressed concrete sections, the stress increase of the prestressing steel – that is the contribution of tendons to the limitation of crack widths – may be disregarded.

For reinforced concrete with an average concrete tensile strength other than $f_{ctm} = 2.9 \text{ N/mm}^2$, the maximum bar diameter according to Table 7.6-4 may be modified as follows:

– for restraint cracking (crack formation stage):

$$\phi = \phi_{s,max} \frac{f_{ctm}(t)}{2.9}$$

(7.6-12)

– for load induced cracking (stabilized cracking stage):

$$\phi_s = \phi_{s,max} \frac{h_t}{7.5(h-d)} > \phi_{s,max}$$

(7.6-13)

where:

- ϕ_s is the adjusted maximum bar diameter;
- $\phi_{s,max}$ is the maximum bar diameter given in the Table 7.6-4;
- h is the overall depth of the section;
- h_t is the depth of the tension zone just before cracking;
- d is the effective depth of the cross-section;
- $f_{ctm}(t)$ is the mean value of the concrete tensile strength at the time t when the first crack appears.

Table 7.6-5: Maximum bar spacing for crack width control without calculation for reinforced sections

Steel stress σ_s (MPa)	Maximum bar spacing (mm)
160	300
200	300
240	250
280	150
320	100
360	60

Note: The values in this table have been obtained under the same assumptions as those described in Table 7.6-4, assuming a height of 400 mm for the cross-section.

7.6.5 Limit states of deformation

7.6.5.1 General

7.6.5.1.1 Requirements

In-service deformations (deflections and rotations) may impair:

- the appearance of the structure;
- the integrity of non-structural parts;
- the proper functioning of the structure or its equipment.

To establish such limits is not within the scope of this Model Code. However, some practical rules are given in subsection 7.6.5.2.3 for some categories of simple buildings.

The deformations must be accommodated by other connected elements such as partitions, glazing, claddings, services or finishes. In some cases, a limitation may be required to ensure the proper functioning of machinery or apparatus supported by the structure, or to avoid ponding on flat roofs. Where applicable, acceptable limit values should be established in agreement with the client or their representative.

In general, the appearance and general utility of the structure can be impaired, if the calculated sag of a beam, slab or cantilever subjected to quasi-permanent loads exceeds span/250. The sag is assessed relative to the supports. Pre-camber may be used to compensate some or all of the deflection, but any upward deflection incorporated in the formwork should not generally exceed span/250.

To avoid excessive deformations, appropriate limiting values should be respected.

Deflections that could damage adjacent parts of the structure should be limited. For the deflection increment after the installation of adjacent construction parts, $\text{span}/500$ is normally an appropriate limit for quasi-permanent loads. Other limits may be considered, depending on the sensitivity of the adjacent parts.

In order to ensure a satisfactory behaviour in the serviceability limit state, deformations should be calculated as follows:

- the long term deformations are calculated for the quasi-permanent load combinations;
- the instantaneous deformations should be calculated for the rare load combinations.

For the calculation of camber, only the quasi-permanent load combinations are considered.

In order to calculate camber, the mean values of the material properties may be used.

The actual deformations may differ considerably from the calculated values, particularly if the values of the applied moments are close to the cracking moment. The difference will depend on the dispersion of the material properties, the ambient conditions, the loading conditions, the previous loading conditions, the restraints at the supports and so on.

Attention must be paid to cases where the basic assumptions of plane sections and uniformly distributed stresses across the section may not be adequate, such as in the case of shear lag effects in large prestressed structures.

For prestressed concrete members it may be necessary to control deflections assuming unfavourable deviations of the prestressing force and the dead load.

Most prestressed concrete flexural members will have a net positive (upward) camber at the time of transfer of prestress, caused by the eccentricity of the prestressing force. This camber may increase or decrease with time, depending on the stress distribution across the member under sustained loads.

In members composed of parts made of different concretes, such as a precast beam with a cast-in-place slab, the sectional properties of the composite transformed section must be obtained by summing the contribution of the different parts taking into account the different moduli of elasticity of each concrete.

7.6.5.1.2 Combination of actions

The combinations of actions to be considered depend on the criteria in question and are defined in subsection 7.6.2.

7.6.5.1.3 Data for the materials

The values of the material properties to be applied depend on the criteria in question.

In order to prevent damage due to deformations, prudent values of the material properties should be used.

7.6.5.1.4 Modelling

Depending on the precision needed, appropriate deformation models should be used, as described in the following subsections.

7.6.5.2 Deformations due to bending with or without axial force

7.6.5.2.1 General methods

The most general method to assess deformations is to perform a non-linear analysis capable of calculating the instantaneous and time-dependent deformations, taking into account the effects of applied loads and prestressing, the non-linear behaviour of concrete and steel and the time-dependent deformations due to creep and shrinkage of concrete and the relaxation of prestressing steel. These methods require, in general, iterative and incremental procedures, necessary to fulfil the equilibrium and compatibility conditions and the materials constitutive properties at any load level and instant of time.

In structures where the construction process requires an accurate estimation of deflections, a step-by-step analysis should be performed, including the sequence of loading and possible changes in geometry, structural scheme, support conditions and the evolution of material properties with time.

Deflections and axial displacements are obtained by the integration of curvatures and axial strains along the member length.

(a) Instantaneous deflections

Members which are not expected to be loaded above the level which would cause the tensile strength of the concrete to be reached should be considered to be uncracked, and to have a linear elastic response.

Members loaded above the cracking load level are expected to have a behaviour between the uncracked and the fully cracked stages. Thus, the instantaneous mean axial strains and mean curvatures due to axial loads and bending moments can be obtained with a non-linear sectional analysis, which includes cracking of concrete and tension stiffening, and assumes plane sections and a perfect bond between the concrete and the reinforcement.

In a cracked section under constant bending moment, changes in the stresses, strains and position of the neutral axis occur due to creep and shrinkage, as shown in Figure 7.6-9.

For uncracked members, it can be assumed that the creep deflections are proportional to the instantaneous deflections due to permanent loads, unless a large amount of reinforcement exists.

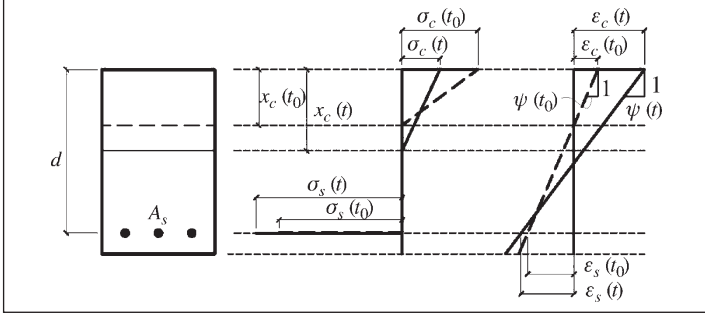


Figure 7.6-9: Stresses and strains at times t_0 and t , due to creep effect, in the presence of a constant bending moment

General methods are described in the CEB Information Bulletin No. 158, "Cracking and Deformations", Lausanne, 1985, and CEB Bulletin d'Information No. 235, "Serviceability Models", Lausanne, 1997. These bulletins contain the equations necessary to find the unknown parameters of the problems stated above.

The deflections of composite structures should consider the curvatures resulting from the differential shrinkage of precast and cast-in-situ components.

The time-dependent deflections are influenced by environmental and curing conditions, the age at time of loading, amount of compression reinforcement, magnitude of the stresses due to sustained loading and prestressing as well as strength gain of concrete after release of prestress. In particular, camber is especially sensitive to the concrete properties at the age of release of prestress, level of stresses, storage method, time of erection, placement of superimposed loads and environmental conditions.

The deformation a_I in the uncracked state in bending is calculated using the member stiffness $(E_c I_c)_I$, where E_c is the E modulus of the concrete for the loading type considered (instantaneous or long term) and I_c is the moment of inertia of the uncracked cross-section.

The deformation a_{II} in the cracked state in bending is calculated using the stiffness of the cracked member, assuming that no tension stiffening occurs. For a member of rectangular section the stiffness in the cracked state is equal to:

$$(EI)_{II} = d^2 \left(1 - \frac{x}{d} \right) \left(1 - \frac{1}{3} \frac{x}{d} \right) A_{s1} E_s + d'^2 \left(\frac{x}{d'} - 1 \right) \left(\frac{x}{d'} \frac{1}{3} - 1 \right) A_{s2} E_s \quad (7.6-15)$$

where:

$$\frac{x}{d} = \alpha_e \rho_1 \left(1 + \frac{\rho_2}{\rho_1} \right) \left(-1 + \sqrt{1 + \frac{2 \left(1 + \frac{\rho_2 d'}{\rho_1 d} \right)}{n \rho_1 \left(1 + \frac{\rho_2}{\rho_1} \right)^2}} \right)$$

(b) Long term deflections

In order to obtain the delayed deflections, the increment of curvatures and axial strains with time must be obtained and integrated along the member length. Thus, a time-dependent sectional analysis is required which incorporates the effects of creep and shrinkage of concrete and the relaxation of prestressing steel, as well as their interaction with cracking of concrete and tension stiffening, among other non-linearities. For the usual level of concrete stresses at service, the principles of linear viscoelasticity are accepted (see subsection 7.2.4).

For uncracked members, it can be assumed that the creep deflections are proportional to the instantaneous deflections due to permanent loads, unless a large amount of reinforcement exists.

7.6.5.2.2 Simplified method for RC structures

Members that are not expected to be loaded above the level which would cause the tensile strength of the concrete to be exceeded anywhere within the member should be considered to be uncracked. Members that are expected to crack, but may not be fully cracked, will behave in a manner intermediate between the uncracked and fully cracked conditions and, for members subjected mainly to flexure, an adequate prediction of the deformation is given by the expression:

$$a = \zeta a_{II} + (1 - \zeta) a_I \quad (7.6-14)$$

where:

a is the deformation parameter considered which may be, for example, a strain, a curvature or a rotation. As a simplification, a may also be taken as deflection;

a_I, a_{II} are the values of the deformation calculated for the uncracked and fully cracked conditions respectively;

ζ is an interpolation coefficient (allowing for the effect of tension stiffening at a section) given by the expression:

$$\zeta = 1 - \beta \left(\frac{\sigma_{sr}}{\sigma_s} \right)^2 \quad (7.6-16)$$

When there is only tensile reinforcement:

$$(EI)_{II} = d^2 \left(1 - \frac{x}{d}\right) \left(1 - \frac{1}{3} \frac{x}{d}\right) A_{st} E_s \quad (7.6-17)$$

where:

$$\frac{x}{d} = -\alpha_e \rho + \sqrt{(\alpha_e \rho)^2 + 2\alpha_e \rho l}$$

if the concrete in the compression area is still in the elastic state.

Alternatively, instead of interpolation, an equivalent stiffness deduced from (Eq. 7.6-14) can be used for direct simplified calculation of deflection:

$$(EI)_{eff} = \frac{(EI)_I \cdot (EI)_{II}}{\zeta \cdot (EI)_I + (1 - \zeta) \cdot (EI)_{II}}$$

For calculating long term deflections due to creep and shrinkage the following simplified procedure can be used:

$$\begin{aligned} a_t &= (1 + \varphi) a_g & \text{for } M_{Ed} < M_{cr} \\ a_t &= a_g + a_\phi + a_{sh} & \text{for } M_{Ed} \geq M_{cr} \end{aligned}$$

where:

- φ is the creep coefficient (see subsection 5.1.9.4.3);
- a_g is the instantaneous deflection due to quasi-permanent loads;
- a_ϕ is the creep deflection;
- a_{sh} is the shrinkage deflection.

β is a coefficient accounting for the influence of the duration of loading or repeated loading on the average strain:
 $\beta = 1.0$ for a single short term loading;
 $\beta = 0.5$ for sustained loads or multiple cycles of repeated loading;

σ_s is the stress in the tension reinforcement calculated on the basis of a cracked section under the load considered;
 σ_{sr} is the stress in the tension reinforcement calculated on the basis of a cracked section under the loading conditions that cause first cracking.

The stresses σ_s and σ_{sr} for the interpolation coefficient ζ are calculated at the most unfavourable section which is usually the section subjected to the maximum bending moment.

σ_{sr}/σ_s in Eq. (7.6-16) may be replaced by M_r/M for flexure and N_r/N for pure tension, where M_r is the cracking moment and N_r is the cracking force. M and N represent the moment and normal force for the load combination considered.

For loads with a duration long enough to cause creep, the total deformation including creep is obtained by using an effective modulus of elasticity of concrete according to:

$$E_{c,ef} = \frac{E_{cm}}{1 + \varphi} \quad (7.6-18)$$

where:

φ is the creep coefficient corresponding to the load and time interval.

Shrinkage curvatures may be assessed by:

$$\frac{1}{r_{cs}} = \varepsilon_{cs} \cdot \alpha_e \cdot \frac{S}{I} \quad (7.6-19)$$

where:

- $1/r_{cs}$ is the curvature due to shrinkage;
- ε_{cs} is the free shrinkage strain (see subsection 5.1.10.7.2);
- S is the first moment of area of the reinforcement about the centroid of the section;
- I second moment of area of the section;
- α_e is the effective modular ratio = $E_s / E_{c,ef}$.

S and I should be calculated for the uncracked and the fully cracked condition. The final curvature is assessed by Eq. (7.6-14).

The most rigorous method for assessing deflections is to compute the curvatures at a number of sections along the member and then calculate the deflection by numerical integration. In most cases, it will be acceptable to compute the deflections twice, assuming the whole member to be in the uncracked condition and in the fully cracked condition, and then interpolate according to Eq. (7.6-14).

7.6.5.2.3 Simplified method for PC structures

Under the quasi-permanent load combination, the structure is considered to be uncracked.

The curvature at time t is defined as:

$$\frac{1}{r}(t) = \frac{\varepsilon_{c,bottom}(t) - \varepsilon_{c,top}(t)}{h} \quad (7.6-20)$$

where $\varepsilon_{c,bottom}(t)$ and $\varepsilon_{c,top}(t)$ are the strains at time t in the bottom and top fibres of the cross-section and h is the height of the section.

The strains at time t are calculated in a way that takes into account the effect of self-weight, the initial prestress, other quasi-permanent actions and the stress variation due to creep, shrinkage and relaxation according to the following expression:

$$\varepsilon_c(t) = \varepsilon_{cs}(t, t_0) + \sigma_c(t_0) \cdot \frac{1 + \varphi(t, t_0)}{E_{cm}} + \frac{\sigma_c(t) - \sigma_c(t_0)}{E_{cm}} \cdot [1 + \chi \cdot \varphi(t, t_0)] \quad (7.6-21)$$

where:

$\sigma_c(t_0)$ is the concrete stress at the initial time t_0 due to prestress and load effects;

$\sigma_c(t)$ is the concrete stress at time t .

For long term loading the ageing coefficient χ can be assumed to be equal to 0.8.

7.6.5.2.4 Cases where calculations may be omitted

The approach given in subsection 7.6.5.2.3 is also followed in Eurocode 2.

Limits to the span/depth ratio may be formulated, which will be adequate for avoiding deflection problems in normal circumstances.

The limiting span to depth ratio l/d may be estimated using the expressions (7.6-22a) and (7.6-22b) and multiplied by correction factors to allow for the type of reinforcement used and other variables (Figure 7.6-10).

$$\frac{l}{d} = K \left[11 + 1.5 \sqrt{f_{ck}} \frac{\rho_0}{\rho} + 3.2 \sqrt{f_{ck}} \left(\frac{\rho_0}{\rho} - 1 \right)^{3/2} \right] \text{ if } \rho \leq \rho_0 \quad (7.6-22a)$$

$$\frac{l}{d} = K \left[11 + 1.5 \sqrt{f_{ck}} \frac{\rho_0}{\rho - \rho'} + \frac{1}{12} \sqrt{f_{ck}} \sqrt{\frac{\rho'}{\rho_0}} \right] \text{ if } \rho > \rho_0 \quad (7.6-22b)$$

where:

l/d is the limit span/depth ratio;

K is the factor depending on the structural system;

ρ_0 is the reference reinforcement ratio = $\sqrt{f_{ck}} \cdot 10^{-3}$;

ρ is the required tension reinforcement ratio at mid-span to resist the moment due to the design loads (at support for cantilevers);

ρ' is the required compression reinforcement ratio at mid-span to resist the moment due to de design loads (at support for cantilevers);

f_{ck} is the characteristic concrete compression strength in N/mm^2 .

If other stress levels apply than $\sigma_s = 250 \text{ N/mm}^2$, for which the Eqs. (7.6-22a) and (7.6-22b) have been derived, the values obtained by those equations should be multiplied by $250/\sigma_s$. It will normally be conservative to assume that:

$$250 / \sigma_s = 500 / (f_{yk} A_{s,req} / A_{s,prov}) \quad (7.6-23)$$

where:

σ_s is the tensile strain at mid-span (at support for cantilevers) under the design load at SLS;

$A_{s,prov}$ is the area of steel provided at this section;

$A_{s,req}$ is the area of steel required at this section for ultimate load.

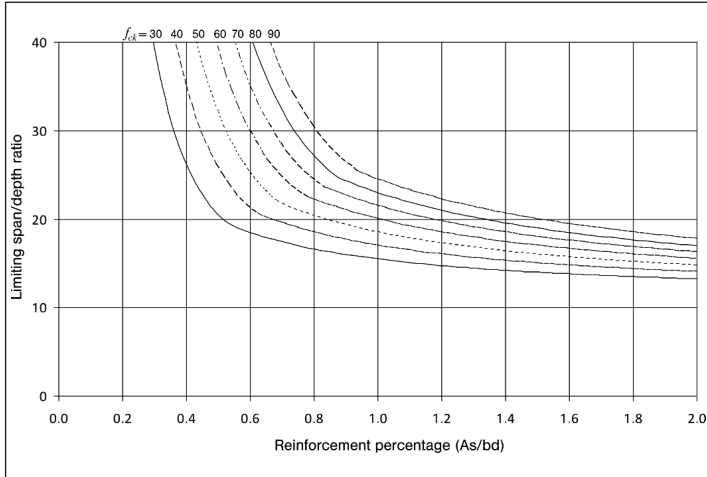


Figure 7.6-10: Graphical representation of Eqs. (7.6-22a) and (7.6-22b) for a simply supported slab bearing in one direction ($K = 1.0$) and $\sigma_s = 250 \text{ N/mm}^2$

The values given by Eqs. (7.6-22a) and (7.6-22b) and Table 7.6-6 have been derived from the results of a parametric study made for a series of beams and slabs simply supported with rectangular cross-section, using the general approach given in subsection 7.6.5.2.2. Different values of the concrete strength class and a 500 MPa characteristic yield strength were considered. For a given area of tension reinforcement the ultimate moment was calculated and the quasi-permanent load was assumed as 50% of the corresponding total design load. The span to depth limits obtained satisfy the limiting deflection span/250 for quasi-permanent loads and span/500 for quasi-permanent loads after construction.

Eqs. (7.6-22a) and (7.6-22b) have been derived under the assumption that the steel stress under the appropriate design load at SLS at a cracked section at the mid span of a beam or slab or at the support of a cantilever, is 250 N/mm^2 (corresponding roughly to $f_{yk} = 500 \text{ N/mm}^2$).

For flanged sections where the ratio of the flange breadth to the rib breadth exceeds 3, the values of l/d given by Eq. (7.6-22) should be multiplied by 0.8.

For beams and slabs, other than flat slabs, with spans exceeding 7 m, which support partitions liable to be damaged by excessive deflections, the values of l/d given by Eq. (7.6-22) should be multiplied by $7/l$ (with l in metres).

For slabs where the greater span exceeds 8.5 m, and which support partitions liable to be damaged by excessive deflections, the values of l/d given by Eq. (7.6-22) should be multiplied by $8.5/l$ (with l in metres).

The values K follow from Table 7.6-6.

Table 7.6-6: Basic ratios of span/effective depth for reinforced members without axial compression

Structural system	K	Concrete highly stressed $\rho = 1.5\%$	Concrete lightly stressed $\rho = 0.5\%$
Simply supported beams, one- or two-way spanning simply supported slab	1.0	14	20
End span of continuous beam or one-way continuous slab continuous over one long side	1.3	18	26
Interior span of beam or one- or two-way spanning slab	1.5	20	30
Slab supported on columns without beams (flat slabs) (based on longer span)	1.2	17	24
Cantilever	0.4	6	8

7.6.6 Vibrations

7.6.6.1 General

Vibrations can be caused by several variable actions, for example:

- rhythmic movements made by people, such as walking, running, jumping and dancing;
- machines;
- waves due to wind and water;
- rail and road traffic;
- construction work such as driving or placing by vibration of sheet;
- piles, compressing soil by means of vibrations as well as blasting work.

Vibrations that endanger the structure, such as very large deflections due to resonance or the loss of resistance due to fatigue, should be included in the verification for ULS of the structure.

Table 7.6-7: Critical frequency in structures subject to vibrations caused by movements of people

Structures	Frequency (s^{-1}) f_{crit}
Gymnasia and sports halls	8.0
Dance rooms and concert halls without permanent seating	7.0
Concert halls with permanent seating	3.4
Critical work areas	1.0
Residence	1.4 – 4.0
Office	4.0
Workshop	8.0
Structures for pedestrians and cyclists	See below*

* Natural frequencies between 1.6 and $2.4 s^{-1}$ and between 3.5 and $4.5 s^{-1}$ are to be avoided in structures for pedestrians and cyclists. Joggers can also cause vibrations in structures with natural frequencies between 2.4 and $3.5 s^{-1}$.

Vibrations of structures may affect the serviceability of a structure as follows:

- functional effects (discomfort to occupants, affecting operation of machines, etc.);
- structural effects (mostly on non-structural elements, as cracks in partition walls, loss of cladding, etc.).

7.6.6.2 Vibrational behaviour

To secure satisfactory behaviour of a structure subject to vibrations, the natural frequency of vibration of the relevant structure should

be kept sufficiently apart from critical values which depend on the function of the corresponding building, see Table 7.6-7. The natural frequency must be far enough from the critical frequency.

The vibrational behaviour of structures can be influenced by the following measures:

- changing the dynamic actions;
- changing the natural frequencies by changing the rigidity of the structure or the vibrating mass;
- increasing the damping features etc.

7.6.7 Verification of serviceability limit state by numerical simulation

7.6.7.1 Fracture mechanics-based models

Analysis of the states of stress and deformation under serviceability conditions can be performed by non-linear finite element analyses based on fracture mechanics. This is a general method and is applicable to structures subjected to general states of stress. However, sufficiently fine numerical models are required in order to capture individual cracks.

In current numerical models there are two types of crack models:

- (1) Discrete crack model. In this model the crack is formed at the interface between two elements. After opening it changes the boundary conditions. The fracture properties are described by the interface constitutive law. This approach requires a re-meshing and mesh refinement near the crack tip during the crack propagation.
- (2) Smeared crack model. According to this approach the crack is modelled by assuming orthotropic damage to the concrete within the area assigned to an element, or an integration point. The crack is modelled by a crack band. During the crack propagation the tensile strain localizes within the crack band. This is caused by the softening nature of the stress-crack opening law. The objectivity of the solution (low mesh sensitivity) is ensured by considering the crack band size as a regularization parameter of the strain localization.

The smeared crack model is more suitable for practical applications, where many cracks occur. The discrete crack model is suitable for cases with one distinct crack (Figure 7.6-11.).

Basically, three parameters describe the constitutive law of crack opening: tensile strength, shape of the softening function and fracture energy.

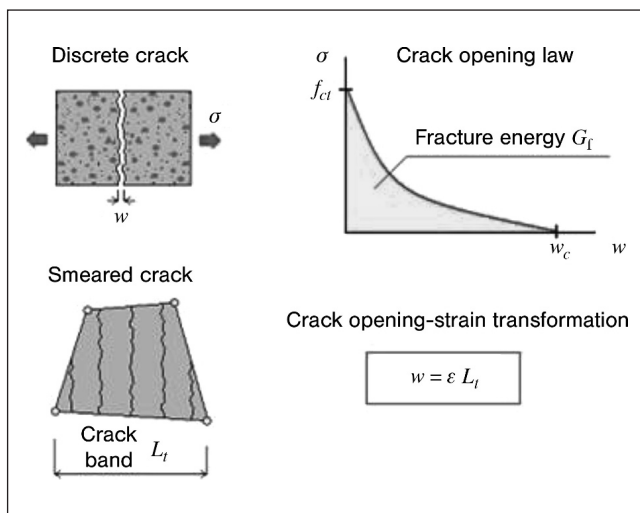


Figure 7.6-11: Constitutive law of crack opening for discrete and smeared crack models

In fracture mechanics based models the cracks are formed as a result of strain localization after reaching the concrete tensile strength. This process is successful only for a sufficiently small size of the finite element mesh.

As a rough estimate, the element size should be smaller than half the crack spacing. Consequently, in reinforced concrete structures with small bar spacing, where small crack spacing occurs, the element size should be small, while in plain concrete structures, or locations without reinforcement, the elements can be larger. The upper limit for the acceptable element size is controlled by the fracture energy of the concrete.

7.6.7.2 Tension stiffening-based models

In cases outside of the range suitable for the fracture mechanics-based models mentioned in 7.6.7.1, the crack effect can be described by tension stiffening. In this approach no discrete cracks are considered and the properties of cracked concrete are regarded in an average sense by tension stiffening. This model offers adequate results for the analysis of deformations and strains required for the analysis of crack widths.

The expressions for tension stiffening, crack spacing and crack width, given in Eqs. (7.6-3) to (7.6-5) can be applied for the smeared crack model.

The rules in this section have been derived predominantly for steel fibre reinforced concrete (FRC). They are based on the assumptions of subsection 5.6.1. Design recommendations, especially focusing on high and ultra-high strength fibre concrete, are in preparation (*fib* Task Group 8.6).

A distinction should be made between structures with linear elements (statically determinate and indeterminate beams or frames), where the stress redistribution is limited to few sections, where strain localization may occur and structures with a higher degree of redundancy, where stress redistribution occurs in multiple cracks (as in slabs).

In linear structures there may be critical zones where a model based on a plane section can be assumed, by spreading the localized relative rotation in defined regions characterized by a prescribed length in which the curvature is assumed constant. In these zones the curvature jumps to larger values (post-peak values) with respect to the values reached in the adjacent regions (where the curvature corresponds to the unloaded elastic values), although the bending moment is the same to respect the rotational equilibrium.

For thin walled structures, fibre orientation may be influenced by the reduced thickness (wall effect), which is significantly dependent on the casting direction. In these structures, material properties can be better determined by performing “structural tests”.

In thin walled structures subjected to bending (shell structures), due to the reduced thickness, the ultimate strain should be significantly reduced in order to satisfy ductility requirements expressed in terms of ultimate strain.

The design rules are not intended to be used for slabs on grade, or for temporary sprayed concrete linings, or to design applications in which increased resistance to plastic shrinkage, increased resistance to abrasion or impact are aimed for. The design rules refer to applications, covered in the scope of EC2.

Due to the nature of FRC, design supported by testing leads to optimized solutions in terms of structural performance versus cost.

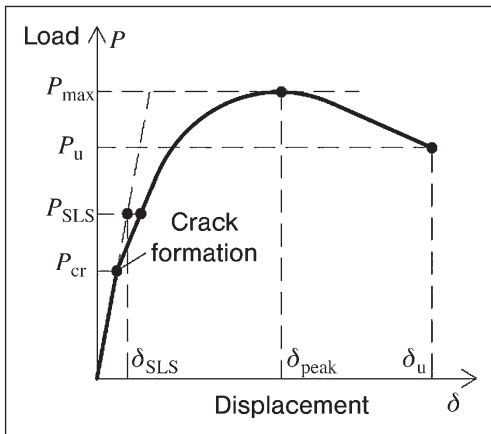


Figure 7.7-1: Typical load (P) – displacement (δ) curve for a FRC structure

In the case of linear elements, without conventional reinforcement, subjected to axial tension with small eccentricity (i.e. without compressive stresses in the section), in addition to the limitations provided by Eqs. (7.7-1) and (7.7-2), FRC should have a hardening behaviour in tension (see section 5.6.1, Figure 5.6-2).

7.7 Verification of safety and serviceability of FRC structures

7.7.1 Classification

FRC structures can be classified as:

- structures with linear elements (beams, and columns);
- walls;
- slabs;
- shells (e.g. thin walled members);
- three-dimensional members.

7.7.2 Design principles

Structural design must satisfy requirements for resistance and serviceability for the expected service life of FRC elements. The ductility requirement in bending can be satisfied by minimum conventional reinforcement (see also section 7.13; Fig.7.7.1).

In all FRC structures without the minimum conventional reinforcement, one of the following conditions must be satisfied:

$$\delta_u \geq 20 \delta_{SLS} \quad (7.7-1)$$

$$\delta_{peak} \geq 5 \delta_{SLS} \quad (7.7-2)$$

where δ_u is the ultimate displacement, δ_{peak} is the displacement at the maximum load and δ_{SLS} is the displacement at maximum service load computed by performing a linear elastic analysis with the assumptions of uncracked concrete and initial elastic Young's modulus.

Usually, δ_u is related to the maximum deformation requirement of the structure.

The ultimate load P_u should always be higher than the load at crack initiation P_{cr} and higher than the maximum service load P_{SLS} .

When the structure is able to significantly redistribute the applied loads at failure, a factor K_{Rd} , that takes into account favourable effects due to redistribution, can be assumed:

$$P_{Rd} = K_{Rd} \cdot P(f_{Fd}) \quad (7.7-3)$$

In FRC structures, K_{Rd} is mainly affected by:

- the volume of the structure involved in the crack propagation process at failure (V), with respect to that used in the material identification procedure of the post-cracking residual strengths (V_0) (Fig.7.7-2);
- the ratio between the maximum load P_{max} reached and the first cracking load P_{cr} , that quantifies the redistribution capability $K_{Rd} = K_{Rd} (V/V_0, P_{max}/P_{cr})$.

$P(f_{Fd})$ is the resistant load computed taking into account the design strength of FRC.

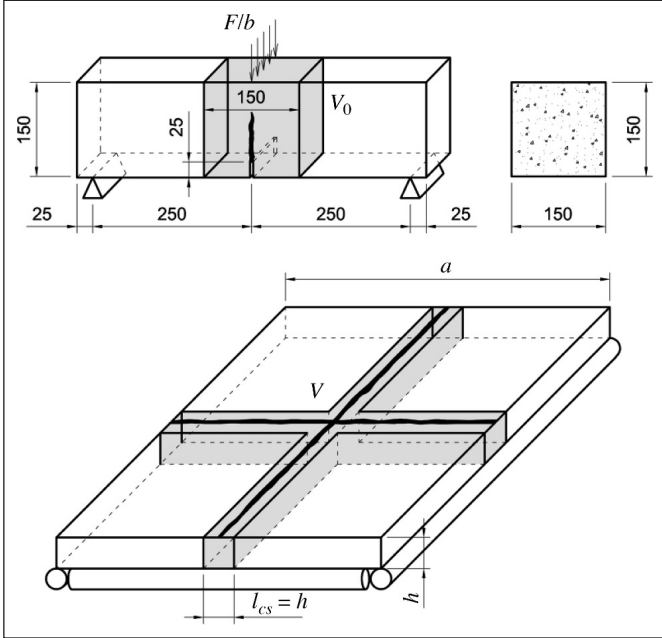


Figure 7.7-2: Volume involved in the failure for the classification test (V_0) and for a structure (V)

In the literature, different methods are proposed for evaluating such a coefficient that considers essentially the experimental evidence of a mechanical global response, which fits the response attainable with the average values of the residual post-cracking strength, when large load redistribution occurs. This is mainly due to the reduction of the standard deviation in the structure's response in relation to that measurable in a standard test, where a limited number of fibres and a specific geometrical location of the notch lead to larger scatter.

K_{Rd} can be computed by a structural analysis that takes into account a random redistribution of the mechanical characteristics. When a statistical distribution of P_{max} is obtained starting from an assumed standard deviation of the mechanical constitutive law, the factor K_{Rd} can be computed as:

$$K_{Rd} = \frac{P_{max,k}}{P_{max,m}} \frac{f_{Ftum}}{f_{Ftuk}} \leq 1.4$$

where:

$P_{max,k}$ is the characteristic value of the maximum load;

$P_{max,m}$ is the mean value of the maximum load;

f_{Ftuk} is the characteristic value of the ultimate residual tensile strength of FRC, determined by considering $w_u = \min(l_{cs}\epsilon_{Fu}, 2.5 \text{ mm})$, according to Eq. (5.6-6), and assuming ϵ_{Fu} equal to 2% for variable strain distribution along the cross-section and 1% for constant tensile strain distribution along the cross-section;

f_{Ftum} is the mean value of the ultimate residual tensile strength of FRC.

Further information on the design equations, as treated in this section, is given by Di Prisco et al. (2013), Recommendations for fibre reinforced concrete in *fib* Model Code 2010. Structural Concrete, 14. doi: 10.1002/suco.201300021.

7.7.3 Verification of safety (ULS)

7.7.3.1 Bending and/or axial compression in linear members

The bending failure stage is supposed to be reached when one of the following conditions applies (Figure 7.7-3); the meaning of the variables is found in subsections 5.6.4 and 5.6.6):

- attainment of the ultimate compressive strain in the FRC, ϵ_{cu} ;
- attainment of the ultimate tensile strain in the steel (if present), ϵ_{su} ;
- attainment of the ultimate tensile strain in the FRC, ϵ_{Fu} .

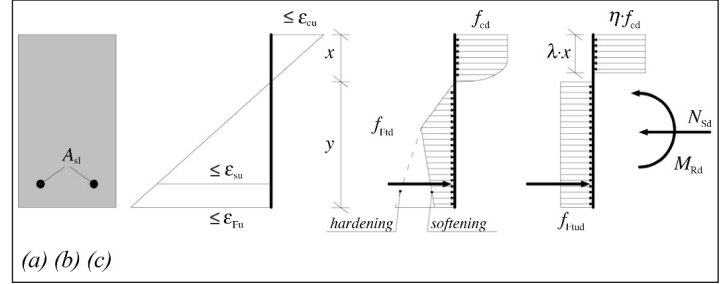


Figure 7.7-3: ULS for bending moment and axial force: use of the simplified stress/strain relationship (λ and η coefficient in accordance Eq. (7.2-15) to (7.2-18) in subsection 7.2.3.1.5)

7.7.3.2 Shear in beams

7.7.3.2.1 Beams without longitudinal and shear reinforcement

When FRC with tensile-hardening behaviour is used and members without both longitudinal and transverse reinforcement are considered, the principal tensile stress, σ_1 , must not be higher than the design tensile strength:

$$\sigma_1 \leq \frac{f_{Ftuk}}{\gamma_F} \quad (7.7-4)$$

where:

f_{Ftuk} is the characteristic value of the ultimate residual tensile strength of FRC determined with Eq. (5.6-6) for $w_u = 1.5$ mm [MPa];

γ_F value found in Table 5.6-1.

7.7.3.2.2 Beams without shear reinforcement

The design value for the shear resistance in members with conventional longitudinal reinforcement and without shear reinforcement is given by (in N):

$$V_{Rd,F} = \left\{ \frac{0.18}{\gamma_c} \cdot k \cdot \left[100 \cdot \rho_l \cdot \left(1 + 7.5 \cdot \frac{f_{Ftuk}}{f_{ctk}} \right) \cdot f_{ck} \right]^{1/3} + 0.15 \cdot \sigma_{cp} \right\} \cdot b_w \cdot d \quad (7.7-5)$$

[stresses and strengths in MPa]

where:

γ_c is the partial safety factor for the concrete without fibres;

k is a factor that takes into account the size effect and is equal to:

$$1 + \sqrt{\frac{200}{d}} \leq 2.0$$

d is the effective depth of the cross-section [mm];

ρ_l is the longitudinal reinforcement ratio defined as:

$$\rho_l = A_{sl} / b_w d$$

f_{Ftuk} is to be determined by an axial-tensile test.

This approach was recently developed and validated, and is now reliable, but is not consistent with the approach described in section 7.3.3. Other approaches are available in the literature; in particular, the following that is aligned to the approach presented in the section 7.3.3 and is more extensively explained in *fib* Bulletin 57 “Shear and punching shear in RC and FRC elements. Workshop proceedings.” (*fib*, 2010), although not yet fully validated.

Eq. (7.7-4) is valid for steel fibre reinforced concrete with conventional strength. It has not been validated for other fibre materials or for non-conventional strength concretes, such as reactive powder concrete.

- A_{sl} is the cross-sectional area of the reinforcement which extends $\geq l_{bd} + d$ beyond the considered section [mm²];
- f_{Ftuk} is the characteristic value of the ultimate residual tensile strength for FRC, by considering $w_u = 1.5$ mm according to Eq. (5.6-6) [MPa];
- f_{ctk} is the characteristic value of the tensile strength for the concrete without fibres [MPa];
- f_{ck} is the characteristic value of cylindrical compressive strength [MPa];
- $\sigma_{cp} = N_{Ed}/A_c < 0.2 f_{cd}$ [MPa] is the average stress acting on the concrete cross-section A_c [mm²] for an axial force N_{Ed} [N], due to loading or prestressing actions ($N_{Ed} > 0$ for compression);
- b_w is the smallest width of the cross-section in the tensile area [mm].

The shear resistance, $V_{Rd,F}$, is assumed to be not smaller than the minimum value, $V_{Rd,Fmin}$, defined as:

$$V_{Rd,Fmin} = (v_{min} + 0.15 \cdot \sigma_{cp}) \cdot b_w \cdot d \quad (7.7-6)$$

where $v_{min} = 0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$.

A recent model that follows the approach to shear described in subsection 7.3.3.4, computes the term $V_{Rd,F}$ as follows:

$$V_{Rd,F} = \frac{1}{\gamma_F} (k_v \sqrt{f_{ck}} + k_f f_{Ftuk} \cot \theta) z b_w \quad (7.7-7)$$

where:

f_{Ftuk} is the characteristic value of the ultimate tensile strength for FRC, as determined by direct tensile tests, corresponding to the crack width at ultimate, w_u ;

$k_f = 0.8$;

and

$$k_v = \frac{0.4}{1 + 1500 \varepsilon_x} \cdot \frac{1300}{1000 + k_{dg} z} \quad \text{for } \rho_w < 0.08 \sqrt{f_{ck}/f_{yk}}$$

$$k_v = \frac{0.4}{1 + 1500 \varepsilon_x} \quad \text{for } \rho_w \geq 0.08 \sqrt{f_{ck}/f_{yk}} \quad (7.7-8)$$

In Eq. (7.7-7) ε_x is the longitudinal strain at the mid depth of the effective shear depth as determined by either Eq. (7.3-14) or (7.3-16), as appropriate, z is the internal lever arm (in mm) between the flexural tensile and compressive forces (Figure 7.3-9) and k_{dg} is an aggregate size influence parameter.

The aggregate size influence parameter in Eq. (7.7-7), k_{dg} , is given by:

$$k_{dg} = \frac{32}{16 + d_g} \geq 0.75 \quad (7.7-9)$$

where d_g is the maximum aggregate size in mm. If the size of the maximum aggregate particles is less than 16 mm, this parameter may be taken as $k_{dg} = 1.0$.

The limits of the angle of the compressive stress field, θ , relative to the longitudinal axis of the member, as shown in Figure 7.3-11, are:

$$\theta_{min} \leq \theta \leq 45^\circ \quad (7.7-10)$$

where the minimum strut inclination angle is:

$$\theta_{min} = 29^\circ + 7000 \varepsilon_x \quad (7.7-11)$$

Where for the determination of f_{Ftuk} , the crack width at ultimate (w_u) is taken as:

$$w_u = 0.2 + 1000 \varepsilon_x \geq 0.125 \text{ mm} \quad (7.7-12)$$

7.7.3.2.3 Beams with shear and longitudinal reinforcement

For the design of members with shear reinforcement, the basic relation Eq. (7.3-23) applies:

$$V_{Rd} = V_{Rd,c} + V_{Rd,s} \quad (7.7-13)$$

For FRC elements this equation becomes:

$$V_{Rd} = V_{Rd,F} + V_{Rd,s} \quad (7.7-14)$$

where:

$V_{Rd,s}$ is to be taken from Eq. (7.3-29);

$V_{Rd,F}$ follows from Eq. (7.7-5).

7.7.3.2.4 Minimum shear reinforcement

The minimum shear reinforcement should be provided by either stirrups (7.3-22), or fibres.

Eq. (7.7-15) is based on steel fibre concrete research, and should be checked for other types of material.

The minimum amount of conventional shear reinforcement (stirrups) is not required if the following condition is fulfilled:

$$f_{Ftuk} \geq 0.08 \sqrt{f_{ck}} \quad (7.7-15)$$

where:

f_{Ftuk} is the characteristic value of the ultimate residual tensile strength for FRC, by considering $w_u = 1.5$ mm according to Eq. (5.6.-6) [MPa].

This allows limiting the development and the diffusion of the inclined cracking and, as a consequence, can ensure sufficient member ductility.

When the above-mentioned limitation is not applied, conventional shear reinforcement (stirrups) have to be introduced according to Eq. (7.7-14).

7.7.3.3 Torsion in beams

7.7.3.3.1 Beams without longitudinal and transverse reinforcement

When FRC with a hardening tensile behaviour is used in a member without both longitudinal reinforcement and transverse reinforcement, the principal tensile stress must not exceed the design tensile strength:

$$\sigma_1 \leq \frac{f_{Ftuk}}{\gamma_F} \quad (7.7-16)$$

where:

f_{Ftuk} is the characteristic value of the ultimate residual tensile strength for FRC at $w_u = 1.5$ mm according to Eq. (5.6.-6) [MPa];

γ_F is the partial safety factor for the FRC, which follows from Table 5.6-1.

7.7.3.3.2 Beams with longitudinal and transverse reinforcement

The presence of fibres increases the torsion capacity, but design models are not currently available. Models should be proven by experiments on real size elements.

7.7.3.4 Walls

7.7.3.4.1 Walls without conventional reinforcement

For 2D elements loaded in their plane, it is possible to check the limit state criteria for SLS and ULS by means of the biaxial domain of failure, where the uniaxial tension strengths are reduced to f_{Fts} or f_{Ftu} , according to the limit state considered.

7.7.3.4.2 Walls with conventional reinforcement

In FRC structures, the fibre contribution can be accounted for by non-linear finite element analyses or strut-and-tie models with the constitutive laws defined in subsection 5.6.4

In FRC structures satisfying minimum requirements (Eqs. (5.6-2) and (5.6-3)), secondary conventional reinforcement can be omitted.

7.7.3.5 Slabs

7.7.3.5.1 Members without reinforcement

For slab elements without conventional reinforcement (Figure 7.7-4) with predominantly bending actions, the strength verification can be done with reference to the resisting moment, m_{Rd} , evaluated by considering a rigid plastic relationship (Figure 7.7-3c):

$$m_{Rd} = \frac{f_{tud} \cdot t^2}{2} \quad (7.7-17)$$

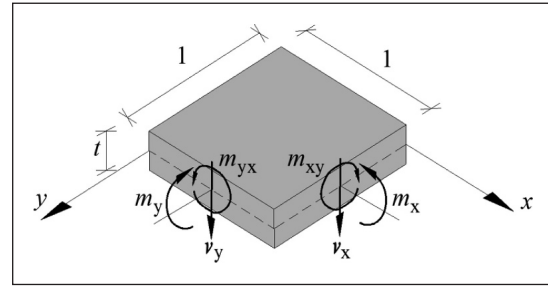


Figure 7.7-4: Actions in a slab element

When a linear analysis is performed, the maximum principal moment should be lower than m_{Rd} .

When a limit analysis is performed, m_{Rd} can be regarded as the reference value.

The design bending moment can be increased, as described in 7.7.3.

Shear in FRC slabs without conventional reinforcement or prestressing is not regarded as dominant unless significant load concentrations occur close to the support.

7.7.3.5.2 Members with reinforcement

The verification of FRC elements with conventional reinforcement can be done with a non-linear analysis (e.g. limit analysis, non-linear finite element analysis).

7.7.3.5.3 Punching

For FRC members, Eq. (7.3-60) can be replaced by:

$$V_{Rd} = V_{Rd,F} + V_{Rd,s} \quad (7.7-18)$$

where $V_{Rd,F}$ represents the fibre reinforced concrete contribution to shear.

The design shear resistance attributed to the fibres may be taken as

$$V_{Rd,F} = V_{Rd,c} + V_{Rd,f} \quad (7.7-19)$$

For redundant structures, such as slabs, the ultimate deformation determines the ultimate limit state condition. From that the corresponding crack opening (w_u) can be computed, which can be in the order of magnitude of the SLS crack opening.

For slabs with longitudinal reinforcement, the ultimate crack opening $w_u = \psi d/6$ is suggested, where ψ is calculated from subsection 7.3.5.4. However, this equation has not been validated for all thicknesses.

$$V_{Rd,f} = \frac{f_{Ftuk}}{\gamma_F} b_0 d_v \quad (7.7-20)$$

f_{Ftuk} is the characteristic value of the ultimate residual tensile strength for FRC, calculated taking into account $w_u = 1.5$ mm [MPa];

b_0 is the shear resisting control perimeter as defined in subsection 7.3.5.2(4) [mm];

d_v is the shear resisting effective depth defined in subsection 7.3.5.2(2) [mm].

When reinforcement is needed for punching resistance, a minimum amount of fibres (and if applicable transverse reinforcement) is required in order to ensure sufficient deformation capacity at failure as follows:

$$V_{Rd,s} + V_{Rd,f} \geq 0.5 \cdot V_{Ed} \quad (7.7-21)$$

7.7.3.5.4 Shear in slabs with longitudinal reinforcement

If shear reinforcement is to be provided according to subsection 7.3.3.3, Eq. (7.7-5) may be applied as an alternative.

7.7.4 Verification of serviceability (SLS)

7.7.4.1 Stress limitation

The compressive stresses at the SLS must be limited in accordance with subsection 7.6.3.3.

Tensile stresses in the rebars must be limited at the SLS in accordance with subsection 7.6.3.4.

In structural FRC elements having a tension softening behaviour after cracking, the tensile stress verification is not necessary if the element is verified at the ULS.

In structural FRC elements having a tension hardening behaviour after cracking, the tensile stress verification must be done by imposing the limitation:

$$\sigma_t \leq 0.6 \cdot f_{Ftsk} \quad (7.7-22)$$

where σ_t is the principal tensile stress and f_{Ftsk} is the characteristic value of f_{Fts} given in Eq. (5.6-5).

7.7.4.2 Crack width in members with conventional reinforcement

The design crack width w_d in FRC elements can be calculated by:

$$w_d = 2 \left\{ k \cdot c + \frac{1}{4} \cdot \frac{\phi_s}{\rho_{s,ef}} \cdot \frac{(f_{ctm} - f_{Ftsm})}{\tau_{bm}} \right\} \cdot \frac{1}{E_s} \cdot (\sigma_s - \beta \cdot \sigma_{sr} + \eta_r \cdot \varepsilon_{sh} \cdot E_s) \quad (7.7-23)$$

where:

f_{Ftsm} follows from Eq. (5.6-5).

For the other symbols see Eq. (7.6-5).

$$\text{With } \sigma_{sr} = (f_{ctm} - f_{Ftsm}) \cdot (1 + \alpha_e \cdot \rho_s) / \rho_{s,eff} \quad (7.7-24)$$

In σ_s the effect of the fibres (f_{Ftsm}) needs to be taken into account.

$$f_{Ftsm} = f_{Ftsk} / 0.7 \quad (7.7-25)$$

where f_{Ftsk} is the characteristic value of f_{Fts} given in Eq. (5.6-5).

7.7.4.3 Minimum reinforcement for crack control

For controlling the crack width in the elements under bending, if needed, a minimum reinforcement should be applied, at least equal to:

Eq. (7.7-23) is derived from Eqs. (7.6-3)–(7.6-5). From these equations, through the modified value of $l_{s,max}$, we get the length over which slip between concrete and steel occurs, necessary to reintroduce the cracking-force $A_c(1 + \alpha_e \rho_{s,ef})f_{ctm}$ back into the concrete by bond.

By virtue of the action of the fibres, which generate a residual tensile strength f_{Ftsm} , the force to be reintroduced by bond is reduced to $A_c(1 + \alpha_e \rho_{s,ef}) \cdot (f_{ctm} - f_{Ftsm})$.

So, the introduction length $l_{s,max}$ is reduced to:

$$l_{s,max} = k \cdot c + \frac{1}{4} \cdot \frac{(f_{ctm} - f_{Ftsm})}{\tau_{bm}} \cdot \frac{\phi_s}{\rho_{s,ef}}$$

$$A_{s,min} = k_c \cdot k \cdot (f_{ctm} - f_{Ftsm}) \frac{A_{ct}}{\sigma_s} \quad (7.7-26)$$

where:

f_{ctm} is the average value of the tensile strength of the concrete matrix;

f_{Ftsm} is the average value of the residual strength of FRC;

A_{ct} is the tensile part of the concrete cross-section, evaluated by considering a stress field at elastic limit;

σ_s is the maximum tensile stress in the reinforcement in the cracked state, that can be considered equal to the yielding stress of the steel;

k_c is a coefficient taking account of the stress distribution in the cross-section just before cracking and the change of the inner lever arm; for rectangular cross-sections $k_c = 1$;

k is a coefficient taking into account of non-uniform self-equilibrating stresses, leading to a reduction of the cracking force:

$k = 1.0$ for webs with $h \leq 300$ mm or flanges with width ≤ 300 mm;

$k = 0.65$ for webs with $h \leq 800$ mm or flanges with width ≥ 800 mm; for intermediate values interpolation can be applied.

When $A_{s,min}$ is negative, the minimum reinforcement can be due only to the fibre reinforcement.

The standard ISO 16204, “Durability – Service life design of concrete structures” is based on the principles given in *fib* MC SLD – see *fib* Bulletin 34, “Model Code for Service Life Design” (*fib*, 2006) and is also fully in line with the provisions given in section 7.8 of this Model Code, see also Helland, S. (2013), Design for service life: implementation of *fib* Model Code 2010 rules in the operational code ISO 16204. Structural Concrete, 14: 10-18. doi: 10.1002/suco.201200021.

Subsection 3.2.2 gives the basis for verification of the design service life in terms of performance requirements (length and target reliability levels).

Traditionally, national and international concrete standards give requirements to achieve the desired design service life based on the “deemed-to-satisfy” and the “avoidance-of-deterioration” approach. Such operative requirements have to be calibrated by the responsible standardization body. This section gives guidance for such calibration

In subsection 4.7.2, the durability related exposure conditions in the design situations are addressed. Reference is made to ISO 22965-1 “Concrete Part 1: Methods of specifying and guidance for the specifier”, which gives an example of how to differentiate the environmental loads with respect to deterioration on the structure by 17 “Exposure classes”. The same classification is adopted by the European CEN standards on design of concrete structures. This classification is qualitative in nature and is by the local standardization body often linked directly to deemed-to-satisfy and avoidance-of-deterioration requirements in operational standards.

If more refined service life designs are to be undertaken by the use of deterioration modelling, this classification of the environmental load must be related to quantified parameters, for instance chloride concentrations for marine structures. When publishing this Model Code, such quantified parameters were not available in any operational standard. Information must therefore be found by measurements on existing structures and in the literature, such as *fib* MC SLD (*fib* Bulletin 34, “Model Code for Service Life Design” (*fib*, 2006) and Concrete Society Technical Report No. 61 (see Bamforth P., Concrete Society Technical Report No. 61: 2004 “Enhancing reinforced concrete durability”).

Another concept related to the verification of limit states associated with durability is described in Guidance Paper F of the Construction Products Directive of the European Community. This is the use of the so-called “torture tests”, where the material is subject to test conditions without doubt harsher than what will be the case in the actual exposure during the design service life. If the material stands the test, it is also accepted that the verification is fulfilled, but with an unknown margin. However, it is not possible to conclude from a failure to withstand such an onerous torture test that the material would underperform during real in-field exposure. Many freeze-thaw tests for concrete are in this category.

At the time of publishing this Model Code, the European standardization body CEN is working on a concept called “Equivalent Durability Concept – EDC”. This implies that a material composition not dealt with in the operational standard may be compared to one reference with a proven long term performance. The comparison is made on the basis of testing. Based on these test results the performance of the candidate material has to be assessed at the end of its design service life and then compared with that of the reference. Such an extrapolation of test results involves the use of modelling. The EDC assumes that this modelling is done by the responsible standardization body and that the user of this concept only applies a fixed ageing factor for the relevant mix composition authorized by the standardization body.

7.8 Verification of limit states associated with durability

7.8.1 General

Verification of limit states associated with durability may be done according to one of the following safety formats given in chapter 4:

- probabilistic safety format;
- partial safety factor format;
- deemed-to-satisfy approach;
- avoidance-of-deterioration approach.

Within section 7.8, the following deterioration mechanisms are addressed:

- carbonation induced corrosion;
- chloride induced corrosion;
- freeze-thaw attack.

For these mechanisms, models with a relatively broad international acceptance exist.

Other deterioration mechanisms such as alkali silica reaction, acid and sulphate attack and delayed ettringite formation are not treated in detail in this Model Code, mainly due to the fact that broadly accepted time-dependent models to be included here are still under discussion.

Requirements to ensure that the design service life is not jeopardized due to these mechanisms are normally based on long term field experience combined with available scientific insight into their nature. The provisions found in operational standards are, based on this, the results of the expert's opinion by the members of the standardization committees.

It is possible that several deterioration mechanisms may occur simultaneously, for instance carbonation with chloride penetration, or freeze-thaw with carbonation with or without chlorides. So there may be a combination of exposure categories (see subsection 4.7.2) leading to the use of the probabilistic safety or partial safety factor formats with deemed-to-satisfy design.

In the following subsections, the new aims of the Model Code related to the durability of concrete structures are presented.

7.8.2 Carbonation induced corrosion – uncracked concrete

7.8.2.1 Probabilistic safety format

7.8.2.1.1 Limit state: depassivation

General equation describing the limit state of depassivation is:

$$P\{\} = P_{dep} = P\{t_{SL} - t_{ini}\} < P_0 \quad (7.8-2)$$

where:

- $P\{\}$ is the probability that depassivation occurs;
- t_{SL} is design service life [years];
- t_{ini} is initiation period [years];
- P_0 is target failure probability.

This general equation describing the limit state of depassivation has been developed by RILEM, see “Durability design of concrete structures, Report of Rilem Technical Committee 130-CSL”, Sarja, A. and Vesikary, E., eds.

When publishing this Model Code, no models with broad international consensus were available for predicting the length of the corrosion period until cracking, spalling or collapse of the structure occurs. For this reason, service life designs are normally done with the limit state of depassivation (reaching a reduced pH of 8–9 at the rebar surface).

The direct consequence of passing this limit state is only that possible future protective measures for repair become more expensive. This rather conservative limit state is therefore normally linked to a corresponding relaxed target reliability level for failure, often in the order of 10^{-1} to 10^{-2} . See also subsection 3.3.3.

To support corrosion of the reinforcement, a certain level of humidity is needed. For structural elements solely exposed to a relative dry indoor environment, a limit state “depassivation” may not be relevant as no significant corrosion will develop.

The following requirement must be fulfilled:

$$P\{\} = P_{dep.} = P\{c - x_c(t_{SL}) < 0\} < P_0 \quad (7.8-1)$$

where:

- $P\{\}$ is the probability that depassivation occurs;
- c is concrete cover [mm];
- $x_c(t_{SL})$ is carbonation depth at time t_{SL} [mm];
- t_{SL} is design service life [years];
- P_0 is target failure probability.

The variables c , and $x_c(t_{SL})$, need to be quantified in a probabilistic approach.

Subsection 7.8.2.1.2 and Eq. (7.8-3) advise the designer to base the design on extrapolated field experience from similar structures to reduce the considerable uncertainty in model and data.

fib Bulletin 34, “Model Code for Service Life Design” (**fib**, 2006) gives further information on the use of the model given as Eq. (7.8-3).

The weather function $W(t)$ may be assumed to obey the function:

$$W(t) = \left(\frac{t_o}{t} \right)^w \quad (7.8-4)$$

where:

- t_o is time of reference [years];
- t is the considered exposure time [years];
- w is weather exponent [-] ($w = 0$ for indoor conditions and increasing to values of $1 > w > 0$ with increasing frequency of wetting events):

$$w = (P_{SR} T_O W)^{b_w} \quad (7.8-5)$$

where:

- $T_O W$ is time of wetness [-];
- P_{SR} is probability of driving rain [-];
- b_w is exponent of regression [-].

fib Bulletin 34, “Model Code for Service Life Design” annex B (**fib**, 2006) gives further information on how these conditions influence the factors governing $W(t)$ and k .

Both the uncertainty of the data and in the model have to be taken into account in the design.

When deriving the product of $W(t)$ and k from existing structures, the influence of these uncertainties will decrease considerably the older the structure is.

Subsection 5.1.13 and Eq. (5.1-141) give guidance in the case the predictions have to be based on accelerated testing on young specimens.

An overview of other models in use, and a database with supporting parameters, is given in Bamforth, P., Concrete Society Technical report no 61: 2004 “Enhancing reinforced concrete durability”.

Reinforcement corrosion leading to cracking, spalling and collapse depend to a large extent on the environment at the concrete surface. The micro environment may vary considerably along the concrete surface of structural elements. The most unfavourable micro environmental conditions are frequent wetting and drying and/or accumulation of aggressive agents (for instance chlorides originating from seawater or de-icing salts). Macro-cell corrosion effects may trigger high corrosion rates in areas with less severe micro environmental condition. For given degrees of corrosion the risk for cracking and spalling depends on the geometry of the cross-section. Most vulnerable cross-sectional areas, for example the edges of beams, should be chosen as decisive for design.

7.8.2.1.2 Design model

The ingress of the carbonation front may be assumed to obey the following equation:

$$x_c(t) = W(t) \cdot k \cdot \sqrt{t} \quad (7.8-3)$$

where:

- k is a factor reflecting aspects such as the execution, basic resistance of the chosen concrete mix (like w/binder-ratio, cement type, additions) under reference conditions and the influence of the basic environmental conditions (such as mean relative humidity and CO_2 concentration) against ingress of carbonation;
- $W(t)$ is a weather function taking the meso-climatic conditions due to wetting events of the concrete surface into account.

For the design of a new structure, the factors $W(t)$ and k , or their product, may be derived from literature data or existing structures where the concrete composition, execution and exposure conditions have been similar to those expected for the new structure.

When assessing the remaining service life of an existing structure, the product of $W(t)$ and k may be derived directly from measurements on the structure.

Alternatively to Eq. (7.8-3), the design may be based on the design model given in subsection 5.1.13.2, Eq. (5.1-141).

Other models may be used, provided that the basic principles formulated in section 4.4 are fulfilled.

7.8.2.1.3 Limit states: corrosion-induced cracking, spalling and collapse

Exemplified with regard to cracking, the following basic limit state function has to be fulfilled:

$$P\{\} = P_{crack} = P\{\Delta r_{(R)} - \Delta r_{(S)}(t_{SL}) < 0\} < P_0 \quad (7.8-6)$$

where:

- $P\{\}$ is the probability that carbonation-induced cracking occurs;
- $\Delta r_{(R)}$ is the maximal corrosion induced increase of the rebar radius which can be accommodated by the concrete without formation of cracks at the concrete surface [μm];
- $\Delta r_{(S)}(t_{SL})$ is increase of the rebar radius due to reinforcement corrosion [μm];
- t_{SL} is design service life [years];
- P_0 is target failure probability.

An alternative design approach is:

$$P\{\} = P_{crack} = P\{t_{SL} - t_{ini} - t_{prop} > 0\} < P_0 \quad (7.8-7)$$

First approaches exist to quantify the variables $\Delta r_{(S)}$, (t_{SL}) and $\Delta r_{(R)}$. Most of the corresponding models are empirically derived, often based on very limited, in consequence insufficient, data basis. The correlation between corrosion rates/concrete quality/micro environment is not yet quantified in detail. The same applies to the limit states spalling and collapse. To get first impressions on the propagation period, *fib* TG 5.6, when preparing *fib* MC SLD, organized a Delphic oracle. One result of the exposure dependent output of this Delphic oracle is given in *fib* Bulletin 34, “Model Code for Service Life Design”, Annex R (*fib*, 2006). Together with existing models describing the initiation period and the herewith overall quantified propagation period, probabilistic calculations with regard to corrosion induced cracking, spalling and collapse of concrete structures may be performed, see Eq. (7.8-6) and Eq. (7.8-7)

where:

$P\{\}$ is the probability that carbonation-induced cracking occurs;

t_{SL} is design service life [years];

t_{ini} is initiation period [years];

t_{prop} is propagation period [years];

P_0 is target failure probability.

The variables $\Delta r_{(R)}$ and $\Delta r_{(S)}(t_{SL})$ or the variables t_{ini} and t_{prop} need to be quantified in a probabilistic approach.

Other methods may be used, provided that the basic principles formulated in section 4.4 are fulfilled.

At the time of publishing this Model Code, no time-dependent model with general international consensus was available for this deterioration process. The time span from initiation to cracking may be estimated from existing structures where the concrete composition, execution and exposure conditions have been similar to those expected for the structure considered.

7.8.2.2 Partial safety factor format

7.8.2.2.1 Limit state: depassivation

The following limit state function has to be fulfilled:

$$c_d - x_{c,d}(t_{SL}) \geq 0 \quad (7.8-8)$$

where:

c_d is design value of the concrete cover [mm];

$x_{c,d}(t_{SL})$ is design value of the carbonation depth at time t_{SL} [mm].

The design value of the concrete cover c_d is calculated as follows:

$$c_d = c_{nom} - \Delta c \quad (7.8-9)$$

where:

c_{nom} is nominal value for the concrete cover [mm];

Δc is safety margin of the concrete cover [mm].

The design value of the carbonation depth at time t_{SL} , $x_{c,d}(t_{SL})$ is calculated as follows:

$$x_{c,d}(t_{SL}) = x_{c,c}(t_{SL}) \cdot \gamma_f \quad (7.8-10)$$

where:

$x_{c,c}(t_{SL})$ is characteristic value of the carbonation depth at time t_{SL} [mm];

γ_f is partial safety factor of the carbonation depth [-].

Other methods may be used, provided that the basic principles formulated in section 4.5 are fulfilled.

The nominal value for the concrete cover is the dimension given to the constructor in the project specification (i.e. on drawings) and is assumed to represent the mean value of the cover depth.

The safety margin, Δc is to ensure that the great majority (in operational standards often assumed as the 95% fractile) of the cover thickness for the reinforcement bars is larger than the minimum cover used as a basis for the service life design.

ISO 22966 “Execution of concrete structures” assumes $\Delta c = 10$ mm if no other values are given in the execution specification.

To exemplify the design procedure and the quantification of the given quantities, an applicable design method is given in Annex C of *fib* Bulletin 34, “Model Code for Service Life Design” (*fib*, 2006).

For given design service lives, basic requirements with regard to minimum cover to the reinforcement, limiting values for concrete composition, such as maximum w/binder ratio, crack width limitation and minimum level of workmanship are given in most operational concrete codes. These sets of requirements should be calibrated according to chapter 4.

An example of such a calibration is given in Maage, M., Smeplass, S., “Carbonation – A probabilistic approach to derive provisions for EN 206-1” DuraNet, Third workshop, Tromsø, Norway, June 2001. Reported in “Betongkonstruksjoners Livsløp” report No. 19, Norwegian Road Administration.

As with carbonation, there were no available models with broad international consensus available for predicting the length of the corrosion period till cracking, spalling or collapse of the structure occurs when this Model Code was published. For this reason service life designs are normally based on the limit state of depassivation (reaching a critical chloride concentration at the rebar surface). As with carbonation, this rather conservative limit state is then normally linked with a corresponding relaxed target reliability level for failing, often in the order of 10^{-1} to 10^{-2} . See also subsection 3.3.3.

For permanently submerged members, lack of oxygen may hinder the corrosion process. However, for submerged members in electrical contact with members above sea level, macro-cell corrosion may occur.

fib Bulletin 34, “Model Code for Service Life Design” (**fib**, 2006) gives further information on the use of the model given as Eq. (7.8-12) and the influence of the various conditions discussed above.

Both the uncertainty of the data and in the model have to be taken into account in the design. Suggestions on how to handle this in some situations by excluding this convection zone (concrete layer, up to which the process of chloride penetration differs from Fick’s second law of diffusion) from the calculations are given in **fib** Bulletin 34.

7.8.2.3 Deemed-to-satisfy design

Within this approach a trading-off of geometrical (concrete cover to reinforcement), material parameters (indirectly linked to diffusion and binding characteristics) and execution aspects (compaction and curing) is applied.

7.8.2.4 Avoidance-of-deterioration design

Generally, avoidance is achieved if depassivation cannot take place due to infinite resistance of the concrete to carbonation or zero environmental load or infinite corrosion resistance of the reinforcement.

7.8.3 Chloride induced corrosion – uncracked concrete

7.8.3.1 Probabilistic safety format

7.8.3.1.1 Limit state: depassivation

The following limit state function must be fulfilled:

$$P\{\} = P_{dep.} = P\{C_{Crit} - C(c, t_{SL}) < 0\} < P_0 \quad (7.8-11)$$

where:

- $P\{\}$ is the probability that depassivation occurs;
- C_{Crit} is the critical chloride content to achieve depassivation of the reinforcement [wt.-%/binder content];
- $C(c, t_{SL})$ is chloride content at depth c and time t [wt.-%/binder content];
- c is concrete cover [mm];
- t_{SL} is design service life [years];
- P_0 is target failure probability.

The variables c , C_{crit} and $C(c, t_{SL})$ must be quantified in a probabilistic approach.

7.8.3.1.2 Design model

The ingress of chlorides in a marine environment may be assumed to obey the modified Fick’s second law of diffusion:

$$C(x, t) = C_s - (C_s - C_i) \cdot \left[\operatorname{erf} \frac{x}{2 \cdot \sqrt{D_{app}(t) \cdot t}} \right] \quad (7.8-12)$$

where:

- $C(x, t)$ is content of chlorides in the concrete at a depth x (structure surface: $x = 0$ m) and at time t [wt.-%/binder content];
- C_s is chloride concentration at the concrete surface [wt.-%/binder content];
- C_i is initial chloride content of the concrete [wt.-%/binder content];
- x is depth with a corresponding content of chlorides $C(x, t)$ [mm];
- $D_{app}(t)$ is apparent coefficient of chloride diffusion through concrete [$\text{m}^2/\text{sec.}$] at time t (see also subsection 5.1.13.3);
- t is time [years] of exposure;
- erf is error function.

When deriving the factor $D_{app}(t_0)$ from existing structures, the influence of these uncertainties will increase considerably the younger the structure is.

This is also the case when the reference diffusion coefficient is derived from a short term laboratory test on young specimens. It will then not reflect execution or ageing effects and if used for design often result in an uneconomical over-design.

The decrease of $D_{app}(t)$ is due to several reasons:

- the continued reaction of the binder, in particular when pozzolans and slag are used;
- the influence of capillary suction of water in the surface zone (decreasing with time);
- the degree of saturation of concrete;
- in this respect, the beneficial effect of penetrated chlorides from the seawater or de-icing salt leading to ion exchange with subsequent pore-blocking in the surface layer.

The uncertainty in determining the ageing-factor α increases if the span in time, represented by the observations, is shorter.

The uncertainty in using an ageing factor α for design for service life periods far beyond that represented by the observations used to derive α , should also be considered.

In the given model, the surface chloride concentration C_s for practical reasons is assumed constant over the time of exposure. This is not correct as a build-up period is obviously needed. However, it is often assumed that this build-up period is relatively short compared with the normal length of a design service life and that the influence of this simplification is limited on the design result. More sophisticated models where the build-up period of C_s is modelled are also sometimes used.

A realistic C_s for new designs should be based on in-field observations on existing structures with similar material composition and exposure. For assessing remaining service life, C_s may be measured directly on the structure in question.

The critical chloride content C_{crit} , when the depassivation of the reinforcement occurs, is also a variable depending on factors such as steel properties, type of cement and additions, moisture content etc. For practical design, a realistic distribution has to be applied. The fundamental knowledge on this property is still limited, and most numbers reported in the literature are based on in-field observations. Often these observations are grouped in tables with qualitatively described ranges as “negligible risk of corrosion”, “possible risk of corrosion” and so on. When performing a probabilistic service life design, it is imperative to make clear if the actual limit state is linked to the occurrence of corrosion (based on the mean and the statistical spread of C_{crit}), or to a conservative deterministic value of for example “negligible risk” or “possible risk”.

For structures subject to cyclic exposure of chlorides, in particular where de-icing salts are used only during the winter season, the chloride ions in the surface layer are not only transported by diffusion. Other effects such as capillary suction and wash-out will also play a role. The depth of this influenced surface layer will be reduced the more impermeable the concrete is and the shorter the cycles are. Suggestions for how to handle such situations are given in *fib* Bulletin 34, “Model Code for Service Life Design” (*fib*, 2006) and in the general literature.

An overview including other models, and a database for supporting parameters, are given in Bamforth, P., Concrete Society Technical Report No. 61: 2004 “Enhancing reinforced concrete durability”.

$$D_{app}(t) = D_{app}(t_0) \left(\frac{t_0}{t} \right)^\alpha \quad (7.8-13)$$

where:

$D_{app}(t_0)$ is the apparent diffusion coefficient measured at a reference time t_0 ;

α is the ageing-factor giving the decrease over time of the apparent diffusion coefficient. Depending on the type of binder and the micro environmental conditions the ageing factor is likely to be between 0.2 and 0.8.

For the design of a new structure, the factors C_s , C_i , α and $D_{app}(t_0)$ may be derived from existing structures where the concrete composition, execution and exposure conditions have been similar to those actual for the new structure.

When assessing the remaining service life of an existing structure, the factors – with the possible exception of α – may be derived directly from measurements on the structure.

Both for design of new structures, and for the assessment of remaining service life of existing structures, the ageing-factor α must be obtained from in-field observations from structures where the concrete composition, execution and exposure conditions have been similar to those for the actual structure. At least observations at two periods of exposure (with a sufficient interval between the observations) are needed for the calculation of the ageing factor.

Other models may also be used, provided that the basic principles formulated in section 4.4 are fulfilled.

Subsection 7.8.2.1.3 applies, with “chloride penetration” substituted for “carbonation”.

Subsection 7.8.2.2 applies, with “chloride penetration” substituted for “carbonation”.

Subsection 7.8.2.3 applies, with “chloride penetration” substituted for “carbonation”.

Subsection 7.8.2.4 applies, with “chloride penetration” substituted for “carbonation”.

The corrosion rates in the region of cracks crossing the reinforcement are extremely dependent on the micro climatic conditions at the concrete surface and the orientation of the concrete surface. Most severe conditions occur in the case of horizontal concrete surfaces and both cracks and chloride ingress from the top. For usual design service lives of more than 10 years and frequent chloride attack (e. g. parking decks in regions where de-icing salts are used) special protective measures are necessary to avoid the rapid penetration of chlorides to the reinforcement (e. g. linings or crack-bridging coatings). In the case of vertical surfaces and soffits with chloride spray or chloride containing water not leaking through cracks, a high quality of concrete cover (cover thickness ≥ 50 mm, low permeability concrete with low water-binder ratio) and ordinary crack width limitation ($w_k \leq 0.3$ mm) ensures sufficiently long service life (≥ 50 years) without extra protection.

In case of carbonation induced corrosion an adequate quality of concrete cover and ordinary crack width limitation ensures sufficiently long service life (≥ 50 years) without extra protection.

fib Bulletin 33 “Durability of post-tensioning tendons” (*fib*, 2006) describes multibarrier systems for the protection of prestressing systems. The bulletin describes three different “protection levels”. To ensure the integrity of the component, more barriers are recommended, the harsher the exposure conditions for the structure. These systems are supposed to satisfy the design criteria with ample margin and may be classified according to chapter 4 as deemed-to-satisfy requirements based on the avoidance-of-deterioration approach.

7.8.3.1.3 Limit states: corrosion-induced cracking, spalling and collapse

See subsection 7.8.2.1.3.

7.8.3.2 Partial safety factor format

See subsection 7.8.2.2.

7.8.3.3 Deemed-to-satisfy design

See subsection 7.8.2.3.

7.8.3.4 Avoidance-of-deterioration design

See subsection 7.8.2.4.

7.8.4 Influence of cracks upon reinforcement corrosion

The minimum structural reliability of a cracked reinforced concrete structure has to be of comparable magnitude to the minimum reliability of a comparable exposed uncracked structure.

A simplified approach is used by most operational standards. This implies that corrosion of the reinforcement is not influenced by crack widths under a certain characteristic value. Depending on the severity of the environment and sensitivity of the structure, this limiting crack width is normally given as a characteristic value (5% upper fractile) in the range of 0.2 to 0.4 mm

Similar to the procedure given in subsections 7.8.2 and 7.8.3, unwanted events with regard to serviceability/functionality have to be identified (SLS). In addition, it must be checked whether ultimate limits are affected by continuously corroding reinforcement within the cracked zone or not.

In harsh exposure conditions (e. g. exposure classes XD3/XS3), if functionality or structural integrity is affected, and if inspection and possible intervention cannot be performed, an avoidance-of-deterioration approach is recommended/needed.

7.8.5 Risk of depassivation with respect to prestressed steel

Relevant application rules given in subsections 7.8.2, 7.8.3 and 7.8.4 to avoid depassivation of prestressed steel on an ULS reliability level must be applied.

Since corrosion of prestressed steel can result in a sudden collapse of the structure, and without previous warning by cracking and spalling associated with ordinary reinforcement, a higher level of reliability for not passing the depassivation limit state is normally needed than for ordinary reinforcement.

7.8.6 Freeze-thaw attack

7.8.6.1 Probabilistic safety format

7.8.6.1.1 Limit state: freeze-thaw damage causing local loss of mechanical properties, cracking, scaling and loss in cross-section – without de-icing agents or seawater

The following limit state function must be fulfilled:

$$P\{\} = P_{\text{freeze-thaw damage}} = P\{T(t < t_{SL}), S_{CR} - S_{ACT}(t < t_{SL}) < 0\} < P_0 \quad (7.8-14)$$

where:

$P\{\}$ is the probability that freeze-thaw damage occurs;

S_{CR} is critical degree of saturation [-];

$S_{ACT}(t)$ is actual degree of saturation at time t [-];

$T(t)$ is concrete temperature in [K];

t_{SL} is design service life [years];

P_0 is target failure probability.

The variables S_{CR} and $S_{ACT}(t)$ need to be quantified in a probabilistic approach.

Other methods may be used, provided that the basic principles formulated in section 4.4 are fulfilled.

7.8.6.1.2 Limit state equation for the salt-freeze-thaw induced surface scaling – with de-icing agents or seawater

The following limit state function must be fulfilled:

$$P\{\} = P_{\text{scaling}} = P\{T(t \leq t_{SL}) - T_R(t, RH, n_{\text{cycles}}, Cl^-, \dots) < 0\} < P_0 \quad (7.8-15)$$

where:

$P\{\}$ is the probability that scaling occurs;

$T(t)$ is concrete temperature in [K];

$T_R(t, \dots)$ is critical freezing temperature for scaling to occur at time t ;

t_{SL} is design service life [years];

Cl^- is chloride content [wt. -%/binder content];

RH is relative humidity [%];

n_{cycles} is number of freeze-thaw cycles [-];

P_0 is target failure probability.

The variables T and T_R must be quantified in a probabilistic approach.

Other methods may be used, provided that the basic principles formulated in section 4.4 are fulfilled.

7.8.6.1.3 Limit states: freeze-thaw induced deflection and collapse

With regard to load-carrying capacity and deformations, the design must include the effect of localized changes in mechanical properties due to freeze-thaw damage.

7.8.6.2 Partial safety factor format

The following limit state function needs to be fulfilled:

$$S_{CR,d} - S_{ACT,d}(t < t_{SL}) \geq 0 \quad (7.8-16)$$

where:

$S_{CR,d}$ is design value of the critical degree of saturation [-];

$S_{ACT,d}(t < t_{SL})$ is design value of the actual degree of saturation at time t [-];

t_{SL} is design service life [years].

To exemplify the design procedure and the quantification of the parameters, an applicable design method is given in Annex B of *fib* Bulletin 34, “Model Code for Service Life Design” (*fib*, 2006).

Equation (7.8-14) is based on the assumption that scaling occurs at the same moment as the concrete surface temperature falls below a certain critical level, the scaling resistance T_R .

It is assumed that this critical level of scaling resistance changes with age, depending on exposure and type of concrete.

To exemplify the design procedure and the quantification of the parameters, an applicable design method is given in Annex B of *fib* Bulletin 34, “Model Code for Service Life Design” (*fib*, 2006).

Instead of partial factors multiplying the actions or dividing the resistance, for the partial safety factor format of the freeze-thaw attack the same effect is obtained by summing or subtracting a value.

For structures exposed to freezing and with the potential for a higher level of moisture than the critical degree of saturation, a design according to the deemed-to-satisfy method is the normal procedure.

As an example, the International standard ISO 22965-1 “Concrete Part 1: Methods of specifying and guidance to the specifier” suggests the following set of requirements on the concrete composition for such structures:

- cement type and maximum w/c-ratio;
- minimum strength and/or minimum cement content;
- freeze-thaw resistant aggregates;
- air entrainment or, if allowed by national provisions, adequate performance demonstrated by freeze-thaw testing (scaling).

Similar sets of requirements, based on long term field experience, are found in most national, regional and international standards.

Due to the complexity in performing design according to the probabilistic method, the calibration of these deemed-to-satisfy requirements is normally based on an avoidance-of-deterioration approach.

This assumes that the freeze-thaw resistance is not reduced over time for a concrete fulfilling the above listed criteria.

Due to absence of freezing, and/or moisture levels below S_{CR} , any detrimental reaction due to freezing will be avoided.

For model for the corrosion depth d caused by acid attack see subsection 5.1.13.6, Eq. (5.1-146).

The design value of the critical degree of saturation must be calculated as follows:

$$S_{CR,d} = S_{CR,min} - \Delta S_{CR} \quad (7.8-17)$$

where:

$S_{CR,min}$ is characteristic value of the critical degree of saturation (minimum value) [-];

ΔS_{CR} is margin of the critical degree of saturation [-].

The design value of the actual degree of saturation at time t , $S_{ACT,d}(t)$ has to be calculated as follows:

$$S_{ACT,d}(t) = S_{ACT,c}(t) + \Delta S_{ACT} \quad (7.8-18)$$

where:

$S_{ACT,c}(t)$ is characteristic value of the actual degree of saturation at time t [-];

ΔS_{ACT} is margin of the actual degree of saturation (load) [-].

Other methods may be used, provided that the basic principles formulated in section 4.5 are fulfilled.

7.8.6.3 Deemed-to-satisfy approach

Within this approach a trading-off of available space for expansion (air entrainment), material (non-freezeable water characteristics) and ageing (carbonation) variables must be established.

7.8.6.4 Avoidance-of-deterioration method

Generally, avoidance is achieved if freeze-thaw deterioration cannot take place due to infinite material resistance or zero environmental load.

7.8.7 Chemical attack

7.8.7.1 Acid attack

7.8.7.1.1 Probabilistic safety format

At the time of publishing this Model Code, models to be included here are still under discussion. A probabilistic approach for SLD is therefore not feasible.

7.8.7.1.2 Partial safety factor format

At the time of publishing this Model Code, models to be included here are still under discussion. A partial safety factor approach for SLD is therefore not feasible.

7.8.7.1.3 Deemed-to-satisfy approach

Within this approach one or more of the following remedies are normally required based on calibration to long term experience:

- limitations to the porosity of the concrete (traditionally expressed by max. w/binder ratio, min. cement content and moisture curing);
- composite cements or the use of additions like silica fume, fly ash and slag;
- type of aggregates.

7.8.7.1.4 Avoidance-of-deterioration approach

The use of linings and protective coatings may avoid contact with acids from the external environment.

Solutions with pH more than about 6.5 are considered as not detrimental to the cement paste based on Portland clinker.

In the case of severe acid attack, linings or durable coatings should be provided to avoid direct contact of the acid with the concrete surface.

7.8.7.2 Sulphate attack

When hardened cement paste is attacked by sulphates of various bases, two deterioration mechanisms may occur: ettringite formation and thaumasite formation.

Ettringite (sulphoaluminate from C_3A) formation of calcium causes expansion and may result in disruption of the hardened paste.

In presence of Calcium Carbonate and low temperatures ($5-8^{\circ}\text{C}$) thaumasite may be formed causing disruption of the hardened cement paste without expansion.

7.8.7.2.1 Probabilistic safety format

At the time of publishing this Model Code, no time-dependent model with general international consensus was available for this deterioration process. A probabilistic approach for SLD is therefore not feasible.

7.8.7.2.2 Partial safety factor format

At the time of publishing this Model Code, no time-dependent model with general international consensus was available for this deterioration process. A partial safety factor approach for SLD is therefore not feasible.

7.8.7.2.3 Deemed-to-satisfy approach

Within this approach, one or more of the following remedies are normally required based on calibration to long term experience:

- maximum amount of C_3A in the cement. Cements with C_3A content less than 3–5% and low C_3S content, and blast furnace cements with more than 60% slag are often regarded as “sulphate resistant”;
- composite cements or the use of additions like silica fume, fly ash and slag;
- limitations to the porosity of the concrete (traditionally expressed by max. w/binder ratio, min. cement content and moisture curing);
- avoidance of limestone aggregates or fillers in case of the risk of thaumasite formation.

In some national standards for Sulphate resistant cements a requirement on maximum content of $(C_4AF + 2(C_3A))$ is given in addition to the C_3A requirement.

7.8.7.2.4 Avoidance-of-deterioration approach

To support ettringite formation, sulphate ions and C_3A need to be present in the concrete member. If one or both of these conditions are below certain concentrations, avoidance of deleterious reactions will be achieved. In the case of thaumasite formation, even low amounts of calcium carbonate may be sufficient to allow for slow deterioration of the hardened cement paste.

In case of severe sulphate attack linings or durable coatings should be provided to avoid direct contact of the acid with the concrete surface.

7.8.8 Alkali–aggregate reactions

Alkali–aggregate reactions involve deleterious chemical reactions between the alkalis in the paste and silica or carbonate constituents in the aggregate.

7.8.8.1 Probabilistic safety format

At the time of publishing this Model Code, models to be included here are still under discussion. A probabilistic approach for SLD is therefore not feasible.

7.8.8.2 Partial safety factor format

At the time of publishing this Model Code, models to be included here were still under discussion. A partial safety factor approach for SLD is therefore not feasible.

7.8.8.3 Deemed-to-satisfy approach

The total alkali is often expressed as Na_2O equivalent = $Na_2O + 0.658 \cdot K_2O$

Within this approach one, or a combination of the following remedies are normally required based on calibration to long term experience:

- maximum amount of alkalis originating from cement and other sources;
- composite cements or the use of additions like silica fume, fly ash and slag;
- use of lithium nitrite/hydroxide and other chemical additions;
- limitations to the alkali-reactivity of the aggregate;
- accelerated testing of concrete samples (e. g. “torture tests”, see commentary to subsection 7.8.1);
- keeping the humidity in the structure below that needed to support the reaction (for instance, dry indoor environment).

Due to possible differences in local geology, care should be taken to transfer experience from one region to another.

7.8.8.4 Avoidance-of-deterioration approach

The use of linings and protective coatings may avoid ingress of alkali from the external environment

To support an alkali–aggregate reaction, reactive aggregate, sufficient humidity and alkalis need to be present in the concrete member. If one or more of these conditions missing, avoidance of reaction will be achieved.

7.8.9 Delayed ettringite formation

If concrete is exposed to high temperature over a certain period in its early life, delayed ettringite formation can occur depending on humidity and concrete mix design (alkali content, chemical composition of cement, use of additions etc.).

7.8.9.1 Probabilistic safety format

At the time of publishing this Model Code, models to be included here are still under discussion. A probabilistic approach for SLD is therefore not feasible.

7.8.9.2 Partial safety factor format

At the time of publishing this Model Code, models to be included here are still under discussion. A partial safety factor approach for SLD is therefore not feasible.

7.8.9.3 Deemed-to-satisfy approach

Unless specified otherwise, the peak temperature of the concrete within a component exposed to wet or cyclic wet environment must not exceed 70°C, unless data are provided to prove that, with the combination of materials used, higher temperatures will have no significant adverse effect on the service performance of the concrete. See ISO 22966.

7.8.9.4 Avoidance-of-deterioration approach

To support delayed ettringite formation, high temperatures at early age, critical contents of SO₃ and C₃A of cement, enough water and sufficient calcium hydroxide in the porous solution need to occur in the concrete member. If one or more of these conditions are missing, avoidance of the reaction will be achieved.

Examples of accidental and/or exceptional events to be considered include loads by a gas explosion, collision loads, impact loads on roofs, loads by extremely high soil water, loads due to fire or earthquakes, consequences of human errors and deterioration effects.

This strategy accepts that some components of the system will fail, but requires that the total damage to the system is limited. When applying this strategy, it has to be verified that the structure has sufficient redundancy and possibilities to mobilize so-called alternative load paths. Redundancy significantly increases system robustness, and is often the only viable means of ensuring sufficient robustness. On the other hand, achieving redundancy may be difficult and/or expensive, and is therefore often limited to critical parts of the system.

This strategy does not accept that some components of the system fail. When applying this strategy, the structure has to be designed so that it is strong enough to withstand every possible load, for example by choosing components whose bearing resistance significantly exceeds the expected actions. In general, this strategy leads to disproportionately expensive solutions.

Non-structural measures are based on the general concept of avoiding actions (prevention) and reactions (protection) and mitigation of consequences. Non-structural measures can be very efficient for specific accidental actions.

According to the approach directed at avoiding actions, one reduces the risk, for example by barriers (for collision), quality assurance (for errors), maintenance (for deterioration) or prohibition (e.g. prohibition of transport of dangerous substances through a tunnel).

According to the approach directed at reducing the effect of the action, one reduces the risk, for example by sprinklers (for fire), relief openings (for explosions) or supplemental damping and/or base isolation (for earthquake).

7.9 Verification of robustness

7.9.1 General

It should be assured with adequate reliability that no accidental and/or exceptional events or failure of a structural component should cause disproportional damage of a large part of the structure or even total collapse of the whole structure. Hence, the robustness of the system (i.e. adequate safety in relation with accidental and/or exceptional events) must be analysed.

In order to attain adequate robustness of the system one or more of the following strategies may be followed:

- structural measures:
 - limiting the amount of structural damage;
 - designing the key structural components to withstand every possible load;
- non-structural measures:
 - reducing the probability that the action occurs or reducing the action intensity (prevention);
 - reducing the effect of the action on the structure (protection);
 - mitigating the consequences of failure.

In the design of conventional structures one might eliminate the use of an explicit robustness requirement and avoid verification of redundancy by taking into the design explicitly as many aspects related to specific accidental actions as possible and applying relevant non-structural measures to deal with them. This approach offers the possibility of reducing the risks in the most economical way. However, if robustness requirements are explicitly considered in design, it must be verified that the structure has sufficient redundancy and possibilities to mobilize alternative load paths. Accordingly, the redundancy of the system has to be verified. For conventional structures, the simplified approach for verification of redundancy is recommended, see subsection 7.9.2.

In the design of unconventional structures (e.g. large structures, new materials, new design concepts) robustness must be achieved by limiting the amount of total damage to the system. Accordingly, redundancy of the system has to be verified. For the verification of redundancy of unconventional structures, the simplified approach is recommended, see subsection 7.9.2.

In the case of existing structures, robustness of the system may be verified by assessing the effectiveness of applied structural and non-structural measures. In particular, redundancy of the system may be assessed according to the simplified procedure given in subsection 7.9.2.

Strategies for retrofitting an existing structure, which has inadequate safety in relation to accidental and/or exceptional events are in principle similar to those recommended for design and depend on the type of structure. However, the cost of improving the redundancy of the system (i.e. costs of bringing the structure into compliance with the alternative load path principle) may be so high as to make this approach impractical. Often the only rational approach for improving the robustness of an existing system is designing the key structural components so that they can resist any potential load.

7.9.2 Specific methods to improve robustness by structural measures

7.9.2.1 Robustness by creating an alternative loading path

The redundancy of a structural system is related to its strength and shape, where particularly the degree of static indeterminacy (number of potential alternative load paths) is important.

Besides, global design strategy based on classical requirements on sufficient ductility and tying of elements can be applied to achieve some degree of redundancy.

The refined verification procedure is extremely complicated and does not work for unforeseeable hazards.

It should be made sure that in the case of an extreme event an alternative loading path can be mobilized around an area of local damage.

Two procedures for progressive collapse analysis can be followed to verify the redundancy of a system: a *refined procedure* and a *simplified procedure*.

According to the *refined procedure*, the occurrence and the effect of an accidental and/or exceptional action (impact, explosion etc.) are simulated for all possible action scenarios. The damage of the structural components is calculated and the stability of the remaining structure is assessed. Next, the consequences are estimated in terms of the number of casualties and economic losses. Various measures can be compared on the basis of economic criteria.

According to the *simplified approach*, the situation is considered that any structural component that has significance for the bearing capacity of the structure can lose its bearing function. It is then verified if the remaining part of the structure can withstand the frequently occurring loads with a defined target reliability for alternative load paths, for a relatively short period of time, necessary to mobilize the users of the structure and/or to carry out necessary repair or strengthening.

This target reliability for alternative load paths depends on the normal safety target for the structure, the period under consideration (hours, days or months) and the probability that the element under consideration is removed (by other causes then already considered in design).

The probability that the element under consideration is removed by some cause, not yet considered in design, depends on the sophistication of the design procedure and on the type of structure.

For unconventional structures the probability of having some unspecified failure cause is substantial. If in addition new materials or new design concepts are used, unexpected failure causes become more likely.

For conventional structures it should be possible to include all relevant collapse origins in the design. Although it will always be possible to think of failure causes not covered by the design, those failure causes will have a remote likelihood and may be disregarded on the basis of theoretical arguments.

In a building with a frame structural system a storey-sway mechanism can be prevented and a beam-sway one enforced, by satisfying around all beam-column joints the condition:

$$\sum M_{Rc} \geq \gamma_{Rd} \sum M_{Rb} \quad (7.9-1)$$

where $\sum M_{Rc}$ and $\sum M_{Rb}$ are the sums of the moment resistances of the columns and the beams framing into the joint and $\gamma_{Rd} \geq 1.0$ a factor for over-strengths not explicitly modelled (e.g. due to steel strain hardening).

7.9.2.2 Capacity design

Capacity design ensures a hierarchy of resistances of structural components and failure modes such that pre-emptive brittle failure modes are prevented and ductile ones promoted. To this end, brittle failure modes are verified for force action effects obtained not by analysis but from equilibrium, assuming that all relevant ductile components develop their force capacities. Ductile structural components are designed then for the plastic deformations induced in them by the design actions.

Capacity design may also be used – in seismic design in particular – to prevent undesirable plastic mechanisms associated with limited global deformation capacity and enforce the intended one(s), involving only ductile structural components.

7.10 Verification of sustainability**7.10.1 Impact on environment****7.10.1.1 General**

The purpose of the verification of environmental performance is to:

- i. ensure that decision-makers consider environmental impacts before deciding whether or not to proceed with new projects;
- ii. differentiate the performance of alternative solutions with regards to their environmental impact.

The LCA consists of four phases:

- Phase I. Goal and scope definition;
- Phase II. Inventory analysis;
- Phase III. Impact assessment;
- Phase IV. Interpretation.

with the boundary of the LCA scoped in Phase I using appropriate assumptions. The LCA of environmental impacts is conducted in the entire life cycles, respective stages of the life cycles, or certain ranges of the life cycles.

The general information on LCA is obtained in following standards:

- ISO 14040 “Environmental Management – Life cycle assessment – Principles and framework”;
- ISO 14041 “Environmental Management – Life cycle assessment – Goal and scope definition and inventory analysis”;
- ISO 14042 “Environmental management – Life cycle assessment – Life cycle impact assessment”;
- ISO 14043 “Environmental management – life cycle assessment – Life cycle interpretation”;
- ISO 14044 “Environmental management – Life cycle assessment – Requirements and guidelines”.

However, the ISO 14000 series was developed for goods and services. The ISO 13315 series will provide the appropriate information for concrete in the future.

When an assessment is conducted, the system boundary must be set according to the purpose of the assessment and reliable inventory data prepared. The general principles, framework and basic rules on environmental management related to concrete and concrete structures are provided in ISO 13315-1.

In the integration of each impact using a unified index, a reasonable weighting among each effect is needed. When using commercially available software for an evaluation, an understanding of the purpose and characteristics of the software is needed in regards to analyses being undertaken.

System boundaries need to be defined with due consideration of each source component and its impact in the final decision. For example, a bridge project may be sensitive to the materials production, execution and transportation but almost nothing to energy cost savings through the life of the structure due to its thermal mass. On the other hand, for a building structure the impact of thermal mass on greenhouse gas (GHG) savings over the life of the structure may be significant and, thus, included within the system boundary.

The range of assessment for consideration of environmental impact should be proportionate with the scale of the structure or project being considered.

Current practice in many countries is assessment through broad-based green building rating schemes. The following are examples of some current national schemes: BREEAM in the UK and BREEAM-NL in the Netherlands, HQE in France,

When conducting design, construction, use, conservation and demolition of a concrete structure, as well as with regard to the process of recycling and disposal, it must be confirmed at the design stage that the environmental performance required for the structure is satisfied. This is undertaken using a life cycle assessment (LCA) analysis.

In conducting an assessment:

- the system boundary must be set with consideration to the contribution of each component to the total and the sensitivity to the result of excluding any component;
- reliable inventory data must be prepared with consideration to the location of the structure and the supply and manufacturing processes of source materials and products.

LEED in the USA, CASBEE in Japan, Green Star in Australia, and Green Star-NZ in New Zealand.

Such approaches, however, may not always lead to more sustainable products as they involve many assumptions. For example, in building structures, potential benefits of embodied energy, higher durability and lower maintenance are not generally considered in current schemes, although this may change in time.

An environmental impact calculation (EIC) through a full or partial LCA at the project, structural or structural element level can be undertaken as a measure of optimization at the level considered. Currently, there are three common measures adopted for such measurements: a measure of the embodied energy (EE), of the CO₂ emissions and of the global warming potential (GWP). One or more measures may be used in combination with results benchmarked against existing or common practice. Examples of such an approach are given in Appendix E of *fib* Bulletin 47 “Environmental design of concrete structures – general principles” (*fib*, 2008). Other measures may also be appropriate; ISO 13315-1 provides the framework and basic rules on environmental management related to concrete and concrete structures.

The GWP equates greenhouse gases and their expected influence on global warming to an equivalent volume of CO₂. It is an indicator of impact over a set time period with time periods typically of 20, 50 or 100 years. For example, the 100 GWP for the three more common greenhouse gas emissions (carbon dioxide, nitrous oxide and methane) is:

$$100\text{-year GWP} = \text{CO}_2 + 298\text{N}_2\text{O} + 25\text{CH}_4$$

and has units of kilograms of CO₂ equivalent per kilogram of material.

To undertake an EIC, an inventory of materials is needed and will vary from country to country and location to location, depending on local materials and practices.

The calculation must consider the total quantity of each material (e.g. concrete, reinforcing bar and strand) used in the structure, or structural component, under consideration with the total calculation being the sum of the volume of each material multiplied by their respective unit quantity as per the inventory of materials.

The inventory data should be appropriately collected and evaluated. The JSCE provides an example of current inventory data related to concrete (JSCE Guidelines for Concrete No.7: Recommendation of environmental performance verification for concrete structures (Draft), 2006). However, it should be noted that the inventory data varies depending on various conditions and may change significantly in the future.

Background information on verification of sustainability, as treated in this section, is given by Sakai, K. (2013), Sustainability in *fib* Model Code 2010 and its future perspective. Structural Concrete, 14. doi: 10.1002/suco.201300012.

Impact on the environment must be evaluated using applicable indexes for each environmental effect to be verified.

Performance requirements must be specified by criteria such as target indices (resources, energy, emission), means indices (reduce, reuse, recycle) and/or combined indices such as embodied energy or global warming potential.

7.10.1.2 Verification

Environmental aspects subject to verification are to be selected and then the set value (*S*) is determined as the performance requirement. Performance requirements may be given to the designer as a limiting value that can be used for verification, according to legal regulations or by instructions from the owner.

The performance (*R*), which is to be achieved by the structure or element, is quantified by using the same index or environmental aspect as that adopted for the regulation, limit or target value (*S*) for which *S* was set.

Environmental performance of a concrete structure must be verified by confirming that the retained performance (*R*), defined by using appropriate indexes with regard to the impact on environment, is larger (or smaller) than the set value (*S*) of the relevant performance requirement.

Whether or not a structure satisfies the performance requirements for the environmental aspects being the subject of verification, is confirmed by whether or not $R \geq S$ or $R \leq S$, as applicable. The direction of the inequality sign varies according to the type of environmental aspect being considered. For example, in cases involving the life cycle carbon dioxide emission, verification is achieved by setting the upper limit (S) and confirming that it is higher than the estimated carbon dioxide emission (R) of the concrete or structure, whichever is being considered.

In cases where the impact on the environment is to be reduced by a factor from a standard value, the verification is conducted by confirming that the ratio of the actual impact to the standard impact is lower than the specified impact reduction factor.

In cases involving recycled product substitution, it may be possible to decide that the requirement is met if the amount used (R) is larger than a predetermined minimum (S).

Where past results and other data have shown that concrete, reinforced concrete or a concrete structure satisfies a certain range of specifications (such that the constituent materials used for the concrete, the shape/size of the structure and the execution method are similar) and the performance is above a certain level, the specifications that are satisfied for the concrete or concrete structure may be used in place of performance verification, without conducting further quantitative evaluations of an individual structure or structural element.

The primary purpose of the verification of impact on society is to bring about a more sustainable and equitable biophysical and human environment.

When the aesthetics as an impact on society is considered in the design of a concrete structure, it must be confirmed that the aesthetics required for the structure are satisfied. The primary purpose of the verification of aesthetics is to create graceful structures in harmony with their environment at a reasonable cost.

A social impact assessment (SIA) is a methodology to review the social effects of infrastructure and building projects and other development interventions.

At present, there are no general criteria for evaluating quantitatively the impact on society of a structure. Therefore, depending on the purpose of the structures and the intents of designers, appropriate rating or target indexes should be set (with the weighting).

When the aesthetics is considered, performance requirements for aesthetics must be specified. At present, there is no general rating index for evaluating quantitatively the aesthetics of a structure. Therefore, depending on the purpose of the structures and the intents of designers, appropriate indexes to evaluate structural beauty, artistic beauty, scenery beauty, elegance and so on, should be set with the weighting

In the absence of a general rating index for evaluating quantitatively the impact on society of a project, it is possible to evaluate retained performance (R) on the impact on society of a project, for example, by a questionnaire to people. The designer will give social aspects to be considered and set the values (S) based on the regulation, the requests from an owner or his/her intents.

For elements where the performance has been previously established to meet the specification criteria, the environmental performance may be deemed to be satisfied.

7.10.2 Impact on society

7.10.2.1 General

When conducting, design, construction, use, conservation and demolition of a concrete structure as well as with regard to the process of recycling and disposal, it must be confirmed at the design stage that social performance required for the structure is satisfied.

The impact on society must be evaluated using appropriate indexes regarding the social effects of objects to be verified.

Performance requirements for impact on society must be specified.

7.10.2.2 Verification

Social performance of a concrete structure must be verified by confirming that the retained performance (R) defined by using appropriate indexes regarding the impact on society is larger (or smaller) than the set value (S) of relevant performance requirements.

Regarding the aesthetics, it must be verified by confirming that the retained performance (R) defined by using appropriate indexes regarding the aesthetic aspects to be verified is larger than the set value (S) of performance requirement.

In the absence of a general rating index for evaluating quantitatively the aesthetics of a concrete structure, it is possible to evaluate retained performance (R) on aesthetics of a concrete structure, for example, by a questionnaire to people. The designer will give aesthetical aspects to be considered and set the values (S) based on the regulation, the requests from an owner or his/her intents.

When a specification, in which the examples for the harmony of ratio, mass, light, perspective, colour and texture between various elements of a structure are provided, is utilized, the verification may be omitted.

Regarding the items for which performance can be ensured by providing specifications, verification may be omitted.

A paper with background information and examples on reliability-based non-linear analysis, as treated in this section, was written by Cervenka (Cervenka, V. (2013), Reliability-based non-linear analysis according to *fib* Model Code 2010. Structural Concrete, 14: 19–28. doi: 10.1002/suco.201200022).

7.11 Verifications assisted by numerical simulations

7.11.1 Purpose

Numerical simulation is performed by numerical methods of structural analysis based on non-linear solution methods and material models reflecting the expected real behaviour under a given type of loading and given environmental conditions.

Numerical simulation is another way of checking structural performance and is an alternative to physical testing in a laboratory or on the site. Unlike in real testing, simulation is not limited by structural size, load magnitude and testing facility. However, the power of computer simulation is limited by the chosen numerical model and its objectivity, so that adequate checks should be exercised to verify the results and conclusions.

The verification assisted by numerical simulations should ensure that the design leads to the reliability level required by this Model Code.

7.11.2 Methods of numerical simulation

7.11.2.1 Numerical model

The model formulation should be based on principles of mechanics, namely it should satisfy requirements on equilibrium of forces, compatibility of displacements and constitutive laws. The inevitable errors due to numerical approximations should be controlled by solution-error criteria. It is reasonable to maintain a balanced approximation between all model parts.

The complexity of the model grows with each new feature. Therefore, it is appropriate to include only those features, which are significant for the given case, in order to keep the model as simple as possible.

7.11.2.2 Finite element method

The finite element method is typically used for the numerical solution of a continuum problem. Depending on the type of formulation (stiffness, compliance and mixed methods) the results are by definition different from the exact solution.

In the most widely used stiffness method, the shape of displacement field is assumed and equilibrium is satisfied only in an integral sense. That means that the best possible equilibrium is found for a given approximation (finite element type and size). The internal stresses are lower, compared with an exact solution.

A finite element formulation should satisfy the requirement of convergence to the exact solution by reducing the element size (and/or increasing the number of degrees of freedom).

Thus, independently of the material model, the approximations introduced by the finite element formulation only, can be a significant source of errors in numerical analysis.

Similarly, in other numerical methods, the errors due to approximations should be adequately checked.

Typical algorithm for non-linear finite element analysis is schematically illustrated in Figure 7.11-1.

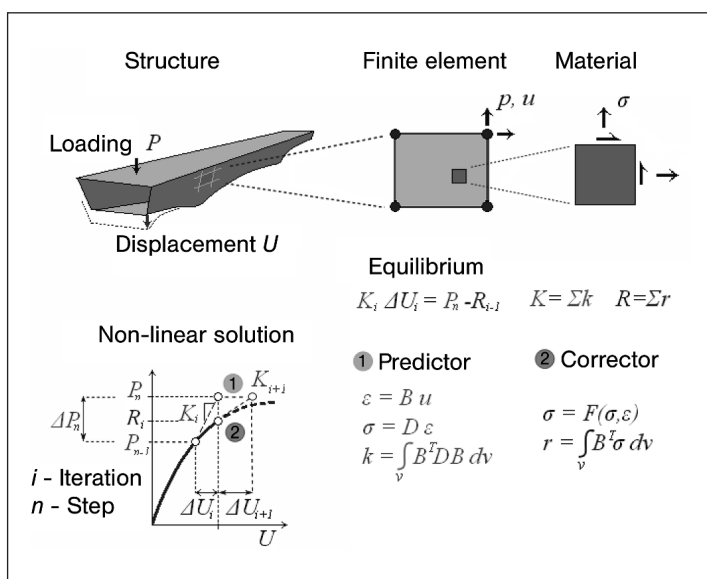


Figure 7.11-1: Typical algorithm for non-linear finite element analysis

The non-linear solution is performed by predictor-corrector iterative process. In the predictor, point 1, the solution is estimated by a linear analysis based on tangent or initial material stiffness D . The solution is improved in the corrector, point 2, based on non-

linear constitutive law F . The iterative process is stopped when the difference between predictor and corrector is acceptably small. A balanced approximation of numerical methods in all model parts – that is, in structural discretization, element formulation and material laws – should be maintained.

Errors of the non-linear solution are controlled by convergence criteria. Solution convergence is satisfied when the error is within prescribed limits. In case of the stiffness method, as illustrated in Figure 7.11-1, the most significant convergence criterion is the error in force equilibrium. In addition, increments of displacements or residual energy can be checked. Choice of adequate error tolerance is an important aspect of non-linear analysis. The admissible errors must be appropriately validated, for example by a convergence study, in which results obtained with different tolerances are compared.

Material models for concrete are usually based on the following theories:

- non-linear fracture mechanics;
- theory of plasticity with plastic flow rule;
- microplane theory;
- damage theory;
- viscoelastic and viscoplastic theory.

The choice of an appropriate theory depends on the specific application.

Basic material tests serve to validate the constitutive relations. Basic material tests are made on simple structures, with the aim of reducing the influence of geometry and boundary conditions under well-defined stress and strain conditions. Examples of such tests are compressive tests of concrete cylinders, fracture tests of concrete prismatic specimens subjected to three-point loading and tension stiffening tests in uniaxial tension of reinforcing bars embedded in concrete members. Each type of behaviour usually requires a special material test. These tests are typically described in codes for material testing, such as those recommended by RILEM.

The aim of structural tests is to validate the ability of algorithm or software to objectively reproduce a certain structural behaviour. This is often performed by means of benchmark calculations. For example, if a shear wall is to be simulated, a validation of the software by shear wall experiments should be made available. Such studies can be considered as a rational basis for choosing adequate material models and software for a given structure.

Mesh sensitivity tests should be performed, or made available, in order to validate the finite element mesh of the numerical model. At least three mesh cases with different element sizes should be tested and their effect on the resistance should be evaluated. In cases significant mesh sensitivity, when at least two different mesh sizes do not provide sufficiently similar results, the results must be considered as not objective.

7.11.2.3 Material models

Material models that are used for concrete, reinforcement and their interaction, must capture all significant and relevant features of material behaviour for the problem under consideration.

Constitutive laws should be based on principles of continuum mechanics and failure, and must ensure objectivity of the solution in the context of numerical methods.

Models for material softening – that is materials exhibiting a decrease of strength after reaching certain ultimate stress value – should include appropriate regularization techniques in order to reduce mesh sensitivity of strain-based formulations of constitutive laws.

7.11.2.4 Validation of numerical models

The procedures used in numerical analysis and the choices made with regard to methods (subsection 7.11.2.2) and material models (subsection 7.11.2.3), should be validated by experiments and benchmarks. This validation should include:

- basic material tests;
- structural tests;
- mesh sensitivity tests.

The objective of non-linear analysis is to simulate the real structural behaviour and to evaluate the representative value of resistance – see subsection 4.6.2.1. Such a task can also be formulated as a prediction of the most probable real resistance, which is the mean value of resistance. Therefore, the mean resistance is chosen as a reference for safety assessment by non-linear analysis. The uncertainty due to random variation of material properties (and possibly to other parameters of resistance) is described by the random variation of resistance. In addition, a model uncertainty must be included separately.

The design condition is compatible with the one applied within the partial factor design method, Eq. (4.5-8). However, the design resistance is in this case considered as a design variable and a global resistance method is applied.

The methods for the design resistance assessment are listed according to the level of their probabilistic justification. In the *probabilistic method* design resistance is explicitly evaluated for required failure probability or reliability index. In the *global resistance method* the design resistance is estimated based on simplified probabilistic approach assuming significant approximations. In the *partial safety factor method* the design resistance is determined directly using design values of random variables without evaluation of global safety. The *partial safety factor method* is comparable with the partial factor design method applied to global resistance. The choice of a relevant method for the calculation of the design resistance depends on engineering judgement while taking into consideration the resistance model and possible uncertainties. In practical design a usage of at least two methods is recommended, which provides independent verifications of limit state.

The probabilistic analysis based on numerical simulation with random sampling can be briefly described as follows:

- (1) Formulation of a numerical model based on non-linear finite element method. Such a model describes the resistance function $r(\mathbf{r})$ and can perform deterministic analysis of resistance for a given set of input variables.
- (2) Randomization of input variables (material properties, dimensions, boundary conditions etc.). This can also include some effects, which are not in the action function $s(\mathbf{s})$ (e.g. prestressing, dead load etc.). Random properties are defined by a random distribution type and its parameters (mean, standard deviation etc.). They describe the uncertainties due to variation of resistance properties. The randomization can be done by two methods: (a) *Random variables*, where the parameter is constant within a sample (structure), but changes between samples. (b) *Random fields*, where the parameter is randomly variable within a sample. A correlation of random variables should be considered appropriately.
- (3) Probabilistic analysis of resistance. This can be performed by the numerical method of Monte Carlo type sampling, such as the Latin hypercube sampling (LHS) method. Results of this analysis provide random parameters of resistance, such as mean, standard deviation and so on and the type of distribution function for resistance.

7.11.3 Safety formats for non-linear analysis

7.11.3.1 General

The design principles based on the global resistance design method described in section 4.6 apply, whereas the character of numerical simulation and non-linear analysis must be considered appropriately.

The design condition is considered in the form of Eq. (4.6-2) and (4.6-4) with separated uncertainties for actions and resistance.

$$F_d \leq R_d, \quad R_d = \frac{R_m}{\gamma_R^* \gamma_{Rd}} \quad (7.11-1)$$

where:

F_d is design value of actions;

R_d is design resistance;

R_m is the mean value of resistance;

γ_R^*, γ_{Rd} are global resistance safety factor and model uncertainty factor, respectively.

The design resistance R_d can be evaluated by various methods based on different levels of implementation of probabilistic theory:

- probabilistic method for required reliability index or failure probability;
- global resistance method;
- partial safety factor method.

7.11.3.2 Probabilistic method

The design value of resistance R_d can be evaluated by probabilistic analysis following the general safety format according to section 4.4. In this approach, the resistance function $r(\mathbf{r})$ is represented by non-linear structural analysis. The safety can be evaluated with the help of the reliability index β , or alternatively by the failure probability P_f taking into account all uncertainties due to random variation of material properties, dimensions and possibly other random effects.

Global design resistance R_d must be obtained as

$$R_d = \frac{1}{\gamma_{Rd}} R(\alpha \beta) \quad (7.11-2)$$

where:

$\gamma_{Rd} > 1$ is the model uncertainty factor;

$R(\alpha \beta)$ is the resistance corresponding to reliability index β , which is reduced by factor α in order to account for a separate safety assessment of resistance, that is only the right side of the design condition (4.6-1).

- (4) Evaluation of design resistance based on the reliability index β or probability of failure.

For evaluation of design resistance, that is only the right side of the design condition (4.6-1) is considered: the reliability index must be considered in the form of the product $\alpha\beta$, where β is the overall reliability index and α is the reduction factor for resistance. In the absence of more detail data, the value $\alpha\beta = 0.8 \cdot 3.8 = 3.04$ can be considered for the reliability index. This corresponds to the failure probability $P_f = 10^{-3}$ and to the average situation in design according to partial safety factor method.

In this method, the mean material properties for the calculation of resistance are considered, whereas different uncertainties of steel and concrete are accounted for. The safety approach can be demonstrated by a simplified reasoning when considering isolated limit states of each material and current partial factors.

The safety factors for the mean in the case of steel failure, γ_{GL}^{steel} is defined as:

$$f_{sd} \gamma_{GL}^{steel} = f_{sd} 1.1 \gamma_s = f_{sm} \quad (7.11-7)$$

$$\gamma_{GL}^{steel} = 1.1 \times 1.15 = 1.27 \quad (7.11-8)$$

where 1.1 is the factor for the mean strength and $\gamma_s = 1.15$ is the partial safety factor for the characteristic strength of the steel.

The safety factor for the mean in the case of concrete failure, $\gamma_{GL}^{concrete}$ is defined as:

$$f_{cd} \gamma_{GL}^{concrete} = f_{cd} 1.1 \gamma_c = f_{cm} \quad (7.11-9)$$

$$\gamma_{GL}^{concrete} = 1.1 \times 1.5 = 1.65 \quad (7.11-10)$$

where 1.1 is the factor for the mean strength and $\gamma_c = 1.5$ is the partial safety factor for the characteristic strength of the concrete.

Since the partial factors are formulated for the same failure probability, they yield different safety factors for the mean of these two materials. In order to prevent two different global factors, the mean strength of concrete is modified to provide for the greater safety required for concrete.

A new reduced mean concrete strength can be derived according to:

$$f_{cm} = \frac{1.1 \gamma_s}{\gamma_c} f_{ck} \cong 0.85 f_{ck} \quad (7.11-11)$$

It yields the new global safety factor of concrete:

$$\gamma_{GL}^{concrete} = \frac{f_{cm}}{f_{cd}} = \frac{0.85 f_{ck}}{\frac{f_{ck}}{1.5}} = 1.27 \quad (7.11-12)$$

The unique safety factor 1.27 for mean for steel and concrete is obtained.

7.11.3.3 Global resistance methods

7.11.3.3.1 Global resistance factor method

Design resistance is calculated from

$$R_d = \frac{r(f_m, \dots)}{\gamma_R \gamma_{Rd}} \quad (7.11-3)$$

The function r represents the non-linear analysis with mean input material.

Reinforcing steel properties:

$$f_{ym} = 1.1 f_{yk} \quad (7.11-4)$$

where f_{ym} is the mean value for the yield stress of the steel

Concrete properties:

$$f_{cf} = 1.1 \alpha_{cf} f_{cfk} \frac{\gamma_s}{\gamma_c} \quad (7.11-5)$$

where f_{cf} is the concrete strength parameter reduced to account for greater random variability of concrete. It can be compressive strength, tensile strength or fracture energy density. By analogy, other parameters entering the numerical model may be considered.

Coefficient α_{cf} accounts for sustained load effect and unfavourable effects of load application and represents the same uncertainty as given by α_{cc} , α_{ct} in subsection 7.2.3.1.4. The range of this parameter is (0.85; 1). Using other than the lower limit must be substantiated by nonexistence of such uncertainties.

f_{cfk} is a characteristic value of the relevant concrete property (compressive strength, tensile strength, fracture energy density).

This can be further simplified by inserting values of the partial factor for steel $\gamma_s = 1.15$ and for concrete $\gamma_c = 1.5$ resulting in:

$$f_{cmd} = 0.85 f_{ck} \quad (7.11-6)$$

The reduction factor 0.85 must be used to obtain the reduced mean values of other material properties of concrete such as tensile strength, fracture energy or bond strength.

The partial factor of resistance has to be $\gamma_R = 1.2$

The model uncertainty factor must be considered by the factor $\gamma_{Rd} = 1.06$

Probabilistic studies indicate that the random distribution of resistance of reinforced concrete members can be described by a two-parameter lognormal distribution with the lower bound at the origin. Therefore, the present method is based on the assumption of a lognormal distribution, which can be identified by two random parameters: R_m – mean resistance and V_R – coefficient of variation of resistance. Other parameters, such as a characteristic resistance, can be derived since they are dependent on the above definition. (The method is referred to as ECOV.)

Based on these assumptions, it is proposed to identify the random distribution of the resistance by mean R_m and characteristic R_k values. The following expression for the characteristic value can be regarded as an acceptable approximation:

$$R_k = R_m \exp(-1.65 V_R) \quad (7.11-17)$$

Assuming the probability for the characteristic value as 0.05 the coefficient of variation can be expressed as:

$$V_R = \frac{1}{1.65} \ln \left(\frac{R_m}{R_k} \right) \quad (7.11-18)$$

Finally, the global factor for the mean resistance γ_R can be calculated from:

$$\gamma_R = \frac{R_m}{R_d} = \exp(\alpha_R \beta V_R) \quad (7.11-19)$$

The values of parameters according to subsection 3.3.3.1 of this Model Code can be considered to be $\alpha = 0.8$, $\beta = 3.8$, corresponding to probability $P_f = 10^{-3}$, which leads to the final expression for the global factor as a function of coefficient of variation of resistance:

$$\gamma_R^m = \exp(3.04 V_R) \quad (7.11-20)$$

It should be emphasized, that this method depends on reliable and justified values for the mean and characteristic values of material parameters used for input of resistance analysis and relevant to the failure mode under consideration.

In this case the structural analysis is based on extremely low material parameters in all locations, which does not correspond to the probabilistic concept of simulation. This may cause deviations in structural response, for example in failure mode. However, the case studies indicate that the partial factor method is practically applicable. The method of partial factors may be used as a safe estimate in the absence of a more refined solution.

The model uncertainty factor γ_{Rd} , as proposed in subsection 4.5.1.3 (b) should be used appropriately. In a standard situation the model uncertainty is included in the partial safety factor $\gamma_M = \gamma_m \gamma_{Rd}$ and reflected in design values of basic material parameters $f_d = f_m / \gamma_m$. Therefore, in the partial factor method the model uncertainty factor need not be applied to the design resistance R_d .

7.11.3.2 Method of estimation of a coefficient of variation of resistance (ECOV)

Estimates of the mean and characteristic values of resistance have to be calculated using corresponding values of material parameters:

$$R_m = r(f_m, \dots), \quad R_k = r(f_k, \dots) \quad (7.11-13)$$

where r represents the non-linear analysis and f_m and f_k are mean and characteristic values of input material parameters, respectively.

The coefficient of variation V_R of resistance is determined from:

$$V_R = \frac{1}{1.65} \ln \left(\frac{R_m}{R_k} \right) \quad (7.11-14)$$

The global resistance factor γ_R for the mean is determined from:

$$\gamma_R = \exp(\alpha_R \beta V_R) \quad (7.11-15)$$

where α_R is a sensitivity (weight) factor for the reliability of resistance and β is a reliability index.

The design resistance is then calculated from:

$$R_d = \frac{R_m}{\gamma_{Rd} \gamma_R} \quad (7.11-16)$$

The global resistance factor γ_R is related to the random variation of resistance due to basic material parameters and failure modes. The model uncertainty factor for well validated numerical models is proposed as $\gamma_{Rd} = 1.06$. For low-level validation, higher model uncertainty values > 1.06 should be used.

7.11.3.4 Partial factor method

The method of partial factors can be applied, in which the design resistance R_d is calculated using the design values as input parameters f_d for the non-linear analysis:

$$R_d = r(f_d, \dots) \quad (7.11-21)$$

The design parameters f_d are obtained by the method described in subsection 4.5.1.3 and $r(f_d, \dots)$ represents the non-linear analysis model.

The advanced identification methods are based on numerical methods of probabilistic methods and optimization techniques. They are utilizing stochastic simulation based on Latin hypercube sampling combined with artificial neural network. This approach reflects the sensitivity of resistance to the input parameters and can offer a best fit set of material parameters.

7.11.4 Resistance parameter identification

Resistance parameter identification is an inverse method to the assessment of resistance. In a standard design situation the assessment of structural resistance is calculated for a given set of materials parameters using structural analysis. The numerical simulation, such as non-linear finite element analysis, is an advanced way of modelling structural resistance.

In an inverse task the structural resistance is known and material properties are sought using a structural model. This can be described as parameter identification for a certain structural model. The aim of identification is to identify the material parameters, which ensure the best fit of the calculated and given resistances for the considered structural model.

This method can be used for identification of material parameters, in basic material tests, or in tests of existing structures. Material parameters determined by identification can be used for simulation of other structures in similar conditions.

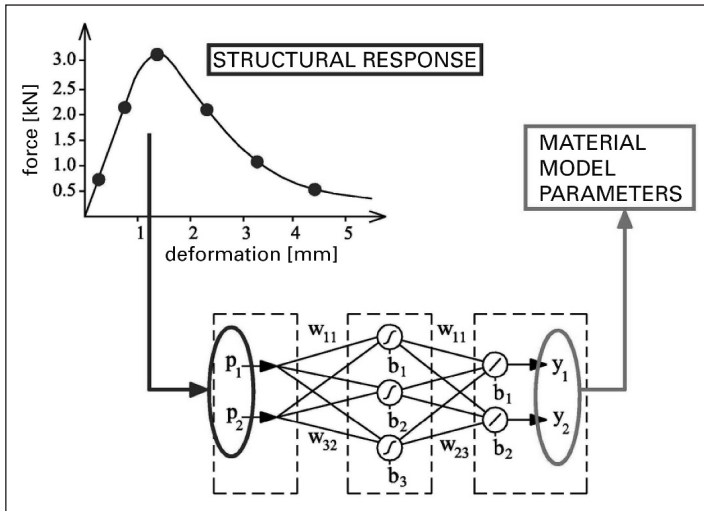


Figure 7.11-2: Typical algorithm for identification of the material model parameters

The weight of most important parameters, which affect the resistance and failure, is automatically recognized by this method. These methods are more rational and superior to the heuristic trial-and-error fitting, especially in the case of many input parameters. Simulation of resistance is performed by non-linear finite element analysis. Virtual simulation is used to create a training set for the neural network model. Finally, the trained model can provide the best set of material parameters for a given experimental load-deflection curve.

In this context, “response” may mean stiffness, strength or durability characteristics.

Examples of such structural members or structural systems may be:

- assemblages of reinforced concrete prefabricated elements;
- units with special shape (polyhedral panels, bunkers).

Examples of special design problems studied experimentally may be:

- prestress losses;
- special production processes (accelerated curing etc.).

Examples of tests which generally cannot be statistically interpreted:

- tests on frictional losses of prestress made by measurements of forces transmitted by tendons on the structure during construction;
- tests on uncertainties related to static equilibrium made by measurements on the smaller reaction of supports;
- single tests on large-dimension elements or existing structures.

When dealing with cases in which a statistical interpretation of results is not possible, a combined procedure may be used to reduce the entity of experimental tests, as follows:

- Experimental tests integrated by “a priori knowledge”, using a Bayesian approach on structures similar both in geometry and age. This kind of approach allows for the reduction of experimental tests and for a more reliable estimation of resistance.
- Numerical simulations with non-linear analyses followed by a sensitivity analysis of the parameters showing a significant influence on the result. This kind of analyses should also be validated by the comparison of different non-linear programs. Either a semi-probabilistic or a completely probabilistic approach may be used, in agreement with section 7.11.
- Numerical simulation integrated by “later knowledge” derived from experimental tests able to allow for the improvement of the relationship between actions and effect of actions. In practice, the numerical models will be calibrated on the actual structural response derived from destructive and/or non-destructive tests, by means of an optimization procedure involving the main mechanical parameters. This procedure may be considered as an improvement of numerical models by “later” experimental information.

In a wide sense, verification assisted by testing may cover all kinds of load tests on either scaled-down models, on special specimens or on the entire structure itself; these tests are expected to give additional information to the designer. In this respect, two particular cases may be distinguished in practice:

- *Small structural elements repeatedly produced under factory conditions:* An improvement of the existing design model is sought; a considerable number of full scale specimens are tested, taking into account all actions and influences during the lifetime of the structure where these elements will be incorporated.
- *Larger structural elements to be constructed under site conditions:* An analytical model may not be initially available or considerable modification of existing models is sought. A comparatively lower number of tests are carried out simulating the real life conditions. This case is meant to be the main case addressed by this section.

7.12 Verification assisted by testing

7.12.1 Scope

This section contains guidance for the experimental assessment of the response of structural members of structural systems not included in the Model Code’s range of application. In very particular cases (see subsection 7.12.3) verification assisted by testing may also be used in order to modify specific provisions in this Model Code’s range of application.

This section covers only those cases in which a statistical interpretation of results is possible.

In cases in which a statistical interpretation of results is not possible, a procedure based on a combination of testing and numerical verifications, as discussed in section 7.11, can be applied.

7.12.2 Definition

Verification assisted by testing is considered here as a procedure where loading tests on limited series of representative specimens are used for the determination of the response of structural members or structural systems.

However, the following cases are not covered by this section:

- non-destructive test-loadings of single finished structures;
- wind tunnel or earthquake simulator tests;
- routine control tests on reinforced concrete industrial products;
- accelerated durability tests;
- proof loading.

Such cases may be deviations from the provisions of this Model Code with respect to, for example:

- geometric shapes;
- combinations of new materials;
- particular detailing and/or connections;
- severe actions.

Such cases may be considered where, for example:

- the economic importance of a project or of an element to be mass produced justifies the effort of verification assisted by testing;
- the codified model, in some particular conditions, is judged to be clearly too approximate.

Special care is necessary for several reasons:

- computational design versus ultimate limit states often covers indirectly serviceability as well and intermediate limit states (which for simplicity are not codified): a modification of the codified provisions for ULS might result in deficiencies in this respect;
- verification assisted by testing is not able to account for some uncertainties (e.g. on structural analysis or on parasitic phenomena due to the environment) which are covered by code provisions, nor to justify a reliability format more precise than the codified one.

This is particularly important when verification assisted by testing is to be permitted for financial reasons (see subsection 7.12.3).

In this respect, see for example:

- the in-time variation of the basic variables (subsection 7.12.5.2);
- the condition affecting the conversion factor η (subsection 7.12.9.1.1).

If some of these variations and conditions are expected to have a systematic effect on the structural response under investigation, their complete reproduction in the laboratory must be secured.

7.12.3 Aims of verification assisted by testing

The aim of the verification assisted by testing is to obtain design values for the parameters governing the response of structural members and structural systems under specified load conditions with respect to a certain limit state.

Verification assisted by testing may be pursued instead of verification by calculation when:

- the calculation models are insufficient or out of this Model Code's range of application;
- the calculation models given in this Model Code are thought to lead to unnecessary uneconomic results or where greater precision is desirable.

Verification assisted by testing must be done with special care and reasonable checks should be exercised to verify the results and conclusions.

7.12.4 Requirements

The verification-assisted-by-test procedure should ensure that the design leads to the same level of reliability as this Model Code.

To this end, all possible critical conditions, actions and possible influences expected during the lifetime of the structure must be appropriately reproduced in the laboratory.

The whole procedure must be developed by the designer and be approved by the relevant counterpart.

The experimental assessment must be performed by qualified institutions with staff experienced in planning, executing and evaluating tests.

7.12.5 Planning

7.12.5.1 Calculation model-limit states

A plan should be drafted by the designer, which should contain the objective of testing and all indications necessary for the sampling

If such a model is not available prior to testing, trial preliminary tests should be carried out in order to facilitate parameter identification.

The response depends in general on a set of measurable quantities. The quantities which are random with respect to an elementary population are referred to as basic variables *X*. The quantities which are considered as deterministic, with respect to an elementary population, are referred to as nominal variables *W*. Nominal variables may be constant within a population or vary in a predetermined manner.

NOTE: The terminology used in this section and the following ones, with regard to the variables, is not exactly the same as elsewhere in this Model Code (e.g. in section 4.3 and subsequent) because the character of some variables (fundamental or not) is not yet established at this stage. Some variables indeed, although random, may have very little influence on the response.

In case of unidentified significant parameters, a larger scattering of results is expected.

Special care is necessary because statistical parameters of basic variables do not have an intrinsic character. They generally depend on the considered statistical population, which may be very different at the level of the Code, at the level of the specimens and at the level of the expected application of the test result.

Special mention should be made of the quasi-deterministic concrete strength of the specimens which exhibit a very low variability of concrete strength. Thus, in the relevant model, the characteristic strength of in-structure concrete should be made equal to the characteristic strength of the specimen concrete.

- Aspects to be considered are for example:
- the in-time development of the concrete strength positive or negative, depending on the concrete composition;
 - the possible decrease of the tensile strength of concrete due to hydrothermal cyclic conditions;
 - the possible decrease of steel strength, ductility or fatigue performance due to minor corrosion conditions.

The necessary number of tests is higher where not only mean values but also standard deviations are to be statistically assessed.

The dimensions of the specimens should, if possible, cover the entire range of the probable variation of the dimensions of the structural element.

or manufacturing of the specimens, the execution of the tests and their evaluation.

The experimental procedure may cover either an ultimate limit state or a serviceability limit state. In either case, minimum knowledge of the relevant response mechanism and parameters is sought.

It is desirable that on the basis of this knowledge, an empirically or physically based calculation model is available prior to testing, in order to evaluate the response. This model is referred to as “prior calculation model” g_R

$$R_t = g_R(X, W, D) \tag{7.12-1}$$

where:

- R_t is the response of the available theoretical model;
- X is the vector of basic variables with known statistical distribution;
- W is the vector of known nominal variables;
- D is the vector of unknown coefficients to be determined by the testing.

All quantities affecting the response should be present in the model either as basic variables or as nominal variables.

7.12.5.2 Information on basic variables

For the basic variables included in the calculation model, statistical parameters have to be known. If these values are not known, it is recommended to estimate them by preliminary tests. Where information is available only from a limited population, variances need to be increased accordingly.

The in-time variation of the basic variables should be taken into account in the model.

Gross errors cannot be covered by this procedure.

7.12.5.3 Number of specimens

The number of tests carried out should be sufficiently large in order to lead to results with satisfactory small statistical uncertainty (see subsection 7.12.8)

7.12.5.4 Scale effects

The specimens should preferably be dimensioned as close to full scale as possible, so that the scale effects do not increase the model uncertainties.

When scaled-down specimens are used, special attention should be paid to the influence of aggregate size, size and number of rebars, concrete workability and so on, on the response.

Other influences may be temperature, humidity and so on, conditioning the behaviour of the specimens and their response.

The actions process (“load path”) should also include data regarding rate effects (e.g. load application velocity, the order of application of each influence).

Specimens taken from production may be used either for preliminary tests or in the case of small structural elements repeatedly produced under factory conditions defined in subsection 7.12.2.

Quantitative and qualitative observations are highly recommended in order to check these assumptions, made implicitly or explicitly, or to allow alternative evaluations afterwards.

If, during testing, a given condition is found to influence considerably the results, this condition must be more systematically studied as a new basic variable of the model g_R .

When a direct measurement is not possible, an indirect measurement is allowed. Then the respective conversions should be used, by introducing the necessary new variables of the conversion factors or by increasing the variance of the relevant variable.

In the case of indirect measurement of a variable, the difference of the statistical properties between the populations in the laboratory and on site must be taken into account.

Since the response of the structural member is in general identified in terms of load intensity at which the limit state is reached, the actual limit value of the limit load should be measured with additional care.

It is recommended that measurements are redundant in order to make a mutual checking of the results possible.

Otherwise, similitude laws, such as on the basis of fracture mechanics, must be appropriately taken into account.

7.12.5.5 Actions

The actions may be direct forces (loads in general), or imposed deformations or other influences varying in space and/or in time.

The scheduled actions process should be fully determined by a set of load (or occasionally of other influence) parameters. The actions process must be selected so that they are representative for the anticipated scope of application of the structural member.

Each specimen may be subjected to the same or different action processes.

7.12.5.6 Origin of specimens

The specimens should be specifically manufactured for the testing.

7.12.6 Testing conditions and measurements

Special care should be taken in order to check that all assumptions made in the planning (see subsection 7.12.5) are satisfied.

7.12.6.1 Basic and nominal variables

The actual values of all basic and nominal variables included in the calculation model should be, as far as practicable, determined by direct measurements for each specimen and each experiment. If not, the resulting uncertainty should be accounted for.

7.12.6.2 Actions

The actual values of the imposed actions have to be recorded during the test and especially at the “critical point” of the limit state considered.

7.12.6.3 Deformation – structural behaviour

During the tests, systematic measurements must be carried out concerning the deformations (elongation, deflection, rotation).

7.12.7 Laboratory report

The laboratory report should contain at least:

- the name of the part asking for the verification assisted by testing;
- the names of the laboratory staff involved in the verification assisted by testing;
- the scope of the verification assisted by testing;
- technical description of the specimens (dimensions, materials, fabrication technique, number of specimens etc.);
- testing procedure;
- all measurements of basic and nominal variables;
- the actions process;
- the deformation of specimens;
- crack pattern;

For details, see:

- ISO 12491:1997, “Statistical methods for quality control of building materials and components”;
- Kersken-Bradley, M., Maier, W., Rackwitz, R. and Vrouwenvelder, A. with contributions by Bijlaard, F., Mathieu, H., Sedlacek G. and Stark, J., “Estimation of structural properties by testing for use in limit state design, general principles for planning, sampling and evaluation for the elementary case”, JCSS Working document, IABSE, Zurich, 1990.

The coefficients $\mathbf{D}$ may also be considered as random variables, but in order to check the validity of the Eq. (7.12-1), in a first step, the coefficient $\mathbf{D}$ should be given the best value $\mathbf{D}_m$.

For simple models of the type

$$R_t = d g_R(\mathbf{X}, \mathbf{W}) \quad (7.12-1)$$

d is single parameter (i.e. model uncertainty) and the distribution of d is a central t-distribution. When elaborating the test a distinction should be made between variable x_i that can be measured in each test and variable x_i that cannot be measured.

Often it is known that coefficients can only take values larger or smaller than unity.

Generally, the characteristic value R_k (usually 5% fractile) is defined as the value for which:

$$P(R_t < R_k) = 0.05 \quad (7.12-2)$$

Accordingly, appropriate statistical analysis must be conducted (see references given in 0) or the following simplified approximation must be applied:

$$R_k \approx d_k g(\mathbf{X}_k, \mathbf{W}) \quad (7.12-3)$$

If relevant, the characteristic values $\mathbf{X}_k$ are substituted in R_k by other representative values depending on the combination and/or the limit state.

- the failure mode and the critical material;
- photos and/or video recordings;
- the prior calculation model used;
- laboratory comments on the results obtained.

7.12.8 Statistical analysis of test results

7.12.8.1 Estimation of the unknown coefficients $\mathbf{D}$

The estimation of the coefficients $\mathbf{D}$ (see Eq. (7.12-1)) may be based on:

- least square methods;
- maximum likelihood methods;
- Bayesian regression analysis.

For linear models or models which can be transformed to linear, coefficients should generally be assumed as normally distributed.

Where no linear models are obtained, a Taylor expansion in the vicinity of the expected design point may be used to avoid non-linear regression analysis.

When coefficients of applied models are to be estimated, prior information on those coefficients should be derived from physical considerations.

7.12.8.2 Characteristic value

The characteristic value R_k may be calculated from test results taking into consideration the statistical uncertainty as a result of a limited number of tests.

The calculation can either be based on a predictive Bayesian estimate or on assuming a specific confidence level (normally the confidence level is taken equal to about 0.75).

7.12.9 Verification procedure

7.12.9.1 Design values

7.12.9.1.1 Design values for ULS

The design value of the response is given by the expression:

$$R_d = \eta R_k / \gamma_{Rd} \gamma_m \quad (7.12-6)$$

where:

R_k is the characteristic response defined statistically on the basis of the test results;

The material safety factor γ_m is to be adopted in accordance with this Model Code as:

$$\gamma_m = \gamma_s \quad (7.12-4)$$

or

$$\gamma_m = \gamma_c \quad (7.12-5)$$

For numerical values of the γ -factors according to the failure mode and to the material decisive for the failure see chapter 4.

The differences covered by the factor η may, for example, be:

- loading time (when concrete strength is critical, a factor equal to 0.85 should be applied at least to account for sustained loads);
- support conditions;
- humidity conditions and their alternations;
- differences in geometry;
- differences in workmanship and/or curing conditions.

Influences of systematic character should be covered by the model and not by η , for example in case of a brittleness factor such as $(1 - f_{ck}/250)$.

Those conversion conditions included in γ_m should not be duplicated when assessing η .

Only influences of secondary importance may be accounted for by means of conversion factors estimated in a more or less empirical way.

γ_{Rd} should be increased according to experience in cases where the scatter of d is high (e. g. its coefficient of variation is greater than 0.15) or the expected variability of the basic variables is significantly higher on site than in specimens (see subsection 7.12.5.2), or the value of the conversion factor on site conditions seems to be less than the already adopted η value.

Experts' opinions may be used in evaluating η and γ_{Rd} provided that all systematic differences are taken into account in the tests.

For details, see ISO 12491:1997, "Statistical methods for quality control of building materials and components".

γ_{Rd} values for SLS are generally different than those for ULS. Some minor aspects of a conversion factor may as well be accounted for by this γ_{Rd} factor.

γ_m is the material partial safety factor adopted according to the failure mode of the material decisive for the bearing capacity;

η is a conversion factor, taking into account the differences between testing conditions and conditions in the actual structure. If a more detailed model is known, accounting separately for the role of both materials, separate conversion and safety factors may be used accordingly;

γ_{Rd} is a complementary model uncertainty factor intended to cover differences between the testing conditions and the actual ones in the structure, which cannot be accounted by d_k , by γ_m or by the conversion factor η .

It is also possible to make a direct estimation of the design value on the basis of the statistical results,

7.12.9.1.2 Design values for SLS

The design values for the serviceability limit states are given by the expression:

$$R_d = R_k / \gamma_{Rd} \quad (7.12-7)$$

with R_k and γ_{Rd} as defined in subsection 4.5.2.5.

7.12.9.2 Verification

The most frequent verification inequality is:

$$E_d \leq R_d \quad (7.12-8)$$

where:

E_d denotes the design value of the load effects;

R_d denotes the design value of the resistance.

7.13 Detailing

7.13.1 Basic principles

The dimensions of structural members must be chosen in relation to the amount and position of the reinforcement and to the properties of the structural materials. The position and composition of the reinforcement must allow a qualitatively perfect casting and compaction of the concrete. Special attention must be paid to the placing of the reinforcement.

The detailing must correspond to the structural model underlying the structural analysis. The influences of any deviations between the structural model and the actual properties of the structure must be taken into account.

If the reinforcement is used for the earthing of construction works, the measures have to be discussed with the responsible electrical engineers, whereby particular attention must be paid to corrosion protection.

7.13.2 Positioning of reinforcement

7.13.2.1 General

The provisions of this subsection apply to reinforcing bars with diameters ≤ 40 mm. Bar diameters above 32 mm must only be used in exceptional cases; the principles of detailing must be adapted if necessary.

7.13.2.2 Cover of reinforcement

The cover of reinforcement c_{nom} refers to minimum values which have to be complied with if no special protective measures are taken or no special circumstances exist. Permissible dimensional tolerances are given in subsection 4.5.1.4.4 (see Tables 4.5-3 to 4.5-4).

The concrete cover must be specified on the drawings. It is defined as an absolute minimum cover, c_{min} plus an allowance in design for deviation Δc_{dev} :

$$c_{nom} = c_{min} + \Delta c_{dev} \quad (7.13-1)$$

The cover is the largest of four values, depending on:

- technological requirements;
- bond and anchorage requirements;
- fire requirements;
- durability requirements.

The highest value applies. The cover chosen should be shown on the drawings.

In order to ensure good casting, the cover should be at least equal to the maximum aggregate particle diameter plus 5 mm.

In order to ensure the transfer of the bond forces between concrete and reinforcing bars, c_{min} must be larger than:

- in case of single bars: the diameter of the reinforcing bar;
- for bundles: the equivalent diameter $\emptyset_n$ – see Eq. (7.6-7).

In case of post-tensioned bars c_{min} must be larger than:

- for circular ducts: the duct diameter;
- for rectangular ducts: the smaller dimension or half the largest dimension.

For pretensioned strands or wires c_{min} must be larger than:

- 1.5 the diameter of the strand or plain wire;
- 2.5 the diameter of an indented wire

Generally Δc_{dev} can be taken equal to 10 mm. Where a quality assurance system is used, including measurement of the concrete cover, the allowance in design for deviation Δc_{dev} may be reduced to a value $10 \text{ mm} \geq \Delta c_{dev} \geq 5 \text{ mm}$. Where a very accurate measuring device is used for monitoring and non-conforming members are rejected (e.g. precast elements) the value Δc_{dev} can be further reduced.

Table 7.13-1 gives indicative values for minimum cover with respect to durability demands, assuming moderate concrete strength (from

In order to fulfil the fire resistance requirements, reference is made to subsection 7.5.1.

C20/25). The values apply for a service life of 50 years. For a service life of 100 years, the values of c_{min} should be increased. The necessary increase of the cover depends on the time dependent nature of the ingress of chemicals with a detrimental effect. In general, the cover thickness should be increased with about 30%. The cover can be reduced if a concrete with smaller permeability is used. This reduction can be more substantial if the water cement ratio decreases and the degree of hydration increases. Also quality control plays an important part. Another possibility is to design the concrete mixture for low permeability.

Table 7.13-1: Minimum cover for reasons of durability (indicative values)

Environmental class	XC2 XC4				XD1 XD2 XD3		
	X0	XC1	XC3		XS1	XS2	XS3
Reinforced concrete	10	15	25	30	35	40	45
Prestressed concrete	10	25	35	40	45	50	55

For durability, the minimum cover depends on the concrete resistance against the ingress of chemical substances in combination with the aggressivity of the environment and the intended service life of the structure. For verification of durability requirements, see section 7.8.

In the case of structural members in which concrete is cast directly against soil, the minimum value of the cover on the reinforcement is $c_{nom} \geq 75$ mm; in the case of soil preparation $c_{nom} \geq 50$ mm.

If the cover of reinforcement exceeds 50 mm, it may be appropriate (e.g. regarding fire action) to provide a skin reinforcement.

In the case of architectural concrete surfaces, in particular those with a treated surface, particular attention has to be paid to the cover of the reinforcement.

7.13.2.3 Minimum bar spacing

The spacing of reinforcing bars and prestressing tendons must be sufficient to allow perfect casting and compaction of the concrete. Splices of reinforcement must be staggered if necessary.

The clear distance between parallel reinforcing bars and prestressing tendons should be at least equal to the maximum aggregate particle diameter +5 mm and the diameter of adjacent bars. It must be at least 20 mm.

In the case of dense reinforcement, the arrangement of bar bundles with a maximum of three contacting bars running in the same direction is permitted.

7.13.2.4 Forms and bends

Compliance with the minimum mandrel diameters (Figure 7.13.1) prevents bending cracks in the reinforcing steel and failure of the concrete enclosed by the bend. The mandrel diameters have to be specified by the design engineers. The following minimum mandrel diameters have to be complied with:

- for bends $d_1 = 15\emptyset$
- for end hooks, angle hooks, loops $d_2 = 6\emptyset$ for bars ≤ 20 mm
- $d_2 = 8\emptyset$ for bars > 20 and ≤ 30 mm
- $d_2 = 10\emptyset$ for bars > 30 and ≤ 40 mm
- for stirrups $d_3 = 4\emptyset$ for bars ≤ 16 mm

If transverse reinforcement is provided, the mandrel diameters may be reduced in special cases (e.g. frame corners or loop anchorages):

- for end hooks, angle hooks, loops $d_2 = 4\emptyset$ for bars ≤ 16 mm
- $d_2 = 7\emptyset$ for bars > 16 mm ≤ 30 mm
- $d_2 = 10\emptyset$ for bars > 30 and ≤ 40 mm

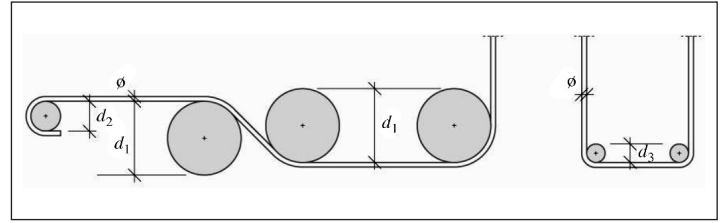


Figure 7.13-1: Mandrel diameters

No welds are permitted in the immediate vicinity of bends.

In the case of welded reinforcement, the minimum distance between the bend and the weld is $10\emptyset$. The distance may be reduced if the weld is not fully loaded.

In the case of bent mesh reinforcement, the cross bars should be located outside the bending zone.

Recommended values for the minimum permissible mandrel diameter for welded reinforcement or welded fabric bent after welding are given in Figure 7.13-2.

Recommended minimum mandrel	
 or 	 or 
$5\emptyset$	$d \geq 3\emptyset : 5\emptyset$ $d > 3\emptyset$ or welding within the bent zone: $20\emptyset$

Figure 7.13-2: Recommended minimum mandrel diameter for welded reinforcement or welded fabric after welding

Specifying the minimum radii of curvature of prestressing tendons the technical documentation of the prestressing system should be considered. In addition, it may be necessary to check the local concrete stresses. See subsection 5.4.9.2 which gives minimum radii of curvature for post-tensioning tendons.

7.13.2.5 Anchorage

This subsection gives rules which are simplified with regard to the relations given in subsection 6.1.3.

The more refined rules of subsection 6.1.3 may permit shorter anchorage lengths.

The contribution of a hook or bend to anchorage of tension reinforcement and the contribution of end bearing to anchorage of compression reinforcement is included in the bond length given by Eq. (7.13-2). Eq. (7.13-2) has been derived for reinforcing bars with a characteristic yield strength of 500 N/mm^2 .

This subsection covers anchorage of ribbed reinforcing bars and meshes of reinforcing steel which satisfy the requirements for classification as “high bond reinforcement” in accordance with subsection 5.2.4.2.

This subsection does not cover bars forming part of a bundle.

The rules here are limited to the following:

- characteristic yield strength of reinforcement $\leq 500 \text{ MPa}$;
- minimum requirements for confining reinforcement and cover in 6.1.3.1 are satisfied; where the diameter $\emptyset$ of the anchored bars is smaller than 20 mm , transverse reinforcement or links provided for other reasons may be assumed sufficient to satisfy the minimum requirements for confining reinforcement without further justification;
- on flexural members, loading is predominantly uniformly distributed along the span;
- no applied tensile stress acts transverse to the axis of the anchored bar.

If any of these conditions is not satisfied the detailed rules of subsection 6.1.3 must be used.

The dimensioning value of the bond length is:

$$l_b = \alpha_1 \cdot \beta_b \cdot \left(\frac{25}{f_{ck}} \right)^{0.5} \emptyset / \eta_3 \geq l_{b,\min} \quad (7.13-2)$$

Values for β_b in Table 7.13-2 include a partial safety factor of 1.5 on bond strength.

Figure 7.13-3 provides illustrations of representative situations for each type of bond zone.

where:

l_b has been derived as a simplification of Eq. (6.1-19);

$l_{b,min}$ minimum bond length, see Eq. (6.1-26);

$\alpha_1 = A_{s,cal} / A_{s,ef}$ for anchorages in zones AS, AB and AC;

where:

$A_{s,cal}$ is the calculated area of reinforcement required by the design;

$A_{s,ef}$ is the area of reinforcement provided;

$\alpha_1 = 1.0$ in other circumstances;

η_3 represents the influence of bar diameter, as given in subsection 6.1.3.1;

$\eta_3 = 1.0$ for $\varnothing \leq 25$ mm;

$\eta_3 = (25/\varnothing)^{0.3}$ for $\varnothing > 25$ mm ($\varnothing$ in mm);

β_b is a factor given in Table 7.13-2 for anchorage length appropriate to bond zones classified as follows:

Type AS : straight bars in tension in zones near the ends of members and in which the support reaction or load from upper storeys is transferred through the anchorage zone in a direction perpendicular to the plane passing through the axes of the anchored bars.

Type AB : tension bars which terminate in a hook or bend in zones near the ends of members and in which the support reaction or load from upper storeys is transferred through the anchorage zone in a direction perpendicular to the plane(s) passing through the axes of the anchored bars.

Type AC : column and wall bars anchored near the middle of footings or pilecaps and acting in compression under all design loadings.

Type RS : straight bars in all other situations.

Type RB : anchorages of bars terminating in a hook or bend in all other situations.

Table 7.13-2: Bond length factors for anchorages

Bond zone	Bond length factor β_b			
	Casting position			
	“Good”		“Poor”	
	$c_{min} / \varnothing \geq 2.5,$	$c_{min} / \varnothing \geq 1.0,$	$c_{min} / \varnothing \geq 2.5,$	$c_{min} / \varnothing \geq 1.0,$
AS	42	42	59	59
AB	29	29	39	39
AC	29	—	39	—
RS	66	95	95	131
RB	41	74	74	95

Casting position is defined in subsection 6.1.3.2. Minimum cover c_{min} is defined in Figure 6.1-2 and Figure 6.1-15 for straight and bent/hooked bars respectively.

$\beta_{b,min}$ must be taken as 15 for type RS and RB bond zones, or 10 for types AS, AB and AC.

Mesh reinforcement with welded cross bars may be considered to lie within types AB or RB as appropriate. l_b may be reduced by 15% for each cross bar positioned within the anchorage area, but not by more than 30%.

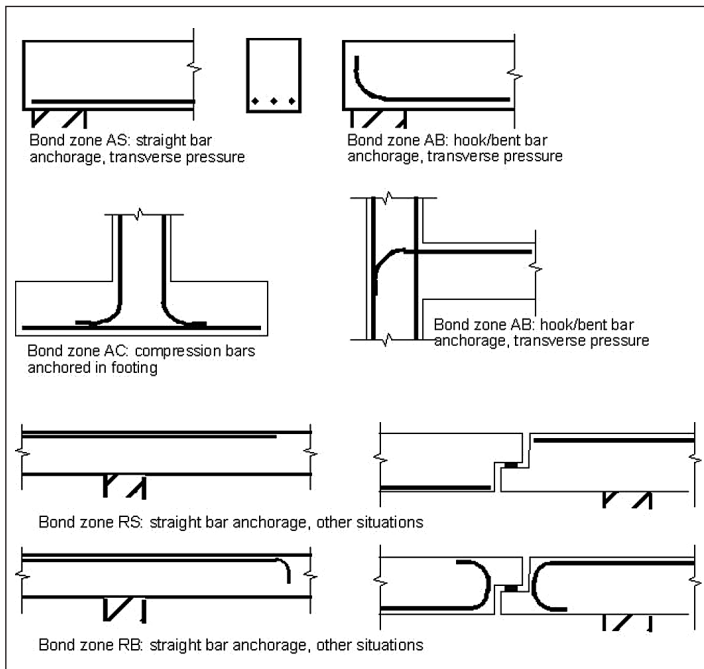


Figure 7.13-3: Classification of anchorage bond zones

This subsection gives rules which are simplified with regard to the relations given in subsection 6.1.3.

The more refined rules of subsection 6.1.3 may permit shorter lap lengths.

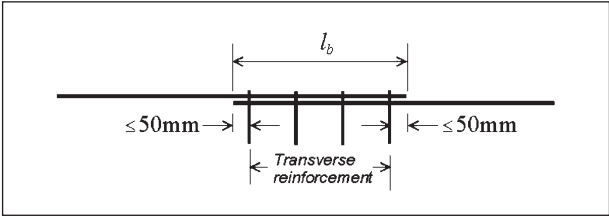


Figure 7.13-4: Location of links near ends of lap

The contribution of end bearing in compression laps and anchorages is included in the average bond strength given by Eq. (7.13-2)

Values for β_b in Table 7.13-3 include a partial safety factor of 1.5 on bond strength.

Figure 7.13-5 provides illustrations of representative situations for each type of bond zone.

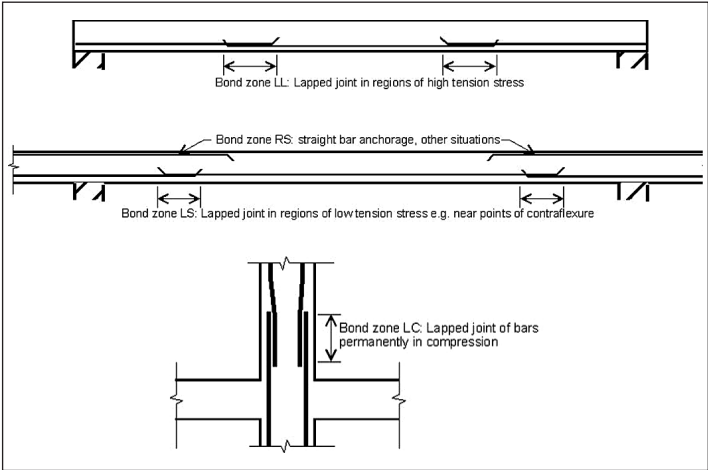


Figure 7.13-5: Classification of lapped splice zones

7.13.2.6 Lapped joints

This subsection covers lapped joints or splices of ribbed reinforcing bars and meshes of reinforcing steel which satisfy the requirements for classification as “high bond” in accordance with requirements of subsection 5.2.4.2.

Wherever possible, splices should be arranged in zones of low stress.

For bar diameters > 12 mm, lap splices should, if possible, be staggered so that, in a cross-section, not more than one-third of the force in the reinforcement needs to be transferred by a lap.

- The rules in this subsection are limited to the following:
- characteristic yield strength of reinforcement ≤ 500 MPa;
 - minimum requirements for confining reinforcement and cover in 6.1.3.1 are satisfied; where the diameter $\varnothing$ of the anchored bars is smaller than 20 mm, transverse reinforcement or links provided for other reasons may be assumed sufficient to satisfy minimum requirements for confining reinforcement without further justification;
 - at least one item of transverse reinforcement should be positioned within a lap length no further than 50 mm from the end of both the lapped bars; other items of transverse reinforcement should be spaced evenly throughout the lap length (Figure 7.13-4);
 - no applied stress transverse to the axis of the lapped bars.

If any of these conditions are not satisfied, the detailed rules of subsection 6.1.3 must be used.

The dimensioning value for the lap length is given by Eq. (7.13-2), where the value of β_b is given in Table 7.13-3.

- Bond zones for lapped splices are classified as follows:
- Type LS : straight bars in tension where the splice will be subject to a stress not exceeding 50% of the characteristic strength of the reinforcement at the ultimate limit state;
- Type LL: straight bars in tension not classified as type LS;
- Type LC: straight bars in compression under all load combinations.

Table 7.13-3: Bond length factors for lapped joints

Bond zone	Bond length factor β_b			
	Casting position			
	“Good”		“Poor”	
	Confinement			
	$c_{min} / \varnothing \geq 2.5$	$c_{min} / \varnothing \geq 1.0$	$c_{min} / \varnothing \geq 2.5$	$c_{min} / \varnothing \geq 1.0$
LS	48	66	66	95
LL	66	95	95	131
LC	41	74	74	95

Mesh reinforcement is not covered by this subsection. See subsections 6.1.4.2 and 6.1.4.3.

7.13.2.7 Deviations and curvatures

In the case of curved or kinked tension or compression chords, the effects of the deviation forces must be considered.

Deviation forces acting toward the surface of the concrete, as shown in Figure 7.13-6 must in general be resisted by means of additional stirrup reinforcement. If this is not provided it must be verified that the deviation forces at yielding of the steel can be resisted by the cover concrete. In this specific case the concrete tensile strength may be taken into account with a maximum value according to:

$$f_{ctd} = \frac{1}{3} \frac{f_{ctk0.05}}{\gamma_c} \quad (7.13-3)$$

The strength according to Eq. (7.13-3) has to be reduced by 50% in case of plastic deformations in the reinforcement.

If the deviation forces have to be resisted by the concrete, the inaccuracies of execution (thinner cover concrete, smaller spacing between reinforcements, non-uniform curvature) must be taken into account for the verification.

For the calculation of the concrete tensile strength the effective width b_u may be assumed to be equal to:

$$b_u = s - \emptyset \leq 2\sqrt{3(c_{nom} + \frac{\emptyset}{2})} \quad (7.13-4)$$

where s = spacing between reinforcing bars and $\emptyset$ = bar diameter.

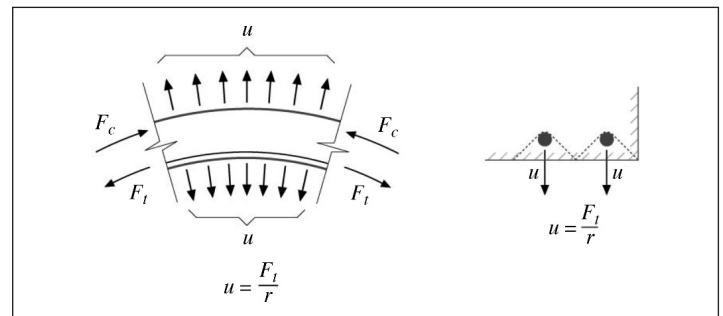


Figure 7.13-6: Equilibrium in curved tension and compression chords (r = radius of bend)

In the case of reinforcing bars which are curved or bent parallel to outer surfaces, the effects of the transverse tensile forces need to be examined.

Transverse tensile forces according to Figure 7.13-7 should be resisted by means of transverse reinforcement (e.g. stirrup or U-shaped reinforcement). If no transverse reinforcement is provided, it must be verified that the transverse tensile forces due to the yield tensile force can be resisted. The concrete tensile strength may be taken into account according to Eq. (7.13-3) at most. No verification is necessary if the lateral cover of reinforcement corresponds to at least three times the bar diameter.

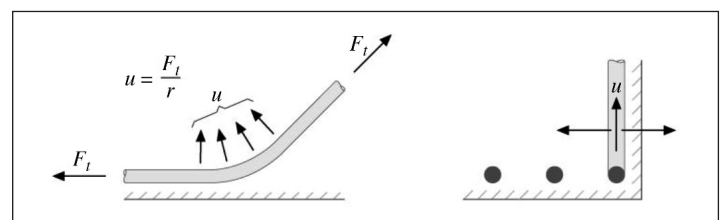


Figure 7.13-7: Transverse tensile forces in the case of bent-up reinforcing bars

7.13.3 Prestressed structures

With regard to the transmission of the prestressing force into the concrete for post-tensioned structures, two zones are distinguished:

- (1) The area immediately around and behind the contact area, where the prestressing force is transmitted to the concrete: dimensions and reinforcement detailing are specified in Technical Approvals (e. g. ETA) of the prestressing system. The reliability for the corresponding detailing is the responsibility of the system holder.
- (2) The zone where the prestressing force is spread over the full cross-section. Design and detailing of the reinforcement in this zone is the responsibility of the designer.

Force spreading zones must be detailed in such a way that the dimensioning value of the prestressing force according to section 5.3 can be introduced into the structure.

The flow of forces should be examined using suitable models, neglecting the tensile strength of the concrete. Splitting forces have to be resisted by means of reinforcement.

A minimum reinforcement should in any case be placed in anchorage and coupling zones to avoid the occurrence of harmful cracks.

Anchorage and couplings must if possible be positioned in areas in which only small stress variations occur.

7.13.3.1 Anchorage of prestressing wires and strands

The equations (7.13-5a) and (7.13-5b) follow from Eq. (6.1-40), subsection 6.1.8.4, for a sudden release of the prestressing force, and a position of the tendon in the good bond area of the member.

A practical expression for the upper bound value of the transmission length l_{bpt} is:

$$l_{bpt,95\%} = \frac{0.10\phi_s\sigma_{pi}}{f_{ctd}} \quad \text{for strands} \quad (7.13-5a)$$

and

$$l_{bpt,95\%} = \frac{0.15\phi_s\sigma_{pi}}{f_{ctd}} \quad \text{for indented wires} \quad (7.13-5b)$$

The design value of the anchorage length l_{bpd} is:

$$l_{bpd} = l_{bpt,95\%} + \frac{A_p(\sigma_{pd} - \sigma_{pcs})}{\pi \cdot \phi_s \cdot f_{bpd}} \quad (7.13-6)$$

where:

σ_{pd} is the tendon stress under the design load;

σ_{pcs} is the tendon stress after all losses;

σ_{pi} is the initial prestress;

f_{bpd} follows from Eq. (6.1-38).

7.13.4 Bearings and joints

In the case of unreinforced concrete structures, the arrangement of expansion joints may be practical in order to compensate for the effects of shrinkage and temperature changes.

In the case of reinforced concrete structures, expansion joints may have to be arranged if the effects of shrinkage, creep, temperature changes and non-uniform settlements cannot adequately be rendered harmless by the provision of crack distributing reinforcement.

In order to reduce the effects of shrinkage, it needs to be examined whether shrinkage joints for the temporary partition of larger structural members or other measures have to be specified.

Bearings of important structural elements must be permanently protected against moisture. They must be easily accessible and easy to replace.

Expansion joints must, if possible, be designed in such a way that moisture cannot reach the joint.

7.13.5 Structural members

7.13.5.1 Unreinforced structural members

Unreinforced concrete is used for solid structural members that are primarily subjected to compression and are not subjected to any significant seismic or dynamic loads. Examples include:

- foundations;
- arches and vaults;
- retaining walls;
- walls;
- compression members.

The dimensioning value of the compressive strength f_{cd} must be reduced by 20% for verifying the structural safety of unreinforced structural members.

When verifying structural safety, a state of equilibrium between the internal and external forces must be assumed, neglecting the concrete tensile strength. Kinematic compatibility has to be ensured. Any constrained deformations and displacements have to be taken into account. External forces (reactions) should be taken into account if these act even in case of small deformations (e.g. frictional forces, abutment forces).

For structural members of secondary importance, the structural safety has to be verified taking into account the concrete tensile strength according to

$$f_{ctd} = 0.5 \frac{f_{ctk,0.05}}{\gamma_c} \quad (7.13-7)$$

In addition, it must be ensured that the formation of cracks does not result in failure of the structural member.

The structural safety of unreinforced compression members may be verified in analogy to subsection 7.3.7. The dimensioning value of the compressive strength must be reduced as indicated above.

7.13.5.2 Beams and T-beams

A beam is defined as a bearing element, spanning in one direction, with a width smaller than five times the cross-sectional depth.

Sections containing less reinforcement than $A_{s,min}$ should be considered as unreinforced.

- In the application to new structures, the rules of this subsection should be complied with in full.
- In the application for the assessment of existing structures, the detailing rules presented in this subsection may be relaxed provided that the assessment on strength in cases of non-compliance is assessed fully.

The minimum amount of shear reinforcement (Eq. 7.13-9) is required to ensure that failure does not occur immediately upon shear cracking and that truss action can develop.

The area of longitudinal reinforcement should not be taken as less than:

$$A_{s,min} = 0.26 \frac{f_{ctm}}{f_{yk}} b_t d \quad (7.13-8)$$

where b_t is the width of the tension zone.

Adequate reinforcement must be provided to cope with restraining effects which have been neglected in the structural analysis.

Where shear reinforcement is required (see subsection 7.3.3), the minimum area of shear reinforcement must be:

$$A_{sw,min} = 0.08 \sqrt{f_{ck}} \frac{b_w s_w}{f_{yk}} \quad (7.13-9)$$

In the case of wide webs, the web width may be taken into account up to a maximum of 400 mm. In the case of web widths > 500 mm, stirrups with more than two legs must be used.

In beams, stirrups must generally be provided. Their spacing s_w must not exceed $0.75d$ or 500 mm.

In beams, at least 50% of the shear reinforcement should consist of closed stirrups, because the inclined struts tend to exert a lateral force on the outer longitudinal reinforcing bars (Figure 7.13-8). In slabs those lateral forces can be accounted for by transverse bars.

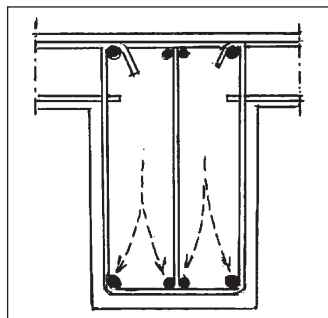


Figure 7.13-8: Lateral forces on outer bars due to strut action

Transverse reinforcement provided for shear and torsion may consist of a combination of:

- stirrups or ties perpendicular to the axis of the member, enclosing the longitudinal tension reinforcement and the compression zone;
- cages, ladders, welded wire fabric and so on which are cast-in without enclosing the longitudinal reinforcement but are properly anchored in the compression and tension zones. Such reinforcement may only be applied in combination with stirrups enclosing the longitudinal reinforcement, carrying at least 50% of the total force in the shear reinforcement;
- longitudinal bars bent to provide an inclined portion having an angle of 30° or more with the longitudinal bars and crossing potential diagonal cracks. However, only the centre three-quarters of the inclined portion of these bars may be considered effective;
- headed shear reinforcement demonstrated to be able to achieve the yield strength of the bar;
- spirals.

The stirrups must enclose the longitudinal tensile reinforcement. They must be anchored in such a way that the necessary stirrup forces can act over the height of the lever arm of the internal forces.

In the support areas, at least 25% of the total chord reinforcement required in the span must be fully anchored.

In the area of negative moments of T-beams, a significant portion of the tensile reinforcement must be concentrated over the web. In order to limit crack widths, a certain proportion must also be distributed over the adjacent slab. The effectiveness of the longitudinal reinforcement distributed outside of the web must be ensured by means of transverse reinforcement.

Adequate transverse reinforcement, the cross-section of which should amount to at least 0.2% of the slab cross-section, must be provided in the compression slab to ensure the shear connection.

7.13.5.3 Slabs

A slab is a bearing element, spanning in one or two directions. A one-way bearing element is considered to be a slab if the width of the cross-section is equal or larger than five times the overall cross-sectional depth.

- In the application to new structures, the rules of this subsection should be complied with in full.
- In the application for the assessment of existing structures, the detailing rules presented in this subsection may be relaxed provided that the bearing resistance in cases of non-compliance is fully assessed.

In the zones of largest moments, the bar spacing of the main reinforcement must not exceed the lesser of 1.2 times the slab thickness and 300 mm.

The minimum reinforcement of slabs must be specified in accordance with the serviceability requirements. Transverse reinforcement must not be less than 20% of the longitudinal reinforcement.

For the case of slabs without shear reinforcement, at least one-half of the bending reinforcement required at the points of maximum moment must be fully anchored beyond the extent of the supports.

Free slab edges must be reinforced with bent-up longitudinal reinforcement or with stirrup reinforcement of at least $\varnothing 10$ mm, in accordance with Figure 7.13-9.

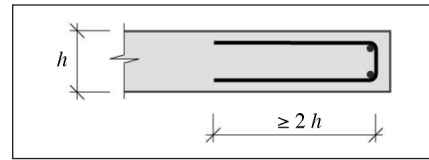


Figure 7.13-9: Free slab edges

The reinforcement details should aim to avoid localization of cracks (anchorage, splices etc.) as well as ensure optimum efficiency of transverse reinforcement.

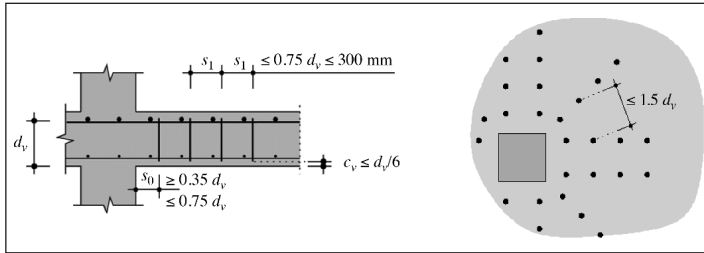


Figure 7.13-10: Detailing rules for punching shear reinforcement

For punching shear, the development length and splices in flexural reinforcement must not be located inside the perimeter defined by the minimum of:

- a line at $2.5d_v$ from the control perimeter of the supported region;
- the position of the line of contraflexure of radial bending moments.

For members with punching shear reinforcement, the design equations of subsection 7.3.5.3 are applicable provided that:

- a minimum of two shear reinforcing elements are provided in the radial direction;
- the geometry and type of shear reinforcement guarantees anchorage at both ends;
- the distance between the first shear reinforcing element and the face of the support is larger than or equal to $0.35d_v$ and smaller than or equal to $0.75d_v$ (Figure 7.13-10); any shear reinforcement closer to the support than $0.35d_v$ must not be considered for resistance;
- the maximum spacing between shear reinforcing elements in the radial direction is not larger than the smaller value of $0.75d_v$ and 300 mm;
- the maximum distance between concentric shear reinforcing elements at the level of the second radial reinforcement does not exceed $1.5d_v$;
- the maximum concrete cover at the compression side of the slab does not exceed $d_v/6$;
- the maximum diameter of the shear reinforcement ϕ_{max} does not exceed the values given in Table 7.13-4.

Table 7.13-4: Maximum diameter of punching shear reinforcement as a function of slab effective depth d_v

d_v [mm]	ϕ_{max} [mm]
< 160	—
160–180	14
181–220	16
221–260	18
261–340	20
341–600	25
> 600	30

7.13.5.4 Compression members

In the case of storey-high compression members, the minimum dimensions according to Table 7.13-5 must generally be complied with.

Table 7.13-5: Minimum dimensions of compression members

	Cast in place [mm]	Precast element [mm]_____
Compression members (side ratio up to 4:1)	200	150
Reinforced walls	150	100
Unreinforced walls	120	—

In compression members, the minimum longitudinal reinforcement ratio is 0.2%. In the case of large column cross-sections, a wall thickness of at least 200 mm according to Figure 7.13-11 may be taken into account in determining the minimum reinforcement ratio.

The minimum reinforcement must be distributed proportionally around the circumference of the column cross-section and must have a diameter of at least 8 mm.

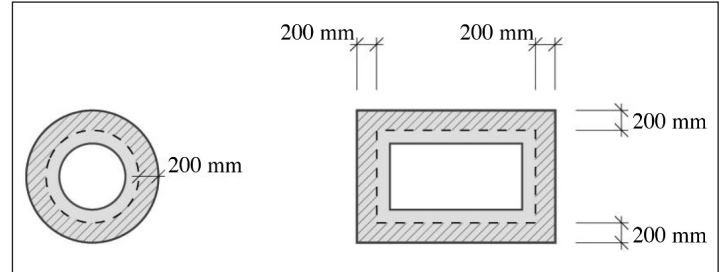


Figure 7.13-11: Minimum wall thickness for determining the minimum reinforcement of compression members

In walls, the minimum required cross-section of the vertical reinforcement is 0.2% of the concrete cross-section required for structural safety. The vertical bar spacing must not be greater than either twice the wall thickness or 300 mm.

Compression members with a reinforcement ratio below that stipulated above must be dimensioned like unreinforced structural members according to subsection 7.13.5.1.

The maximum longitudinal reinforcement ratio in compression members must in general not exceed 0.04 outside lap regions, unless it is shown that the integrity of the concrete is not affected, and the full strength is achieved at the ULS. This limit should be increased to a maximum of 0.08 at laps. In the case of large reinforcement ratios, special detailing and execution measures have to be taken and the stirrup reinforcement must be strengthened.

The longitudinal reinforcing bars of compression members must be prevented from local buckling by means of stirrups. If the reinforcement of the compression zone reaches the yield strength in the ultimate limit state, the corner bars as well as every second longitudinal bar must be enclosed with hooks or additional stirrups.

The spacing s of the stirrups and hooks must fulfil the following requirements:

$$\begin{aligned} s &\leq 15\phi_{sl,min} \\ s &\leq a_{min} \\ s &\leq 300 \text{ mm} \end{aligned} \tag{7.13-10}$$

where a_{min} is the minimum cross-sectional dimension and $\phi_{sl,min}$ is the minimum diameter of the longitudinal reinforcing bars.

In compression members with a polygonal cross-section, a longitudinal bar must be provided at least at every corner.

In the zones where forces are applied, in the area of splices and where the cross-section of compression members changes, additional stirrups must be provided in order to resist transverse tensile forces. The bar diameter of the stirrups must be at least 1/3 of the diameter of the thickest longitudinal bar.

The provisions for compression members apply analogously to the stirrup reinforcement in walls.

The horizontal reinforcement of walls must be dimensioned according to the requirements of section 7.6 but must not be less than 25% of the vertical reinforcement.

In the hinge regions of columns constructed with concrete of strength classes of C60 or higher, and without steel fibres, stirrups must be detailed such that an equivalent minimum core confining pressure of $0.01f_{ck}$ is achieved where the confining pressure is determined in accordance with subsection 7.2.3.1.6. Confining reinforcement must be continued beyond $1.2h$ to each side of sections with maximum moments. Alternatively, 2% by weight of end-hooked fibres may be used to provide sufficient sectional ductility to compression members of high strength concrete.

7.13.6 Special aspects of precast concrete elements and composite structural members

7.13.6.1 General

In designing precast concrete elements and structures assembled out of these, the final state as well as the construction states during transport and assembly must be taken into consideration. The development of the concrete strength over time must also be taken into account.

In the case of precast concrete elements and their connections, the technical requirements relating to production and assembly as well as the static requirements must be taken into account when specifying the dimensional tolerances. In particular, the effects of deformations due to creep and shrinkage of the concrete as well as temperature changes must be adequately taken into account.

Panels of facade plates with a multilayered structure must, as far as possible, be able to move freely within their plane. The panels must be connected by means of fatigue-proof and corrosion-resistant fastenings. The connecting elements must, as far as possible, be:

- chemically and physically compatible;
- protected against chemical and physical influences;
- fireproof in accordance with the structure.

7.13.6.2 Bearings

The nominal length of a simple bearing as shown in Figure 7.13-12 may be calculated as:

$$a = a_1 + a_2 + a_3 + \sqrt{(\Delta a_2^2 + \Delta a_3^2)} \quad (7.13-11)$$

where:

a_1 is the net bearing length with regard to the bearing stress, being:

$$a_1 = F_{Ed} / (b_1 f_{Rd});$$

F_{Ed} is design value of support reaction;

b_1 is net bearing width;

f_{Rd} is design value of bearing strength, with:

$$f_{Rd} = 0.4 f_{cd} \text{ for dry connections;}$$

$$f_{Rd} = f_{bed} \leq 0.85 f_{cd};$$

where:

f_{cd} is lowest design strength between supporting and supported member;

f_{bed} is design strength of bedding material;

a_2 is distance between edge of bearing and end of supporting member (Figure 7.13-12);

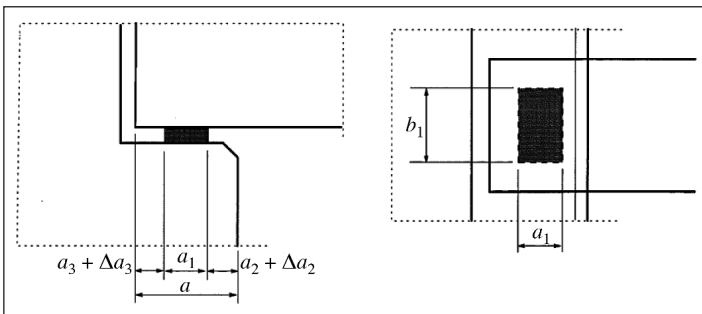


Figure 7.13-12: Definitions of bearing geometry

- a_3 is distance between most remote edge of bearing and end of supported member (Figure 7.13-12);
 - Δa_2 is allowance for tolerances for distance between supporting members (Figure 7.13-12);
 - Δa_3 is allowance for tolerances for the length of the supported member, with: $\Delta a_3 = l_n / 2500$, where l_n = length of supported member (Figure 7.13-12).
- Values for a_1 , a_2 , a_3 and Δa_2 are given in Tables 7.13-6 to 7.13-9.

Table 7.13-6: Minimum value of a_1 (Figure 7.13-12), in mm

Relative bearing stress σ_{Ed} / f_{cd}	≤ 0.15	0.15–0.4	> 0.4
Line support (floors, roofs)	25	30	40
Ribbed floors and purlins	55	70	80
Concentrated supports	90	110	140

Table 7.13-7: Minimum distance a_2 (Figure 7.13-12), in mm

Support material	σ_{Ed} / f_{cd}	≤ 0.15	0.15–0.4	> 0.4
Steel	line	0	0	10
	concentrated	5	10	15
Reinforced Concrete $\geq C30$	line	5	10	15
	concentrated	10	15	20
Plain concrete Reinf. Concrete $< C30$	line	10	15	15
	conc.	20	25	25
Brickwork	line	10	15	—
	concentrated	20	25	—

Table 7.13-8: Minimum distance a_3 (Figure 7.13-11), in mm

Detailing of reinforcement	Support type	
	Line	Concentrated
Continuous bars over support (restrained or not)	0	0
Straight bars, horizontal, close to end of member	5	15, but not less than end cover
Tendons or straight bars exposed at end of member		15
Vertical loop reinforcement	15	End cover + inner radius of bend

Table 7.13-9: Allowance Δa_2 (Figure 7.13-11), in mm

Support material	Δa_2
Steel or precast concrete	$10 \leq 1/1200 \leq 30$ mm
Brickwork or cast in situ	$15 \leq 1/1200 + 5 \leq 40$ mm

7.13.6.3 Mortar joints

Joints filled with mortar are used as a connection between precast members transmitting compression forces. The joint bearing capacity is:

$$N_{Rd,j} = f_{cd,j} \cdot A_j = \beta \cdot f_{cd,w} \cdot a_1 \cdot l \quad (7.13-12)$$

where:

$f_{cd,w}$ is design compressive strength of wall concrete;

$f_{cd,m}$ is design compressive strength of joint mortar;

$f_{cd,j}$ is design compressive strength of joint;

$\beta_0 = f_{cd,m} / f_{cd,w}$;

$\beta = f_{cd,j} / f_{cd,w}$.

The values β can be read from Figure 7.13-13.

The strength of the mortar should be at least 50% of the strength of the adjacent members.

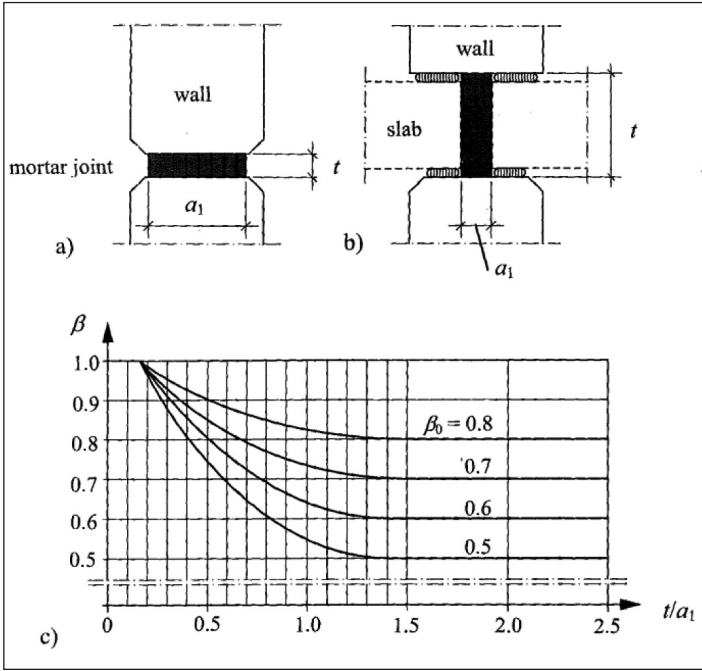


Figure 7.13-13: Capacity of mortar joint (fib Bulletin 43)

7.13.6.4 Loop connections

Loop connections as exemplified in Figure 7.3-14 can be used to transfer tensile forces, bending moments and shear forces. Loop connections are used between solid slabs where continuity is demanded.

The radius of the loop should satisfy the following demands:

$$r \geq \frac{\pi \cdot \phi}{4} \frac{f_{yd}}{\sigma_{c,rad}} \quad (7.13-13)$$

$$R \geq 8\phi$$

where:

$$\sigma_{c,rad} \leq f_{cd} \sqrt{b_i / \phi} \quad (7.13-14a)$$

$$\sigma_{c,rad} \leq 3f_{cd} \quad (7.13-14b)$$

in which:

$$b_i = 2 \cdot (c_e + \frac{\phi}{2}) \text{ but not smaller than } t \text{ (see Figure 7.13-14);}$$

$\sigma_{c,rad}$ = radial compressive strength at inside bend of loop (Figure 7.13-14);

c_e = concrete cover between U-bar and edge of element;

ϕ = diameter of reinforcing bar.

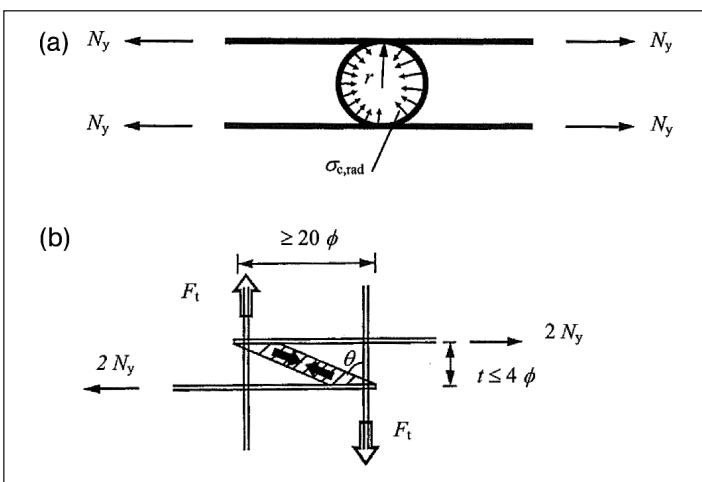


Figure 7.13-14: Loop connection (N_y = yield force in longitudinal bar, F_t = force in transverse bar)

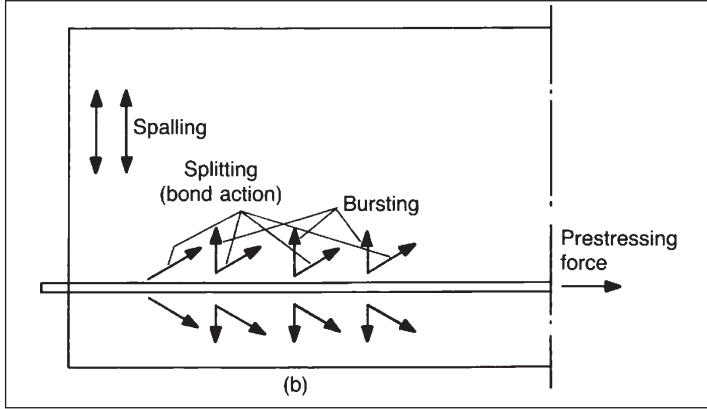


Figure 7.13-15: Bursting, spalling and splitting in the anchorage zone of a member, prestressed with steel with direct bond.

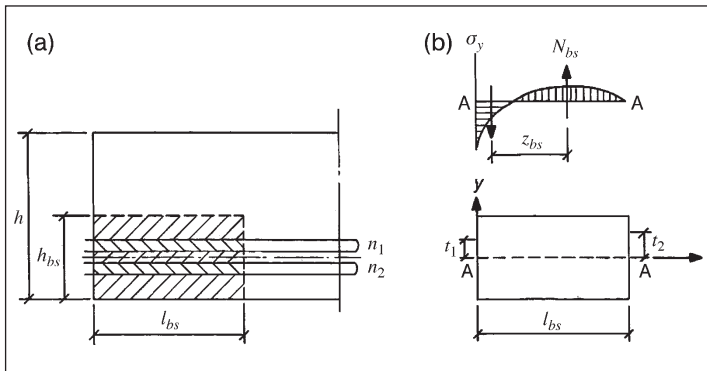


Figure 7.13-16: Calculation of the bursting force: (a) dimensions of the symmetrical prism; (b) moment equilibrium along section A-A

7.13.6.5 Transverse stresses in the anchorage zone of prestressed tendons

7.13.6.5.1 General

In the case of prestressing with pretensioned tendons or wires, three potential ways of cracking can be distinguished:

- bursting can occur due to tensile stresses which are generated as a result of spreading of the prestressing forces over the cross-section;
- spalling can occur at the end of the member, especially in the case of thin webs;
- splitting can occur along the transmission length as a result of the effect of internal pressure exerted by the prestressing steel during shortening (wedging effect).

Figure 7.13-15 shows the various mechanisms in the anchorage zone of a prestressed member.

7.13.6.5.2 Bursting

For the calculation of the bursting force the symmetric prism analogy may be used (Figure 7.13-16). The calculation is based on a virtual prismatic element, defined in order to describe the bursting forces. The prism is shown in Figure 7.13-16(a) (shaded area). The length of the prism is:

$$l_{bs} = \sqrt{h_{bs}^2 + (0.6l_{bpt})^2} < l_{bpt} \quad (7.13-15)$$

where l_{bpt} is the transmission length, where now the 5% value is used: this is equal to 50% of the values given in Eq. (7.13-5a) or (7.13-5b).

The internal lever arm for the bursting force is:

$$z_{bs} = 0.5l_{bs}$$

The bursting force N_{bs} follows from the moment equilibrium along section A-A (Figure 7.13-16):

$$N_{bs} = \frac{1/2(n_1 + n_2)t_2 - n_1t_1}{z_{bs}} \gamma_1 F_{sd} \quad (7.13-16)$$

where:

t_1 is distance between the centroid of tendons above section A-A to the centroid of the prism;

t_2 is distance between the centroid of the concrete stress block above section A-A to the centroid of the prism;

n_1, n_2 are numbers of the tendons above and below section A-A, respectively;

F_{sd} is design prestressing force per tendon;

$\gamma_1 = 1.1$, being the supplementary safety factor against overstressing.

The maximum bursting stress follows from:

$$\sigma_{bs} = N_{bs} / (b_{bs}l_{bs}) \quad (7.13-17)$$

where b_{bs} = width of the prism.

If $\sigma_{bs} < f_{ctd}$ no bursting reinforcement is required.

7.13.6.5.3 Spalling

The spalling stresses can be read from Figure 7.13-7, based on elastic analysis. If $\sigma_{spl} < f_{ctd}$ no spalling reinforcement is necessary.

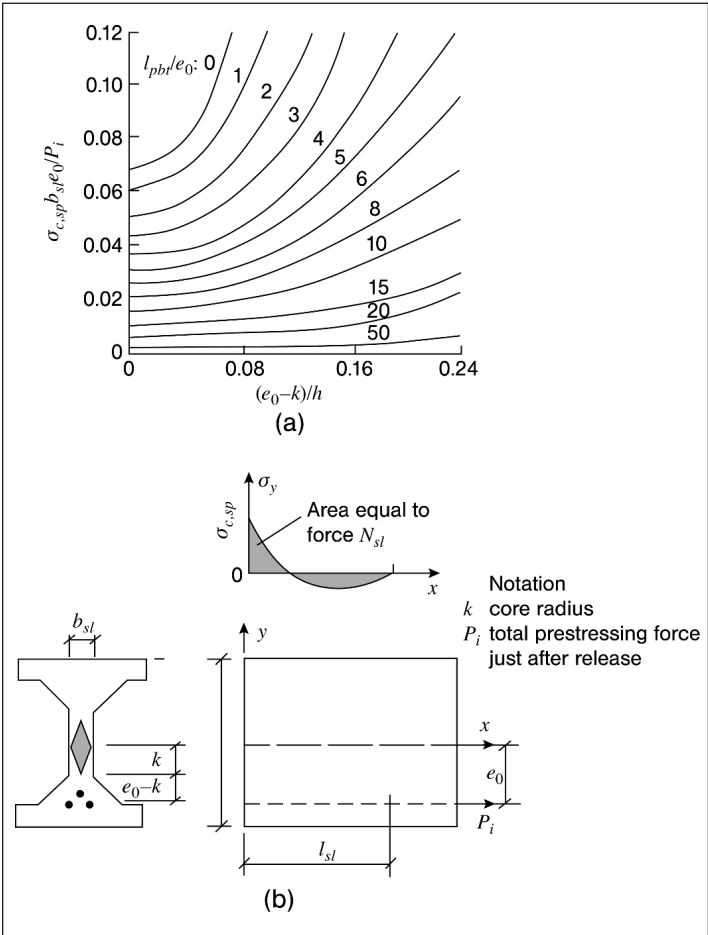


Figure 7.13-17: Maximum spalling stress as a function of eccentricity and transmission length (based on linear elastic analysis) for members with $h < 400$ mm

7.13.6.5.4 Splitting

No reinforcement against splitting forces is necessary if the distance between the strands/wires (with diameter $\varnothing$) and the cover satisfies the minimum values given in Table 7.13-10:

Table 7.13-10: Minimum cover as a function of the clear spacing to resist the splitting stresses around pretensioned strands

Concrete strength	Clear spacing	Cover
C20/25 to C50/60	$\geq 3\varnothing$	$\geq 3\varnothing$
	$= 2.5\varnothing$	$\geq 4\varnothing$
$\geq C55/67$	$\geq 2.5\varnothing$	$\geq 2.5\varnothing$
	$= 2\varnothing$	$\geq 3\varnothing$

For creating an envelope of allowable covers, interpolation between the clear spacings $2.5\varnothing$ – $3.0\varnothing$ and $2.0\varnothing$ – $2.5\varnothing$ is possible.

7.14 Verification of anchorages in concrete

The *fib* Bulletin 58 “Design of anchorages in concrete. Guide to good practice” (*fib*, 2011) covers the design of anchorages using the following types of anchors:

- cast-in anchors, such as headed anchors and anchor channels;
- post-installed anchors, such as torque-controlled and deformation controlled expansion anchors, undercut anchors, screw anchors, bonded anchors and torque-controlled bonded expansion anchors;
- post-installed deformed reinforcing bars.

The suitability of an anchor for the intended use should be demonstrated by a standard or a technical approval or by the results of suitable prequalification tests.

The following design situations are covered:

- verification of ultimate limit state for predominantly static loading;
- verification of ultimate limit state for non-static loading (fatigue, seismic);
- verification of ultimate limit state for extreme thermal conditions (fire);
- verification of serviceability;
- verification of limit states associated with durability.

The *fib* Bulletin 58 “Design of anchorages in concrete. Guide to good practice” (*fib*, 2011) includes the local transmission of loads into the structure by anchorages without or with special anchor reinforcement. In the design of the structure the anchor loads and additional design requirements given in section 8 of the *fib* Bulletin 58 “Design of anchorages in concrete. Guide to good practice” (*fib*, 2011) should be taken into account.

The local and structural effects of anchorages should be considered in design. The design of anchorages in concrete should be performed according to the *fib* Bulletin 58 “Design of anchorages in concrete. Guide to good practice” (*fib*, 2011). Only anchors that have been prequalified for the intended use should be used.

8 Construction

8.1 General

The execution of concrete works at the construction site must comply with the requirements given in ISO 22966 or a technical specification that fulfils the minimum requirements of this standard.

In the following subsections, key-elements from ISO 22966 are given together with some amendments.

The “execution specification” is the total sum of documents required for the execution of the works as provided by the designer to the constructor.

It includes ISO 22966 and the “project specification” prepared to supplement and qualify the requirements of ISO 22966 for a particular project, as well as referring to the national provisions relevant in the place of use.

Annex A of ISO 22966 lists issues in this standard where information for inclusion in the project specification, where relevant, is needed.

8.2 Execution management

8.2.1 Assumptions

These assumptions correspond with the assumptions of ISO 22966.

The following assumptions apply for this chapter:

This Model Code assumes that there is a site management which will take charge of the organization of the works and enable the correct and safe use of the equipment and machinery, the required quality of materials, the execution of a conforming structure and its safe use up to the delivery of the works.

This Model Code presupposes that the work is carried out with the necessary skill and adequate equipment and resources to perform the work in accordance with ISO 22966 and the requirements of the execution specification.

Health, safety and environmental aspects of construction are not within the scope of this Model Code.

8.2.2 Documentation

Before commencement of execution of any part of the works, the execution specification relevant to that part of the works must be complete and available.

The following items must be included in the execution specification:

- a reference to ISO 22966 and, if published, its national annex;
- a reference to other relevant international standards and national technical approvals;
- a reference to relevant national regulations and standards;
- a project specification giving information and requirements for the particular project prepared to supplement and qualify the requirements of the above listed documents;
- drawings and other technical documents needed for the execution.

“As-built” documentation of the direct input parameters to the service life design models might confirm the design assumptions and possibly give the basis for corrective measures. It might also serve as a basis for the condition assessments of the structure during its service life. Such extracts of the “as-built” documentation is sometimes named the structure’s “birth certificate”.

In execution class 2 and 3 (see subsection 8.2.3), an inspection report for the materials and products brought to the construction site as well of the completed works is required.

8.2.3 Quality management

Supervision and inspection are parts of the quality management.

Quality management forms an integrated part of the reliability differentiation for the project, see subsection 3.3.3.

The three execution classes give the option to specify the required level of quality management based on the importance of the component/structure and the criticality of the execution for its ability to fulfil its function.

Execution class 1 should only be used for structures where consequences in case of failure are small or negligible.

The execution classes comprise requirements for inspection and, depending on the relevant national annex or the execution specification, requirements for quality planning focusing on organizational measures and allocation of resources and personnel.

The three execution classes are connected to the three levels of reliability differentiation given in ISO 2394:1998, subsection 4.2.3.

The extent of inspection to be applied must be according to national regulations and must be stated in the execution specification by the selection of the appropriate execution class.

The types of inspection are normally differentiated in three categories:

- self-inspection by the operator;
- inspection in accordance with the procedures of the constructor;
- possible inspection by another company, that is an independent inspection.

If the inspection of the completed works reveals that the original design assumptions are not met, a partial or full re-design may be needed to get an overview of the possible consequences for the future performance of the structure. The new calculation must be supplemented with data from the inspection.

An example of such a situation may be if the inspection document mentions a too small concrete cover of the reinforcement. The actual cover should then be implemented in a recalculation of the service life and form the basis for conclusions regarding possible rectifications.

Updates from the inspection should also result in updates of the “maintenance plan” and the “inspection plan” for the structure, if relevant.

Prefabrications include cut reinforcement, cut and bent reinforcement and assemblies of these elements using ties, mechanical splices or welds manufactured off-site by a reinforcement fabricator and delivered to site or manufactured on the construction site.

Light surface rust, of a normal atmospheric nature, should be acceptable. However, if reinforcing steel is stored in an environment in which salt is present in the atmosphere – for example in a coastal zone – then rust formed may be considered harmful.

Passivation of the zinc coating can be achieved by storing the products outdoors for a sufficient period of time (4 weeks is thought to be a sufficient time) or by application of an appropriate passivation treatment.

Quality management aims to eliminate failures due to gross errors, and to ensure the resistance assumed in the design.

Requirements for quality management are specified using one of the following three classes, for which the required strictness increases from class 1 to class 3:

- execution class 1;
- execution class 2;
- execution class 3.

The execution class may refer to the complete structure, to components of the structure or to certain materials/technologies used for the execution.

The execution class being used must be stated in the execution specification.

The needed inspection to perform a conformity evaluation of the completed work must be carried out and the results documented.

Where inspection reveals a non-conformity, appropriate action must be taken to ensure that the structure remains able to perform as designed.

The following aspects must be investigated in the listed order:

- the implications of the non-conformity on further execution and fitness for intended design purpose;
- the measures necessary to make the component acceptable;
- the necessity of rejection and replacement of the non-repairable component.

8.3 Reinforcing steel works

The following subsections apply to both prefabrications and site fabricated reinforcement.

8.3.1 Transportation and storage

Reinforcing steel bars, coils, welded fabric and prefabrications must not be damaged (i.e. by mechanical damage and/or corrosion) during transportation and storage.

The surface condition of the reinforcing steel must be examined prior to use on site to ensure that it is free from mechanical damage (e.g. notches) and from loose rust and deleterious substances which may adversely affect the steel, concrete or the bond between them.

When galvanized reinforcement is used, either the zinc coating must be sufficiently passive to avoid chemical reactions with the cement, or the concrete must be made with cement that has no detrimental effect on the bond to the galvanized reinforcement.

8.3.2 Identification

The reinforcing steel delivered on site must be identified to steel grade and manufacturer, using either the rolling marks on the steel

Mechanical cutting is an appropriate method.

Recommended values for the minimum permissible mandrel diameter for bending reinforcing steel are given in Table 8.3-1.

Table 8.3-1: Recommended minimum mandrel diameter for bending bars

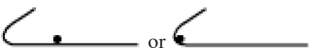
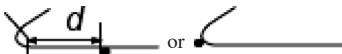
Bar diameter (mm)	Recommended minimum mandrel diameter for bends, hooks and loops
$\varnothing \leq 16 \text{ mm}$	$4\varnothing$
$16 \text{ mm} < \varnothing \leq 32 \text{ mm}$	$7\varnothing$
$32 \text{ mm} < \varnothing \leq 40 \text{ mm}$	$10\varnothing$

The recommended minimum mandrel diameters for bends, hooks and loops given in Table 8.3-1 are minimum values for reinforcing bars in accordance with product standards, which specify mandrels for bending tests which are $3\varnothing$, $6\varnothing$ and $7\varnothing$, respectively. Larger minimum mandrel diameters in the order of $10\varnothing$ to $15\varnothing$ should be used if re-bending or straightening of bars is intended.

For the sake of good detailing, larger minimum mandrel diameters may be required – see also subsection 7.13.2.4

Recommended values for the minimum permissible mandrel diameter for welded reinforcement or welded fabric bent after welding are given in Table 8.3-2.

Table 8.3-2: Recommended minimum mandrel diameter for welded reinforcement or welded fabric bent after welding

Recommended minimum mandrel	
 or 	 or 
$5\varnothing$	$5\varnothing$ if $d \geq 3\varnothing$ $20\varnothing$ if $d < 3\varnothing$ or welding within the bent zone

when available or the accompanying documentation in order to check that it conforms to the appropriate standard referred to in the order.

Anchorage devices and couplers must be identified using the marks on the devices or on its packaging/documentation in order to check that they conform to the appropriate standard or approval referred to in the order.

If means of identification has been lost, acceptance tests on samples may be required.

8.3.3 Cutting and bending

Cutting and bending of reinforcing steel must be made in accordance with accepted standard practice and must conform to the execution specification.

Bending must be carried out by mechanical methods using equipment designed for the purpose and in one operation at a uniform rate with the aid of mandrels. The diameter of the mandrel used must be suitable for the actual type of reinforcement and must not be less than values that could induce bending cracks in the reinforcement or failure of the concrete inside the bend of the reinforcement.

Bending of steel reinforcement at temperatures lower than -5°C should only be permitted if allowed by the execution specification and provided that the bending procedure conforms to accepted additional precautionary procedures.

Unless otherwise agreed, bending by heating must not be permitted.

Some countries require that the decoiled and straightened reinforcement must have the same properties as specified in the product standard for straight reinforcement.

Suitability for welding is defined in the appropriate product standards and is normally defined by limiting the carbon equivalent value of the steel (CEV).

Table 8.3-3 gives examples of appropriate welding processes and applications.

Table 8.3-3: Examples of appropriate welding processes and applications

Loading case	Welding method	Bars in tension ¹	Bars in compression ¹
Predominantly static	flash-welding	butt joint	
	manual metal arc welding and metal arc welding with filling electrode	butt joint with Ø ≥ 20 mm, splice, lap, cruciform joints ³ , joint with other steel members	
	metal arc active welding	splice, lap, cruciform joints ³ , joint with other steel members	
		—	butt joint with Ø ≥ 20 mm
	friction welding	butt joint, joint with other steels	
Not predominantly static	resistance spot welding	lap joint ⁴ , cruciform joint ^{2, 4}	
	flash-welding	butt joint	
	manual metal arc welding		butt joint with Ø ≥ 14 mm
	metal arc active welding		butt joint with Ø ≥ 14 mm
	resistance spot welding	lap joint ⁴ , cruciform joint ^{2, 4}	

Notes

1. Only bars with approximately the same nominal diameter may be welded together.

2. Permitted ratio of mixed diameter bars ≥ 0.57.

3. For loadbearing joints Ø ≤ 16 mm.

4. For loadbearing joints Ø ≤ 28 mm.

ISO 17660-1, “Welding – Welding of reinforcement steel – Part 1: Loadbearing welded joints” and “Part 2: Non-loadbearing welded joints” give requirements for welding.

When permitted by the execution specification, straightening of bent reinforcement must be allowed only if:

- special equipment to limit local stresses is used;
- appropriate straightening procedures are available;
- visual inspection of the straightened reinforcement is carried out for any cracks and other damage.

Reinforcement produced from coils will only be acceptable if an appropriate decoiling and straightening equipment is used and appropriate procedures in accordance with the equipment manufacturer’s instructions are applied.

The decoiled and straightened reinforcement must meet the requirements of the relevant specification.

8.3.4 Welding

Welding must only be carried out on reinforcing steel that is suitable for welding and by the use of approved equipment, personnel and approved welding procedures.

Welding of reinforcing steel and welding of reinforcing steel to structural steel in loadbearing joints must be performed according to appropriate procedures and as specified in the execution specification.

In general, reinforcing steel should not be welded at or near bends of the bars.

Spot welding of non-loadbearing welds performed according to an appropriate procedure should be permitted unless otherwise specified.

Spot welding of non-loadbearing joints for assembly of reinforcement using reinforcing steel with established suitability may be permitted without production weld testing unless otherwise specified.

Where a risk of fatigue exists for the concrete structure, welding of reinforcement should conform to special requirements.

The production and control of the welded connections should comply with the relevant requirements.

8.3.5 Joints

Bars must be jointed by laps, couplers or welding in accordance with the execution specification.

Overlapping bars should be placed in contact; in beams and columns the laps should be tied. If points of overlapping are not shown on the drawings, they should be evenly distributed and the longitudinal distance between two adjacent laps should not be less than 0.3 times the lap length.

Joints made with mechanical connecting devices must be carried out according to the manufacturer's instructions.

8.3.6 Assembly and placing of the reinforcement

The reinforcement must be placed according to the execution specification.

The reinforcement must be fixed and secured so that its final position after placing and concreting lies within the tolerances given in the execution standard or specification.

The assembly of reinforcement may be done with tie wire or spot welding (see subsection 8.3.4).

The specified cover to the reinforcement must be maintained by the use of suitable chairs and spacers.

8.3.7 Construction documents – reinforcement

Reinforcing steel works must be defined with details and required quality of works in a execution specification containing as a minimum: (i) construction drawings (shop drawings) giving all necessary information such as geometry of the structure and amount and position of reinforcing steel; (ii) description of all products to be used with any requirement for the application of the products; (iii) works description (method statements) including inspection requirements.

Actual construction works must be documented with sufficient detail and in sufficient number of construction records/certificates as specified in the construction specification.

8.4 Prestressing works

8.4.1 General

To ensure safety, reliability and durability of tendons it is necessary to provide:

- skilled supervisors and personnel, trained for the purpose and the prestressing system to be used;
- robust and safe equipment (regular maintenance and calibration) compatible with the prestressing system to be used.

Suitable safety precautions must be taken on site during all prestressing works.

The engineer must provide detailed execution specifications to sufficiently specify the requirements for prestressing works to be done on site including at least the following:

- specifications for the tensile elements and the prestressing system to be used;
- drawings to specify the tendon geometry;
- minimum concrete strength for stage or final stressing of tendons;

It is assumed that the execution specification gives detailed information on the layout and spacing of reinforcement as well as any precautions to be taken in areas of congested reinforcement.

Acceptable installation tolerances are given in ISO 22966, section 10. See also subsection 8.6.5 of this Model Code.

Concrete and cementitious spacers should have at least the same strength and should give at least the same corrosion protection as the concrete in the structure. Steel spacers should ensure that the correct concrete cover is achieved.

It is recommended that the prestressing works should be done by a specialized prestressing company.

High amounts of energy are stored in stressed tendons. Sudden unexpected release of the tendons is therefore dangerous.

- prestressing forces for each tendon and each tendon end, and sequence of stressing of tendons;
- permissible tolerances for prestressing force and tendon elongations;
- requirement for any hold point during prestressing works needing approval of the engineer before continuation of works;
- requirements for any special testing to be performed (e.g. measurement of transmission, i.e. friction losses in tendon, or grouting tests);
- requirements for traceability of materials.

The company responsible for the installation of the prestressing system must provide information to site to define the specific methods for the prestressing works to be done including at least the following:

- site requirements for storage and handling of materials and equipment;
- shop drawings showing tendon layout (support heights), anchorage assembly, and grouting details (vent locations);
- method statements for tendon installation, tensioning and grouting, adapted to the complexity of the site;
- theoretical tendon elongations and permissible tolerance for each tendon at specified stressing force;
- required quality control and reporting with optional hold points as specified by the engineer.

8.4.2 Packaging, transportation, storage and handling of materials and components

Materials and components must be packed such that they are protected from excessive moisture and/or dirt. Temporary corrosion protection may need to be provided on materials and components where required for the duration of storage and/or environmental conditions (e.g. humidity).

Materials and components must be protected from contact with any harmful substances.

Materials and components must be packed, transported, stored and handled such that they are not damaged and do not deteriorate (e.g. rust). Any damage or deterioration which is considered unacceptable must be repaired or the materials or components must be rejected.

Certificates of materials received on site must be compared against the execution specifications. Materials not complying must be rejected or placed in quarantine.

Materials must be used before their shelf life expires.

8.4.3 Prestressing works for post-tensioning tendons

8.4.3.1 Installation of tendons

Installation of the tendons and components must be in accordance with the shop drawings and method statements relevant for the project.

The placing of anchorages, coupling devices and ducts must be according to the specified geometry and tolerances. Care must be taken to avoid kinks in the tendon profile and damage to the components. Anchorages, coupling devices and ducts must be sealed against ingress of water and other foreign substances into the tendons during construction.

Tensile elements must be installed in the ducts as specified in the relevant procedures in the workshop (prefabricated tendons) or on site either before or after placing of the concrete.

Tensile elements must be protected from welding sparks.

The tensile elements must obtain temporary corrosion protection before installation if the expected time between installation/stressing and grouting of the duct exceeds acceptable time limits. Alternatively, the installed tensile elements may be protected by blowing of dry air or other equivalent methods. Temporary

Tendon installation tolerances are provided in national or international standards; see for example EN 13670, ISO 22966 or ACI 318.

No particular temporary corrosion protection is required if tendons made of prestressing steel are provided with permanent corrosion protection within 2 and 6 weeks after stressing for aggressive and benign environments, respectively, or if the prestressing steel is supplied with a factory-applied corrosion protection.

If the above periods cannot be satisfied, temporary corrosion protection with a suitable water soluble oil may be provided. Only oils must be used which do not need to be removed before providing permanent corrosion protection (see *fib* Bulletin 20, “FIP Recommendations for Corrosion Protection of Prestressing Steels”). Tendons should not be flushed with water.

Refer to the FIP State of the Art Report: “Tensioning of tendons: force-elongation relationship”, Thomas Telford, London, 1986.

In general, the tendon force is determined with the pressure in the stressing equipment. In exceptional cases, direct force measurement with load cells or similar may be done.

The calibration certificates must not be older than 6 months, in general.

Typically post-tensioning tendons are stressed in the following stages:

- stressing from 0 to 100% of the specified stressing force in one stage for tendons with little tendon deviation and little tendon slack, and recording of the tendon elongations only at the 100% mark. Such tendons are typically found in building slabs as single strand tendons or tendons in flat ducts;
- stressing in stages of 25, 50, 75 and 100% of the specified stressing force for tendons with considerable tendon deviation and tendon slack, and recording of the tendon elongations at each stage. With this procedure, the effect of tendon slack may be deleted from the full force-elongation graph. Such tendons are typically found in bridge construction as multistrand/wire tendons in round ducts.

If the actual tendon force and/or elongations are below acceptable values, the following measures may be considered:

- if the available prestressing steels and prestressing system allow, to apply a higher stress at the stressing end(s) of the tendons. This stress should never exceed the value of $0.95 f_{p0.1k}$;
- lubricate prestressing steel of tendons to reduce the friction losses;
- install and stress additional internal tendons into provided spare anchorages and ducts or additional external tendons into provided spare anchorages and deviators as may have been specified in the execution specification.

Elongation measurements are only a rough indication for the friction losses during stressing. If these friction losses are considered critical, the measurement of the force transmission with lift-off tests is recommended.

corrosion protection materials applied to the tendon must allow sufficient bond after grouting and must have no detrimental effect on the tensile elements or on the grouting material. Only products approved by the supplier of the prestressing system must be used.

Tendons, ducts and vents must be suitably marked for identification and recording during stressing and grouting operations.

Tendon components must be inspected before placing of the concrete for compliance with the relevant project and system documents.

Traceability of tendon components must comply with the execution specification.

8.4.3.2 Tensioning operations

General

Stressing of tendons must be in accordance with the project specific method statements.

Activities before stressing

- check concrete strength to have reached or exceeded the specified minimum strength for stressing;
- check that structure is free to move as may be needed during stressing;
- inspect exposed tendon ends and surfaces on which anchorages and stressing equipment will bear;
- check calibration certificates of stressing jacks to be recent.

Activities during stressing

- increase tendon force gradually and steadily in the stages indicated in the method statement;
- record tendon force (pressure) and tendon elongation at the stages specified in the method statement.

- compare specified force and elongation with measured and calculated values;

Suggested values for permissible deviations from expected elongations are:

- for tendons of 15 m length or less: 15% for a particular tendon, but not more than 7% for the sum of all values of tendons in the same section;
- for tendons of more than 15 m length: 10% for a particular tendon, but not more than 5% for the sum of all values of tendons in the same section.

Tendon elongations should be calculated with the properties of the prestressing steel as given in the material certificates.

Where tendon tails are burned off, the anchorage devices should be protected from excessive heat development.

For additional information, see the *fib* Bulletin 20, Guide to Good Practice “Grouting of tendons in prestressed concrete”, 2002, and EN standards 445-447(2007).

In hot or cold climates, only grouting materials must be used which have been confirmed to provide the specified performance and to remain sufficiently fluid over the expected period of time at the expected range of temperatures.

If measuring equipment (temperature gauges) is available, the 5°C value may be referred to the temperature of the structure in the vicinity of the tendon.

Grouting of a duct should be done without interruption. The speed at which the grouting material is pumped through the duct can vary from 5 to 15 m/min.

- take corrective actions if deviation between specified and measured values is greater than permitted tolerances;
- maintain tendon tails to allow re-stressing; only cut tendon tails and grout after final approval.

Activities after stressing

- visual inspection of concrete and anchorages;
- supervision of temporary protection of tendons;
- cutting of tendon tails once the stressing records have been approved.

Records

- pressure in stressing jack;
- elongation of tendons with adjustment for seating in tendon anchorages, as applicable, and for effects of measuring procedure (e.g. elongation inside stressing jack, if applicable);
- any particular observations during tensioning;
- site stressing records to be transmitted to and approved by the design engineer in view of any necessary adaptation of above instructions.

8.4.3.3 Grouting of prestressing ducts

General

Grouting of prestressing ducts must be in accordance with the project specific method statements. Grouting materials must be in accordance with subsection 5.4.2.3.

Activities before grouting

- equipment must be confirmed to be capable of adequately mixing and pumping the specified grouting material and must be operational;
- permanent supplies of water under pressure and of compressed air;
- grouting material to be confirmed for use under conditions expected on site (temperature, time required for grouting operations, tendon profile) with suitability tests or similar;
- instructions for actions in case of incidents (e.g. fault during injection) and harmful climatic conditions (e.g. after and during periods with temperature lower than 5°C);
- materials batched (excess to allow for overflow);
- blowing of air through the ducts, or using other approved methods, to ensure that ducts are free of harmful material (e.g. water, ice);
- vents prepared and marked with identification;
- preparation of moulds and other tools for quality control tests for grouting material.

Activities during grouting

- grouting to be carried out in a continuous operation and at the rate and in the sequence specified in the method statement;
- grouting must continue until the density of the grout flowing from the tendon ends and the vent openings is about the same as that of the injected grout; after closing the last vent, pressure may be held for about one minute to verify that there are no unacceptable leaks in the system.

Provisional inspection ports in the anchorages are recommended to check the grouting of the anchorages behind the anchor head.

Small voids which do not leave the prestressing steel unprotected do not need to be grouted.

To obtain a good seal of anchorage recesses, a preparatory treatment to the surface around the anchorages may be applied; in large recesses, connecting reinforcement should be provided.

Tendon installation tolerances are provided in national or international standards – see for example ISO 22966, EN 13670 or ACI 318. See also subsection 8.6.5 of this Model Code.

Activities after grouting

- immediately after grouting, check all accessible tendon areas for complete grouting, for example by light tapping. If voids are suspected, pump grouting material again until voids are filled;
- all equipment to be thoroughly washed shortly after grouting, followed by thorough draining of pump, mixer and hoses;
- record all operations, measurements, tests and particular events and submit to the engineer;
- after setting of grouting material check all accessible tendon areas (caps, vents, inspection ports) for complete grouting. If unacceptable voids are found, top up with fresh grouting material, or with vacuum grouting as may be suitable.

Records

- records to be established on all operations, measures, tests and particular events; to be transmitted to the design engineer.

Sealing of anchorages

Once the grouting material has set, all openings, grouting tubes and vents must be hermetically sealed and the anchorage recesses filled as specified in the execution specification to prevent penetration of water and harmful products (e.g. de-icing agents).

8.4.4 Prestressing works for pretensioning tendons

8.4.4.1 Installation of tendons

Installation of the tendons and components such as hold-down devices at deviators must be in accordance with the shop drawings and method statements relevant for the project.

The placing of anchorages and tensile elements must be to the specified geometry and tolerances. Care must be taken to avoid kinks in the tendon profile. Installation must be in accordance with the relevant procedures.

Details of local debonding of tendons must be provided as specified in the relevant shop drawings.

Tendons must be protected from welding sparks.

The tensile elements must be suitably protected from adverse environmental conditions. The time between installation and stressing of the tendons and pouring of the concrete providing permanent corrosion protection must be limited in accordance with subsection 8.4.3.1.

Tendons must be suitably marked for identification and recording during stressing and release operations.

Tendon components must be inspected before placing of the concrete for compliance with the relevant project and system documents.

Traceability of tendon components must comply with the execution specification.

8.4.4.2 Tensioning operations

General

Stressing of tendons must be in accordance with the project specific method statements.

Activities before stressing

- inspect tendon ends and abutments on which anchorages and stressing equipment will bear;
- check calibration certificates of stressing jacks to be recent.

Refer to the FIP State of the Art Report: “Tensioning of tendons: force-elongation relationship”, Thomas Telford, London, 1986.

In general, the tendon force is determined by the pressure in the stressing equipment. In exceptional cases, direct force measurement with load cells or similar may be done.

The calibration certificates must not be older than 6 months, in general.

If the actual tendon forces and/or elongations are below acceptable values, the following measures may be considered:

- if the available prestressing steels and prestressing system allow, to apply a higher stress at the stressing end(s) of the tendons. This stress should never exceed the value of $0.95 f_{p0.1k}$;
- install and stress additional internal tendons as may have been specified in the execution specification.

Suggested values for permissible deviations from expected elongations for deviated tendons are:

- for tendons of 15 m length or less: 15% for a particular tendon, but not more than 7% for the sum of all values of tendons in the same section;
- for tendons of more than 15 m length: 10% for a particular tendon, but not more than 5% for the sum of all values of tendons in the same section.

Suggested values for permissible deviations from expected elongations for straight tendons are:

- 10% for a particular tendon, but not more than 5% for the sum of all values of tendons in the same section.

Tendon elongations should be calculated with the properties of the prestressing steel as given in the material certificates.

Sudden release of the tendons may cause significant increase of the transmission length.

Activities during stressing

- stress tendons in the specified sequence gradually to the specified force and/or elongation;
- record pressure at jack or load cell and elongations of prestressing steel;
- compare measured and calculated values;
- take corrective actions if deviation between specified and measured values is greater than permitted tolerances.

Release of tendons

- check that concrete strengths have reached or exceeded the specified minimum strength for release;
- release the tendons with the approved method and in the specified sequence;
- record the tendon draw-in as specified for the project;
- compare measured and expected draw-in;
- observe instructions if deviation is greater than expected.

Records

- pressure in stressing jack;
- elongations of tendons with adjustment for seating in tendon anchorages, as applicable, and for effects of measuring procedure (e.g. elongation inside stressing jack, if applicable);
- records of draw-in of tensile elements;
- records to be established on all operations, measures, tests and particular events; to be transmitted to the design engineer;
- any particular observations to be recorded during tensioning;
- site stressing records to be transmitted to and approved by the design engineer in view of any necessary adaptation of above instructions.

8.4.4.3 Sealing

All tendon ends, debonded tendon areas and any other penetration (e.g. hold-down devices at deviators) must be hermetically sealed as specified for the project to prevent penetration of water and harmful products (e.g. de-icing agents).

8.4.5 Replacement of tendons

Any potential future replacement of internal unbonded or external tendons must be carefully planned and carried out by an experienced specialist contractor.

Particular procedures have to be specified and applied. When replacing the tendons by cutting individual elements, special procedures have to be applied to avoid the potential transfer of load from already cut individual prestressing elements to the remaining

ISO 22966 provides details of required construction documents.

fib Bulletin 48, “Formwork and falsework for heavy construction”, gives guidance for the selection of falsework and formwork.

The actual upper sieve size “ D ” represents the sieve size of which at least 80% to 85% of the aggregate fraction, but not more than 99% of the sample pass.

The term d_{max} normally defines the “maximum nominal upper aggregate size” which is selected to ensure a proper casting taking into account the cover and free spacing between the reinforcement bars.

In principle, any value of D less than d_{max} satisfies a requirement related to d_{max} .

ISO 22965-1 defines concrete as “...mixing cement, coarse and fine aggregate and water...”

Standards on aggregates may define coarse aggregates as aggregate with $D \geq 4$ mm.

Concrete with aggregates with actual upper size D in the range 4 – 16 mm might in some cases not support the design assumptions, for example concerning aggregate interlock, shear capacity, stiffness and fracture energy.

ISO 22965-1 therefore states in a note that in concrete for general purpose the coarse aggregate should normally have a D of at least 16 mm.

To ensure compatibility with the design assumptions, the acceptable range of D for the coarse aggregate, given with an upper and a lower value, should therefore be given in the execution specification, in particular if D smaller than 16 mm is allowed.

elements due to bond or clamping effects at deviation points which could overload the tendon and lead to an unintended sudden failure of the tendon.

Care must be taken to avoid unacceptably high unbalanced loads on deviators or diaphragms with tendon anchorages.

8.4.6 Construction documents – prestressing

Prestressing works must be defined with details and required quality of works in a execution specification containing as a minimum: (i) construction drawings (shop drawings) giving all necessary information such as geometry of the structure, number and position of prestressing tendons and anchorages; (ii) description of all prestressing systems to be used with any requirement for the application of the systems; (iii) works description (method statements) including inspection requirements.

Actual construction works must be documented with sufficient detail and in sufficient number of construction records/certificates as specified in the construction specification. Actual prestressing systems hardware must be documented with assembly drawings.

8.5 Falsework and formwork

Falsework and formwork including their supports and foundations must be designed and constructed so that they are:

- capable of resisting any foreseeable action to which they are submitted during the construction process;
- stiff enough to ensure that the tolerances specified for the structure are satisfied and the integrity of the structural member is not affected.

8.6 Concreting

8.6.1 Specification of concrete

Concrete and its specification must comply with ISO 22965.

The concrete specification must include requirements given in the execution specification and requirements related to the actual method of execution.

The actual upper sieve size D of the aggregate to be used in the concrete must be within the range given in the execution specification.

SCC mix design should comply with specific requirements in the fresh state depending on the type of application, and especially on:

- confinement conditions related to the concrete element geometry and the quantity, type and location of reinforcement, inserts and recesses;
- placing equipment (pump, truck-mixer, skip, etc.);
- placing methods (number of delivery points);
- finishing method.

Those requirements might be expressed and justified in terms of:

- flowability and grouting ability;
- viscosity (measure of the speed of flow);
- passing ability (flow without blocking);
- segregation stability.

The following methods are suitable for curing used separately or in sequence:

- keeping the formwork in place;
- covering the concrete surface with vapour-proof sheets which are secured at the edges and joints to prevent draughts;
- placing of wet coverings on the surface and protection of these coverings against drying out;
- keeping the concrete surface visibly wet with suitable water;
- application of a curing compound of established suitability.

Other curing methods of equal effectiveness may be used.

At the time of publishing this Model Code, standardized test methods characterizing the properties of curing compound are not available.

The quality of the curing will influence the properties of the concrete cover. Good curing will realize the potential of the mixed concrete, while less good curing will not. Deemed-to-satisfy requirements for service life design normally end up with tabulated requirements to cover thickness and cover quality. The tradition differs from nation to nation on how the quality of curing is handled. In some national provisions a relatively high w/c ratio (proxy characteristic for the potential quality) is specified together with strict requirements for the curing regime. In other provisions lower w/c ratios are given with a more relaxed demand for the curing.

ISO 22966 gives requirements for the construction operations involving structural precast elements from their reception at the site or in the case of site-manufactured elements from removal from the forms, until the completion of their installation and final acceptance.

The placing of any additional reinforcement and in-situ concreting for the completion of the structure must conform to the relevant sections of ISO 22966.

ISO 22966 gives two sets of permitted geometrical deviations.

The requirements given in section 10 of ISO 22966 are considered as achieving the design assumptions for the normally required level of safety, while those given in annex G of ISO 22966 are assumed to have relevance for service performance.

8.6.2 Placing and compaction

The concrete must be placed and compacted in order to ensure that all reinforcement and cast-in items are properly embedded and that the concrete achieves its intended strength and durability.

By the use of concrete described as self compacting concrete (SCC), the compaction of the fluid concrete is achieved due to the effect of gravity. Working procedures for the actual cast must be established, based on the constructor's experience and/or pretesting, to enable the required compaction to be obtained.

Additional requirements to those given in ISO 22965 to the fresh concrete properties and its conformity criteria, if any, must be agreed with the producer.

8.6.3 Curing

Concrete in its early life must be cured and protected:

- to minimize plastic shrinkage;
- to ensure adequate surface strength;
- to ensure adequate surface zone durability;
- from harmful weather conditions;
- from freezing;
- from harmful vibration, impact or damage.

ISO 22966 defines four “curing classes” for the duration of applied curing ranging from 12 hours to 70% of specified characteristic 28 days compressive strength.

The curing class to be applied must be stated in the execution specification.

8.6.4 Execution with precast concrete elements

Precast elements must be used as specified in the execution specification and the design coordination between them and the structural performance of the overall structure must be verified.

Handling, storage and protection of the precast elements must be carried out in accordance with the execution specification.

Requirements for the placing and adjustment of the precast elements must be given in the erection specification.

Structural connections including other materials, for instance steel, must conform to the relevant standards for execution like international welding standards.

8.6.5 Geometrical tolerances

The completed structure must be within the maximum allowable deviations to avoid detrimental effects in terms of:

- mechanical resistance and stability in transient and in service stages;
- service performance during the use of the construction works;

The tolerances given in subsection 4.5.2 are slightly different with those given in ISO 22966, but may be considered as equivalent.

Any requirements for special tolerances must be identified in the execution specification and the following information must then be given:

- any amendments to the permitted deviations given in ISO 22966;
- any further type of deviation to be controlled, together with defined parameters and permitted values;
- whether these special tolerances apply to all relevant components or to particular components which are identified;
- if the “box principle” must be applied, and what deviation is permitted.

The box principle will require that all points of the structure are within the specified theoretical position with a margin in any direction corresponding to the permitted deviation.

- placing compatibility for the erection of the structure and its non-structural components.

Deviations from the specified tolerance range must be handled in accordance with subsection 8.2.3. Small deviations which have no significant consequence on the performance of the finished structure may be ignored.

9 Conservation

Special performance requirements may be defined for special structures, such as historical structures. Special performance requirements, such as preservation of structural and architectural features in the structures, may restrict the type of intervention.

Information obtained via conservation activities can provide feedback to the design employed (in particular to the verification of limit states associated to durability – see section 7.8), facilitating assessment of compliance with the original performance requirements or revised performance requirements, if appropriate.

Further background information on conservation of concrete structures, as treated in this section, is given by Matthews et al. (2013), *Conservation of concrete structures according to fib Model Code 2010*. Structural Concrete, 14. doi: 10.1002/suco.201300046.

These are activities to achieve the intended service life of a structure.

These are activities to achieve an extended service life of a structure or revised performance requirements.

While one conservation strategy might be chosen for the management of an entire structure, in other circumstances different conservation strategies could rationally be adopted for the management of different groups of component parts of a structure.

Current engineering practice for conservation management for new structures is typically based on minimizing of costs to achieve and

9.1 General

In this Model Code, conservation is defined to mean all activities aimed at maintaining or returning a structure to a state which satisfies the defined performance requirements.

This chapter provides a general basis for the conservation of concrete structures and the components thereof, during their planned and/or extended service life, so as to ensure that the performance level of the structure/its components remain above that required for structural safety and serviceability.

Typically, two different conservation objectives are distinguished:

- conservation activities concerned with enabling a structure to meet its intended service life, as envisaged at the time of design;
- conservation activities concerned with extending the planned service life of a structure or enabling it to meet revised performance requirements (e.g. revised loading or functionality needs).

Condition control activities required for the conservation of a concrete structure must be planned when designing the structure. These plans must be developed on the basis of the design assumptions and a prognosis for the behaviour under the envisaged environmental and loading conditions over its intended service life, considering the performance requirements that need to be met.

Condition control activities required for the conservation of a concrete structure to extend its intended service life or accommodate revised performance requirements must be developed when re-designing the structure on the basis of knowledge of the current condition of the structure, a prognosis for future behaviour under the envisaged environmental and loading conditions and recognizing the implications of the revised performance requirements.

9.2 Conservation strategies and tactics

9.2.1 General

The necessity for conservation works varies widely and is influenced by many factors including the design and detailing of the structure, the environmental and loading conditions, along with the performance requirements.

Appropriate conservation strategies (one or possibly more) must be chosen for the management of the structure, with suitable conservation tactics being chosen for implementation.

For new structures, the conservation strategy and tactics must be defined during design stage. Both the chosen strategy and tactics must be confirmed/revised on the basis of the outcome of the condition evaluation, plus any associated intervention works, performed after construction of the structure.

For existing structures without a condition control plan in place, the preliminary conservation strategy and tactics must be set out during the re-design stage. Both the chosen strategy and tactic must be confirmed/revised on the basis of the outcome of the current condition evaluation and current performance requirements.

In the case of revised performance requirements, the conservation strategy and tactics must be revised to meet the new performance requirements. Both the chosen strategy and tactics must be set out/confirmed/revised during the re-design stage on the basis of the outcome of the current condition evaluation.

The goal will usually be to minimize life cycle costs for the structure, as discussed in section 3.5.

maintain a minimum performance/quality requirements (e.g. minimizing the costs of design, construction and conservation while satisfying appropriate performance/quality requirements).

For existing structures, optimization of conservation schemes is often similarly based on minimizing of the costs of conservation, while satisfying minimum performance/quality requirements.

Rational optimization requires an approach which weights the direct and associated indirect through-life cost of conservation against the benefits such as a longer service life, fewer faults or greater safety. Increasingly, attempts are being made to deal with conservation management on the basis of rational risk analysis. For more details, see “Probability in Civil Engineering” CUR Report 190, 1997.

Conservation activities are categorized on the basis of whether they:

- adopt proactive or reactive activities;
- adopt planned or unplanned activities;
- amend the reference performance level of the structure or component part; whether the activities are to maintain the intended performance level (i.e. a maintenance or repair activity) or are to enhance the current performance level (i.e. strengthening activity).

The reasons for adopting a proactive approach might include consideration of factors relating to technical, functional, aesthetic or economic issues. For example, this might be where interventions will be expensive and/or technically difficult if undertaken once appreciable deterioration has occurred. A proactive approach is likely to be particularly appropriate for structures having a long design service life (e.g. monumental, important or sensitive structures).

A proactive conservation strategy relates to an assumed use of condition control level/inspection regime CCL3. For a new structure these requirements would be defined during the service life design (see section 7.8).

A proactive conservation strategy should be applied to structures or parts thereof where deterioration needs to be kept within acceptable and predetermined bounds.

This approach is where appropriate limit states are defined beforehand that lead to the decision to increase the frequency of condition control (warning limits) and those that lead to the decision to carry out other conservation activities (activity limits). Condition based conservation may also be referred to as state dependent conservation.

Condition based conservation may be applied when it is difficult to make a prognosis of the future condition of a structure in the absence of information about the actions on the structure.

The choice of conservation strategy and tactic to be used depends on various factors including:

- consequences of a potential failure, which are related to the social and economic importance of the structure, the impact on third parties, the function of the structure or components thereof, the design and so on;
- feasibility of evaluating the condition of the structure; that is, how easy it will be to undertake a condition survey and assessment (which is related to the ease of inspections, measurements, the nature of the design etc.);
- predictability of the service life of the structure or its components, which is related to factors such as the design service life and the environmental conditions;
- recording and quantification of the actions that occur during the service life of the structure (e.g. mechanical loadings, physical, chemical and biological actions on the structure);
- feasibility of preventative or remedial interventions (e.g. repairs, replacements etc.);
- cost of conservation activities, which are related to matters such as the direct cost of the works; plus the frequency of inspections, repairs, replacements, indirect costs and so on, when a through-life perspective is being taken.

The available conservation strategies and respective tactics are classified on the basis of their proactive versus reactive characteristics as follows:

- proactive conservation activities;
- reactive conservation activities;
- situations where conservation activities are not feasible.

9.2.2 Strategy using proactive conservation measures

A proactive conservation strategy is based on preventative (or protective) measures and interventions aimed at avoiding and minimizing future deterioration or loss by applying some form of treatment or taking action prior to damage becoming visible.

A proactive approach to conservation of concrete structures is desirable as it should enable early identification of problems and possible risk issues affecting the condition of the structure, potentially enabling early preventative action being taken to minimize the overall cost of ownership. A proactive conservation strategy should be applied to structures or parts thereof where it is necessary or desirable that the performance be kept above a specified minimum performance requirement.

Tactics for delivering a proactive conservation strategy may involve various approaches including the following, which are illustrated below:

- condition based conservation;
- time based conservation.

9.2.2.1 Condition based conservation

With this approach the condition of the structure or its components is determined regularly and the evaluated condition must be registered, together with changes over time. The frequency at which the data is gathered and the associated evaluations are made may vary with time, potentially being related to the perceived criticality of the condition of the structure.

This approach requires that an appropriate parameter (or parameters) exist which can be evaluated and used to reflect the condition of the structure or its components. The condition based

Alternatively, condition based conservation may be adopted when regular determination of the condition of the structure by direct inspection is simple and cheap.

An example of this approach is the monitoring of the number of load or action cycles applied to a structure. Conservation works are undertaken after the registration of a certain number of load units (i.e. after registering an extreme action or after registering a predetermined measure of cumulative actions). Thus the application of a load dependent conservation tactic involves a load unit norm. The criteria and threshold values (limit states) are established beforehand during the service life design stage. Similar criteria might be applied for other types of actions.

Registration of the actions which cause deterioration is often difficult. In such situations, a conservation tactic that involves a time based norm may be more appropriate and practical to implement.

Time dependent conservation may be adopted when regular determination of the condition of the structure is not feasible or is expensive compared with the anticipated intervention costs. Different preventative interventions could be undertaken at different times/frequencies.

Reactive conservation strategy may be applied to structures or components thereof when apparent deterioration does not compromise meeting critical performance requirements. A reactive approach may be appropriate where direct condition control by regular inspections and interventions is not practical (e.g. because it is not feasible, too expensive and/or technically too difficult).

A reactive conservation strategy relates to the following assumed condition control level/inspection regimes. For a new structure these requirements would be defined during the service life design (see section 7.8):

- CCL2 is a reactive approach that utilizes a planned inspection but this is only visual in nature;
- CCL1 is a reactive approach utilizing an ad-hoc inspection and testing regime.

The failure dependent conservation tactic involves a predetermined failure norm/non-compliant performance condition. Thus on this basis a remedial intervention, perhaps involving a repair or replacement, to a structural component is not undertaken until it no longer meets the specified performance requirements. The failure dependent conservation tactic might potentially be applied to those parts of the structure that do not contribute to the probability of failure of the structural system as a whole. However, proper judgment of the consequences of failure is required because use of a failure dependent conservation tactic would often not be acceptable because of the potentially large consequences of failure.

There are parts of structures, such as foundations, where it may not be feasible to apply direct condition control and conservation measures.

The situation where there is no conservation strategy relates to the condition control level/inspection regime CCL0, which assumes that no direct physical inspection or monitoring is feasible.

For a new structure these requirements would be defined during the service life design (refer section 7.8).

tactic involves norms related to limit states. Exceeding the predetermined limit states invokes a preventative intervention to extend the service life of the structure.

Condition based conservation might involve consideration of factors such as:

- the number of load or action cycles applied;
- the development of cracking and/or spalling;
- loss of structural strength or stiffness;
- aesthetic effects (i.e. unacceptable deterioration in appearance).

9.2.2.2 Time dependent conservation

This approach involves the use of a time based norm to define when a preventative intervention should be undertaken to extend the service life of a structure.

This involves estimating an appropriate timing of preventative interventions or the time period between such interventions. The goal is to estimate a time period (life span) corresponding to a sufficiently low probability of failure/non-compliance for the actions being considered, typically taking them as a random variable over the period of time in question.

9.2.3 Strategy using reactive conservation measures

A reactive conservation strategy is based on remedial (or corrective) measures and interventions, aimed at arresting currently active processes which are causing deterioration or damage. Typically, remedial interventions involve some form of treatment or the taking of measures after damage has become apparent, presumably involving visual indications (e.g. cracking or spalling of concrete).

Tactics for delivering a reactive conservation strategy may include waiting for a specified failure/non-compliant condition to develop.

9.2.4 Situations where conservation measures are not feasible

There are parts of structures where direct condition control, inspection and the application of conservation measures is very difficult or not practical, or where it would be difficult economically and/or technically for preventative or remedial measures to be undertaken.

However, it may be feasible to obtain indirect indications of condition or performance of these elements via activities such as

In such circumstances it may be possible to gather indirect indications of structural performance through ancillary behaviours. For example, the performance of foundations might be inferred from related indirect parameters, such as differential displacements and/or relative movements within the various parts of structure.

These activities will require proper organization of the through-life conservation process, including development of the condition control plan (CCP) for chosen conservation strategy and tactics (see subsection 9.3.2).

Relationships between the activities described in this section and related activities within the Model Code for service life design are drawn out.

Condition survey, condition assessment and condition evaluation are condition control processes, which are performed as a part of the overall conservation process. For a chosen conservation strategy and conservation tactic, an appropriate condition control plan (CCP) must be developed that facilitates conservation management – see subsection 9.3.2

Figure 9.3-1 shows an idealized representation of the through-life performance of a structure requiring a remedial intervention to meet its intended design service life.

After completing the construction, the first condition control is performed and the birth certificate document is recorded. During the post-construction lifetime, knowledge of the performance of the structure is accumulated by means of through-life condition survey and/or monitoring.

Evaluation of the performance records against the performance criteria supports decision-making as to whether there is a need for an intervention. After carrying out the (required) intervention, the condition control is performed and a re-birth certificate for a structure is recorded. During the post-intervention lifetime the knowledge of the performance of the (possibly revised) structure is accumulated by means of through-life condition survey and/or monitoring.

land surveying, structure or ground movement measurement, or other indirect but related activities such as leakage detection. Before this approach is adopted a proper judgment should be made of the consequences of failure and the feasibility of condition control.

Should significant damage or deterioration occur and be detected, a potentially difficult and expensive reactive or remedial intervention could be required.

9.3 Conservation management

9.3.1 Through-life conservation process

To allow effective and efficient life cycle management, the conservation process must be adequately managed and documented. In particular, the conservation planning must be prepared in accordance with the selected conservation classification for the structure or its component parts, as appropriate.

In general, the through-life conservation process for a structure involves the following types and sequence of activities:

- condition survey, which is gathering information on the current condition and the previous development of that condition – see section 9.4;
- condition assessment, which is assessing available information to obtain an indication of current performance and to make a prognosis of future performance, including identification of deterioration mechanisms and prediction of damage – see section 9.5;
- condition evaluation and decision-making activities which are concerned with establishing the implications for the conservation of an asset (see section 9.6), evaluating potential conservation options, the selection of an appropriate intervention option(s) and the timing of such works – see section 9.7;
- execution of preventative or remedial works – see subsections 9.7.2 and 9.7.3;
- undertaking through-life condition survey and/or monitoring including recording of the information required for life cycle management – see section 9.8.

The general flow of conservation process procedures for new structures and for existing structures is shown in Figure 9.3-2. In parallel, it is indicated which information must be recorded to meet the requirements of life cycle quality management, as discussed in section 3.6 with regard to development of the life cycle file.

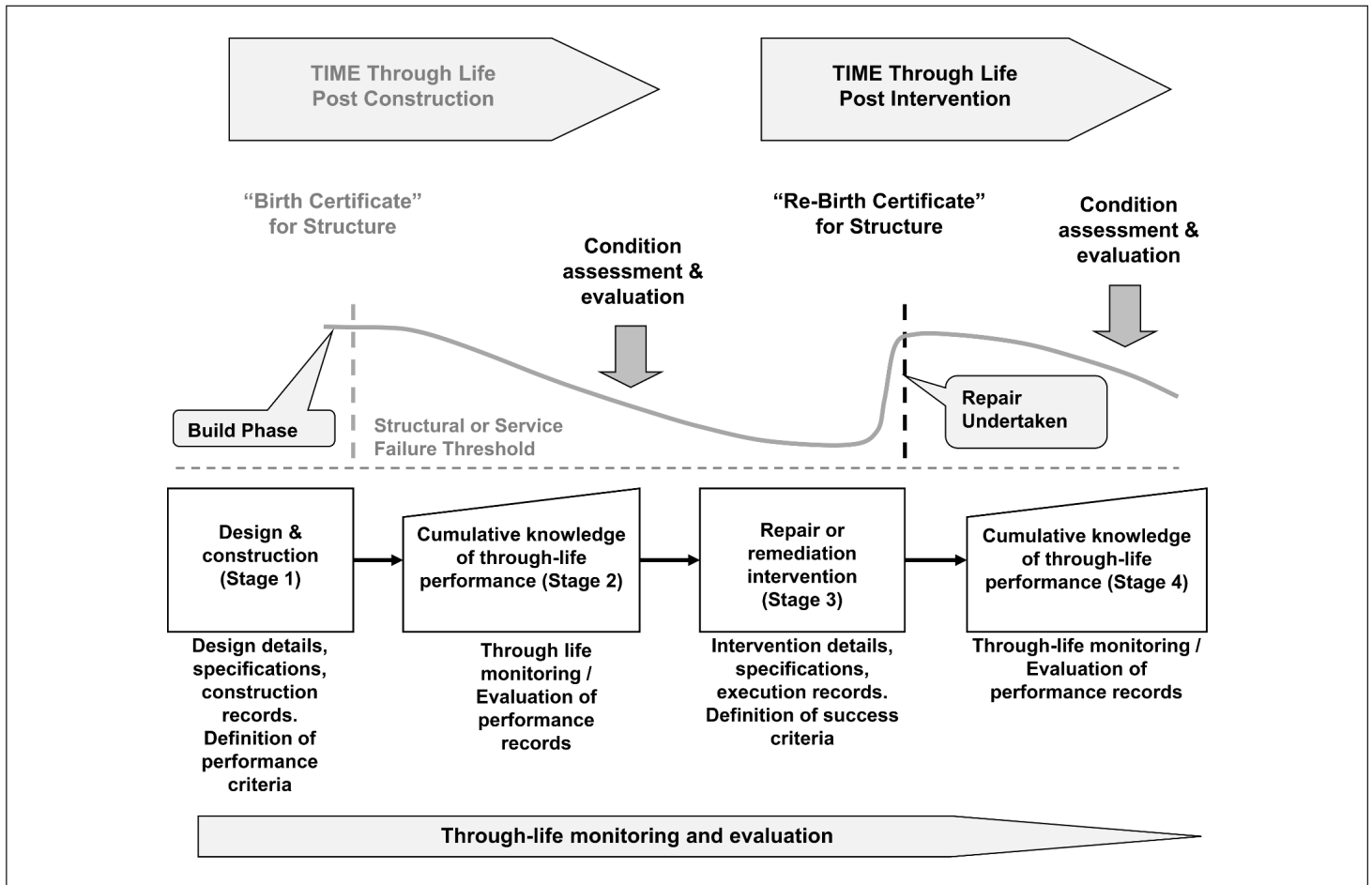


Figure 9.3-1: Idealized representation of the through-life performance of a structure requiring a remedial intervention to meet its intended design service life

Notes to Figure 9.3-1:

- Stage 1. Collection of details about the design concepts and execution of the original structure along with the expectations for its performance, including material specifications and information upon the quality on execution, to allow the birth certificate document to be prepared.
- Stage 2. Gathers information upon/monitors through-life environmental and loading influences, as well as performance post-construction and prior to a preventative or remedial intervention.
- Stage 3. As Stage 1 but for preventative or remedial intervention(s), plus the definition of appropriate success criteria (upon which to evaluate the subsequent performance of the intervention) to allow the re-birth certificate document to be prepared.
- Stage 4. Monitoring of through-life performance post-intervention, using defined performance indicators and environmental parameters and related factors establishing the context within which this data should be evaluated against the success criteria defined in stage 3 (and recorded in the re-birth certificate document).

Condition evaluation: Use of an agreed methodology to assess the overall condition of the structure (i.e. condition control is performed); see section 9.6.

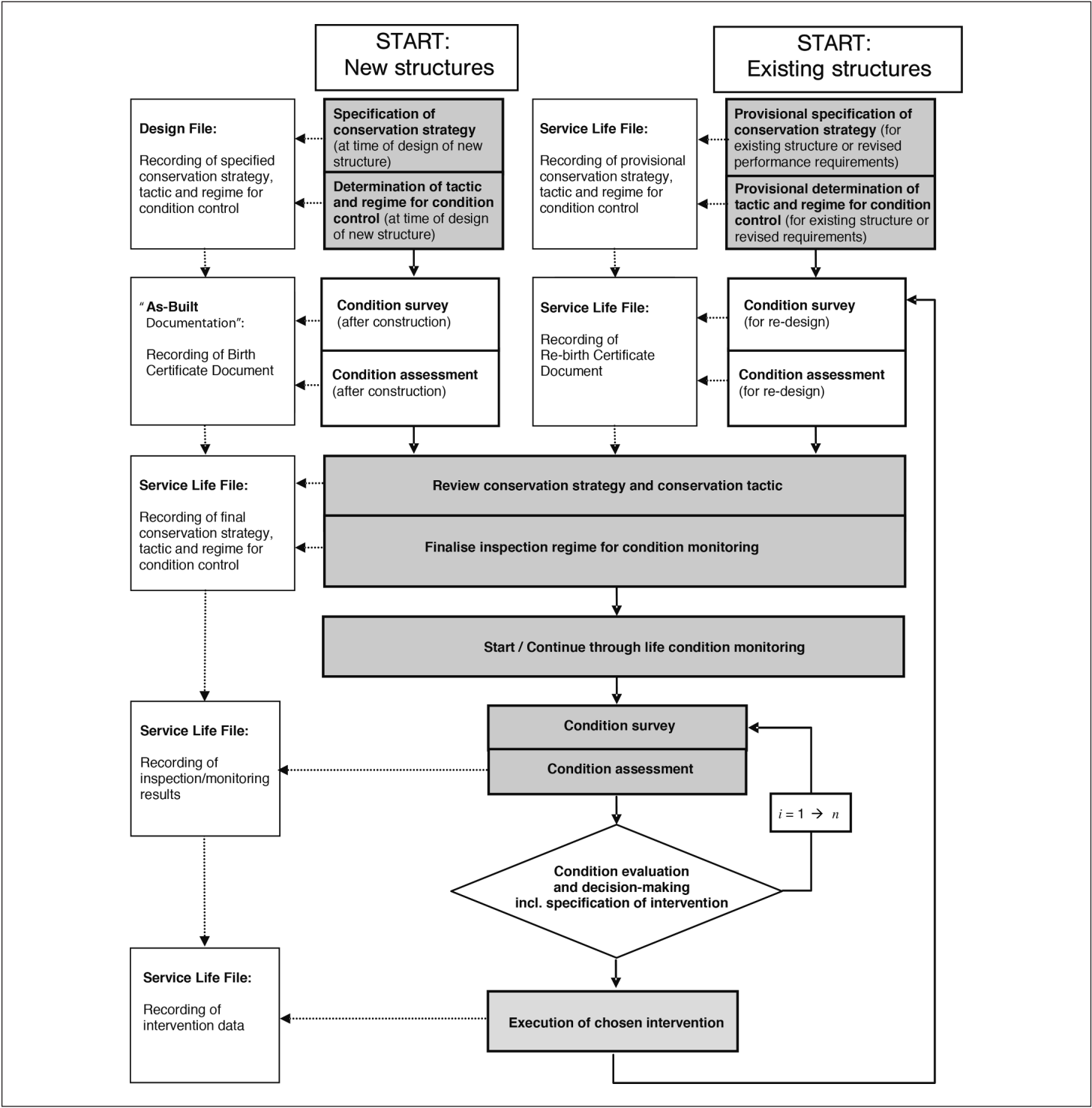


Figure 9.3-2: Conservation management: through-life conservation process and recording of information

The necessity for condition control, conservation and maintenance works varies widely, depending on factors such as the social and economic importance of the structure, the nature of its design and that of its components, potential threats posed to third parties and the intended (planned) service life of the structure. Account needs to be taken of the potential difficulties of undertaking inspections and associated condition control, conservation and maintenance works, along with prediction of deterioration and execution of intervention works.

The condition control and conservation planning adopted in chapter 9 must accord with the selected conservation classification for the structure or its component parts, as defined during the service life design process (as defined below).

Examples of structures within the three categories of condition control and conservation are as follows:

Category A:

- structures whose deterioration must not be apparent or would be technically unacceptable;
- monumental, important or sensitive buildings and structures.

Category B:

- structures for which remedial measures can be taken after deterioration becomes apparent (presumably visually);
- buildings and other common structures.

Category C:

- structures where it would be difficult economically and/or technically for preventative or remedial measures to be taken, such as foundations, for which direct inspection is so difficult and/or costly that assessment and judgment may need to be made based on information gathered on indirect performance indicators.

The Conservation Plan must state the types of inspection, testing and condition monitoring that have to take place, what components of the structure are to be inspected/monitored, what the frequency of the inspections should be and so on.

The provision for planned activities in the conservation plan must address the deterioration processes that are relevant for a structure during its service life.

This process leads to the final conservation plan to be adopted for the through-life management of the structure.

This process leads to a conservation plan which is to be adopted for the future through-life management of the structure.

The results are usually compared to specified requirements and standards for determining whether the item or activity is in line with these targets.

Surveys may involve visual inspections, measurements, tests or some form of gauging applied to certain characteristics in regards

Different categories are defined depending on factors such as the importance of the structure, its function, design service life, impact on third parties, environmental conditions, ease of maintenance and cost. Requirements for condition control should be defined at the time of design. The categories are as follows:

Category A: structures or parts thereof which are to be managed by planned proactive condition control and conservation.

Category B: structures or parts thereof which are managed by reactive condition control.

Category C: structures or parts thereof for which condition control and conservation measures are extremely difficult or are not practical.

9.3.2 Conservation plan

The conservation plan must state the type and frequency of the foreseen conservation activities and may include:

- specification of the regime for inspection, testing and condition monitoring activities – see section 9.4;
- methods of estimating the degree and rate of deterioration – see section 9.5;
- methods of evaluating structural performance against the performance requirements (criteria to be met) – see section 9.6;
- interventions which may be required as a result of non-conformity with the performance requirements – see section 9.7;
- methods of documentation of condition control – see section 9.8.

For new structures, the conservation strategy and conservation tactics must be defined and the conservation plan must be developed during the design stage. Both strategy and tactics must be reviewed on the basis of the outcome of the condition evaluation performed after construction of the structure.

For existing structures without a conservation plan in place, a conservation plan must be developed during the re-design stage on the basis of the current condition evaluation and consideration of the current performance requirements.

In case of the revised performance requirements, the conservation strategy and tactics, along with the conservation plan, must be revised to meet new performance requirements/criteria to be met. These must be reviewed during the re-design stage, on the basis of the outcome of the current condition evaluation and consideration of the revised performance requirements.

9.4 Condition survey

9.4.1 Condition survey and monitoring activities

In this Model Code, condition survey is understood to mean activities performed to gather information regarding the form and nature of the structure, current or potential deterioration mechanisms and/or change in performance of the structure or service conditions which are needed to evaluate conformity with the design (data) for actions and/or material and/or product properties. Condition survey is a part of condition control, which also involves condition assessment (see section 9.5) and condition evaluation (see section 9.6).

Accordingly, condition surveying is understood to mean an organized examination by a range of appropriate techniques and

to an object or activity. Testing undertaken is usually non-invasive or non-destructive in nature. Monitoring often involves the automatic recording of performance related data for the structure and possibly some degree of associated data processing. Strictly this does not need to be so, as there are a variety of means of gathering appropriate data.

Engineering evaluation and interpretation of the data obtained is central to the process of converting data into useful context related information.

It may be possible to gather indirect indications of performance through ancillary behaviours. For example, the performance of foundations might be inferred from the measurement of differential structural movements.

Proactive conservation requires more than just periodic visual inspections because environmental factors slowly and subtly change the internal conditions within the outer zone of the concrete (e.g. typically within the concrete providing the cover to the embedded reinforcing bars). As a result, concrete structures tend to experience changes in their condition which initially do not provide visible indications. Thus these changes do not manifest themselves as deterioration. Accordingly these changes cannot be recognized on the basis of visual observations alone.

It is only perhaps many years later that physical deterioration becomes sufficiently developed to be visibly evident (e.g. cracking and spalling of concrete arising from the corrosion of embedded reinforcing bars). As these changes and deterioration processes generally develop progressively from the surface of the concrete, and often over considerable periods of time, information on the progress of these actions is generally gathered using appropriate forms of non-destructive testing or sampling of the concrete. Appropriate data can also be obtained by suitable monitoring techniques.

While similar inspection, testing and monitoring activities may be employed, criteria for evaluating the results may differ where a structure is to meet either:

- its intended service life as envisaged at the time of design; or
- revised performance requirements, which might include an extension of service life, revised loading and/or other actions or functionality needs.

Information acquired by inspection, testing and monitoring the condition of the structure can greatly improve the accuracy of performance prediction by more accurately assessing the variability of the input parameters, which are typically assumed to be random variables. This can help reduce the uncertainty associated with the input parameters and aid characterization of their variability, as illustrated in Figure 9.4-1.

procedures performed on a periodic basis during the service life of the structure, while condition monitoring is understood to mean the process of observing such parameters on a quasi-continuous basis.

Significant changes in the data relating to the condition of the structure may be indicative of a developing failure. However, such deductions can only be established through an adequate process of data assessment and evaluation, which enables an appropriate context-based engineering interpretation to be made of the indications obtained.

Surveys, inspections, testing and condition monitoring activities must be carried out from an early stage in the service life of the structure.

Regimes for survey and/or monitoring the condition must be devised on the basis of the requirements of the conservation strategies selected for the structures or their components.

Potential approaches are shown in Table 9.4-1.

Table 9.4-1: Inspection, testing and monitoring regimes for classes of condition control/conservation strategy

Conservation strategy/ Class of condition control	Inspection, testing and monitoring regime (associated condition control level)
Proactive conservation measures (category A structures or structural elements)	Planned periodic inspection and systematic monitoring of parameters relevant to the design, in particular to the deterioration processes that are critical for the verification of the limit states associated to durability. (condition control level: CCL3)
Reactive conservation measures (category B structures or structural elements)	Planned periodic inspection (i.e. visual inspection by qualified personnel). No systematic testing or monitoring. (condition control level: CCL2)
	Ad hoc inspection and testing/investigation. No systematic inspection, testing or monitoring. (condition control level: CCL1)
No conservation measures (category C structures or structural elements)	No direct inspection, testing or monitoring. (condition control level: CCL0)

9.4.2 Locations for surveys and monitoring activities

Locations where inspection, testing and condition monitoring activities are to be undertaken must be carefully selected so that the desired information about the deterioration of materials and/or structural performance can be obtained, keeping in mind factors such as:

- the likely mechanism(s) and rate of deterioration;
- the environmental conditions;
- the conservation strategy and tactics, and the inspection testing and monitoring regimes defined at the time of design or re-design.

9.4.3 Tools and techniques for surveys and monitoring

Tools and techniques for inspection, testing and monitoring could involve a wide range of procedures. Typically they are likely to include a combination of visual observations, material sampling and possibly selected non-destructive and non-invasive testing methods.

Gathering the data necessary to establish the form and current condition of a structure may involve numerous techniques including inspections, measurements, sampling, as well as various forms of local condition and global response testing.

The approaches chosen must be appropriate to:

- the required reliability of the information to be obtained;
- the conservation strategy and tactics, and the inspection testing and monitoring regimes defined at the time of design or re-design.

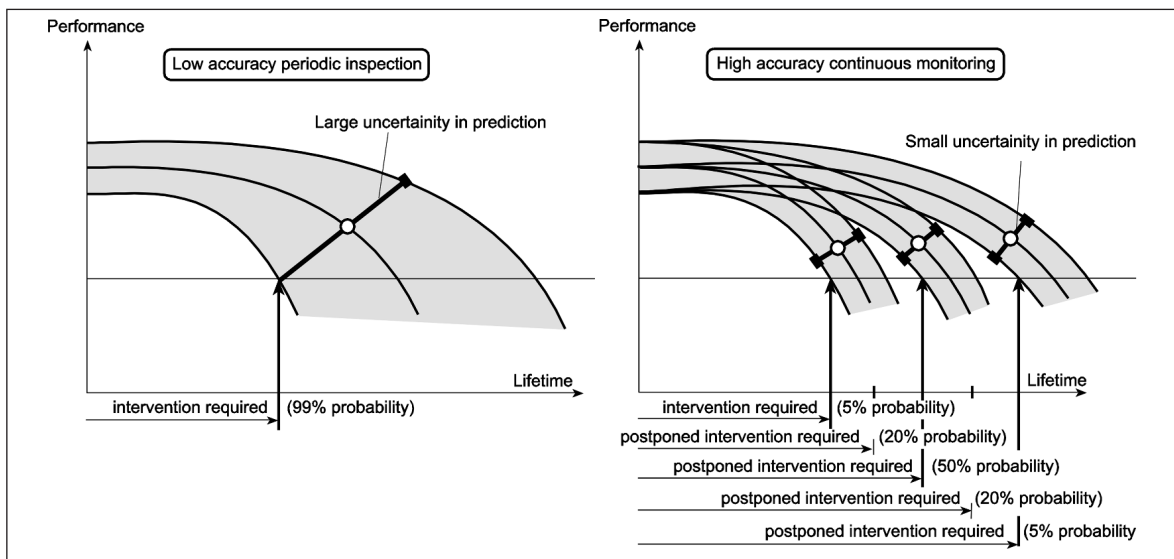


Figure 9.4-1: Estimating the timing of an intervention – illustrating the influence of the uncertainty of the input data

Figure 9.4-1 illustrates how the reliability of the data gathered and the assessment made using it can influence the estimated timing of required interventions, which can change significantly when the level of uncertainty in the data and assessed outcome are taken into account.

For more information regarding specific tools and techniques for inspection, testing and monitoring, see *fib* Bulletin 17: “Management, maintenance and strengthening of concrete structures” (*fib*, 2002), and *fib* Bulletin 22: “Monitoring and safety evaluation of existing concrete structures” (*fib*, 2003).

An inspection is a primarily visual examination, often at close range, of a structure or its components with the objective of gathering information about their form, current condition, service environment and general circumstances.

A survey is a process, often involving visual examination but which may utilize various forms of sampling and localized condition testing (e.g. measurement of depth of cover to reinforcement), whereby information is gathered about the form and current condition of a structure or its components. The testing undertaken is usually non-invasive or non-destructive in nature.

An investigation is a systematic process of inquiry undertaken to establish the specific details of a situation without restriction upon the methods used. For example, it may involve inspections, measurements, sampling and various forms of local condition and global response testing to establish the form and current condition of a structure. The process may also provide data for a prognosis of future condition and behaviour of the structure.

For more information regarding regimes for inspection, testing and monitoring, see *fib* Bulletin 44: “Concrete structure management – Guide to ownership and good practice”, (*fib*, 2008).

9.4.4 Gathering data for condition control purposes

Data may be gathered from the structure for condition control purposes by a variety of techniques which are used for undertaking inspections, measurements, testing and monitoring activities.

Inspection, surveys and investigations can be classified on the basis of their nature and the methods to be used, their frequency, the planning and timing of inspections/surveys, as shown in Table 9.4-2.

Table 9.4-2: Gathering data for condition control purposes for various conservation strategies

Gathering procedure (inspection/survey type)	Proactive conservation measures	Reactive conservation measures	No conservation measures
Initial inspection or survey	planned	planned	none
Routine (or regular) inspection or survey	planned	planned or ad hoc	none
Detailed investigation	planned	planned or ad hoc	none
Extraordinary inspection or survey	ad hoc	ad hoc	none

Initial inspection/survey should include not only the obvious inspection of external appearance of the structure, but potentially should address other “out of sight” parameters (e.g. cover to reinforcing bars) and workmanship issues. Data from the initial inspection/survey can be of use for subsequent routine/regular, detailed and extraordinary inspections/surveys.

An initial inspection/survey related to the development of a birth certificate for the structure should be conducted as the first step in collecting information on the structure.

In the case of an existing structure requiring a remedial intervention, such a process would lead to the development of a re-birth certificate for the structure. This would be expected to lead to the adoption of a proactive conservation strategy for condition control in the structure. Thus such an initial inspection/survey should be conducted as the first step in collecting information on the structure.

The specific tools and techniques to be used and the frequency of such inspections/surveys must be decided on the basis of factors such as the likely mechanism(s) of deterioration, environmental conditions and the importance of the structure. It is anticipated that these issues would have been considered at the time of design or re-design, when an appropriate conservation strategy and condition control level would be selected.

It is usually desirable that the structure or the components concerned be inspected at close range (arm's length). Some form of non-destructive testing, such as estimating the depth of carbonation or the penetration of chlorides, is likely to be required in order to understand the progress of initiation of deterioration. Not all inspections/survey need to have the same objectives and coverage. Thus the focus of particular inspections/surveys may differ and the scope and methods used would be expected to vary depending upon the nature of the structure or the component being inspected. Not all of these will need to be inspected/surveyed at the same frequency or in the same degree of detail.

This type of activity is intended to obtain detailed data regarding the deterioration and performance degradation of the structure or the component(s) concerned. The approach will often be hypothesis driven on the basis of the deterioration mechanism(s) acting or suspected. These activities are likely to be guided by records of past inspections/surveys, as well as those of previous maintenance, preventative and remedial works interventions. This information would typically be reviewed in conjunction with the results of deterioration prediction modelling.

The detailed quantitative data gathered might include characterization of the concrete surface, the quality of concrete, the depth of carbonation and chloride concentration in the concrete, the degree of corrosion of the reinforcing bars and/or similar parameters. Furthermore, the detailed inspection will need to include the data on the environmental conditions, such as temperature, humidity, airborne chlorides, loads and actions acting on the structure and so on.

The investigation will require use of appropriate tools and techniques, such as non-invasive and non-destructive testing and core drilling for material sampling, with the aim of locating and defining deterioration, damage and defects. Provision for close access to parts of the structure will be required.

Initial inspection or survey

An initial inspection/survey is carried out using appropriate tools and techniques in order to examine whether or not the structure is adequately constructed or whether previously specified preventative/remedial actions have been implemented satisfactorily.

It may also provide the means of collecting basic data about the as-built or repaired structure in order to initiate appropriate maintenance or preventative intervention works, or as part of a previously unplanned inspection/survey regime.

The following three types of inspection/survey are assumed as the initial inspection:

- inspection/survey carried out prior to the use of a newly constructed structure upon completion of construction (birth certificate);
- the first inspection/survey of an existing structure which has been subjected to a major preventative or remedial intervention, or substantially renewed (re-birth certificate);
- the first inspection/survey of an existing structure carried out for the purpose of an intervention activity in the absence of previously collected data.

Routine (or regular) inspections or surveys

Routine (or regular) inspections/surveys must be carried out periodically at intervals defined by the service life design to identify changing internal conditions within the concrete, indications of deterioration and the time of their first appearance.

These inspections/surveys must be carried out using appropriate tools and techniques to meet the objectives of the particular inspection.

Detailed investigation

A detailed investigation must be carried out to obtain sufficient specific information concerning the deterioration of materials and/or structural performance to facilitate detailed evaluation or re-design of the structure or components concerned. There are various circumstances when a detailed investigation may be required including when:

- signs of significant deterioration or a change in the performance level are observed during a routine/regular inspection/survey;
- a routine/regular inspection/survey is unable to provide the information required;
- it is suspected that the structural integrity of the structure could have been adversely affected by the extent of deterioration.

More information is required to decide whether a major remedial intervention is necessary and upon its potential scope. In a detailed investigation the items for inspection and/or the locations where testing and/or measurements are to be made, must be appropriate both in number and characteristic to the mechanism(s) of deterioration acting or suspected.

Extraordinary inspections/surveys are usually carried out when the structure is subjected to some form of extreme event such as a natural disaster perhaps involving an earthquake or typhoon, fire or collision involving vehicles or ships. Such an inspection is intended to accurately ascertain the state of the structure and judge if a remedial intervention is necessary. Extraordinary inspections/surveys may also be carried out after an accident associated with the deterioration of a structure, perhaps caused by a piece of spalled concrete falling causing injury to a passer-by.

An extraordinary inspection/survey would usually use a similar methodology to a routine or regular inspection. If further information upon the form or condition of a structure is judged to be necessary, a detailed investigation might then follow. Extraordinary inspections/surveys may also be carried out for a particular class of structure after an accident due to the deterioration of such a structure, as noted above. Such inspections/surveys may be carried out for similar structures, where the same accident could occur.

Judgment criteria are further discussed in section 9.6 on condition evaluation and decision-making.

When making inspections/surveys or investigations of a deteriorated structure, consideration should be given to whether the structure poses an immediate threat to the environment, its users, any third party, or the inspection/investigation team.

In such a situation, suitable emergency measures and temporary works may need to be taken immediately, in order to minimize risks and to attempt to avoid potential collapse and so on.

Depending upon the circumstances, it may be appropriate to take a number of immediate actions to minimize the risk of injury or worse. These measures might include closing the structure to further use, evacuating the structure and the surrounding area, setting up an exclusion zone to control access to the structure and the area which might be affected by any collapse and/or the release of stored/contained materials and shutting-off services such as piped gas which might be damaged and create secondary hazards.

Detail investigations may be required once the structure has been “stabilized” and put into a “safe” condition.

Figure 9.3-1 illustrates the time dimension associated with the inspection/surveys processes portrayed in Figure 9.4-2. Figure 9.3-1 shows an idealized representation of the through-life performance of a structure from the time of its construction up to the time when the first remedial intervention was undertaken so that the structure would meet its intended service life. The various inspections/surveys portrayed in Figure 9.4-2 would correspond to the through-life monitoring activities (stages 2 and 4) shown in Figure 9.3-1. The intervention activities portrayed in Figure 9.4-2 would correspond to stage 3 in Figure 9.3-1.

Extraordinary inspection or survey

An extraordinary inspection/survey will usually be carried out after the structure has been subjected to an accidental load or some form of extreme event, such as that caused by an earthquake, storm, flood, fire, impact by a vehicle or ship, to assess the extent of the damage and the need for a remedial intervention.

9.4.5 General flow of condition survey process

The general flow of the through-life conservation management process is shown in Figure 9.3-2. This is complemented by that in Figure 9.4-2, which presents a flow diagram illustrating the decision-making process for inspections/surveys and interventions; with particular attention being given to circumstances where judgments need to be made.

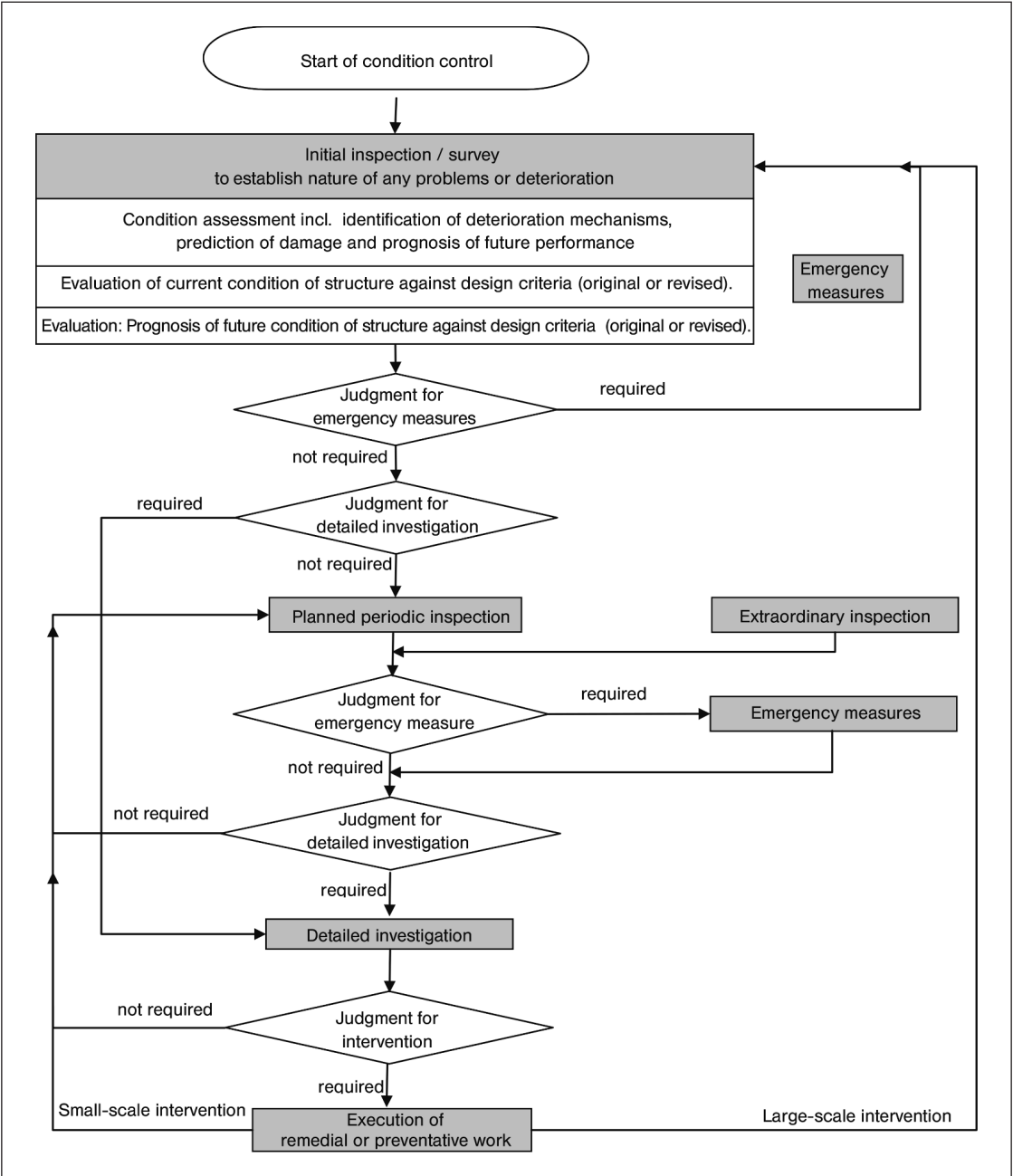


Figure 9.4-2: General flow of condition survey for structures taken into condition control following construction

9.5 Condition assessment

9.5.1 Identification of deterioration mechanisms and prediction of damage

This process would be expected to draw upon information obtained for and detailed in the birth certificate, and potentially also in the re-birth certificate, assuming that these were available.

The deterioration mechanism, present deterioration level and deterioration rate of materials and/or structural performance must be determined using appropriate models on the basis of the information obtained during from inspection, testing and monitoring activities, from the design and construction records, information upon previous interventions and the environmental conditions.

9.5.2 Identification of deterioration mechanism

As activities required for conservation of a concrete structure to meet its intended service life must be planned when designing the structure, this process would be expected to lead to the identification of the deterioration mechanism which might potentially affect the structure or its components.

It is necessary to identify the deterioration mechanisms which are affecting, or might potentially affect, the structure or its components. These must be identified from consideration of factors such as the environment to which the structure is exposed, variations in the

Similarly the process of evaluation and re-design required to extend the intended service life of a concrete structure or to accommodate revised performance requirements would be expected to lead to the identification of the deterioration mechanism(s) which might potentially affect the structure or its components.

The following list illustrates some potential deterioration mechanisms, but this should not be considered to include all possible mechanisms:

- carbonation induced corrosion of reinforcement;
- chloride induced corrosion of reinforcement;
- corrosion induced cracking and spalling of concrete;
- scaling, surface spalling or internal damage (disintegration) due to freezing-thawing with or without de-icing-agents;
- alkali silica reaction (ASR) induced cracking and deformation (expansion);
- cracking, deformation (expansion, swelling) or dissolution due to chemical attack (sulfate attack, magnesium attack, acid attack, salt attack etc.);
- fatigue induced cracking or internal damage;
- sustained load induced cracking or internal damage.

Among the above-mentioned deterioration mechanisms, corrosion and corrosion induced effects are the most common in many countries. Deterioration involving the combined effects of a number of the above mechanisms is also commonly observed.

This information would be used to make a prognosis of the behaviour of the structure under the envisaged environmental and loading conditions over the remainder of its intended or extended service life, considering the performance requirements that need to be met.

This process would be expected to draw upon information obtained for and detailed in the birth certificate, and potentially also in the re-birth certificate assuming that these were available.

It is assumed that the inspection and monitoring process outlined above will be able to furnish sufficient information to permit decisions to be made in respect of options for the management of the structure. If there is not sufficient information it may be necessary to initiate a detailed inspection.

exposure conditions, previous condition and performance data, if available, and any sign of deterioration that may be observed or detected.

9.5.3 Factors influencing deterioration

Factors influencing deterioration may be classified into the following :

- actions upon the structure (external factors) such as temperature, humidity and any other characteristics of the environment, including physical loading;
- factors related to the nature of the structure (internal factors) such as designed parameters (e.g. material characteristics, concrete cover for the reinforcing bars, geometry of sections etc.) and quality management during construction.

More than one factor may influence the development and progress of deterioration arising in a structure/member. For this case the combined (synergetic) effects of the deterioration factors must be considered.

9.5.4 Determination of deterioration level and rate

The deterioration level and the rate of change of material properties and/or structural performance must be determined from the results of the inspections/surveys and/or monitoring carried out and by using appropriate models for the mechanism(s) of deterioration.

9.6 Condition evaluation and decision-making

9.6.1 General

An overall evaluation of the extent of material and/or structural performance deterioration must be based on:

- records in the design file and the as-built-documentation, see chapter 4;
- results of inspection and monitoring outlined above;
- assessment of the nature, severity, significance and rate of deterioration.

Decision-making can be made either on a proactive (i.e. preventative intervention) or reactive (i.e. remedial intervention) basis, depending upon the underlying philosophy adopted for the management of the structure. It may be sufficient to re-design/evaluate the structure to address the performance non-conformity regarding limit states associated to durability. Consideration may also need to be given to changed performance requirements.

Account would need to be taken of the requirements for condition control and the conservation strategy and tactics defined at the time of design and/or re-design.

The structural performance is evaluated and verified for the performance requirement by following the structural analysis and verification provided in chapters 6 and 7 with the information related to materials and structural configurations, which have been obtained through the condition survey.

The process of selecting the most appropriate intervention option for the given circumstances might involve:

- confirmation of the conservation strategy and tactics to be applied to the asset concerned, and whether there is any intentional differentiation of the performance standards across the asset (e.g. a higher standard of aesthetic appearance may be required on facades which are in public view, or a higher performance standard may be required on some facades to minimize the risk of spalled concrete falling into public zones);
- confirmation of appropriate indicators of through-life performance;
- establishing the threshold levels for the performance indicators to be used in the design and evaluation of the intervention, together with the subsequent post-intervention monitoring;
- identification of intervention methods appropriate to the desired result (e.g. the proposed improvement in the condition of the

Also, condition evaluation must take account of the conservation strategy and tactics defined at the time of design or re-design of the structure.

Depending on how this evaluation compares with a minimum required performance level, a decision must be taken on how these findings and prognoses impact upon the current and any proposed manner of managing the structure or its components. It may be necessary to undertake an appropriate preventative (proactive) or remedial (reactive) intervention.

Suitable preventative or remedial activities, including repair or strengthening, must be initiated as necessary.

9.6.2 Threshold levels for deterioration of material and/or structural performance

The minimum level of performance for the (deteriorated) structure must be determined at the time of design and/or re-design.

The level set will take account of the required performance, such as the load carrying capacity to achieve the required functional performance, as well as issues such as serviceability, and possibly other factors such as the appearance of the structure.

9.6.3 Judgement criteria

When the results of the inspections undertaken (initial inspections, planned periodic inspections etc.) show defects/damage/changes in condition (expected or otherwise), a detailed investigation should be undertaken to evaluate the effects of those defects/damage/changes in condition upon structural performance.

If the defects/damage/changes obviously put structural safety and/or asset value in immediate jeopardy, the emergency measures should be undertaken immediately before detailed investigation is undertaken.

If the detailed investigation indicates that the structural performance does not satisfy the performance requirement or may not do so at some future time during the intended service life, it will be necessary to undertake an appropriate and timely intervention to rectify this situation.

9.6.4 Selection of interventions

Selection of an intervention option, including the materials to be used, must be made with consideration of the reasons for undertaking the intervention, the conditions under which the execution will have to be carried out and the conservation strategy/condition control level required after the intervention is made, so that the purpose of the intervention can be achieved in the most reliable way. Ease of conservation after intervention must also be taken into account.

The selection of the optimum intervention option can operate at a number of different levels, from making decisions upon the treatment of individual members up to that of the overall structure. Thus the selection perspective may be on the basis of a particular element, a zone or the overall structure. Thus different preventative or remedial intervention methods could potentially be used in conjunction to meet the various objectives and performance goals. The steps involved might include:

structure), which may or may not involve restoration back to the initial condition of the structure or fulfilment of the design requirements.

The process of choosing an intervention option may involve:

- defining the nature of the acceptance testing and verifications required;
- confirming the means for monitoring through-life performance post-intervention;
- benchmarking or comparison of the available intervention methods to establish the most appropriate (best) method for the particular circumstances and requirements.

The final published report for the CONREPNET project (EC Growth project GTC1-2001-43067) discusses the issues associated with the selection of appropriate interventions options when taking a through-life perspective.

Figure 9.7-1 shows schematically the performance of alternative intervention options. The evaluations made in this process need to consider a range of factors such as the importance of structure, the significance of non-compliance with the structure's performance requirement, and an extension of service life in future.

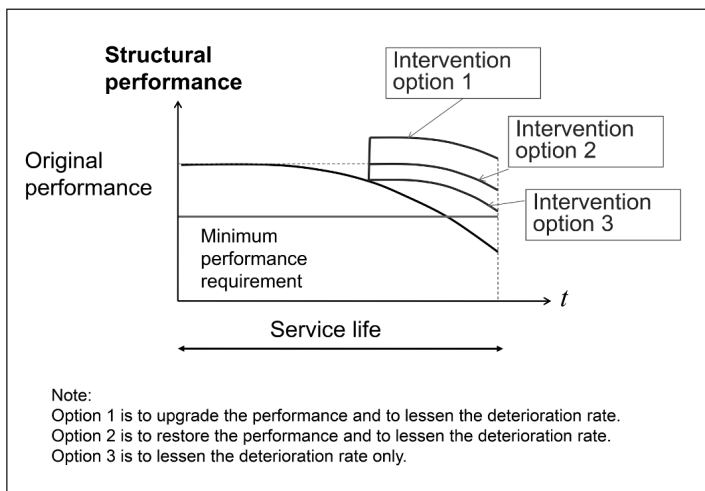


Figure 9.7-1: Schematic for intervention option selection

Planned interventions are typical for a proactive conservation strategy. A reactive conservation strategy may require both planned and unplanned interventions.

In the case of situations where conservation measures are not feasible, should significant damage or deterioration occur to the structure and be detected, a potentially difficult and expensive reactive or remedial intervention could be required.

For more information regarding interventions, see *fib* Bulletin 17: “Management, maintenance and strengthening of concrete structures” (*fib*, 2002) and *fib* Bulletin 44: “Concrete structure management – Guide to ownership and good practice”, (*fib*, 2008).

- performance requirements review, work analysis and work statement development;
- intervention design: goals and requirements/acceptance criteria;
- intervention design: specification development;
- execution of the selected intervention option(s);
- monitoring of the effectiveness of the intervention work.

9.7 Interventions

Undertaking interventions is one facet of the overall strategy for the through-life management of a structure, which necessarily involves consideration of time as well as cost in the evaluation of possible options.

In future, cost may need to include issues such as the carbon burden and environmental impacts associated with the structure and the proposed works, as well as their monetary cost (i.e. their first-cost and their whole-life cost).

Sufficient diagnostic tests must be carried out before execution to ensure that the intervention option selected is appropriate to the circumstances and future performance requirements.

Once the most appropriate preventative or remedial intervention option(s) has been selected it is still necessary to get the work carried out. To do this, the owner will usually need the support of a team of technical advisers, depending upon the expertise held by the owner. Together they should be able to address the various technical and process matters that may arise. Intervention activities need particular knowledge and experience. This group should be concerned with the performance of the repaired concrete structure. In essence what is required is an integrated team approach, wherever the necessary expertise actually resides.

Potential interventions can be classified based on planning as:

- planned interventions: maintenance, preventative and remedial interventions which are planned at the time of design for the construction of new structures or the design of repairs to an existing structure;
- unplanned interventions: remedial or strengthening interventions which are not planned at the time of design but become necessary due to unexpected damage or revision of performance requirements.

Appropriate steps must also be taken to ensure the use of proper materials, methods and quality management. A plan for the execution of the intervention including details of the materials to be used, the method of undertaking the proposed works and quality control tests to ensure quality of the work, must be drawn up before the work commences; see subsection 9.7.7.

A detailed record of the intervention work must be maintained for future reference; see section 9.8.

This term is commonly applied in the context of building fabric components with a limited life (e.g. coatings, surface finishes), components associated with water management and rainwater run-off, items where regular intervention is required to maintain their effective operation by removing any build-up of detritus (e.g. gutters, gullies, drains etc.), to overcome the effects of ageing or fatigue upon materials (paints, coatings, sealants, movement joints, etc.), to maintain an acceptable appearance by cleaning of surfaces, etc.

There are various methods and materials used for making preventative interventions including the following:

Methods:

- cathodic prevention;
- chloride extraction;
- re-alkalization;
- electrochemical dehumidification;
- surface coating, surface barriers, sealants and so on;
- overlays, wearing courses and such like.

Materials:

- various materials available.

There are various methods and materials for undertaking repairs including the following:

Methods:

- restoration of affected concrete section (patch and sprayed repairs);
- restoration of reinforcement provision (replacement of reinforcing bars);
- surface coating, surface barriers, overlays, sealants and such like.

Materials:

- various materials available including cast and sprayed concretes and mortars, reinforced concrete, fibre reinforced concrete, polymer concrete, proprietary repair mortars.

There are various methods and materials for undertaking these types of works which could include all the methods and techniques used in original construction. There is a wide range of various materials available including cast and sprayed concretes, reinforced concrete, fibre reinforced concrete and polymer concrete.

9.7.1 Maintenance interventions

Maintenance interventions involve various recurrent activities performed to either prevent or correct the effects of minor deterioration, degradation or mechanical wear of the structure or its components in order to keep their future performance at the level anticipated by the designer.

9.7.2 Preventative interventions

Preventative interventions are a proactive conservation activity concerned with applying some form of treatment or taking action prior to a change in a material property (e.g. such as that caused by the influence of carbonation or chlorides) adversely affecting the ability of the structure, or parts thereof, to meet the required performance levels because of deterioration.

In particular, proactive maintenance interventions seek to utilize actions which will result in an extension of the period of initiation, prior to the onset of the propagation of deterioration, or slow down the further development of deterioration.

The situation may include circumstances where the performance requirements have changed over time or where the planned service life has been extended. It is implied that the treatment or action will be taken prior to deterioration and damage becoming apparent/visible on the structure, for example cracking or spalling of concrete.

9.7.3 Remedial interventions

Remedial interventions are undertaken to reinstate to an acceptable level the current and future performance of a structure or its components which are defective, deteriorated, degraded or damaged in some way so their performance level is below that anticipated by the designer; generally without restriction upon the materials or methods employed.

Remedial interventions are typically required after a change in a material property (e.g. such as that caused by the influence of carbonation or chlorides) has adversely affected the ability of the structure, or parts thereof, to meet the required performance levels because of deterioration (e.g. corrosion of embedded reinforcing bars). The situation may include circumstances where the performance requirements have changed over time or where the planned service life has been extended.

Remedial interventions might be instigated for the purposes of repair, rehabilitation, remediation, restoration or retrofitting of the structure concerned, or of elements thereof.

9.7.4 Rebuild, reconstruction and replacement

Rebuild, reconstruction and replacement activities are intended to provide a new structure or structural components to replace the original damaged, defective or deteriorated entity, generally without restriction upon the materials or methods employed.

Commonly, this type of intervention would be undertaken to restore the performance level of the structure or specific structural components to the level anticipated by the designer. In this case, these activities would be deemed to be a remedial intervention. However, in other circumstances these works might provide the opportunity to revise the performance requirements to a higher level (i.e. above the level anticipated by the designer). In this case, these activities might be deemed to be a strengthening or upgrading intervention.

There are various methods and materials used for making strengthening or upgrading interventions including the following:

Methods:

- external bonding (plate or sheet bonding, over or under-laying, jacketing, etc.);
- introduction of external prestressing and addition of tension cables;
- addition of a member (girder, brace, support, etc.);
- increase in member cross-sectional size;
- replacement of member.

Materials:

- concrete (reinforced concrete, prestressed concrete, fibre reinforced concrete, polymer concrete, etc.);
- steel (plate, cable, bars, etc.);
- non-metallic materials (plate, sheet, cable, etc.).

This approach is only valid on the basis that the additional information gathered is able to contribute directly to the judgments to be made on the performance of the structure, including safety considerations, and the remaining design service life.

For example, where shear failure of a beam was the main concern, it would not be appropriate to rely on the monitoring of flexural deflections because it is unlikely that this parameter would give adequate indication of distress or developing failure prior to a catastrophe collapse of the beam occurring.

In some situations the aesthetic appearance of a structure or building may be paramount and it may be necessary to assign this parameter a particularly high importance and to ensure that this aspect is always kept in order.

Plans for further action must be drawn up on the basis of the known extent of deterioration, taking account of what further information is required in order to make a satisfactory assessment of its current condition and safety. Additional investigations may be required. Once the immediate threat and concerns have been addressed, consideration should turn to aspects of the remaining design service

Thus in some instances this approach may be considered to be an extreme form of remedial intervention, which involves the creation of a new structure or structural component to replace the original damaged, defective or deteriorated entity after its destruction or demolition. These activities require careful planning and management in order not to compromise the safety and/or performance of other parts of the structure while the works are carried out.

9.7.5 Strengthening or upgrading interventions

Strengthening or upgrading interventions are undertaken to improve the strength or some other performance property of a structure or structural elements (e.g. stiffness) above the level anticipated by the designer.

The revised performance requirements would be defined during the re-design process. Thus after these works the structure or structural elements would be expected to meet the revised performance requirements.

9.7.6 Other activities and measures

Intensified inspection, survey or monitoring

Intensified inspection/survey refers to inspections/surveys which may be carried out by suitably increasing one or more of: frequency of inspection/survey, number of inspected/surveyed items and the locations to be inspected/surveyed.

It is not appropriate to rely on an intensified inspection, survey or monitoring if the data or information gathered does not contribute directly to the decision-making procedure associated with the safety management of the structure.

Usage restriction

Usage restriction is a structure management strategy which imposes some restriction on usage (e.g. restriction on the maximum traffic load or speed, maximum imposed floor load in a building, etc.), so that the maximum effects of specified actions on the structure would be reduced.

The extent of restriction must be determined depending on the level of deterioration observed or suspected and a prognosis for the (near) future.

Monitoring and/or other measures, such as detailed investigations, may also be necessary in addition to imposing the restriction.

Emergency measures

When a deteriorated structure poses an immediate threat to the environment, its user or any third party, suitable emergency action and temporary works must be taken immediately. Detailed investigations may be required once the structure has been “stabilized” and put into a “safe” condition.

life, actual and desired structural performance and functionality, along with conservation strategy, tactics and any previous remedial actions taken.

See also observations made in subsection 9.4.5.

Dismantling and removal of a severely deteriorated structure is generally the last option. It must be carried out only after taking into account of all relevant factors including hazards to the environment, public safety and disposal of the debris. Details are provided in chapter 10. Special considerations may be required for prestressed concrete structures.

See also observations made in subsection 9.4.5.

The term execution is used to refer to all the physical activities undertaken for the physical completion of the intervention works. These activities include procurement, scaffolding, falsework, formwork, reinforcement, concreting and curing, as well as the related inspection and documentation of those activities.

The certainty that the desired outcomes will be achieved can be improved by “thinking intervention” long before the intervention process actually starts on site. In this way, those involved develop a process-focused approach which potentially leads not only to a reduction in problems on site and associated costs, but also to related improvements in health and safety for those involved.

Quality assurance activities: ensure that effective preventive actions are incorporated in relevant process control procedures at all stages of the definition and execution of the intervention so as to minimize the identified problems and/or reduce the level of risk to an acceptable value.

The process control procedures, and the preventive actions incorporated in them, must be documented so that the responsible operators and supervisors are clear not only what is to be done during the course of each process, but also how the process is to be undertaken.

Quality control activities: ensure that verification procedures are prepared and implemented which define how, when, where and by whom each intervention related process will be monitored. The observations need to be recorded by appropriately qualified, experienced and independent staff, dedicated to surveillance activities, with adequate time and resources to conduct the required surveillance activities.

Appendix G of *fib* Bulletin 44: “Concrete structure management – Guide to ownership and good practice” (*fib*, 2008) sets out steps for the development of supplementary documents to support the preparation of a project specification for the execution works for concrete structures, linking with the requirements of the ISO Execution Standard (ISO 22966, *Execution of concrete structures*, ISO, 2009).

Further information on quality management during execution is also given in the Model Code chapter on life cycle management – see section 3.5.

Dismantling and removal

Dismantling and removal of a severely deteriorated structure requires careful planning and evaluation before embarking upon the process as there are a wide range of potential complications which may arise.

9.7.7 Execution of interventions

Execution of interventions must be carried out with minimum disturbance to the surrounding environment and the service condition of the structure.

A detailed project specification for the execution of the intervention works must be developed clarifying and extending the general requirements of the ISO Execution Standard as necessary with project specific needs to meet the performance and durability requirements for the structure, together with the necessary supporting documents.

As part of the activities associated with selecting an appropriate intervention option, all intervention processes should be effectively analysed with a view to identifying significant potential problems and risks of non-conformance with the project requirements, including assessing the level of risk involved in each case.

The performance of the structure after intervention is strongly affected by the quality of execution. Therefore, the execution of intervention must be properly planned and carried out under an approved quality management plan based on the selected intervention option(s) and the requirements set down in the project specification. These activities must be supported by appropriate quality assurance and quality control activities.

Appropriate verification tests must be carried out as part of the execution process for quality control purposes. A detailed record of the execution must be maintained for future reference.

After execution, an initial inspection/survey must be carried out to examine the structural performance after intervention and to collect necessary data for future condition control.

This process would be expected to draw upon information contained in the *birth certificate* document. There is the expectation that these activities would draw upon/contribute to the preparation of a *re-birth certificate*.

The conservation records should be kept for the life of the structure. It is also important that an appropriate way is established to transfer the knowledge and understanding of the structure from the present generation of engineers/persons with responsibility for the management of structure to the following generation who will take over responsibility for its care and management.

9.8 Recording

To allow effective and efficient life cycle management, the conservation records must be kept and preserved for future reference in the service-life file. Conservation records must be included in the life cycle file. The conservation records for a structure must be preserved while it remains in service/in existence.

Details of recording during conservation are given in section 3.5, along with other information regarding quality management.

10 Dismantlement

10.1 General

In design for service life, due consideration should be given to a potential future need for refurbishment, adaptation to changing needs and dismantlement, considering re-use of parts of the structure involved. It should be noted that already in the design phase of buildings, the possibility of using the building for another function than the original one should be regarded. As an example, buildings with large span floors with residual bearing capacity, and facades allowing the entrance of ample daylight are often more suitable for a change of function than buildings with small rooms and heavy separation walls. A good conceptual design of structures can therefore prevent later dismantlement, since the structure can be re-used for another function.

In the past, structures have rarely been designed taking into account the possibility of adaptation or re-use in future for other aims. The increasing need to demolish structures has therefore led to an accelerated development of all types of demolition techniques. However, those new technologies often go along with other non-satisfactory phenomena resulting from insufficient analysis of their influence on the stability of the structure and with other negative effects for environment.

Structures can reach the end of their service life for different reasons:

- the structure no longer meets the demands of safety;
- the structure is no longer suitable to fulfil its function;
- the structure should be upgraded, but this is more expensive than demolishing and building another one.

The dismantlement of large structures requires careful preparation. The aim of dismantlement should be defined with regard to its extent (partial or full), requirements for eventual re-use of the material or components, and the time available for demolition. The structure to be dismantled should be investigated with regard to its bearing structure and capacity, the materials used, its stability in the various stages of dismantlement and the effects of eventual contamination.

An appropriate method of dismantlement should be chosen, taking into account economy and boundary conditions of its eventual continued use during the (partial) dismantlement process. Measures should be taken to minimize the hindrance for the environment with regard to noise, dust, water, contamination, vibration and shocks. Every stage of the dismantlement should be carefully planned with regard to the stability of the remaining part of the structure, including the eventual effect of dynamic loads. The release of eventual dangerous substances and corresponding measures should be given due attention. Finally, a plan should be made for the eventual re-use of the materials, their separation and handling or their removal or storage.

For an eventual re-use it should be also considered that re-used materials might have a shorter service life than new materials. Maintenance might also differ between re-used and new materials.

Dismantlement can be subdivided into two types:

- conventional dismantlement: demolition of a structure without previous demounting of structural members or structural materials;
- controlled dismantlement: systematic selective deconstruction with planned recycling. In this case separation of material fractions is strived after.

The consequence classes are connected to the levels of reliability differentiation. For more details, see ISO 13822 “Bases for design of structures – Assessment of existing structures” and ISO 2394 “General principles on reliability for structures”.

Especially the structural system (framework, stability by stiffening walls or cores, mixed systems) should have a significant influence on the choice of the dismantlement procedure.

Moreover the materials used have to be the subject of inventory. This especially applies to materials which can be health-damaging or can disturb the environment (types, masses, integration in the structure).

The history of the building has to be regarded (previous damage, for instance by fire or earthquake and the method of repair).

Particular consideration should be given to the use of parts of the structure (eventually already in the partially dismantled stage) for supporting demolition tools or equipment for dismantlement. Especially the use of rigorous demolition techniques, for example breakdown by explosion, deserves thorough analysis and experience.

The inspection of a building in the preparation stage of dismantlement can be carried out according to Figure 10.2-1.

10.2 Preparing dismantlement

10.2.1 General

Aspects to be considered in the planning stage of dismantlement are:

- consequence class of the structure;
- technical data;
- site conditions;
- time constraints;
- waste disposal;
- environmental protection;
- labour safety.

10.2.2 Consequence class of the structure

Structures can be subdivided into consequence classes;

- consequence class 1: low consequence for loss of human life; economic, social or environmental consequences small or negligible;
- consequence class 2: moderate consequence for loss of human life; economic, social or environmental consequences considerable;
- consequence class 3: serious consequences for loss of human life, or for economic, social or environmental concerns.

At least for the consequence classes 2 and 3, a detailed plan for dismantlement is required. In this respect, the following aspects should be regarded: function of the structure, size, structural system including stability provisions, age and possible deterioration, previous repair or structural alteration and structural materials used.

10.2.3 Structural analysis for dismantlement

Dismantlement is a transient situation. For the phase of dismantlement, basically the same safety requirements apply as for the construction stage. By means of a structural analysis, unstable situations during dismantlement should be prevented. Intermediate stages have to be considered and may require temporary measures for stabilization.

10.2.4 Investigation of potential contamination

The structure should be investigated with regard to the eventual presence of harmful substances. These substances, which have a negative effect on human health or environment, can be of organic or inorganic origin. Moreover the presence of disturbing substances should be traced. These are components of demolished material which can negatively influence the properties in case of re-use (gypsum, isolation material, foam, wood).

10.2.5 Waste disposal concept

The plan for waste disposal should regard the following aspects:

- volume of waste and quality;
- local spatial conditions;
- possibilities of regional re-use;
- possibilities for regional disposal.

10.2.6 Preparation report

The preparation phase should end with a number of conclusions:

- choice of dismantlement technique;
- measures for coping with harmful substances and disturbing material components before and during dismantlement;
- definition of necessary safety provisions;
- choice for waste disposal;
- definition of necessary capacities for transport, intermediate storage and further processing.

10.3 Health and safety provisions

A plan should be made to deal with the following safety and health aspects:

- Threats for the environment: this aspect regards protection against dynamic effects, flying material fractions and dust. This can be reached by providing various types of barriers and moistening.
- Threats of injury to persons due to: falling parts, splitters and larger scale collapse. This can be avoided by the right sequence of dismantlement and a reliable structural analysis of all intermediate stages of dismantlement. Moreover, reliable scaffolding has to be used and information on the presence and position of electric ducts should be supplied.

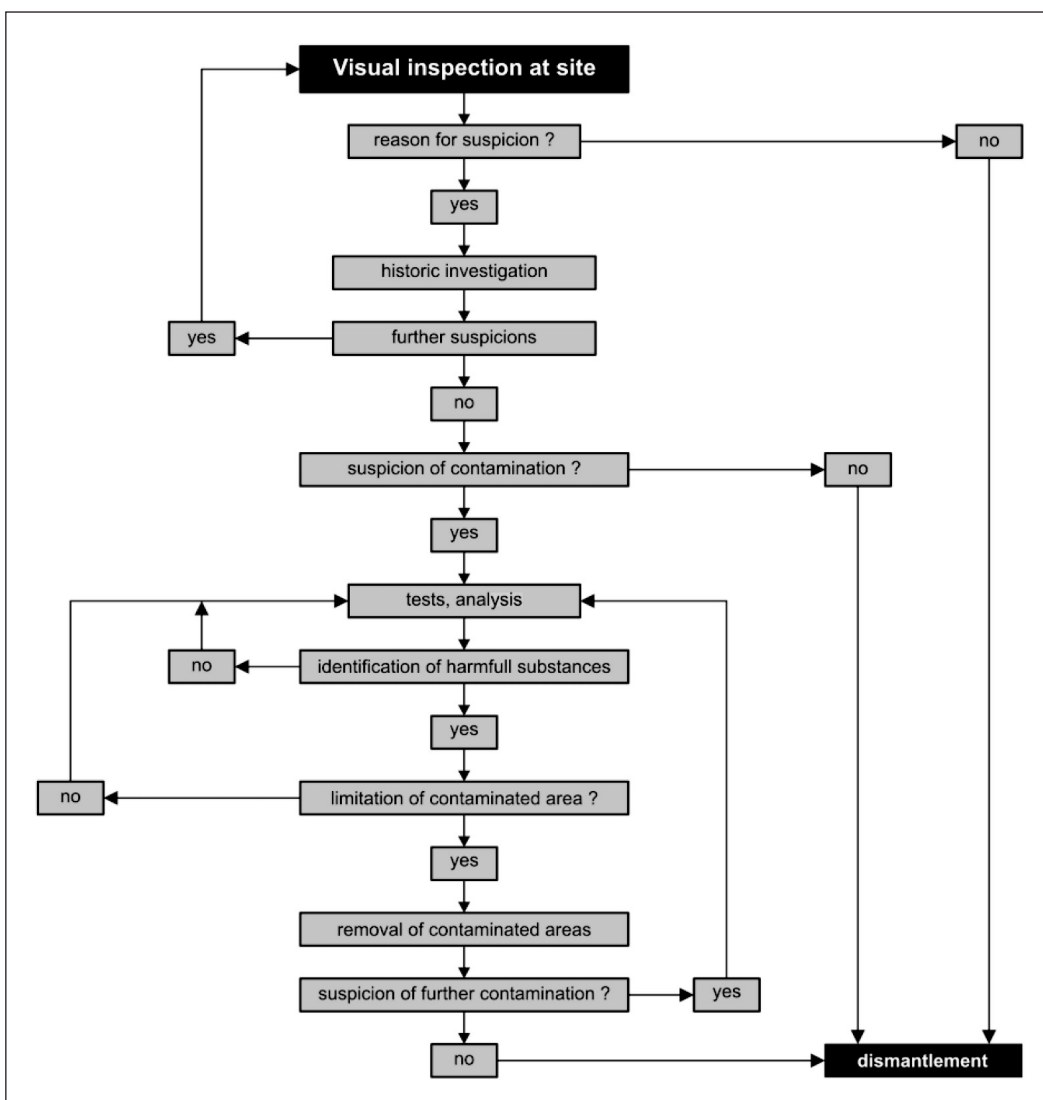


Figure 10.2-1: The inspection of a building in the preparation stage of dismantlement (Toppel, C. O. "Technische und ökonomische Bewertung verschiedener Abbruch-verfahren im Industriebau", Selbstverlag, 2003)

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